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Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026

A COGNITIVE-TAXONOMY HANDBOOK

Bloom's Taxonomy for Rasashastra & Bhaishajya Kalpana

Topic-by-topic domain and level mapping, objective design and usage guide for AyUG-RB — Second Professional BAMS
Ayurvediya Aushadhi Nirmana & Prayoga Vigyana

Curriculum	NCISM, Board of Ayurveda — applicable from the 2021–22 batch, academic year 2023–24 onwards
Coverage	2 papers · 26 theory topics · 10 practical groups · 268 objectives coded in both taxonomies
Framework	Bloom (1956) and the Anderson & Krathwohl revision (2001) carried in parallel, with the psychomotor and affective domains, read against NCISM Table 3 Column C3 and Table 4 Column C4

268	203	33	32
OBJECTIVES CODED	COGNITIVE	PSYCHOMOTOR	AFFECTIVE

Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026 · For departmental use in curriculum delivery, objective writing, assessment design and paper setting.

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Unit 6 Kalpana Nirmana I (Primary & Secondary dosage 11 obj · 10 marks 29

Unit 7 Kalpana Nirmana-II (Method of Preparation of 4 obj · 10 marks 32

Unit 8 Rasa Dravya Parichaya- I 9 obj · 10 marks 35

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PART A

Understanding the taxonomy

Eight short chapters. The first four explain what Bloom's taxonomy is, what its two versions are, what its three domains cover, and where it already sits inside the AyUG-RB curriculum. The next three turn it into practice: choosing verbs, choosing how to teach, choosing how to check. The last explains how to read a Unit in Part B. Read straight through once; after that, use the Units and come back here only when something needs settling.

1. **What Bloom's taxonomy is, and why this book carries two versions**
2. **The three domains** — thinking, doing, and how a student goes about the work
3. **The second axis** — what kind of knowledge an objective works on
4. **Where Bloom already sits in your curriculum** — reading Columns C3 and C4
5. **Verbs** — the working end of the taxonomy
6. **Choosing how to teach**, level by level
7. **Choosing how to check**, level by level
8. **How to use a Unit in Part B**

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PART A · CHAPTER 1

What Bloom's taxonomy is, and why this book carries two versions

In 1956 a committee of American examiners led by Benjamin Bloom published a way of sorting examination questions. They had noticed something every teacher notices sooner or later: two questions on the same topic can be worlds apart in difficulty, not because one covers more ground, but because one asks the student to *do* something harder with what they know. "Name the four types" and "decide which of the four applies here, and say why" sit on the same syllabus line. They are not the same question.

What the committee produced was a ladder of six levels, each asking more of the student than the one below: **Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation**. The ladder is cumulative. You cannot apply what you do not understand, and you cannot understand what you cannot recall. That single idea — that the levels stack — is what makes the taxonomy useful for planning rather than merely for labelling.

THE 2001 REVISION

Forty-five years later a group led by Lorin Anderson and David Krathwohl — Krathwohl had worked on the original — published a revision. Three things changed.

- The levels became **verbs** rather than nouns: *Remember, Understand, Apply, Analyse, Evaluate, Create*. A taxonomy of thinking should be named by what the thinker does.
- **The top two levels swapped.** Bloom put Synthesis at level 5 and Evaluation at level 6. The revision reversed them: Evaluate is level 5, Create is level 6. The reasoning was that making something new is a larger act than judging something that already exists.
- A **second axis** was added, for the kind of knowledge being worked on. Chapter 3 covers it.

> **This is the one thing to carry away from this chapter.** When a colleague says "that question is at the top level", ask which scheme they mean. In the 1956 scheme the top level is Evaluation — judging. In the 2001 scheme it is Create — producing. Those are different questions, different teaching and different marking. Most confusion between two well-informed teachers about Bloom comes down to this.

WHY BOTH, IN THIS BOOK

The AyUG-RB curriculum uses the **1956 scheme**. Its codes are CK, CC, CAP, CAN, CS and CE — Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation. If this book used only the 2001 names, you could not match it against your own syllabus document.

But most faculty development programmes now teach the 2001 scheme, most recent writing on assessment uses it, and younger colleagues will have met it first. If this book used only the 1956 names, it would be teaching a vocabulary its readers are being trained out of.

So every level in this book is named twice, in a fixed order: the 1956 name first, because that is what your curriculum prints, then the 2001 name. *Comprehension / Understand. Evaluation / Evaluate.* Where the two schemes disagree about position — only at the top two levels — the book says so on the page rather than quietly picking one.

PART A · CHAPTER 2

The three domains — thinking, doing, and how a student goes about the work

Bloom's six levels describe thinking only. The original committee knew that thinking is not all of learning, and planned three volumes. The **cognitive** volume — thinking — appeared in 1956. The **affective** volume — attitudes, values, how a student goes about the work — appeared in 1964, edited by Krathwohl. The **psychomotor** volume — physical skill — was never written by that committee at all; others filled the gap later, and the version your curriculum uses comes from that later work.

All three matter in Rasashastra and Bhaishajya Kalpana, and a pharmacy subject probably needs all three more than most. A student who can recite a procedure flawlessly, cannot carry it out, and does not much care whether the material is genuine has met one domain out of three.

THINKING — THE COGNITIVE DOMAIN, SIX LEVELS

Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation, as Chapter 1 set out. This is the domain the theory papers examine, and it accounts for most of the objectives in the curriculum.

DOING — THE PSYCHOMOTOR DOMAIN, FIVE LEVELS AS USED HERE

Physical skill has its own ladder, and it does not map onto the thinking one. Your curriculum uses five levels:

- **Set** — ready to begin. Knows what is needed and can lay it out.
- **Guided response** — performs while being shown or guided.
- **Mechanism** — performs reliably alone, without prompting.
- **Adaptation** — changes the method when conditions change.
- **Origination** — devises a procedure that was not given.

The published psychomotor schemes have more levels than five; these are the five the curriculum uses. Note where the real work sits: getting a student from *Guided response* to *Mechanism* is most of what practical hours are for, and it is slow. Demonstrating well takes one session. Withdrawing your help gradually until the student is reliable alone takes many.

GOING ABOUT IT — THE AFFECTIVE DOMAIN, FIVE LEVELS

- **Receiving** — pays attention when it matters.
- **Responding** — joins in willingly.
- **Valuing** — treats it as worth doing, unprompted.
- **Organisation** — fits it together with other things they value.
- **Characterisation** — it has become how they work.

> **The affective domain needs a different kind of evidence, and this is where it is usually lost.** You cannot see an attitude in a single answer on a single afternoon. Valuing shows up as what a student does when nobody is checking — whether they verify the material, whether they record the reading they actually took. That evidence is gathered by watching over time: conduct in the practical, a short reflection, a checklist applied more than once across a term. An examination hall cannot produce it. So affective objectives have to be planned for in advance, or they go unassessed while appearing to have been assessed.

THE DOMAINS DO NOT LINE UP

An objective at Mechanism — performing reliably alone — may need only level 1 thinking, if the procedure is followed by rote. Another at Set — merely getting ready — may need level 4 thinking, if choosing the right equipment requires judgement. Do not assume the three ladders move together. The Units in Part B show all three separately for exactly this reason.

PART A · CHAPTER 3

The second axis — what kind of knowledge an objective works on

The 2001 revision added a question the 1956 scheme did not ask. The first question is: what thinking must the student do? The second is: *what kind of knowledge is it about?* Cross the two and you get a grid rather than a ladder, and the grid explains something teachers meet constantly — that two objectives at the same level can need completely different teaching.

THE FOUR KINDS

- **Factual** — terminology, names, lists, specific details. The vocabulary of the subject. Learned by rehearsal and self-testing, and by very little else.
- **Conceptual** — classifications, principles, relationships, the reason a thing is so. Learned by explaining it to somebody, which is why a student who has only read it can rarely produce it.
- **Procedural** — methods, techniques, criteria for deciding when to use which. Learned by carrying it out. Reading a procedure is not learning it.
- **Metacognitive** — knowing about your own learning: what you find hard, which approach works for you, when to change tack. Learned by being made to look back at how you worked.

WHY THIS CHANGES THE TEACHING

Take two objectives, both at level 1. One asks a student to recall the names in a group; the other asks them to recall the steps of a procedure in order. Same level, and the first is Factual while the second is Procedural. Rehearsal will fix the first. The second will not hold until the student has done it, because a remembered list of steps and a performable procedure are not the same possession.

This is also why "they know it but they cannot do it" is such a common complaint in practical subjects. Often the objective was Procedural and the teaching was pitched as though it were Factual.

> **How the Units read this.** Each objective in Part B is labelled with its kind of knowledge, worked out from the words of the objective itself — its verb, and what the verb acts on. Where an objective genuinely works on two kinds at once, both are named: "Factual + Procedural" means part of it is learned by rehearsal and part by doing. Where the wording supports a reading but does not settle it, the label carries a small circle (°), so you can see that a judgement was made and overrule it if you disagree. Nothing is guessed silently.

ONE THING WORTH NOTICING

The Metacognitive row is empty across the whole of AyUG-RB, as is the Create level. That is simply what the curriculum asks for at this stage, and no individual teacher needs to fix it. It is recorded here because it is the kind of thing worth having in front of you when objectives are next revised, and because a teacher who wants to build reflective habit in students should know they are going beyond what the syllabus requires rather than filling a gap in it.

PART A · CHAPTER 4

Where Bloom already sits in your curriculum

Bloom is not something this book adds to AyUG-RB. It is already there, and has been since the curriculum was published. Most teachers have simply never had it pointed out.

COLUMN C3 AND COLUMN C4

Open the curriculum at Table 3, the theory objectives. The third column is headed *Domain/sub*. Every objective carries a code there: CK, CC, CAP, CAN, CS, CE for thinking; PSY-SET, PSY-GUD, PSY-MEC, PSY-ADT, PSY-ORG for doing; AFT-REC, AFT-RES, AFT-VAL, AFT-SET, AFT-CHR for how the student goes about it. Table 4, the practical objectives, carries the same codes in Column C4.

Those sixteen codes are Bloom, the psychomotor scheme, and Krathwohl's affective scheme. The full key is in Appendix C2. What matters here is that the curriculum has already decided, for each of its 268 objectives, what kind of learning it is asking for — and that this decision is available to you before you plan a single session.

WHAT THE OTHER COLUMNS GIVE YOU

- **Column D3 / D4 — Must know, Desirable to know, Nice to know.** An independent judgement about priority. It is not part of Bloom and does not move with level: an objective can be Must know at level 1 and Nice to know at level 6, or the reverse.
- **Column F3 / F4 — teaching and learning method.** The curriculum names methods for each objective. Chapter 6 sets these against the levels.
- **Column G3 / G4 — assessment.** The tools named for each objective. Chapter 7 sets these against the levels.
- **Column I3 / I4 — term.** When in the year the objective falls.

> **A word about Column E3, which this book does not use.** Column E3 is also headed *Level*, but it carries a different scheme — Knows, Knows how, Shows how, Does — which is Miller's pyramid, not Bloom. It answers a different question: not what kind of thinking is needed, but how authentic the evidence has to be. It is worth knowing well, and it has its own companion volume. In *this* book, "level" always means the Bloom level given in Column C3 or C4. Where you see "level 5" here, it never means "Shows how".

READING A CODE IN PRACTICE

Suppose an objective carries **CC**, **MK**, and a teaching method of group discussion. Read that as: this is level 2 thinking, Comprehension in the 1956 scheme and Understand in the 2001 one; it is Must know, so no student may leave it behind; and the curriculum expects students to handle the material themselves rather than receive it again. That is already most of a lesson plan, and it came out of three columns.

PART A · CHAPTER 5

Verbs — the working end of the taxonomy

A level is an idea. A verb is what a student actually meets. This is why the 2001 revision renamed the levels as verbs, and it is the most practical thing in the whole taxonomy: **an objective is taught and examined at the level of its verb**, whatever else is written beside it.

Consider two objectives on identical content. *Enlist the types. Justify the choice between the types.* The first is level 1: a student meets it by rehearsal, and answers it in a line. The second is level 6: a student meets it by making a judgement and defending it, and answers it in a page. A student who prepared for the first and met the second in the examination has not been unlucky. They read the objective and it did not tell them what they needed to know.

A WORKING LIST

- **Level 1, Knowledge / Remember** — define, enlist, list, name, recall, recite, state, identify, recognise, label.
- **Level 2, Comprehension / Understand** — describe, explain, discuss, illustrate, classify, summarise, differentiate, distinguish, elaborate, outline.
- **Level 3, Application / Apply** — apply, demonstrate, use, perform, prepare, calculate, solve, employ, conduct, measure.
- **Level 4, Analysis / Analyse** — analyse, compare, contrast, examine, determine, categorise, correlate, differentiate between, select, review.
- **Level 5 in 1956, Synthesis** — design, formulate, develop, construct, compose, devise, plan, produce. (In the 2001 scheme these belong to level 6, Create.)
- **Level 6 in 1956, Evaluation** — evaluate, justify, appraise, critique, assess, judge, defend, validate, interpret. (In the 2001 scheme, level 5.)

The affective domain has its own verbs, and they are not thinking verbs: *appreciate, value, accept, comply, respond, defend, advocate, adhere, display, practise.* If an affective objective opens with *explain*, it will be taught and marked as level 2 thinking, whatever its code says, because *explain* is what the student will be asked to do.

> **Where a verb and a code point in different directions, follow the verb — and know that you are choosing.** The verb is what appears on the question paper, so in practice it wins by default. That is fine, provided it is a decision rather than an accident. If the verb asks for less than the code, you have room to teach higher than the syllabus requires. If it asks for more, plan for the verb, or your students will meet a question they were never prepared for. The Units in Part B tell you, objective by objective, which case you are in.

WHEN YOU WRITE YOUR OWN OBJECTIVES

Pick the verb last and pick it deliberately. Decide the level first, then choose a verb from that level, then check that the verb can actually be assessed by the tool you have in mind. An objective that says *appreciate* and is assessed by a written paper has a mismatch built into it before the term begins. One verb, one level, one objective: where you find yourself writing “define, recite and interpret” in a single line, you have three objectives at three levels, and they will need three different questions.

PART A · CHAPTER 6

Choosing how to teach, level by level

The commonest planning error in a crowded syllabus is to choose the method by the hour rather than by the level — a lecture because there is a lecture slot. Lectures build level 1 well and level 2 poorly, and above level 2 they build almost nothing on their own. What follows is what each level needs.

LEVEL 1 — KNOWLEDGE / REMEMBER

Exposure, then rehearsal, then retrieval practice. A lecture puts the facts in front of students; only repeated recall makes them stay. The strongest tool is the cheapest: make students retrieve without looking, then check. Quizzes, word puzzles and game-based methods — all named in this curriculum — are retrieval practice in a form students will actually do. Spread it across weeks; a single long session before the examination produces recall that lasts until the examination.

LEVEL 2 — COMPREHENSION / UNDERSTAND

Students must handle the material rather than receive it again. Have them sort it, compare it, put it in an order of their own, and explain it to a classmate. Discussion and flipped classroom do this work. Repeating the lecture more slowly does not: the difficulty is not that they did not hear it.

LEVEL 3 — APPLICATION / APPLY

Work examples through together, then hand over a situation the student has not seen. The hand-over is the teaching. Without it students learn the example rather than the method, and it is easy to mistake one for the other, because a class that can complete the worked example looks exactly like a class that has understood.

LEVEL 4 — ANALYSIS / ANALYSE

Give students two things and ask how they differ and why. Analysis needs something to set against something else; a single object cannot be analysed. Where the curriculum already distinguishes sub-categories, the material for this is sitting there.

LEVEL 5 IN 1956, SYNTHESIS — CREATE IN 2001

Not asked for anywhere in this curriculum. If you choose to reach for it, it means asking students to assemble something that was not handed to them, and marking it against criteria set out in advance.

LEVEL 6 — EVALUATION / EVALUATE

Give students something to judge, and tell them what to judge it against. Then require the criteria to be stated, not just the conclusion. Small-group debate works well because a student has to defend a position and hear it challenged, which is the whole of the skill. This is the level most often taught as though it were level 2 — explaining the accepted view rather than reaching a view.

> **The practical hours belong mostly to one level.** In the practical Units, the bulk of the time should go to moving students from *Guided response* to *Mechanism* — from performing with your help to performing reliably without it. That transition is invisible if you only watch the best student in the group, because they arrive at it early. Watch the middle of the group to know where the class actually is.

PART A · CHAPTER 7

Choosing how to check, level by level

A check is fair when it can only be passed by a student who has reached the level, and can be passed by any student who has. Most unfair checks fail the first half: they can be passed by recall when the objective asked for judgement.

WHAT EACH LEVEL CAN BE CHECKED WITH

- **Level 1** — multiple choice, and short answers that ask for a list, a name or a definition. Many short questions, spread out. Long-answer marks spent here are wasted, because a long answer cannot distinguish a student who knows the list from one who knows it better.
- **Level 2** — short answers whose stem requires the material rearranged: sorted, compared, put in the student's own sequence. **Test your stem this way:** if it can be answered word-for-word from the notes, it has slipped to level 1 whatever you intended.
- **Level 3** — a short or structured long answer built on a case the student has not seen. If the case is familiar, the question measures memory.
- **Level 4** — structured questions that ask for the parts and for how the parts relate.
- **Level 6** — long answers. Give the marks for the reasoning and the criteria, not for the facts repeated inside the answer. A student who reaches a sound conclusion for stated reasons has met the level; one who reaches it by recalling what the book concluded has not.

DOING OBJECTIVES CAN ONLY BE CHECKED BY WATCHING

This is worth stating plainly, because the curriculum's prescribed tool for practical objectives is very often a viva. A viva is a good tool. It shows whether a student knows a procedure and can explain it. It cannot show whether they can carry it out, and a fluent student who has never handled the material will pass it comfortably.

Where an objective is about doing, the evidence has to be seen: a checklist you tick while the student works, an OSPE station, or an observed performance. This need not be elaborate. A checklist of the steps you will look for, written before the session, is enough — and it is the only record that shows the procedure was actually performed.

ATTITUDE OBJECTIVES NEED EVIDENCE OVER TIME

Conduct across the term, a short written reflection, a checklist applied on more than one occasion. Decide in advance when you will collect it. An affective objective that reaches the end of term uncollected has not been assessed, however carefully it was written.

> **Three questions before you set any paper.** First: for each question, what level does it actually reach — not what level was intended? Second: could a student pass it by recall alone? Third: does the balance of marks match the balance of the syllabus, or have the easy-to-mark levels quietly taken the marks? Appendix C6 gives the level and priority profile of every Unit, which is the fastest way to answer the third.

PART A · CHAPTER 8

How to use a Unit in Part B

Part B has thirty-six Units: twenty-six theory Units following the topics of Papers 1 and 2, and ten practical Units following Table 4. Each begins on its own page, and each is laid out the same way, so that once you have read one you can use any of them.

WHAT IS ON THE PAGE

1. **The heading strip** — hours, term, marks, how many objectives, and the highest level the Unit asks for. Read the last of these first. It tells you whether this is a Unit that stops at recall or one that reaches judgement, and that single fact shapes everything else.
2. **Four counters** — how many objectives fall in each domain, how many are Must know, and which kind of knowledge dominates.
3. **What this topic asks students to do** — the levels as a picture, in both schemes side by side. Empty levels are shown as empty rather than left out, because knowing what a topic does *not* ask for is as useful as knowing what it does.
4. **What kind of knowledge is involved** — the four kinds, with how each is learned.
5. **The objectives, grouped by level** — the curriculum's own objectives, sorted into their levels so that a session can be planned around one level at a time. The last column tells you how to pitch each one.
6. **Teaching and checking, level by level** — what the teaching has to accomplish at each level this Unit uses, and what a fair check looks like.
7. **Two panels** — one for the teacher, one to give to students.

THREE WAYS TO USE IT

Planning a session. Go to the grouped objectives and take one level at a time. Teach upward: the lower levels are the ground the higher ones stand on. Then check the method against Chapter 6, and the check you have in mind against Chapter 7.

Setting a paper. Use the level picture to decide where the marks should go, and the grouped objectives to see which ones can carry a long question. A Unit whose highest level is Comprehension cannot honestly carry a question that asks for judgement, however many marks it is worth in the syllabus.

Giving students their bearings. The student panel is written to be read by students, not about them. The single most useful thing in it is the first line: read the verb in every objective. Students who do that stop preparing for the wrong question.

> **What this book does not do.** It does not rewrite the curriculum, and it does not tell you what to teach in Rasashastra and Bhaishajya Kalpana — that is your subject, and every objective in Part B appears in the curriculum's own words. What the book adds is the taxonomy layer: what kind of learning each objective asks for, and what follows for teaching and checking. Where the wording of the curriculum and its own code point in different directions, the Unit shows you both and leaves the decision with you.

PART B

The Units

Thirty-six Units, each on its own pages: twenty-six following the theory topics of Papers 1 and 2, and ten following the practical topics of Table 4. Every objective appears in the curriculum's own words, sorted into its level so that a session can be planned around one level at a time. Chapter 8 explains the layout; after that the Units are meant to be opened at whichever one you are about to teach.

1. **Paper 1** — Ayurvediya Aushadhi Nirmana Vigyana, Units 1 to 14
2. **Paper 2** — Ayurvediya Aushadhi Prayoga Vigyana, Units 1 to 12
3. **Practical** — NCISM Table 4, Units 1 to 10

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PAPER 1 · UNIT 1

Chronological development of Ayurvediya Aushadhi Nirmana

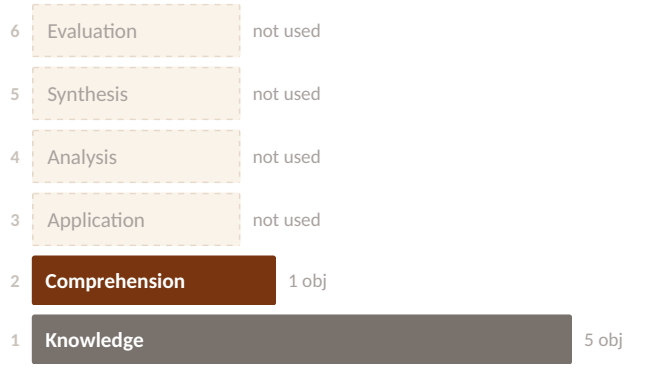
Hours: 2 lecture · 1 non-lecture | Term 1 | Theory marks: 5 | **7 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
6	**1**	**3**	**Conceptual**
levels 1–6	Krathwohl domain	3 desirable · 1 nice to know	kind of knowledge

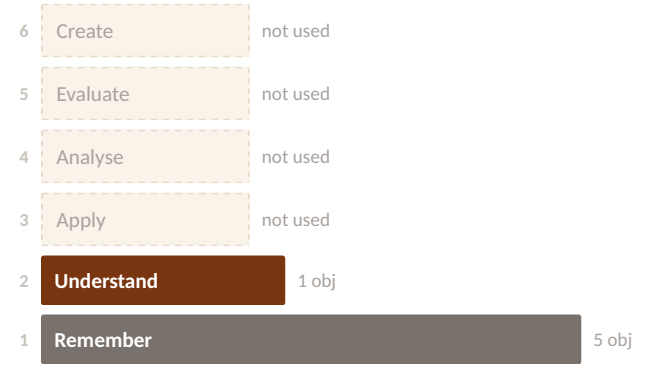
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	**5**	**1**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 5 OBJECTIVES				
Explain historical evolution of Ayurvediya aushadhi nirman and Rasashastra.	*explain*	Conceptual ◦	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Describe about contribution of Nagarjuna Acharya to Rasashastra.	*describe*	Conceptual ◦	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Enlist important classical texts of Rasashastra and describe their unique features in short.	*enlist*	Conceptual ◦	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Describe Recent development in Ayurvedic Pharmaceuticals viz new dosage forms, pharmaceutical modification techniques.	*describe*	Procedural °	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Define Rasa and Rasayana and describe difference between Rasa and Rasayana.	*define*	Factual °	Desirable	Teach at this level.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Describe structure of Pharmacy and enlist formulations prepared in pharmacy, after visiting the unit of teaching pharmacy of own campus.	*describe*	Conceptual °	Must know	Teach at this level.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Justify design of ancient Rasashala.	*justify*	Conceptual	Nice to know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	5	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	1	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 6 of these 7 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 1 · UNIT 2

Paribhasha (Terminology)

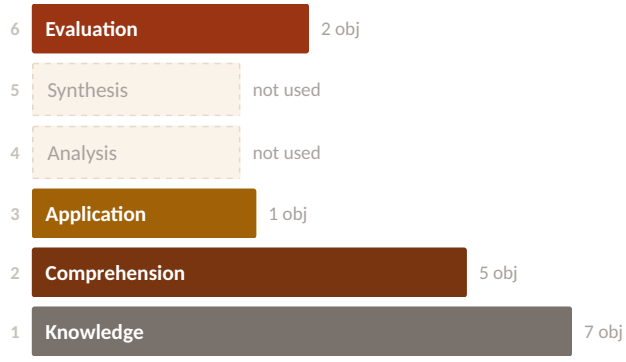
Hours: 8 lecture · 4 non-lecture | Term 1 | Theory marks: 10 | **19 objectives** | Highest level asked for: **Evaluation / Evaluate**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
15	**4**	**12**	**Factual**
levels 1–6	Krathwohl domain	5 desirable · 2 nice to know	kind of knowledge

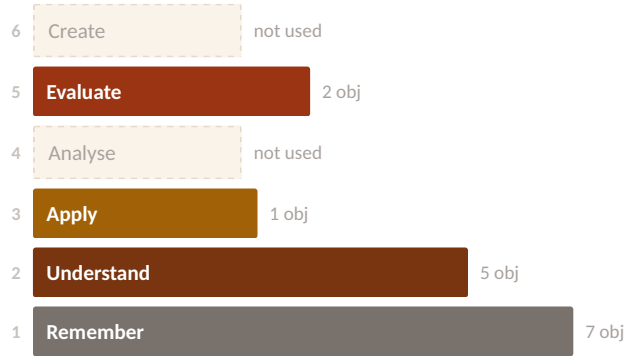
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
11	**8**	**3**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 7 OBJECTIVES				
Explain the term Paribhasha and its importance in Ayurvediya Aushadhi Nirmana.	*explain*	Factual + Conceptual °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Enlist and discuss Sandigdha, Krutrima, Pratinidhi and Anupalabdha dravya.	*enlist*	Factual °	Desirable	Teach at this level.
Explain the term Marana and describe its types with examples.	*explain*	Factual + Conceptual °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Enlist all relevant prakriya paribhasha of Amrutikarana Lohitikarana, Sattvapatana, druti and discuss with examples.	*enlist*	Factual °	Desirable	Teach at this level.
Identify names of Parada Bandhas.	*identify*	Factual	Nice to know	Teach at this level.
Describe Grahyagrahya parameters used for selection of Rasadravyas.	*describe*	Procedural °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Explain importance of Mana-paribhasha and classical types of Mana. Recall Charts of Mana.	*explain*	Conceptual	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 5 OBJECTIVES				
Describe the terms Aushadha, Bheshaja, Kalpana, Kashaya Kashaya yoni, Samskara- based on their word derivations.	*describe*	Factual °	Must know	Teach at this level.
Enlist sequentially - names of all drugs classified in the varga(group). Recite shlokas of Maharasa, Uprasa, Sadharana Rasa Varga from Rasaratnasamuchchaya.	*enlist*	Factual	Must know	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Recite sequentially names of Parada Ashta samskara. Explain the terms Jarana Murchchhana and cite types with examples. Compare Jarana and Murchchhana.	*recite*	Factual	Must know	Teach at **level 1**. The verb *recite* is what the exam will ask for.
Illustrate all Bhasma pariksha as per classical description.	*illustrate*	Conceptual °	Must know	Teach at this level.
Recognise and discuss Dhanvantari Bhaga and Rudra bhaga.	*recognise*	Factual °	Nice to know	Teach at **level 1**. The verb *recognise* is what the exam will ask for.
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Describe the definition of the term Shodhana. Explain with examples different techniques used for the procedures of Shodhana.	*describe*	Factual + Procedural °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
LEVEL 6 · EVALUATION / EVALUATE — 2 OBJECTIVES				
Define the word Siddhilakshani. Recite examples of classical siddhilakshani. Interpret its rationality.	*define*	Factual °	Must know	Teach at **level 1**. The verb *define* is what the exam will ask for.
Categorize parameters of Drug Standardization and develop a checklist for assessment of quality of rasadravyas.	*categorize*	Procedural	Desirable	Teach at **level 4**. The verb *categorize* is what the exam will ask for.
AFFECTIVE DOMAIN — 4 OBJECTIVES				
Explain Value of selection of genuine raw material.	*explain*	Conceptual	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.
Explain importance of ethical practices for drug processing(Shodhan, Marana).	*explain*	Conceptual	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.
Discuss and justify importance of Bhasma Pariksha.	*discuss*	Conceptual	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.
Explain value of keen and accurate application of weights and measures in Ayurvediya Aushadhi nirmana.	*explain*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	7	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	5	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	2	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 12 of these 19 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Factual* — recall it until it sticks; *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 6, repeating it does not mean you know it.

PAPER 1 · UNIT 3

Adharbhuta Siddhanta (Application of fundamental principles)

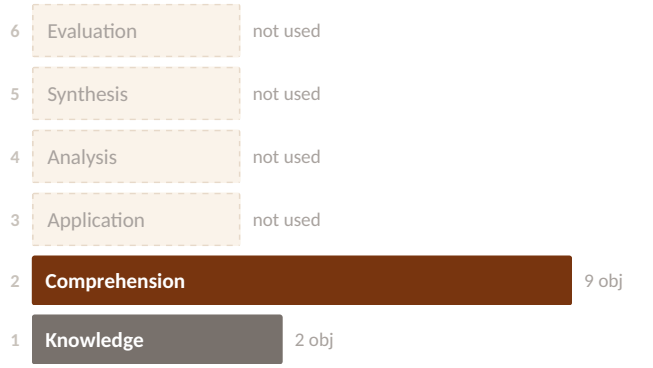
Hours: 4 lecture · 2 non-lecture | Term 1 | Theory marks: 5 | **11 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
11	**0**	**9**	**Conceptual**
levels 1–6	Krathwohl domain	2 desirable · 0 nice to know	kind of knowledge

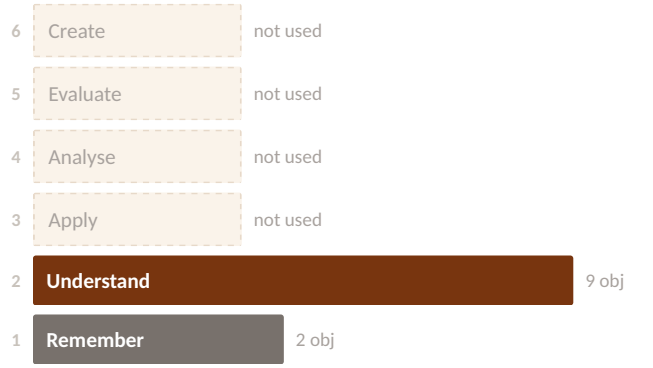
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	**10**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Explain Saveeryata Avadhi(Shelf life) of Ayurvedic dosage forms.	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Explain importance of Kala (Time) Samskara in Ayurvediya Aushadhi Nirmana. Elaborate Aushadhi sevana kala mentioned in Sharangdhara samhita. Describe chrono- Therapeutics.	*explain*	Conceptual	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 9 OBJECTIVES				
Elaborate fundamental principles of Ayurvediya Aushadhi Nirmana alongwith their classical references and discuss their application with classical examples of various kalpas.	*elaborate*	Conceptual	Must know	Teach at this level.
Explain Dravya Samgraha vidhi. Explain types of Bhumi desha and types of drugs to be collected from particular place & places from where Dravya should not be collected.	*explain*	Conceptual	Must know	Teach at this level.
Describe time of Dravya collection and explain rationality behind.	*describe*	Conceptual °	Must know	Teach at this level.
Enlist parts of Sthavara & Jangama Dravyas used for Ayurvediya Aushadhi nirmana.	*enlist*	Factual °	Must know	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Define Samskara. Discuss role of Samskara in Ayurvediya Aushadhi Nirmana alongwith various examples mhita.	*define*	Conceptual °	Must know	Teach at **level 1**. The verb *define* is what the exam will ask for.
Describe Ardra - Shushka Dravya & Anukta – Visheshokta Dravya collection principles.	*describe*	Conceptual	Must know	Teach at this level.
Describe importance of kalpa sevan matra. Explain factors considered for deciding dosage of any drug (Ayurvedic as well as modern medicine principles). Describe Posology.	*describe*	Conceptual	Must know	Teach at this level.
Discuss Yogika Dravya Sidhdhanta(Drug combination).	*discuss*	Conceptual °	Must know	Teach at this level.
Justify Aushadhi kalpa namakarana siddhanta with examples.	*justify*	Conceptual °	Desirable	Teach at **level 6**. The verb *justify* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	2	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	9	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 11 of these 11 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
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FOR THE STUDENT

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- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 1 · UNIT 4

Yantropakaranani- I (Equipments and machineries)

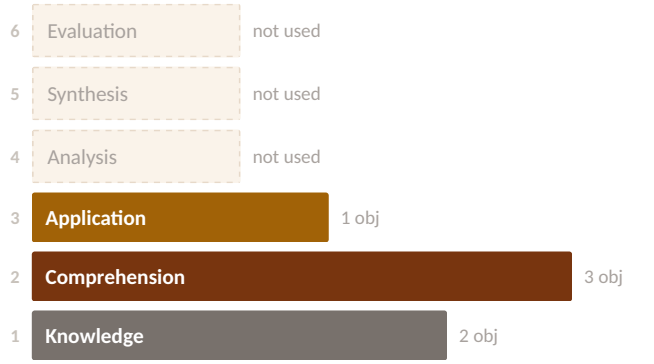
Hours: 6 lecture · 4 non-lecture | Term 1 | Theory marks: 5 | **6 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
6	**0**	**1**	**Conceptual**
levels 1–6	Krathwohl domain	5 desirable · 0 nice to know	kind of knowledge

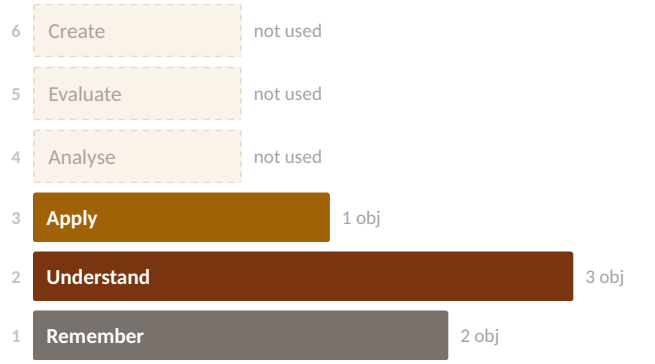
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BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	**3**	**2**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Enlist categorical information about the following Yantras in the charts- Ulukhal Yantra, Patan Yantra, Jarana Yantra, Patala Yantra and Swedana Yantra.	*enlist*	Factual	Desirable	Teach at this level.
Enlist Ayurvediya aushadhi kalpana and equipments/yantras/ machines used for preparation of each kalpana.	*enlist*	Procedural °	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 3 OBJECTIVES				
Choose and record contemporary machines used in Ayurvedic drug preparation.	*choose*	Procedural °	Desirable	Teach at **level 4**. The verb *choose* is what the exam will ask for.
Discuss the pharmaceutical use of Distillation apparatus, Ball Mill, Pulveriser, End Runner, Edge Runner, Tablet compression machine, Capsule filling machine, Pouch filling machine, Liquid filling machine in Ayurvediya Aushadhi Nirmana.	*discuss*	Conceptual °	Desirable	Teach at this level.
Interpret the mechanism and effect of Yantras / machines on the Physical and Chemical properties of the drug material.	*interpret*	Conceptual °	Desirable	Teach at **level 6**. The verb *interpret* is what the exam will ask for.
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Describe the principles behind construction and working of the classical Yantras used for Ayurvediya Aushadhi nirmana.	*describe*	Conceptual	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	2	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	3	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 5 of these 6 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 1 · UNIT 5

Yantropakaranani -II (Equipments, fuel and Heating Devices)

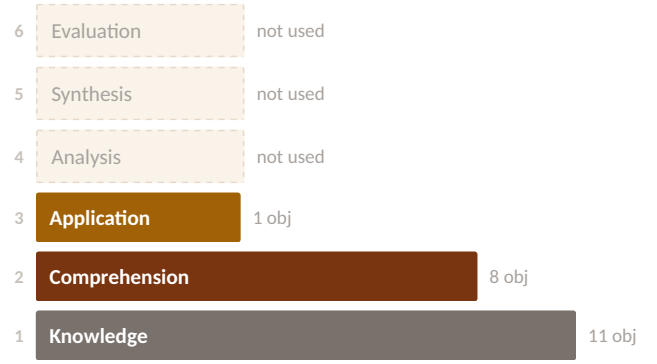
Hours: 5 lecture · 4 non-lecture | Term 1 | Theory marks: 5 | **21 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
20	**1**	**3**	**Conceptual**
levels 1–6	Krathwohl domain	14 desirable · 4 nice to know	kind of knowledge

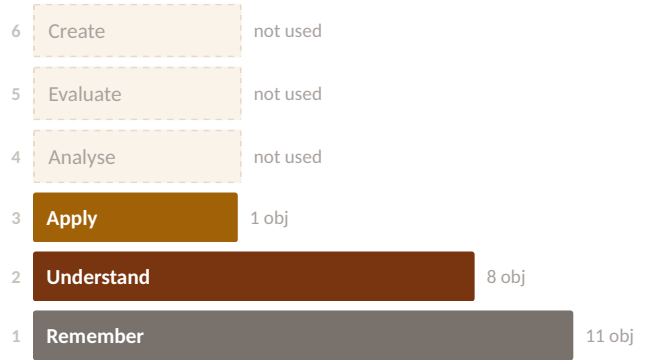
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
9	**11**	**3**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 11 OBJECTIVES				
Describe the term Yantra and enlist yantras described in classics useful for aushadhi nirmana.	*describe*	Factual ◦	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Define the term Puta and recite its classical explanation.	*define*	Factual	Must know	Teach at this level.
Produce categorical information(Size of Pit, Number of cowdunkcakes, use etc.) about following Putas viz. Mahaputa Gajaputa, Varahputa, Kukkutaputa, Kapotputa, Lavakputa Kumbhaputa, Bhandaputa, Valukaputa and Bhudharputa.	*produce*	Procedural ◦	Must know	Teach at **level 5**. The verb *produce* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Compile names and significance of temperature measuring devices with reference to Puta.	*compile*	Factual + Conceptual °	Desirable	Teach at **level 4**. The verb *compile* is what the exam will ask for.
Enlist various devices used for heating during Ayurvediya Aushadhi nirmana.	*enlist*	Factual °	Desirable	Teach at this level.
Enlist constituents needed for Samanya Musha Nirmana.	*enlist*	Factual °	Desirable	Teach at this level.
Define the term Musha and describe uses of various types of Musha.	*define*	Factual + Conceptual °	Desirable	Teach at this level.
Enlist and discuss the material used for Sandhi Bandhan (while using Sharava and Kachakupi).	*enlist*	Factual °	Nice to know	Teach at this level.
Identify and record the types of Crucibles.	*identify*	Conceptual °	Nice to know	Teach at this level.
Describe types of Koshthi and their uses.	*describe*	Conceptual	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Enlist various fuels used for heating for estimation of their heat value.	*enlist*	Conceptual °	Nice to know	Teach at this level.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 8 OBJECTIVES				
Recognise the principles behind construction and working of the classical Putas.	*recognise*	Conceptual °	Must know	Teach at **level 1**. The verb *recognise* is what the exam will ask for.
Identify and enlist contemporary devices used in the preparation of Bhasma. T.	*identify*	Procedural °	Desirable	Teach at **level 1**. The verb *identify* is what the exam will ask for.
Review the temperature pattern of various Putas by referring research articles.	*review*	Conceptual °	Desirable	Teach at **level 4**. The verb *review* is what the exam will ask for.
Enlist the specific Puta used for a particular Bhasma Nirmana.	*enlist*	Factual °	Desirable	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Recall Pakaj Utpatti Siddhant and interpret it for Agni Sannikarsha Sanskara.	*recall*	Factual °	Desirable	Teach at **level 1**. The verb *recall* is what the exam will ask for.
Explain the Pharmaceutical use of the Hot plate, heating mantle induction stove, hot air oven, muffle furnace (horizontal and vertical type).	*explain*	Conceptual °	Nice to know	Teach at this level.
Describe the Current trends in heat transfer device e.g. steam jacketed heating device, programmed muffle furnace programmed furnace for Parpati / Pottali preparation.	*describe*	Procedural °	Desirable	Teach at this level.
Assess and interpret the effect of fuel in quantum of heat given (time and temperature).	*assess*	Conceptual °	Desirable	Teach at **level 6**. The verb *assess* is what the exam will ask for.
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Interpret the effect of heat transformation in the material subjected to heating through Musha.	*interpret*	Conceptual °	Desirable	Teach at **level 6**. The verb *interpret* is what the exam will ask for.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Analyse and appraise use of proper heating device and fuels used for Ayurvediya Aushadhi Nirmana.	*analyse*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	11	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	8	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 19 of these 21 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 1 · UNIT 6

Kalpana Nirmana I (Primary & Secondary dosage forms)

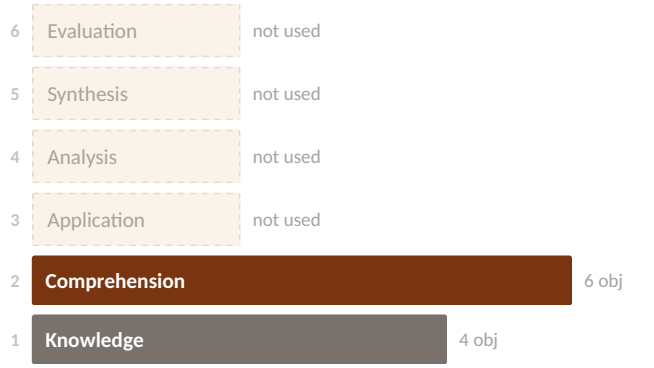
Hours: 6 lecture · 4 non-lecture | Term 1 | Theory marks: 10 | **11 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
10	**1**	**7**	**Conceptual**
levels 1–6	Krathwohl domain	4 desirable · 0 nice to know	kind of knowledge

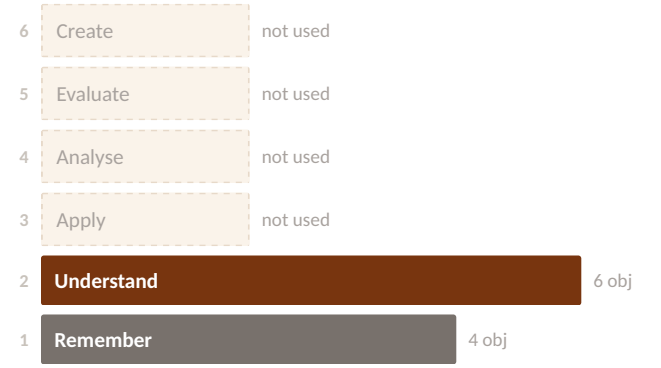
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
3	**7**	**3**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 4 OBJECTIVES				
Define with synonyms and classify different varieties of the dosage form in Ayurvediya Aushadhi nirmana.	*define*	Factual ◦	Must know	Teach at this level.
Describe method of preparation of dosage form along with the principle involved.	*describe*	Conceptual + Procedural ◦	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Describe meaning of the term Upakalpana. Enlist panchavidha kashaya kalpana and their respective Upakalpana.	*describe*	Factual ◦	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Explain various kalpas belonging to Various dosage forms - Ingredients, proportion, matra, sevan vidhi if any, therapeutic.	*explain*	Conceptual °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 6 OBJECTIVES				
1. Explain Kashaya kalpana 2. Enlist the dosage form come under Panchavidha Kashaya Kalpana and their Upakalpana 3. Justify Primary, secondary and tertiary dosage form under Panchavidha kashaya Kalpana and their Upakalpana.	*explain*	Conceptual °	Must know	Teach at this level.
Describe methods of preparation of the dosage form along with the principles of extraction, concentration and dilution etc.	*describe*	Conceptual + Procedural °	Must know	Teach at this level.
Explain in details about Standard Operating Procedure (SOP) of each kalpana.	*explain*	Procedural	Must know	Teach at this level.
Enlist details of the applications/administration (samanya Matra Anupana or sahapana, indications and contraindications) of the dosage form with various examples.	*enlist*	Factual °	Must know	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Assess and discuss modern day development and market trend of the dosage form.	*assess*	Conceptual °	Desirable	Teach at **level 6**. The verb *assess* is what the exam will ask for.
Determine the advantages and disadvantages of the dosage form.	*determine*	Conceptual °	Desirable	Teach at **level 4**. The verb *determine* is what the exam will ask for.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Critically evaluate rationality behind different proportion of water used in various formulations.	*critically*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	4	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	6	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 10 of these 11 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 1 · UNIT 7

Kalpana Nirmana-II (Method of Preparation of different dosage forms& Dietary Supplements))

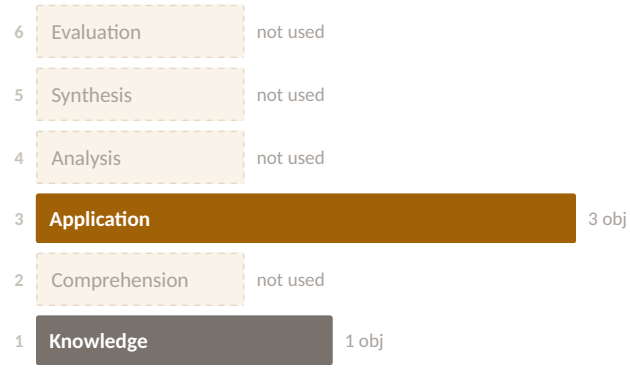
Hours: 5 lecture · 4 non-lecture | Term 1 | Theory marks: 10 | **4 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
4	**0**	**4**	**Procedural**
levels 1–6	Krathwohl domain	0 desirable · 0 nice to know	kind of knowledge

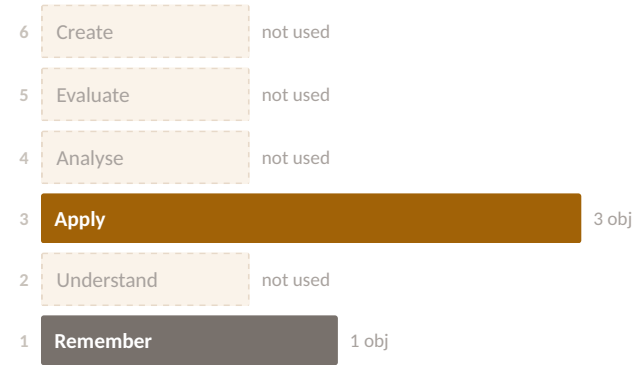
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**3**	**4**	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 1 OBJECTIVES				
Explain the basic principles involved, processing techniques, quality control parameters used/ involved in the preparation of Sneha Kalpana, Kalpana with examples instruments used in small and Large scale production, Concept of Avartana, Research updates on Snehakalpana, Market survey.	*explain*	Conceptual + Procedural °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Explain the basic principles involved, processing techniques, quality control parameters used/ involved in the preparation of Avaleha, Kalpana with examples, instruments used in small and Large scale production, Research updates.	*explain*	Procedural	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Describe the Significance of sandhana kalpana, classification Difference between Madya and Shukta Kalpana, general method of preparation, essential ingredients, Anukta Mana, Sandhana Vidhi, observations, Burning candle test, Lime water Test difference between Asava and Arishta, essential knowledge of sale and clinical practice, Research updates.	*describe*	Conceptual + Procedural °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Explain Definition, significance of Pathya, types, general method of preparation Manda, Peya, Yavagu, Vilepi, Anna or Odana Kalpana, Krushara, Yusha, Takra, Khada, Kambalika Raga, Shadava, Related Research updates, Market survey of Dietary Supplements and Nutraceuticals.	*explain*	Conceptual + Procedural °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	1	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	3	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 1 of these 4 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Procedural* — do it, do not only read it; *Conceptual* — explain it out loud without notes.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 1 · UNIT 8

Rasa Dravya Parichaya- I

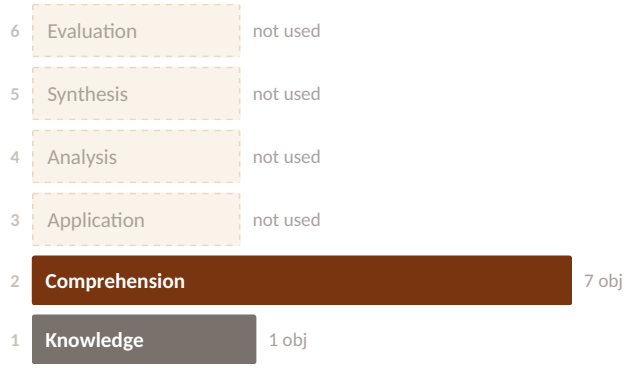
Hours: 12 lecture · 4 non-lecture | Term 2 | Theory marks: 10 | **9 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
8	**1**	**7**	**Conceptual**
levels 1–6	Krathwohl domain	2 desirable · 0 nice to know	kind of knowledge

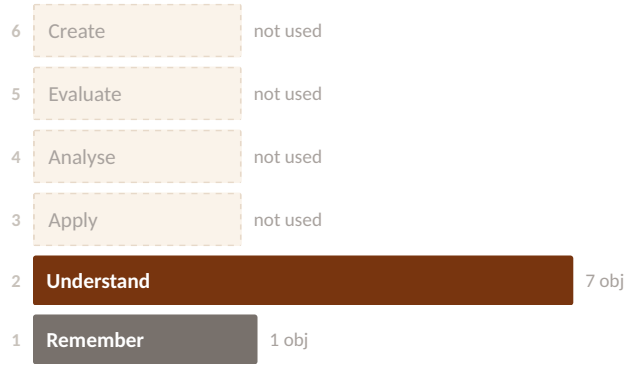
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**5**	**4**	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 1 OBJECTIVES				
Enlist Kalpas prepared from these Rasdravyas and therapeutic importance of that Rasadravya.	*enlist*	Conceptual ◦	Desirable	Teach at this level.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 7 OBJECTIVES				
Enlist synonyms of Rasadravya and explain significance of that T.	*enlist*	Conceptual ◦	Must know	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Explain classical & Mineralogical identification and Sources of Rasadravyas.	*explain*	Conceptual ◦	Must know	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Discriminate types, Grahya-Agrahyata and Dosha of Rasadravya.	*discriminate*	Conceptual	Must know	Teach at **level 4**. The verb *discriminate* is what the exam will ask for.
Explain Shodhan and Marana and other processing techniques of the Rasa- Dravya.	*explain*	Procedural °	Must know	Teach at this level.
Discuss probable physical, chemical changes occurring during process & after Shodhana and Marana of Parada, Haratala Tamra and Shankha.	*discuss*	Procedural °	Must know	Teach at this level.
Explain research updates about Shilajatu, Naga, Mukta and Kapardika.?	*explain*	Conceptual °	Desirable	Teach at this level.
Discuss probable physical, chemical changes occurring after Bhavana, Puta & whole process of Bhasma nirmana from Rasadravyas.	*discuss*	Procedural °	Must know	Teach at this level.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Develop checklist for identification of genuine rasadravyas.	*develop*	Procedural	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	1	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	7	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 8 of these 9 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 1 · UNIT 9

Rasa Dravya Parichaya II

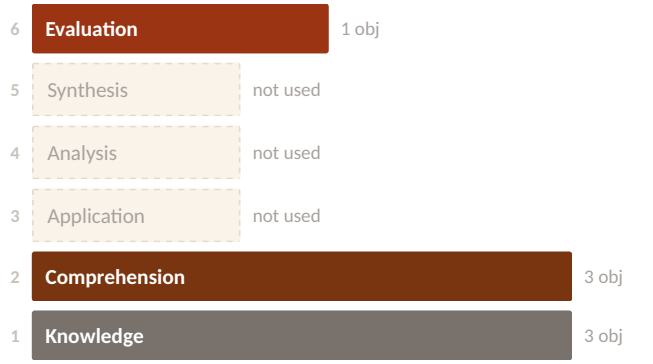
Hours: 7 lecture · 6 non-lecture | Term 2 | Theory marks: 5 | **7 objectives** | Highest level asked for: **Evaluation / Evaluate**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
7	**0**	**1**	**Conceptual**
levels 1–6	Krathwohl domain	6 desirable · 0 nice to know	kind of knowledge

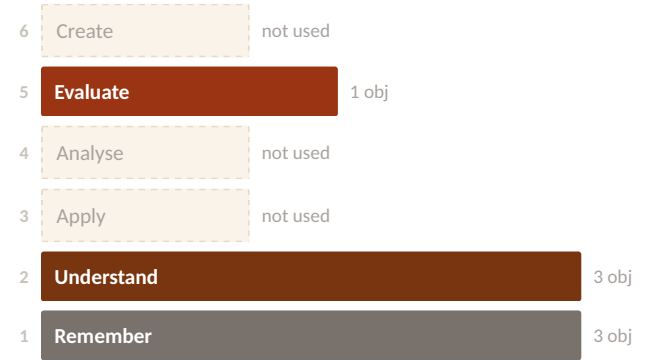
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
2	**4**	**2**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 3 OBJECTIVES				
Explain classical & Mineralogical identification and their Sources.	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Elaborate types, Grahya-Agrahyata and Dosha of Rasadravyas.	*elaborate*	Conceptual	Desirable	Teach at **level 2**. The verb *elaborate* is what the exam will ask for.
Enlist names of rasadravyas and important kalpas prepared from respective dravya. Describe therapeutic importance of that Rasadravya.	*enlist*	Factual + Conceptual °	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 3 OBJECTIVES				
Enlist the important synonyms of Rasadravyas.	*enlist*	Factual °	Desirable	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Explain Shodhana, Marana & and other processing techniques with probable chemical reactions.	*explain*	Procedural °	Must know	Teach at this level.
Explain research updates of Kasisa, Gauripashana and Akika.	*explain*	Conceptual °	Desirable	Teach at this level.
LEVEL 6 · EVALUATION / EVALUATE — 1 OBJECTIVES				
Explain Shodhana, Marana & and other processing techniques with probable chemical reactions.	*explain*	Procedural °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	3	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	3	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	1	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 6 of these 7 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
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FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Evaluation* — decide something, write down the reasons you used, then argue the opposite side.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 6, repeating it does not mean you know it.

PAPER 1 · UNIT 10

Rasadravya Parichaya III

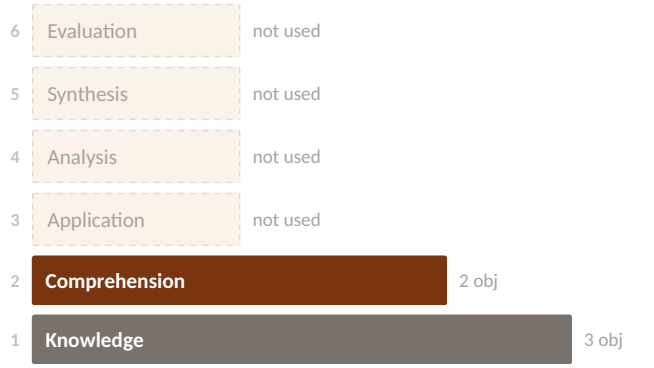
Hours: 3 lecture · 6 non-lecture | Term 2 | Theory marks: 5 | **5 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
5	**0**	**0**	**Conceptual**
levels 1–6	Krathwohl domain	2 desirable · 3 nice to know	kind of knowledge

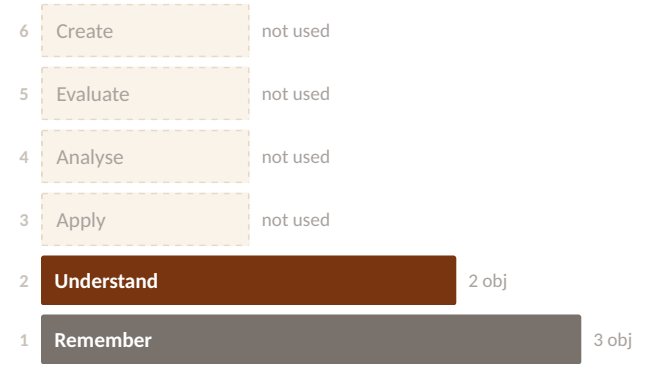
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
2	**3**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 3 OBJECTIVES				
Enlist Synonyms and sources.	*enlist*	Factual ◦	Nice to know	Teach at this level.
Explain research update of Badarashma.	*explain*	Conceptual ◦	Nice to know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Enlist Rasadravyas from this group and describe important kalpas with their therapeutic utility.	*enlist*	Factual ◦	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 2 OBJECTIVES				
Determine types, Grahya-Agrahya, properties & classical as well as Mineralogical Identification of Rasadravyas.	*determine*	Conceptual	Nice to know	Teach at **level 4**. The verb *determine* is what the exam will ask for.
Explain Shodhana, Marana and Probable Chemical Changes.	*explain*	Conceptual °	Desirable	Teach at this level.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	3	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	2	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 5 of these 5 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 1 · UNIT 11

Kalpana Nirman -III (Method of Preparation of different dosage forms)

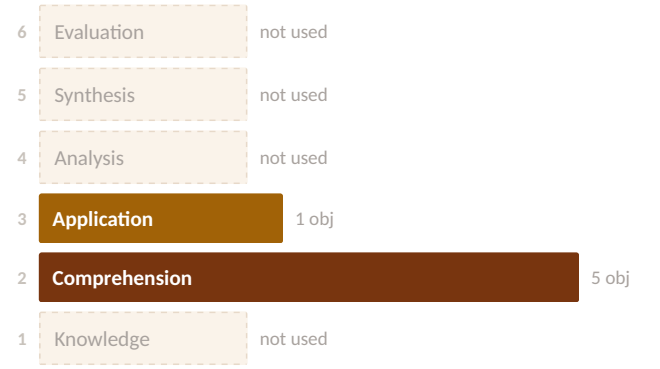
Hours: 8 lecture · 6 non-lecture | Term 2 | Theory marks: 10 | **6 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
6	**0**	**2**	**Procedural**
levels 1–6	Krathwohl domain	1 desirable · 3 nice to know	kind of knowledge

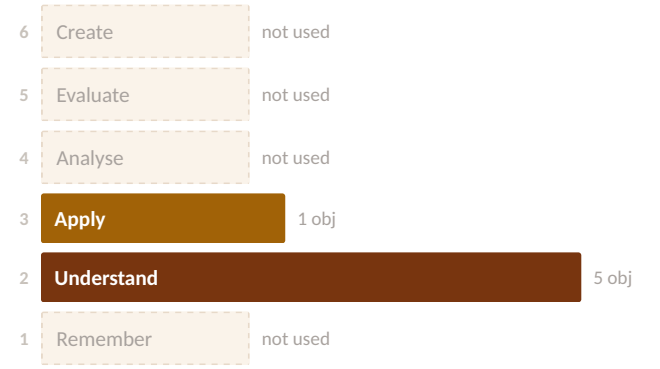
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**2**	**6**	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 5 OBJECTIVES				
Describe sharkara kalpana along with preparation method of Syrup. Explain therapeutic importance of prepared sharkara kalpana along with its shelf life.	*describe*	Conceptual + Procedural °	Nice to know	Teach at this level.
Describe Gudapaka preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Gudapaka along with its shelf life.	*describe*	Conceptual + Procedural °	Nice to know	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Describe Lavana kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lavana kalpana along with its shelf life and packaging techniques.	*describe*	Procedural	Desirable	Teach at this level.
Describe Ayaskriti kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Ayaskriti kalpana along with its shelf life and packaging techniques.	*describe*	Procedural	Nice to know	Teach at this level.
Describe Lepa kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lepa kalpana, integration with Kayachikitsa and Panchakarma department along with its shelf life and packaging techniques.	*describe*	Procedural	Must know	Teach at this level.
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Describe Kshara kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Kshara kalpana and Ksharasutra integration with Shalya tantra department along with its shelf life and packaging techniques.	*describe*	Procedural	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	—	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	5	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 5 of these 6 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Procedural* — do it, do not only read it; *Conceptual* — explain it out loud without notes.
- **Work upward, and check yourself as you go.** *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 1 · UNIT 12

Chaturvidha Rasayana

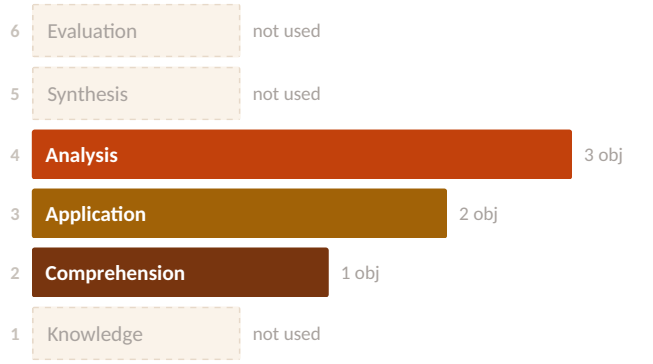
Hours: 4 lecture · 4 non-lecture | Term 2 | Theory marks: 10 | **6 objectives** | Highest level asked for: **Analysis / Analyse**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
6	0	2	Conceptual
levels 1–6	Krathwohl domain	4 desirable · 0 nice to know	kind of knowledge

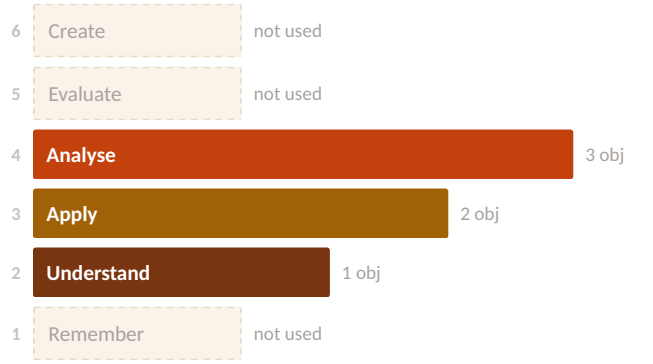
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	4	4	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Describe importance and unique features of Chaturvidha Rasayana - Kharaliya Rasayana, Parpati Rasayana, Kupipakwa Rasayana & Pottali Rasayana.	*describe*	Conceptual	Must know	Teach at this level.
LEVEL 3 · APPLICATION / APPLY — 2 OBJECTIVES				
Explain definition, types and method of preparation of Chaturvidha Rasayana.	*explain*	Conceptual + Procedural °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Explain law of Definite proportion and proportion of Mercury and sulphur needed for preparation of sulphide of Mercury. in Chaturvidha rasayana.	*explain*	Procedural °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
LEVEL 4 · ANALYSIS / ANALYSE — 3 OBJECTIVES				
Determine role of Yantra, Agni & Kala for pharmaceutical process of chaturvidha rasayana.	*determine*	Conceptual + Procedural °	Desirable	Teach at this level.
Explain shelf-life of Chaturvidha Rasayana.	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Describe chemical changes occurring during the preparation of chaturvidha rasayana and its impact on Pharmacological action of Chaturvidha Rasayana.	*describe*	Procedural °	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	—	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	1	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	2	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	3	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 1 of these 6 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it; *Analysis* — put two things side by side and write down what differs, and why.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 4, repeating it does not mean you know it.

PAPER 1 · UNIT 13

Current and emerging trend in Ayurvedic pharmaceuticals

Hours: 3 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **6 objectives** | Highest level asked for: **Knowledge / Remember**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
5	**1**	**0**	**Conceptual**
levels 1–6	Krathwohl domain	5 desirable · 1 nice to know	kind of knowledge

WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3

6	Evaluation	not used
5	Synthesis	not used
4	Analysis	not used
3	Application	not used
2	Comprehension	not used
1	**Knowledge**	5 obj

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6	Create	not used
5	Evaluate	not used
4	Analyse	not used
3	Apply	not used
2	Understand	not used
1	**Remember**	5 obj

The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	**4**	**1**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 5 OBJECTIVES				
Discuss the classification of different dosage forms.	*discuss*	Conceptual	Desirable	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Explain the need of different dosage forms.	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Enlist categorical information about the cosmetics used in daily routine. Appreciate Ayurvedic cosmetic products.	*enlist*	Factual °	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Choose and record the contemporary machineries used in the manufacture of cosmetics.	*choose*	Conceptual °	Nice to know	Teach at **level 4**. The verb *choose* is what the exam will ask for.
Enlist Quality Control parameters of cosmetics preparations.	*enlist*	Procedural °	Desirable	Teach at this level.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Appreciate Ayurvedic cosmetic products.	*appreciate*	Conceptual °	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	5	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 5 of these 6 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
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FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 1, repeating it does not mean you know it.

PAPER 1 · UNIT 14

GMP(Schedule T) & Regulatory aspects of Ayurvedic drugs

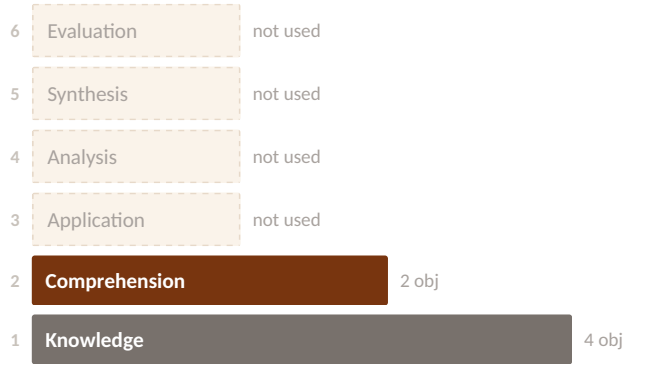
Hours: 2 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **7 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
6	**1**	**2**	**Conceptual**
levels 1–6	Krathwohl domain	4 desirable · 1 nice to know	kind of knowledge

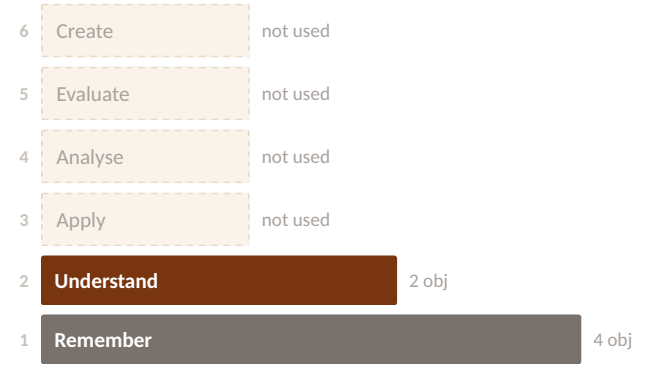
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**6**	**1**	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 4 OBJECTIVES				
Discuss the guidelines of Food Safety and Standards Authority of India (FSSAI) and FDA.	*discuss*	Conceptual °	Desirable	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Determine the principles and practice of establishment of Ayurvedic pharmacy.	*determine*	Conceptual	Desirable	Teach at **level 4**. The verb *determine* is what the exam will ask for.
Discuss the NABL guidelines for testing laboratory (Chemical).	*discuss*	Procedural °	Nice to know	Teach at **level 2**. The verb *discuss* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Explain long forms of these - (FSSAI) and (FDA), (CCRAS), (DCGI), (CDSCO).	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 2 OBJECTIVES				
Explain the legal and regulatory aspects of manufacturing, and sale of Ayurvedic drugs.	*explain*	Conceptual °	Must know	Teach at this level.
Describe acts and rules mentioned in Drug & Cosmetic Act 1940 & Rule 1945 and their relevance to Ayurvedic, Siddha, Unani (ASU) drugs.	*describe*	Conceptual °	Must know	Teach at this level.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Discuss ethical aspect of large scale drug preparation in Ayurvedic Pharmacies.	*discuss*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	4	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	2	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 6 of these 7 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 2 · UNIT 1

Aushadhi Prayoga Vigyana

Hours: 1 lecture · 2 non-lecture | Term 1 | Theory marks: 5 | **1 objectives** | Highest level asked for: **Knowledge / Remember**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
1	**0**	**1**	**Factual**
levels 1–6	Krathwohl domain	0 desirable · 0 nice to know	kind of knowledge

WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3

6	Evaluation	not used
5	Synthesis	not used
4	Analysis	not used
3	Application	not used
2	Comprehension	not used
1	**Knowledge**	1 obj

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6	Create	not used
5	Evaluate	not used
4	Analyse	not used
3	Apply	not used
2	Understand	not used
1	**Remember**	1 obj

The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	—	—	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 1 OBJECTIVES				
Define Aushadhi Prayoga Vigyana and its scope and enlist prashasta bheshaja laxana.	*define*	Factual °	Must know	Teach at this level.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	1	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 1 of these 1 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 1, repeating it does not mean you know it.

PAPER 2 · UNIT 2

Single drug (Herbal & Mineral)

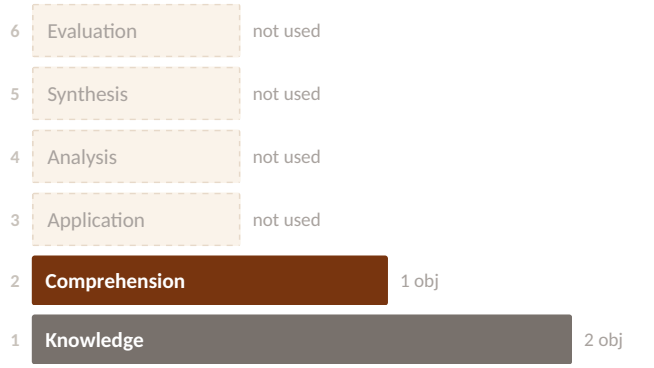
Hours: 8 lecture · 2 non-lecture | Term 1 | Theory marks: 10 | **5 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
3	**2**	**3**	**Conceptual**
levels 1–6	Krathwohl domain	1 desirable · 1 nice to know	kind of knowledge

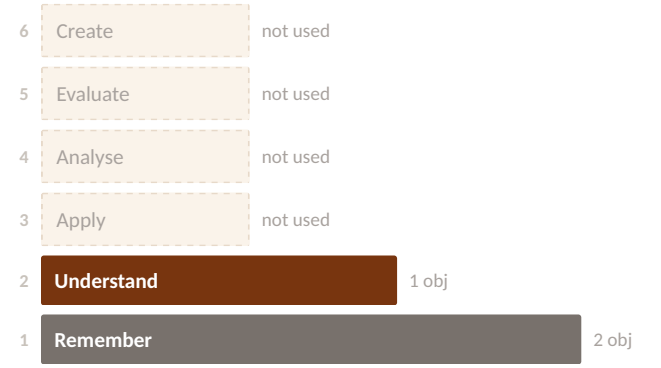
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	**5**	**1**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Describe different dosage forms prepared out of a single herb or mineral.	*describe*	Conceptual ◦	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Describe therapeutic efficacy of different formulations of Vishadrvya (Bhallataka), with its toxic effects and remedy.	*describe*	Conceptual ◦	Nice to know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Describe therapeutic efficacy of different formulations of Guduchi, Gairika, Gandhaka.	*describe*	Conceptual ◦	Must know	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
AFFECTIVE DOMAIN — 2 OBJECTIVES				
Appreciate how the therapeutic efficacy varies depending on the dosage form.	*appreciate*	Conceptual °	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.
Appraise multiple factors considered for preparation of various dosage forms from a single drug and creat list of more such examples.	*appraise*	Factual + Conceptual + Procedural °	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

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TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	2	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	1	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 5 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
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FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
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PAPER 2 · UNIT 3

Single drug(Bhasma, Shuddha & Pishti)

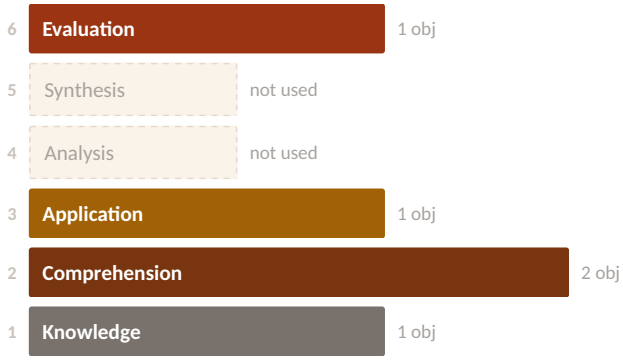
Hours: 12 lecture · 6 non-lecture | Term 2 | Theory marks: 15 | **5 objectives** | Highest level asked for: **Evaluation / Evaluate**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
5	**0**	**2**	**Conceptual**
levels 1–6	Krathwohl domain	3 desirable · 0 nice to know	kind of knowledge

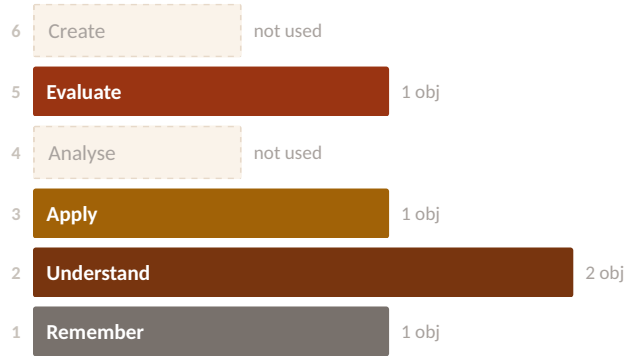
WHAT THIS TOPIC ASKS STUDENTS TO DO

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REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

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FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**5**	—	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 1 OBJECTIVES				
Describe Apakwa Ashuddha Avidhi Bhasma Sevanajanya Vyadhi and their Shanti-upaya.	*describe*	Conceptual ◦	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 2 OBJECTIVES				
Explain chemical form/composition, Pharmacodynamics and pharmacokinetics of Abhraka, Loha and Godanti Bhasma.	*explain*	Conceptual ◦	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Explain therapeutic properties, dosage, Anupana, Pathyapathya duration of treatment, Sevana Kala, shelf life, important Yogas of each Bhasma.	*explain*	Conceptual °	Must know	Teach at this level.
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Describe in detail Amayika Prayoga (Therapeutic uses) of each Bhasma/Pishti with given references.	*describe*	Conceptual °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
LEVEL 6 · EVALUATION / EVALUATE — 1 OBJECTIVES				
Explain research updates and clinical evidences of Swarna Bhasma, Makshika Bhasma and Shankha Bhasma.	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	1	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	2	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	1	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 5 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 6, repeating it does not mean you know it.

PAPER 2 · UNIT 4

Aushadhi Kalpa -I (Compound formulations)

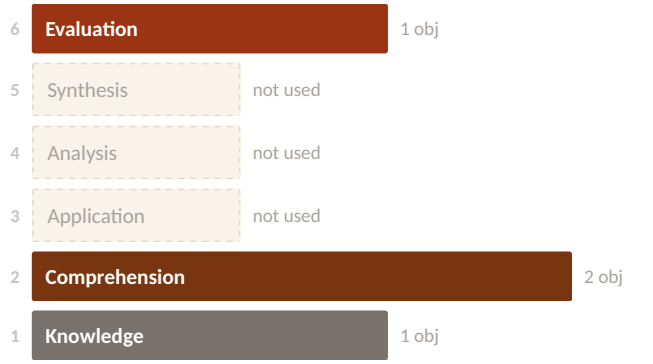
Hours: 16 lecture · 4 non-lecture | Term 2 | Theory marks: 15 | **5 objectives** | Highest level asked for: **Evaluation / Evaluate**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
4	**1**	**2**	**Conceptual**
levels 1–6	Krathwohl domain	3 desirable · 0 nice to know	kind of knowledge

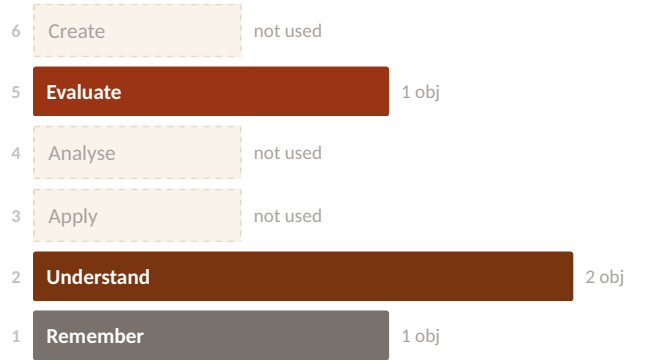
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**5**	—	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 1 OBJECTIVES				
Describe reference, ingredients, therapeutic properties dosage and all administration details alongwith Anupana of each formulation.	*describe*	Conceptual ◦	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 2 OBJECTIVES				
Explain Pathya Apathya, Sevana Avadhi (duration of treatment) and shelf-life of each formulation.	*explain*	Conceptual ◦	Must know	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Interpret probable mode of action of each formulation as per Ayurveda.	*interpret*	Conceptual °	Desirable	Teach at **level 6**. The verb *interpret* is what the exam will ask for.
LEVEL 6 · EVALUATION / EVALUATE — 1 OBJECTIVES				
Explain research updates and clinical evidences of Arogyavardhini Gutika and Gandhaka Rasayana.	*explain*	Conceptual °	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Critically analyse compound drugs for their therapeutic actions mentioned in the classics.	*critically*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	1	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	2	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	1	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 5 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Evaluation* — decide something, write down the reasons you used, then argue the opposite side.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 6, repeating it does not mean you know it.

PAPER 2 · UNIT 5

Aushadhi Kalpa-II (Compound Drugs/Formulations)

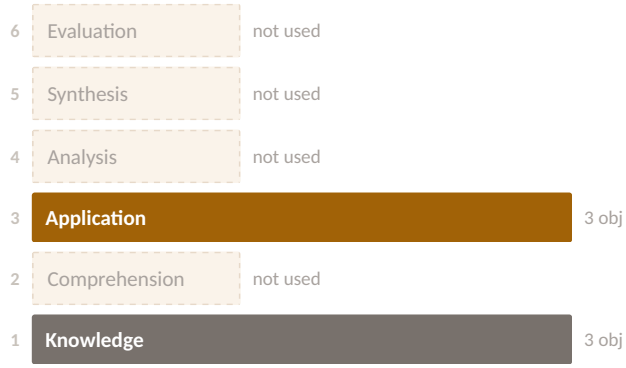
Hours: 14 lecture · 2 non-lecture | Term 3 | Theory marks: 15 | **7 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
6	**1**	**3**	**Conceptual**
levels 1–6	Krathwohl domain	2 desirable · 2 nice to know	kind of knowledge

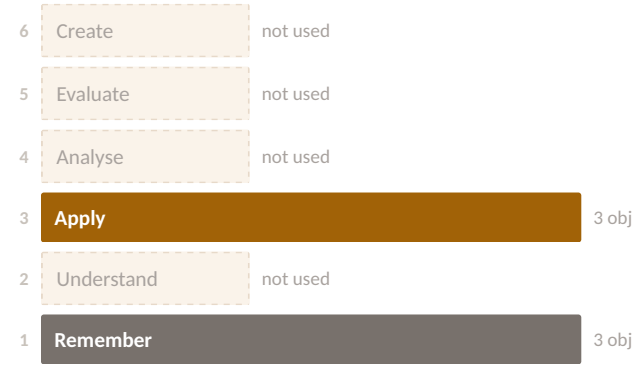
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
2	**5**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 3 OBJECTIVES				
Describe Ingredients, indications, dose, anupana, pathyapathya duration of treatment, sevana kala, shelf life, and clinical evidences of Panchaguna Taila, Dadimavaleha, Bramhi Ghrita.	*describe*	Conceptual °	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Describe ingredients, with its dose, therapeutic importance anupana and pathya-apathya along with its clinical evidences and Research updates.	*describe*	Conceptual	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Enlist the formulations with its dose, anupana and indications.	*enlist*	Factual °	Nice to know	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Describe Ingredients, therapeutic importance, dose, anupana pathya-pathya, duration of treatment, sevana kala, shelf life research updates and clinical evidences of Dashamoola Kwatha Pushyanuga Churna, Sanjivani Vati, Chitrakadi Gutika Simhanada Guggulu, Yogaraja Guggulu, Chyavanaprashavaleha Gandhakadya Malahara, Ashokarishta, Kutajarishta Panchagavya Ghrita, Bilvadi Gutika.	*describe*	Conceptual	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Describe Ingredients, indications, dose, anupana, pathyapathya duration of treatment, sevana kala, shelf life, and clinical evidences of Mahamanjistadi kwatha, Sudarshana Churna Vyoshadi Vati, Bala Chaturbhadra Rasa, Lavana Bhaskara Churna, Narayana Taila,, Neeliringadi Taila Aravindasava Kumaryasava.	*describe*	Conceptual °	Nice to know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Understand & Recite following classical kalpas. Describe Amayika prayoga of each formulation. Understand probable mode of action as per AyurvedaSearch and. Record relevant reaserch articles. Create charts describing details of these drugs. Review market avaibality & poplularities of these drugs among practicing vaidyas & near by population.	*understand*	Factual °	Must know	Teach at **level 2**. The verb *understand* is what the exam will ask for.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Explain ethical aspect of administration of compound drugs mainly regarding duration of the treatment given.	*explain*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	3	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	3	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 7 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 2 · UNIT 6

Dosage Forms & Cosmetic Products

Hours: 5 lecture · 2 non-lecture | Term 3 | Theory marks: 5 | **12 objectives** | Highest level asked for: **Knowledge / Remember**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
11	**1**	**5**	**Conceptual**
levels 1–6	Krathwohl domain	6 desirable · 1 nice to know	kind of knowledge

WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3

6	Evaluation	not used
5	Synthesis	not used
4	Analysis	not used
3	Application	not used
2	Comprehension	not used
1	**Knowledge**	11 obj

REVISED 2001 — Anderson & Krathwohl

6	Create	not used
5	Evaluate	not used
4	Analyse	not used
3	Apply	not used
2	Understand	not used
1	**Remember**	11 obj

The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
3	**9**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 11 OBJECTIVES				
Define the term- Dosage forms.	*define*	Factual	Must know	Teach at this level.
Discuss the need of different dosage forms.	*discuss*	Conceptual ◦	Must know	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Explain the classification of different dosage forms.	*explain*	Conceptual	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Enlist the solid / liquid / semisolid dosage forms.	*enlist*	Factual ◦	Must know	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
Discuss the routes of administration of different dosage forms.	*discuss*	Conceptual °	Must know	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Describe the advantages and disadvantages of currently available dosage forms.	*describe*	Conceptual °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Discuss the research updates about modification of classical Ayurvedic dosage forms.	*discuss*	Conceptual °	Desirable	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Discuss the relevant case studies of different dosage forms.	*discuss*	Conceptual °	Nice to know	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Define the term Cosmetics.	*define*	Factual	Desirable	Teach at this level.
Explain the classification of cosmetics based on their application on the specific body parts.	*explain*	Conceptual	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Differentiate between the ancient cosmetics and modern day cosmetics.	*differentiate*	Conceptual °	Desirable	Teach at **level 4**. The verb *differentiate* is what the exam will ask for.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
compare and appraise herbal cosmetics vs synthetic products.	*compare*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	11	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 11 of these 12 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
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FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 1, repeating it does not mean you know it.

PAPER 2 · UNIT 7

Nutraceuticals

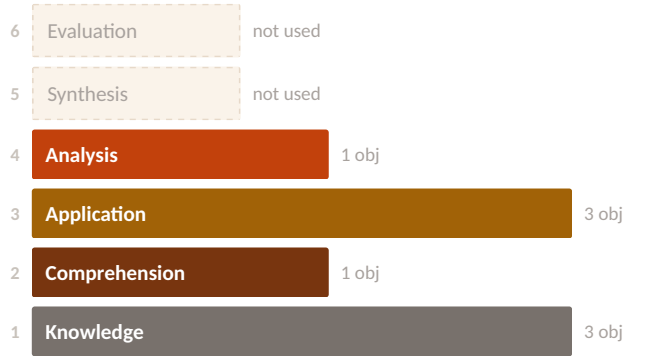
Hours: 6 lecture · 1 non-lecture | Term 3 | Theory marks: 5 | **9 objectives** | Highest level asked for: **Analysis / Analyse**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
8	**1**	**4**	**Conceptual**
levels 1–6	Krathwohl domain	4 desirable · 1 nice to know	kind of knowledge

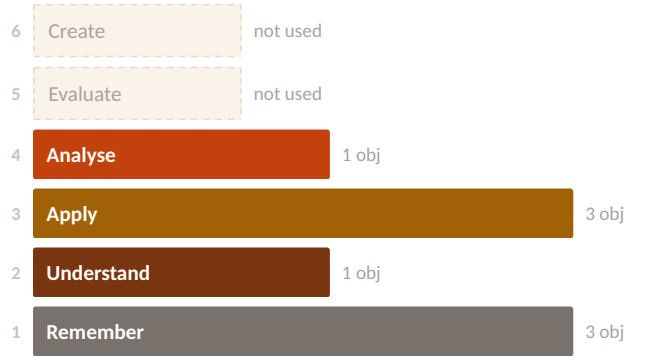
WHAT THIS TOPIC ASKS STUDENTS TO DO

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BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



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WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
3	**7**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 3 OBJECTIVES				
Describe importance of combination of aahara and aushadha.	*describe*	Conceptual	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Explain Nutraceuticals with its types.	*explain*	Conceptual	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Explain research updates about nutraceuticals.	*explain*	Conceptual °	Nice to know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Enlist atleast two examples of Aushadhi siddha aahara with its indications for each category-->(Aharadravya belonging to following category and kalpa prepared from that aahara dravya and other aushadhi ingredients)1) Jala varga2) Dugdha varga3) Ikshu varga4) Suka-dhanya varga5) Shimbi Dhanya Varga6) Shaka Varga7) Phala Varga8) Lavana Varga9) Spices mhita.	*enlist*	Factual °	Desirable	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Describe with examples mode of action of Ayurvedic nutraceuticals mhita.	*describe*	Conceptual °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Enlist at least two dietary preparations from Ayurvedic classics which can serve as rasayana/ nutraceutical in-1) General Health2) Pediatric Health3) Geriatric Health4) Women(Garbhini/ Sutika) Health5) Cardio-protection6) Chronic illness-recovery stage.	*enlist*	Factual °	Must know	Teach at **level 1**. The verb *enlist* is what the exam will ask for.
Describe potential of Ayurvedic diet and Rasayana in sports endeavor.	*describe*	Conceptual °	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
LEVEL 4 · ANALYSIS / ANALYSE — 1 OBJECTIVES				
Corelate rasayana and nutraceutical with examples.	*correlate*	Conceptual °	Desirable	Teach at this level.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
creat list of unique features of Ayurvedic neutraceuticals and appraise their current need.	*create*	Factual + Conceptual °	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	3	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	1	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	3	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	1	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 4 of these 9 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 4, repeating it does not mean you know it.

PAPER 2 · UNIT 8

Anupana Prayoga for Aushadhi Kalpa

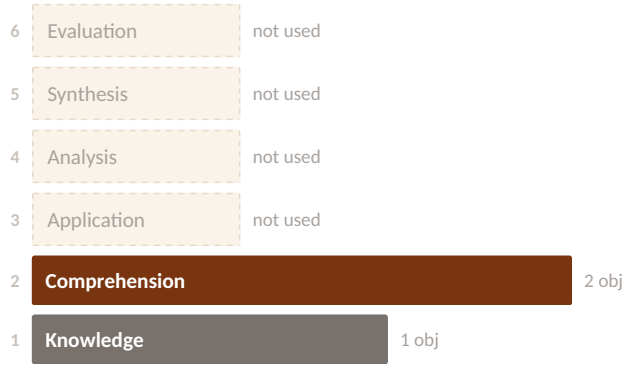
Hours: 4 lecture · 1 non-lecture | Term 3 | Theory marks: 5 | **4 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
3	**1**	**3**	**Factual**
levels 1–6	Krathwohl domain	1 desirable · 0 nice to know	kind of knowledge

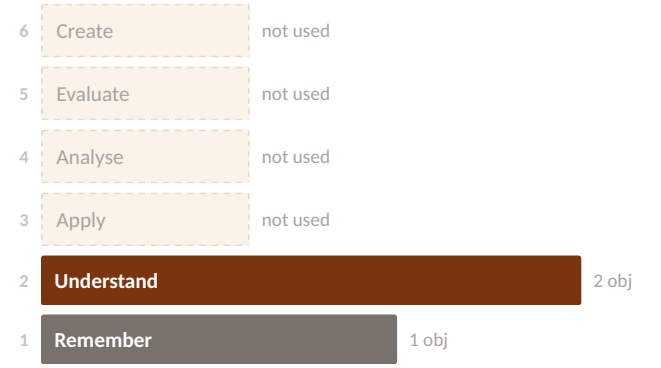
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
2	**2**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 1 OBJECTIVES				
Define and classify Anupana and Sahapana.	*define*	Factual ◦	Must know	Teach at this level.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 2 OBJECTIVES				
Explain different factors to be considered for selection of Anupana as per Disease and Patients.	*explain*	Conceptual ◦	Must know	Teach at this level.
Enlist different anupana for a single drug based on the condition of patient and disease. Elaborate with the help of examples of kalpas.	*enlist*	Factual ◦	Must know	Teach at **level 1**. The verb *enlist* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Describe value of proper selection of Anupanas with examples.	*describe*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	1	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	2	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 4 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Factual* — recall it until it sticks; *Conceptual* — explain it out loud without notes.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 2 · UNIT 9

Aushadhi Prayoga Marga

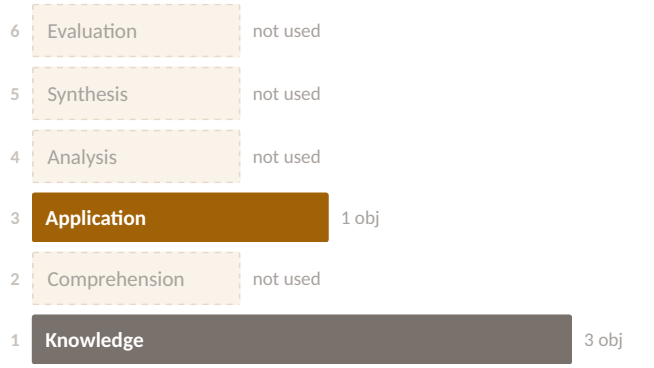
Hours: 5 lecture · 1 non-lecture | Term 3 | Theory marks: 10 | **5 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
4	**1**	**2**	**Conceptual**
levels 1–6	Krathwohl domain	2 desirable · 1 nice to know	kind of knowledge

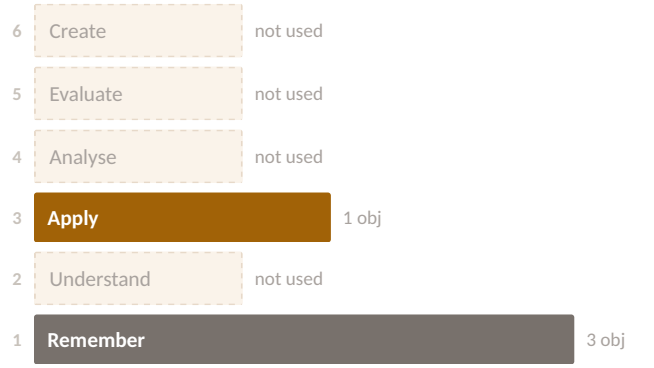
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
1	**4**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 3 OBJECTIVES				
Discuss the various types of Aushadhi Prayog Marga (route of drug administration).	*discuss*	Conceptual	Must know	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
Describe the advantages and disadvantages of each Aushadhi Prayog Marga.	*describe*	Conceptual °	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Enlist the dosage forms used in the specific Aushadhi Prayog Marga.	*enlist*	Factual °	Desirable	Teach at this level.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Discuss the nature of drug (s) administered in various routes of drug administration.	*discuss*	Conceptual °	Nice to know	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Appraise administration of drugs through various routes mentioned in Ayurvedic treatment.	*appraise*	Conceptual	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	3	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 5 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 2 · UNIT 10

Rational prescription along with safe dispensing of Ayurvedic formulations.

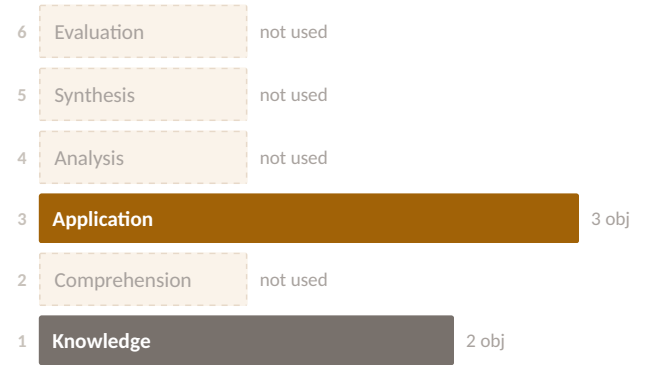
Hours: 1 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **6 objectives** | Highest level asked for: **Application / Apply**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
5	**1**	**3**	**Conceptual**
levels 1–6	Krathwohl domain	3 desirable · 0 nice to know	kind of knowledge

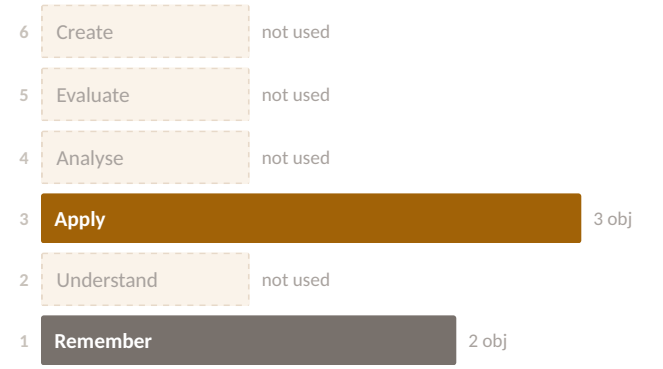
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
—	**4**	**2**	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Explain the importance of rational prescribing of drugs and the concept of essential drugs.	*explain*	Conceptual	Desirable	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Describe the standard protocol for safe dispensing of Ayurvedic drugs.	*describe*	Procedural °	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Describe and write demo ideal prescription.	*describe*	Conceptual °	Must know	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Explain the safe dispensing and efficacious use of Ayurvedic drugs.	*explain*	Conceptual °	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.
Demonstrate and educate home remedies to small group of population.	*demonstrate*	Procedural °	Desirable	Teach at this level.
AFFECTIVE DOMAIN — 1 OBJECTIVES				
explain ethical aspects related to prescription writing.	*explain*	Conceptual	Must know	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	2	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	3	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 2 of these 6 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Procedural* — do it, do not only read it.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Application* — find a case you were not given in class and work through it.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 3, repeating it does not mean you know it.

PAPER 2 · UNIT 11

Traditional & Local health Practices

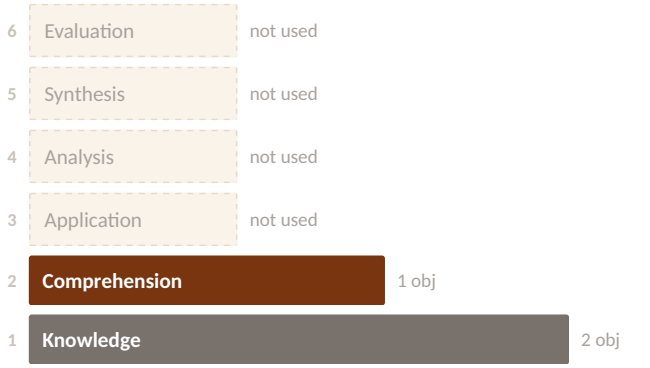
Hours: 2 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **4 objectives** | Highest level asked for: **Comprehension / Understand**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
3	**1**	**0**	**Factual**
levels 1–6	Krathwohl domain	4 desirable · 0 nice to know	kind of knowledge

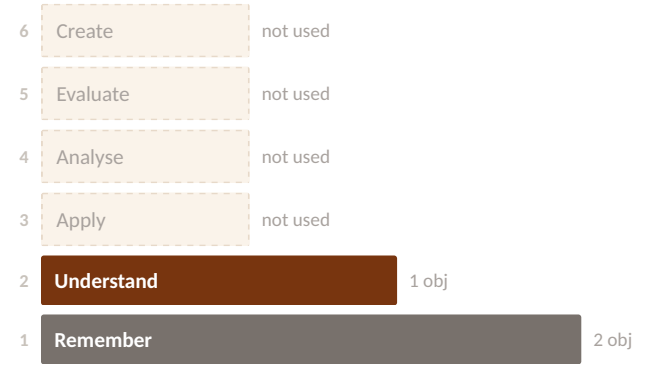
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
3	1	—	—
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Recognise the possible potential of product development and research based on Traditional knowledge.	*recognise*	Factual ◦	Desirable	Teach at this level.
Identify the factors responsible for grant of patent and erroneous grant of patent on indian traditional knowledge.?	*identify*	Factual ◦	Desirable	Teach at this level.
LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Identify Local Health Traditions and Healing Knowledge.	*identify*	Factual ◦	Desirable	Teach at **level 1**. The verb *identify* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
AFFECTIVE DOMAIN — 1 OBJECTIVES				
Appraise traditional knowledge of Ayurvedic medicines.	*appraise*	Conceptual	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	2	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	1	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	—	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	—	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 3 of these 4 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
- **Read the verb before you plan the class.** The verb is what students will meet in the exam. If it asks for more than the code, plan for the verb. If it asks for less, you have room to go higher.
- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for “enlist” and for “justify” in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Factual* — recall it until it sticks; *Conceptual* — explain it out loud without notes.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Comprehension* — explain it out loud to someone else without looking at your notes.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 2, repeating it does not mean you know it.

PAPER 2 · UNIT 12

Pharmacovigilance for Ayurveda drugs

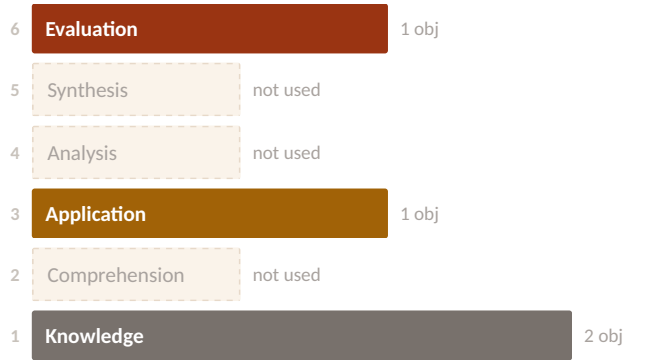
Hours: 1 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **6 objectives** | Highest level asked for: **Evaluation / Evaluate**

COGNITIVE	AFFECTIVE	MUST KNOW	MAINLY
4	**2**	**1**	**Conceptual**
levels 1–6	Krathwohl domain	5 desirable · 0 nice to know	kind of knowledge

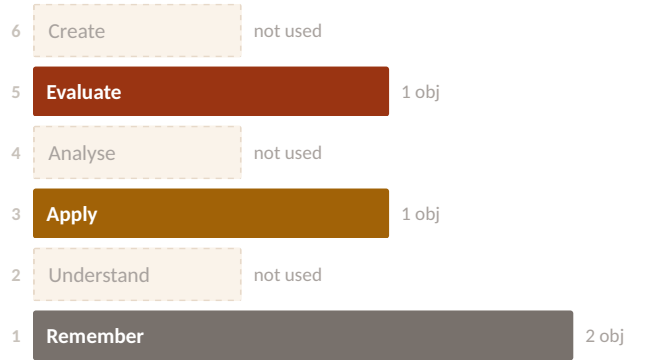
WHAT THIS TOPIC ASKS STUDENTS TO DO

Every objective in Table 3 has a code in Column C3. That code says what kind of thinking the student has to do. Put all the codes together and you see the shape of the topic: where most of the work sits, and how far up it goes. Both schemes are shown, because different departments and examiners use different ones.

BLOOM 1956 — the scheme printed in Column C3



REVISED 2001 — Anderson & Krathwohl



The top two levels swap between the two schemes. In the 1956 scheme, *Synthesis* is level 5 and *Evaluation* is level 6. In the 2001 scheme they are the other way round: *Evaluate* is level 5 and *Create* is level 6. The colours show this. The left column breaks its colour order at the top; the right one does not. Keep this in mind when you talk to colleagues, because "the top level" means different things in the two schemes.

WHAT KIND OF KNOWLEDGE IS INVOLVED

The 2001 scheme asks a second question. The first is: what thinking must the student do? The second is: what kind of knowledge is it about? This matters, because each kind is learned in a different way. The counts below come from the words of each objective - its verb, and what the verb acts on. Some objectives work on two kinds at once, and are counted under both.

FACTUAL	CONCEPTUAL	PROCEDURAL	METACOGNITIVE
2	**5**	**—**	**—**
learned by recall it until it sticks	learned by explain it out loud without notes	learned by do it, do not only read it	learned by look back at how you worked

THE OBJECTIVES, GROUPED BY LEVEL

These are the same objectives as Table 3, but sorted by level. This lets you plan one level at a time. Bloom builds upward, so teach the lower levels first: the higher ones stand on them. The last column tells you which level to teach each objective at. Sometimes the verb asks for a different level from the code. When that happens, follow the verb, because the verb is what the exam paper will use.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Describe the term Pharmacovigilance and explain importance of Pharmacovigilance for Ayurvedic drugs.	*describe*	Factual + Conceptual ◦	Desirable	Teach at **level 2**. The verb *describe* is what the exam will ask for.
Explain the status and central sector scheme of Pharmacovigilance for Ayurveda, Siddha, Unani, and Homeopathy (ASU & H) drugs.	*explain*	Conceptual ◦	Must know	Teach at **level 2**. The verb *explain* is what the exam will ask for.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	HOW TO PITCH IT
LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Discuss and make critical comments on the safe and efficacious use of Ayurvedic drugs.	*discuss*	Conceptual °	Desirable	Teach at **level 2**. The verb *discuss* is what the exam will ask for.
LEVEL 6 · EVALUATION / EVALUATE — 1 OBJECTIVES				
Define Adverse Drug Reactions (ADR) and its types QZ, TT-Theor.	*define*	Conceptual °	Desirable	Teach at **level 1**. The verb *define* is what the exam will ask for.
AFFECTIVE DOMAIN — 2 OBJECTIVES				
Identify and monitor ADRs.	*identify*	Factual °	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.
Debate on ADR of ASU drugs.	*debate*	Conceptual °	Desirable	Watch this over time: how they behave in the practical, a short reflection, a checklist. A written answer cannot show an attitude.

° the kind of knowledge is likely, but not certain from the wording alone. · **Level** here always means the Bloom level given in Column C3.

TEACHING AND CHECKING, LEVEL BY LEVEL

For each level: what your teaching has to do, and what a fair test looks like. The methods named are the ones the curriculum sets for this topic in Column F3.

LEVEL	1956 / 2001	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Knowledge** Remember	2	Show it, then make students recall it again and again. A lecture puts the facts in front of them. Only repeated recall makes the facts stay. Quizzes, word puzzles and games are recall practice in a form students will actually do.	Ask for the list, the name, the definition. Use many short questions spread over several weeks, not one big test at the end.
Level 2	**Comprehension** Understand	—	Students must work with the material, not just hear it again. Ask them to sort it, compare it, put it in their own order, and explain it to a classmate. Discussion does this. More lecturing does not.	Ask for the material in a new arrangement - sorted, compared or reordered. If a student can answer straight from the notes, the question has slipped down to level 1.
Level 3	**Application** Apply	1	Do examples together first. Then give them a situation they have not seen before. That hand-over is the teaching. Without it, students learn the example but not the method.	Give a case they have not seen and ask what they would do. A familiar case only tests memory.
Level 4	**Analysis** Analyse	—	Give students two things and ask how they differ, and why. You cannot analyse one thing on its own.	Questions that ask for the parts, and for how the parts fit together.
Level 5	**Synthesis** Create	—	This curriculum does not ask for it. If you want to use it, it means asking students to build something new that you did not hand them.	Something the student made, marked against rules you set out in advance.
Level 6	**Evaluation** Evaluate	1	Give them something to judge, and tell them what to judge it against. Make them state their reasons. Small-group debate works well: students defend a view and hear it challenged.	Long questions. Give marks for the reasons and the standards used, not for facts repeated inside the answer.

FOR THE TEACHER

- **Teach from the bottom up.** 2 of these 6 objectives are at levels 1 and 2. Students need those first, because everything higher is built on them. But tell the class on day one that the topic does not stop there.
- **Pick the method to fit the level.** A lecture is good for level 1 and poor for level 2. From level 2 upward, students have to work with the material themselves. The discussion hours and self-study hours are where real understanding is built.
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- **Attitude objectives need a different kind of proof.** Watch how students behave in the practical. Ask for a short written reflection. Use a checklist across the term. You cannot test an attitude in an exam hall, so decide in advance when you will look for it.

FOR THE STUDENT

- **Read the verb in every objective.** The verb tells you what you will have to do. You prepare for "enlist" and for "justify" in completely different ways. Knowing which is which saves more time than any revision trick.
- **Study each kind of knowledge in its own way.** *Conceptual* — explain it out loud without notes; *Factual* — recall it until it sticks.
- **Work upward, and check yourself as you go.** *Knowledge* — cover the answer, recall it, then check. Repeat over several days; *Application* — find a case you were not given in class and work through it; *Evaluation* — decide something, write down the reasons you used, then argue the opposite side.
- **Test yourself at the level the objective asks for.** Repeating the words of an objective is only level 1. If the objective asks for level 6, repeating it does not mean you know it.

PRACTICAL · UNIT 1

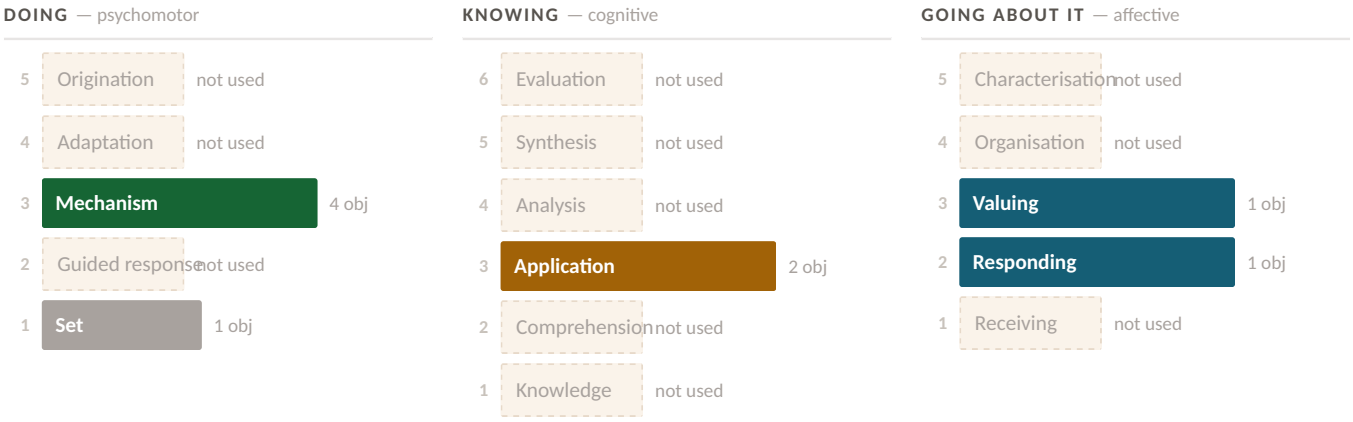
Paribhasha concept based Practicals

NCISM Table 4 | **9 objectives** | Psychomotor 5 · Cognitive 2 · Affective 2 | Highest doing level: **Mechanism**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
5	**2**	**2**	**6**
what they can do	what they must know	how they work	3 desirable · 0 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
DOING — LEVEL 1 · SET — 1 OBJECTIVES				
Identify the drugs and recite classical name, English name and chemical composition and varga(class). Recite Maharasa uparasa, Sadharana rasa shlokas from Rasaratna samuchchaya.	*identify*	Factual	Must know	**Watch them do it.** Checklist or OSPE station. You may guide while watching. The verb *identify* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
DOING — LEVEL 3 · MECHANISM — 4 OBJECTIVES				
Demonstrate the correct procedure / SOP and assess the changes observed after the procedure. Discuss the relevant samskara and its role in that procedure. Interpret the Physical, Chemical and Biological alterations.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Observe and describe the pharmaceutical preparation of Gandhaka druti. Explain the term alotropism. Recite melting boiling and evaporating temperature of Gandhaka.	*observe*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. They work unaided. The verb *observe* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
Demonstrate preparation of Kajjali. Recite its classical shloka. Prepare a checklist of parameters for its siddhilakshana. Interpret law of definite proportion to explain and calculate amount of free sulfur in the prepared amount of kajjali. Calculate the expected weight of rasasindura from the given amount of kajjali.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Identify relevant Bhasmapariksha and demonstrate samanya and vishesha Bhasma pariksha. Interpret physical/ chemical laws relevant to classical bhasma pariksha.	*identify*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided. The verb *identify* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
KNOWING — LEVEL 3 · APPLICATION / APPLY — 2 OBJECTIVES				
Observe the procedure of Vanga Jarana. Explain the difference between Jarana and Jaranaa. Discuss chemical process of oxidation and reduction. Recall the information about melting points of all the metals.	*observe*	Procedural	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.
Identify various weights and recite essential measures from conversion chart of AFI. Recognise weighing machines and their weighing capacity.	*identify*	Factual	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.
GOING ABOUT IT — LEVEL 2 · RESPONDING — 1 OBJECTIVES				
Appraise ancient indications mentioned in classics for confirmation of end point of the pharmaceutical process(Siddhi lakshana).	*appraise*	Conceptual + Procedural °	Must know	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.
GOING ABOUT IT — LEVEL 3 · VALUING — 1 OBJECTIVES				
Identify and value SOP of each procedure of the practical conducted.	*identify*	Procedural	Must know	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

TEACHING AND CHECKING THE DOING LEVELS

For each doing level this unit uses: what the teaching has to accomplish, and what counts as fair evidence.

LEVEL	NAME	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Set** ready to begin - knows what is needed and can lay it out	1	Show the setup and name every item. Then have students lay it out themselves before anything else happens. A student who cannot assemble what is needed cannot begin.	Identification, and a setup checklist you tick while they work.
Level 2	**Guided response** performs while being shown or guided	—	Demonstrate the whole thing once at normal speed, then again slowly, then let them copy it while you stand beside them and correct as they go.	Watch them do it with you present. Correct in the moment - the record here is your own observation, not a written answer.
Level 3	**Mechanism** performs reliably alone, without prompting	4	Repeated supervised practice, with your help withdrawn a little each time, until they can complete it unaided. This is where most practical hours should go.	Observe them working alone against a checklist, or run an OSPE station. The test is whether it holds up without prompting.
Level 4	**Adaptation** changes the method when conditions change	—	Change one condition and let them work out the adjustment - a different quantity, a different starting material, equipment that is not available.	Give an altered situation and watch what they change and why. Ask them afterwards for their reason; the reason is part of the evidence.
Level 5	**Origination** devises a procedure that was not given	—	Not asked for in this curriculum. It would mean a student devising a procedure that was never demonstrated to them.	—

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Practical viva P-VIVA	Spoken	5	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Checklist CHK	Observed	2	Shows performance directly. The only kind of evidence that settles a doing objective.
P-EXAM P-EXAM	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Identification P-ID	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.
Practical performance P-PRF	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.
P-REC P-REC	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-SUR P-SUR	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
PRN PRN	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.

This unit has **5 doing objectives** but observed tools are set **4 times**. Spoken tools are set 5 times. Where a doing objective is checked only by viva, add your own observation - a checklist you tick while the student works is enough, and it is the only record that shows the procedure was actually performed.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 2

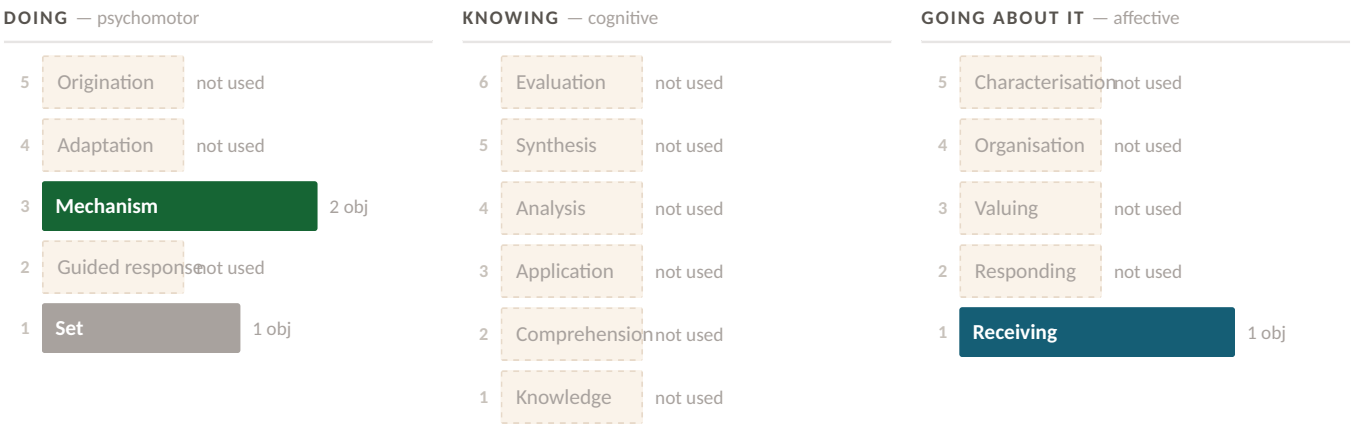
Panchavidha Kashaya Kalpana & their Upakalpana Practicals

NCISM Table 4 | **4 objectives** | Psychomotor 3 · Cognitive 0 · Affective 1 | Highest doing level: **Mechanism**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
3	**0**	**1**	**4**
what they can do	what they must know	how they work	0 desirable · 0 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



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Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
DOING — LEVEL 1 · SET — 1 OBJECTIVES				
Recall, Identify and authenticate the raw materials required as per the dosage form/ formulations.	*recall*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. You may guide while watching. The verb *recall* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
DOING — LEVEL 3 · MECHANISM — 2 OBJECTIVES				
Demonstrate the method of preparation to get the desired dosage form following SOP.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Assess and explain different parameters to achieve desired characters/ end points (siddhi lakshana) as per classical and contemporary parameters.	*assess*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
GOING ABOUT IT — LEVEL 1 · RECEIVING — 1 OBJECTIVES				
Describe ethical responsibility expected during Pharmaceutical preparation of Ayurvedic Formulations.	*describe*	Conceptual	Must know	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

TEACHING AND CHECKING THE DOING LEVELS

For each doing level this unit uses: what the teaching has to accomplish, and what counts as fair evidence.

LEVEL	NAME	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Set** ready to begin - knows what is needed and can lay it out	1	Show the setup and name every item. Then have students lay it out themselves before anything else happens. A student who cannot assemble what is needed cannot begin.	Identification, and a setup checklist you tick while they work.
Level 2	**Guided response** performs while being shown or guided	—	Demonstrate the whole thing once at normal speed, then again slowly, then let them copy it while you stand beside them and correct as they go.	Watch them do it with you present. Correct in the moment - the record here is your own observation, not a written answer.
Level 3	**Mechanism** performs reliably alone, without prompting	2	Repeated supervised practice, with your help withdrawn a little each time, until they can complete it unaided. This is where most practical hours should go.	Observe them working alone against a checklist, or run an OSPE station. The test is whether it holds up without prompting.
Level 4	**Adaptation** changes the method when conditions change	—	Change one condition and let them work out the adjustment - a different quantity, a different starting material, equipment that is not available.	Give an altered situation and watch what they change and why. Ask them afterwards for their reason; the reason is part of the evidence.
Level 5	**Origination** devises a procedure that was not given	—	Not asked for in this curriculum. It would mean a student devising a procedure that was never demonstrated to them.	—

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Practical viva P-VIVA	Spoken	3	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Practical performance P-PRF	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.

This unit has **3 doing objectives** but observed tools are set **1 time**. Spoken tools are set 3 times. Where a doing objective is checked only by viva, add your own observation - a checklist you tick while the student works is enough, and it is the only record that shows the procedure was actually performed.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 3

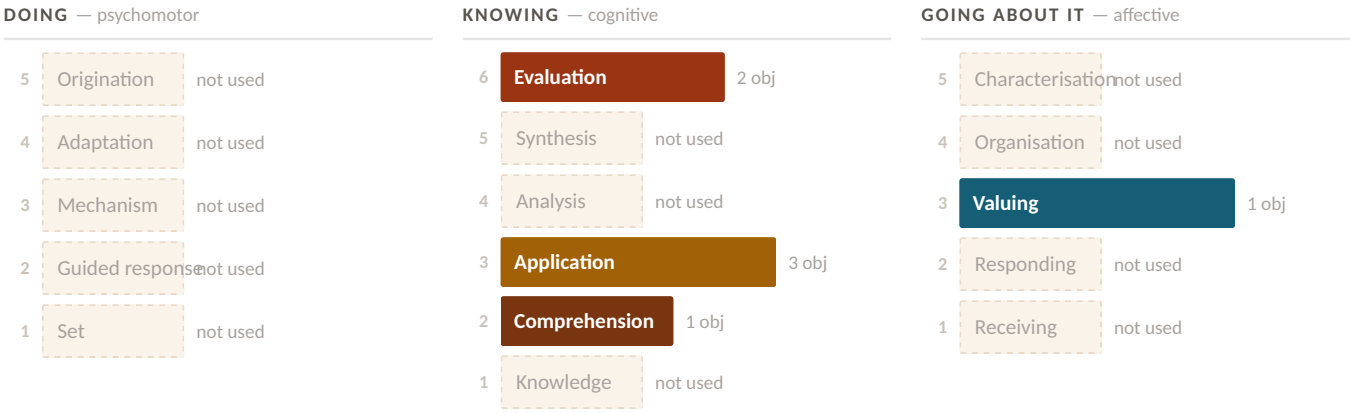
Rasa Dravya Aushadhi Nirmana Practicals

NCISM Table 4 | **7 objectives** | Psychomotor 0 · Cognitive 6 · Affective 1 | Highest doing level: **none — no psychomotor objective here**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
0	**6**	**1**	**6**
what they can do	what they must know	how they work	1 desirable · 0 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
KNOWING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Explain reference Shloka of kalpas & write with interpretation -list of ingredients, their proportion, principles of yoga - yoga samyojana dravya.	*explain*	Factual + Conceptual °	Must know	Spoken or written answer is fair here.
KNOWING — LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Identify raw Dravya, used part with classical Mana and interpret in metric system.	*identify*	Factual °	Must know	Spoken or written answer is fair here.
Identify, Utilize & know mechanism of Yantra used for drug preparation.	*identify*	Procedural °	Must know	Spoken or written answer is fair here.
Demonstrate Sidhdhi Lakshna.	*demonstrate*	Procedural °	Must know	Spoken or written answer is fair here.
KNOWING — LEVEL 6 · EVALUATION / EVALUATE — 2 OBJECTIVES				
Observe & Record various parameters responsible for good manufacturing which are done during process of pharmaceutical preparation.	*observe*	Procedural	Must know	Spoken or written answer is fair here.
Assess finished product as per classics.	*assess*	Conceptual °	Must know	Spoken or written answer is fair here.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
GOING ABOUT IT — LEVEL 3 · VALUING — 1 OBJECTIVES				
Appraise peculiarities of chaturvidha rasayana preparation.	*appraise*	Conceptual	Desirable	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Practical viva P-VIVA	Spoken	3	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-EXAM P-EXAM	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Viva VV-Viva	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Poster M-POS	Product	1	Shows the result. Say in advance what you will mark it against.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 4

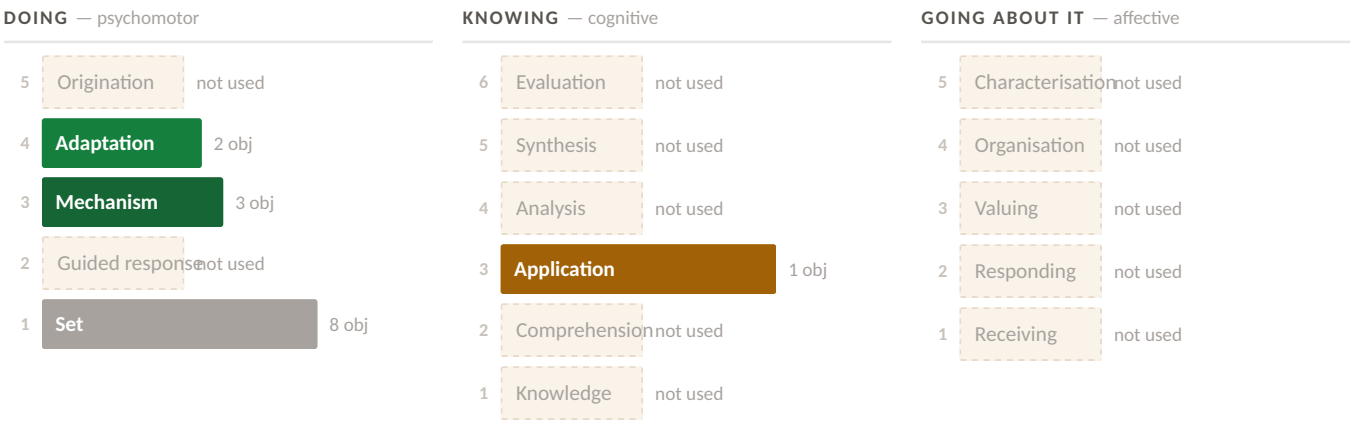
Bheshaja Kalpana Practicals -I

NCISM Table 4 | **14 objectives** | Psychomotor 13 · Cognitive 1 · Affective 0 | Highest doing level: **Adaptation**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
13	**1**	**0**	**6**
what they can do	what they must know	how they work	5 desirable · 3 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
DOING — LEVEL 1 · SET — 8 OBJECTIVES				
Demonstrate Arka preparation method, along with its packaging technique and therapeutic importance.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. You may guide while watching.
Identify the genuine sample of Guggulu and demonstrate preparation of different types of Guggulu along with drying techniques.	*identify*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. You may guide while watching. The verb *identify* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
Identify the useful part of the raw material and demonstrate Sattva preparation method.	*identify*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. You may guide while watching. The verb *identify* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
Demonstrate preparation of Varti along with drying and packaging techniques.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. You may guide while watching.
Demonstrate Lavana preparation methods, along with its packaging techniques.	*demonstrate*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. You may guide while watching.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
Demonstrate preparations of Sikta taila along with identify the genuine sample of Sikta. Perform and. Describe Malahara kalpana preparation method with its confirmatory tests of end point.	*demonstrate*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. You may guide while watching.
Demonstrate preparation of different types of Malahara with its applied aspect.	*demonstrate*	Procedural	Nice to know	**Watch them do it.** Checklist or OSPE station. You may guide while watching.
Demonstrate preparation of different types of Lepa and preparation of Shatadhouta ghrita.	*demonstrate*	Procedural	Nice to know	**Watch them do it.** Checklist or OSPE station. You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 3 OBJECTIVES				
Demonstrate preparation of Vati along with drying techniques.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate preparation and application aspects of Upanaha kalpana.	*demonstrate*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. They work unaided.
Perform and. Describe Lepa kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lepa kalpana, integration with Kayachikitsa and Panchakarma department along with its shelf life and packaging techniques.	*perform*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. They work unaided.
DOING — LEVEL 4 · ADAPTATION — 2 OBJECTIVES				
Identify the internal and external applications of different kshara kalpana along with importance of ksharasutra through surgical aspects.	*identify*	Conceptual + Procedural °	Must know	**Watch them do it.** Checklist or OSPE station. Alter a condition and watch the adjustment. The verb *identify* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
Demonstrate and. Explain preparation methods of Masi kalpana for its internal or external application in clinical practice.	*demonstrate*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. Alter a condition and watch the adjustment.
KNOWING — LEVEL 3 · APPLICATION / APPLY — 1 OBJECTIVES				
Observe demonstration of formulation of Danta manjana.	*observe*	Factual °	Nice to know	Spoken or written answer is fair here, but pair it with the doing objective it supports.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

TEACHING AND CHECKING THE DOING LEVELS

For each doing level this unit uses: what the teaching has to accomplish, and what counts as fair evidence.

LEVEL	NAME	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Set** ready to begin - knows what is needed and can lay it out	8	Show the setup and name every item. Then have students lay it out themselves before anything else happens. A student who cannot assemble what is needed cannot begin.	Identification, and a setup checklist you tick while they work.
Level 2	**Guided response** performs while being shown or guided	—	Demonstrate the whole thing once at normal speed, then again slowly, then let them copy it while you stand beside them and correct as they go.	Watch them do it with you present. Correct in the moment - the record here is your own observation, not a written answer.
Level 3	**Mechanism** performs reliably alone, without prompting	3	Repeated supervised practice, with your help withdrawn a little each time, until they can complete it unaided. This is where most practical hours should go.	Observe them working alone against a checklist, or run an OSPE station. The test is whether it holds up without prompting.
Level 4	**Adaptation** changes the method when conditions change	2	Change one condition and let them work out the adjustment - a different quantity, a different starting material, equipment that is not available.	Give an altered situation and watch what they change and why. Ask them afterwards for their reason; the reason is part of the evidence.
Level 5	**Origination** devises a procedure that was not given	—	Not asked for in this curriculum. It would mean a student devising a procedure that was never demonstrated to them.	—

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Practical viva P-VIVA	Spoken	9	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Viva, knowledge check V-KC	Spoken	4	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
PRN PRN	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-EXAM P-EXAM	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-SUR P-SUR	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.

This unit has **13 doing objectives** but observed tools are set **0 times**. Spoken tools are set 13 times. Where a doing objective is checked only by viva, add your own observation - a checklist you tick while the student works is enough, and it is the only record that shows the procedure was actually performed.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 5

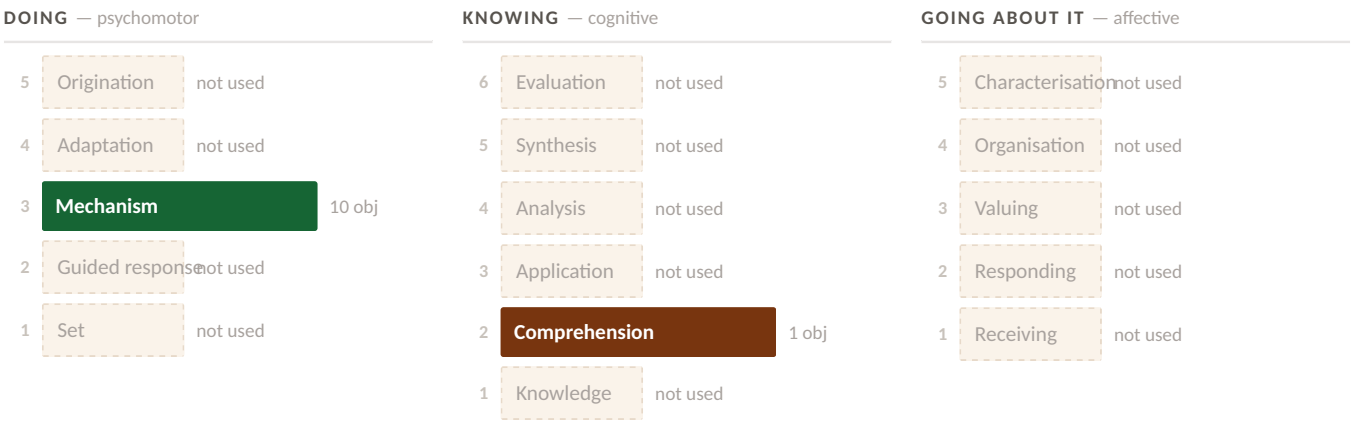
Bheshaja Kalpana Practicals -II

NCISM Table 4 | **11 objectives** | Psychomotor 10 · Cognitive 1 · Affective 0 | Highest doing level: **Mechanism**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
10	**1**	**0**	**7**
what they can do	what they must know	how they work	2 desirable · 2 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
DOING — LEVEL 3 · MECHANISM — 10 OBJECTIVES				
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Ghrita Murchana and Taila Murchana.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Jatyadi Gritha/ Triphala Gritha/ Ksheera Shatphala Gritha/Phala Gritha.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage Ksheera Bala Taila/Kasisadi Taila/ Panchaguna Taila/ Arka Taila/Kutajasuryapaki taila.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage Bhallataka Taila Patana/ Jayapala Taila Patana/Vishvamitra Kalpa Sneha.	*demonstrate*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. They work unaided.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Vasavaleha/ Chavana Prasha Avaleha/ Kushmanda Avaleha.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Nimbu Sharkara.	*demonstrate*	Procedural	Nice to know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Daruharidra Rasakriya.	*demonstrate*	Procedural	Nice to know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Haridra Khanda/ Narikela Khanada.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Kumaryasava/Drakshasava.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Kutajarista/Ashokarishta/Takrarishta.	*demonstrate*	Procedural	Must know	**Watch them do it.** Checklist or OSPE station. They work unaided.
KNOWING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlist the results, packing techniques and storage of unique formulations for e.g. Kanji/ Madushukta.	*demonstrate*	Procedural	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

TEACHING AND CHECKING THE DOING LEVELS

For each doing level this unit uses: what the teaching has to accomplish, and what counts as fair evidence.

LEVEL	NAME	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Set** ready to begin - knows what is needed and can lay it out	—	Show the setup and name every item. Then have students lay it out themselves before anything else happens. A student who cannot assemble what is needed cannot begin.	Identification, and a setup checklist you tick while they work.
Level 2	**Guided response** performs while being shown or guided	—	Demonstrate the whole thing once at normal speed, then again slowly, then let them copy it while you stand beside them and correct as they go.	Watch them do it with you present. Correct in the moment - the record here is your own observation, not a written answer.
Level 3	**Mechanism** performs reliably alone, without prompting	10	Repeated supervised practice, with your help withdrawn a little each time, until they can complete it unaided. This is where most practical hours should go.	Observe them working alone against a checklist, or run an OSPE station. The test is whether it holds up without prompting.
Level 4	**Adaptation** changes the method when conditions change	—	Change one condition and let them work out the adjustment - a different quantity, a different starting material, equipment that is not available.	Give an altered situation and watch what they change and why. Ask them afterwards for their reason; the reason is part of the evidence.
Level 5	**Origination** devises a procedure that was not given	—	Not asked for in this curriculum. It would mean a student devising a procedure that was never demonstrated to them.	—

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
OSPE station OSPE	Observed	6	Shows performance directly. The only kind of evidence that settles a doing objective.
P-EXAM P-EXAM	Spoken	6	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Practical viva P-VIVA	Spoken	5	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-REC P-REC	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
PRN PRN	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Compilation COM	Record over time	1	Shows habit and consistency. The right tool for how a student goes about the work.

This unit has **10 doing objectives** but observed tools are set **6 times**. Spoken tools are set 5 times. Where a doing objective is checked only by viva, add your own observation - a checklist you tick while the student works is enough, and it is the only record that shows the procedure was actually performed.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 6

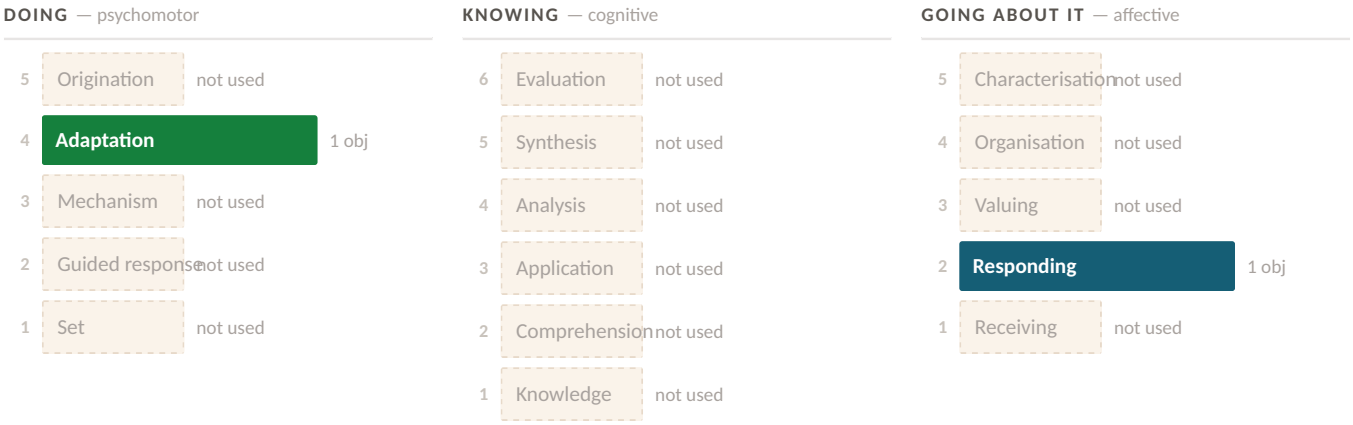
Dosage Forms & Self-care Products Practicals

NCISM Table 4 | **2 objectives** | Psychomotor 1 · Cognitive 0 · Affective 1 | Highest doing level: **Adaptation**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
1	**0**	**1**	**0**
what they can do	what they must know	how they work	2 desirable · 0 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
DOING — LEVEL 4 · ADAPTATION — 1 OBJECTIVES				
Observe instruments used to prepare with one example for each their method of preparation, ingredients used with their quantityand Quality control Parameters.of following self care productsHair care: Shampoo Body care: Soap, Perfume Face care: Face Pack/Talcum Powder Lip Care: Lip Balm, Lipstick Oral care: Tooth Paste/ Mouth Wash Foot Care: Foot Cream Hand Disinfectant: Hand SanitizerSkin Care: Moisturizer, Sunscreen Lotion following self care products.	*observe*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. Alter a condition and watch the adjustment. The verb *observe* asks only for knowing, while the code asks for doing. Decide which you will assess, and say so to the class.
GOING ABOUT IT — LEVEL 2 · RESPONDING — 1 OBJECTIVES				
Observe instruments used to prepare soliddosage form, liquid dosage form and Semisolid Dosage Form with one example for each, their method of preparation, ingredients used with their quantity and Quality control Parameters.	*observe*	Procedural	Desirable	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

TEACHING AND CHECKING THE DOING LEVELS

For each doing level this unit uses: what the teaching has to accomplish, and what counts as fair evidence.

LEVEL	NAME	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Set** ready to begin - knows what is needed and can lay it out	—	Show the setup and name every item. Then have students lay it out themselves before anything else happens. A student who cannot assemble what is needed cannot begin.	Identification, and a setup checklist you tick while they work.
Level 2	**Guided response** performs while being shown or guided	—	Demonstrate the whole thing once at normal speed, then again slowly, then let them copy it while you stand beside them and correct as they go.	Watch them do it with you present. Correct in the moment - the record here is your own observation, not a written answer.
Level 3	**Mechanism** performs reliably alone, without prompting	—	Repeated supervised practice, with your help withdrawn a little each time, until they can complete it unaided. This is where most practical hours should go.	Observe them working alone against a checklist, or run an OSPE station. The test is whether it holds up without prompting.
Level 4	**Adaptation** changes the method when conditions change	1	Change one condition and let them work out the adjustment - a different quantity, a different starting material, equipment that is not available.	Give an altered situation and watch what they change and why. Ask them afterwards for their reason; the reason is part of the evidence.
Level 5	**Origination** devises a procedure that was not given	—	Not asked for in this curriculum. It would mean a student devising a procedure that was never demonstrated to them.	—

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Practical viva P-VIVA	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Checklist CHK	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.
P-EXAM P-EXAM	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 7

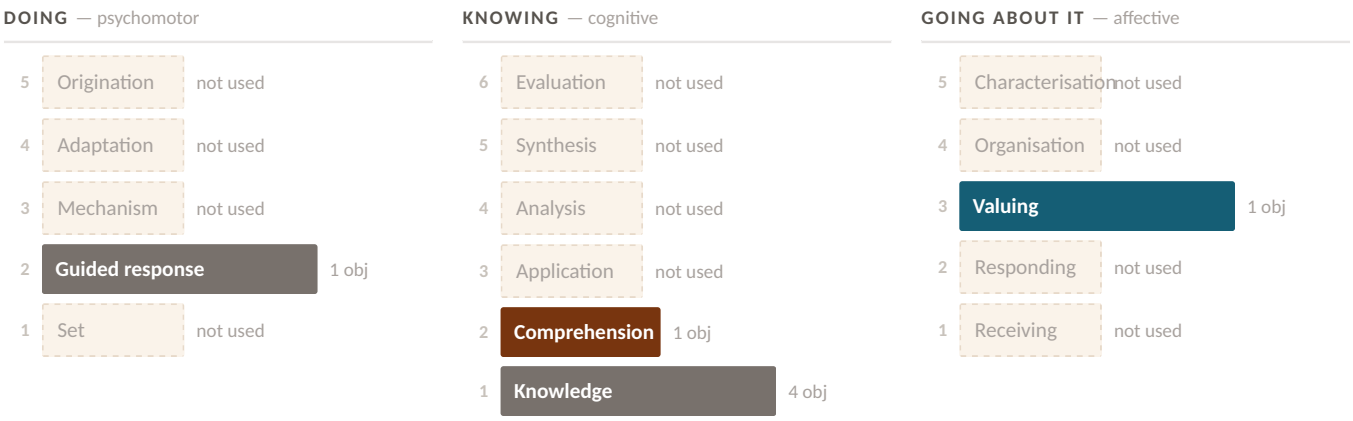
Field Visit/ Study Tour

NCISM Table 4 | **7 objectives** | Psychomotor 1 · Cognitive 5 · Affective 1 | Highest doing level: **Guided response**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
1	**5**	**1**	**0**
what they can do	what they must know	how they work	6 desirable · 1 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
DOING — LEVEL 2 · GUIDED RESPONSE — 1 OBJECTIVES				
Prepare own products as per prevalent FDA guidelines.	*prepare*	Procedural	Desirable	**Watch them do it.** Checklist or OSPE station. You may guide while watching.
KNOWING — LEVEL 1 · KNOWLEDGE / REMEMBER — 4 OBJECTIVES				
Field Visit - Record the storage condition of the raw / in-process / finished goods prepared in the approved sections.	*field*	Procedural	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.
Determine the role of various Yantras / machineries used in the approved sections of the Pharmacy.	*determine*	Conceptual	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.
Enlist the documents required in Batch Manufacturing of Ayurvedic medicines.	*enlist*	Factual °	Nice to know	Spoken or written answer is fair here, but pair it with the doing objective it supports.
Identify the role of Quality Control instruments and equipments in ensuring a quality Ayurvedic product.	*identify*	Conceptual °	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
KNOWING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Differentiate between the ancient and contemporary methods of drug preparation.	*differentiate*	Procedural	Desirable	Spoken or written answer is fair here, but pair it with the doing objective it supports.
GOING ABOUT IT — LEVEL 3 · VALUING — 1 OBJECTIVES				
Appreciate the importance of Good Manufacturing Practices and Good Packaging Practices required in Ayurvedic Drug manufacturing.	*appreciate*	Conceptual	Desirable	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

TEACHING AND CHECKING THE DOING LEVELS

For each doing level this unit uses: what the teaching has to accomplish, and what counts as fair evidence.

LEVEL	NAME	N	WHAT THE TEACHING HAS TO DO	WHAT A FAIR CHECK LOOKS LIKE
Level 1	**Set** ready to begin - knows what is needed and can lay it out	—	Show the setup and name every item. Then have students lay it out themselves before anything else happens. A student who cannot assemble what is needed cannot begin.	Identification, and a setup checklist you tick while they work.
Level 2	**Guided response** performs while being shown or guided	1	Demonstrate the whole thing once at normal speed, then again slowly, then let them copy it while you stand beside them and correct as they go.	Watch them do it with you present. Correct in the moment - the record here is your own observation, not a written answer.
Level 3	**Mechanism** performs reliably alone, without prompting	—	Repeated supervised practice, with your help withdrawn a little each time, until they can complete it unaided. This is where most practical hours should go.	Observe them working alone against a checklist, or run an OSPE station. The test is whether it holds up without prompting.
Level 4	**Adaptation** changes the method when conditions change	—	Change one condition and let them work out the adjustment - a different quantity, a different starting material, equipment that is not available.	Give an altered situation and watch what they change and why. Ask them afterwards for their reason; the reason is part of the evidence.
Level 5	**Origination** devises a procedure that was not given	—	Not asked for in this curriculum. It would mean a student devising a procedure that was never demonstrated to them.	—

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
PRN PRN	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
DEB DEB	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Compilation COM	Record over time	1	Shows habit and consistency. The right tool for how a student goes about the work.
Poster M-POS	Product	1	Shows the result. Say in advance what you will mark it against.
P-SUR P-SUR	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-EXAM P-EXAM	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.

This unit has **1 doing objectives** but observed tools are set **0 times**. Spoken tools are set 0 times. Where a doing objective is checked only by viva, add your own observation - a checklist you tick while the student works is enough, and it is the only record that shows the procedure was actually performed.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 8

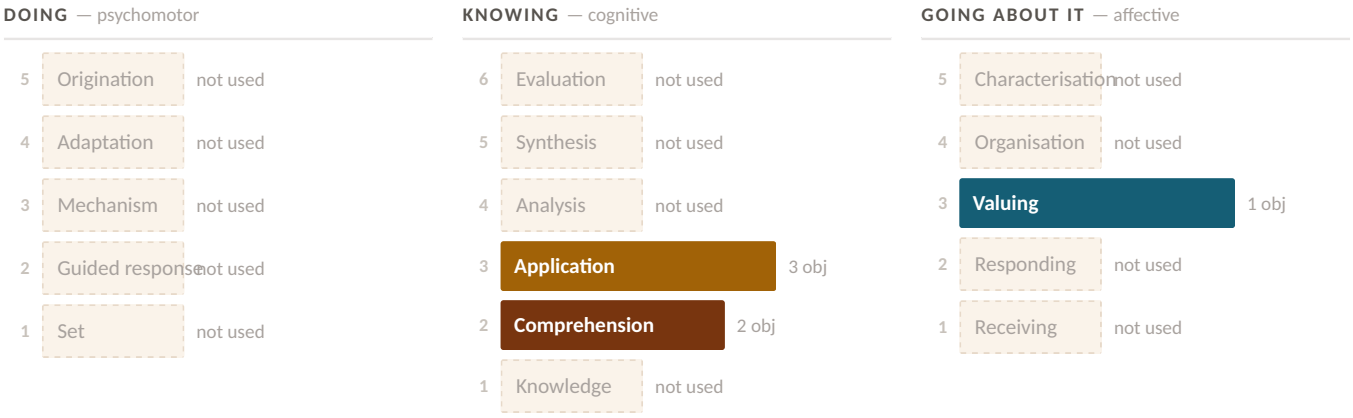
Hospital IPD Practical

NCISM Table 4 | **6 objectives** | Psychomotor 0 · Cognitive 5 · Affective 1 | Highest doing level: **none — no psychomotor objective here**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
0	**5**	**1**	**2**
what they can do	what they must know	how they work	3 desirable · 1 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
KNOWING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2 OBJECTIVES				
Review & Observe the case Sheets.	*review*	Conceptual °	Nice to know	Spoken or written answer is fair here.
Discuss for formulation prescription manner.	*discuss*	Conceptual °	Desirable	Spoken or written answer is fair here.
KNOWING — LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Demonstrate & Explain prescription method.	*demonstrate*	Procedural	Desirable	Spoken or written answer is fair here.
Measure dose of prescribed drug.	*measure*	Procedural °	Must know	Spoken or written answer is fair here.
Enlist Time, Duration, Pathya, Apathya while prescribing drugs.	*enlist*	Factual °	Desirable	Spoken or written answer is fair here.
GOING ABOUT IT — LEVEL 3 · VALUING — 1 OBJECTIVES				
value details and methodical writing of case records of hospital IPD patients.	*value*	Conceptual	Must know	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Viva, knowledge check V-KC	Spoken	3	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-SUR P-SUR	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Case study T-CS	Written	1	Shows reasoning. Pair it with observation where the objective is about doing.
P-PS P-PS	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Practical performance P-PRF	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 9

Drug Dispensing Practical

NCISM Table 4 | **4 objectives** | Psychomotor 0 · Cognitive 4 · Affective 0 | Highest doing level: **none — no psychomotor objective here**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
0	4	0	0
what they can do	what they must know	how they work	3 desirable · 1 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.

DOING — psychomotor

5	Origination	not used
4	Adaptation	not used
3	Mechanism	not used
2	Guided response	not used
1	Set	not used

KNOWING — cognitive

6	**Evaluation**	1 obj
5	Synthesis	not used
4	Analysis	not used
3	**Application**	2 obj
2	**Comprehension**	1 obj
1	Knowledge	not used

GOING ABOUT IT — affective

5	Characterisation	not used
4	Organisation	not used
3	Valuing	not used
2	Responding	not used
1	Receiving	not used

The five doing levels, in plain words. Set — ready to begin - knows what is needed and can lay it out · **Guided response** — performs while being shown or guided · **Mechanism** — performs reliably alone, without prompting · **Adaptation** — changes the method when conditions change · **Origination** — devises a procedure that was not given. The full psychomotor scheme has more levels than this; these are the five the curriculum uses.

THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
KNOWING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Determine and identify the cause of common errors occurred during dispensing of Ayurvedic drugs.	*determine*	Conceptual °	Desirable	Spoken or written answer is fair here.
KNOWING — LEVEL 3 · APPLICATION / APPLY — 2 OBJECTIVES				
Assess the arrangement of Ayurvedic drugs according to nature and type in dispensing room or pharmacy.	*assess*	Conceptual °	Nice to know	Spoken or written answer is fair here.
Explain use of correct drug supply to the right patients, in the required dosage, quantities and clear drug information.	*explain*	Conceptual °	Desirable	Spoken or written answer is fair here.
KNOWING — LEVEL 6 · EVALUATION / EVALUATE — 1 OBJECTIVES				
Assess practice of prescription processing and labelling of the drugs.	*assess*	Conceptual °	Desirable	Spoken or written answer is fair here.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Viva, knowledge check V-KC	Spoken	2	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Practical performance P-PRF	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.
PRN PRN	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PRACTICAL · UNIT 10

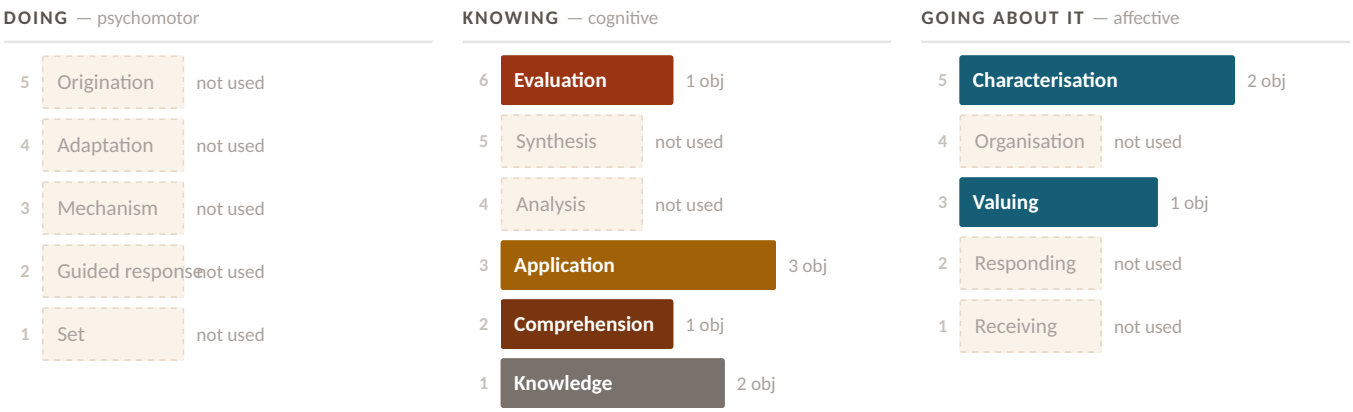
Quality Control Practicals

NCISM Table 4 | **10 objectives** | Psychomotor 0 · Cognitive 7 · Affective 3 | Highest doing level: **none — no psychomotor objective here**

PSYCHOMOTOR	COGNITIVE	AFFECTIVE	MUST KNOW
0	7	3	7
what they can do	what they must know	how they work	3 desirable · 0 nice to know

WHAT THIS PRACTICAL ASKS STUDENTS TO DO

A practical works on three fronts at once: what a student can *do*, what they need to *know* to do it, and how they *go about* it. Each front has its own set of levels, and they do not line up with one another. Read the doing column first - it is what a practical is for.



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THE OBJECTIVES, GROUPED BY LEVEL

Table 4 sorted into its three domains, and within each into levels, so a session can be planned around one level at a time. The last column says what kind of evidence the objective needs. A doing objective needs to be watched; no written or spoken answer can stand in for it.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
KNOWING — LEVEL 1 · KNOWLEDGE / REMEMBER — 2 OBJECTIVES				
Develop analytical skills for understanding Identity, Purity and Strength of raw materials and finished products as per the standard guidelines of the Ayurvedic Pharmacopoeia of India.	*develop*	Procedural °	Must know	Spoken or written answer is fair here.
Identify the geological description of minerals: physical parameters.	*identify*	Procedural °	Desirable	Spoken or written answer is fair here.
KNOWING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1 OBJECTIVES				
Demonstrate the calibration techniques used for Weighing Balance (chemical and physical), pH Meter, Hot Air Oven and Electric Muffle Furnace to ensure the accuracy of the instrument what it is intended to measure.	*demonstrate*	Procedural	Desirable	Spoken or written answer is fair here.
KNOWING — LEVEL 3 · APPLICATION / APPLY — 3 OBJECTIVES				
Describe ayurvedic perspectives of quality control parameters applying for solid, semisolid and liquid dosage forms.	*describe*	Procedural °	Must know	Spoken or written answer is fair here.
Demonstrate bulk density, loss on drying, total ash, pH and water soluble extractives of Churna A book.	*demonstrate*	Procedural °	Must know	Spoken or written answer is fair here.
Determine iodine value, acid value and saponification value of oils / ghee.	*determine*	Conceptual	Must know	Spoken or written answer is fair here.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	PRIORITY	EVIDENCE IT NEEDS
KNOWING — LEVEL 6 · EVALUATION / EVALUATE — 1 OBJECTIVES				
Calculate the dose of various dosage forms and their dispensing methods.	*calculate*	Procedural	Must know	Spoken or written answer is fair here.
GOING ABOUT IT — LEVEL 3 · VALUING — 1 OBJECTIVES				
Appraise quality control procedures done for Ayurvedic formulations.	*appraise*	Conceptual + Procedural °	Desirable	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.
GOING ABOUT IT — LEVEL 5 · CHARACTERISATION — 2 OBJECTIVES				
Perform the analytical tests of tablets/ vati/ gutika for hardness uniformity of weight and friability.	*perform*	Conceptual °	Must know	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.
Estimate specific gravity, alcohol content and total solids of Asava & Arishta.	*estimate*	Conceptual °	Must know	**Watch over time.** Conduct across the term, a short reflection, a checklist applied more than once.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both.

THE ASSESSMENT TOOLS SET FOR THIS UNIT

Column G4 names the tools. What each one can show is worth knowing before you plan the session, because a spoken tool cannot show whether a student can carry out a procedure.

TOOL	KIND	TIMES SET	WHAT IT CAN AND CANNOT SHOW
Practical viva P-VIVA	Spoken	4	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-EXAM P-EXAM	Spoken	3	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
Identification P-ID	Observed	1	Shows performance directly. The only kind of evidence that settles a doing objective.
PRN PRN	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.
P-SUR P-SUR	Spoken	1	Shows whether a student knows and can explain. It cannot show whether they can perform the procedure.

FOR THE TEACHER

- **Most of the hours belong to level 3.** Getting a student from guided to unaided is slow, and it is the whole point. Demonstrating well is quick; withdrawing help gradually is what takes time.
- **Do not let a viva stand in for watching.** A student can describe a procedure fluently and still not be able to carry it out. Where the objective is about doing, the evidence has to be seen.
- **Write the checklist before the session, not after.** Decide the steps you will look for, and how many attempts a student gets. Then the mark is about the student rather than about the day.
- **Name the knowing objectives that support each doing one.** Students who cannot say why a step exists tend to drop it the moment they are unsupervised.

FOR THE STUDENT

- **Reading a procedure is not learning it.** These objectives are met by doing the thing, with your hands, more than once.
- **Aim to work unaided.** Being able to do it while someone guides you is level 2. Doing it alone, reliably, is level 3, and that is what is expected.
- **Know why each step is there.** When something changes, only the reason tells you what to change too.
- **Keep your record honestly as you go.** A log book filled in at the end is a piece of writing; filled in as you work, it is evidence.

PART C

Reference

Nine appendices. Everything here is worked out from the curriculum's own objectives, so the counts in these tables and the counts in the Units are the same numbers. Use C1 and C2 to decode anything in the syllabus document; C3 to compare Units at a glance when planning a term or setting a paper; C4 and C5 when writing or rewording objectives of your own.

1. **C1** The sixteen curriculum codes, decoded
2. **C2** The two schemes side by side
3. **C3** Level profile of every Unit
4. **C4** Verb index — every verb the curriculum uses, and its level
5. **C5** Kinds of knowledge across the curriculum
6. **C6** Level against priority
7. **C7** Teaching methods named in the curriculum
8. **C8** Assessment tools named in the curriculum
9. **C9** Sources and further reading

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PART C · APPENDIX C1

The sixteen curriculum codes, decoded

Every code that appears in Column C3 of Table 3 and Column C4 of Table 4, with the domain it belongs to, the level it sits at, and how many objectives carry it. This is the table to keep beside the syllabus document.

CODE	DOMAIN	LEVEL	NAME	TIMES USED IN AYUG-RB
CK	Thinking	Level 1	Knowledge / Remember	86
CC	Thinking	Level 2	Comprehension / Understand	68
CAP	Thinking	Level 3	Application / Apply	35
CAN	Thinking	Level 4	Analysis / Analyse	4
CS	Thinking	Level 5	Synthesis / Create	not used
CE	Thinking	Level 6	Evaluation / Evaluate	10
PSY-SET	Doing	Level 1	Set	10
PSY-GUD	Doing	Level 2	Guided response	1
PSY-MEC	Doing	Level 3	Mechanism	19
PSY-ADT	Doing	Level 4	Adaptation	3
PSY-ORG	Doing	Level 5	Origination	not used
AFT-REC	Going about it	Level 1	Receiving	1
AFT-RES	Going about it	Level 2	Responding	5
AFT-VAL	Going about it	Level 3	Valuing	21
AFT-SET	Going about it	Level 4	Organisation	1
AFT-CHR	Going about it	Level 5	Characterisation	4

Sixteen codes are defined; fourteen are used. **CS** (Synthesis / Create) and **PSY-ORG** (Origination) appear in the key but against no objective.

PART C · APPENDIX C2

The two schemes side by side

The six thinking levels in both schemes, with what a student actually does at each. Where a colleague speaks of "the top level", this table settles which one is meant.

LEVEL	BLOOM 1956, AS PRINTED IN THE CURRICULUM	REVISED 2001, ANDERSON & KRATHWOHL	WHAT THE STUDENT DOES
Level 1	**Knowledge**	**Remember**	recalls it
Level 2	**Comprehension**	**Understand**	restates it in their own arrangement
Level 3	**Application**	**Apply**	uses it on something new
Level 4	**Analysis**	**Analyse**	takes it apart and shows how the parts relate
Level 5	**Synthesis**	**Evaluate** ← the schemes differ here	makes something that was not given
Level 6	**Evaluation**	**Create** ← and here	judges it against stated criteria

The top two levels are the only place the two schemes disagree. In 1956, **Synthesis** is level 5 and **Evaluation** level 6. In 2001, **Evaluate** is level 5 and **Create** level 6. This book follows the 1956 numbering, because that is what the curriculum prints.

PART C · APPENDIX C3

Level profile of every Unit

All thirty-six Units in one view. This is the fastest way to see where a term's weight sits, and to check that the marks on a paper are going where the thinking actually is.

UNIT	TOPIC	OBJ	MK	LEVELS USED	HIGHEST	MUST KNOW
PAPER 1 — AYURVEDIYA AUSHADHI NIRMANA VIGYANA						
1	Chronological development of Ayurvediya Aush	7	5	1 2	Comprehension	3 of 7
2	Paribhasha (Terminology)	19	10	1 2 3 6	Evaluation	12 of 19
3	Adharbhuta Siddhanta (Application of fundame	11	5	1 2	Comprehension	9 of 11
4	Yantropakaranani- I (Equipments and machiner	6	5	1 2 3	Application	1 of 6
5	Yantropakaranani -II (Equipments, fuel and H	21	5	1 2 3	Application	3 of 21
6	Kalpana Nirmana I (Primary & Secondary dosag	11	10	1 2	Comprehension	7 of 11
7	Kalpana Nirmana-II (Method of Preparation of	4	10	1 3	Application	4 of 4
8	Rasa Dravya Parichaya- I	9	10	1 2	Comprehension	7 of 9
9	Rasa Dravya Parichaya II	7	5	1 2 6	Evaluation	1 of 7
10	Rasadravya Parichaya III	5	5	1 2	Comprehension	0 of 5
11	Kalpana Nirman -III (Method of Preparation o	6	10	2 3	Application	2 of 6
12	Chaturvidha Rasayana	6	10	2 3 4	Analysis	2 of 6
13	Current and emerging trend in Ayurvedic	6	5	1	Knowledge	0 of 6
14	GMP(Schedule T) & Regulatory aspects of Ayur	7	5	1 2	Comprehension	2 of 7
PAPER 2 — AYURVEDIYA AUSHADHI PRAYOGA VIGYANA						
1	Aushadhi Prayoga Vigyana	1	5	1	Knowledge	1 of 1
2	Single drug (Herbal & Mineral)	5	10	1 2	Comprehension	3 of 5
3	Single drug(Bhasma, Shuddha & Pishti)	5	15	1 2 3 6	Evaluation	2 of 5
4	Aushadhi Kalpa -I (Compound formulations)	5	15	1 2 6	Evaluation	2 of 5
5	Aushadhi Kalpa-II (Compound Drugs/ Formulatio	7	15	1 3	Application	3 of 7
6	Dosage Forms & Cosmetic Products	12	5	1	Knowledge	5 of 12
7	Nutraceuticals	9	5	1 2 3 4	Analysis	4 of 9
8	Anupana Prayoga for Aushadhi Kalpa	4	5	1 2	Comprehension	3 of 4
9	Aushadhi Prayoga Marga	5	10	1 3	Application	2 of 5
10	Rational prescription along with safe dispen	6	5	1 3	Application	3 of 6
11	Traditional & Local health Practices	4	5	1 2	Comprehension	0 of 4
12	Pharmacovigilance for Ayurveda drugs	6	5	1 3 6	Evaluation	1 of 6
PRACTICAL UNITS — NCISM TABLE 4						
1	1.Paribhasha concept based Practicals	9	—	1 3	Mechanism	6 of 9

UNIT	TOPIC	OBJ	MK	LEVELS USED	HIGHEST	MUST KNOW
2	2.Panchavidha Kashaya Kalpana & their Upakal	4	—	1 3	Mechanism	4 of 4
3	3.Rasa Dravya Aushadhi Nirmana Practicals	7	—	—	no doing objective	6 of 7
4	4.Bheshaja Kalpana Practicals -I	14	—	1 3 4	Adaptation	6 of 14
5	5.Bheshaja Kalpana Practicals -II	11	—	3	Mechanism	7 of 11
6	6.Dosage Forms & Self-care Products Practica	2	—	4	Adaptation	0 of 2
7	7.Field Visit/ Study Tour	7	—	2	Guided response	0 of 7
8	8.Hospital IPD Practical	6	—	—	no doing objective	2 of 6
9	9. Drug Dispensing Practical	4	—	—	no doing objective	0 of 4
10	10.Quality Control Practicals	10	—	—	no doing objective	7 of 10

For theory Units, **Levels used** and **Highest** refer to the thinking levels. For practical Units they refer to the doing levels, since that is what a practical is for. **Mk** is the theory marks the curriculum allots.

PART C · APPENDIX C4

Verb index — every verb the curriculum uses, and its level

The opening verb of all 268 objectives, sorted by the level that verb belongs to. The verb is what a student meets in the examination, which makes this the most practical table in the book.

LEVEL	NAME	VERBS THE CURRICULUM ACTUALLY USES, WITH HOW OFTEN
Level 1	**Knowledge** Remember	enlist 29, identify 17, define 11, observe 6, recognise 3, recall 2, recite 1
Level 2	**Comprehension** Understand	explain 49, describe 47, discuss 17, elaborate 2, illustrate 1, understand 1
Level 3	**Application** Apply	demonstrate 28, perform 2, prepare 1, measure 1, calculate 1
Level 4	**Analysis** Analyse	determine 7, choose 2, review 2, differentiate 2, categorize 1, compile 1, analyse 1, discriminate 1, compare 1, correlate 1
Level 5	**Synthesis** Create	develop 2, produce 1, create 1
Level 6	**Evaluation** Evaluate	assess 6, appraise 6, interpret 3, justify 2, critically 2
Affective	**Attitude verbs**	appreciate 3, debate 1, value 1

Use this when writing an objective: choose the level first, then take a verb from that row. Where an objective's verb sits at a different level from its code, the Unit in Part B says so and tells you which to pitch to.

PART C · APPENDIX C5

Kinds of knowledge across the curriculum

The second axis of the 2001 scheme, counted across all 268 objectives. An objective that works on two kinds is counted under both, which is why the totals here exceed 268.

KIND OF KNOWLEDGE	OBJECTIVES	HOW IT IS LEARNED, AND WHAT GOES WRONG WHEN IT IS TAUGHT AS SOMETHING ELSE
Factual	56	Rehearsal and self-testing: cover, retrieve, check, repeat across days. Taught as concept, students can discuss it but cannot produce the list under pressure.
Conceptual	147	Explaining it aloud to somebody without notes. Taught as fact, students can recite the principle and cannot use it on anything new.
Procedural	88	Carrying it out, more than once. Taught as fact or concept, this is the origin of "they know it but they cannot do it".
Metacognitive	not used	Being made to look back at how they worked. Not asked for anywhere in this curriculum.

OBJECTIVES WORKING ON MORE THAN ONE KIND

COMBINATION	OBJECTIVES	WHAT IT MEANS FOR TEACHING
Conceptual + Procedural	12	Part of the objective is learned one way and part another. Split the session accordingly rather than choosing one.
Factual + Conceptual	8	Part of the objective is learned one way and part another. Split the session accordingly rather than choosing one.
Factual + Procedural	1	Part of the objective is learned one way and part another. Split the session accordingly rather than choosing one.
Factual + Conceptual + Procedural	1	Part of the objective is learned one way and part another. Split the session accordingly rather than choosing one.

The kind of knowledge is read from the wording of each objective — its verb and what the verb acts on. Where the wording supports a reading without settling it, the Unit marks the label with a small circle (°).

PART C · APPENDIX C6

Level against priority

How Must know, Desirable and Nice to know are spread across the levels. Useful when deciding what can be moved to self-directed learning and what cannot.

LEVEL	NAME	MUST KNOW	DESIRABLE	NICE TO KNOW
Level 1	**Knowledge** / Remember	27	46	13
Level 2	**Comprehension** / Understand	35	26	7
Level 3	**Application** / Apply	21	10	4
Level 4	**Analysis** / Analyse	—	4	—
Level 5	**Synthesis** / Create	—	—	—
Level 6	**Evaluation** / Evaluate	4	6	—
Doing	all levels	20	9	4
Going about it	all levels	13	18	1

Priority is an independent judgement, not part of Bloom. It does not rise with level, and it should not be assumed to: an objective can be Must know at level 1 and Nice to know at level 6.

PART C · APPENDIX C7

Teaching methods named in the curriculum

The methods that appear in Column F3 and F4, grouped by the level each serves best. Expand the abbreviations here when reading the syllabus document.

BEST SUITED TO	METHODS THE CURRICULUM NAMES
Level 1	Lecture (L), Lecture with PowerPoint (L&PP), Lecture with video clips (L_VC), Recitation (REC), Game-based learning (GBL), Word puzzle (WP), Quiz (QZ), Puzzle (PUZ), Self-directed learning (SDL), Library session (LS)
Level 2	Discussion (DIS), Lecture and group discussion (L&GD), Flipped classroom (FC), Brainstorming (BS), Peer learning (PER), Peer learning (PL), Group interaction (GRI), Blended learning (BL), Mobile learning (ML)
Level 3 and above	Problem-based learning (PBL), Case-based learning (CBL), Project-based learning (PrBL), Team-based learning (TBL), Inquiry-based (IBL), Inquiry based (IB), Team project work (TPW), Practical training (PT), Demonstration (D), Demonstration with lecture (D_L), Simulation (SIM), Role play (RP), Field visit (FV), Real-life experience (RLE), Educational visit (EDU), Early clinical exposure (ECE), Debate (DEB), Presentation (PRN)

A method is not fixed to one level, but each has a level it serves best. Lecture builds level 1 well and level 2 poorly. From level 2 upward students must handle the material themselves, which is what the discussion and self-directed hours are for.

PART C · APPENDIX C8

Assessment tools named in the curriculum

The tools that appear in Column G3 and G4, grouped by what kind of evidence each produces. Match the kind to the objective before choosing the tool.

KIND	TOOLS THE CURRICULUM NAMES	WHAT THIS KIND CAN SHOW
Written	Theory test (TT-Theory, 16), Open book test (T-OBT, 10), Extended matching items (T-EMI, 5), Constructed response questions (T-CRQs, 3), Problem solving (P-PS, 2), Case study (T-CS, 2)	Reasoning, and thinking at any level if the stem is built for it.
Spoken	Practical viva (P-VIVA, 57), Viva, knowledge check (V-KC, 19), M-POSF (M-POSF, 9), QZ (QZ, 7), DEB (DEB, 7), Interview (C-INT, 2), M-CHTF (M-CHTF, 2), Viva (VV-Viva, 2), PUZ (PUZ, 1), WP (WP, 1), RK (RK, 1)	Whether a student knows and can explain. **It cannot show whether they can carry out a procedure.**
Observed	Practical examination (P-EXAM, 21), Checklist (CHK, 7), OSPE station (OSPE, 6), Identification (P-ID, 4), Practical performance (P-PRF, 4)	Performance directly. The only evidence that settles a doing objective.
Over time	Survey (P-SUR, 11), Compilation (COM, 6), Practical record (P-REC, 4)	Habit and consistency. The right evidence for attitude objectives.
Product	Presentation (PRN, 25), Poster (M-POS, 14), Chart (M-CHT, 2), Poster (P-POS, 1)	The result. Say in advance what it will be marked against.

Counts are how many objectives name each tool. Where a doing objective is checked only by a spoken tool, add your own observation: a checklist ticked while the student works is enough, and it is the only record showing the procedure was performed.

PART C · APPENDIX C9

Sources and further reading

The primary sources for the taxonomies used in this book, the curriculum it is built on, and a note on what was and was not altered.

THE TAXONOMIES

- Bloom, B. S. (ed.), *Taxonomy of Educational Objectives: The Classification of Educational Goals. Handbook I: Cognitive Domain.* New York: Longmans, Green, 1956. The original, and the scheme the AyUG-RB curriculum uses.
- Anderson, L. W. and Krathwohl, D. R. (eds.), *A Taxonomy for Learning, Teaching and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives.* New York: Longman, 2001. The revision: verb names, the swapped top two levels, and the knowledge axis.
- Krathwohl, D. R., Bloom, B. S. and Masia, B. B., *Taxonomy of Educational Objectives, Handbook II: Affective Domain.* New York: David McKay, 1964. The source of the five affective levels.
- Simpson, E. J., *The Classification of Educational Objectives in the Psychomotor Domain.* Washington: Gryphon House, 1972; and Dave, R. H., 'Psychomotor levels', in R. J. Armstrong (ed.), *Developing and Writing Behavioral Objectives*, 1970. The psychomotor levels used in the curriculum come from this later work rather than from Bloom's committee.
- Krathwohl, D. R., 'A Revision of Bloom's Taxonomy: An Overview', *Theory into Practice*, 41(4), 2002. A short account of what changed in 2001 and why.

THE CURRICULUM

- National Commission for Indian System of Medicine, Board of Ayurveda. *Course curriculum for Second Professional BAMS — Rasashastra evam Bhaishajyakalpana (AyUG-RB).* New Delhi: NCISM. Applicable from the 2021–22 batch, academic year 2023–24 onwards. Tables 2, 3, 4 and 4a are the source of every objective, code, priority, method and tool in this book.

A NOTE ON HOW THIS BOOK WAS MADE

Every objective in Part B is reproduced in the curriculum's own words, with two exceptions, both mechanical. Plain-English spelling errors in the source have been corrected — *Discribe* to *Describe*, *Expalin* to *Explain*, and about thirty others — and sentence punctuation lost to the source document's line wrapping has been restored. No Sanskrit or technical term has been altered anywhere, and no subject content has been added, removed or reworded.

The counts, level assignments and tables in Part C are generated from the same data as the Units, so the two can never disagree. Where a judgement had to be made — chiefly in reading the kind of knowledge from an objective's wording — it is made from the verb and its grammatical object, marked with a small circle (°) where the wording supports a reading without settling it, and left visible for the reader to overrule.

A COMPETENCY-ASSESSMENT HANDBOOK

Miller's Pyramid for Rasashastra & Bhaishajya Kalpana

Topic-by-topic competency mapping, assessment design and usage guide for AyUG-RB — Second Professional BAMS
Ayurvediya Aushadhi Nirmana & Prayoga Vigyana

Curriculum	NCISM, Board of Ayurveda — applicable from the 2021–22 batch, academic year 2023–24 onwards
Coverage	2 papers · 26 theory topics · 10 practical groups · 268 mapped learning objectives
Framework	Miller (1990) four tiers, with the Cruess *Is* tier (2016), mapped onto NCISM Table 3 Column E3 and Table 4 Column E4

450	240	210	400
TEACHING HOURS	THEORY HOURS	PRACTICAL HOURS	TOTAL MARKS

Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026 · For departmental use in curriculum delivery, assessment design and paper setting.

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Unit 1 Chronological development of Ayurvediya Aushadhi 7 obj · Knows how 11

Unit 2 Paribhasha (Terminology) 19 obj · Shows how 13

Unit 3 Adharbhuta Siddhanta (Application of fundamental 11 obj · Knows how 15

Unit 4 Yantropakaranani- I (Equipments and machineries) 6 obj · Knows how 17

Unit 5 Yantropakaranani -II (Equipments, fuel and Heating 21 obj · Does 19

Unit 6 Kalpana Nirmana I (Primary & Secondary dosage 11 obj · Knows how 21

Unit 7 Kalpana Nirmana-II (Method of Preparation of 4 obj · Knows how 23

Unit 8 Rasa Dravya Parichaya- I 9 obj · Shows how 25

Unit 9 Rasa Dravya Parichaya II 7 obj · Knows how 27

Unit 10 Rasadravya Parichaya III 5 obj · Knows 29

Unit 11 Kalpana Nirman -III (Method of Preparation of 6 obj · Knows how 31

Unit 12 Chaturvidha Rasayana 6 obj · Does 33

Unit 13 Current and emerging trend in Ayurvedic 6 obj · Knows how 35

Unit 14 GMP(Schedule T) & Regulatory aspects of Ayurvedic 7 obj · Knows 37

PART B — PAPER 2, AYURVEDIYA AUSHADHI PRAYOGA VIGYANA

Unit 1 Aushadhi Prayoga Vigyana 1 obj · Knows 39

Unit 2 Single drug (Herbal & Mineral) 5 obj · Knows how 40

Unit 3 Single drug(Bhasma, Shuddha & Pishti) 5 obj · Shows how 42

Unit 4 Aushadhi Kalpa -I (Compound formulations) 5 obj · Knows how 44

Unit 5 Aushadhi Kalpa-II (Compound Drugs/Formulations) • Chemical/phytochemical composition 7 obj · Knows how 46

Unit 6 Dosage Forms & Cosmetic Products 12 obj · Knows how 48

Unit 7 Nutraceuticals 9 obj · Knows how 50

Unit 8 Anupana Prayoga for Aushadhi Kalpa 4 obj · Knows how 52

Unit 9 Aushadhi Prayoga Marga 5 obj · Knows how 54

Unit 10 Rational prescription along with safe dispensing of 6 obj · Shows how 56

Unit 11 Traditional & Local health Practices 4 obj · Knows how 58

Unit 12 Pharmacovigilance for Ayurveda drugs 6 obj · Knows how 60

PART B — PRACTICAL UNITS, NCISM TABLE 4

Unit 1 Paribhasha concept based Practicals 9 obj · Does 62

Unit 2 Panchavidha Kashaya Kalpana & their Upakalpana Practicals 4 obj · Shows how 64

Unit 3 Rasa Dravya Aushadhi Nirmana Practicals 7 obj · Does 66

Unit 4 Bheshaja Kalpana Practicals -I 14 obj · Shows how 68

Unit 5 Bheshaja Kalpana Practicals -II 11 obj · Does 70

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Unit 6 Dosage Forms & Self-care Products Practicals 2 obj · Knows how 72
Unit 7 Field Visit/ Study Tour 7 obj · Does 74
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PART C — REFERENCE

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PART A

Foundations

Five short chapters. The first explains what Miller's pyramid is and, just as importantly, what it is not. The second shows that it is already inside your curriculum, in a column most teachers have never had pointed out. The third gives the three rules of constructive alignment, the fourth sets out which instrument can certify which tier, and the fifth explains how to read a unit in Part B. Read straight through once; after that, use the units.

1. **Miller's pyramid** — what it is, and what it is not
2. **The pyramid is already inside your curriculum** — reading Column E3
3. **Constructive alignment** — the three rules
4. **The assessment toolkit**, tier by tier
5. **How to use a unit in Part B**

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PART A · CHAPTER 1

Miller's pyramid — what it is, and what it is not

In 1990 George Miller, a medical educator, published a single diagram that has shaped assessment ever since. He was answering a question every examiner faces: *what does this test actually prove?*

His answer was a pyramid of four tiers, each demanding evidence more authentic than the one below.

- **Knows** — the student can recall it.
- **Knows how** — the student can reason with it, apply it to a case on paper.
- **Shows how** — the student can do it when someone is watching, in a set-up arranged for the purpose.
- **Does** — the student does it in real practice, unprompted, when nobody is checking.

The tiers are cumulative but not interchangeable. A student who reaches *Does* necessarily passed through the others; a student who has only been tested at *Knows* has been told nothing about the tiers above. That asymmetry is the whole point. **Evidence from a lower tier never certifies a higher one.**

WHAT THE PYRAMID IS NOT

It is not a taxonomy of thinking. It says nothing about whether a task is easy or hard, or whether it calls for recall or judgement. A *Knows* objective can be intellectually demanding and a *Does* objective can be routine. Bloom's taxonomy answers "what kind of thinking is this?"; Miller answers "how authentic must the evidence be?" The two are independent axes, and this curriculum carries both — Bloom in one column, Miller in another. Each has its own companion volume.

It is also not a ranking of worth. A curriculum that stops at *Shows how* for a dangerous procedure is not being lazy; it is making a judgement about supervision and safety. Reading the ceilings as a scoreboard is the commonest misuse of the pyramid.

> **The Cruess addition.** In 2016 Richard and Sylvia Cruess proposed a fifth tier above *Does*: **Is** — professional identity. Not what the student does but who they have become. It cannot be examined in a hall, and it is not coded anywhere in your curriculum. It is worth naming because some of what you certify at *Does* — a safety briefing recorded honestly, a result reported when it was inconvenient — is really evidence of *Is*. Where a unit in Part B says so, that is what it means.

WHY IT MATTERS HERE

This curriculum is weighted towards the lower tiers — 127 of its 268 objectives sit at *Knows* and a further 107 at *Knows how*. Only 34 reach *Shows how* or *Does*. For a second-year undergraduate that is a defensible shape, but it means the practical hours carry almost all of the evidence that cannot be gathered on paper, and they are easy to let slip.

PART A · CHAPTER 2

The pyramid is already inside your curriculum

Miller is not something this book adds to your curriculum. It has been there since publication, in a column most teachers have never had pointed out.

COLUMN E3

Open Table 3 (theory) and Table 4 (practical). The column headed **Level** carries one of four values against every objective: *Know*, *Knows how*, *Shows how* or *Does*. Those are Miller's four tiers, and they are the curriculum's statement of how authentic the evidence must be before that objective can be called met.

Across the 268 objectives of this curriculum the distribution is:

TIER	OBJECTIVES	SHARE
Knows	127	47.4%
Knows how	107	39.9%
Shows how	23	8.6%
Does	11	4.1%

Total 268 objectives.

Two thirds of this curriculum can be certified on paper. The remaining third cannot, and it is concentrated in the practical groups. If the practical record is treated as an administrative task, that third goes uncertified while the paperwork looks complete.

> **A word about the other column headed "Domain".** Beside the Level column sits another carrying codes like CK, CC, CAP, PSY-MEC and AFT-VAL. Those are Bloom's taxonomy and the psychomotor and affective domains — a different question entirely, and the subject of the companion volume. In *this* book, "level" and "tier" always mean Miller.

THE CEILING

Every unit in this curriculum has a **ceiling**: the highest tier any of its objectives reaches. That single fact governs assessment more than anything else on the page. It tells you what evidence you must gather, and it tells you what you may not ask for. Part B prints it for every unit.

PART A · CHAPTER 3

Constructive alignment — the three rules

Constructive alignment is the principle that what you teach, what you ask students to do, and what you accept as proof must all sit at the same tier. Stated as rules, it comes to three.

RULE 1 — THE TIER DECIDES THE INSTRUMENT

Not the topic, not the hours, not the marks. Once you know an objective sits at *Shows how*, the set of instruments that can certify it is fixed, and a written paper is not among them. Chapter 4 gives the toolkit tier by tier.

RULE 2 — NEVER ASSESS ABOVE THE CEILING

If the curriculum certifies a unit to *Knows how*, do not build an OSPE station for it. You would be examining something the students were never taught to that standard, and the marks would measure prior exposure rather than your teaching. This rule protects students.

RULE 3 — NEVER CERTIFY A HIGHER TIER WITH LOWER-TIER EVIDENCE

This is the rule that gets broken, and it is broken in one direction. A *Does* objective assessed by viva has not been assessed. A *Shows how* objective assessed by a long answer has not been assessed. The student may pass, the record may look complete, and nothing has been established.

> **The asymmetry is the thing to remember.** You may always gather evidence from a tier *above* the one required — a student observed doing the thing has plainly also shown how. You may never accept evidence from a tier *below*. Every alignment error worth catching is a violation of that one sentence.

WORKING THE THREE RULES BACKWARDS

In practice the fastest way to plan a session is to start at the end. Ask what evidence you will hold at the finish — a script, a checklist, a log entry. That names the tier. The tier names the instrument, and the instrument names the teaching that must precede it. A session planned forwards from the syllabus tends to arrive at the assessment it can most easily set, which is almost always a written one.

PART A · CHAPTER 4

The assessment toolkit, tier by tier

Each tier admits a particular set of instruments. Using one from the wrong tier is the single commonest fault in assessment design, and it is invisible unless you look for it, because the paperwork is complete either way.

TIER	THE QUESTION IT ANSWERS	TEACHING THAT BUILDS IT	INSTRUMENTS THAT CERTIFY IT	EVIDENCE YOU KEEP
Knows	**Can the student recall it?**	Lecture, reading, recitation, retrieval practice.	MCQ, short answer, oral quiz, spotting.	Written script, quiz sheet.
Knows how	**Can the student reason with it?**	Worked cases, discussion, problem-based work; the student must reason aloud and be corrected.	Structured long answer, case-based questions, structured viva.	Reasoned written answer, viva score sheet.
Shows how	**Can the student do it when watched?**	Demonstration, then supervised practice with your help withdrawn a little at a time.	OSPE or OSCE station, observed performance against a checklist.	Checklist score, annotated artefact, station record.
Does	**Does the student do it unprompted, in real practice?**	Real work with real consequences, supervised at a distance rather than at the elbow.	Log book, record book, portfolio, observation over time, presentation with peer feedback.	Dated log entry, countersigned record, portfolio piece.

THE TRAP AT EACH TIER

- **Knows** — Spending long-answer marks here. A long answer cannot separate a student who knows the list from one who knows it better.
- **Knows how** — Setting a question answerable from memory. If the stem can be met by recall, the objective has silently dropped to Knows.
- **Shows how** — Certifying by viva. A fluent student who has never handled the material will pass a viva comfortably.
- **Does** — Certifying with a written test. Nothing written can show that a student does something unprompted when nobody is watching.

> **The instruments this curriculum names.** This curriculum names its instruments in Column G3 and G4 — TT-Theory, T-EW, T-OBT, P-VIVA, P-ID, OSPE, checklists, the log book and the practical record. Appendix C3 sorts every one of them by the tier it can honestly reach.

ON THE LOG BOOK AND RECORD BOOK

These are the only instruments in routine use that can carry *Does*-tier evidence, and they are usually treated as paperwork. A log filled in at the end of term is a piece of writing. The same log filled in as the work happens, dated and countersigned, is the only proof you will have that a student did something unprompted. Decide at the start of each unit what artefact each *Does* objective will leave behind, and the record becomes evidence rather than an administrative task.

PART A · CHAPTER 5

How to use a unit in Part B

Part B has twenty-six theory units and ten practical units. Each begins on its own page and is laid out the same way, so once you have read one you can use any of them.

WHAT IS ON THE PAGE

1. **The heading and counters** — hours, objectives, marks, and the **ceiling**: the highest tier this unit reaches. Read the ceiling first; it governs everything else.
2. **The session map** — the curriculum's own objectives with the tier each is rated to, so you can see at a glance where the weight sits.
3. **The pyramid for this unit** — four rows, one per tier. Each names what a student must be able to do at that tier here, then how to build it, how to assess it, and what evidence you keep. Tiers the curriculum does not use are shown greyed, because knowing what is *not* certified matters as much as knowing what is.
4. **The ceiling statement** — a single line naming the highest tier and warning against assessing above it or certifying it with the wrong instrument.
5. **Two panels** — one for the teacher, one to give to students.

THREE WAYS TO USE IT

Planning a session. Start at the ceiling row of the pyramid and work down: that row tells you what the students must end up able to do, and the rows beneath tell you what has to be in place first.

Setting a paper. A written paper can honestly reach *Knows* and *Knows how* and no further. Use the session map to see which objectives those are; anything above must be certified elsewhere, and the unit says where.

Auditing what you already do. Take your existing assessment for a unit and ask which tier it can certify. If it is a written test and the ceiling is *Shows how* or *Does*, the gap is your finding.

> **What this book does not do.** It does not rewrite the curriculum and does not tell you what to teach — every objective in Part B appears in the curriculum's own words. What it adds is the Miller layer: what standard of evidence each objective demands, and which instruments can supply it.

PART B

The units

Thirty-six units: twenty-six following the theory topics of Papers 1 and 2, and ten following the practical topics of Table 4. Each carries the curriculum's own objectives gathered by tier, then the pyramid for that unit — what each tier asks for here, how to build it, how to assess it, and what evidence you keep. Chapter 5 explains the layout.

1. **Paper 1** — Ayurvediya Aushadhi Nirmana Vigyana, units 1 to 14
2. **Paper 2** — Ayurvediya Aushadhi Prayoga Vigyana, units 1 to 12
3. **Practical** — NCISM Table 4, units 1 to 10

↑ Contents

PAPER 1 · UNIT 1

Chronological development of Ayurvediya Aushadhi Nirmana

Hours: 2 lecture · 1 non-lecture | Term 1 | Theory marks: 5 | **7 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	2	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 5 OBJECTIVES		
Explain historical evolution of Ayurvediya aushadhi nirman and Rasashastra.	Must know	TT-Theory
Describe about contribution of Nagarjuna Acharya to Rasashastra.	Must know	TT-Theory
Enlist important classical texts of Rasashastra and describe their unique features in short.	Desirable	TT-Theory
Describe Recent development in Ayurvedic Pharmaceuticals viz new dosage forms, pharmaceutical modification techniques.	Desirable	TT-Theory
Define Rasa and Rasayana and describe difference between Rasa and Rasayana.	Desirable	—
KNOWS HOW — 2 OBJECTIVES		
Describe structure of Pharmacy and enlist formulations prepared in pharmacy, after visiting the unit of teaching pharmacy of own campus.	Must know	—
Justify design of ancient Rasashala.	Nice to know	PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Describe structure of Pharmacy and enlist formulations prepared in pharmacy, after visiting the unit of teaching pharmacy of own campus; Justify design of ancient Rasashala.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Explain historical evolution of Ayurvediya aushadhi nirman and Rasashastra; Describe about contribution of Nagarjuna Acharya to Rasashastra; Enlist important classical texts of Rasashastra and describe their unique features in short; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 2

Paribhasha (Terminology)

Hours: 8 lecture · 4 non-lecture | Term 1 | Theory marks: 10 | **19 objectives** | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
10	8	1	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 10 OBJECTIVES		
Explain the term Paribhasha and its importance in Ayurvediya Aushadhi Nirmana.	Must know	T-CRQs
Describe the terms Aushadha, Bheshaja, Kalpana, Kashaya Kashaya yoni, Samskara- based on their word derivations.	Must know	—
Enlist sequentially - names of all drugs classified in the varga(group). Recite shlokas of Maharasa, Uprasa, Sadharana Rasa Varga from Rasaratnasamuchchaya.	Must know	P-ID, P-REC, PUZ
Enlist and discuss Sandigdha, Krutrima, Pratinidhi and Anupalabdha dravya.	Desirable	P-ID
Explain the term Marana and describe its types with examples.	Must know	T-OBT
Enlist all relevant prakriya paribhasha of Amrutikarana Lohitikarana, Sattvapatana, druti and discuss with examples.	Desirable	PRN
Identify names of Parada Bandhas.	Nice to know	T-OBT
Recognise and discuss Dhanvantari Bhaga and Rudra bhaga.	Nice to know	C-INT
Explain importance of ethical practices for drug processing(Shodhan, Marana).	Must know	P-POS
Explain value of keen and accurate application of weights and measures in Ayurvediya Aushadhi nirmana.	Desirable	PRN
KNOWS HOW — 8 OBJECTIVES		
Describe the definition of the term Shodhana. Explain with examples different techniques used for the procedures of Shodhana.	Must know	QZ
Recite sequentially names of Parada Ashta samskara. Explain the terms Jarana Murchchhana and cite types with examples. Compare Jarana and Murchchhana.	Must know	M-CHT, PRN, T-EMI
Illustrate all Bhasma pariksha as per classical description.	Must know	P-EXAM
Define the word Siddhilakshani. Recite examples of classical siddhilakshani. Interpret its rationality.	Must know	—
Explain importance of Mana-paribhasha and classical types of Mana. Recall Charts of Mana.	Desirable	T-CRQs
Categorize parameters of Drug Standardization and develop a checklist for assessment of quality of rasadravyas.	Desirable	WP
Explain Value of selection of genuine raw material.	Must know	DEB
Discuss and justify importance of Bhasma Pariksha.	Must know	—
SHOWS HOW — 1 OBJECTIVE		
Describe Grahyagrahya parameters used for selection of Rasadravyas.	Must know	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Describe Grahyagrahya parameters used for selection of Rasadravyas.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
8 objectives

Describe the definition of the term Shodhana. Explain with examples different techniques used for the procedures of Shodhana; Recite sequentially names of Parada Ashta samskara. Explain the terms Jarana Murchchhana and cite types with examples. Compare Jarana and Murchchhana; Illustrate all Bhasma pariksha as per classical description; and 5 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
10 objectives

Explain the term Paribhasha and its importance in Ayurvediya Aushadhi Nirmana; Describe the terms Aushadha, Bheshaja, Kalpana, Kashaya Kashaya yoni, Samskara- based on their word derivations; Enlist sequentially - names of all drugs classified in the varga(group). Recite shlokas of Maharasa, Uprasa, Sadharana Rasa Varga from Rasaratnasamuchchaya; and 7 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 3

Adharbhuta Siddhanta (Application of fundamental principles)

Hours: 4 lecture · 2 non-lecture | Term 1 | Theory marks: 5 | **11 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	9	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 2 OBJECTIVES		
Explain Saveeryata Avadhi(Shelf life) of Ayurvedic dosage forms.	Desirable	—
Explain importance of Kala (Time) Samskara in Ayurvediya Aushadhi Nirmana. Elaborate Aushadhi sevana kala mentioned in Sharangdhara samhita. Describe chrono- Therapeutics.	Must know	TT-Theory
KNOWS HOW — 9 OBJECTIVES		
Elaborate fundamental principles of Ayurvediya Aushadhi Nirmana alongwith their classical references and discuss their application with classical examples of various kalpas.	Must know	M-POSF
Explain Dravya Samgraha vidhi. Explain types of Bhumi desha and types of drugs to be collected from particular place & places from where Dravya should not be collected.	Must know	—
Describe time of Dravya collection and explain rationality behind.	Must know	—
Enlist parts of Sthavara & Jangama Dravyas used for Ayurvediya Aushadhi nirmana.	Must know	—
Define Samskara. Discuss role of Samskara in Ayurvediya Aushadhi Nirmana alongwith various examples mhita.	Must know	—
Describe Ardra - Shushka Dravya & Anukta – Visheshokta Dravya collection principles.	Must know	—
Describe importance of kalpa sevan matra. Explain factors considered for deciding dosage of any drug (Ayurvedic as well as modern medicine principles). Describe Posology.	Must know	—
Discuss Yogika Dravya Sidhdhanta(Drug combination).	Must know	—
Justify Aushadhi kalpa namakarana siddhanta with examples.	Desirable	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Elaborate fundamental principles of Ayurvediya Aushadhi Nirmana alongwith their classical references and discuss their application with classical examples of various kalpas; Explain Dravya Samgraha vidhi. Explain types of Bhumi desha and types of drugs to be collected from particular place & places from where Dravya should not be collected; Describe time of Dravya collection and explain rationality behind; and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Explain Saveeryata Avadhi(Shelf life) of Ayurvedic dosage forms; Explain importance of Kala (Time) Samskara in Ayurvediya Aushadhi Nirmana. Elaborate Aushadhi sevana kala mentioned in Sharangdhara samhita. Describe chrono-Therapeutics.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 4

Yantropakaranani- I (Equipments and machineries)

Hours: 6 lecture · 4 non-lecture | Term 1 | Theory marks: 5 | **6 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
4	2	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 4 OBJECTIVES		
Choose and record contemporary machines used in Ayurvedic drug preparation.	Desirable	—
Discuss the pharmaceutical use of Distillation apparatus, Ball Mill, Pulveriser, End Runner, Edge Runner, Tablet compression machine, Capsule filling machine, Pouch filling machine, Liquid filling machine in Ayurvediya Aushadhi Nirmana.	Desirable	—
Enlist categorical information about the following Yantras in the charts- Ulukhal Yantra, Patan Yantra, Jarana Yantra, Patala Yantra and Swedana Yantra.	Desirable	—
Enlist Ayurvediya aushadhi kalpana and equipments/yantras/ machines used for preparation of each kalpana.	Desirable	—
KNOWS HOW — 2 OBJECTIVES		
Describe the principles behind construction and working of the classical Yantras used for Ayurvediya Aushadhi nirmana.	Must know	—
Interpret the mechanism and effect of Yantras / machines on the Physical and Chemical properties of the drug material.	Desirable	M-POS

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Describe the principles behind construction and working of the classical Yantras used for Ayurvediya Aushadhi nirmana; Interpret the mechanism and effect of Yantras / machines on the Physical and Chemical properties of the drug material.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
4 objectives

Choose and record contemporary machines used in Ayurvedic drug preparation; Discuss the pharmaceutical use of Distillation apparatus, Ball Mill, Pulveriser, End Runner, Edge Runner, Tablet compression machine, Capsule filling machine, Pouch filling machine, Liquid filling machine in Ayurvediya Aushadhi Nirmana; Enlist categorical information about the following Yantras in the charts- Ulukhal Yantra, Patan Yantra, Jarana Yantra, Patala Yantra and Swedana Yantra; and 1 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 5

Yantropakaranani -II (Equipments, fuel and Heating Devices)

Hours: 5 lecture · 4 non-lecture | Term 1 | Theory marks: 5 | **21 objectives** | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
13	7	—	1
certified here	certified here	not used	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 13 OBJECTIVES		
Describe the term Yantra and enlist yantras described in classics useful for aushadhi nirmana.	Desirable	TT-Theory
Define the term Puta and recite its classical explanation.	Must know	M-POSF
Produce categorical information(Size of Pit, Number of cowdunkcakes, use etc.) about following Putas viz. Mahaputa Gajaputa, Varahputa, Kukkutaputa, Kapotputa, Lavakputa Kumbhaputa, Bhandaputa, Valukaputa and Bhudharputa.	Must know	M-POS
Identify and enlist contemporary devices used in the preparation of Bhasma. T.	Desirable	M-POS, QZ
Review the temperature pattern of various Putas by referring research articles.	Desirable	M-POS, QZ
Compile names and significance of temperature measuring devices with reference to Puta.	Desirable	M-POSF
Enlist various devices used for heating during Ayurvediya Aushadhi nirmana.	Desirable	P-SUR
Enlist constituents needed for Samanya Musha Nirmana.	Desirable	—
Enlist and discuss the material used for Sandhi Bandhan (while using Sharava and Kachakupi).	Nice to know	T-OBT
Identify and record the types of Crucibles.	Nice to know	M-POS
Describe types of Koshthi and their uses.	Desirable	COM
Describe the Current trends in heat transfer device e.g. steam jacketed heating device, programmed muffle furnace programmed furnace for Parpati / Pottali preparation.	Desirable	COM, M-POS
Enlist various fuels used for heating for estimation of their heat value.	Nice to know	M-POS
KNOWS HOW — 7 OBJECTIVES		
Recognise the principles behind construction and working of the classical Putas.	Must know	M-POSF
Enlist the specific Puta used for a particular Bhasma Nirmana.	Desirable	M-POSF
Interpret the effect of heat transformation in the material subjected to heating through Musha.	Desirable	PRN
Recall Pakaj Utpatti Siddhant and interpret it for Agni Sannikarsha Sanskara.	Desirable	—
Explain the Pharmaceutical use of the Hot plate, heating mantle induction stove, hot air oven, muffle furnace (horizontal and vertical type).	Nice to know	M-POSF, P-SUR
Assess and interpret the effect of fuel in quantum of heat given (time and temperature).	Desirable	M-POS
Analyse and appraise use of proper heating device and fuels used for Ayurvediya Aushadhi Nirmana.	Desirable	DEB
DOES — 1 OBJECTIVE		
Define the term Musha and describe uses of various types of Musha.	Desirable	T-OBT

THE PYRAMID FOR THIS UNIT

Does 1 objective	Define the term Musha and describe uses of various types of Musha. **Build:** Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.
Shows how not used here	This unit does not ask for Shows how. Nothing here needs to be certified at that standard. **Build:** Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.
Knows how 7 objectives	Recognise the principles behind construction and working of the classical Putas; Enlist the specific Puta used for a particular Bhasma Nirmana; Interpret the effect of heat transformation in the material subjected to heating through Musha; and 4 more. **Build:** Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.
Knows 13 objectives	Describe the term Yantra and enlist yantras described in classics useful for aushadhi nirmana; Define the term Puta and recite its classical explanation; Produce categorical information(Size of Pit, Number of cowdunkcakes, use etc.) about following Putas viz. Mahaputa Gajaputa, Varahputa, Kukkutaputa, Kapotputa, Lavakputa Kumbhaputa, Bhandaputa, Valukaputa and Bhudharputa; and 10 more. **Build:** Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 6

Kalpana Nirmana I (Primary & Secondary dosage forms)

Hours: 6 lecture · 4 non-lecture | Term 1 | Theory marks: 10 | **11 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
7	4	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 7 OBJECTIVES		
Define with synonyms and classify different varieties of the dosage form in Ayurvediya Aushadhi nirmana.	Must know	—
Enlist details of the applications/administration (samanya Matra Anupana or sahapana, indications and contraindications) of the dosage form with various examples.	Must know	—
Assess and discuss modern day development and market trend of the dosage form.	Desirable	—
Determine the advantages and disadvantages of the dosage form.	Desirable	—
Describe method of preparation of dosage form along with the principle involved.	Must know	—
Describe meaning of the term Upakalpana. Enlist panchavidha kashaya kalpana and their respective Upakalpana.	Desirable	—
Explain various kalpas belonging to Various dosage forms -Ingredients, proportion, matra, sevan vidhi if any, therapeutic.	Must know	P-EXAM
KNOWS HOW — 4 OBJECTIVES		
1. Explain Kashaya kalpana 2. Enlist the dosage form come under Panchavidha Kashaya Kalpana and their Upakalpana 3. Justify Primary, secondary and tertiary dosage form under Panchavidha kashaya Kalpana and their Upakalpana.	Must know	—
Describe methods of preparation of the dosage form along with the principles of extraction, concentration and dilution etc.	Must know	—
Explain in details about Standard Operating Procedure (SOP) of each kalpana.	Must know	—
Critically evaluate rationality behind different proportion of water used in various formulations.	Desirable	P-VIVA

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

1. Explain Kashaya kalpana 2. Enlist the dosage form come under Panchavidha Kashaya Kalpana and their Upakalpana 3. Justify Primary, secondary and tertiary dosage form under Panchavidha kashaya Kalpana and their Upakalpana; Describe methods of preparation of the dosage form along with the principles of extraction, concentration and dilution etc; Explain in details about Standard Operating Procedure (SOP) of each kalpana; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
7 objectives

Define with synonyms and classify different varieties of the dosage form in Ayurvediya Aushadhi nirmana; Enlist details of the applications/administration (samanya Matra Anupana or sahapana, indications and contraindications) of the dosage form with various examples; Assess and discuss modern day development and market trend of the dosage form; and 4 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 7

Kalpana Nirmana-II (Method of Preparation of different dosage forms& Dietary Supplements))

Hours: 5 lecture · 4 non-lecture | Term 1 | Theory marks: 10 | **4 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
3	1	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 3 OBJECTIVES		
Explain the basic principles involved, processing techniques, quality control parameters used/ involved in the preparation of Avaleha, Kalpana with examples, instruments used in small and Large scale production, Research updates.	Must know	T-OBT
Explain the basic principles involved, processing techniques, quality control parameters used/ involved in the preparation of Sneha Kalpana, Kalpana with examples instruments used in small and Large scale production, Concept of Avartana, Research updates on Snehakalpana, Market survey.	Must know	—
Describe the Significance of sandhana kalpana, classification Difference between Madya and Shukta Kalpana, general method of preparation, essential ingredients, Anukta Mana, Sandhana Vidhi, observations, Burning candle test, Lime water Test difference between Asava and Arishta, essential knowledge of sale and clinical practice, Research updates.	Must know	T-OBT
KNOWS HOW — 1 OBJECTIVE		
Explain Definition, significance of Pathya, types, general method of preparation Manda, Peya, Yavagu, Vilepi, Anna or Odana Kalpana, Krushara, Yusha, Takra, Khada, Kambalika Raga, Shadava, Related Research updates, Market survey of Dietary Supplements and Nutraceuticals.	Must know	P-EXAM

THE PYRAMID FOR THIS UNIT

Does not used here	This unit does not ask for Does. Nothing here needs to be certified at that standard. **Build:** Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.
Shows how not used here	This unit does not ask for Shows how. Nothing here needs to be certified at that standard. **Build:** Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.
Knows how 1 objective	Explain Definition, significance of Pathya, types, general method of preparation Manda, Peya, Yavagu, Vilepi, Anna or Odana Kalpana, Krushara, Yusha, Takra, Khada, Kambalika Raga, Shadava, Related Research updates, Market survey of Dietary Supplements and Nutraceuticals. **Build:** Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.
Knows 3 objectives	Explain the basic principles involved, processing techniques, quality control parameters used/ involved in the preparation of Avaleha, Kalpana with examples, instruments used in small and Large scale production, Research updates; Explain the basic principles involved, processing techniques, quality control parameters used/ involved in the preparation of Sneha Kalpana, Kalpana with examples instruments used in small and Large scale production, Concept of Avartana, Research updates on Snehakalpana, Market survey; Describe the Significance of sandhana kalpana, classification Difference between Madya and Shukta Kalpana, general method of preparation, essential ingredients, Anukta Mana, Sandhana Vidhi, observations, Burning candle test, Lime water Test difference between Asava and Arishta, essential knowledge of sale and clinical practice, Research updates. **Build:** Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 8

Rasa Dravya Parichaya- I

Hours: 12 lecture · 4 non-lecture | Term 2 | Theory marks: 10 | **9 objectives** | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
3	5	1	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 3 OBJECTIVES		
Enlist synonyms of Rasadravya and explain significance of that T.	Must know	—
Enlist Kalpas prepared from these Rasdravyas and therapeutic importance of that Rasadravya.	Desirable	PRN
Develop checklist for identification of genuine rasadravyas.	Must know	P-PS
KNOWS HOW — 5 OBJECTIVES		
Discriminate types, Grahya-Agrahyata and Dosha of Rasadravya.	Must know	P-VIVA
Explain Shodhan and Marana and other processing techniques of the Rasa- Dravya.	Must know	P-VIVA
Discuss probable physical, chemical changes occurring during process & after Shodhana and Marana of Parada, Haratala Tamra and Shankha.	Must know	P-EXAM
Explain research updates about Shilajatu, Naga, Mukta and Kapardika.?	Desirable	P-VIVA
Discuss probable physical, chemical changes occurring after Bhavana, Puta & whole process of Bhasma nirmana from Rasadravyas.	Must know	T-OBT
SHOWS HOW — 1 OBJECTIVE		
Explain classical & Mineralogical identification and Sources of Rasadravyas.	Must know	—

THE PYRAMID FOR THIS UNIT

Does not used here	This unit does not ask for Does. Nothing here needs to be certified at that standard. **Build:** Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.
Shows how 1 objective	Explain classical & Mineralogical identification and Sources of Rasadravyas. **Build:** Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.
Knows how 5 objectives	Discriminate types, Grahya-Agrahyata and Dosha of Rasadravya; Explain Shodhan and Marana and other processing techniques of the Rasa- Dravya; Discuss probable physical, chemical changes occurring during process & after Shodhana and Marana of Parada, Haratala Tamra and Shankha; and 2 more. **Build:** Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.
Knows 3 objectives	Enlist synonyms of Rasadravya and explain significance of that T; Enlist Kalpas prepared from these Rasdravyas and therapeutic importance of that Rasadravya; Develop checklist for identification of genuine rasadravyas. **Build:** Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 9

Rasa Dravya Parichaya II

Hours: 7 lecture · 6 non-lecture | Term 2 | Theory marks: 5 | **7 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
4	3	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 4 OBJECTIVES		
Explain classical & Mineralogical identification and their Sources.	Desirable	P-VIVA
Elaborate types, Grahya-Agrahyata and Dosha of Rasadravyas.	Desirable	P-VIVA
Explain Shodhana, Marana & and other processing techniques with probable chemical reactions.	Desirable	P-VIVA
Enlist names of rasadravyas and important kalpas prepared from respective dravya. Describe therapeutic importance of that Rasadravya.	Desirable	M-CHTF
KNOWS HOW — 3 OBJECTIVES		
Enlist the important synonyms of Rasadravyas.	Desirable	P-VIVA
Explain Shodhana, Marana & and other processing techniques with probable chemical reactions.	Must know	—
Explain research updates of Kasisa, Gauripashana and Akika.	Desirable	PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
3 objectives

Enlist the important synonyms of Rasadravyas; Explain Shodhana, Marana & and other processing techniques with probable chemical reactions; Explain research updates of Kasisa, Gauripashana and Akika.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
4 objectives

Explain classical & Mineralogical identification and their Sources; Elaborate types, Grahya-Agrahyata and Dosha of Rasadravyas; Explain Shodhana, Marana & and other processing techniques with probable chemical reactions; and 1 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 10

Rasadravya Parichaya III

Hours: 3 lecture · 6 non-lecture | Term 2 | Theory marks: 5 | **5 objectives** | Ceiling: **Knows**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	—	—	—
certified here	not used	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 5 OBJECTIVES		
Enlist Synonyms and sources.	Nice to know	M-CHTF, T-OBT
Determine types, Grahya-Agrahya, properties & classical as well as Mineralogical Identification of Rasadravyas.	Nice to know	CHK, P-VIVA
Explain Shodhana, Marana and Probable Chemical Changes.	Desirable	PRN
Explain research update of Badarashma.	Nice to know	P-VIVA
Enlist Rasadravyas from this group and describe important kalpas with their therapeutic utility.	Desirable	PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
not used here

This unit does not ask for Knows how. Nothing here needs to be certified at that standard.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Enlist Synonyms and sources; Determine types, Grahya-Agrahya, properties & classical as well as Mineralogical Identification of Rasadravyas; Explain Shodhana, Marana and Probable Chemical Changes; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows. Everything here can be certified on paper. Do not build practical stations for it — there is nothing above recall to observe. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 11

Kalpana Nirman -III (Method of Preparation of different dosage forms)

Hours: 8 lecture · 6 non-lecture | Term 2 | Theory marks: 10 | **6 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
4	2	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 4 OBJECTIVES		
Describe sharkara kalpana along with preparation method of Syrup. Explain therapeutic importance of prepared sharkara kalpana along with its shelf life.	Nice to know	P-VIVA
Describe Gudapaka preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Gudapaka along with its shelf life.	Nice to know	P-VIVA
Describe Lavana kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lavana kalpana along with its shelf life and packaging techniques.	Desirable	T-CRQs
Describe Lepa kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lepa kalpana, integration with Kayachikitsa and Panchakarma department along with its shelf life and packaging techniques.	Must know	V-KC
KNOWS HOW — 2 OBJECTIVES		
Describe Kshara kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Kshara kalpana and Ksharasutra integration with Shalya tantra department along with its shelf life and packaging techniques.	Must know	P-VIVA
Describe Ayaskriti kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Ayaskriti kalpana along with its shelf life and packaging techniques.	Nice to know	P-VIVA

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Describe Kshara kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Kshara kalpana and Ksharasutra integration with Shalya tantra department along with its shelf life and packaging techniques; Describe Ayaskriti kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Ayaskriti kalpana along with its shelf life and packaging techniques.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
4 objectives

Describe sharkara kalpana along with preparation method of Syrup. Explain therapeutic importance of prepared sharkara kalpana along with its shelf life; Describe Gudapaka preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Gudapaka along with its shelf life; Describe Lavana kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lavana kalpana along with its shelf life and packaging techniques; and 1 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 12

Chaturvidha Rasayana

Hours: 4 lecture · 4 non-lecture | Term 2 | Theory marks: 10 | **6 objectives** | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	4	1	1
not used	certified here	certified here	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS HOW — 4 OBJECTIVES		
Describe importance and unique features of Chaturvidha Rasayana - Kharaliya Rasayana, Parpati Rasayana, Kupipakwa Rasayana & Pottali Rasayana.	Must know	—
Determine role of Yantra, Agni & Kala for pharmaceutical process of chaturvidha rasayana.	Desirable	—
Describe chemical changes occurring during the preparation of chaturvidha rasayana and its impact on Pharmacological action of Chaturvidha Rasayana.	Desirable	—
Explain law of Definite proportion and proportion of Mercury and sulphur needed for preparation of sulphide of Mercury. in Chaturvidha rasayana.	Desirable	—
SHOWS HOW — 1 OBJECTIVE		
Explain shelf-life of Chaturvidha Rasayana.	Desirable	P-VIVA
DOES — 1 OBJECTIVE		
Explain definition, types and method of preparation of Chaturvidha Rasayana.	Must know	—

THE PYRAMID FOR THIS UNIT

Does
1 objective

Explain definition, types and method of preparation of Chaturvidha Rasayana.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Explain shelf-life of Chaturvidha Rasayana.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Describe importance and unique features of Chaturvidha Rasayana - Kharaliya Rasayana, Parpati Rasayana, Kupipakwa Rasayana & Pottali Rasayana; Determine role of Yantra, Agni & Kala for pharmaceutical process of chaturvidha rasayana; Describe chemical changes occurring during the preparation of chaturvidha rasayana and its impact on Pharmacological action of Chaturvidha Rasayana; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This unit does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **2 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 13

Current and emerging trend in Ayurvedic pharmaceuticals

Hours: 3 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **6 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	1	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 5 OBJECTIVES		
Discuss the classification of different dosage forms.	Desirable	M-POSF, P-VIVA
Explain the need of different dosage forms.	Desirable	M-POSF, P-VIVA
Enlist categorical information about the cosmetics used in daily routine. Appreciate Ayurvedic cosmetic products.	Desirable	M-POS, PRN
Choose and record the contemporary machineries used in the manufacture of cosmetics.	Nice to know	M-POS
Enlist Quality Control parameters of cosmetics preparations.	Desirable	M-POSF, P-VIVA
KNOWS HOW — 1 OBJECTIVE		
Appreciate Ayurvedic cosmetic products.	Desirable	P-SUR, PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
1 objective

Appreciate Ayurvedic cosmetic products.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Discuss the classification of different dosage forms; Explain the need of different dosage forms; Enlist categorical information about the cosmetics used in daily routine. Appreciate Ayurvedic cosmetic products; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 1 · UNIT 14

GMP(Schedule T) & Regulatory aspects of Ayurvedic drugs

Hours: 2 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **7 objectives** | Ceiling: **Knows**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
7	—	—	—
certified here	not used	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 7 OBJECTIVES		
Explain the legal and regulatory aspects of manufacturing, and sale of Ayurvedic drugs.	Must know	—
Describe acts and rules mentioned in Drug & Cosmetic Act 1940 & Rule 1945 and their relevance to Ayurvedic, Siddha, Unani (ASU) drugs.	Must know	QZ
Discuss the guidelines of Food Safety and Standards Authority of India (FSSAI) and FDA.	Desirable	—
Determine the principles and practice of establishment of Ayurvedic pharmacy.	Desirable	—
Discuss the NABL guidelines for testing laboratory (Chemical).	Nice to know	CHK
Explain long forms of these - (FSSAI) and (FDA), (CCRAS), (DCGI), (CDSCO).	Desirable	T-EMI
Discuss ethical aspect of large scale drug preparation in Ayurvedic Pharmacies.	Desirable	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
not used here

This unit does not ask for Knows how. Nothing here needs to be certified at that standard.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
7 objectives

Explain the legal and regulatory aspects of manufacturing, and sale of Ayurvedic drugs; Describe acts and rules mentioned in Drug & Cosmetic Act 1940 & Rule 1945 and their relevance to Ayurvedic, Siddha, Unani (ASU) drugs; Discuss the guidelines of Food Safety and Standards Authority of India (FSSAI) and FDA; and 4 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows. Everything here can be certified on paper. Do not build practical stations for it — there is nothing above recall to observe. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 1

Aushadhi Prayoga Vigyana

Hours: 1 lecture · 2 non-lecture | Term 1 | Theory marks: 5 | **1 objectives** | Ceiling: **Knows**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	—	—	—
certified here	not used	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 1 OBJECTIVE		
Define Aushadhi Prayoga Vigyana and its scope and enlist prashasta bheshaja laxana.	Must know	TT-Theory

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
not used here

This unit does not ask for Knows how. Nothing here needs to be certified at that standard.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Define Aushadhi Prayoga Vigyana and its scope and enlist prashasta bheshaja laxana.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows. Everything here can be certified on paper. Do not build practical stations for it — there is nothing above recall to observe. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 2

Single drug (Herbal & Mineral)

Hours: 8 lecture · 2 non-lecture | Term 1 | Theory marks: 10 | **5 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
3	2	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 3 OBJECTIVES		
Appreciate how the therapeutic efficacy varies depending on the dosage form.	Must know	DEB
Describe therapeutic efficacy of different formulations of Vishadrvya (Bhallataka), with its toxic effects and remedy.	Nice to know	P-VIVA
Describe therapeutic efficacy of different formulations of Guduchi, Gairika, Gandhaka.	Must know	P-VIVA
KNOWS HOW — 2 OBJECTIVES		
Describe different dosage forms prepared out of a single herb or mineral.	Must know	COM, P-VIVA, TT-Theory
Appraise multiple factors considered for preparation of various dosage forms from a single drug and creat list of more such examples.	Desirable	COM

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Describe different dosage forms prepared out of a single herb or mineral; Appraise multiple factors considered for preparation of various dosage forms from a single drug and creat list of more such examples.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
3 objectives

Appreciate how the therapeutic efficacy varies depending on the dosage form; Describe therapeutic efficacy of different formulations of Vishadrvya (Bhallataka), with its toxic effects and remedy; Describe therapeutic efficacy of different formulations of Guduchi, Gairika, Gandhaka.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 3

Single drug(Bhasma, Shuddha & Pishti)

Hours: 12 lecture · 6 non-lecture | Term 2 | Theory marks: 15 | **5 objectives** | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	3	1	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 1 OBJECTIVE		
Explain research updates and clinical evidences of Swarna Bhasma, Makshika Bhasma and Shankha Bhasma.	Desirable	PRN
KNOWS HOW — 3 OBJECTIVES		
Explain chemical form/composition, Pharmacodynamics and pharmacokinetics of Abhraka, Loha and Godanti Bhasma.	Desirable	P-VIVA
Explain therapeutic properties, dosage, Anupana, Pathyapathya duration of treatment, Sevana Kala, shelf life, important Yogas of each Bhasma.	Must know	—
Describe Apakwa Ashuddha Avidhi Bhasma Sevanajanya Vyadhi and their Shanti-upaya.	Desirable	P-VIVA
SHOWS HOW — 1 OBJECTIVE		
Describe in detail Amayika Prayoga (Therapeutic uses) of each Bhasma/Pishti with given references.	Must know	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Describe in detail Amayika Prayoga (Therapeutic uses) of each Bhasma/Pishti with given references.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
3 objectives

Explain chemical form/composition, Pharmacodynamics and pharmacokinetics of Abhraka, Loha and Godanti Bhasma; Explain therapeutic properties, dosage, Anupana, Pathyapathya duration of treatment, Sevana Kala, shelf life, important Yogas of each Bhasma; Describe Apakwa Ashuddha Avidhi Bhasma Sevanajanya Vyadhi and their Shanti-upaya.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Explain research updates and clinical evidences of Swarna Bhasma, Makshika Bhasma and Shankha Bhasma.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 4

Aushadhi Kalpa -I (Compound formulations)

Hours: 16 lecture · 4 non-lecture | Term 2 | Theory marks: 15 | **5 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	4	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 1 OBJECTIVE		
Explain research updates and clinical evidences of Arogyavardhini Gutika and Gandhaka Rasayana.	Desirable	P-VIVA
KNOWS HOW — 4 OBJECTIVES		
Describe reference, ingredients, therapeutic properties dosage and all administration details alongwith Anupana of each formulation.	Must know	—
Explain Pathya Apathya, Sevana Avadhi (duration of treatment) and shelf-life of each formulation.	Must know	—
Interpret probable mode of action of each formulation as per Ayurveda.	Desirable	—
Critically analyse compound drugs for their therapeutic actions mentioned in the classics.	Desirable	PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Describe reference, ingredients, therapeutic properties dosage and all administration details alongwith Anupana of each formulation; Explain Pathya Apathya, Sevana Avadhi (duration of treatment) and shelf-life of each formulation; Interpret probable mode of action of each formulation as per Ayurveda; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Explain research updates and clinical evidences of Arogyavardhini Gutika and Gandhaka Rasayana.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 5

Aushadhi Kalpa-II (Compound Drugs/Formulations)

Hours: 14 lecture · 2 non-lecture | Term 3 | Theory marks: 15 | **7 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
3	4	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 3 OBJECTIVES		
Describe Ingredients, indications, dose, anupana, pathyapathya duration of treatment, sevana kala, shelf life, and clinical evidences of Panchaguna Taila, Dadimavaleha, Bramhi Ghrita.	Desirable	P-VIVA
Explain ethical aspect of administration of compound drugs mainly regarding duration of the treatment given.	Desirable	—
Enlist the formulations with its dose, anupana and indications.	Nice to know	T-EMI
KNOWS HOW — 4 OBJECTIVES		
Describe Ingredients, therapeutic importance, dose, anupana pathya-pathya, duration of treatment, sevana kala, shelf life research updates and clinical evidences of Dashamoola Kwatha Pushyanuga Churna, Sanjivani Vati, Chitrakadi Gutika Simhanada Guggulu, Yogaraja Guggulu, Chyavanaprashavaleha Gandhakadya Malahara, Ashokarishta, Kutajarishta Panchagavya Ghrita, Bilvadi Gutika.	Must know	—
Describe Ingredients, indications, dose, anupana, pathyapathya duration of treatment, sevana kala, shelf life, and clinical evidences of Mahamanjistadi kwatha, Sudarshana Churna Vyoshadi Vati, Bala Chaturbhadra Rasa, Lavana Bhaskara Churna, Narayana Taila,, Neeliringadi Taila Aravindasava Kumaryasava.	Nice to know	P-VIVA
Understand & Recite following classical kalpas. Describe Amayika prayoga of each formulation. Understand probable mode of action as per AyurvedaSearch and. Record relevant reaserch articles. Create charts describing details of these drugs. Review market avaibality & poplularities of these drugs among practicing vaidyas & near by population.	Must know	P-EXAM, TT-Theory, V-KC
Describe ingredients, with its dose, therapeutic importance anupana and pathya-apathya along with its clinical evidences and Research updates.	Must know	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Describe Ingredients, therapeutic importance, dose, anupana pathya-pathya, duration of treatment, sevana kala, shelf life research updates and clinical evidences of Dashamoola Kwatha Pushyanuga Churna, Sanjivani Vati, Chitrakadi Gutika Simhanada Guggulu, Yogaraja Guggulu, Chyavanaprashavaleha Gandhakadya Malahara, Ashokarishta, Kutajarishta Panchagavya Ghrita, Bilvadi Gutika; Describe Ingredients, indications, dose, anupana, pathyapathya duration of treatment, sevana kala, shelf life, and clinical evidences of Mahamanjistadi kwatha, Sudarshana Churna Vyoshadi Vati, Bala Chaturbhadra Rasa, Lavana Bhaskara Churna, Narayana Taila,, Neeliringadi Taila Aravindasava Kumaryasava; Understand & Recite following classical kalpas. Describe Amayika prayoga of each formulation. Understand probable mode of action as per AyurvedaSearch and. Record relevant reaserch articles. Create charts describing details of these drugs. Review market avaibality & poplularities of these drugs among practicing vaidyas & near by population; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
3 objectives

Describe Ingredients, indications, dose, anupana, pathyapathya duration of treatment, sevana kala, shelf life, and clinical evidences of Panchaguna Taila, Dadimavaleha, Bramhi Ghrita; Explain ethical aspect of administration of compound drugs mainly regarding duration of the treatment given; Enlist the formulations with its dose, anupana and indications.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 6

Dosage Forms & Cosmetic Products

Hours: 5 lecture · 2 non-lecture | Term 3 | Theory marks: 5 | **12 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
11 certified here	1 certified here	— not used	— not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 11 OBJECTIVES		
Define the term- Dosage forms.	Must know	TT-Theory
Discuss the need of different dosage forms.	Must know	TT-Theory
Explain the classification of different dosage forms.	Desirable	TT-Theory
Enlist the solid / liquid / semisolid dosage forms.	Must know	T-EMI
Discuss the routes of administration of different dosage forms.	Must know	—
Describe the advantages and disadvantages of currently available dosage forms.	Must know	TT-Theory
Discuss the research updates about modification of classical Ayurvedic dosage forms.	Desirable	—
Discuss the relevant case studies of different dosage forms.	Nice to know	P-SUR
Define the term Cosmetics.	Desirable	TT-Theory
Explain the classification of cosmetics based on their application on the specific body parts.	Desirable	—
Differentiate between the ancient cosmetics and modern day cosmetics.	Desirable	TT-Theory
KNOWS HOW — 1 OBJECTIVE		
compare and appraise herbal cosmetics vs synthetic products.	Desirable	PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
1 objective

compare and appraise herbal cosmetics vs synthetic products.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
11 objectives

Define the term- Dosage forms; Discuss the need of different dosage forms; Explain the classification of different dosage forms; and 8 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 7

Nutraceuticals

Hours: 6 lecture · 1 non-lecture | Term 3 | Theory marks: 5 | **9 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	4	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 5 OBJECTIVES		
Describe importance of combination of aahara and aushadha.	Must know	—
Explain Nutraceuticals with its types.	Must know	TT-Theory
Enlist at least two dietary preparations from Ayurvedic classics which can serve as rasayana/ nutraceutical in-1) General Health2) Pediatric Health3) Geriatric Health4) Women(Garbhini/ Sutika) Health5) Cardio-protection6) Chronic illness-recovery stage.	Must know	—
Explain research updates about nutraceuticals.	Nice to know	M-POS
Enlist atleast two examples of Aushadhi siddha aahara with its indications for each category-->(Aharadravya belonging to following category and kalpa prepared from that aahara dravya and other aushadhi ingredients)1) Jala varga2) Dugdha varga3) Ikshu varga4) Suka-dhanya varga5) Shimbi Dhanya Varga6) Shaka Varga7) Phala Varga8) Lavana Varga9) Spices mhita.	Desirable	—
KNOWS HOW — 4 OBJECTIVES		
Corelate rasayana and nutraceutical with examples.	Desirable	T-OBT
Describe with examples mode of action of Ayurvedic nutraceuticals mhita.	Must know	PRN
Describe potential of Ayurvedic diet and Rasayana in sports endeavor.	Desirable	DEB
creat list of unique features of Ayurvedic neutraceuticals and appraise their current need.	Desirable	PRN

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Corelate rasayana and nutraceutical with examples; Describe with examples mode of action of Ayurvedic nutraceuticals mhita; Describe potential of Ayurvedic diet and Rasayana in sports endeavor; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Describe importance of combination of aahara and aushadha; Explain Nutraceuticals with its types; Enlist at least two dietary preparations from Ayurvedic classics which can serve as rasayana/ nutraceutical in-1) General Health2) Pediatric Health3) Geriatric Health4) Women(Garbhini/ Sutika) Health5) Cardio-protection6) Chronic illness-recovery stage; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 8

Anupana Prayoga for Aushadhi Kalpa

Hours: 4 lecture · 1 non-lecture | Term 3 | Theory marks: 5 | **4 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	2	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 2 OBJECTIVES		
Define and classify Anupana and Sahapana.	Must know	—
Describe value of proper selection of Anupanas with examples.	Desirable	M-CHT
KNOWS HOW — 2 OBJECTIVES		
Explain different factors to be considered for selection of Anupana as per Disease and Patients.	Must know	—
Enlist different anupana for a single drug based on the condition of patient and disease. Elaborate with the help of examples of kalpas.	Must know	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Explain different factors to be considered for selection of Anupana as per Disease and Patients; Enlist different anupana for a single drug based on the condition of patient and disease. Elaborate with the help of examples of kalpas.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Define and classify Anupana and Sahapana; Describe value of proper selection of Anupanas with examples.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 9

Aushadhi Prayoga Marga

Hours: 5 lecture · 1 non-lecture | Term 3 | Theory marks: 10 | **5 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
4	1	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 4 OBJECTIVES		
Discuss the various types of Aushadhi Prayog Marga (route of drug administration).	Must know	—
Describe the advantages and disadvantages of each Aushadhi Prayog Marga.	Desirable	P-SUR
Enlist the dosage forms used in the specific Aushadhi Prayog Marga.	Desirable	T-EMI
Appraise administration of drugs through various routes mentioned in Ayurvedic treatment.	Must know	—
KNOWS HOW — 1 OBJECTIVE		
Discuss the nature of drug (s) administered in various routes of drug administration.	Nice to know	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
1 objective

Discuss the nature of drug (s) administered in various routes of drug administration.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
4 objectives

Discuss the various types of Aushadhi Prayog Marga (route of drug administration); Describe the advantages and disadvantages of each Aushadhi Prayog Marga; Enlist the dosage forms used in the specific Aushadhi Prayog Marga; and 1 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 10

Rational prescription along with safe dispensing of Ayurvedic formulations.

Hours: 1 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **6 objectives** | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	3	2	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 1 OBJECTIVE		
Explain the importance of rational prescribing of drugs and the concept of essential drugs.	Desirable	CHK, QZ, V-KC
KNOWS HOW — 3 OBJECTIVES		
Explain the safe dispensing and efficacious use of Ayurvedic drugs.	Must know	T-CS, V-KC
Describe the standard protocol for safe dispensing of Ayurvedic drugs.	Desirable	T-OBT
explain ethical aspects related to prescription writing.	Must know	—
SHOWS HOW — 2 OBJECTIVES		
Describe and write demo ideal prescription.	Must know	CHK, V-KC
Demonstrate and educate home remedies to small group of population.	Desirable	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
2 objectives

Describe and write demo ideal prescription; Demonstrate and educate home remedies to small group of population.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
3 objectives

Explain the safe dispensing and efficacious use of Ayurvedic drugs; Describe the standard protocol for safe dispensing of Ayurvedic drugs; explain ethical aspects related to prescription writing.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Explain the importance of rational prescribing of drugs and the concept of essential drugs.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **2 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 11

Traditional & Local health Practices

Hours: 2 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **4 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
3	1	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 3 OBJECTIVES		
Identify Local Health Traditions and Healing Knowledge.	Desirable	P-SUR
Identify the factors responsible for grant of patent and erroneous grant of patent on indian traditional knowledge.?	Desirable	—
Appraise traditional knowledge of Ayurvedic medicines.	Desirable	—
KNOWS HOW — 1 OBJECTIVE		
Recognise the possible potential of product development and research based on Traditional knowledge.	Desirable	—

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
1 objective

Recognise the possible potential of product development and research based on Traditional knowledge.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
3 objectives

Identify Local Health Traditions and Healing Knowledge; Identify the factors responsible for grant of patent and erroneous grant of patent on indian traditional knowledge.?; Appraise traditional knowledge of Ayurvedic medicines.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PAPER 2 · UNIT 12

Pharmacovigilance for Ayurveda drugs

Hours: 1 lecture · 4 non-lecture | Term 3 | Theory marks: 5 | **6 objectives** | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
4	2	—	—
certified here	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 4 OBJECTIVES		
Describe the term Pharmacovigilance and explain importance of Pharmacovigilance for Ayurvedic drugs.	Desirable	C-INT, M-POS, V-KC
Explain the status and central sector scheme of Pharmacovigilance for Ayurveda, Siddha, Unani, and Homeopathy (ASU & H) drugs.	Must know	V-KC
Define Adverse Drug Reactions (ADR) and its types QZ, TT-Theor.	Desirable	QZ, V-KC, VV-Viva
Debate on ADR of ASU drugs.	Desirable	DEB
KNOWS HOW — 2 OBJECTIVES		
Identify and monitor ADRs.	Desirable	PRN, RK, V-KC
Discuss and make critical comments on the safe and efficacious use of Ayurvedic drugs.	Desirable	QZ, V-KC

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Identify and monitor ADRs; Discuss and make critical comments on the safe and efficacious use of Ayurvedic drugs.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
4 objectives

Describe the term Pharmacovigilance and explain importance of Pharmacovigilance for Ayurvedic drugs; Explain the status and central sector scheme of Pharmacovigilance for Ayurveda, Siddha, Unani, and Homeopathy (ASU & H) drugs; Define Adverse Drug Reactions (ADR) and its types QZ, TT-Theor; and 1 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 1

Paribhasha concept based Practicals

9 objectives | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	5	—	3
certified here	certified here	not used	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 1 OBJECTIVE		
Appraise ancient indications mentioned in classics for confirmation of end point of the pharmaceutical process(Siddhi lakshana).	Must know	P-VIVA
KNOWS HOW — 5 OBJECTIVES		
Identify the drugs and recite classical name, English name and chemical composition and varga(class). Recite Maharasa uparasa, Sadharana rasa shlokas from Rasaratna samuchchaya.	Must know	P-ID, P-VIVA
Observe and describe the pharmaceutical preparation of Gandhaka druti. Explain the term alotropism. Recite melting boiling and evaporating temperature of Gandhaka.	Desirable	P-VIVA
Observe the procedure of Vanga Jarana. Explain the difference between Jarana and Jaranaa. Discuss chemical process of oxidation and reduction. Recall the information about melting points of all the metals.	Desirable	P-VIVA
Identify various weights and recite essential measures from conversion chart of AFI. Recognise weighing machines and their weighing capacity.	Desirable	P-SUR, PRN
Identify and value SOP of each procedure of the practical conducted.	Must know	P-EXAM
DOES — 3 OBJECTIVES		
Demonstrate the correct procedure / SOP and assess the changes observed after the procedure. Discuss the relevant samskara and its role in that procedure. Interpret the Physical, Chemical and Biological alterations.	Must know	CHK, P-PRF
Demonstrate preparation of Kajjali. Recite its classical shloka. Prepare a checklist of parameters for its siddhilakshana. Interpret law of definite proportion to explain and calculate amount of free sulfur in the prepared amount of kajjali. Calculate the expected weight of rasasindura from the given amount of kajjali.	Must know	P-EXAM, P-REC
Identify relevant Bhasmapariksha and demonstrate samanya and vishesha Bhasma pariksha. Interpret physical/ chemical laws relevant to classical bhasma pariksha.	Must know	CHK, P-VIVA

THE PYRAMID FOR THIS UNIT

Does
3 objectives

Demonstrate the correct procedure / SOP and assess the changes observed after the procedure. Discuss the relevant samskara and its role in that procedure. Interpret the Physical, Chemical and Biological alterations; Demonstrate preparation of Kajjali. Recite its classical shloka. Prepare a checklist of parameters for its siddhilakshana. Interpret law of definite proportion to explain and calculate amount of free sulfur in the prepared amount of kajjali. Calculate the expected weight of rasasindura from the given amount of kajjali; Identify relevant Bhasmapariksha and demonstrate samanya and vishesha Bhasma pariksha. Interpret physical/ chemical laws relevant to classical bhasma pariksha.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Identify the drugs and recite classical name, English name and chemical composition and varga(class). Recite Maharasa uparasa, Sadharana rasa shlokas from Rasaratna samuchchaya; Observe and describe the pharmaceutical preparation of Gandhaka druti. Explain the term alotropism. Recite melting boiling and evaporating temperature of Gandhaka; Observe the procedure of Vanga Jarana. Explain the difference between Jarana and Jaranaa. Discuss chemical process of oxidation and reduction. Recall the information about melting points of all the metals; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Appraise ancient indications mentioned in classics for confirmation of end point of the pharmaceutical process(Siddhi lakshana).

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **3 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 2

Panchavidha Kashaya Kalpana & their Upakalpana Practicals

4 objectives | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	2	1	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 1 OBJECTIVE		
Describe ethical responsibility expected during Pharmaceutical preparation of Ayurvedic Formulations.	Must know	P-PRF
KNOWS HOW — 2 OBJECTIVES		
Recall, Identify and authenticate the raw materials required as per the dosage form/ formulations.	Must know	P-VIVA
Assess and explain different parameters to achieve desired characters/ end points (siddhi lakshana) as per classical and contemporary parameters.	Must know	P-VIVA
SHOWS HOW — 1 OBJECTIVE		
Demonstrate the method of preparation to get the desired dosage form following SOP.	Must know	P-VIVA

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Demonstrate the method of preparation to get the desired dosage form following SOP.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Recall, Identify and authenticate the raw materials required as per the dosage form/ formulations; Assess and explain different parameters to achieve desired characters/ end points (siddhi lakshana) as per classical and contemporary parameters.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Describe ethical responsibility expected during Pharmaceutical preparation of Ayurvedic Formulations.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 3

Rasa Dravya Aushadhi Nirmana Practicals

7 objectives | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	3	3	1
not used	certified here	certified here	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS HOW — 3 OBJECTIVES		
Explain reference Shloka of kalpas & write with interpretation -list of ingredients, their proportion, principles of yoga - yoga samyojana dravya.	Must know	P-VIVA
Identify, Utilize & know mechanism of Yantra used for drug preparation.	Must know	VV-Viva
Appraise peculiarities of chaturvidha rasayana preparation.	Desirable	M-POS
SHOWS HOW — 3 OBJECTIVES		
Observe & Record various parameters responsible for good manufacturing which are done during process of pharmaceutical preparation.	Must know	P-EXAM
Demonstrate Sidhdhi Lakshna.	Must know	P-VIVA
Assess finished product as per classics.	Must know	P-VIVA
DOES — 1 OBJECTIVE		
Identify raw Dravya, used part with classical Mana and interpret in metric system.	Must know	P-EXAM

THE PYRAMID FOR THIS UNIT

Does
1 objective

Identify raw Dravya, used part with classical Mana and interpret in metric system.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
3 objectives

Observe & Record various parameters responsible for good manufacturing which are done during process of pharmaceutical preparation; Demonstrate Sidhdhi Lakshna; Assess finished product as per classics.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
3 objectives

Explain reference Shloka of kalpas & write with interpretation -list of ingredients, their proportion, principles of yoga - yoga samyojana dravya; Identify, Utilize & know mechanism of Yantra used for drug preparation; Appraise peculiarities of chaturvidha rasayana preparation.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This unit does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **4 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 4

Bheshaja Kalpana Practicals -I

14 objectives | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
7	4	3	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 7 OBJECTIVES		
Identify the genuine sample of Guggulu and demonstrate preparation of different types of Guggulu along with drying techniques.	Must know	P-VIVA
Identify the useful part of the raw material and demonstrate Sattva preparation method.	Must know	P-VIVA
Identify the internal and external applications of different kshara kalpana along with importance of ksharasutra through surgical aspects.	Must know	P-VIVA
Demonstrate and. Explain preparation methods of Masi kalpana for its internal or external application in clinical practice.	Desirable	P-VIVA
Demonstrate preparation and application aspects of Upanaha kalpana.	Desirable	P-EXAM, V-KC
Demonstrate preparations of Sikta taila along with identify the genuine sample of Sikta. Perform and. Describe Malahara kalpana preparation method with its confirmatory tests of end point.	Desirable	P-VIVA
Demonstrate preparation of different types of Malahara with its applied aspect.	Nice to know	P-SUR, V-KC
KNOWS HOW — 4 OBJECTIVES		
Demonstrate preparation of Vati along with drying techniques.	Must know	P-VIVA
Demonstrate preparation of Varti along with drying and packaging techniques.	Must know	P-VIVA
Demonstrate preparation of different types of Lepa and preparation of Shatadhouta ghrita.	Nice to know	PRN, V-KC
Observe demonstration of formulation of Danta manjana.	Nice to know	PRN
SHOWS HOW — 3 OBJECTIVES		
Demonstrate Arka preparation method, along with its packaging technique and therapeutic importance.	Must know	—
Demonstrate Lavana preparation methods, along with its packaging techniques.	Desirable	P-VIVA
Perform and. Describe Lepa kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lepa kalpana, integration with Kayachikitsa and Panchakarma department along with its shelf life and packaging techniques.	Desirable	P-VIVA, V-KC

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
3 objectives

Demonstrate Arka preparation method, along with its packaging technique and therapeutic importance; Demonstrate Lavana preparation methods, along with its packaging techniques; Perform and. Describe Lepa kalpana preparation method with its confirmatory tests and precautions Explain therapeutic importance of prepared Lepa kalpana, integration with Kayachikitsa and Panchakarma department along with its shelf life and packaging techniques.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Demonstrate preparation of Vati along with drying techniques; Demonstrate preparation of Varti along with drying and packaging techniques; Demonstrate preparation of different types of Lepa and preparation of Shatadhouta ghrita; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
7 objectives

Identify the genuine sample of Guggulu and demonstrate preparation of different types of Guggulu along with drying techniques; Identify the useful part of the raw material and demonstrate Sattva preparation method; Identify the internal and external applications of different kshara kalpana along with importance of ksharasutra through surgical aspects; and 4 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **3 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 5

Bheshaja Kalpana Practicals -II

11 objectives | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	5	3	3
not used	certified here	certified here	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS HOW — 5 OBJECTIVES		
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage Bhallataka Taila Patana/ Jayapala Taila Patana/Vishvamitra Kalpa Sneha.	Desirable	OSPE, P-EXAM, P-VIVA
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Vasavaleha/ Chavana Prasha Avaleha/ Kushmanda Avaleha.	Must know	P-REC, P-VIVA
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Kumaryasava/Drakshasava.	Must know	—
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Kutajarista/Ashokarishta/Takrarishta.	Must know	P-VIVA
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlist the results, packing techniques and storage of unique formulations for e.g. Kanji/ Madushukta.	Desirable	COM, P-VIVA
SHOWS HOW — 3 OBJECTIVES		
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage Ksheera Bala Taila/Kasisadi Taila/ Panchaguna Taila/ Arka Taila/Kutajasuryapaki taila.	Must know	OSPE, P-EXAM
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Nimbu Sharkara.	Nice to know	OSPE, P-EXAM
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Daruharidra Rasakriya.	Nice to know	PRN
DOES — 3 OBJECTIVES		
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Ghrita Murchana and Taila Murchana.	Must know	OSPE, P-EXAM, P-REC
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Jatyadi Gritha/ Triphala Gritha/ Ksheera Shatphala Gritha/Phala Gritha.	Must know	OSPE, P-EXAM
Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Haridra Khanda/ Narikela Khanada.	Must know	OSPE, P-EXAM, P-VIVA

THE PYRAMID FOR THIS UNIT

Does
3 objectives

Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Ghrita Murchana and Taila Murchana; Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Jatyadi Gritha/ Triphala Gritha/ Ksheera Shatphala Gritha/Phala Gritha; Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Haridra Khanda/ Narikela Khanada.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
3 objectives

Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage Ksheera Bala Taila/Kasisadi Taila/ Panchaguna Taila/ Arka Taila/Kutajasuryapaki taila; Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Nimbu Sharkara; Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Daruharidra Rasakriya.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage Bhallataka Taila Patana/ Jayapala Taila Patana/Vishvamitra Kalpa Sneha; Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Vasavaleha/ Chavana Prasha Avaleha/ Kushmanda Avaleha; Demonstrate the SoP involved in the drug selection, measuring the appropriate quantity of ingredients, preparation, observe the Physical Changes, Siddhi lakshana, enlisting the results, packing and storage of Kumaryasava/Drakshasava; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This unit does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **6 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 6

Dosage Forms & Self-care Products Practicals

2 objectives | Ceiling: **Knows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	2	—	—
not used	certified here	not used	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS HOW — 2 OBJECTIVES		
Observe instruments used to prepare soliddosage form, liquid dosage form and Semisolid Dosage Form with one example for each, their method of preparation, ingredients used with their quantity and Quality control Parameters.	Desirable	CHK, P-VIVA
Observe instruments used to prepare with one example for each their method of preparation, ingredients used with their quantityand Quality control Parameters.of following self care productsHair care: Shampoo Body care: Soap, Perfume Face care: Face Pack/Talcum Powder Lip Care: Lip Balm, Lipstick Oral care: Tooth Paste/ Mouth Wash Foot Care: Foot Cream Hand Disinfectant: Hand SanitizerSkin Care: Moisturizer, Sunscreen Lotion following self care products.	Desirable	P-EXAM, P-VIVA

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Observe instruments used to prepare soliddosage form, liquid dosage form and Semisolid Dosage Form with one example for each, their method of preparation, ingredients used with their quantity and Quality control Parameters; Observe instruments used to prepare with one example for each their method of preparation, ingredients used with their quantityand Quality control Parameters.of following self care productsHair care: Shampoo Body care: Soap, Perfume Face care: Face Pack/Talcum Powder Lip Care: Lip Balm, Lipstick Oral care: Tooth Paste/ Mouth Wash Foot Care: Foot Cream Hand Disinfectant: Hand SanitizerSkin Care: Moisturizer, Sunscreen Lotion following self care products.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This unit does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Knows how. A written paper can carry all of this, provided the questions require reasoning rather than recall. Do not assess above this tier. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Knows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **0 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 7

Field Visit/ Study Tour

7 objectives | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	1	—	1
certified here	certified here	not used	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 5 OBJECTIVES		
Field Visit - Record the storage condition of the raw / in-process / finished goods prepared in the approved sections.	Desirable	COM, M-POS
Differentiate between the ancient and contemporary methods of drug preparation.	Desirable	PRN
Enlist the documents required in Batch Manufacturing of Ayurvedic medicines.	Nice to know	P-SUR
Identify the role of Quality Control instruments and equipments in ensuring a quality Ayurvedic product.	Desirable	DEB
Appreciate the importance of Good Manufacturing Practices and Good Packaging Practices required in Ayurvedic Drug manufacturing.	Desirable	DEB, PRN
KNOWS HOW — 1 OBJECTIVE		
Determine the role of various Yantras / machineries used in the approved sections of the Pharmacy.	Desirable	—
DOES — 1 OBJECTIVE		
Prepare own products as per prevalent FDA guidelines.	Desirable	P-EXAM

THE PYRAMID FOR THIS UNIT

Does
1 objective

Prepare own products as per prevalent FDA guidelines.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
not used here

This unit does not ask for Shows how. Nothing here needs to be certified at that standard.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
1 objective

Determine the role of various Yantras / machineries used in the approved sections of the Pharmacy.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Field Visit - Record the storage condition of the raw / in-process / finished goods prepared in the approved sections; Differentiate between the ancient and contemporary methods of drug preparation; Enlist the documents required in Batch Manufacturing of Ayurvedic medicines; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 8

Hospital IPD Practical

6 objectives | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	5	1	—
not used	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS HOW — 5 OBJECTIVES		
Review & Observe the case Sheets.	Nice to know	P-SUR, V-KC
Discuss for formulation prescription manner.	Desirable	T-CS, V-KC
Measure dose of prescribed drug.	Must know	—
Enlist Time, Duration, Pathya, Apathya while prescribing drugs.	Desirable	P-PRF
value details and methodical writing of case records of hospital IPD patients.	Must know	—
SHOWS HOW — 1 OBJECTIVE		
Demonstrate & Explain prescription method.	Desirable	P-PS, V-KC

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Demonstrate & Explain prescription method.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Review & Observe the case Sheets; Discuss for formulation prescription manner; Measure dose of prescribed drug; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This unit does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **1 objective here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 9

Drug Dispensing Practical

4 objectives | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	2	1	1
not used	certified here	certified here	certified here

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS HOW — 2 OBJECTIVES		
Assess practice of prescription processing and labelling of the drugs.	Desirable	—
Determine and identify the cause of common errors occurred during dispensing of Ayurvedic drugs.	Desirable	—
SHOWS HOW — 1 OBJECTIVE		
Explain use of correct drug supply to the right patients, in the required dosage, quantities and clear drug information.	Desirable	PRN, V-KC
DOES — 1 OBJECTIVE		
Assess the arrangement of Ayurvedic drugs according to nature and type in dispensing room or pharmacy.	Nice to know	P-PRF, V-KC

THE PYRAMID FOR THIS UNIT

Does
1 objective

Assess the arrangement of Ayurvedic drugs according to nature and type in dispensing room or pharmacy.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Explain use of correct drug supply to the right patients, in the required dosage, quantities and clear drug information.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
2 objectives

Assess practice of prescription processing and labelling of the drugs; Determine and identify the cause of common errors occurred during dispensing of Ayurvedic drugs.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This unit does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this unit is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **2 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PRACTICAL · UNIT 10

Quality Control Practicals

10 objectives | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	3	5	—
certified here	certified here	certified here	not used

SESSION MAP — THE CURRICULUM'S OBJECTIVES BY TIER

Every objective this unit lists, gathered under the tier its own Level column gives it. The tier tells you what standard of evidence the objective demands, and therefore which instruments can honestly certify it.

LEARNING OBJECTIVE, AS WORDED BY NCISM	PRIORITY	ASSESSMENT NAMED
KNOWS — 2 OBJECTIVES		
Identify the geological description of minerals: physical parameters.	Desirable	P-ID
Appraise quality control procedures done for Ayurvedic formulations.	Desirable	—
KNOWS HOW — 3 OBJECTIVES		
Develop analytical skills for understanding Identity, Purity and Strength of raw materials and finished products as per the standard guidelines of the Ayurvedic Pharmacopoeia of India.	Must know	P-VIVA
Demonstrate the calibration techniques used for Weighing Balance (chemical and physical), pH Meter, Hot Air Oven and Electric Muffle Furnace to ensure the accuracy of the instrument what it is intended to measure.	Desirable	P-VIVA, PRN
Determine iodine value, acid value and saponification value of oils / ghee.	Must know	P-VIVA
SHOWS HOW — 5 OBJECTIVES		
Describe ayurvedic perspectives of quality control parameters applying for solid, semisolid and liquid dosage forms.	Must know	P-VIVA
Demonstrate bulk density, loss on drying, total ash, pH and water soluble extractives of Churna A book.	Must know	P-EXAM
Perform the analytical tests of tablets/ vati/ gutika for hardness uniformity of weight and friability.	Must know	P-EXAM
Estimate specific gravity, alcohol content and total solids of Asava & Arishta.	Must know	P-EXAM
Calculate the dose of various dosage forms and their dispensing methods.	Must know	P-SUR

THE PYRAMID FOR THIS UNIT

Does
not used here

This unit does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
5 objectives

Describe ayurvedic perspectives of quality control parameters applying for solid, semisolid and liquid dosage forms; Demonstrate bulk density, loss on drying, total ash, pH and water soluble extractives of Churna A book; Perform the analytical tests of tablets/ vati/ gutika for hardness uniformity of weight and friability; and 2 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
3 objectives

Develop analytical skills for understanding Identity, Purity and Strength of raw materials and finished products as per the standard guidelines of the Ayurvedic Pharmacopoeia of India; Demonstrate the calibration techniques used for Weighing Balance (chemical and physical), pH Meter, Hot Air Oven and Electric Muffle Furnace to ensure the accuracy of the instrument what it is intended to measure; Determine iodine value, acid value and saponification value of oils / ghee.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Identify the geological description of minerals: physical parameters; Appraise quality control procedures done for Ayurvedic formulations.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this unit cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **5 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PART C

Reference

Four appendices, generated from the same data as Part B so the two cannot disagree. C1 decodes the tiers and names the instruments that certify each. C2 gives the ceiling of every unit in one view — the fastest way to check that an assessment matches what it is meant to certify. C3 lists the instruments the curriculum itself names, sorted by the tier each can honestly reach. C4 records the sources and what was and was not altered.

1. **C1** The four tiers, and what certifies each
2. **C2** Ceiling of every unit
3. **C3** The instruments this curriculum names, by tier
4. **C4** Sources, and a note on how this book was made

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PART C · APPENDIX C1

The four tiers, and what certifies each

Miller's four tiers with the count of objectives at each, the instruments that can certify them, and the evidence you keep. Nothing below a tier can certify it.

TIER	OBJECTIVES	INSTRUMENTS	EVIDENCE
Knows	127	MCQ, short answer, oral quiz, spotting.	Written script, quiz sheet.
Knows how	107	Structured long answer, case-based questions, structured viva.	Reasoned written answer, viva score sheet.
Shows how	23	OSPE or OSCE station, observed performance against a checklist.	Checklist score, annotated artefact, station record.
Does	11	Log book, record book, portfolio, observation over time, presentation with peer feedback.	Dated log entry, countersigned record, portfolio piece.

TIER	OBJECTIVES	SHARE
Knows	127	47.4%
Knows how	107	39.9%
Shows how	23	8.6%
Does	11	4.1%

Total 268 objectives.

PART C · APPENDIX C2

Ceiling of every unit

The highest tier each unit reaches. Read across: the ceiling names the evidence you must gather, and the tier above it names what you may not ask for.

NO	TITLE	OBJ	CEILING
PAPER 1 — AYURVEDIYA AUSHADHI NIRMANA VIGYANA			
1	Chronological development of Ayurvediya Aushadhi	7	Knows how
2	Paribhasha (Terminology)	19	Shows how
3	Adharbhuta Siddhanta (Application of fundamental	11	Knows how
4	Yantropakaranani- I (Equipments and machineries)	6	Knows how
5	Yantropakaranani -II (Equipments, fuel and Heating	21	Does
6	Kalpana Nirmana I (Primary & Secondary dosage	11	Knows how
7	Kalpana Nirmana-II (Method of Preparation of	4	Knows how
8	Rasa Dravya Parichaya- I	9	Shows how
9	Rasa Dravya Parichaya II	7	Knows how
10	Rasadravya Parichaya III	5	Knows
11	Kalpana Nirman -III (Method of Preparation of	6	Knows how
12	Chaturvidha Rasayana	6	Does
13	Current and emerging trend in Ayurvedic	6	Knows how
14	GMP(Schedule T) & Regulatory aspects of Ayurvedic	7	Knows
PAPER 2 — AYURVEDIYA AUSHADHI PRAYOGA VIGYANA			
1	Aushadhi Prayoga Vigyana	1	Knows
2	Single drug (Herbal & Mineral)	5	Knows how
3	Single drug(Bhasma, Shuddha & Pishti)	5	Shows how
4	Aushadhi Kalpa -I (Compound formulations)	5	Knows how
5	Aushadhi Kalpa-II (Compound Drugs/Formulations) • Ch	7	Knows how
6	Dosage Forms & Cosmetic Products	12	Knows how
7	Nutraceuticals	9	Knows how
8	Anupana Prayoga for Aushadhi Kalpa	4	Knows how
9	Aushadhi Prayoga Marga	5	Knows how
10	Rational prescription along with safe dispensing of	6	Shows how
11	Traditional & Local health Practices	4	Knows how
12	Pharmacovigilance for Ayurveda drugs	6	Knows how
PRACTICAL UNITS — NCISM TABLE 4			
1	1.Paribhasha concept based Practicals	9	Does
2	2.Panchavidha Kashaya Kalpana & their Upakalpana Pra	4	Shows how
3	3.Rasa Dravya Aushadhi Nirmana Practicals	7	Does
4	4.Bheshaja Kalpana Practicals -I	14	Shows how

NO	TITLE	OBJ	CEILING
5	5.Bheshaja Kalpana Practicals -II	11	Does
6	6.Dosage Forms & Self-care Products Practicals	2	Knows how
7	7.Field Visit/ Study Tour	7	Does
8	8.Hospital IPD Practical	6	Shows how
9	9. Drug Dispensing Practical	4	Does
10	10.Quality Control Practicals	10	Shows how

A written paper can certify Knows and Knows how. Anything above must be certified in the practical, the record book or the log book.

PART C · APPENDIX C3

The instruments this curriculum names, by tier

Every assessment abbreviation that appears in the curriculum, sorted by the highest tier it can honestly certify.

HIGHEST TIER IT REACHES	INSTRUMENTS THE CURRICULUM NAMES	WHAT IT CAN AND CANNOT SHOW
Knows	TT-Theory 16, T-OBT 10, T-EMI 5, T-CRQs 3, P-ID 4, V-KC 19, M-CHT 2	Recall only. Cannot show reasoning or performance.
Knows how	T-EW, T-MEQs, T-CS 2, P-VIVA 57, VV-Viva 2, C-INT 2, M-POS 14, P-POS 1	Reasoning on paper or aloud. Cannot show that the student can perform.
Shows how	OSPE 6, P-PRF 4, CHK 7, P-MOD	Performance when watched. The lowest instrument that settles a doing objective.
Does	Log book, COM 6, P-REC 4	Independent practice over time. The only evidence that reaches the top tier.

Counts are how many objectives name each instrument. An instrument may always be used below its tier; it may never be used above it.

PART C · APPENDIX C4

Sources, and a note on how this book was made

The primary sources for the framework, the curriculum this book is built on, and a note on what was and was not altered.

THE FRAMEWORK

- Miller, G. E., 'The assessment of clinical skills / competence / performance', *Academic Medicine*, 65(9), 1990. The original four-tier paper.
- Cruess, R. L., Cruess, S. R. and Steinert, Y., 'Amending Miller's Pyramid to Include Professional Identity Formation', *Academic Medicine*, 91(2), 2016. The proposed fifth tier, *Is*.
- Biggs, J., 'Enhancing teaching through constructive alignment', *Higher Education*, 32, 1996. The source of the alignment principle in Chapter 3.

A NOTE ON HOW THIS BOOK WAS MADE

Every objective in Part B is reproduced in the curriculum's own words. Plain-English spelling errors have been corrected and sentence punctuation lost to the source document's line wrapping has been restored. No Sanskrit or technical term has been altered, and no subject content has been added, removed or reworded.

National Commission for Indian System of Medicine, Board of Ayurveda. *Course curriculum for Second Professional BAMS — Rasashastra evam Bhaishajyakalpana (AyUG-RB).* New Delhi: NCISM. Tables 2, 3, 4 and 4a are the source of every objective, tier, priority and instrument in this book. All 268 objectives and the 100 marks of each paper reconcile against the curriculum's own summary.

The counts and tables in Part C are generated from the same parsed data as Part B, so the two cannot disagree.

A COGNITIVE-TAXONOMY HANDBOOK

Bloom's Taxonomy for Rasashastra & Bhaishajya Kalpana

Module-by-module domain and level mapping, objective design and usage guide for AYPG-AB-RB — Applied Basics of Rasashastra and Bhaishajya Kalpana
MD/MS Ayurveda, Semester II

Curriculum	NCISM, Board of Ayurveda — applicable from the 2024–25 batch
Coverage	8 modules · 42 units · 171 session learning objectives · 47 practical training and 44 experiential learning activities
Framework	Bloom (1956) and the Anderson & Krathwohl revision (2001) carried in parallel, with the psychomotor and affective domains, read against NCISM Table 3 Column 3E

171
OBJECTIVES CODED

105
THINKING

65
DOING

16
CREDITS · 480 HOURS

Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026 · For departmental use in curriculum delivery, objective writing, assessment design and paper setting.

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Chapter 1 What Bloom's taxonomy is, and why this book carries two versions 5

Chapter 2 The three domains 6

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Chapter 7 Choosing how to teach, level by level 11

Chapter 8 Choosing how to check, level by level, and the modular assessment 12

Chapter 9 How to use a module and a unit in Part B 13

PART B — THE MODULES AND UNITS

Module 1 Pharmaceutics and Pharmaceutical Technology 24 obj · 56 h 15

Unit 1 Introduction to Pharmaceutics 6 obj

Unit 2 Pharmaceutical Dosage Forms 4 obj

Unit 3 Pharmaceutical Formulation 5 obj

Unit 4 Pharmaceutical Manufacturing and Quality Control 5 obj

Unit 5 Biopharmaceutics 4 obj

Module 2 Databases and Information Systems for Metal, Mineral, and Plant Research 22 obj · 56 h 17

Unit 1 Introduction to Databases and Information Systems 5 obj

Unit 2 Metal and Mineral Databases 6 obj

Unit 3 Plant Databases 4 obj

Unit 4 Database application 4 obj

Unit 5 Challenges and limitation of Databases 3 obj

Module 3 Essential Chemistry :- Unlocking the Rasashastra Techniques 21 obj · 56 h 19

Unit 1 Atomic, Molecular Structure, and Periodic Table Elements 3 obj

Unit 2 Specialized Chemistry Fields-An Outline 2 obj

Unit 3 Organic Chemistry 3 obj

Unit 4 Phytochemistry 7 obj

Unit 5 Metallurgy 6 obj

Module 4 Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures 15 obj · 56 h 21

Unit 1 Shodhana and Marana Procedures 3 obj

Unit 2 SOP of Shodhana Processes 4 obj

Unit 3 SOP of Marana process 4 obj

Unit 4 Chemical Changes involved in Shodhana and Marana processes 1 obj

Unit 5 Shodhana and Marana- Case Studies, troubleshooting and emerging trends 3 obj

Module 5 Principles, specification, SOPs of handling of classical and advanced instruments 29 obj · 84 h 23

Unit 1 Puta and its modern aspects 4 obj

Unit 2 Musha and its modern aspects 4 obj

Unit 3 Chromatography 5 obj

Unit 4 Spectroscopy 4 obj

Unit 5 Microscopy 4 obj
Unit 6 Calibration techniques 3 obj
Unit 7 Overview of characterization instruments 5 obj
Module 6 Animal house and Animal handling technique 13 obj · 28 h 25
Unit 1 Introduction to animal house management 3 obj
Unit 2 Study of animal behaviour 3 obj
Unit 3 Species-specific handling techniques 2 obj
Unit 4 Animal health and disease prevention 3 obj
Unit 5 Animal welfare and ethics 2 obj
Module 7 Classical and Advanced quality control techniques for Ayurveda Formulations 26 obj · 56 h 27
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Unit 3 Quality Control parameter for solid dosage forms 9 obj
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Unit 5 Quality Control parameter for liquid and gaseous dosage forms 6 obj
Module 8 Nano-science, nano-technology and its application in Ayurveda 21 obj · 56 h 29
Unit 1 Introduction to Nanotechnology 3 obj
Unit 2 Ayurvedic Formulations and Nanotechnology 10 obj
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PART A

Understanding the taxonomy

Nine short chapters. The first five explain what Bloom's taxonomy is, what its two versions are, what its three domains cover, where it already sits inside this curriculum, and how verbs turn it into practice. Chapter 6 deals with something specific to postgraduate study here — the fixed one-to-two-to-three split between lecture, practical training and experiential learning, which decides more about what a student reaches than any other single feature of the course. The last three cover teaching, checking, and how to read Part B.

1. **What Bloom's taxonomy is, and why this book carries two versions**
2. **The three domains** — thinking, doing, and how a student goes about the work
3. **The second axis** — what kind of knowledge an objective works on
4. **Where Bloom already sits in your curriculum** — reading Column 3E
5. **Verbs** — the working end of the taxonomy
6. **The one-to-two-to-three split** — lecture, practical training, experiential learning
7. **Choosing how to teach**, level by level
8. **Choosing how to check**, level by level, and the modular assessment
9. **How to use a module and a unit in Part B**

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PART A · CHAPTER 1

What Bloom's taxonomy is, and why this book carries two versions

In 1956 a committee of American examiners led by Benjamin Bloom published a way of sorting examination questions. They had noticed what every teacher notices sooner or later: two questions on the same content can be worlds apart, not because one covers more ground, but because one asks the student to *do* something harder with what they know.

What they produced was a ladder of six levels, each asking more than the one below: **Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation**. The ladder is cumulative. You cannot apply what you do not understand, and you cannot understand what you cannot recall. That the levels stack is what makes the taxonomy useful for planning rather than merely for labelling.

THE 2001 REVISION

- The levels became **verbs**: *Remember, Understand, Apply, Analyse, Evaluate, Create.*
- **The top two swapped.** Bloom put Synthesis at level 5 and Evaluation at level 6. The revision reversed them: Evaluate is 5, Create is 6, on the reasoning that making something new is a larger act than judging something that exists.
- A **second axis** was added for the kind of knowledge involved. Chapter 3 covers it.

> **The one thing to carry away.** When a colleague says a question is "at the top level", ask which scheme they mean. In 1956 the top is Evaluation — judging. In 2001 it is Create — producing. Different questions, different teaching, different marking. Most disagreement between two well-informed teachers about Bloom comes down to this.

WHY BOTH, IN THIS BOOK

Your curriculum uses the **1956 scheme**: the codes in Column 3E are CK, CC, CAP, CAN, CS and CE. If this book used only the 2001 names you could not match it against your syllabus document. But most faculty development now teaches the 2001 scheme. So every level here is named twice, 1956 first: *Comprehension / Understand.* Where the two schemes disagree about position — only at the top two — the book says so on the page.

> **This curriculum reaches the top, and that is unusual.** The undergraduate AyUG-RB syllabus does not use Synthesis at all. Semester II uses it four times, and it uses the top of the doing ladder twice. If you have taught only the undergraduate course, the ceiling here is genuinely higher and the teaching has to change accordingly.

PART A · CHAPTER 2

The three domains — thinking, doing, and how a student goes about the work

Bloom's six levels describe thinking only. The original committee planned three volumes. The **cognitive** volume appeared in 1956; the **affective** volume — attitudes and values — in 1964; the **psychomotor** volume was never written by that committee, and the version your curriculum uses comes from later work by others.

THINKING — SIX LEVELS

Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation. In this semester, 105 of the 171 objectives are cognitive.

DOING — FIVE LEVELS

- **Set** — ready to begin; knows what is needed and can lay it out.
- **Guided response** — performs while being shown or guided.
- **Mechanism** — performs reliably alone, without prompting.
- **Adaptation** — changes the method when conditions change.
- **Origination** — devises a procedure that was not given.

Sixty-five of the 171 objectives are here — 38% of the semester. That is the single most important fact about this curriculum. It is not a course that can be delivered from the front of a room. The largest single group is *Mechanism*, and getting a student from guided to unaided is what most of the practical hours are for.

GOING ABOUT IT — FIVE LEVELS

Receiving, Responding, Valuing, Organisation, Characterisation — from paying attention when it matters, through treating something as worth doing unprompted, to it having become how the student works.

> **This semester is almost silent here.** One objective in 171 carries an affective code, in Module 2 Unit 5. That is worth knowing rather than working around: professional habits — verifying material, recording the reading actually taken, handling animals properly — are being built in this semester whether or not the curriculum codes them. If you want evidence of them, you will have to plan for it yourself, because the syllabus does not ask. Part B therefore shows two ladders per unit rather than three; the single affective objective appears in its own band in the objective table of the unit that holds it.

THE DOMAINS DO NOT MOVE TOGETHER

An objective at Mechanism may need only level 1 thinking if the procedure is followed by rote. Another at Set may need level 4 thinking if choosing the right equipment requires judgement. Module 3 in this semester reaches only Analysis on the thinking ladder but *Origination* on the doing ladder — the highest doing level in the whole course. Read both ladders.

PART A · CHAPTER 3

The second axis — what kind of knowledge an objective works on

The 2001 revision added a second question. The first is: what thinking must the student do? The second is: *what kind of knowledge is it about?*

- **Factual** — terminology, names, specific details. Learned by rehearsal and self-testing.
- **Conceptual** — classifications, principles, relationships, the reason a thing is so. Learned by explaining it to somebody.
- **Procedural** — methods, techniques, criteria for choosing between them. Learned by carrying them out.
- **Metacognitive** — knowing about your own learning. Learned by being made to look back at how you worked.

> **This semester is overwhelmingly Procedural.** Of the 171 objectives, 99 work on procedural knowledge, 70 on conceptual and only 10 on factual. That is the profile of a laboratory and workshop course rather than a reading course, and it lines up with the 38% psychomotor figure. The practical consequence is blunt: **reading a procedure is not learning it**, and a student who has only read will look identical to one who has practised right up until the moment they are asked to perform.

WHY IT CHANGES THE TEACHING

Two objectives at the same level can need entirely different sessions. One asks for the names in a group; another asks for the steps of a procedure in order. Rehearsal fixes the first. The second will not hold until the student has done it, because a remembered list of steps and a performable procedure are not the same possession. This is why “they know it but they cannot do it” is such a common complaint — usually the objective was Procedural and the teaching was pitched as though it were Factual.

In Part B every objective is labelled with its kind of knowledge, read from the words of the objective itself. Where an objective works on two kinds, both are named. Where the wording supports a reading without settling it, the label carries a small circle (°), so you can see a judgement was made and overrule it.

PART A · CHAPTER 4

Where Bloom already sits in your curriculum

Bloom is not something this book adds. It has been in the curriculum since publication, in a column most teachers have never had pointed out.

COLUMN 3E

Open Table 3. The fifth column is headed *Domain/Sub Domain*. Every one of the 171 session learning objectives carries a code there: CK, CC, CAP, CAN, CS, CE for thinking; PSY-SET, PSY-GUD, PSY-MEC, PSY-ADT, PSY-ORG for doing; AFT-REC through AFT-CHR for how the student goes about the work. The full key is Appendix C1.

THE COLUMNS AROUND IT

- **3A — Course outcome.** Which of the course-level outcomes the objective serves.
- **3C — Notional learning hours.** How long the objective is expected to take.
- **3D — Lecture, practical training or experiential learning.** The mode, and for the last two a reference into the activity tables. Chapter 6 is about this column.
- **3G — Teaching and learning methods.**

> **A word about Column 3F, which this book does not use.** Column 3F is headed *Level*, but it carries a different scheme — Knows, Knows how, Shows how, Does — which is Miller's pyramid, not Bloom. It answers a different question: not what kind of thinking is needed, but how authentic the evidence has to be. It has its own companion volume. In *this* book, "level" always means the Bloom level in Column 3E. Where you read "level 5" here, it never means "Shows how".

MODULE LEARNING OBJECTIVES SIT ABOVE SESSION OBJECTIVES

Each module states its own learning objectives, and these are what the modular assessment tests. They are pitched at their own levels, which do not always match the session objectives beneath them. Part B shows each module's objectives with the level of their verb, so you can see at a glance whether the module assessment is aimed higher or lower than the sessions that feed it.

PART A · CHAPTER 5

Verbs — the working end of the taxonomy

A level is an idea; a verb is what a student meets. **An objective is taught and examined at the level of its verb**, whatever else is written beside it.

- **Level 1, Knowledge / Remember** — define, enlist, list, name, recall, recite, state, identify, recognise, observe, record.
- **Level 2, Comprehension / Understand** — describe, explain, discuss, illustrate, classify, differentiate, outline, present.
- **Level 3, Application / Apply** — apply, demonstrate, perform, prepare, calculate, measure, conduct, practise, operate, handle.
- **Level 4, Analysis / Analyse** — analyse, compare, examine, determine, categorise, correlate, review, explore, investigate, search.
- **Level 5 in 1956, Synthesis** — design, formulate, develop, construct, devise, synthesise, integrate, optimise. (Level 6, Create, in the 2001 scheme.)
- **Level 6 in 1956, Evaluation** — evaluate, justify, appraise, critique, assess, judge, defend, interpret. (Level 5 in the 2001 scheme.)

The affective domain has its own verbs, and they are not thinking verbs: *appreciate, value, accept, comply, respond, defend, advocate, adhere, display, practise.*

> **Where a verb and a code disagree, follow the verb — and know you are choosing.** The verb is what appears on the paper, so it wins by default. That is fine provided it is a decision rather than an accident. If the verb asks less than the code, you have room to teach higher than the syllabus requires. If it asks more, plan for the verb. Part B tells you, objective by objective, which case you are in.

WRITING YOUR OWN

Decide the level first, then take a verb from that level, then check the verb can be assessed by the tool you have in mind. One verb, one level, one objective. Where you find yourself writing “identify, demonstrate and evaluate” on a single line, you have three objectives at three levels and they will need three different checks.

PART A · CHAPTER 6

The one-to-two-to-three split — lecture, practical training, experiential learning

This is the feature of postgraduate study that most decides what students actually reach, and it has no equivalent in the undergraduate course. Every credit in this semester is thirty hours, and the curriculum fixes how those hours are spent: **five hours of lecture, ten of practical training, thirteen of experiential learning, and two of modular assessment** — a ratio of roughly one to two to three.

Across the semester that comes to **80 lecture hours, 160 practical and 240 experiential**. Two thirds of the course happens with the teacher not talking.

THE THREE KINDS OF HOUR DO NOT REACH THE SAME LEVELS

- **Lecture** puts material in front of students and lets them retrieve it. It builds level 1 well and level 2 poorly, and on its own it builds nothing above level 2. Eighty of the 171 objectives are set as lecture.
- **Practical training** moves a student from performing with help to performing unaided. That single transition — Guided response to Mechanism — is what these hours are for, and it is slow. Forty-seven objectives are set here.
- **Experiential learning** hands over an unfamiliar situation and requires a judgement, a decision or a product. This is where the upper levels are actually built, and where the modular assessment lives. Forty-four objectives are set here.

> **The commonest failure is silent demotion.** An objective set as experiential learning and then delivered as a lecture has been quietly moved down two levels, and nothing in the timetable will show it. The students will have been told about the thing instead of doing it. Part B prints the kind of hour against every objective for this reason: it is the clearest statement the curriculum makes about how a thing is meant to be learned.

THE ACTIVITY TABLES

Every practical and experiential objective carries a reference — 1.1, 8.2 and so on — into the curriculum's activity tables, which describe what the student is to do. There are 47 practical training activities and 44 experiential ones. Appendices C6 and C7 reproduce all 91 with the level each serves, so you can see what a reference points at without turning back to the syllabus.

PART A · CHAPTER 7

Choosing how to teach, level by level

The commonest planning error is to choose the method by the hour rather than by the level. What follows is what each level needs.

LEVEL 1 — KNOWLEDGE / REMEMBER

Exposure, then rehearsal, then retrieval practice. Make students retrieve without looking, then check. Spread it across weeks. In this semester level 1 is small — eight objectives — so it should not be taking large amounts of lecture time.

LEVEL 2 — COMPREHENSION / UNDERSTAND

Students must handle the material, not receive it again: sort it, contrast it, put it in their own order, explain it to a classmate. Fifty-five objectives sit here, the largest single group in the semester, and lecture alone will not deliver them.

LEVEL 3 — APPLICATION / APPLY

Work examples through together, then hand over a situation the student has not seen. The hand-over is the teaching.

LEVEL 4 — ANALYSIS / ANALYSE

Give students two things and ask how they differ and why. Analysis needs something to set against something else.

LEVEL 5 IN 1956, SYNTHESIS

Used four times in this semester, in Modules 4 and 5. It means the student assembles something that was not handed to them — a protocol, a specification, a procedure. Set the criteria in advance and mark against them.

LEVEL 6 — EVALUATION

Give something to judge and criteria to judge it against, then require the criteria to be stated. Small-group critique works well. This is the level most often taught as though it were level 2 — explaining the accepted view rather than reaching one.

THE DOING LADDER

- **Set** — show the setup, name every item, then have students lay it out themselves.
- **Guided response** — demonstrate once at speed, once slowly, then let them copy while you correct in the moment.
- **Mechanism** — repeated supervised practice with your help withdrawn a little each time. **Most practical hours belong here.**
- **Adaptation** — change one condition and let them work out the adjustment.
- **Origination** — used twice, in Module 3. The student devises a procedure that was never demonstrated.

PART A · CHAPTER 8

Choosing how to check, level by level, and the modular assessment

A check is fair when it can only be passed by a student who has reached the level, and can be passed by any student who has. Most unfair checks fail the first half.

- **Level 1** — short questions asking for a list, a name, a definition. Many of them, spread out.
- **Level 2** — questions requiring the material rearranged. If it can be answered word-for-word from notes, it has slipped to level 1.
- **Level 3** — an unseen case. A familiar case measures memory.
- **Level 4** — questions asking for the parts and how they relate.
- **Level 5** — something the student produced, marked against criteria set in advance.
- **Level 6** — give the marks for the reasoning and the criteria, not for the facts repeated inside the answer.

DOING OBJECTIVES CAN ONLY BE CHECKED BY WATCHING

A viva shows whether a student knows a procedure and can explain it. It cannot show whether they can carry it out, and a fluent student who has never handled the material will pass it comfortably. With 65 doing objectives in this semester, this matters more here than anywhere. The evidence has to be seen: a checklist ticked while the student works, an OSPE station, an observed performance. A checklist of the steps you will look for, written before the session, is enough.

THE MODULAR ASSESSMENT

Each module carries a modular assessment worth 25 marks per credit — 400 marks across the semester — using two of the module's hours per credit. The curriculum requires a **different method in each module**, with a record of the pattern used.

> **Choose the method against the module's ceiling, not by rotation.** A module whose ceiling is Origination or Synthesis needs an assessment that lets a student produce something. A module that stops at Application does not, and using an elaborate instrument there wastes everyone's time. The modules in Part B print each ceiling for exactly this decision.

PART A · CHAPTER 9

How to use a module and a unit in Part B

Part B has eight modules, each followed by its units — fifty entries in all. Every one begins on its own page.

THE MODULE

1. **The strip** — credits, units, thinking and doing counts, formative marks.
2. **Where the hours go** — the lecture, practical and experiential split for this module, with what each kind of hour can reach and what it must deliver here.
3. **What the module asks students to do** — both ladders, across all its units.
4. **Module learning objectives**, each with the level of its verb. These are what the modular assessment tests.
5. **The units**, with each unit's highest thinking and highest doing level side by side.

THE UNIT

1. **The heading** — objectives, hours, the mix of lecture, practical and experiential, and the course outcomes served.
2. **The syllabus points** as the curriculum lists them.
3. **Both ladders** for this unit.
4. **The objectives grouped by level**, each with its verb, its kind of knowledge, its kind of hour, and what evidence it needs.
5. **Two panels** — one for the teacher, one to give to students.

THREE WAYS TO USE IT

Planning a session. Take one level at a time from the grouped objectives and teach upward. Check the method against Chapter 7 and the kind of hour against Chapter 6.

Designing the modular assessment. Start from the module's ceilings and the module learning objectives, then pick an instrument that can actually reach them.

Giving students their bearings. The student panel is written to be read by students. Its first line is the useful one: read the verb, and read the kind of hour.

> **What this book does not do.** It does not rewrite the curriculum and it does not tell you what to teach — every objective in Part B appears in the curriculum's own words. What it adds is the taxonomy layer: what kind of learning each objective asks for, and what follows for teaching and checking.

PART B

The modules and units

Fifty entries: eight module overviews, each followed by its units. Every objective appears in the curriculum's own words, sorted into its level so a session can be planned around one level at a time, with the kind of hour printed against each — lecture, practical training or experiential learning. Chapter 9 explains the layout; after that they are meant to be opened at whichever one you are about to teach.

1. **Modules** — hours, ladders, module learning objectives, and the units within
2. **Units** — syllabus points, both ladders, and the objectives grouped by level

↑ Contents

MODULE 1

Pharmaceutics and Pharmaceutical Technology

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	14	10	50
60 notional hours	24 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module Pharmaceutics and Pharmaceutical Technology provides a comprehensive foundation in pharmaceutics and pharmaceutical technology,covering the entire spectrum of pharmaceutical design, formulation, manufacture, and evaluation. It also delves into the pharmaceutical industry, regulatory frameworks, and product development processes. Additionally, the module explores various pharmaceutical

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

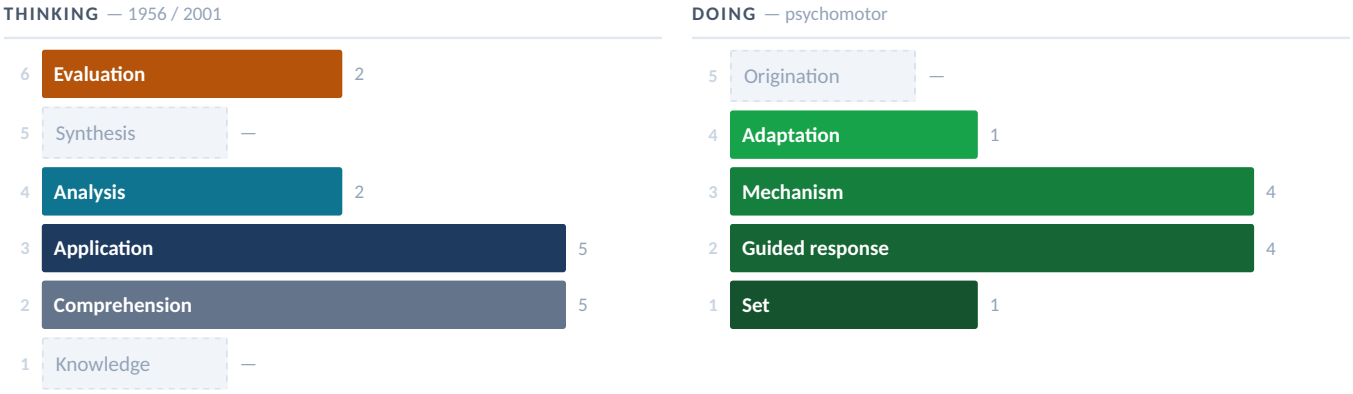
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	10 h	10	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	20 h	8	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	30 h	6	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 4 hours of the 30 shown, worth 50 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.



MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Discuss the fundamental principles and scope of pharmaceutics and pharmaceutical technology.	*discuss*	Level 2 Comprehension / Understand

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
2	Describe the different types of pharmaceutical dosage forms and their formulation principles.	*describe*	Level 2 Comprehension / Understand
3	Discuss the importance of Good Manufacturing Practices (GMP) and quality control in pharmaceutical manufacturing.	*discuss*	Level 2 Comprehension / Understand
4	Apply biopharmaceutic principles to pharmaceutical product development and optimization.	*apply*	Level 3 Application / Apply
5	Illustrate the pharmaceutical product development process and regulatory frameworks governing the pharmaceutical industry.	*illustrate*	Level 2 Comprehension / Understand

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Introduction to Pharmaceutics	6	12 h	Evaluation	Adaptation
2	Pharmaceutical Dosage Forms	4	8 h	Analysis	Mechanism
3	Pharmaceutical Formulation	5	16 h	Application	Mechanism
4	Pharmaceutical Manufacturing and Quality Control	5	10 h	Evaluation	Mechanism
5	Biopharmaceutics	4	10 h	Application	Mechanism

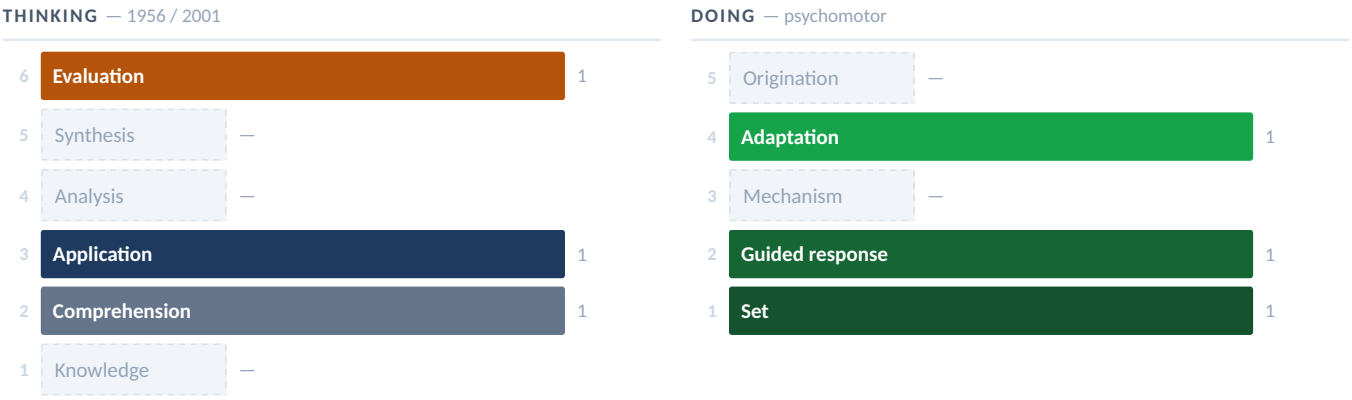
MODULE 1 · UNIT 1

Introduction to Pharmaceutics

6 objectives | 12 hours | Lecture 2 · Practical 2 · Experiential 2 | Outcomes: CO 1, CO 2, CO 4, CO 5, CO 7

- Definition and scope of pharmaceutics
- Historical development of pharmaceutics
- Pharmaceutical industry overview
- Regulatory frameworks (FDA, etc.)
- Pharmaceutical product development process

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Define pharmaceutics and describe its scope and historical development.	*define*	Factual °	Lecture 1h	Written or spoken. Pitch at **level 1** — the verb *define* is what will be asked.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Discuss Pharmaceutical industry overview, market size (national and international) and regulatory framework.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1				
Enlist and evaluate challenges and solutions in the pharmaceutical product development process.	*enlist*	Procedural °	Experiential 3h	Written or spoken. Pitch at **level 1** — the verb *enlist* is what will be asked.
DOING — LEVEL 1 · SET — 1				
Demonstrate regulatory frameworks (FDA, etc.).	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate pharmaceutical product development process.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 4 · ADAPTATION — 1				
Identify applications suiting various regulatory frameworks.	*identify*	Procedural	Experiential 3h	**Watch them do it.** Alter a condition and watch what they change.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

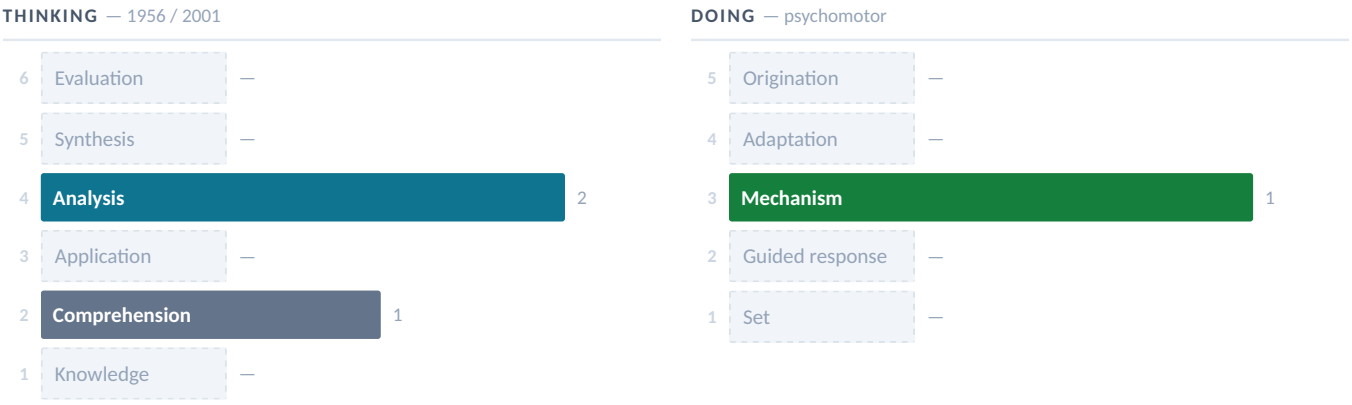
MODULE 1 · UNIT 2

Pharmaceutical Dosage Forms

4 objectives | 8 hours | Practical 2 · Lecture 1 · Experiential 1 | Outcomes: CO 1, CO 5

- Classification of dosage forms
- Solid dosage forms (tablets, capsules, etc.)
- Liquid dosage forms (solutions, suspensions, etc.)
- Semi-solid dosage forms (ointments, creams, etc.)
- Parenteral dosage forms (injections, infusions, etc.)

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Classify various pharmaceutical dosage forms.	*classify*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2				
Identify and categorise classical Ayurvedic dosage forms.	*identify*	Factual °	Practical 2h	Written or spoken. Pitch at **level 1** — the verb *identify* is what will be asked.
Identify and categorise modified Ayurvedic dosage forms.	*identify*	Factual °	Practical 2h	Written or spoken. Pitch at **level 1** — the verb *identify* is what will be asked.
DOING — LEVEL 3 · MECHANISM — 1				
Explore, identify, and collect various classical and modified pharmaceutical dosage forms.	*explore*	Procedural	Experiential 3h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 1 · UNIT 3

Pharmaceutical Formulation

5 objectives | 16 hours | Lecture 2 · Practical 2 · Experiential 1 | Outcomes: CO 1, CO 5

- Principles of pharmaceutical formulation
- Excipients and their functions
- Solid dosage forms
- Liquid and semi-solid dosage forms
- Parenteral dosage forms

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss formulation of parenteral dosage forms.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Describe the principles of pharmaceutical formulations, excipients, and their functions.	*describe*	Conceptual	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 2				
Formulate Solid Dosage Forms selecting appropriate excipients.	*formulate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
Formulate liquid Dosage Forms selecting appropriate excipients.	*formulate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Identify an Appropriate Excipient According to Solid and Liquid Dosage Form.	*identify*	Procedural	Experiential 6h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

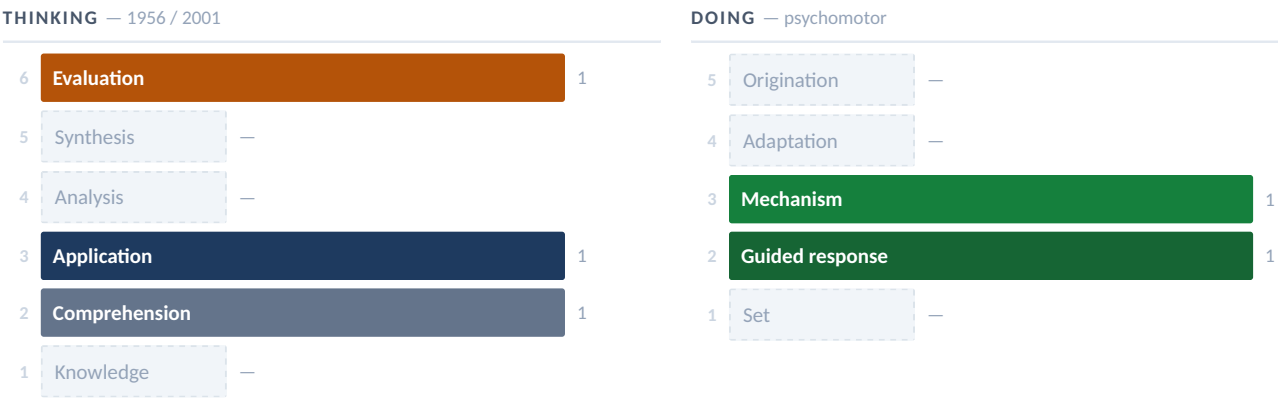
MODULE 1 · UNIT 4

Pharmaceutical Manufacturing and Quality Control

5 objectives | 10 hours | Lecture 2 · Practical 2 · Experiential 1 | Outcomes: CO 3, CO 4

Good Manufacturing Practices (GMP& cGMP)
Quality control and quality assurance
Packaging and labelling
Portability and storage

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss Good Manufacturing Practices (GMP& cGMP).	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Discuss pharmaceutical stability and shelf life.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1				
Evaluate GMP audit and compliance.	*evaluate*	Conceptual °	Practical 2h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate, design and evaluate pharmaceutical packing and labelling.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Design and develop Pharmaceutical Packaging systems and labelling.	*design*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 1 · UNIT 5

Biopharmaceutics

4 objectives | 10 hours | Lecture 3 · Experiential 1 | Outcomes: CO 4, CO 5, CO 6, CO 7, CO 8

- Biopharmaceutics and pharmacokinetics
- Bioavailability and bioequivalence
- Pharmaceutical dissolution and absorption
- Pharmaceutical product development and optimization

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Describe biopharmaceutics and pharmacokinetics.	*describe*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 2				
Describe bioavailability and bioequivalence.	*describe*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
Discuss pharmaceutical product development and optimization.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
DOING — LEVEL 3 · MECHANISM — 1				
Design and Optimize Pharmaceutical Formulation for Enhanced Bioavailability.	*design*	Procedural	Experiential 7h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
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MODULE 2

Databases and Information Systems for Metal, Mineral, and Plant Research

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	17	4	50
60 notional hours	22 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module, Databases and Information Systems for Metal, Mineral, and Plant Research, provides a comprehensive introduction to databases and information systems, focusing on metal, mineral, and plant research. This module will provide the fundamental principles of databases, data retrieval languages, and various databases relevant to the field, including metal, mineral, and plant databas

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

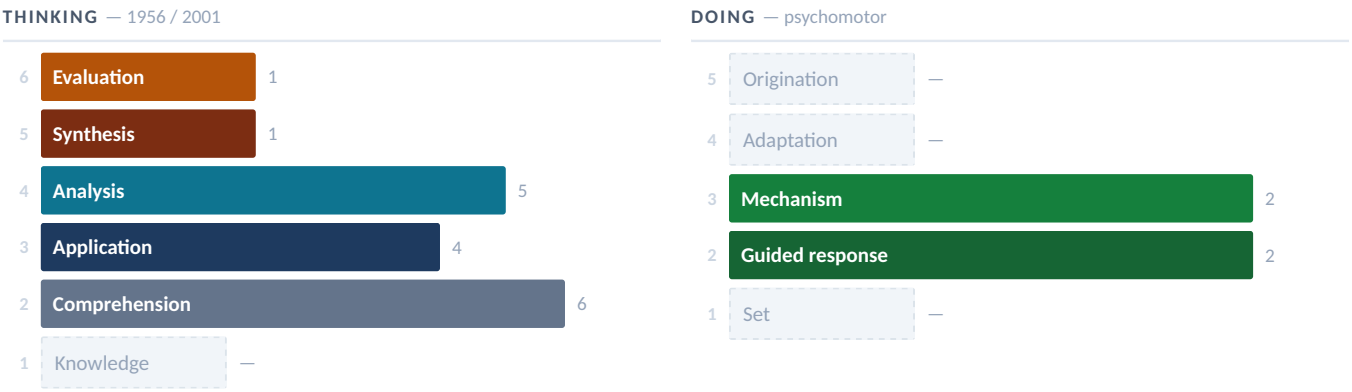
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	10 h	10	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	20 h	7	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	30 h	5	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 4 hours of the 30 shown, worth 50 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.



MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Describe the fundamental concepts of databases and information systems, including types of databases and database management systems.	*describe*	Level 2 Comprehension / Understand
2	Acess metal, mineral, and plant databases for research and development, including advanced searching techniques and data analysis.	*acess*	Level ? —
3	Integrate multiple databases and apply database applications in research, industry, and decision-making.	*integrate*	Level 5 Synthesis / Create
4	Evaluate the challenges and limitations of databases in metal, mineral, and plant research.	*evaluate*	Level 6 Evaluation / Evaluate
5	Design and develop a database application for metal, mineral, or plant research, incorporating emerging trends and technologies.	*design*	Level 5 Synthesis / Create

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Introduction to Databases and Information Systems	5	12 h	Application	Mechanism
2	Metal and Mineral Databases	6	16 h	Synthesis	—
3	Plant Databases	4	12 h	Analysis	—
4	Database application	4	12 h	Evaluation	Mechanism
5	Challenges and limitation of Databases	3	4 h	Analysis	—

MODULE 2 · UNIT 1

Introduction to Databases and Information Systems

5 objectives | 12 hours | Lecture 2 · Practical 2 · Experiential 1 | Outcomes: CO 1

- Overview of databases and information systems
- Types of databases (relational, object-oriented, etc.)
- Database management systems (DBMS)
- Data retrieval and querying languages (SQL, etc.)
- Introduction to metal, mineral, and plant databases

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Describe overview of databases and information systems, types of databases (relational object-oriented, etc.), and database management systems (DBMS).	*describe*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Describe data retrieval and querying languages (SQL, etc.) and an overview of metal mineral, and plant databases.	*describe*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 2				
Demonstrate various searching techniques of databases.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
Demonstrate Modelling, Querying, and Analytics of Data.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Design and Implement a Simple Database for a Plant Nursery.	*design*	Procedural	Experiential 6h	**Watch them do it.** They work unaided.

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FOR THE TEACHER

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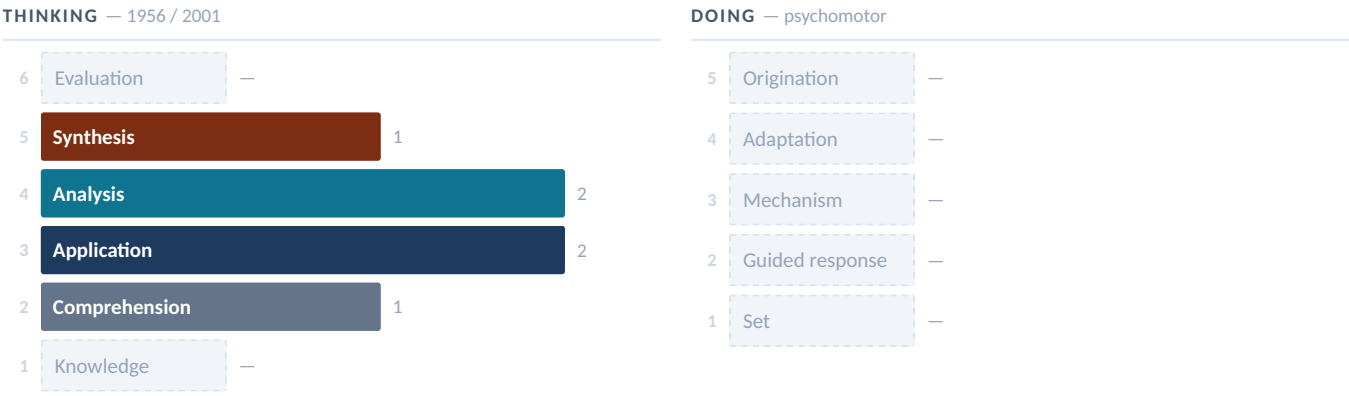
MODULE 2 · UNIT 2

Metal and Mineral Databases

6 objectives | 16 hours | Practical 3 · Lecture 2 · Experiential 1 | Outcomes: CO 3, CO 4

- Overview of metal and mineral databases
- USGS Mineral Database
- Mineral Database
- Webmineral Mineral Database
- Introduction to Metal Databases

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

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OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Describe overview of metal and mineral databases, USGS (U.S. Geological Survey).	*describe*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 2				
Discuss Mineral and metal Databases, including their applications and significance.	*discuss*	Conceptual	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
Search, explore and analyze Metal Database and Applications.	*search*	Conceptual °	Practical 3h	Written or spoken. Pitch at **level 4** — the verb *search* is what will be asked.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2				
Analyze USGS Mineral Database.	*analyze*	Conceptual °	Practical 2h	Written or spoken answer, pitched at the coded level.
Analyse Web mineral Database and Metal Database.	*analyse*	Conceptual °	Experiential 6h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1				
Analysis and integration of Web mineral Database.	*analyse*	Procedural °	Practical 3h	Written or spoken. Pitch at **level 4** — the verb *analyse* is what will be asked.

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FOR THE TEACHER

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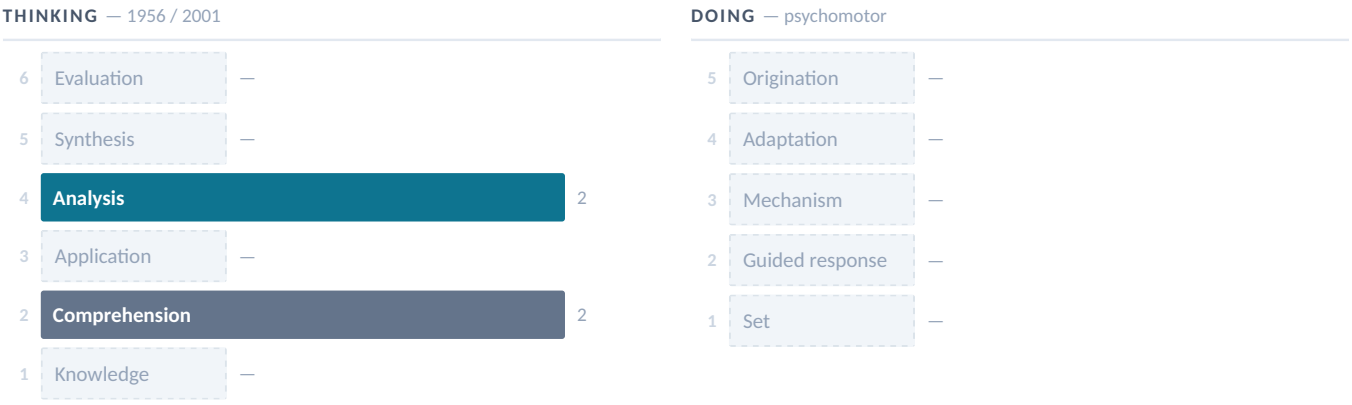
MODULE 2 · UNIT 3

Plant Databases

4 objectives | 12 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 3, CO 4

- Overview of plant databases
- The Plant List (TPL)
- International Plant Names Index (IPNI)
- Royal Botanic Gardens, Kew (RBG Kew) databases
- Introduction to plant genetic databases (e.g., PlantGDB)

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Illustrate overview of plant databases, The Plant List (TPL), and International Plant Names Index (IPNI).	*illustrate*	Factual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Describe Royal Botanic Gardens, Kew (RBG Kew) databases and introduction to plant genetic databases (e.g., PlantGDB).	*describe*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2				
Demonstrate techniques to explore Plant Databases.	*demonstrate*	Procedural	Practical 4h	Written or spoken. Pitch at **level 3** — the verb *demonstrate* is what will be asked.
Explore and analyse plant database.	*explore*	Conceptual °	Experiential 6h	Written or spoken answer, pitched at the coded level.

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FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
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MODULE 2 · UNIT 4

Database application

4 objectives | 12 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 3, CO 4

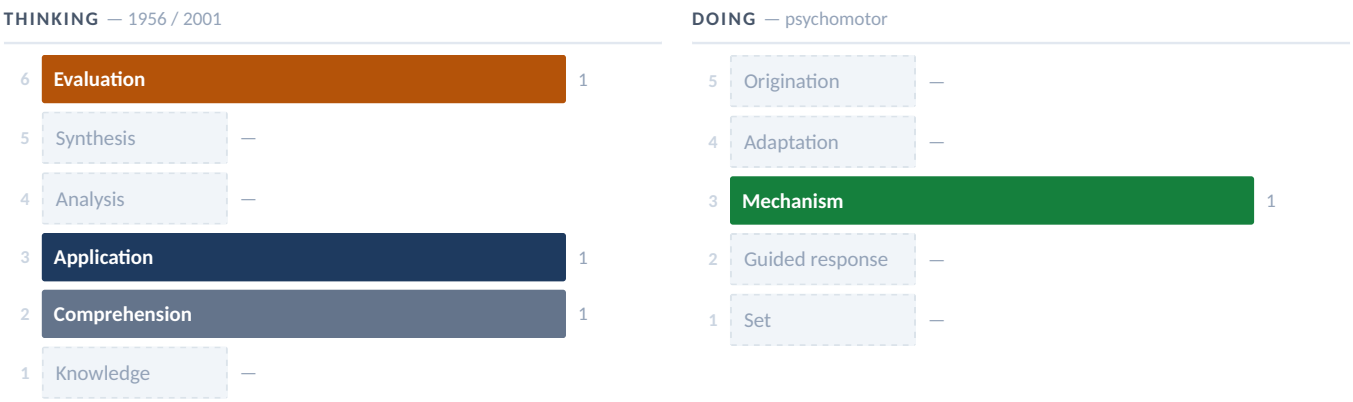
Database applications in research and industry

Using databases for decision-making and policy development

Future directions: emerging trends and technologies in database development

Group project: developing a database application for metal, mineral, or plant research

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

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OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss database applications in research and industry, and use databases for decision-making and policy development.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Analyze future directions in database development and applications, including emerging trends and technologies, and group project guidelines.	*analyze*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 4** — the verb *analyze* is what will be asked.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1				
Analyse case studies of Database Applications in Metal, Mineral, or Plant Research.	*analyse*	Conceptual °	Practical 4h	Written or spoken. Pitch at **level 4** — the verb *analyse* is what will be asked.
DOING — LEVEL 3 · MECHANISM — 1				
Apply and justify case studies of Database Applications in Metal, Mineral, or Plant Research.	*apply*	Procedural	Experiential 6h	**Watch them do it.** They work unaided.

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FOR THE TEACHER

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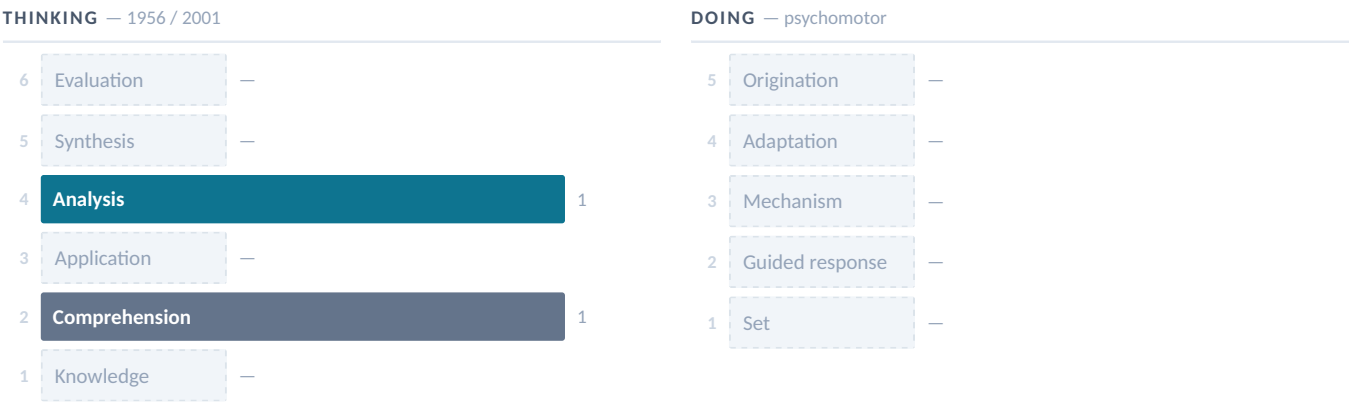
MODULE 2 · UNIT 5

Challenges and limitation of Databases

3 objectives | 4 hours | Lecture 2 · Experiential 1 | Outcomes: CO 3, CO 4

Challenges in Pharmaceutical Formulation Development
Limitations of In Vitro and In Vivo Testing
Regulatory Challenges in Pharmaceutical Product Approval
Scalability and Manufacturing Challenges

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

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OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Identify Database-Related Challenges in Pharmaceutical Development.	*identify*	Factual °	Lecture 1h	Written or spoken. Pitch at **level 1** — the verb *identify* is what will be asked.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Discuss data Quality, Integration, Security, Scalability, and Regulatory Compliance.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
GOING ABOUT IT — LEVEL 2 · RESPONDING — 1				
Perform a role play related to the challenges and limitations of databases.	*perform*	Conceptual	Experiential 2h	**Watch over time.** Conduct across the module, not a single answer.

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FOR THE TEACHER

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MODULE 3

Essential Chemistry :- Unlocking the Rasashastra Techniques

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	12	9	50
60 notional hours	21 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module Essential Chemistry - Unlocking the Rasashastra Techniques integrates fundamental chemistry principles with traditional Ayurvedic practices,enhancing understanding and application of Rasashastra techniques. By exploring the connections between chemistry and Ayurveda, learners will gain adeeper 2 10 20 30 60 understanding of the processing methods and transformations involved in Rasasha

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

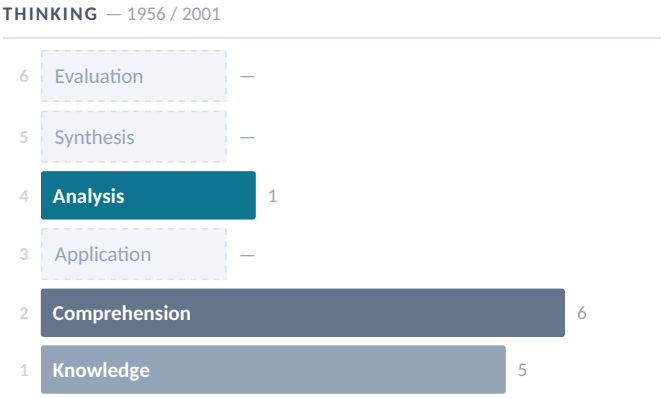
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

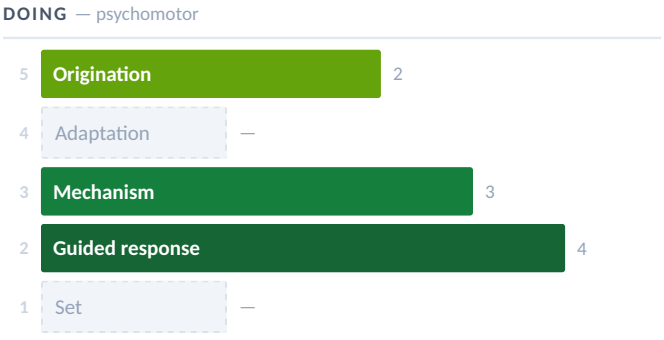
KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	10 h	10	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	20 h	5	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	30 h	6	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 4 hours of the 30 shown, worth 50 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.





THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Atomic, Molecular Structure, and Periodic Table Elements	3	12 h	Knowledge	Mechanism
2	Specialized Chemistry Fields-An Outline	2	6 h	Comprehension	—
3	Organic Chemistry	3	8 h	Analysis	—
4	Phytochemistry	7	16 h	Comprehension	Mechanism
5	Metallurgy	6	14 h	Comprehension	Origination

MODULE 3 · UNIT 1

Atomic, Molecular Structure, and Periodic Table Elements

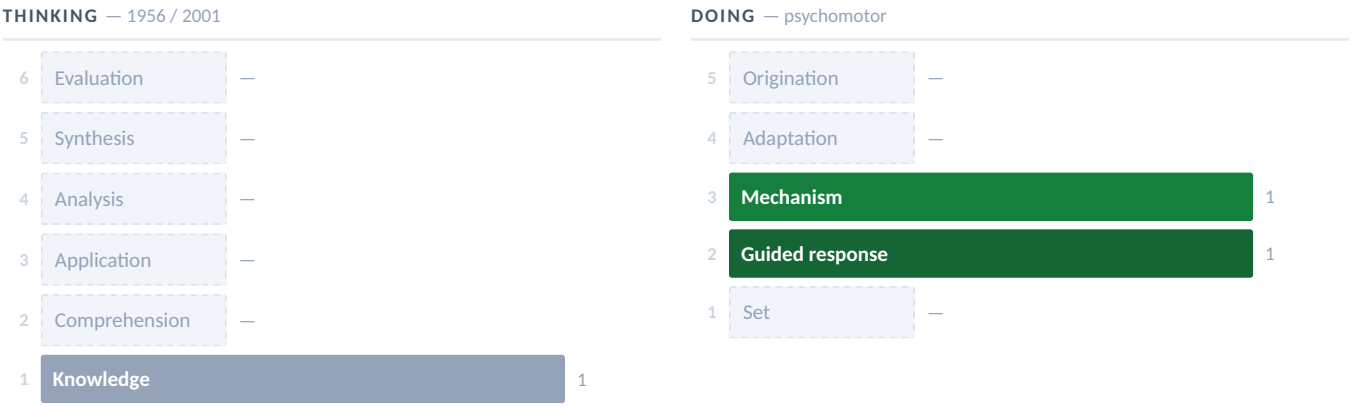
3 objectives | 12 hours | Lecture 1 · Practical 1 · Experiential 1 | Outcomes: CO 8

Importance of knowledge of chemistry for understanding various processing methods of Rasashastra and Oushadha nirmana.

Atomic structure, Periodic table, Chemical bonding, Isomerism - Application of these principles in Shodhana, Marana, Amruteekarana and Chaturvidha Rasayana

Periodic Table, Main group elements, Transition metals: understanding of chemical properties of non-metals, metals, variable oxidation states, thermal stability,

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1				
Describe the role of basic chemistry principles and their application in Rasashastra.	*describe*	Conceptual	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Analyse chemical bonding types and principles and isomerism and their significance in better understanding of different Rasashastra processes such as Shodhana, and Marana.	*analyse*	Conceptual	Practical 5h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Apply the knowledge Periodic Table for Chemical Properties of Main Group and Transition Metals.	*apply*	Procedural	Experiential 6h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 3 · UNIT 2

Specialized Chemistry Fields-An Outline

2 objectives | 6 hours | Lecture 1 · Practical 1 | Outcomes: CO 8

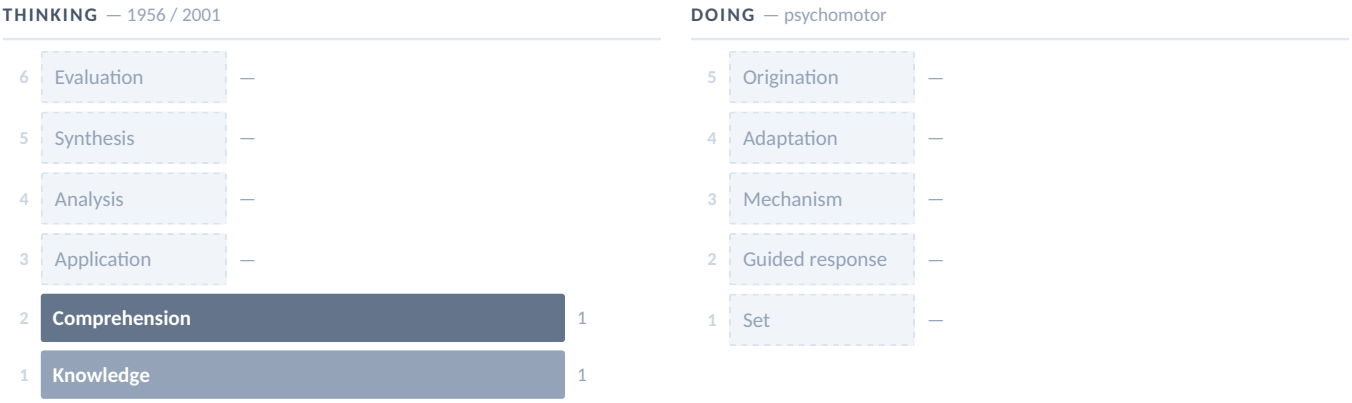
Crystallography- detoxification, transformation of compound

Organometallic chemistry- Chaturvidha Rasayana

Bioinorganic chemistry- Amruteekarana (Metal-Carbon bond-stable compound formation)

Application in Shodhana, Marana, Amruteekarana, Chaturvidha Rasayana

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1				
Describe the principles of crystallography and an overview of organometallic and bioinorganic chemistry.	*describe*	Conceptual	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Demonstrate the essential fundamental principles of crystallography and organometallic principles with special reference to Rasashastra procedures.	*demonstrate*	Conceptual + Procedural °	Practical 5h	Written or spoken. Pitch at **level 3** — the verb *demonstrate* is what will be asked.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 3 · UNIT 3

Organic Chemistry

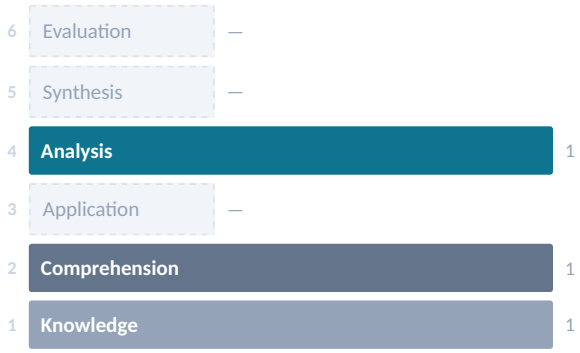
3 objectives | 8 hours | Lecture 2 · Experiential 1 | Outcomes: CO 8

Structure of organic molecules: understanding of Herbal components, Interaction of herbs and animal products with metals for detoxification, compound formation, and compound stability.

Functional groups: Reactivity and properties of organic molecules influencing compound action

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1				
Describe the structure of organic molecules.	*describe*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss the interaction of herbs and animal products with metals.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Explore and identify different functional groups and discuss their influence on molecular interactions and therapeutic efficacy.	*explore*	Conceptual °	Experiential 6h	Written or spoken answer, pitched at the coded level.

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FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

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- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 3 · UNIT 4

Phytochemistry

7 objectives | 16 hours | Lecture 3 · Practical 2 · Experiential 2 | Outcomes: CO 8

- Definition and significance of phytochemistry
- Overview of plant-based bioactive compounds
- Classes of Phytochemicals
- Extraction methods
- Phytochemical screening

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3				
Describe the significance of phytochemistry.	*describe*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Describe the overview of plant-based bioactive components.	*describe*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss different types of extraction methods including solvent selection.	*discuss*	Conceptual + Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 2				
Demonstrate different types of extraction methods including solvent selection.	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
Demonstrate qualitative identification of phytoconstituents in single herbs and herbal formulations.	*demonstrate*	Procedural	Practical 3h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 2				
Demonstrate extraction of different herbal formulations using various extraction techniques and solvents.	*demonstrate*	Procedural	Experiential 3h	**Watch them do it.** They work unaided.
Identify phytoconstituents in Herbs and Herbal Formulations: A.	*identify*	Procedural	Experiential 3h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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FOR THE STUDENT

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MODULE 3 · UNIT 5

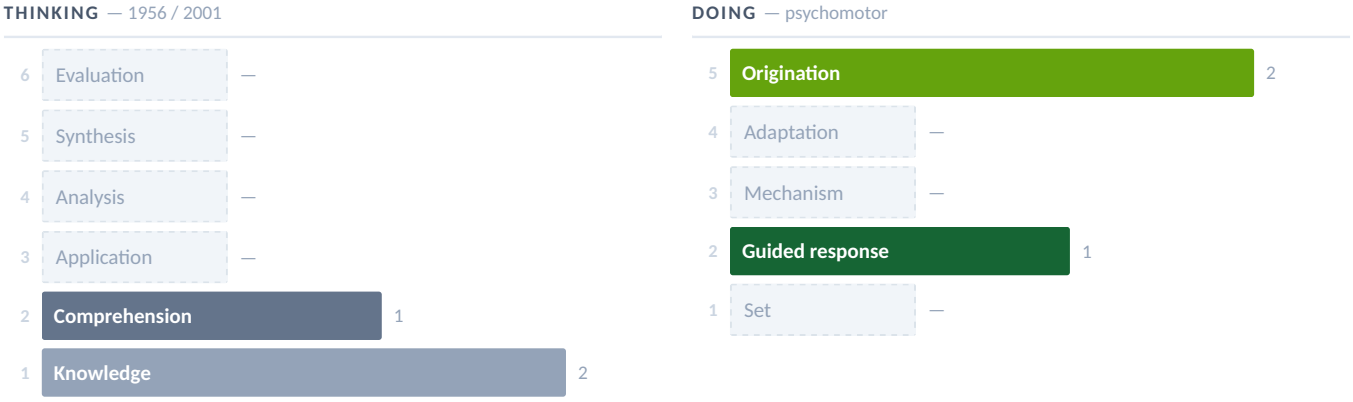
Metallurgy

6 objectives | 14 hours | Lecture 3 · Experiential 2 · Practical 1 | Outcomes: CO 8

Introduction to metallurgy and general physicochemical methods of extraction of metals from ores

Common ores used in Rasashastra and their Satvapatana from metallurgical point of view

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 2				
Define metallurgy and describe its scope.	*define*	Factual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Describe general physicochemical methods of extraction of metals from ores.	*describe*	Procedural °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss common ores used in Rasashastra and their Satvapatana with metallurgical point of view.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate different types of metals and their properties.	*demonstrate*	Procedural	Practical 3h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 5 · ORIGINATION — 2				
Differentiate various types of metals based on their properties.	*differentiate*	Conceptual + Procedural °	Experiential 4h	**Watch them do it.** Alter a condition and watch what they change.
Differentiate metals using basic metal identification techniques.	*differentiate*	Procedural	Experiential 4h	**Watch them do it.** Alter a condition and watch what they change.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
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FOR THE STUDENT

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- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 4

Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	11	4	50
60 notional hours	15 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module, Concept, understanding, application, SOPs, and chemical changes of Shodhana and Marana procedures provides a comprehensive understanding of Shodhana (purification/ Detoxification/ modification/Potentiating) and Maran (incineration) procedures in Ayurvedic pharmaceutical manufacturing. It covers the fundamental principles, applications, and standard opera

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

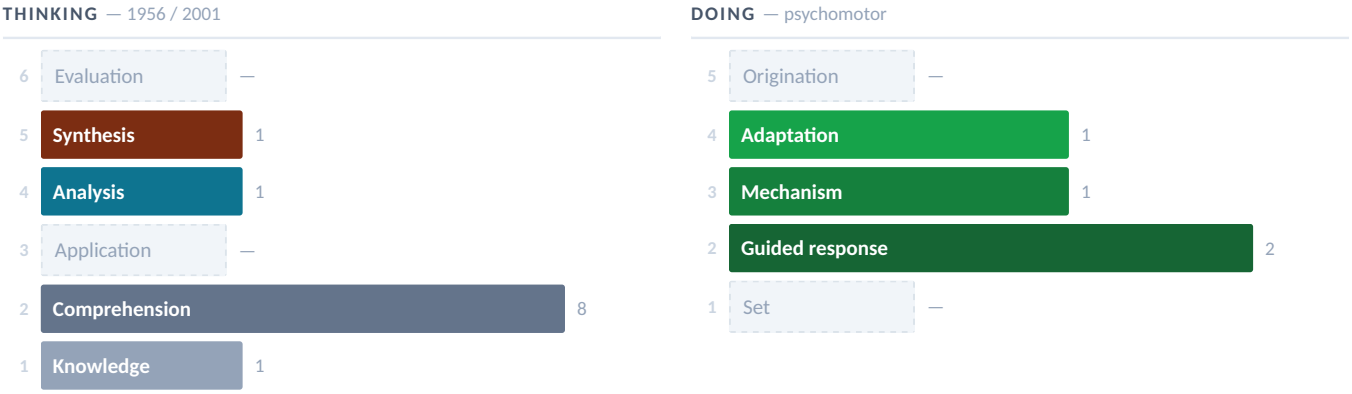
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	10 h	10	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	20 h	2	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	30 h	3	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 4 hours of the 30 shown, worth 50 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.



MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Demonstrate a comprehensive understanding of Shodhana (purification/ Detoxification/ modification/Potentiating) and Maran (incineration) procedures.	*demonstrate*	Level 3 Application / Apply

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
2	Describe the fundamental principles, applications, and standard operating procedures (SOPs) of Shodhana and Marana processes.	*describe*	Level 2 Comprehension / Understand
3	Describe the chemical changes involved in the Shodhana and Marana processes.	*describe*	Level 2 Comprehension / Understand
4	Analyze real-world case studies of Shodhana and Marana procedures.	*analyze*	Level 4 Analysis / Analyse
5	Apply troubleshooting techniques for common issues in Shodhana and Marana processes.	*apply*	Level 3 Application / Apply
6	Plan future directions in Shodhana and Marana procedures.	*plan*	Level 5 Synthesis / Create

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Shodhana and Marana Procedures	3	3 h	Analysis	—
2	SOP of Shodhana Processes	4	22 h	Comprehension	Adaptation
3	SOP of Marana process	4	22 h	Comprehension	Mechanism
4	Chemical Changes involved in Shodhana and Marana processes	1	1 h	Comprehension	—
5	Shodhana and Marana- Case Studies, troubleshooting and emerging trends	3	8 h	Synthesis	—

MODULE 4 · UNIT 1

Shodhana and Marana Procedures

3 objectives | 3 hours | Lecture 3 | Outcomes: CO 1, CO 2, CO 7

Overview of Shodhana and Marana procedures

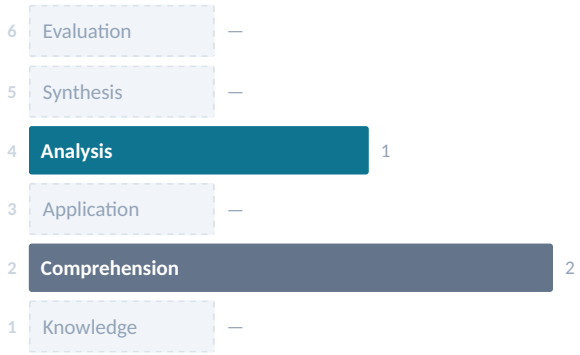
Importance of purification and incineration in Ayurvedic pharmaceutical manufacturing

Principles of Shodhana and Marana procedures

Applications of Shodhan and Maran procedures in Ayurvedic pharmaceutical manufacturing

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Discuss Shodhana and Marana procedures.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss the importance of purification and incineration in Ayurvedic pharmaceutical manufacturing.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Analyse principles of Shodhana and Marana procedures with their application in Ayurvedic pharmaceutical manufacturing.	*analyse*	Conceptual + Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

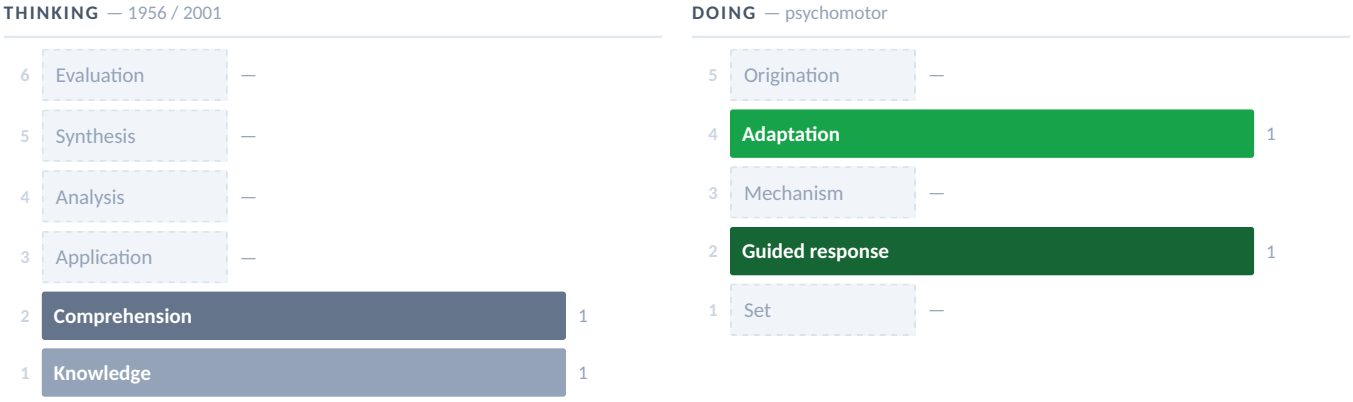
MODULE 4 · UNIT 2

SOP of Shodhana Processes

4 objectives | 22 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 1, CO 7, CO 8

Types of Shodhana procedures (physical, chemical, biological)
Methods of Shodhana (filtration, sedimentation, distillation, etc.)
SOPs for Shodhana procedures

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1				
Describe types of Shodhana procedures (physical, chemical, biological).	*describe*	Conceptual + Procedural °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss methods of Shodhana (filtration, sedimentation, distillation, etc.) and its SOPs.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate Shodhana procedure of Rasadravyas using various methods.	*demonstrate*	Procedural	Practical 10h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 4 · ADAPTATION — 1				
Perform Shodhana of Rasadravyas with various methods.	*perform*	Procedural	Experiential 10h	**Watch them do it.** Alter a condition and watch what they change.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
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- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 4 · UNIT 3

SOP of Marana process

4 objectives | 22 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 1, CO 7

Types of Marana procedures (incineration, calcination, etc.)
Methods of Marana (oxidation, thermal decomposition, etc.)
SOPs for Marana procedures

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Describe types of Marana procedures (incineration, calcination, etc.).	*describe*	Conceptual + Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss methods of Marana (oxidation, thermal decomposition, etc.) and its SOPs.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate Marana procedure of Rasadravyas with different methods.	*demonstrate*	Procedural	Practical 10h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Perform Marana procedure of Rasadravyas with different methods.	*perform*	Procedural	Experiential 10h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 4 · UNIT 4

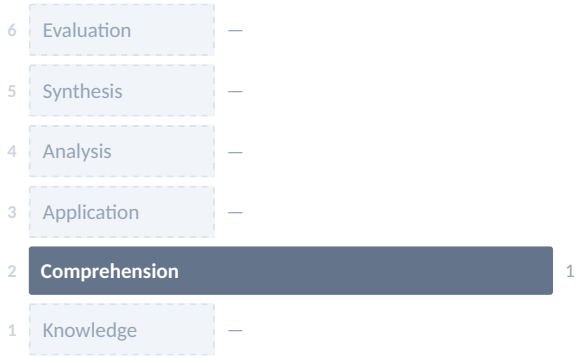
Chemical Changes involved in Shodhana and Marana processes

1 objectives | 1 hours | Lecture 1 | Outcomes: CO 1, CO 7

Chemical changes involved in Shodhan procedures	Chemical Changes involved in Marana procedures

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss chemical changes involved in Shodhana and Marana procedures.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 4 · UNIT 5

Shodhana and Marana- Case Studies, troubleshooting and emerging trends

3 objectives | 8 hours | Lecture 2 · Experiential 1 | Outcomes: CO 1, CO 7, CO 8

Case studies of Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing

Troubleshooting and optimization of Shodhana and Marana procedures

Future directions and emerging trends in Shodhana and Marana procedures

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Describe Troubleshooting and optimization of Shodhana and Marana procedures.	*describe*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss future directions and emerging trends in Shodhana and Marana procedures.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1				
Analyse and write Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing through case studies.	*analyse*	Procedural °	Experiential 6h	Written or spoken. Pitch at **level 4** — the verb *analyse* is what will be asked.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 5

Principles, specification, SOPs of handling of classical and advanced instruments

CREDITS	UNITS	THINKING	DOING	FORMATIVE
3	7	20	9	75
90 notional hours	29 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module Principles, specifications, SOPs for handling classical and advanced instruments describes the principles, specifications, and SOPs for handling classical instruments like Puta and Musha, and their modern aspects. It also deals with the principles, types, and applications of chromatography, spectroscopy, and microscopy in the standardization of Ayurvedic formulations, demon

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

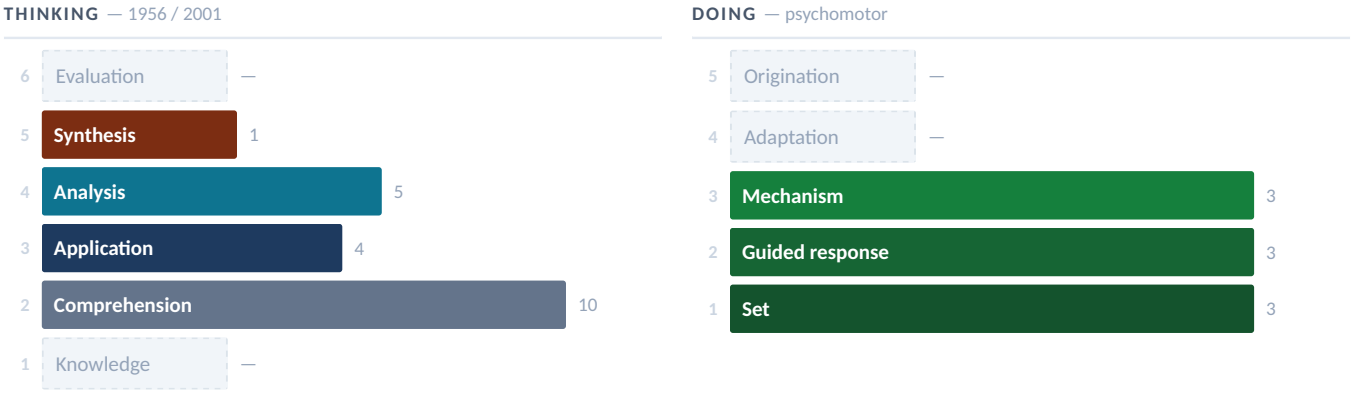
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	15 h	15	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	30 h	7	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	45 h	7	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 6 hours of the 45 shown, worth 75 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.



MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Describe the principles, specifications, and SOPs for handling classical instruments like Puta and Musha, and their modern aspects.	*describe*	Level 2 Comprehension / Understand
2	Demonstrate the use of advanced characterization instruments, such as PSA, XRD, XRF, SEM-EDAX, TEM, TGA, DTA, and VSM analysis.	*demonstrate*	Level 3 Application / Apply
3	Develop SOPs for calibration, troubleshooting, and maintenance of various instruments.	*develop*	Level 5 Synthesis / Create
4	Integrate the knowledge of instrumental analysis for the standardization and quality control of Ayurvedic medicines	*integrate*	Level 5 Synthesis / Create

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Puta and its modern aspects	4	14 h	Synthesis	Guided response
2	Musha and its modern aspects	4	14 h	Analysis	Mechanism
3	Chromatography	5	17 h	Application	Mechanism
4	Spectroscopy	4	9 h	Analysis	Set
5	Microscopy	4	12 h	Comprehension	Mechanism
6	Calibration techniques	3	7 h	Analysis	Set
7	Overview of characterization instruments	5	11 h	Analysis	—

MODULE 5 · UNIT 1

Puta and its modern aspects

4 objectives | 14 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 1, CO 2, CO 7

Introduction, principles, and specifications of Puta
SOPs for handling Puta
Modern aspects of Puta (automation, computerization, etc.)
Troubleshooting and maintenance of Puta

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss the principles and specifications of Puta.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Analyse modern aspects of Puta.	*analyse*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1				
Develop Standard Operative Procedures (SOPs)for given Puta.	*develop*	Procedural	Experiential 8h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate selected Puta to give the knowledge of Standard Operative Procedure.	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

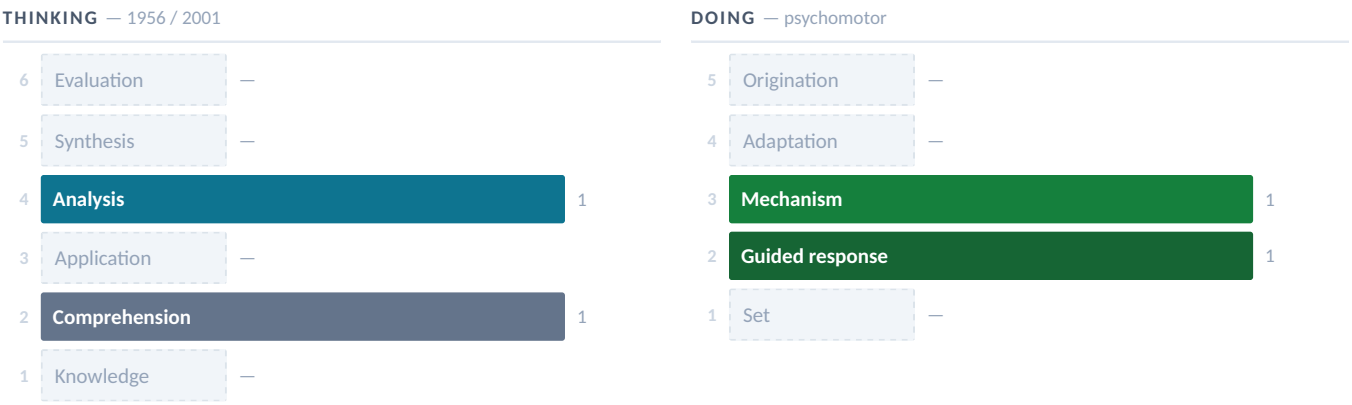
MODULE 5 · UNIT 2

Musha and its modern aspects

4 objectives | 14 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 1, CO 2, CO 7, CO 8

Introduction, principles, and specifications of Puta.
SOPs for handling Musha
Modern aspects of Musha (automation, computerization, etc.)
Troubleshooting and maintenance of Musha

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss the principles and specifications of Musha.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Analyse modern aspects of Musha with special reference to crucibles.	*analyse*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate the handling of Musha following the Standard Operating Procedure (SOP).	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Develop Standard Operative Procedure for handling Musha.	*develop*	Procedural	Experiential 8h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
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MODULE 5 · UNIT 3

Chromatography

5 objectives | 17 hours | Lecture 3 · Practical 1 · Experiential 1 | Outcomes: CO 4, CO 7, CO 8

Scope and applications of chromatography in herbal drug standardization (identification of active compounds, adulterant detection, phytochemical analysis, and quality as

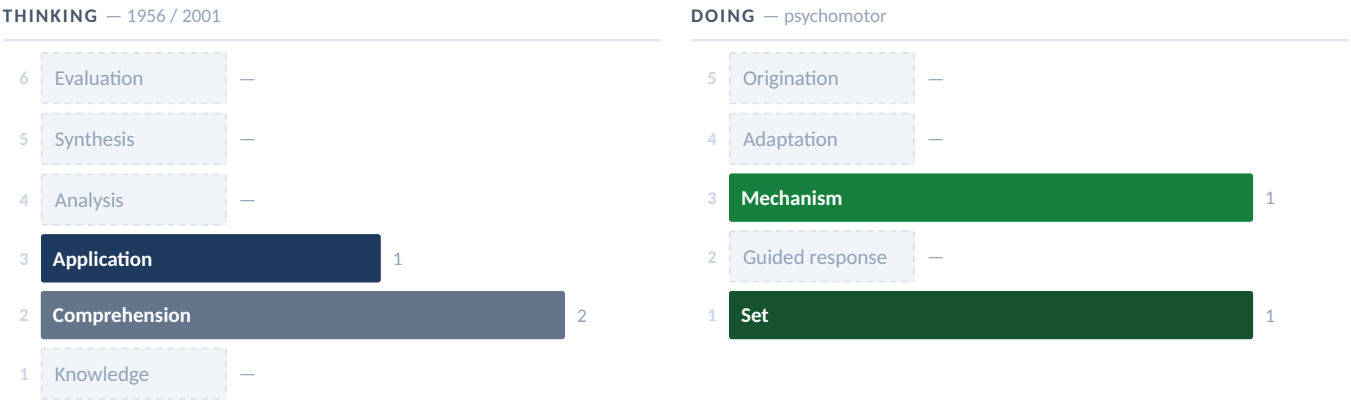
Principle and types of chromatography, including TLC, HPTLC, HPLC, LC-MS, and GC-MS.

Instrumentation of chromatography, including equipment and tools used in different types of chromatography.

Data analysis in chromatography, including interpretation of chromatograms and spectra.

Chromatography's Limitations, challenges, and prospects in pharmaceutical and phytochemical research

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Describe the instrumentation of chromatography, including equipment and tools used in different types of chromatography.	*describe*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss chromatography's Limitations, challenges, and prospects in pharmaceutical and phytochemical research.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Discuss the scope and applications of chromatography in herbal drug standardization.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
DOING — LEVEL 1 · SET — 1				
Demonstrate chromatography techniques to separate, and identify components and interpretation of results.	*demonstrate*	Procedural	Practical 6h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Perform chromatography to separate and identify components and interpret the results.	*perform*	Procedural	Experiential 8h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 5 · UNIT 4

Spectroscopy

4 objectives | 9 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 4, CO 8

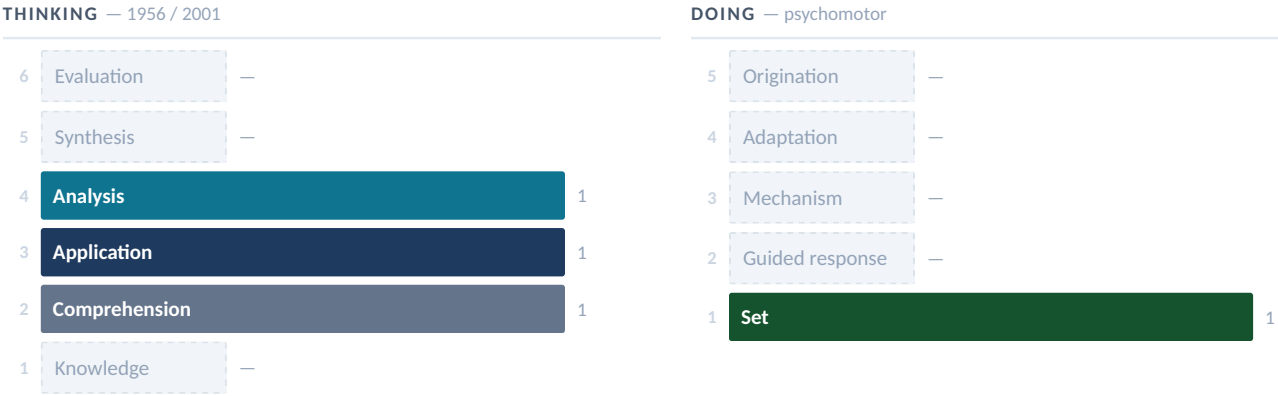
Introduction, principles, and types (IR, UV, NMR, AAS, ICP, etc.) of spectroscopy.

Specifications of spectroscopy instruments

SOPs for handling spectroscopy instruments

Troubleshooting and maintenance of spectroscopy instruments

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss spectroscopy's principles and types (IR, UV, NMR, AAS, ICP, etc.).	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Describe the specifications of spectroscopy instruments.	*describe*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Explore and analyze published articles on spectroscopic instruments.	*explore*	Conceptual °	Experiential 4h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 1 · SET — 1				
Demonstration of spectroscopic instruments through Audiovisual aids.	*demonstrate*	Procedural	Practical 3h	**Watch them do it.** You may guide while watching.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 5 · UNIT 5

Microscopy

4 objectives | 12 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 4, CO 8

Introduction, principles, and types of microscopy (optical, petrological, electron, etc.)

Specifications of microscopy instruments

SOPs for handling microscopy instruments

Use of micrometer in microscopy

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Describe the principles, types, and specifications of microscopy instruments.	*describe*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Describe the Specifications of microscopy instruments.	*describe*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate handling of petrological and trinocular microscopes.	*demonstrate*	Procedural	Practical 6h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Perform hands-on practice on the trinocular microscope.	*perform*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 5 · UNIT 6

Calibration techniques

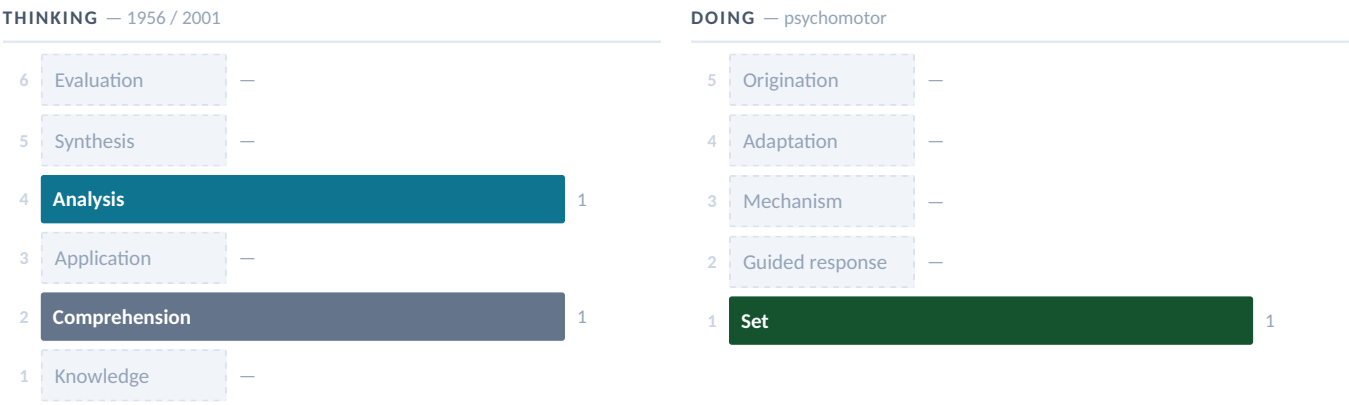
3 objectives | 7 hours | Lecture 1 · Practical 1 · Experiential 1 | Outcomes: CO 4, CO 7, CO 8

Introduction, principles, and types of calibration (instrument calibration, method calibration, etc.)

SOPs for calibration

Troubleshooting and maintenance of calibration instruments

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss principles and types of calibration.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Analyse articles related to challenges in the calibration of equipment.	*analyse*	Conceptual °	Experiential 3h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 1 · SET — 1				
Demonstrate calibration of various instruments present in the department.	*demonstrate*	Procedural	Practical 3h	**Watch them do it.** You may guide while watching.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
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- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 5 · UNIT 7

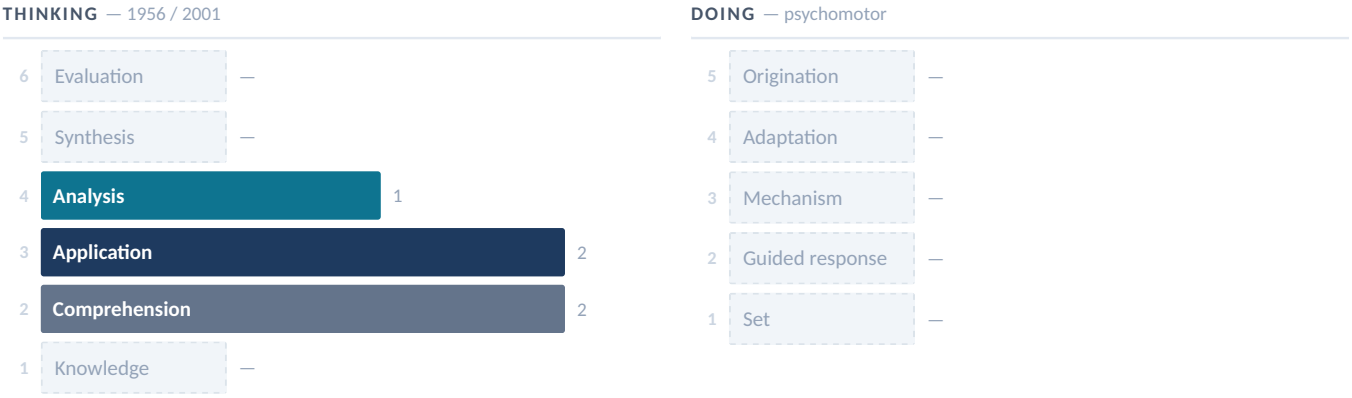
Overview of characterization instruments

5 objectives | 11 hours | Lecture 3 · Practical 1 · Experiential 1 | Outcomes: CO 4, CO 7, CO 8

Overview of advanced characterization instruments and their significance in the standardization of Ayurvedic medicines.

Principles and applications of working mechanism and Applications of instrumental methods: PSA, XRD, XRF, SEM-E-Dax, TEM, TGA, DTA, VSM analysis

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Discuss overview of Advanced Characterization Instruments and their significance in the standardization of Ayurvedic medicines.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss principles and Applications of Spectroscopic and X-ray techniques (PSA, XRD and XRF techniques).	*discuss*	Conceptual + Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 2				
Describe the Principles and Applications of Microscopic and Thermal Techniques(SEM-EDAX, TEM, TGA, DTA, and VSM).	*describe*	Conceptual + Procedural °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
Demonstrate characterization instruments with Audiovisual aids.	*demonstrate*	Procedural °	Practical 4h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Analyse published articles on advanced instruments used for characterization.	*analyse*	Conceptual °	Experiential 4h	Written or spoken answer, pitched at the coded level.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 6

Animal house and Animal handling technique

CREDITS	UNITS	THINKING	DOING	FORMATIVE
1	5	7	6	25
30 notional hours	13 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

Tehniques, emphasizing animal care and welfare. It covers the principles of animal house design, sanitation, and hygiene practices, as well as species-specific handling techniques, animal health, and disease prevention. The module also explores advanced animal handling techniques, animal welfare, and research ethics. 1 5 10 15 30

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	5 h	5	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	10 h	5	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	15 h	3	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 2 hours of the 15 shown, worth 25 marks.

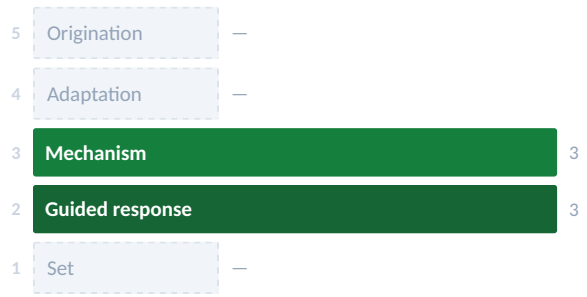
WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Describe the principles of animal care and welfare, and apply them to animal house management.	*describe*	Level 2 Comprehension / Understand

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
2	Demonstrate handling techniques for various animal species, including small and large animals.	*demonstrate*	Level 3 Application / Apply
3	Identify and prevent common diseases in laboratory animals, and implement disease control measures.	*identify*	Level 1 Knowledge / Remember
4	Apply advanced animal handling techniques and promote animal welfare and ethics in research.	*apply*	Level 3 Application / Apply
5	Develop skills in animal behavior observation, enrichment techniques, and humane endpoints and euthanasia methods.	*develop*	Level 5 Synthesis / Create

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Introduction to animal house management	3	7 h	Comprehension	Mechanism
2	Study of animal behaviour	3	8 h	Synthesis	Mechanism
3	Species-specific handling techniques	2	3 h	Comprehension	Guided response
4	Animal health and disease prevention	3	7 h	Comprehension	Mechanism
5	Animal welfare and ethics	2	3 h	Evaluation	—

MODULE 6 · UNIT 1

Introduction to animal house management

3 objectives | 7 hours | Lecture 1 · Practical 1 · Experiential 1 | Outcomes: CO 7, CO 8

- Overview of animal house management
- Animal house design and layout
- Record-keeping and documentation

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Describe the principles of Animal House Management with its design and layout.	*describe*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Design and Set Up an Animal House.	*design*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Present structure and functioning of an animal house.	*present*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 6 · UNIT 2

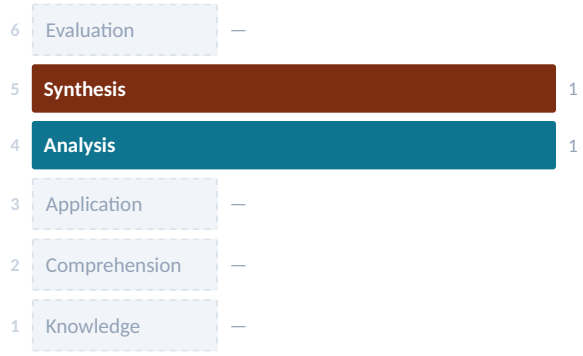
Study of animal behaviour

3 objectives | 8 hours | Lecture 1 · Practical 1 · Experiential 1 | Outcomes: CO 7, CO 8

Introduction to Animal Behaviour	Types of Animal behaviour.

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1				
Identify behavioral patterns and body language in animals.	*identify*	Factual °	Lecture 1h	Written or spoken. Pitch at **level 1** — the verb *identify* is what will be asked.
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1				
Observe and. Interpret Animal Body Language through demonstration.	*observe*	Factual °	Practical 2h	Written or spoken. Pitch at **level 1** — the verb *observe* is what will be asked.
DOING — LEVEL 3 · MECHANISM — 1				
Develop critical thinking and research skills by analyzing online resources and studies on experimental animal behavior.	*develop*	Procedural	Experiential 5h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 6 · UNIT 3

Species-specific handling techniques

2 objectives | 3 hours | Lecture 1 · Practical 1 | Outcomes: CO 7, CO 8

Non-human primates

Aquatic animals

Other specialized species (e.g., birds, reptiles)

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss the handling techniques for various laboratory animal species, ensuring handler safety.	*discuss*	Procedural	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate Species-Specific Handling Techniques for Laboratory Animals.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 6 · UNIT 4

Animal health and disease prevention

3 objectives | 7 hours | Lecture 1 · Practical 1 · Experiential 1 | Outcomes: CO 7, CO 8

Common diseases in laboratory animals
Vaccination and immunization programs
Parasite control and management

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Describe Common diseases in laboratory animals and their management.	*describe*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate microscopic Examination of Faecal Samples for Parasites.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Prepare and present a parasite Control in Laboratory Animals.	*prepare*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
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- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 6 · UNIT 5

Animal welfare and ethics

2 objectives | 3 hours | Lecture 1 · Practical 1 | Outcomes: CO 7, CO 8

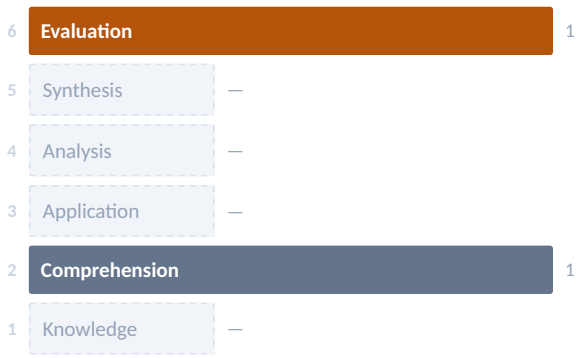
Animal welfare principles and ethics

Humane endpoints and euthanasia methods

Case studies and group discussions on animal welfare and ethics

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss Animal Welfare Principles, Ethics, Humane Endpoints, Euthanasia Methods, and Case Studies in Animal Research.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1				
Evaluate Animal Welfare and Ethics in a Research Setting.	*evaluate*	Conceptual	Practical 2h	Written or spoken answer, pitched at the coded level.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 7

Classical and Advanced quality control techniques for Ayurveda Formulations

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	14	12	50
60 notional hours	26 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module, Classical and Advanced Quality Control Techniques for Ayurveda Formulations, provides a comprehensive understanding of quality control and quality assurance in Ayurveda formulations. It covers classical and advanced quality control techniques, including quality control parameters, regulatory requirements, and documentation. The module also explores quality control paramet

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

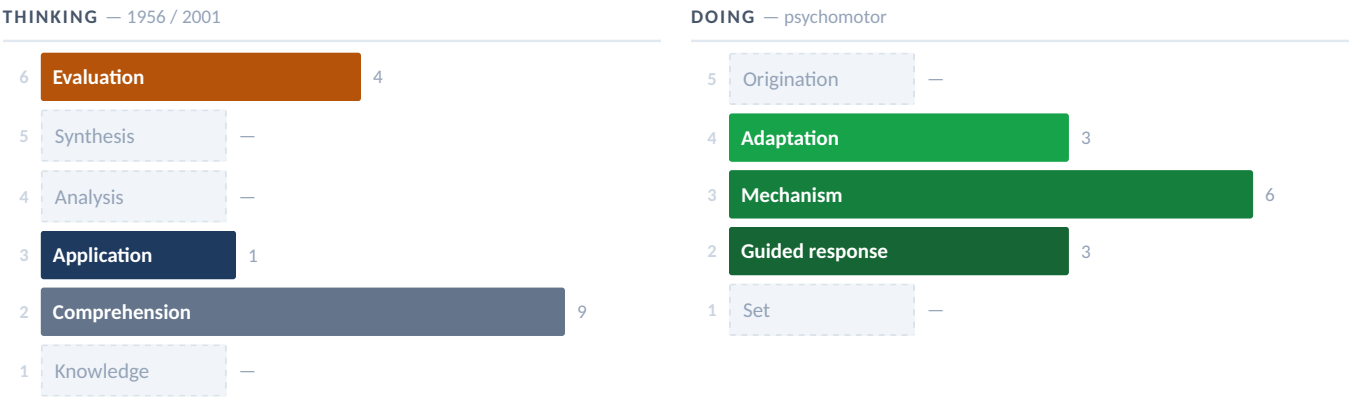
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	10 h	10	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	20 h	8	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	30 h	8	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 4 hours of the 30 shown, worth 50 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.



MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Describe the importance of quality control and quality assurance in Ayurveda product manufacturing.	*describe*	Level 2 Comprehension / Understand
2	Describe classical quality control parameters for Ayurvedic formulations, including Mana, Avaleha Siddhi Lakshan, and Bhasma Pariksha.	*describe*	Level 2 Comprehension / Understand
3	Apply quality control parameters for various Ayurvedic dosage forms, including solid, semi-solid, and liquid.	*apply*	Level 3 Application / Apply
4	Identify regulatory requirements for quality control and ensure documentation and record-keeping compliance.	*identify*	Level 1 Knowledge / Remember
5	Integrate classical and advanced quality control techniques to ensure the quality and safety of Ayurvedic formulations.	*integrate*	Level 5 Synthesis / Create

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Quality Control and Quality Assurance	4	6 h	Comprehension	Adaptation
2	Classical aspects of the quality control parameter for Ayurvedic formulations	4	6 h	Application	Mechanism
3	Quality Control parameter for solid dosage forms	9	26 h	Evaluation	Mechanism
4	Quality Control parameter for semi-solid dosage forms	3	6 h	Evaluation	Mechanism
5	Quality Control parameter for liquid and gaseous dosage forms	6	12 h	Evaluation	Adaptation

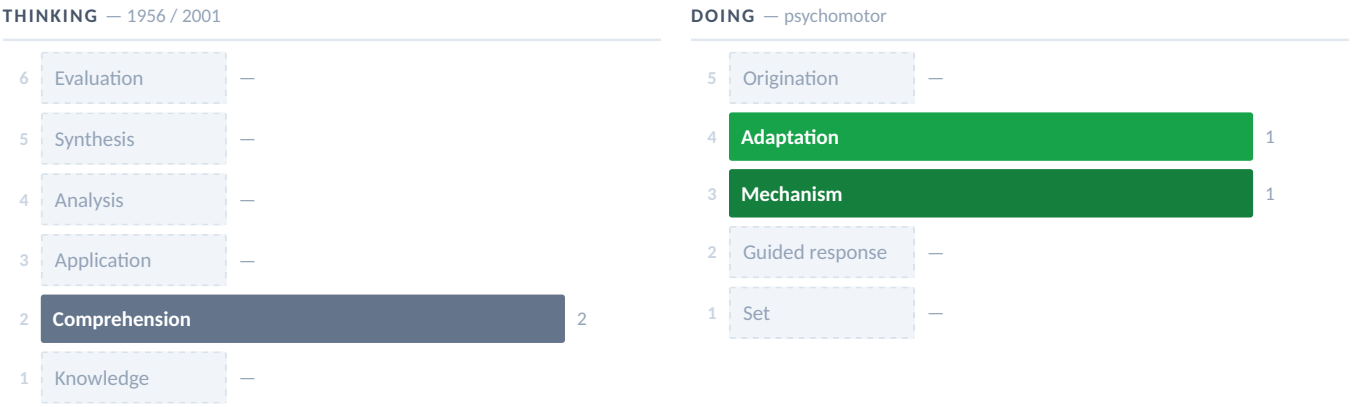
MODULE 7 · UNIT 1

Quality Control and Quality Assurance

4 objectives | 6 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 4, CO 5

- Importance of quality control in Ayurveda
- Quality Assurance of Ayurveda Products
- Quality control parameters (identity, purity, etc.)
- Regulatory requirements for quality control
- Documentation and record-keeping

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Describe the importance of Quality Control and Quality Assurance in Ayurveda.	*describe*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss Quality Control Parameters, Regulatory Requirements, and Documentation.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 3 · MECHANISM — 1				
Practice documentation and record-keeping processes in the quality control section.	*practice*	Procedural	Experiential 2h	**Watch them do it.** They work unaided.
DOING — LEVEL 4 · ADAPTATION — 1				
Demonstrate documentation and record-keeping processes in the quality control section.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** Alter a condition and watch what they change.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 7 · UNIT 2

Classical aspects of the quality control parameter for Ayurvedic formulations

4 objectives | 6 hours | Lecture 2 · Practical 1 · Experiential 1 | Outcomes: CO 1, CO 2, CO 4

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss Siddhi Lakshana of Avaleha, Snehapaka, Sandhana and Guggulu Paka.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Discuss Mana and their application in quality control and Bhasma/Pishti pariksha.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma).	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 1				
Assess Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma).	*assess*	Procedural	Experiential 2h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 7 · UNIT 3

Quality Control parameter for solid dosage forms

9 objectives | 26 hours | Lecture 3 · Practical 3 · Experiential 3 | Outcomes: CO 4

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3				
Discuss quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.	*discuss*	Procedural	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss Quality control parameters for solid dosage forms like tablets, pills, and Varti.	*discuss*	Procedural	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss Quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana Lauha/Mandura.	*discuss*	Procedural	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 2				
Demonstrate Quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.	*demonstrate*	Procedural	Practical 3h	Written or spoken. Pitch at **level 3** — the verb *demonstrate* is what will be asked.
Demonstrate quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana Lauha/Mandura.	*demonstrate*	Procedural	Practical 4h	Written or spoken. Pitch at **level 3** — the verb *demonstrate* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate quality control parameters for solid dosage forms like tablets, pills, and Varti.	*demonstrate*	Procedural	Practical 3h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 3				
Perform Quality control procedures for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.	*perform*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.
Perform quality control procedures for solid dosage forms.	*perform*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
Perform quality control procedures for Rasoushadhi.	*perform*	Procedural	Experiential 5h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 7 · UNIT 4

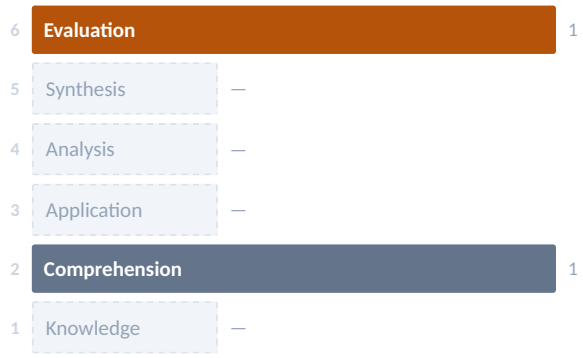
Quality Control parameter for semi-solid dosage forms

3 objectives | 6 hours | Lecture 1 · Practical 1 · Experiential 1 | Outcomes: CO 4

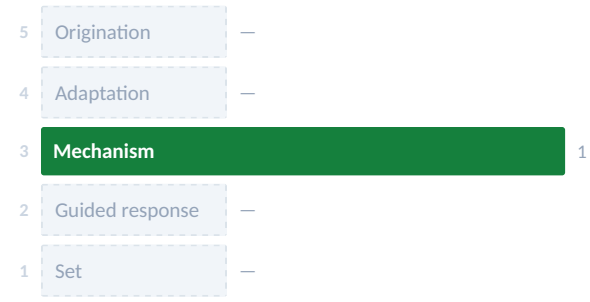
Rasakriya/Avaleha/Paka	Anjana

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Describe Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/Paka/Anjana as per Pharmacopeial standards.	*describe*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1				
Demonstration of Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/Paka/Anjana as per Pharmacopeial standards.	*demonstrate*	Procedural	Practical 2h	Written or spoken. Pitch at **level 3** — the verb *demonstrate* is what will be asked.
DOING — LEVEL 3 · MECHANISM — 1				
Perform quality control tests for semisolid dosage forms.	*perform*	Procedural	Experiential 3h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

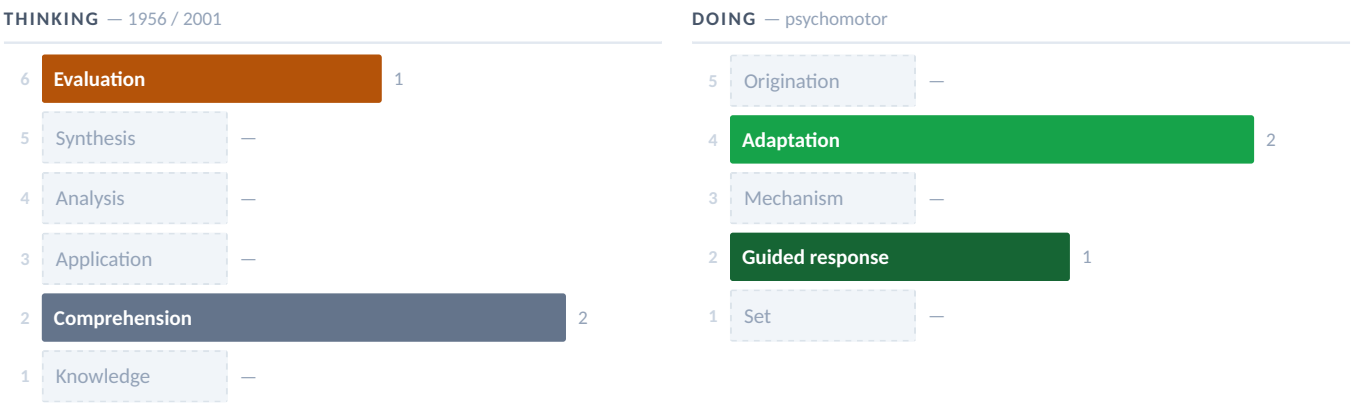
MODULE 7 · UNIT 5

Quality Control parameter for liquid and gaseous dosage forms

6 objectives | 12 hours | Lecture 2 · Practical 2 · Experiential 2 | Outcomes: CO 1, CO 4

Kashaya	Asava-arishta
Arka	Netra Bindu/Karna Bindu
Sharkara	Aerosols/sprays

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Discuss Quality control parameters for liquid dosage forms like Kashaya, Arka Sharakara, Asava-arishta.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss Quality control parameters for Netra Bindu/Karna Bindu and aerosols/sprays.	*discuss*	Procedural °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1				
Demonstrate quality control parameters for liquid dosage forms like Kashaya, Arka Sharakara, Asava-arishta.	*demonstrate*	Procedural	Practical 2h	Written or spoken. Pitch at **level 3** — the verb *demonstrate* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate quality control parameters for Netra Bindu/ Karna Bindu and aerosols/sprays.	*demonstrate*	Procedural	Practical 2h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 4 · ADAPTATION — 2				
Perform quality control tests for liquid dosage forms.	*perform*	Procedural	Experiential 3h	**Watch them do it.** Alter a condition and watch what they change.
Perform quality control tests for Netra Bindu/Karna Bindu and aerosols/sprays.	*perform*	Procedural	Experiential 3h	**Watch them do it.** Alter a condition and watch what they change.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 8

Nano-science, nano-technology and its application in Ayurveda

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	10	11	50
60 notional hours	21 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

This module Understanding Nano-Science, Nano-Technology, and its Application in 2 10 20 30 60 Ayurveda covers integration of nano-science and nano-technology withAyurveda, exploring its potential to revolutionize the field of traditional medicine. This module explores the fundamentals of nanotechnology, its applications in Ayurvedic formulations, and the regulatory and safety aspects of this emerg

WHERE THE HOURS GO, AND WHAT EACH KIND OF HOUR CAN REACH

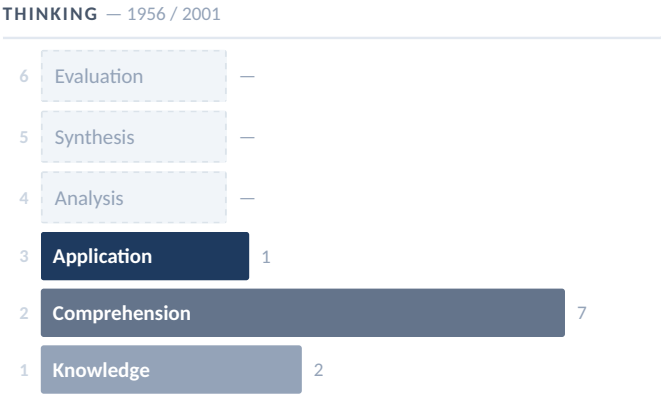
This curriculum splits every module three ways in a fixed 1:2:3 ratio — one part lecture, two parts practical training, three parts experiential learning. The three kinds of hour do not reach the same levels, so the split is itself a statement about where the learning is meant to happen.

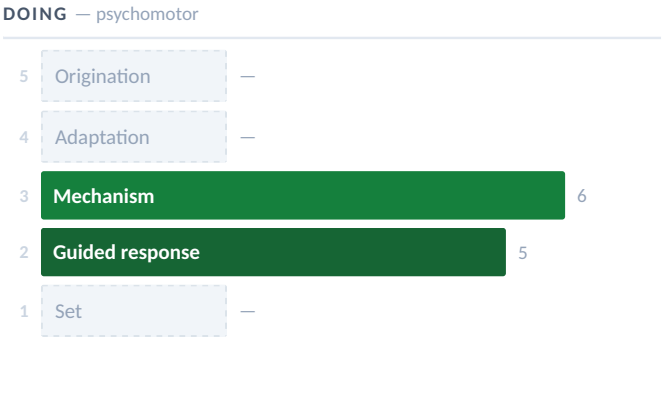
KIND OF HOUR	HOURS	OBJECTIVES	LEVELS IT CAN REACH	WHAT IT MUST DELIVER HERE
Lecture	10 h	10	levels 1 and 2 - exposure and first understanding	Put the material in front of students and make them retrieve it. It cannot, on its own, take them past level 2.
Practical	20 h	5	the doing levels, and level 3 thinking	Move students from performing with help to performing unaided. That transition is what these hours are for.
Experiential	30 h	6	levels 4 to 6, and the upper doing levels	Hand over an unfamiliar situation and require a judgement. This is where the top levels are actually built.

Experiential hours also carry the modular assessment: 4 hours of the 30 shown, worth 50 marks.

WHAT THIS MODULE ASKS STUDENTS TO DO

Every objective in Table 3 carries a code in Column 3E. Gathered across the module, those codes show which ladders this module climbs and how far. Read all three: a module can be demanding on one and undemanding on another.





MODULE LEARNING OBJECTIVES

What the curriculum states a student should be able to do at the end of the module, with the level each is written to. These sit above the session objectives in the units, and are what the modular assessment tests.

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
1	Define nanotechnology, its history, principles, and applications.	*define*	Level 1 Knowledge / Remember

#	MODULE LEARNING OBJECTIVE	VERB	LEVEL OF THAT VERB
2	Describe Ayurveda Formulations with respect to Nanotechnology	*describe*	Level 2 Comprehension / Understand
3	Discuss the potential of nanotechnology in Ayurvedic formulations.	*discuss*	Level 2 Comprehension / Understand
4	Identify and describe various types of nanocarriers for Ayurvedic formulations.	*identify*	Level 1 Knowledge / Remember
5	Discuss the regulatory frameworks, safety concerns, and toxicity issues related to nanotechnology-based Ayurvedic formulations.	*discuss*	Level 2 Comprehension / Understand
6	Describe recent advances and future directions in nanotechnology-based Ayurvedic formulations.	*describe*	Level 2 Comprehension / Understand

THE UNITS IN THIS MODULE

UNIT	TITLE	OBJ	HOURS	HIGHEST THINKING	HIGHEST DOING
1	Introduction to Nanotechnology	3	6 h	Application	Guided response
2	Ayurvedic Formulations and Nanotechnology	10	34 h	Comprehension	Mechanism
3	Nanocarriers for Ayurvedic Formulations	3	6 h	Comprehension	Guided response
4	Regulatory Aspects and Safety Concerns	2	2 h	Comprehension	—
5	Recent Advances and Future Directions	3	8 h	Comprehension	Mechanism

MODULE 8 · UNIT 1

Introduction to Nanotechnology

3 objectives | 6 hours | Lecture 2 · Practical 1 | Outcomes: CO 4, CO 8

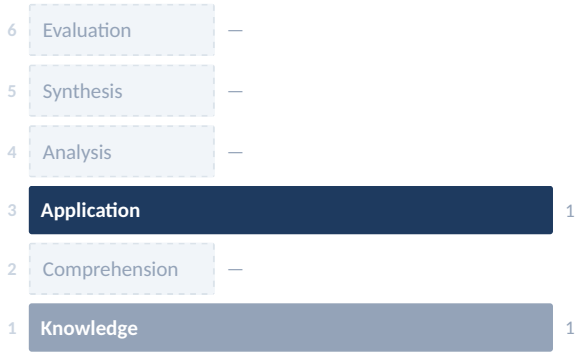
Definition and history of nanotechnology

Basic principles and applications of nanotechnology

Understanding Nanoscale

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1				
Define nanotechnology and describe the history of nanotechnology.	*define*	Factual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
THINKING — LEVEL 3 · APPLICATION / APPLY — 1				
Describe the basic principles and applications of nanotechnology.	*describe*	Conceptual	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *describe* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate the measurement of micro-particles using relevant equipment.	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 8 · UNIT 2

Ayurvedic Formulations and Nanotechnology

10 objectives | 34 hours | Experiential 5 · Practical 3 · Lecture 2 | Outcomes: CO 4, CO 5, CO 8

- Overview of Ayurvedic formulations and their limitations
- Potential of nanotechnology in Ayurvedic formulations
- Characterization micro/nanoparticles in Bhasma
- Characterization micro/nanoparticles in Kupipakwa Rasayana and Pottali
- Characterization of micro-nano sized particles generated during Bhavana
- Synthesize silver nanoparticles (AgNPs) using the chemical precipitation method and grasp the fundamental principles underlying nanoparticle synthesis
- Market Survey of Nano-Based Herbal Formulations

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Discuss an overview of Ayurvedic Nano formulations and their limitations.	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Identify the potential of nanotechnology in Ayurvedic formulations.	*identify*	Factual	Lecture 1h	Written or spoken. Pitch at **level 1** — the verb *identify* is what will be asked.
DOING — LEVEL 2 · GUIDED RESPONSE — 3				
Demonstrate particle size of Bhasma prepared with different numbers of Puta.	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
Demonstrate particle size of various Kupipakwa Rasayana samples.	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
Synthesize silver nanoparticles (AgNPs) using the chemical precipitation method.	*synthesize*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.
DOING — LEVEL 3 · MECHANISM — 5				
Measure particle size of Bhasma prepared with different numbers of Puta.	*measure*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.
Measure particle size of various Kupipakwa Rasayana samples.	*measure*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
Review articles regarding the green synthesis of metallic nanoparticles.	*review*	Procedural	Experiential 2h	**Watch them do it.** They work unaided.
Analyze the impact of Bhavana by measuring the particle size of samples.	*analyze*	Procedural	Experiential 4h	**Watch them do it.** They work unaided.
Conduct a market Survey of Nano-Based Ayurvedic Formulations.	*conduct*	Procedural	Experiential 6h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
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MODULE 8 · UNIT 3

Nanocarriers for Ayurvedic Formulations

3 objectives | 6 hours | Lecture 2 · Practical 1 | Outcomes: CO 4, CO 5

Types of nanocarriers (liposomes, nanoparticles, etc.)

Advantages and limitations of nanocarriers in Ayurvedic formulations

Conversion of Taila into Oil in water nano emulsion and its characterization

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Discuss types of nanocarriers (liposomes, nanoparticles, etc.).	*discuss*	Conceptual	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss the advantages and limitations of nanocarriers in Ayurvedic formulations.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 2 · GUIDED RESPONSE — 1				
Demonstrate converting Taila into an oil-in-water nanoemulsion and its characterization.	*demonstrate*	Procedural	Practical 4h	**Watch them do it.** You may guide while watching.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
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- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 8 · UNIT 4

Regulatory Aspects and Safety Concerns

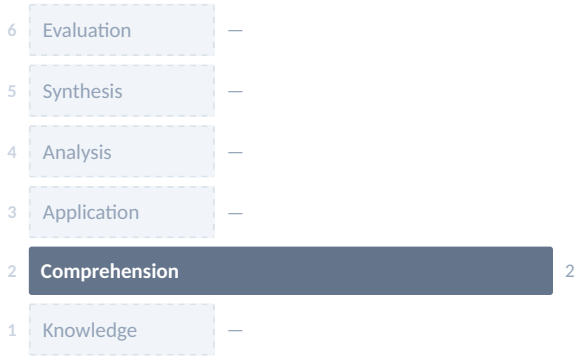
2 objectives | 2 hours | Lecture 2 | Outcomes: CO 5, CO 8

Regulatory frameworks for nanotechnology-based Ayurvedic formulations

Safety concerns and toxicity issues related to nanotechnology-based Ayurvedic formulations

WHAT THIS UNIT ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2				
Discuss regulatory frameworks for nanotechnology-based Ayurvedic formulations.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
Discuss safety concerns and toxicity issues related to nanotechnology-based Ayurvedic formulations.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

MODULE 8 · UNIT 5

Recent Advances and Future Directions

3 objectives | 8 hours | Lecture 2 · Experiential 1 | Outcomes: CO 5, CO 8

> Recent advances in nanotechnology-based Ayurvedic formulations
> Future directions and potential applications
>
> Exploring the Future of Nanomedicine with Ayurvedic Formulations

WHAT THIS UNIT ASKS STUDENTS TO DO



THE OBJECTIVES, GROUPED BY LEVEL

Table 3 sorted into its domains and levels, so a session can be planned around one level at a time. The kind of hour is shown against each: it is the strongest clue to what the curriculum intends, because an objective set as experiential learning is meant to be reached by doing rather than by telling.

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR	EVIDENCE IT NEEDS
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1				
Discuss recent advances in nanotechnology-based Ayurvedic formulations.	*discuss*	Conceptual °	Lecture 1h	Written or spoken. Pitch at **level 2** — the verb *discuss* is what will be asked.
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1				
Discuss future directions and potential applications of nanotechnology in Ayurvedic formulations.	*discuss*	Conceptual °	Lecture 1h	Written or spoken answer, pitched at the coded level.
DOING — LEVEL 3 · MECHANISM — 1				
Integrate nanomedicine with Ayurvedic formulations and their prospects.	*integrate*	Procedural	Experiential 6h	**Watch them do it.** They work unaided.

° the kind of knowledge is likely, but not certain from the wording alone. Where two kinds are named, the objective works on both. **Level** always means the Bloom level of Column 3E.

FOR THE TEACHER

- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective in this unit. An objective set as experiential learning and then delivered as a lecture has been quietly demoted.
- **Teach the thinking that the doing needs, first.** Students who cannot say why a step exists drop it the moment they are unsupervised.
- **Write the checklist before the session.** Decide the steps you will look for and how many attempts a student gets, so the mark is about the student and not about the day.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will be asked to produce.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder; doing it alone and reliably is level 3, and that is the expectation here.
- **Keep the record as you go.** A log filled in at the end is a piece of writing; filled in as you work, it is evidence.

PART C

Reference

Nine appendices, all worked out from the curriculum's own objectives, so the counts here and the counts in Part B are the same numbers. C1 and C2 decode anything in the syllabus document. C3 compares every module and unit at a glance. C6 and C7 reproduce all ninety-one practical and experiential activities with the level each serves, so a reference like "Experiential Learning 8.2" can be looked up without turning back to the syllabus.

1. **C1** The sixteen curriculum codes, decoded
2. **C2** The two schemes side by side
3. **C3** Level profile of every module and unit
4. **C4** Verb index — every verb the curriculum uses, and its level
5. **C5** Kinds of knowledge across the semester
6. **C6** All 47 practical training activities, mapped
7. **C7** All 44 experiential learning activities, mapped
8. **C8** Where the hours go, module by module
9. **C9** Sources and further reading

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PART C · APPENDIX C1

The sixteen curriculum codes, decoded

Every code that appears in Column 3E of Table 3, with the domain it belongs to, the level it sits at, and how many objectives carry it. Keep this beside the syllabus document.

CODE	DOMAIN	LEVEL	NAME	TIMES USED THIS SEMESTER
CK	Thinking	1	Knowledge / Remember	8
CC	Thinking	2	Comprehension / Understand	55
CAP	Thinking	3	Application / Apply	15
CAN	Thinking	4	Analysis / Analyse	15
CS	Thinking	5	Synthesis / Create	4
CE	Thinking	6	Evaluation / Evaluate	8
PSY-SET	Doing	1	Set	4
PSY-GUD	Doing	2	Guided response	26
PSY-MEC	Doing	3	Mechanism	28
PSY-ADT	Doing	4	Adaptation	5
PSY-ORG	Doing	5	Origination	2
AFT-REC	Going about it	1	Receiving	not used
AFT-RES	Going about it	2	Responding	1
AFT-VAL	Going about it	3	Valuing	not used
AFT-SET	Going about it	4	Organisation	not used
AFT-CHR	Going about it	5	Characterisation	not used

Sixteen codes are defined; 12 are used here. Unlike the undergraduate curriculum, this semester **does** use Synthesis and Origination.

PART C · APPENDIX C2

The two schemes side by side

The six thinking levels in both schemes, with what a student actually does at each.

LEVEL	BLOOM 1956, AS PRINTED IN THE CURRICULUM	REVISED 2001	WHAT THE STUDENT DOES
Level 1	**Knowledge**	**Remember**	recalls it
Level 2	**Comprehension**	**Understand**	restates it in their own arrangement
Level 3	**Application**	**Apply**	uses it on something new
Level 4	**Analysis**	**Analyse**	takes it apart and shows how the parts relate
Level 5	**Synthesis**	**Evaluate** — the schemes differ here	makes something that was not given
Level 6	**Evaluation**	**Create** — the schemes differ here	judges it against stated criteria

The top two levels are the only place the schemes disagree. This book follows the 1956 numbering, because that is what the curriculum prints in Column 3E.

PART C · APPENDIX C3

Level profile of every module and unit

All eight modules and forty-two units in one view. The fastest way to see where a semester's weight sits and to check that an assessment matches its module's ceiling.

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
MODULE 1 · PHARMACEUTICS AND PHARMACEUTICAL TECHNOLOGY — 24 OBJECTIVES, 56 H, CEILING EVALUATION / ADAPTATION						
1	1	Introduction to Pharmaceutics	6	12	Evaluation	Adaptation
1	2	Pharmaceutical Dosage Forms	4	8	Analysis	Mechanism
1	3	Pharmaceutical Formulation	5	16	Application	Mechanism
1	4	Pharmaceutical Manufacturing and Quality Control	5	10	Evaluation	Mechanism
1	5	Biopharmaceutics	4	10	Application	Mechanism
MODULE 2 · DATABASES AND INFORMATION SYSTEMS FOR METAL, MINERAL, AND PLANT RESEARCH — 22 OBJECTIVES, 56 H, CEILING EVALUATION / MECHANISM						
2	1	Introduction to Databases and Information Systems	5	12	Application	Mechanism
2	2	Metal and Mineral Databases	6	16	Synthesis	—
2	3	Plant Databases	4	12	Analysis	—
2	4	Database application	4	12	Evaluation	Mechanism
2	5	Challenges and limitation of Databases	3	4	Analysis	—
MODULE 3 · ESSENTIAL CHEMISTRY :- UNLOCKING THE RASASHASTRA TECHNIQUES — 21 OBJECTIVES, 56 H, CEILING ANALYSIS / ORIGINATION						
3	1	Atomic, Molecular Structure, and Periodic Table Elements	3	12	Knowledge	Mechanism
3	2	Specialized Chemistry Fields-An Outline	2	6	Comprehension	—
3	3	Organic Chemistry	3	8	Analysis	—
3	4	Phytochemistry	7	16	Comprehension	Mechanism
3	5	Metallurgy	6	14	Comprehension	Origination
MODULE 4 · CONCEPT, UNDERSTANDING, APPLICATION, SOPS AND CHEMICAL CHANGES OF SHODHANA AND MARANA PROCEDURES — 15 OBJECTIVES, 56 H, CEILING SYNTHESIS / ADAPTATION						
4	1	Shodhana and Marana Procedures	3	3	Analysis	—
4	2	SOP of Shodhana Processes	4	22	Comprehension	Adaptation
4	3	SOP of Marana process	4	22	Comprehension	Mechanism
4	4	Chemical Changes involved in Shodhana and Marana processes	1	1	Comprehension	—
4	5	Shodhana and Marana- Case Studies, troubleshooting and emerging trends	3	8	Synthesis	—
MODULE 5 · PRINCIPLES, SPECIFICATION, SOPS OF HANDLING OF CLASSICAL AND ADVANCED INSTRUMENTS — 29 OBJECTIVES, 84 H, CEILING SYNTHESIS / MECHANISM						
5	1	Puta and its modern aspects	4	14	Synthesis	Guided response
5	2	Musha and its modern aspects	4	14	Analysis	Mechanism
5	3	Chromatography	5	17	Application	Mechanism
5	4	Spectroscopy	4	9	Analysis	Set

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
5	5	Microscopy	4	12	Comprehension	Mechanism
5	6	Calibration techniques	3	7	Analysis	Set
5	7	Overview of characterization instruments	5	11	Analysis	—
MODULE 6 · ANIMAL HOUSE AND ANIMAL HANDLING TECHNIQUE — 13 OBJECTIVES, 28 H, CEILING EVALUATION / MECHANISM						
6	1	Introduction to animal house management	3	7	Comprehension	Mechanism
6	2	Study of animal behaviour	3	8	Synthesis	Mechanism
6	3	Species-specific handling techniques	2	3	Comprehension	Guided response
6	4	Animal health and disease prevention	3	7	Comprehension	Mechanism
6	5	Animal welfare and ethics	2	3	Evaluation	—
MODULE 7 · CLASSICAL AND ADVANCED QUALITY CONTROL TECHNIQUES FOR AYURVEDA FORMULATIONS — 26 OBJECTIVES, 56 H, CEILING EVALUATION / ADAPTATION						
7	1	Quality Control and Quality Assurance	4	6	Comprehension	Adaptation
7	2	Classical aspects of the quality control parameter for Ayurvedic formulations	4	6	Application	Mechanism
7	3	Quality Control parameter for solid dosage forms	9	26	Evaluation	Mechanism
7	4	Quality Control parameter for semi-solid dosage forms	3	6	Evaluation	Mechanism
7	5	Quality Control parameter for liquid and gaseous dosage forms	6	12	Evaluation	Adaptation
MODULE 8 · NANO-SCIENCE, NANO-TECHNOLOGY AND ITS APPLICATION IN AYURVEDA — 21 OBJECTIVES, 56 H, CEILING APPLICATION / MECHANISM						
8	1	Introduction to Nanotechnology	3	6	Application	Guided response
8	2	Ayurvedic Formulations and Nanotechnology	10	34	Comprehension	Mechanism
8	3	Nanocarriers for Ayurvedic Formulations	3	6	Comprehension	Guided response
8	4	Regulatory Aspects and Safety Concerns	2	2	Comprehension	—
8	5	Recent Advances and Future Directions	3	8	Comprehension	Mechanism

Both ceilings are shown because they move independently. Module 3 reaches only Analysis on the thinking ladder but Origination on the doing ladder, the highest doing level in the semester.

PART C · APPENDIX C4

Verb index — every verb the curriculum uses, and its level

The opening verb of all 171 objectives, sorted by the level that verb belongs to. The verb is what a student meets, which makes this the most practical table in the book.

LEVEL	NAME	VERBS THE CURRICULUM USES, WITH HOW OFTEN
Level 1	**Knowledge** Remember	identify 8, define 3, enlist 1, observe 1
Level 2	**Comprehension** Understand	discuss 42, describe 26, classify 1, illustrate 1, present 1
Level 3	**Application** Apply	demonstrate 34, perform 11, apply 2, measure 2, prepare 1, conduct 1
Level 4	**Analysis** Analyse	analyse 10, explore 4, analyze 3, differentiate 2, search 1, review 1
Level 5	**Synthesis** Create	design 4, develop 3, formulate 2, synthesize 1, integrate 1
Level 6	**Evaluation** Evaluate	evaluate 2, assess 1
Affective	**Attitude verbs**	practice 1

Choose the level first, then take a verb from that row. Where an objective's verb sits at a different level from its code, Part B says which to pitch to.

PART C · APPENDIX C5

Kinds of knowledge across the semester

The second axis of the 2001 scheme, counted across all 171 objectives.

KIND OF KNOWLEDGE	OBJECTIVES	HOW IT IS LEARNED
Factual	10	Rehearsal and self-testing. Only ten objectives here — this is not a memorisation course.
Conceptual	73	Explaining it aloud to somebody without notes.
Procedural	96	Carrying it out, more than once. **The dominant kind in this semester**, and the reason reading cannot substitute for the practical hours.
Metacognitive	not used	Being made to look back at how you worked. Not asked for here.

An objective working on two kinds is counted under both, so the total exceeds 171. The kind is read from the wording of each objective; where the wording does not settle it, the label carries a small circle (°).

PART C · APPENDIX C6

All 47 practical training activities, mapped

Table 4 of the curriculum, with the level each activity is cited to serve. Practical training is where a student moves from performing with help to performing unaided.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 1 — PHARMACEUTICS AND PHARMACEUTICAL TECHNOLOGY			
1.1	Demonstration of regulatory frameworks (FDA,etc.)	Doing 1 · Set	The teacher will demonstrate a scenario where a pharmaceutical company is developing a new drug for a specific disease condition. He will identify and explain key regulations and guidelines the company must follow for IND (Investigational New Drug) application submission. Students are asked to explore specific regulatory guidelines applicable to different regulatory bodies, construct the process of application, and present
1.2	Demonstration of pharmaceutical product development process	Doing 2 · Guided response	The teacher will demonstrate a video presentation (15 to 25 minutes) on the pharmaceutical product development process, covering introduction to a selected drug candidate, discovery and target identification, preclinical testing and formulation development, clinical trials, regulatory submission and approval, and commercialization and post-marketing surveillance.
1.3	Identify and categorise Classical Ayurvedic dosage forms	Thinking 4 · Analysis	Each student will identify and classify different classical dosage forms mentioned in Ayurvedic classics into solid, liquid, and semi-solid categories under the guidance of the teacher. Students will record observations and characteristics of each dosage form.
1.4	Identification and categorisation of modified Ayurvedic dosage forms	Thinking 4 · Analysis	Each student will identify and classify different modified Ayurvedic dosage forms into solid, semi-solid, liquid, and gaseous categories under the guidance of the teacher. Students will examine and sort samples of different dosage forms, such as tablets, capsules, solutions, suspensions, ointments, creams, and infusions. Students will record observations and characteristics of each dosage form.
1.5	Formulation of Solid Dosage Forms selecting appropriate excipients	Doing 2 · Guided response	The teacher will design and prepare a simple tablet formulation using a model drug and excipients. Measure and mix the ingredients, compress the tablets and evaluate their physical characteristics. Students will be instructed to prepare the tablet following the same steps.
1.6	Formulation of liquid Dosage Forms selecting appropriate excipients	Doing 2 · Guided response	The teacher will prepare a simple syrup formulation using a model drug and excipients. Measure and mix the ingredients, dissolve the solids, and evaluate the physical and chemical characteristics of the syrup. Students are instructed to prepare a syrup following the same steps.
1.7	GMP audit and compliance evaluation	Thinking 6 · Evaluation	The teacher will explain the importance of GMP and cGMP in drug manufacturing. He will provide the checklist to conduct an audit. Under his guidance students will conduct a mock audit to evaluate compliance with Good Manufacturing Practices (GMP) and current Good Manufacturing Practices (cGMP) in a pharmaceutical manufacturing setting. Assess documentation, personnel training, equipment maintenance, and facility cleanliness. In the case of non-GMP certified teaching pharmacy, the student can conduct a mock audit and note down the observations comparing with checklist and present his findings mentioning the shortcomings if any, and required components to match GMP standards

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
1.8	Demonstration, designing, and evaluation of pharmaceutical packaging and labelling	Doing 2 · Guided response	The teacher will demonstrate a standard packaging and labeling design with a suitable example. He will then assign each student the task of designing and evaluating a pharmaceutical packaging and labeling system for a specific product, considering factors such as container closure systems, labeling requirements, and tamper-evident features. The teacher will assess the system's compliance with regulatory requirements and industry standards.
MODULE 2 — DATABASES AND INFORMATION SYSTEMS FOR METAL, MINERAL, AND PLANT RESEARCH			
2.1	Demonstration of various searching techniques for databases	Doing 2 · Guided response	The teacher will demonstrate selected searching techniques for databases and demonstrate. Students will be informed to follow the techniques using various MeSH(Medical Subject Headings) terms
2.2	Demonstration of Modelling, Querying, and Analytics of Data	Doing 2 · Guided response	Teacher will Create a data model for a metal, mineral, or plant-related database. Students will be asked to write complex SQL (Structured Querry Language) queries to retrieve and analyze data. Perform data analytics tasks such as data visualization, data mining, and predictive modelling
2.3	Demonstration of techniques to search and extract data from the USGS Mineral Database	Thinking 4 · Analysis	The teacher will show how to search and extract data from the USGS Mineral Database on a specific mineral or metal. Students will be asked to follow the technique and extract the data for the assigned mineral. They are then asked to analyze and present the data in a report.
2.4	Analyse and integrate Web mineral Database	Thinking 5 · Synthesis	The teacher will demonstrate searching data for minerals from multiple databases and show how to integrate data from multiple metal and mineral databases to solve a real-world problem. Students will be advised to follow the technique to present the integrated database and discuss the challenges and limitations.
2.5	Search and explore Metal Database and Applications	Thinking 3 · Application	The teacher will demonstrate the technique to search and explore metal databases, including their applications and significance. Students will be advised to follow the technique and search metal databases and present a case study on the use of metal databases in industry or research
2.6	Demonstration of techniques to explore	Thinking 4 · Analysis	The teacher will demonstrate the techniques to search and extract data from The Plant List (TPL) and International Plant Names Index (IPNI) on a specific plant species or family. Students will be instructed to follow the technique, gather the data, analyze Plant Databases
2.7	Analysis of case studies of Database Applications in Metal, Mineral, or Plant Research	Thinking 6 · Evaluation	The teacher will demonstrate selected case studies of database applications in metal, mineral, or plant research and will explain how to evaluate the effectiveness of the database applications, identify best practices, and discuss potential improvements and future directions.
MODULE 3 — ESSENTIAL CHEMISTRY :- UNLOCKING THE RASASHASTRA TECHNIQUES			
3.1	Analysis of chemical bonding types, principles, and isomerism	Doing 2 · Guided response	The teacher will assign students video resources on chemical bonding and isomerism. Students will watch videos and take notes. They will be asked to present their notes on a basic understanding of the relevant chemistry concept with an interpretation relating to the Shodhana and Marana process.
3.2	Demonstration of the essential fundamental principles of crystallography and organometalli	Thinking 2 · Comprehension	—
3.3	Demonstration of different types of extraction methods including solvent selection	Doing 2 · Guided response	The teacher will select an herb and a formulation and demonstrate extraction following different methods and solvents. The teacher will explain the process of recording the percentage yield and physical status of the respective extracts. Different herbs and formulations will be selected for each session to avoid repetition.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
3.4	Demonstration of qualitative identification of phytoconstituents in single herbs and herbal formulations	Doing 2 · Guided response	The teacher will select an herb and a formulation and extract the same in specific solvents. Extracts will be tested for phytoconstituents by Test Tube methods. Different herbs and formulations will be selected for each session to avoid repetition.
3.5	Demonstration of different types of metals and their properties. The teacher will select d	Doing 2 · Guided response	—
MODULE 4 — CONCEPT, UNDERSTANDING, APPLICATION, SOPS AND CHEMICAL CHANGES OF SHODHANA AND MARANA PROCEDURES			
4.1	Demonstration of the Shodhana procedure of Rasadravyas using various methods	Doing 2 · Guided response	The teacher will demonstrate a set of SOPs of Various Shodhana procedures like Nimajjana (Immersion), Bhavana (Levegation), Swedana (heating under a liquid bath), Pachana(Boiling),Patana(Sublimation/ distillation),Nirvapa(Quenching),Dhalana (Melting and pouring), Bharjana(Roasting/frying), Nirjaleekarana(Dehydration) etc. with a selected suitable drug, explain scientifically applying different theories, and discuss the rationale behind the procedure and Shodhana Dravya used
4.2	Demonstration of the Marana procedure of Rasadravyas with different methods	Doing 2 · Guided response	The teacher will demonstrate any three Marana procedures along with SOPs for each procedure with a selected suitable drug and type of Puta, analyzing each step of the Marana scientifically applying different theories and discussing the rationale behind Marana Dravya used in the procedure.
MODULE 5 — PRINCIPLES, SPECIFICATION, SOPS OF HANDLING OF CLASSICAL AND ADVANCED INSTRUMENTS			
5.1	Demonstration of selected Puta to give the knowledge of Standard Operative Procedure	Doing 2 · Guided response	The teacher will select any one Puta and one type of fuel and demonstrate the Standard Operative Procedure of the selected Puta and explain the recording and documentation of the temperature pattern, and time frame. Different Puta/fuel should be selected for every batch to avoid repetition of the Puta/fuel.
5.2	Demonstration of the handling of Musha following the Standard Operating Procedure (SOP). T	Doing 2 · Guided response	—
5.3	Demonstration of the chromatography techniques to separate, The teacher will demonstrate t	Doing 1 · Set	—
5.4	Demonstration of spectroscopic instruments through Audiovisual aids	Doing 1 · Set	The teacher will show Audio-Visual Content related to the instrumentation and functioning of various spectroscopy instruments. After the demonstrations, the students will discuss the key points they observed. The teacher will summarize the contents. Further students are advised to observe the working of these instruments during their visit to the characterization lab as a part of module 8
5.5	Demonstration of handling of petrological and trinocular microscopes	Doing 2 · Guided response	The teacher will select any mineral and demonstrate the method of identifying rocks/minerals based on optical characters. For every batch different sample will be selected for demonstration (avoiding repetition).The teacher will select a suitable sample and demonstrate the microscopic characteristics of the sample like morphology and particle size with the help of projection and discuss the findings. For every batch different sample will be selected for demonstration (avoiding repetition)
5.6	Demonstration of calibration of various instruments present in the department	Doing 1 · Set	The concerned expert will demonstrate the methods of calibration of various equipment present in the department.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
5.7	Demonstration of characterization instruments with Audiovisual aids	Thinking 3 · Application	Each student will be instructed to search and collect articles related to specific instruments used for the characterization of Ayurvedic formulations like, PSA, XRD, XRF, SEM-EDAX, TEM, TGA, DTA, and VSM. Articles published in peer-reviewed journals will be selected. Findings in the articles will be presented and discussed (Collected articles should be unique and different for each student and the same will not be accepted in the next sessions)
MODULE 6 — ANIMAL HOUSE AND ANIMAL HANDLING TECHNIQUE			
6.1	Demonstration of animal house model	Doing 2 · Guided response	The teacher will demonstrate an animal house model and describe its compartments considering space requirements, ventilation, temperature control, lighting, noise levels, sanitation, and hygiene including record-keeping and documentation.
6.2	Demonstration of Animal Body Language	Thinking 5 · Synthesis	Teacher will assign each student a small laboratory animal group. They will be provided a checklist of behavioural patterns and body language cues to observe, including posture, facial expressions, and vocalizations. They will be instructed to observe the animals for 30 minutes, recording their observations on the checklist. After the observation period, students will discuss and interpret their findings. Encourage students to present their findings to the class, facilitating a discussion on the importance of recognizing animal behaviour and body language. Note: Before visiting virtual observation using Computer Assisted Learning using dedicated soft ware – Ex Pharm may be carried out
6.3	Demonstration of Species-Specific Handling Techniques for Laboratory Animals	Doing 2 · Guided response	Under the guidance of experienced instructors, students will engage in hands-on handling sessions with various small laboratory animal species, including rodents, rabbits, and guinea pigs. Students will learn and practice species-specific handling techniques, focusing on animal behavior, body language, and welfare considerations. The session will conclude with a debriefing discussion, addressing challenges, best practices, and future applications.
6.4	Demonstration of microscopic Examination of Faecal Samples for Parasites	Doing 2 · Guided response	Under the supervision of the teacher or animal house in-charge, students will collect and label fecal samples from laboratory animals, wearing gloves and lab coats. They will prepare slides with staining solution and examine them under the microscope, starting at 40x and increasing to 100x. They will record parasite presence, morphology, and size. They will compare their findings with reference materials and participate in a debriefing discussion
6.5	Evaluation of Animal Welfare and Ethics in a Research Setting. The teacher will prepare ca	Thinking 6 · Evaluation	—
MODULE 7 — CLASSICAL AND ADVANCED QUALITY CONTROL TECHNIQUES FOR AYURVEDA FORMULATIONS			
7.1	Demonstration of documentation and record-keeping processes in the quality control section	Doing 4 · Adaptation	The teacher will take the students to the quality control section and demonstrate the documentation and record-keeping processes for different dosage forms, from input materials to finished products.
7.2	Demonstration of assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma)	Doing 2 · Guided response	The teacher will select at least one Bhasma from each category (ore/ mineral, metal, and Sudha Varga) and demonstrate classical Bhasma Pareeksha. Additionally, special Bhasma Pareeksha tests for Rasadravyas (such as Amla Pareeksha for copper-containing compounds) will be performed. The scientific basis of these tests will be discussed. To illustrate the application of each test, the teacher will demonstrate Bhasma Pareeksha using both positive and negative samples.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
7.3	Demonstration of Quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva	Thinking 6 · Evaluation	The teacher will select at least one formulation from each category of powder dosage form and will demonstrate the quality testing methods as per the parameters prescribed in Pharmacopoeia.
7.4	Demonstrate quality control parameters for solid dosage forms like tablets, pills, and Varti	Doing 2 · Guided response	The teacher will select at least one formulation from each category of solid dosage form and will demonstrate the quality testing methods as per the parameters prescribed in Pharmacopoeia.
7.5	Demonstration of quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana, Lauha/ Mandura	Thinking 6 · Evaluation	The teacher will select at least one formulation from each category of Rasoushadhi (Chaturvidha Rasayana, Louha/Mandura) and will demonstrate the quality testing methods as per the parameters prescribed in Pharmacopoeia.
7.6	Demonstrate Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/ Paka/Anjana as per	Thinking 6 · Evaluation	The teacher will select at least one formulation from each category of semisolid dosage form and will demonstrate the quality testing methods as per the parameters prescribed in Pharmacopoeia. Pharmacopeial standards
7.7	Demonstration of quality control parameters for liquid dosage forms like Kashaya, Arka, Sharakara, Asava-arishta	Thinking 6 · Evaluation	The teacher will select at least one formulation from each category of liquid dosage form and will demonstrate the quality testing methods as per the parameters prescribed in Pharmacopoeia.
7.8	Demonstration of quality control parameters for Netra Bindu/Karna Bindu and aerosols/sprays	Doing 2 · Guided response	The teacher will select at least one formulation from aerosols/sprays and will demonstrate the quality testing methods as per the parameters prescribed in Pharmacopoeia.
MODULE 8 — NANO-SCIENCE, NANO-TECHNOLOGY AND ITS APPLICATION IN AYURVEDA			
8.1	Demonstration of the measurement of micro-particles using relevant equipment	Doing 2 · Guided response	The teacher will demonstrate the application of microscale measurement using relevant equipment. Various samples of differing dimensions, ranging from micro to centimeter levels, will be selected and measured using equipment like micrometers and vernier calipers. Under the teacher's guidance, students will identify microparticles from the selected samples.
8.2	Demonstration of particle size of Bhasma prepared with different numbers of Puta	Doing 2 · Guided response	The teacher will select Bhasma samples prepared with varying numbers of Puta and demonstrate the particle size measurement in these samples using a relevant microscope.
8.3	Demonstration of particle size of various Kupipakwa Rasayana samples	Doing 2 · Guided response	The teacher will select Kupipakwa Rasayana samples and demonstrate the measurable particle size in the samples using a relevant microscope.
8.4	Synthesiis of silver nano particles (AgNPs) using the chemical precipitatio n method	Doing 2 · Guided response	The teacher will demonstrate the synthesis of silver nanoparticles using the chemical method in the laboratory, utilizing silver nitrate solution, and compare the results with those of Rajata Bhasma
8.5	Conversion of Taila into oil-in-water nanoemulsion and its characterization. The teacher w	Doing 2 · Guided response	—

47 activities. The level shown is that of the objective in Table 3 which cites the activity, so it tells you what the activity is being asked to certify.

PART C · APPENDIX C7

All 44 experiential learning activities, mapped

Table 5 of the curriculum. These are the largest block of hours in the semester and where the upper levels are built; the modular assessment is drawn from them.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 1 — PHARMACEUTICS AND PHARMACEUTICAL TECHNOLOGY			
1.1	Recognising applications suiting various regulatory	Doing 4 · Adaptation	The teacher will assign individual students a scenario where a pharmaceutical company is developing a new drug (different dosage forms) for different health conditions. Identify key regulations and guidelines the company must follow for IND (Investigational New Drug) application submission. Review a case study of a company receiving an FDA warning letter for GMP violations and frameworks propose corrective actions. Prepare a short presentation summarizing findings and recommendations.
1.2	Listing out challenges and solutions in the pharmaceutical product development process	Thinking 6 · Evaluation	Each student will be assigned to identify challenges in pharmaceutical product development of different dosage forms, such as poor solubility and regulatory compliance issues. Design potential solutions and analyze case studies of companies that overcame similar challenges. Present solutions and receive peer feedback, then reflect on key takeaways and implications for real-world scenarios
1.3	Exploration, identification and collection of various classical and modified pharmaceutical dosage forms	Doing 3 · Mechanism	Each student will be informed to visit a pharmacy or hospital dispensary. The department obtains prior permission from the pharmacy or head of the institution as applicable. In the case of a hospital dispensary Students are made in batches depending on the size of the dispensary and post. Students are sent one by one and asked to collect samples from specific dosage forms. Search and collect samples of different dosage forms, such as tablets, capsules, solutions, suspensions, ointments, creams, injections, and infusions. Samples manufactured by different pharmacies may be collected and compared. Record the name, category, and characteristics of each dosage form. Return to the classroom and create a display or presentation showcasing the collected dosage forms
1.4	Selection of an Appropriate Excipient According to Solid and Liquid Dosage Form	Doing 3 · Mechanism	Students will perform an experiential exercise to select and justify appropriate excipients for solid and liquid dosage forms. They will choose an active pharmaceutical ingredient (API) and a specific dosage form, and then determine the required characteristics of the dosage form. Students will then select an excipient to help achieve the desired characteristics and justify their selection. This exercise will help students understand the role of excipients in pharmaceutical formulation and apply theoretical knowledge to practical problems. By completing this exercise, students will develop their critical thinking and problem-solving skills in pharmaceutical formulation.
1.5	Designing and development of Pharmaceutical Packaging systems and labelling	Doing 3 · Mechanism	The teacher will assign each student to design and develop a pharmaceutical packaging system for a specific product, considering factors such as container closure systems, labelling requirements (based on dosage forms and ingredients), and tamper-evident features. Students will be asked to review the labels of various dosage forms available in the dispensary, evaluate them, and present their observations

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
1.6	Optimization of Pharmaceutical Formulation for Enhanced Bioavailability	Doing 3 · Mechanism	The teacher will allot each student a poorly soluble drug and students will be instructed to design a novel pharmaceutical formulation to enhance its bioavailability. Students will conduct literature research, formulate hypotheses, design experiments, and test the formulation. Evaluate the results using pharmacokinetic parameters and present findings in a written report and oral presentation Modular Assessment Assessment method Hour IInstructions—Conduct a structured Modular assessment. The assessment will be for 50 marks for this module. Keep a structured marking pattern. Use different assessment methods in each module for the semester. Keep a record of the structured pattern used for assessment. Calculate the Modular grade point as per Table 6C. Select any two activities for 25 marks each (2x25=50 Marks) 1. Create a database of bioequivalence and bioavailability studies for different drug products. 2. Select a pharmaceutical product and research its stability and storage requirements. 3. Investigate the role of excipients in the bioavailability of a pharmaceutical product. 4. Develop a manufacturing process for the dosage form, including selecting equipment, developing a batch record, and evaluating process validation. 5. Provide each student with a sample pharmaceutical product that is missing some information. Discuss the importance of each missing label and how it could impact patient safety or compliance. OR Any practical in converted form can be taken for assessment.(25 Marks) and Any experiential learning method in converted form can be taken for assessment (25 marks) 4
MODULE 2 — DATABASES AND INFORMATION SYSTEMS FOR METAL, MINERAL, AND PLANT RESEARCH			
2.1	Design and Implement a Simple Database for a Plant Nursery	Doing 3 · Mechanism	Students will be assigned to design a simple database for a plant nursery. Create three tables: Plant/metal/mineral, Customer, and Order. Define the relationships between the tables. Use a DBMS to create the database and insert sample data. Write SQL queries to retrieve and manipulate data. Perform basic operations like adding a new plant, updating customer addresses, and deleting an order.
2.2	Analyse Web mineral Database and Metal Database	Thinking 4 · Analysis	Each student will be assigned a different metal/mineral (from the groups mentioned in Rasashastra like, Maharasa, Uparasa, Sadharana Rasa, Sudhavarga, Loha etc)and asked to conduct an in-depth analysis of the Webmineral, Mineral Database, and research metal databases to identify trends, patterns, and correlations. Present their findings and insights.
2.3	Exploration and analysis of plant database	Thinking 4 · Analysis	Each student will be asked to carry out an independent research project using The Plant List (TPL) and the International Plant Names Index (IPNI) to investigate a specific plant-related question or hypothesis. Collect and analyze data, and present findings in a research paper or presentation.
2.4	Apply and justify case studies of Database Applications in Metal, Mineral, or Plant Research	Doing 3 · Mechanism	Each student will analyze different case studies of database applications in selected metal, mineral, or plant research. Students will evaluate the effectiveness of the database applications, identify best practices, present the same, and get feedback from peers.
2.5	Perform a role play related to the challenges and limitations of Participate in a team rol	Affective · Responding	—
MODULE 3 — ESSENTIAL CHEMISTRY :- UNLOCKING THE RASASHASTRA TECHNIQUES			
3.1	Exploration of the Periodic Table: Chemical Properties of Main Group and Transition Metals	Doing 3 · Mechanism	—
3.2	Exploration and identification of different functional groups and discuss their influence	Thinking 4 · Analysis	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
3.3	Extraction Techniques in Herbal Formulations: A Hands-on Approach	Doing 3 · Mechanism	Each student will be provided with different samples of herbal and polyherbal formulations. They will then extract the samples using various extraction techniques and solvents. The yield will be recorded, and the physical status of the extracts will be noted. To ensure a diverse learning experience, different herbs and formulations will be selected for each session.
3.4	Identification of phytoconstituents in Herbs and Herbal Formulations: A Practical Approach	Doing 3 · Mechanism	The teacher will assign a specific herb and formulation to each student. Students will then extract the allotted herb using designated solvents. The resulting extracts will be tested for phytoconstituents using qualitative test tube methods. To ensure a diverse learning experience, different herbs and formulations will be selected for each session.
3.5	Test different types of metals for their properties. Each student will be provided with va	Doing 5 · Origination	—
3.6	Hands-on experience with basic metal	Doing 5 · Origination	Each student will be provided with various metal samples and will perform hands-on experiments to identify the metals using basic techniques. They will observe and record the physical properties and perform simple chemical tests to identify the metals. The identification techniques observations and results will be noted Modular Assessment Assessment method Hour Instructions: Conduct a structured Modular assessment. Assessment will be for 50 marks for this module. Keep a structured marking pattern. Use different assessment methods in each module for the semester. Keep a record of the structured pattern used for assessment. Calculate the Modular grade point as per Table 6C. Select any two activities for 25 marks each (2x25=50 Marks) 1. Investigate chemical bonding in Ayurvedic herb-drug interactions, Study isomerism in Ayurvedic compounds, and develop a chemical bonding-based model for Ayurvedic formulation design. 2. Ask the students to design and develop Ayurvedic formulations using the principles of atomic structure, periodic table, chemical bonding, and isomerism. 3. Conduct a cost-benefit analysis of the process of Satvapatana of iron ore, including the costs of extraction, purification, and conversion. 4. Research and present on the bioinorganic chemistry of metal ions used in Ayurvedic formulations, including their coordination chemistry and biological activity. OR Any practical in converted form can be taken for assessment.(25 marks) AND Any experiential learning method in converted form can be taken for assessment (25 marks) 4
MODULE 4 — CONCEPT, UNDERSTANDING, APPLICATION, SOPS AND CHEMICAL CHANGES OF SHODHANA AND MARANA PROCEDURES			
4.1	Performing Shodhana of Rasadravyas with various methods	Doing 4 · Adaptation	Each Student will be allotted different drugs for each procedure to perform and experience the process. Students will present their observations, followed by critical appraisal/summarization by the teacher for comprehensive understanding. Each student may be allotted a set of processes, such as: 1. Swedana, Dhalana, and Nirvapa 2. Bhavana 3. Bharjana and Patana 4. Nimajjana, Pachana, and Nirjaleekarana 5. Bhavana, Swedana, and Nirvapa 6. Dhalana, Bharjana, and Nirjaleekarana 7. Nimajjana, Pachana, and Patana
4.2	Performing Marana procedure of Rasadravyas with different methods	Doing 3 · Mechanism	Each student will be allotted different drugs and asked to carry out Marana as per the SOP laid down in classical texts/published literature and experience the procedure. Each student will present their observations. This will be followed by critical appraisal/summarisation by the teacher for a comprehensive understanding

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
4.3	Analysis of Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing through case studies	Thinking 5 · Synthesis	Each student will be allotted different articles published in peer-reviewed journals on Shodhana and Marana procedures and asked to analyze the contents and present their observations. Presentation of observation by each student will be followed by critical appraisal/ summarisation by the teacher for a comprehensive understanding Modular Assessment Assessment method Hour Instructions: Conduct a structured Modular assessment. Assessment will be for 50 marks for this module. Keep a structured marking pattern. Use different assessment methods in each module for the semester. Keep a record of the structured pattern used for assessment. Calculate the Modular grade point as per Table 6C. Select any two activities for 25 marks each (2x25=50 Marks) 1. Develop a detailed chart comparing the effectiveness of various media used in the Shodhana process of Rasa Dravyas, including their chemical composition, pharmacological properties, and therapeutic application. 2. Make a study on the effect of temperature and pressure on Shodhana and Marana, including the rate of reaction and the yield of product. 3. Conduct a comparative analysis of Shodhana and Marana, highlighting their similarities and differences at the physical and chemical levels. OR Any practical in converted form can be taken for assessment(25 Marks) AND 4 Any experiential learning method in converted form can be taken for assessment (25 marks)
MODULE 5 — PRINCIPLES, SPECIFICATION, SOPS OF HANDLING OF CLASSICAL AND ADVANCED INSTRUMENTS			
5.1	Developing Standard Operative Procedures (SOPs)for given Puta	Thinking 5 · Synthesis	The teacher will allot any one Puta and one type of fuel to each student to develop the Standard Operative Procedure of the given Puta and the student will gain experiential learning by carrying out this practical and documenting the temperature required, and time frame. Different Puta/fuel will be allotted for each student (repetition of the Puta/fuel should be avoided in the next session)
5.2	Developing Standard Operative Procedure for handling Musha	Doing 3 · Mechanism	The teacher will allot a different drug to each student to determine its ash value using a Musha/ following a Standard Operative Procedure. The student will gain experiential learning by carrying out this practical and documenting the temperature required, time frame, and confirmatory test. Different drugs will be allotted for each student (repetition of the drug should be avoided in the next session)
5.3	Performance of chromatography to separate, and identify components and interpretation of r	Doing 3 · Mechanism	—
5.4	Analysis of published articles on spectroscopic instruments	Thinking 4 · Analysis	Each student will be instructed to search and collect articles related to specific spectroscopic instruments used for the characterization of Ayurvedic formulations. Articles published in peer-reviewed journals will be selected. Findings in the articles will be presented and discussed (Collected articles should be unique and different for each student and the same will not be accepted in the next sessions)
5.5	Hands-on practice on the trinocular microscope	Doing 3 · Mechanism	The teacher will allot separate samples to each student and the student will examine the given sample using the trinocular microscope and experience the microscopic characteristics of the given sample like morphology and particle size with the help of projection and discuss the findings.
5.6	Analysis of articles related to challenges in the calibration of equipment	Thinking 4 · Analysis	Each student will be instructed to search and collect articles related to the challenges in the calibration of equipment. Findings in the articles will be presented and discussed (Collected articles should be unique and different for each student and the same will not be accepted in the next sessions). The teacher will encourage the students to develop a Standard Operative Procedure for the calibration of classical instruments

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
5.7	Analysis of published articles on advanced instruments used for characterization	Thinking 4 · Analysis	Each student will be instructed to search and collect articles related to specific instruments used for the characterization of Ayurvedic formulations like, PSA, XRD, XRF, SEM-EDAX, TEM, TGA, DTA, and VSM. Articles published in peer-reviewed journals will be selected. Findings in the articles will be presented and discussed (Collected articles should be unique and different for each student and the same will not be accepted in the next sessions) Modular Assessment Assessment method Hour Instructions: Conduct a structured Modular assessment. Assessment will be for 75 marks. Keep a structured marking pattern. Use different assessment methods in each module for the semester. Keep a record of the structured pattern used for assessment. Calculate the Modular grade point as per Table 6C. 1. Design a quality control protocol for Ayurvedic formulation using chromatography. 20 Marks AND 6 Make a chart on the dimensions of putas and temperature given for Rasa dravyas. - 20 Marks OR Approach to conduct a comparative study of different analytical methods for standardization of Ayurvedic medicine.- 20 Marks AND Investigate the application of calibration techniques in emerging technologies, such as nanotechnology. - 20 marks 2. Create a presentation that explores advanced microscopy techniques, such as super-resolution microscopy and single-molecule microscopy. 35 Marks Or 1. Any practical in converted form can be taken for assessment.(40 Marks) AND 2. Any experiential learning method in converted form can be taken for assessment (35 marks)
MODULE 6 — ANIMAL HOUSE AND ANIMAL HANDLING TECHNIQUE			
6.1	Visit to an Animal House	Doing 3 · Mechanism	Students will visit a local animal house, with prior permission from the concerned authority, to critically observe and thoughtfully learn about animal house management practices. Beforehand, they will research and review relevant principles, prepare insightful questions, and create a comprehensive observation checklist. During the visit, students will carefully observe animal enclosures, feeding, and care practices. They will interact collaboratively with staff, take detailed notes and photographs, and reflect deeply on their observations and learning. After the visit, students will discuss their experiences empathetically with peers and the facilitator. They will then develop a well-informed action plan for improving animal house management practices.
6.2	Analysis of online resources to develop critical thinking and research skills, and the ana	Doing 3 · Mechanism	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
6.3	Preparation and presentation of a parasite Control in Laboratory Animals	Doing 3 · Mechanism	Students will be instructed to search video programs that provide an overview of parasite control in laboratory animals, covering common parasites, best practices, and preventions. Best practices for parasite control are then presented, including sanitation and hygiene protocols, fecal examination and testing, and treatment and prevention strategies. Modular Assessment Assessment method Hour Instructions: Conduct a structured Modular assessment. Assessment will be for 25 marks. Keep a structured marking pattern. Use different assessment methods in each module for the semester. Keep a record of the structured pattern used for assessment. Calculate the Modular grade point as per Table 6C. 25 marks may be allotted to any one of the following assessment methods: 1. Students are advised to select different animal species and conduct a study on the behaviour of a specific animal species, including their habitat, diet, and social structure. 2. Develop an SOP for laboratory animal care and management, including information on animal handling and restraint, housing and enrichment, and health monitoring. 3. Investigate a case of animal cruelty and analyze the factors that contributed to cruelty, make a critical evaluation on existing animal welfare policies, and recommend improvements. OR Any practical in converted form can be taken for assessment.(25 Marks) OR Any experiential learning method in converted form can be taken for assessment (25 marks) 2
MODULE 7 — CLASSICAL AND ADVANCED QUALITY CONTROL TECHNIQUES FOR AYURVEDA FORMULATIONS			
7.1	Practice of documentation and record-keeping processes in the quality control section	Doing 3 · Mechanism	Each student will be allotted a different dosage form and tasked with developing documentation and record-keeping processes according to an established proforma. Students will then present their work and engage in peer discussion.
7.2	Assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma)	Doing 3 · Mechanism	The teacher will allot different samples of Bhasma from the categories of ore/mineral, metal, and Sudha Varga to each student. Students will then conduct Bhasma Pareeksha according to classical guidelines. To facilitate a comprehensive understanding, the teacher may provide both positive and negative samples for testing.
7.3	Performance of Quality control procedures for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva	Doing 3 · Mechanism	The teacher will allot different powder dosage form samples to each student. Students will then conduct quality testing on their assigned samples according to standard pharmacopoeial protocols. The results obtained will be compared to pharmacopoeial standards and discussed.
7.4	Performance of quality control procedures for solid dosage forms	Doing 3 · Mechanism	The teacher will allot different solid dosage form samples like tablets, pills, and Varti to each student. Students will then conduct quality testing on their assigned samples following standard pharmacopoeial protocols. The results obtained will be compared to pharmacopoeial standards and discussed.
7.5	Performance of quality control procedures for Rasoushadhi	Doing 3 · Mechanism	The teacher will allot different samples of Rasoushadhi (representing Chaturvidha Rasayana, Loha, and Mandura categories) to each student. Students will then conduct quality testing on their assigned samples following standard pharmacopoeial protocols. The results obtained will be compared to pharmacopoeial standards and discussed.
7.6	Performance of quality control tests for semisolid dosage forms	Doing 3 · Mechanism	The teacher will select at least one formulation from each category of semisolid dosage form(like Rasakriya/Avaleha/Paka/Anjana) and will demonstrate the quality testing methods as per the parameters prescribed in the Pharmacopoeia. The students will then conduct quality testing on their assigned samples as per the standard pharmacopoeial protocol. The results obtained will be compared to pharmacopoeial standards and discussed.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
7.7	Performance of quality control tests for liquid dosage forms	Doing 4 · Adaptation	The teacher will allot different liquid dosage form samples like Kashaya, Arka, Sharakara, Asava-arishta to each student. Students will then conduct quality testing on their assigned samples following standard pharmacopoeial protocols. The results obtained will be compared to pharmacopoeial standards and discussed.
7.8	Performance of quality control tests for Netra Bindu/Karna Bindu and aerosols/sprays. The	Doing 4 · Adaptation	—
MODULE 8 — NANO-SCIENCE, NANO-TECHNOLOGY AND ITS APPLICATION IN AYURVEDA			
8.1	Measurement of particle	Doing 3 · Mechanism	The teacher will allot different Bhasma samples prepared with different numbers of Puta and instruct students to measure the size size of Bhasma prepared with different numbers of Puta of particles measurable in the samples using the relevant microscope.
8.2	Measurement of particle size of various Kupipakwa Rasayana samples	Doing 3 · Mechanism	The teacher will allot different samples of Kupipakwa Rasayana to students and instruct them to measure the size of particles measurable in the samples using the relevant microscope
8.3	Review of articles regarding the green synthesis of metallic nanoparticles	Doing 3 · Mechanism	Students will be instructed to explore relevant articles related to the green synthesis of metallic nanoparticles, analyze the contents, and discuss with peer
8.4	Experiencing the impact of Bhavana	Doing 3 · Mechanism	The teacher will instruct students to collect samples prepared at different numbers/duration of Bhavana and they will be instructed to analyze the particle size of the samples.
8.5	Market Survey of Nano-Based Ayurvedic Formulations	Doing 3 · Mechanism	The teacher will instruct students to conduct a market survey of nano-based Ayurvedic formulations and present the summary of their observations in the departmental seminar
8.6	Integration of nanomedicine with Ayurvedic formulations and its prospects. The department	Doing 3 · Mechanism	—

44 activities. The level shown is that of the objective in Table 3 which cites the activity, so it tells you what the activity is being asked to certify.

PART C · APPENDIX C8

Where the hours go, module by module

The lecture, practical and experiential split for each module, from Table 2. Read alongside Chapter 6.

M	MODULE	CR	LECTURE	PRACTICAL	EXPERIENTIAL	TOTAL	MARKS
1	Pharmaceutics and Pharmaceutical Technology	2	10 h	20 h	30 h	60 h	50
2	Databases and Information Systems for Metal, Mineral, and Plant Research	2	10 h	20 h	30 h	60 h	50
3	Essential Chemistry :- Unlocking the Rasashastra Techniques	2	10 h	20 h	30 h	60 h	50
4	Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures	2	10 h	20 h	30 h	60 h	50
5	Principles, specification, SOPs of handling of classical and advanced instruments	3	15 h	30 h	45 h	90 h	75
6	Animal house and Animal handling technique	1	5 h	10 h	15 h	30 h	25
7	Classical and Advanced quality control techniques for Ayurveda Formulations	2	10 h	20 h	30 h	60 h	50
8	Nano-science, nano-technology and its application in Ayurveda	2	10 h	20 h	30 h	60 h	50
Semester		**16**	**80 h**	**160 h**	**240 h**	**480 h**	**400**

The split is fixed by the credit framework at one part lecture to two parts practical to three parts experiential. Experiential hours include the modular assessment, at two hours per credit. Two thirds of this semester happens with the teacher not talking.

PART C · APPENDIX C9

Sources and further reading

The primary sources for the taxonomies, the curriculum this book is built on, and a note on what was and was not altered.

THE TAXONOMIES

- Bloom, B. S. (ed.), *Taxonomy of Educational Objectives. Handbook I: Cognitive Domain.* New York: Longmans, Green, 1956. The original, and the scheme Column 3E uses.
- Anderson, L. W. and Krathwohl, D. R. (eds.), *A Taxonomy for Learning, Teaching and Assessing.* New York: Longman, 2001. The revision: verb names, the swapped top two levels, and the knowledge axis.
- Krathwohl, D. R., Bloom, B. S. and Masia, B. B., *Taxonomy of Educational Objectives, Handbook II: Affective Domain.* New York: David McKay, 1964.
- Simpson, E. J., *The Classification of Educational Objectives in the Psychomotor Domain.* Washington: Gryphon House, 1972; and Dave, R. H., in R. J. Armstrong (ed.), *Developing and Writing Behavioral Objectives*, 1970.
- Krathwohl, D. R., 'A Revision of Bloom's Taxonomy: An Overview', *Theory into Practice*, 41(4), 2002.

THE CURRICULUM

- National Commission for Indian System of Medicine, Board of Ayurveda. *Curriculum for MD/MS Ayurveda, Semester II — Applied Basics of Rasashastra and Bhaishajya Kalpana (AYPG-AB-RB).* New Delhi: NCISM. Applicable from the 2024–25 batch. Tables 2, 3, 4 and 5 are the source of every objective, code, hour, mode and activity in this book.

A NOTE ON HOW THIS BOOK WAS MADE

Every objective in Part B is reproduced in the curriculum's own words. Plain-English spelling errors have been corrected and sentence punctuation lost to the source document's line wrapping has been restored. No Sanskrit or technical term has been altered, and no subject content has been added, removed or reworded.

Extracting this curriculum needed several repairs peculiar to its layout: domain codes split across line breaks and interleaved with the level beside them, hours joined to the end of an objective's own full stop, activity references torn apart mid-word, and the activity description tables which sit inside Table 3 and had to be separated from the objectives they follow. All 171 objectives, 91 activities and 480 notional hours reconcile against the curriculum's own credit framework.

The counts and tables in Part C are generated from the same parsed data as Part B, so the two cannot disagree. Where a judgement had to be made — chiefly in reading the kind of knowledge from an objective's wording — it is made from the verb and its grammatical object, and marked with a small circle (°) where the wording supports a reading without settling it.

A COMPETENCY-ASSESSMENT HANDBOOK

Miller's Pyramid for Rasashastra & Bhaishajya Kalpana

Module-by-module competency mapping, assessment design and usage guide for AYPG-AB-RB — Applied Basics of Rasashastra and Bhaishajya Kalpana
MD/MS Ayurveda, Semester II

Curriculum	NCISM, Board of Ayurveda — applicable from the 2024–25 batch
Coverage	8 modules · 42 units · 171 session learning objectives · 47 practical training and 44 experiential learning activities
Framework	Miller (1990) four tiers, with the Cruess *Is* tier (2016), mapped onto NCISM Table 3, Column 3F

16	480	400	300
CREDITS	NOTIONAL HOURS	FORMATIVE MARKS	SUMMATIVE MARKS

Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026 · For departmental use in curriculum delivery, assessment design and paper setting.

⌂ HOME

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PART B — THE MODULES

Module 1 Pharmaceutics and Pharmaceutical Technology 24 obj · Does 11

Unit 1 Introduction to Pharmaceutics 6 obj 12

Unit 2 Pharmaceutical Dosage Forms 4 obj 12

Unit 3 Pharmaceutical Formulation 5 obj 12

Unit 4 Pharmaceutical Manufacturing and Quality Control 5 obj 13

Unit 5 Biopharmaceutics 4 obj 13

Module 2 Databases and Information Systems for Metal, Mineral, and Plant Research 22 obj · Does 14

Unit 1 Introduction to Databases and Information Systems 5 obj 14

Unit 2 Metal and Mineral Databases 6 obj 15

Unit 3 Plant Databases 4 obj 15

Unit 4 Database application 4 obj 15

Unit 5 Challenges and limitation of Databases 3 obj 16

Module 3 Essential Chemistry :- Unlocking the Rasashastra Techniques 21 obj · Does 17

Unit 1 Atomic, Molecular Structure, and Periodic Table Elements 3 obj 17

Unit 2 Specialized Chemistry Fields-An Outline 2 obj 18

Unit 3 Organic Chemistry 3 obj 18

Unit 4 Phytochemistry 7 obj 19

Unit 5 Metallurgy 6 obj 19

Module 4 Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures 15 obj · Does 21

Unit 1 Shodhana and Marana Procedures 3 obj 22

Unit 2 SOP of Shodhana Processes 4 obj 22

Unit 3 SOP of Marana process 4 obj 22

Unit 4 Chemical Changes involved in Shodhana and Marana processes 1 obj 22

Unit 5 Shodhana and Marana- Case Studies, troubleshooting and emerging trends 3 obj 23

Module 5 Principles, specification, SOPs of handling of classical and advanced instruments 29 obj · Does 24

Unit 1 Puta and its modern aspects 4 obj 25

Unit 2 Musha and its modern aspects 4 obj 25

Unit 3 Chromatography 5 obj 25

Unit 4 Spectroscopy 4 obj 26

Unit 5 Microscopy 4 obj 26

Unit 6 Calibration techniques 3 obj 26

Unit 7 Overview of characterization instruments 5 obj 27

Module 6 Animal house and Animal handling technique 13 obj · Does 28

Unit 1 Introduction to animal house management 3 obj 29
Unit 2 Study of animal behaviour 3 obj 29
Unit 3 Species-specific handling techniques 2 obj 29
Unit 4 Animal health and disease prevention 3 obj 29
Unit 5 Animal welfare and ethics 2 obj 30
Module 7 Classical and Advanced quality control techniques for Ayurveda Formulations 26 obj · Does 31
Unit 1 Quality Control and Quality Assurance 4 obj 32
Unit 2 Classical aspects of the quality control parameter for Ayurvedic formulations 4 obj 32
Unit 3 Quality Control parameter for solid dosage forms 9 obj 32
Unit 4 Quality Control parameter for semi-solid dosage forms 3 obj 33
Unit 5 Quality Control parameter for liquid and gaseous dosage forms 6 obj 33
Module 8 Nano-science, nano-technology and its application in Ayurveda 21 obj · Does 35
Unit 1 Introduction to Nanotechnology 3 obj 36
Unit 2 Ayurvedic Formulations and Nanotechnology 10 obj 36
Unit 3 Nanocarriers for Ayurvedic Formulations 3 obj 37
Unit 4 Regulatory Aspects and Safety Concerns 2 obj 37
Unit 5 Recent Advances and Future Directions 3 obj 37

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PART A

Foundations

Five short chapters. The first explains what Miller's pyramid is and, just as importantly, what it is not. The second shows that it is already inside your curriculum, in a column most teachers have never had pointed out. The third gives the three rules of constructive alignment, the fourth sets out which instrument can certify which tier, and the fifth explains how to read a unit in Part B. Read straight through once; after that, use the modules.

1. **Miller's pyramid** — what it is, and what it is not
2. **The pyramid is already inside your curriculum** — reading Column 3F
3. **Constructive alignment** — the three rules
4. **The assessment toolkit**, tier by tier
5. **How to use a module in Part B**

↑ Contents

PART A · CHAPTER 1

Miller's pyramid — what it is, and what it is not

In 1990 George Miller, a medical educator, published a single diagram that has shaped assessment ever since. He was answering a question every examiner faces: *what does this test actually prove?*

His answer was a pyramid of four tiers, each demanding evidence more authentic than the one below.

- **Knows** — the student can recall it.
- **Knows how** — the student can reason with it, apply it to a case on paper.
- **Shows how** — the student can do it when someone is watching, in a set-up arranged for the purpose.
- **Does** — the student does it in real practice, unprompted, when nobody is checking.

The tiers are cumulative but not interchangeable. A student who reaches *Does* necessarily passed through the others; a student who has only been tested at *Knows* has been told nothing about the tiers above. That asymmetry is the whole point. **Evidence from a lower tier never certifies a higher one.**

WHAT THE PYRAMID IS NOT

It is not a taxonomy of thinking. It says nothing about whether a task is easy or hard, or whether it calls for recall or judgement. A *Knows* objective can be intellectually demanding and a *Does* objective can be routine. Bloom's taxonomy answers "what kind of thinking is this?"; Miller answers "how authentic must the evidence be?" The two are independent axes, and this curriculum carries both — Bloom in one column, Miller in another. Each has its own companion volume.

It is also not a ranking of worth. A curriculum that stops at *Shows how* for a dangerous procedure is not being lazy; it is making a judgement about supervision and safety. Reading the ceilings as a scoreboard is the commonest misuse of the pyramid.

> **The Cruess addition.** In 2016 Richard and Sylvia Cruess proposed a fifth tier above *Does*: **Is** — professional identity. Not what the student does but who they have become. It cannot be examined in a hall, and it is not coded anywhere in your curriculum. It is worth naming because some of what you certify at *Does* — a safety briefing recorded honestly, a result reported when it was inconvenient — is really evidence of *Is*. Where a unit in Part B says so, that is what it means.

WHY IT MATTERS HERE

This semester sits high on the pyramid: 78 of its 171 objectives are at *Shows how* or *Does* — 46%. Nearly half of what this course certifies cannot be certified on paper at all, and the credit framework says so openly by spending two thirds of every credit outside the lecture room.

PART A · CHAPTER 2

The pyramid is already inside your curriculum

Miller is not something this book adds to your curriculum. It has been there since publication, in a column most teachers have never had pointed out.

COLUMN 3F

Open Table 3. The column headed **Level** carries one of four values against every objective: *Know*, *Knows how*, *Shows how* or *Does*. Those are Miller's four tiers, and they are the curriculum's statement of how authentic the evidence must be before that objective can be called met.

Across the 171 objectives of this semester the distribution is:

TIER	OBJECTIVES	SHARE
Knows	14	8.2%
Knows how	79	46.2%
Shows how	35	20.5%
Does	43	25.1%

Total 171 objectives.

Read that against the hours. Every credit is 5 hours of lecture, 10 of practical training and 13 of experiential learning. The upper two tiers and the lower two are almost evenly split, and so are the hours meant to build them.

> **A word about the other column headed "Domain".** Beside the Level column sits another carrying codes like CK, CC, CAP, PSY-MEC and AFT-VAL. Those are Bloom's taxonomy and the psychomotor and affective domains — a different question entirely, and the subject of the companion volume. In *this* book, "level" and "tier" always mean Miller.

THE CEILING

Every module in this curriculum has a **ceiling**: the highest tier any of its objectives reaches. That single fact governs assessment more than anything else on the page. It tells you what evidence you must gather, and it tells you what you may not ask for. Part B prints it for every module.

PART A · CHAPTER 3

Constructive alignment — the three rules

Constructive alignment is the principle that what you teach, what you ask students to do, and what you accept as proof must all sit at the same tier. Stated as rules, it comes to three.

RULE 1 — THE TIER DECIDES THE INSTRUMENT

Not the topic, not the hours, not the marks. Once you know an objective sits at *Shows how*, the set of instruments that can certify it is fixed, and a written paper is not among them. Chapter 4 gives the toolkit tier by tier.

RULE 2 — NEVER ASSESS ABOVE THE CEILING

If the curriculum certifies a module to *Knows how*, do not build an OSPE station for it. You would be examining something the students were never taught to that standard, and the marks would measure prior exposure rather than your teaching. This rule protects students.

RULE 3 — NEVER CERTIFY A HIGHER TIER WITH LOWER-TIER EVIDENCE

This is the rule that gets broken, and it is broken in one direction. A *Does* objective assessed by viva has not been assessed. A *Shows how* objective assessed by a long answer has not been assessed. The student may pass, the record may look complete, and nothing has been established.

> **The asymmetry is the thing to remember.** You may always gather evidence from a tier *above* the one required — a student observed doing the thing has plainly also shown how. You may never accept evidence from a tier *below*. Every alignment error worth catching is a violation of that one sentence.

WORKING THE THREE RULES BACKWARDS

In practice the fastest way to plan a session is to start at the end. Ask what evidence you will hold at the finish — a script, a checklist, a log entry. That names the tier. The tier names the instrument, and the instrument names the teaching that must precede it. A session planned forwards from the syllabus tends to arrive at the assessment it can most easily set, which is almost always a written one.

PART A · CHAPTER 4

The assessment toolkit, tier by tier

Each tier admits a particular set of instruments. Using one from the wrong tier is the single commonest fault in assessment design, and it is invisible unless you look for it, because the paperwork is complete either way.

TIER	THE QUESTION IT ANSWERS	TEACHING THAT BUILDS IT	INSTRUMENTS THAT CERTIFY IT	EVIDENCE YOU KEEP
Knows	**Can the student recall it?**	Lecture, reading, recitation, retrieval practice.	MCQ, short answer, oral quiz, spotting.	Written script, quiz sheet.
Knows how	**Can the student reason with it?**	Worked cases, discussion, problem-based work; the student must reason aloud and be corrected.	Structured long answer, case-based questions, structured viva.	Reasoned written answer, viva score sheet.
Shows how	**Can the student do it when watched?**	Demonstration, then supervised practice with your help withdrawn a little at a time.	OSPE or OSCE station, observed performance against a checklist.	Checklist score, annotated artefact, station record.
Does	**Does the student do it unprompted, in real practice?**	Real work with real consequences, supervised at a distance rather than at the elbow.	Log book, record book, portfolio, observation over time, presentation with peer feedback.	Dated log entry, countersigned record, portfolio piece.

THE TRAP AT EACH TIER

- **Knows** — Spending long-answer marks here. A long answer cannot separate a student who knows the list from one who knows it better.
- **Knows how** — Setting a question answerable from memory. If the stem can be met by recall, the objective has silently dropped to Knows.
- **Shows how** — Certifying by viva. A fluent student who has never handled the material will pass a viva comfortably.
- **Does** — Certifying with a written test. Nothing written can show that a student does something unprompted when nobody is watching.

> **The instruments this curriculum names.** This curriculum sets a modular assessment in every module and requires a different method each time, with a record of the pattern used. It also runs a university examination at the end of the semester: one theory paper of 100 marks and a practical examination of 200.

ON THE LOG BOOK AND RECORD BOOK

These are the only instruments in routine use that can carry *Does*-tier evidence, and they are usually treated as paperwork. A log filled in at the end of term is a piece of writing. The same log filled in as the work happens, dated and countersigned, is the only proof you will have that a student did something unprompted. Decide at the start of each module what artefact each *Does* objective will leave behind, and the record becomes evidence rather than an administrative task.

PART A · CHAPTER 5

How to use a module in Part B

Part B has eight modules, each carrying its units inside it. Each begins on its own page and is laid out the same way, so once you have read one you can use any of them.

WHAT IS ON THE PAGE

1. **The heading and counters** — hours, objectives, marks, and the **ceiling**: the highest tier this module reaches. Read the ceiling first; it governs everything else.
2. **The session map** — the curriculum's own objectives with the tier each is rated to, so you can see at a glance where the weight sits.
3. **The pyramid for this module** — four rows, one per tier. Each names what a student must be able to do at that tier here, then how to build it, how to assess it, and what evidence you keep. Tiers the curriculum does not use are shown greyed, because knowing what is *not* certified matters as much as knowing what is.
4. **The ceiling statement** — a single line naming the highest tier and warning against assessing above it or certifying it with the wrong instrument.
5. **Two panels** — one for the teacher, one to give to students.

THREE WAYS TO USE IT

Planning a session. Start at the ceiling row of the pyramid and work down: that row tells you what the students must end up able to do, and the rows beneath tell you what has to be in place first.

Setting a paper. A written paper can honestly reach *Knows* and *Knows how* and no further. Use the session map to see which objectives those are; anything above must be certified elsewhere, and the module says where.

Auditing what you already do. Take your existing assessment for a module and ask which tier it can certify. If it is a written test and the ceiling is *Shows how* or *Does*, the gap is your finding.

> **What this book does not do.** It does not rewrite the curriculum and does not tell you what to teach — every objective in Part B appears in the curriculum's own words. What it adds is the Miller layer: what standard of evidence each objective demands, and which instruments can supply it.

PART B

The modules

Eight modules, one per module, each carrying its five units inside it. Every objective appears in the curriculum's own words, gathered under the tier its Level column gives it, with the kind of hour beside it. The pyramid block on each module names what every tier asks for in that module, how to build it, how to assess it and what evidence to keep. Chapter 5 explains the layout.

1. **Modules** — hours, tier counts, the pyramid, and the ceiling
2. **Unit strips** — each unit's objectives, grouped by tier

↑ Contents

MODULE 1

Pharmaceutics and Pharmaceutical Technology

2 credits | 60 notional hours | 5 units | **24 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	13	5	6
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	8	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	6	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
6 objectives

Identify applications suiting various regulatory frameworks; Explore, identify, and collect various classical and modified pharmaceutical dosage forms; Identify an Appropriate Excipient According to Solid and Liquid Dosage Form; and 3 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
5 objectives

Demonstrate regulatory frameworks (FDA, etc.); Demonstrate pharmaceutical product development process; Formulate Solid Dosage Forms selecting appropriate excipients; and 2 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
13 objectives

Define pharmaceutics and describe its scope and historical development; Discuss Pharmaceutical industry overview, market size (national and international) and regulatory framework; Enlist and evaluate challenges and solutions in the pharmaceutical product development process; and 10 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Introduction to Pharmaceutics

6 obj · 12 h · ceiling Does

Definition and scope of pharmaceutics
Historical development of pharmaceutics
Pharmaceutical industry overview
Regulatory frameworks (FDA, etc.)
Pharmaceutical product development process

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Define pharmaceutics and describe its scope and historical development.	Lecture	1	Knows how
Discuss Pharmaceutical industry overview, market size (national and international) and regulatory framework.	Lecture	1	Knows how
Enlist and evaluate challenges and solutions in the pharmaceutical product development process.	Experiential Learning	3	Knows how
SHOWS HOW — 2			
Demonstrate regulatory frameworks (FDA, etc.).	Practical Training	2	Shows how
Demonstrate pharmaceutical product development process.	Practical Training	2	Shows how
DOES — 1			
Identify applications suiting various regulatory frameworks.	Experiential Learning	3	Does

Unit 2 · Pharmaceutical Dosage Forms

4 obj · 8 h · ceiling Does

Classification of dosage forms
Solid dosage forms (tablets, capsules, etc.)
Liquid dosage forms (solutions, suspensions, etc.)
Semi-solid dosage forms (ointments, creams, etc.)
Parenteral dosage forms (injections, infusions, etc.)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Classify various pharmaceutical dosage forms.	Lecture	1	Knows how
Identify and categorise classical Ayurvedic dosage forms.	Practical Training	2	Knows how
Identify and categorise modified Ayurvedic dosage forms.	Practical Training	2	Knows how
DOES — 1			
Explore, identify, and collect various classical and modified pharmaceutical dosage forms.	Experiential Learning	3	Does

Unit 3 · Pharmaceutical Formulation

5 obj · 16 h · ceiling Does

Principles of pharmaceutical formulation
Excipients and their functions
Solid dosage forms
Liquid and semi-solid dosage forms
Parenteral dosage forms

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the principles of pharmaceutical formulations, excipients, and their functions.	Lecture	1	Knows how
Discuss formulation of parenteral dosage forms.	Lecture	1	Knows how
SHOWS HOW — 2			
Formulate Solid Dosage Forms selecting appropriate excipients.	Practical Training	4	Shows how
Formulate liquid Dosage Forms selecting appropriate excipients.	Practical Training	4	Shows how

DOES — 1			
Identify an Appropriate Excipient According to Solid and Liquid Dosage Form.	Experiential Learning	6	Does

Unit 4 · Pharmaceutical Manufacturing and Quality Control

5 obj · 10 h · ceiling Does

Good Manufacturing Practices (GMP& cGMP)
Quality control and quality assurance
Packaging and labelling
Portability and storage

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss Good Manufacturing Practices (GMP& cGMP).	Lecture	1	Knows how
Discuss pharmaceutical stability and shelf life.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate, design and evaluate pharmaceutical packing and labelling.	Practical Training	2	Shows how
DOES — 2			
Evaluate GMP audit and compliance.	Practical Training	2	Does
Design and develop Pharmaceutical Packaging systems and labelling.	Experiential Learning	4	Does

Unit 5 · Biopharmaceutics

4 obj · 10 h · ceiling Does

Biopharmaceutics and pharmacokinetics
Bioavailability and bioequivalence
Pharmaceutical dissolution and absorption
Pharmaceutical product development and optimization

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe biopharmaceutics and pharmacokinetics.	Lecture	1	Knows how
Describe bioavailability and bioequivalence.	Lecture	1	Knows how
Discuss pharmaceutical product development and optimization.	Lecture	1	Knows how
DOES — 1			
Design and Optimize Pharmaceutical Formulation for Enhanced Bioavailability.	Experiential Learning	7	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **11 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 2

Databases and Information Systems for Metal, Mineral, and Plant Research

2 credits | 60 notional hours | 5 units | **22 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	9	2	10
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	7	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
10 objectives

Demonstrate various searching techniques of databases; Demonstrate Modelling, Querying, and Analytics of Data; Analyze USGS Mineral Database; and 7 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
2 objectives

Design and Implement a Simple Database for a Plant Nursery; Analyse case studies of Database Applications in Metal, Mineral, or Plant Research.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Describe overview of databases and information systems, types of databases (relational object-oriented, etc.), and database management systems (DBMS); Describe data retrieval and querying languages (SQL, etc.) and an overview of metal mineral, and plant databases; Describe overview of metal and mineral databases, USGS (U.S. Geological Survey); and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Identify Database-Related Challenges in Pharmaceutical Development.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Introduction to Databases and Information Systems

5 obj · 12 h · ceiling Does

- Overview of databases and information systems
- Types of databases (relational, object-oriented, etc.)
- Database management systems (DBMS)
- Data retrieval and querying languages (SQL, etc.)

Introduction to metal, mineral, and plant databases

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe overview of databases and information systems, types of databases (relational object-oriented, etc.), and database management systems (DBMS).	Lecture	1	Knows how
Describe data retrieval and querying languages (SQL, etc.) and an overview of metal mineral, and plant databases.	Lecture	1	Knows how
SHOWS HOW — 1			
Design and Implement a Simple Database for a Plant Nursery.	Experiential Learning	6	Shows how
DOES — 2			
Demonstrate various searching techniques of databases.	Practical Training	2	Does
Demonstrate Modelling, Querying, and Analytics of Data.	Practical Training	2	Does

Unit 2 · Metal and Mineral Databases

6 obj · 16 h · ceiling Does

Overview of metal and mineral databases
USGS Mineral Database
Mineral Database
Webmineral Mineral Database
Introduction to Metal Databases

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe overview of metal and mineral databases, USGS (U.S. Geological Survey).	Lecture	1	Knows how
Discuss Mineral and metal Databases, including their applications and significance.	Lecture	1	Knows how
DOES — 4			
Analyze USGS Mineral Database.	Practical Training	2	Does
Analysis and integration of Web mineral Database.	Practical Training	3	Does
Search, explore and analyze Metal Database and Applications.	Practical Training	3	Does
Analyse Web mineral Database and Metal Database.	Experiential Learning	6	Does

Unit 3 · Plant Databases

4 obj · 12 h · ceiling Does

Overview of plant databases
The Plant List (TPL)
International Plant Names Index (IPNI)
Royal Botanic Gardens, Kew (RBG Kew) databases
Introduction to plant genetic databases (e.g., PlantGDB)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Illustrate overview of plant databases, The Plant List (TPL), and International Plant Names Index (IPNI).	Lecture	1	Knows how
Describe Royal Botanic Gardens, Kew (RBG Kew) databases and introduction to plant genetic databases (e.g., PlantGDB).	Lecture	1	Knows how
DOES — 2			
Demonstrate techniques to explore Plant Databases.	Practical Training	4	Does
Explore and analyse plant database.	Experiential Learning	6	Does

Unit 4 · Database application

4 obj · 12 h · ceiling Does

Database applications in research and industry

Using databases for decision-making and policy development

Future directions: emerging trends and technologies in database development

Group project: developing a database application for metal, mineral, or plant research

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss database applications in research and industry, and use databases for decision-making and policy development.	Lecture	1	Knows how
Analyze future directions in database development and applications, including emerging trends and technologies, and group project guidelines.	Lecture	1	Knows how
SHOWS HOW — 1			
Analyse case studies of Database Applications in Metal, Mineral, or Plant Research.	Practical Training	4	Shows how
DOES — 1			
Apply and justify case studies of Database Applications in Metal, Mineral, or Plant Research.	Experiential Learning	6	Does

Unit 5 · Challenges and limitation of Databases

3 obj · 4 h · ceiling Does

Challenges in Pharmaceutical Formulation Development

Limitations of In Vitro and In Vivo Testing

Regulatory Challenges in Pharmaceutical Product Approval

Scalability and Manufacturing Challenges

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Identify Database-Related Challenges in Pharmaceutical Development.	Lecture	1	Knows
KNOWS HOW — 1			
Discuss data Quality, Integration, Security, Scalability, and Regulatory Compliance.	Lecture	1	Knows how
DOES — 1			
Perform a role play related to the challenges and limitations of databases.	Experiential Learning	2	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **12 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 3

Essential Chemistry :- Unlocking the Rasashastra Techniques

2 credits | 60 notional hours | 5 units | **21 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	7	4	5
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	6	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
5 objectives

Apply the knowledge Periodic Table for Chemical Properties of Main Group and Transition Metals; Demonstrate extraction of different herbal formulations using various extraction techniques and solvents; Identify phytoconstituents in Herbs and Herbal Formulations: A; and 2 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
4 objectives

Analyse chemical bonding types and principles and isomerism and their significance in better understanding of different Rasashastra processes such as Shodhana, and Marana; Demonstrate different types of extraction methods including solvent selection; Demonstrate qualitative identification of phytoconstituents in single herbs and herbal formulations; and 1 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
7 objectives

Demonstrate the essential fundamental principles of crystallography and organometallic principles with special reference to Rasashastra procedures; Discuss the interaction of herbs and animal products with metals; Explore and identify different functional groups and discuss their influence on molecular interactions and therapeutic efficacy; and 4 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Describe the role of basic chemistry principles and their application in Rasashastra; Describe the principles of crystallography and an overview of organometallic and bioinorganic chemistry; Describe the structure of organic molecules; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Atomic, Molecular Structure, and Periodic Table Elements

3 obj · 12 h · ceiling Does

Importance of knowledge of chemistry for understanding various processing methods of Rasashastra and Oushadha nirmana.

Atomic structure, Periodic table, Chemical bonding, Isomerism -Application of these principles in Shodhana, Marana, Amruteekarana and Chaturvidha Rasayana

Periodic Table, Main group elements, Transition metals: understanding of chemical properties of non-metals, metals, variable oxidation states, thermal stability,

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the role of basic chemistry principles and their application in Rasashastra.	Lecture	1	Knows
SHOWS HOW — 1			
Analyse chemical bonding types and principles and isomerism and their significance in better understanding of different Rasashastra processes such as Shodhana, and Marana.	Practical Training	5	Shows how
DOES — 1			
Apply the knowledge Periodic Table for Chemical Properties of Main Group and Transition Metals.	Experiential Learning	6	Does

Unit 2 · Specialized Chemistry Fields-An Outline

2 obj · 6 h · ceiling Knows how

Crystallography- detoxification, transformation of compound

Organometallic chemistry- Chaturvidha Rasayana

Bioinorganic chemistry- Amruteekarana (Metal-Carbon bond-stable compound formation)

Application in Shodhana, Marana, Amruteekarana, Chaturvidha Rasayana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the principles of crystallography and an overview of organometallic and bioinorganic chemistry.	Lecture	1	Knows
KNOWS HOW — 1			
Demonstrate the essential fundamental principles of crystallography and organometallic principles with special reference to Rasashastra procedures.	Practical Training	5	Knows how

Unit 3 · Organic Chemistry

3 obj · 8 h · ceiling Knows how

Structure of organic molecules: understanding of Herbal components,

Interaction of herbs and animal products with metals for detoxification, compound formation, and compound stability.

Functional groups: Reactivity and properties of organic molecules influencing compound action

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the structure of organic molecules.	Lecture	1	Knows
KNOWS HOW — 2			
Discuss the interaction of herbs and animal products with metals.	Lecture	1	Knows how
Explore and identify different functional groups and discuss their influence on molecular interactions and therapeutic efficacy.	Experiential Learning	6	Knows how

Unit 4 · Phytochemistry

7 obj · 16 h · ceiling Does

Definition and significance of phytochemistry
Overview of plant-based bioactive compounds
Classes of Phytochemicals
Extraction methods
Phytochemical screening

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe the significance of phytochemistry.	Lecture	1	Knows how
Describe the overview of plant-based bioactive components.	Lecture	1	Knows how
Discuss different types of extraction methods including solvent selection.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate different types of extraction methods including solvent selection.	Practical Training	4	Shows how
Demonstrate qualitative identification of phytoconstituents in single herbs and herbal formulations.	Practical Training	3	Shows how
DOES — 2			
Demonstrate extraction of different herbal formulations using various extraction techniques and solvents.	Experiential Learning	3	Does
Identify phytoconstituents in Herbs and Herbal Formulations: A.	Experiential Learning	3	Does

Unit 5 · Metallurgy

6 obj · 14 h · ceiling Does

Introduction to metallurgy and general physicochemical methods of extraction of metals from ores
Common ores used in Rasashastra and their Satvapatana from metallurgical point of view

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 2			
Define metallurgy and describe its scope.	Lecture	1	Knows
Describe general physicochemical methods of extraction of metals from ores.	Lecture	1	Knows
KNOWS HOW — 1			
Discuss common ores used in Rasashastra and their Satvapatana with metallurgical point of view.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate different types of metals and their properties.	Practical Training	3	Shows how
DOES — 2			
Differentiate various types of metals based on their properties.	Experiential Learning	4	Does
Differentiate metals using basic metal identification techniques.	Experiential Learning	4	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **9 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 4

Concept, understanding, application, SOPs and chemical changes of Shodhana and Marana procedures

2 credits | 60 notional hours | 5 units | **15 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	11	3	1
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	2	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Demonstrate Shodhana procedure of Rasadravyas using various methods.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
3 objectives

Perform Shodhana of Rasadravyas with various methods; Demonstrate Marana procedure of Rasadravyas with different methods; Perform Marana procedure of Rasadravyas with different methods.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
11 objectives

Discuss Shodhana and Marana procedures; Discuss the importance of purification and incineration in Ayurvedic pharmaceutical manufacturing; Analyse principles of Shodhana and Marana procedures with their application in Ayurvedic pharmaceutical manufacturing; and 8 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Shodhana and Marana Procedures

3 obj · 3 h · ceiling Knows how

Overview of Shodhana and Marana procedures

Importance of purification and incineration in Ayurvedic pharmaceutical manufacturing

Principles of Shodhana and Marana procedures

Applications of Shodhan and Maran procedures in Ayurvedic pharmaceutical manufacturing

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss Shodhana and Marana procedures.	Lecture	1	Knows how
Discuss the importance of purification and incineration in Ayurvedic pharmaceutical manufacturing.	Lecture	1	Knows how
Analyse principles of Shodhana and Marana procedures with their application in Ayurvedic pharmaceutical manufacturing.	Lecture	1	Knows how

Unit 2 · SOP of Shodhana Processes

4 obj · 22 h · ceiling Does

Types of Shodhana procedures (physical, chemical, biological)

Methods of Shodhana (filtration, sedimentation, distillation, etc.)

SOPs for Shodhana procedures

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe types of Shodhana procedures (physical, chemical, biological).	Lecture	1	Knows how
Discuss methods of Shodhana (filtration, sedimentation, distillation, etc.) and its SOPs.	Lecture	1	Knows how
SHOWS HOW — 1			
Perform Shodhana of Rasadravyas with various methods.	Experiential Learning	10	Shows how
DOES — 1			
Demonstrate Shodhana procedure of Rasadravyas using various methods.	Practical Training	10	Does

Unit 3 · SOP of Marana process

4 obj · 22 h · ceiling Shows how

Types of Marana procedures (incineration, calcination, etc.)

Methods of Marana (oxidation, thermal decomposition, etc.)

SOPs for Marana procedures

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe types of Marana procedures (incineration, calcination, etc.).	Lecture	1	Knows how
Discuss methods of Marana (oxidation, thermal decomposition, etc.) and its SOPs.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate Marana procedure of Rasadravyas with different methods.	Practical Training	10	Shows how
Perform Marana procedure of Rasadravyas with different methods.	Experiential Learning	10	Shows how

Unit 4 · Chemical Changes involved in Shodhana and Marana processes

1 obj · 1 h · ceiling Knows how

Chemical changes involved in Shodhan procedures

Chemical Changes involved in Marana procedures

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss chemical changes involved in Shodhana and Marana procedures.	Lecture	1	Knows how

Unit 5 · Shodhana and Marana- Case Studies, troubleshooting and emerging trends

3 obj · 8 h · ceiling Knows how

Case studies of Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing

Troubleshooting and optimization of Shodhana and Marana procedures

Future directions and emerging trends in Shodhana and Marana procedures

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe Troubleshooting and optimization of Shodhana and Marana procedures.	Lecture	1	Knows how
Discuss future directions and emerging trends in Shodhana and Marana procedures.	Lecture	1	Knows how
Analyse and write Shodhana and Marana procedures in Ayurvedic pharmaceutical manufacturing through case studies.	Experiential Learning	6	Knows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **4 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 5

Principles, specification, SOPs of handling of classical and advanced instruments

3 credits | 90 notional hours | 7 units | **29 objectives** | 75 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	18	6	4
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	15 h	15	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	30 h	7	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	39 h	7	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 6 hours carry the modular assessment, worth 75 marks.

THE PYRAMID FOR THIS MODULE

Does
4 objectives

Demonstrate selected Puta to give the knowledge of Standard Operative Procedure; Develop Standard Operative Procedures (SOPs)for given Puta; Demonstrate chromatography techniques to separate, and identify components and interpretation of results; and 1 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate the handling of Musha following the Standard Operating Procedure (SOP); Develop Standard Operative Procedure for handling Musha; Perform chromatography to separate and identify components and interpret the results; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
18 objectives

Discuss the principles and specifications of Puta; Analyse modern aspects of Puta; Discuss the principles and specifications of Musha; and 15 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Describe the specifications of spectroscopy instruments.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Puta and its modern aspects

4 obj · 14 h · ceiling Does

Introduction, principles, and specifications of Puta

SOPs for handling Puta

Modern aspects of Puta (automation, computerization, etc.)

Troubleshooting and maintenance of Puta

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the principles and specifications of Puta.	Lecture	1	Knows how
Analyse modern aspects of Puta.	Lecture	1	Knows how
DOES — 2			
Demonstrate selected Puta to give the knowledge of Standard Operative Procedure.	Practical Training	4	Does
Develop Standard Operative Procedures (SOPs)for given Puta.	Experiential Learning	8	Does

Unit 2 · Musha and its modern aspects

4 obj · 14 h · ceiling Shows how

Introduction, principles, and specifications of Puta.

SOPs for handling Musha

Modern aspects of Musha (automation, computerization, etc.)

Troubleshooting and maintenance of Musha

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the principles and specifications of Musha.	Lecture	1	Knows how
Analyse modern aspects of Musha with special reference to crucibles.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the handling of Musha following the Standard Operating Procedure (SOP).	Practical Training	4	Shows how
Develop Standard Operative Procedure for handling Musha.	Experiential Learning	8	Shows how

Unit 3 · Chromatography

5 obj · 17 h · ceiling Does

Scope and applications of chromatography in herbal drug standardization (identification of active compounds, adulterant detection, phytochemical analysis, and quality as

Principle and types of chromatography, including TLC, HPTLC, HPLC, LC-MS, and GC-MS.

Instrumentation of chromatography, including equipment and tools used in different types of chromatography.

Data analysis in chromatography, including interpretation of chromatograms and spectra.

Chromatography's Limitations, challenges, and prospects in pharmaceutical and phytochemical research

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the scope and applications of chromatography in herbal drug standardization.	Lecture	1	Knows how
Describe the instrumentation of chromatography, including equipment and tools used in different types of chromatography.	Lecture	1	Knows how
Discuss chromatography's Limitations, challenges, and prospects in pharmaceutical and phytochemical research.	Lecture	1	Knows how
SHOWS HOW — 1			
Perform chromatography to separate and identify components and interpret the results.	Experiential Learning	8	Shows how

DOES — 1			
Demonstrate chromatography techniques to separate, and identify components and interpretation of results.	Practical Training	6	Does

Unit 4 · Spectroscopy

4 obj · 9 h · ceiling Knows how

Introduction, principles, and types (IR, UV, NMR, AAS, ICP, etc.) of spectroscopy.

Specifications of spectroscopy instruments

SOPs for handling spectroscopy instruments

Troubleshooting and maintenance of spectroscopy instruments

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the specifications of spectroscopy instruments.	Lecture	1	Knows
KNOWS HOW — 3			
Discuss spectroscopy's principles and types (IR, UV, NMR, AAS, ICP, etc.).	Lecture	1	Knows how
Demonstration of spectroscopic instruments through Audiovisual aids.	Practical Training	3	Knows how
Explore and analyze published articles on spectroscopic instruments.	Experiential Learning	4	Knows how

Unit 5 · Microscopy

4 obj · 12 h · ceiling Shows how

Introduction, principles, and types of microscopy (optical, petrological, electron, etc.)

Specifications of microscopy instruments

SOPs for handling microscopy instruments

Use of micrometer in microscopy

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the principles, types, and specifications of microscopy instruments.	Lecture	1	Knows how
Describe the Specifications of microscopy instruments.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate handling of petrological and trinocular microscopes.	Practical Training	6	Shows how
Perform hands-on practice on the trinocular microscope.	Experiential Learning	4	Shows how

Unit 6 · Calibration techniques

3 obj · 7 h · ceiling Does

Introduction, principles, and types of calibration (instrument calibration, method calibration, etc.)

SOPs for calibration

Troubleshooting and maintenance of calibration instruments

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss principles and types of calibration.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate calibration of various instruments present in the department.	Practical Training	3	Shows how
DOES — 1			
Analyse articles related to challenges in the calibration of equipment.	Experiential Learning	3	Does

Unit 7 · Overview of characterization instruments

5 obj · 11 h · ceiling Knows how

Overview of advanced characterization instruments and their significance in the standardization of Ayurvedic medicines.

Principles and applications of working mechanism and Applications of instrumental methods: PSA, XRD, XRF, SEM-E-Dax, TEM, TGA, DTA, VSM analysis

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Discuss overview of Advanced Characterization Instruments and their significance in the standardization of Ayurvedic medicines.	Lecture	1	Knows how
Discuss principles and Applications of Spectroscopic and X-ray techniques (PSA, XRD and XRF techniques).	Lecture	1	Knows how
Describe the Principles and Applications of Microscopic and Thermal Techniques(SEM-EDAX, TEM, TGA, DTA, and VSM).	Lecture	1	Knows how
Demonstrate characterization instruments with Audiovisual aids.	Practical Training	4	Knows how
Analyse published articles on advanced instruments used for characterization.	Experiential Learning	4	Knows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **10 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 75 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 6

Animal house and Animal handling technique

1 credits | 30 notional hours | 5 units | **13 objectives** | 25 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	5	2	5
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	5 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	10 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	13 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 2 hours carry the modular assessment, worth 25 marks.

THE PYRAMID FOR THIS MODULE

Does
5 objectives

Present structure and functioning of an animal house; Develop critical thinking and research skills by analyzing online resources and studies on experimental animal behavior; Demonstrate Species-Specific Handling Techniques for Laboratory Animals; and 2 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
2 objectives

Design and Set Up an Animal House; Observe and. Interpret Animal Body Language through demonstration.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Describe the principles of Animal House Management with its design and layout; Identify behavioral patterns and body language in animals; Describe Common diseases in laboratory animals and their management; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Discuss the handling techniques for various laboratory animal species, ensuring handler safety.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Introduction to animal house management

3 obj · 7 h · ceiling Does

Overview of animal house management
Record-keeping and documentation
Animal house design and layout

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe the principles of Animal House Management with its design and layout.	Lecture	1	Knows how
SHOWS HOW — 1			
Design and Set Up an Animal House.	Practical Training	2	Shows how
DOES — 1			
Present structure and functioning of an animal house.	Experiential Learning	4	Does

Unit 2 · Study of animal behaviour

3 obj · 8 h · ceiling Does

Introduction to Animal Behaviour
Types of Animal behaviour.

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Identify behavioral patterns and body language in animals.	Lecture	1	Knows how
SHOWS HOW — 1			
Observe and. Interpret Animal Body Language through demonstration.	Practical Training	2	Shows how
DOES — 1			
Develop critical thinking and research skills by analyzing online resources and studies on experimental animal behavior.	Experiential Learning	5	Does

Unit 3 · Species-specific handling techniques

2 obj · 3 h · ceiling Does

Non-human primates
Other specialized species (e.g., birds, reptiles)
Aquatic animals

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss the handling techniques for various laboratory animal species, ensuring handler safety.	Lecture	1	Knows
DOES — 1			
Demonstrate Species-Specific Handling Techniques for Laboratory Animals.	Practical Training	2	Does

Unit 4 · Animal health and disease prevention

3 obj · 7 h · ceiling Does

Common diseases in laboratory animals
Parasite control and management
Vaccination and immunization programs

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe Common diseases in laboratory animals and their management.	Lecture	1	Knows how
Prepare and present a parasite Control in Laboratory Animals.	Experiential Learning	4	Knows how

DOES — 1			
Demonstrate microscopic Examination of Faecal Samples for Parasites.	Practical Training	2	Does

Unit 5 · Animal welfare and ethics

2 obj · 3 h · ceiling Does

Animal welfare principles and ethics

Humane endpoints and euthanasia methods

Case studies and group discussions on animal welfare and ethics

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss Animal Welfare Principles, Ethics, Humane Endpoints, Euthanasia Methods, and Case Studies in Animal Research.	Lecture	1	Knows how
DOES — 1			
Evaluate Animal Welfare and Ethics in a Research Setting.	Practical Training	2	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **7 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 25 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 7

Classical and Advanced quality control techniques for Ayurveda Formulations

2 credits | 60 notional hours | 5 units | **26 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	9	6	10
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	8	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	8	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
10 objectives

Practice documentation and record-keeping processes in the quality control section; Assess Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma); Demonstrate quality control parameters for solid dosage forms like tablets, pills, and Varti; and 7 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate documentation and record-keeping processes in the quality control section; Demonstrate assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma); Demonstrate Quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Describe the importance of Quality Control and Quality Assurance in Ayurveda; Discuss Quality Control Parameters, Regulatory Requirements, and Documentation; Discuss Mana and their application in quality control and Bhasma/ Pishti pariksha; and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Discuss quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Quality Control and Quality Assurance

4 obj · 6 h · ceiling Does

Importance of quality control in Ayurveda
Quality Assurance of Ayurveda Products
Quality control parameters (identity, purity, etc.)
Regulatory requirements for quality control
Documentation and record-keeping

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the importance of Quality Control and Quality Assurance in Ayurveda.	Lecture	1	Knows how
Discuss Quality Control Parameters, Regulatory Requirements, and Documentation.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate documentation and record-keeping processes in the quality control section.	Practical Training	2	Shows how
DOES — 1			
Practice documentation and record-keeping processes in the quality control section.	Experiential Learning	2	Does

Unit 2 · Classical aspects of the quality control parameter for Ayurvedic formulations

4 obj · 6 h · ceiling Does

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss Mana and their application in quality control and Bhasma/Pishti pariksha.	Lecture	1	Knows how
Discuss Siddhi Lakshana of Avaleha, Snehapaka, Sandhana and Guggulu Paka.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate assessment of Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma).	Practical Training	2	Shows how
DOES — 1			
Assess Bhasma Siddhi Lakshana (Quality tests to confirm the perfect preparation of Bhasma).	Experiential Learning	2	Does

Unit 3 · Quality Control parameter for solid dosage forms

9 obj · 26 h · ceiling Does

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.	Lecture	1	Knows
KNOWS HOW — 2			
Discuss Quality control parameters for solid dosage forms like tablets, pills, and Varti.	Lecture	1	Knows how
Discuss Quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana Lauha/ Mandura.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate Quality control parameters for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.	Practical Training	3	Shows how
Demonstrate quality control parameters for Khalveeya, Parpati and Kupipakwa Rasayana Lauha/Mandura.	Practical Training	4	Shows how

DOES — 4			
Demonstrate quality control parameters for solid dosage forms like tablets, pills, and Varti.	Practical Training	3	Does
Perform Quality control procedures for powder dosage forms like Churna (including Kwatha Churna and Lepa Churna), Kshara, Lavana, Satva.	Experiential Learning	4	Does
Perform quality control procedures for solid dosage forms.	Experiential Learning	4	Does
Perform quality control procedures for Rasoushadhi.	Experiential Learning	5	Does

Unit 4 · Quality Control parameter for semi-solid dosage forms

3 obj · 6 h · ceiling Does

Rasakriya/Avaleha/Paka
Anjana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/Paka/Anjana as per Pharmacopeial standards.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstration of Quality control parameters for semisolid dosage forms like Rasakriya/Avaleha/Paka/Anjana as per Pharmacopeial standards.	Practical Training	2	Shows how
DOES — 1			
Perform quality control tests for semisolid dosage forms.	Experiential Learning	3	Does

Unit 5 · Quality Control parameter for liquid and gaseous dosage forms

6 obj · 12 h · ceiling Does

Kashaya
Arka
Sharkara
Asava-arishta
Netra Bindu/Karna Bindu
Aerosols/sprays

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss Quality control parameters for liquid dosage forms like Kashaya, Arka Sharakara, Asava-arishta.	Lecture	1	Knows how
Discuss Quality control parameters for Netra Bindu/Karna Bindu and aerosols/sprays.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate quality control parameters for liquid dosage forms like Kashaya, Arka Sharakara, Asava-arishta.	Practical Training	2	Shows how
DOES — 3			
Demonstrate quality control parameters for Netra Bindu/Karna Bindu and aerosols/sprays.	Practical Training	2	Does
Perform quality control tests for liquid dosage forms.	Experiential Learning	3	Does
Perform quality control tests for Netra Bindu/Karna Bindu and aerosols/sprays.	Experiential Learning	3	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **16 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 8

Nano-science, nano-technology and its application in Ayurveda

2 credits | 60 notional hours | 5 units | **21 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
5	7	7	2
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	6	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Analyze the impact of Bhavana by measuring the particle size of samples; Conduct a market Survey of Nano-Based Ayurvedic Formulations.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate the measurement of micro-particles using relevant equipment; Demonstrate particle size of Bhasma prepared with different numbers of Puta; Demonstrate particle size of various Kupipakwa Rasayana samples; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
7 objectives

Describe the basic principles and applications of nanotechnology; Discuss an overview of Ayurvedic Nano formulations and their limitations; Identify the potential of nanotechnology in Ayurvedic formulations; and 4 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
5 objectives

Define nanotechnology and describe the history of nanotechnology; Review articles regarding the green synthesis of metallic nanoparticles; Discuss regulatory frameworks for nanotechnology-based Ayurvedic formulations; and 2 more.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Introduction to Nanotechnology

3 obj · 6 h · ceiling Shows how

Definition and history of nanotechnology

Understanding Nanoscale

Basic principles and applications of nanotechnology

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Define nanotechnology and describe the history of nanotechnology.	Lecture	1	Knows
KNOWS HOW — 1			
Describe the basic principles and applications of nanotechnology.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the measurement of micro-particles using relevant equipment.	Practical Training	4	Shows how

Unit 2 · Ayurvedic Formulations and Nanotechnology

10 obj · 34 h · ceiling Does

Overview of Ayurvedic formulations and their limitations

Potential of nanotechnology in Ayurvedic formulations

Characterization micro/nanoparticles in Bhasma

Characterization micro/nanoparticles in Kupipakwa Rasayana and Pottali

Characterization of micro-nano sized particles generated during Bhavana

Synthesize silver nanoparticles (AgNPs) using the chemical precipitation method and grasp the fundamental principles underlying nanoparticle synthesis

Market Survey of Nano-Based Herbal Formulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Review articles regarding the green synthesis of metallic nanoparticles.	Experiential Learning	2	Knows
KNOWS HOW — 2			
Discuss an overview of Ayurvedic Nano formulations and their limitations.	Lecture	1	Knows how
Identify the potential of nanotechnology in Ayurvedic formulations.	Lecture	1	Knows how
SHOWS HOW — 5			
Demonstrate particle size of Bhasma prepared with different numbers of Puta.	Practical Training	4	Shows how
Demonstrate particle size of various Kupipakwa Rasayana samples.	Practical Training	4	Shows how
Synthesize silver nanoparticles (AgNPs) using the chemical precipitation method.	Practical Training	4	Shows how
Measure particle size of Bhasma prepared with different numbers of Puta.	Experiential Learning	4	Shows how
Measure particle size of various Kupipakwa Rasayana samples.	Experiential Learning	4	Shows how
DOES — 2			
Analyze the impact of Bhavana by measuring the particle size of samples.	Experiential Learning	4	Does
Conduct a market Survey of Nano-Based Ayurvedic Formulations.	Experiential Learning	6	Does

Unit 3 · Nanocarriers for Ayurvedic Formulations

3 obj · 6 h · ceiling Shows how

Types of nanocarriers (liposomes, nanoparticles, etc.)
Advantages and limitations of nanocarriers in Ayurvedic formulations
Conversion of Taila into Oil in water nano emulsion and its characterization

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss types of nanocarriers (liposomes, nanoparticles, etc.).	Lecture	1	Knows how
Discuss the advantages and limitations of nanocarriers in Ayurvedic formulations.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate converting Taila into an oil-in-water nanoemulsion and its characterization.	Practical Training	4	Shows how

Unit 4 · Regulatory Aspects and Safety Concerns

2 obj · 2 h · ceiling Knows

Regulatory frameworks for nanotechnology-based Ayurvedic formulations
Safety concerns and toxicity issues related to nanotechnology-based Ayurvedic formulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 2			
Discuss regulatory frameworks for nanotechnology-based Ayurvedic formulations.	Lecture	1	Knows
Discuss safety concerns and toxicity issues related to nanotechnology-based Ayurvedic formulations.	Lecture	1	Knows

Unit 5 · Recent Advances and Future Directions

3 obj · 8 h · ceiling Knows how

Recent advances in nanotechnology-based Ayurvedic formulations
Future directions and potential applications
Exploring the Future of Nanomedicine with Ayurvedic Formulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss recent advances in nanotechnology-based Ayurvedic formulations.	Lecture	1	Knows
KNOWS HOW — 2			
Discuss future directions and potential applications of nanotechnology in Ayurvedic formulations.	Lecture	1	Knows how
Integrate nanomedicine with Ayurvedic formulations and their prospects.	Experiential Learning	6	Knows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **9 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PART C

Reference

Four appendices, generated from the same data as Part B so the two cannot disagree. C1 decodes the tiers and names the instruments that certify each. C2 gives the ceiling of every module in one view — the fastest way to check that an assessment matches what it is meant to certify. C3 lists the instruments the curriculum itself names, sorted by the tier each can honestly reach. C4 records the sources and what was and was not altered.

1. **C1** The four tiers, and what certifies each
2. **C2** Ceiling of every module
3. **C3** The instruments this curriculum names, by tier
4. **C4** Sources, and a note on how this book was made

↑ Contents

PART C · APPENDIX C1

The four tiers, and what certifies each

Miller's four tiers with the count of objectives at each, the instruments that can certify them, and the evidence you keep. Nothing below a tier can certify it.

TIER	OBJECTIVES	INSTRUMENTS	EVIDENCE
Knows	14	MCQ, short answer, oral quiz, spotting.	Written script, quiz sheet.
Knows how	79	Structured long answer, case-based questions, structured viva.	Reasoned written answer, viva score sheet.
Shows how	35	OSPE or OSCE station, observed performance against a checklist.	Checklist score, annotated artefact, station record.
Does	43	Log book, record book, portfolio, observation over time, presentation with peer feedback.	Dated log entry, countersigned record, portfolio piece.

TIER	OBJECTIVES	SHARE
Knows	14	8.2%
Knows how	79	46.2%
Shows how	35	20.5%
Does	43	25.1%

Total 171 objectives.

PART C · APPENDIX C2

Ceiling of every module

The highest tier each module reaches. Read across: the ceiling names the evidence you must gather, and the tier above it names what you may not ask for.

M	U	UNIT	OBJ	CEILING
MODULE 1 — PHARMACEUTICS AND PHARMACEUTICAL TECHNOLOGY — 24 OBJECTIVES, CEILING DOES				
1	1	Introduction to Pharmaceutics	6	Does
1	2	Pharmaceutical Dosage Forms	4	Does
1	3	Pharmaceutical Formulation	5	Does
1	4	Pharmaceutical Manufacturing and Quality Control	5	Does
1	5	Biopharmaceutics	4	Does
MODULE 2 — DATABASES AND INFORMATION SYSTEMS FOR METAL, MINERAL, AND PLANT RESEARCH — 22 OBJECTIVES, CEILING DOES				
2	1	Introduction to Databases and Information Systems	5	Does
2	2	Metal and Mineral Databases	6	Does
2	3	Plant Databases	4	Does
2	4	Database application	4	Does
2	5	Challenges and limitation of Databases	3	Does
MODULE 3 — ESSENTIAL CHEMISTRY :- UNLOCKING THE RASASHASTRA TECHNIQUES — 21 OBJECTIVES, CEILING DOES				
3	1	Atomic, Molecular Structure, and Periodic Table Elements	3	Does
3	2	Specialized Chemistry Fields-An Outline	2	Knows how
3	3	Organic Chemistry	3	Knows how
3	4	Phytochemistry	7	Does
3	5	Metallurgy	6	Does
MODULE 4 — CONCEPT, UNDERSTANDING, APPLICATION, SOPS AND CHEMICAL CHANGES OF SHODHANA AND MARANA PROCEDURES — 15 OBJECTIVES, CEILING DOES				
4	1	Shodhana and Marana Procedures	3	Knows how
4	2	SOP of Shodhana Processes	4	Does
4	3	SOP of Marana process	4	Shows how
4	4	Chemical Changes involved in Shodhana and Marana processes	1	Knows how
4	5	Shodhana and Marana- Case Studies, troubleshooting and emerging trends	3	Knows how
MODULE 5 — PRINCIPLES, SPECIFICATION, SOPS OF HANDLING OF CLASSICAL AND ADVANCED INSTRUMENTS — 29 OBJECTIVES, CEILING DOES				
5	1	Puta and its modern aspects	4	Does
5	2	Musha and its modern aspects	4	Shows how
5	3	Chromatography	5	Does
5	4	Spectroscopy	4	Knows how
5	5	Microscopy	4	Shows how
5	6	Calibration techniques	3	Does
5	7	Overview of characterization instruments	5	Knows how

M	U	UNIT	OBJ	CEILING
MODULE 6 — ANIMAL HOUSE AND ANIMAL HANDLING TECHNIQUE — 13 OBJECTIVES, CEILING DOES				
6	1	Introduction to animal house management	3	Does
6	2	Study of animal behaviour	3	Does
6	3	Species-specific handling techniques	2	Does
6	4	Animal health and disease prevention	3	Does
6	5	Animal welfare and ethics	2	Does
MODULE 7 — CLASSICAL AND ADVANCED QUALITY CONTROL TECHNIQUES FOR AYURVEDA FORMULATIONS — 26 OBJECTIVES, CEILING DOES				
7	1	Quality Control and Quality Assurance	4	Does
7	2	Classical aspects of the quality control parameter for Ayurvedic formulations	4	Does
7	3	Quality Control parameter for solid dosage forms	9	Does
7	4	Quality Control parameter for semi-solid dosage forms	3	Does
7	5	Quality Control parameter for liquid and gaseous dosage forms	6	Does
MODULE 8 — NANO-SCIENCE, NANO-TECHNOLOGY AND ITS APPLICATION IN AYURVEDA — 21 OBJECTIVES, CEILING DOES				
8	1	Introduction to Nanotechnology	3	Shows how
8	2	Ayurvedic Formulations and Nanotechnology	10	Does
8	3	Nanocarriers for Ayurvedic Formulations	3	Shows how
8	4	Regulatory Aspects and Safety Concerns	2	Knows
8	5	Recent Advances and Future Directions	3	Knows how

Module ceilings are shown in the coloured band; unit ceilings in the rows beneath. A unit may sit below its module's ceiling, and most do.

PART C · APPENDIX C3

The instruments this curriculum names, by tier

Every assessment abbreviation that appears in the curriculum, sorted by the highest tier it can honestly certify.

HIGHEST TIER IT REACHES	INSTRUMENTS THE CURRICULUM NAMES	WHAT IT CAN AND CANNOT SHOW
Knows	MCQ, Spotting, T-OBT	Recall only. Cannot show reasoning or performance.
Knows how	LAQ, SAQ, Structured viva, ABQ, Case study	Reasoning on paper or aloud. Cannot show that the student can perform.
Shows how	OSPE, Practical examination, Checklist	Performance when watched. The lowest instrument that settles a doing objective.
Does	Log book, Record book, Portfolio, Presentation	Independent practice over time. The only evidence that reaches the top tier.

Counts are how many objectives name each instrument. An instrument may always be used below its tier; it may never be used above it.

PART C · APPENDIX C4

Sources, and a note on how this book was made

The primary sources for the framework, the curriculum this book is built on, and a note on what was and was not altered.

THE FRAMEWORK

- Miller, G. E., 'The assessment of clinical skills / competence / performance', *Academic Medicine*, 65(9), 1990. The original four-tier paper.
- Cruess, R. L., Cruess, S. R. and Steinert, Y., 'Amending Miller's Pyramid to Include Professional Identity Formation', *Academic Medicine*, 91(2), 2016. The proposed fifth tier, *Is*.
- Biggs, J., 'Enhancing teaching through constructive alignment', *Higher Education*, 32, 1996. The source of the alignment principle in Chapter 3.

A NOTE ON HOW THIS BOOK WAS MADE

Every objective in Part B is reproduced in the curriculum's own words. Plain-English spelling errors have been corrected and sentence punctuation lost to the source document's line wrapping has been restored. No Sanskrit or technical term has been altered, and no subject content has been added, removed or reworded.

National Commission for Indian System of Medicine, Board of Ayurveda. *Curriculum for MD/MS Ayurveda, Semester II — Applied Basics of Rasashastra and Bhaishajya Kalpana (AYPG-AB-RB).* New Delhi: NCISM. Tables 2, 3, 4, 5 and 6 are the source of every objective, tier, hour and mark in this book. All 171 objectives, 91 activities and 480 notional hours reconcile against the curriculum's own credit framework.

The counts and tables in Part C are generated from the same parsed data as Part B, so the two cannot disagree.

A COGNITIVE-TAXONOMY HANDBOOK

Bloom's Taxonomy for Rasashastra & Bhaishajya Kalpana

Module-by-module domain and level mapping, objective design and usage guide for AYPG-RB — Rasashastra and Bhaishajya Kalpana
MD/MS Ayurveda, Semesters III to VI

Curriculum	NCISM, Board of Ayurveda — four semesters, four papers, applicable from the 2024–25 batch
Coverage	32 modules · 124 units · 577 session learning objectives · 141 practical training and 163 experiential learning activities
Framework	Bloom (1956) and the Anderson & Krathwohl revision (2001) carried in parallel, with the psychomotor and affective domains, read against NCISM Table 3 Column 3E

577
OBJECTIVES CODED

277
THINKING

294
DOING

64
CREDITS · 1920 HOURS

Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026 · For departmental use in curriculum delivery, objective writing, assessment design and paper setting.

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Unit 2 Taxonomy of Rasadravyas 5 obj 19

Unit 3 Geological/mineralogical identification of Rasadravyas 6 obj 20

Unit 4 Overview of Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya 3 obj 20

Module 2 Puta and Samskara (Transformative processes of drug manufacturing) Paper 1 · 21 obj · 56 h 22

Unit 1 Puta and Law of thermodynamics 6 obj 23

Unit 2 Concept, understanding, application, SOPs and chemical changes of Jarana, Amritikarana and Lohitikarana 4 obj 23

Unit 3 Concept, understanding, application, SOPs (Standard Operating Procedures) and chemical changes of Satva and Druti procedures 4 obj 24

Unit 4 Samskara (Principles of Transformative processes of drug manufacturing) 2 obj 24

Unit 5 Influence of Desha (place) and Kala(time and seasonal factors) of drug qualities 2 obj 24

Unit 6 Kalaprakarsha- Processing with/Effect of specific time exposure and Bhajana (processing vessel) 3 obj 25

Module 9 Scientific fundamentals of Ayurvedic Pharmaceutical preparations Paper 2 · 15 obj · 56 h 26

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Unit 2 Identity, Purity and Quality of Raw materials used in Ayurveda Pharmaceutics 3 obj 27

Unit 3 Pharmaceutical processing of drugs with its rationality 3 obj 27

Unit 4 Ayurvedic Classical Dosage Forms and their integration with modern scientific validation 6 obj 27

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Unit 3 Principles and designing of various dosage forms 2 obj 30

Unit 4 Principles of Prakshep dravyas (Pharmaceutical Additives) 3 obj 30

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PART A

Understanding the taxonomy

Nine short chapters. The first five explain what Bloom's taxonomy is, what its two versions are, what its three domains cover, where it already sits inside this curriculum, and how verbs turn it into practice. Chapter 6 covers the fixed one-to-two-to-three split between lecture, practical training and experiential learning. Chapter 8 covers both kinds of assessment this course carries — the modular assessment in every module, and the 800-mark university examination at the end of Semester VI. Read straight through once; after that use the module modules and come back only when something needs settling.

1. **What Bloom's taxonomy is, and why this book carries two versions**
2. **The three domains** — thinking, doing, and how a student goes about the work
3. **The second axis** — what kind of knowledge an objective works on
4. **Where Bloom already sits in your curriculum** — reading Column 3E
5. **Verbs** — the working end of the taxonomy
6. **The one-to-two-to-three split** — lecture, practical training, experiential learning
7. **Choosing how to teach**, level by level
8. **Choosing how to check** — the modular assessment and the university examination
9. **How to use a module in Part B**

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PART A · CHAPTER 1

What Bloom's taxonomy is, and why this book carries two versions

In 1956 a committee of American examiners led by Benjamin Bloom published a way of sorting examination questions. They had noticed what every teacher notices sooner or later: two questions on the same content can be worlds apart, not because one covers more ground, but because one asks the student to *do* something harder with what they know.

What they produced was a ladder of six levels, each asking more than the one below: **Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation**. The ladder is cumulative. You cannot apply what you do not understand, and you cannot understand what you cannot recall. That the levels stack is what makes the taxonomy useful for planning rather than merely for labelling.

THE 2001 REVISION

- The levels became **verbs**: *Remember, Understand, Apply, Analyse, Evaluate, Create.*
- **The top two swapped.** Bloom put Synthesis at level 5 and Evaluation at level 6. The revision reversed them: Evaluate is 5, Create is 6, on the reasoning that making something new is a larger act than judging something that exists.
- A **second axis** was added for the kind of knowledge involved. Chapter 3 covers it.

> **The one thing to carry away.** When a colleague says a question is "at the top level", ask which scheme they mean. In 1956 the top is Evaluation — judging. In 2001 it is Create — producing. Different questions, different teaching, different marking. Most disagreement between two well-informed teachers about Bloom comes down to this.

WHY BOTH, IN THIS BOOK

Your curriculum uses the **1956 scheme**: the codes in Column 3E are CK, CC, CAP, CAN, CS and CE. If this book used only the 2001 names you could not match it against your syllabus document. But most faculty development now teaches the 2001 scheme. So every level here is named twice, 1956 first: *Comprehension / Understand.* Where the two schemes disagree about position — only at the top two — the book says so on the page.

> **This curriculum reaches the top of both ladders.** Synthesis appears five times and Origination four, across Semesters III, V and VI. Every one of the four semesters reaches Evaluation on the thinking ladder. If you have taught only the undergraduate course, where Synthesis is never used at all, the ceiling here is genuinely higher and the teaching has to change with it.

PART A · CHAPTER 2

The three domains — thinking, doing, and how a student goes about the work

Bloom's six levels describe thinking only. The original committee planned three volumes. The **cognitive** volume appeared in 1956; the **affective** volume — attitudes and values — in 1964; the **psychomotor** volume was never written by that committee, and the version your curriculum uses comes from later work by others.

THINKING — SIX LEVELS

Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation. Across the four semesters, 277 of the 577 objectives are cognitive — 48%.

DOING — FIVE LEVELS

- **Set** — ready to begin; knows what is needed and can lay it out.
- **Guided response** — performs while being shown or guided.
- **Mechanism** — performs reliably alone, without prompting.
- **Adaptation** — changes the method when conditions change.
- **Origination** — devises a procedure that was not given.

Two hundred and ninety-four of the 577 objectives are here — half the course. That is the single most important fact about this curriculum, and it is more pronounced than in any other year of the programme: 12% of the undergraduate syllabus is psychomotor, 38% of Semester II, and 50% here.

The distribution inside the ladder matters as much as the total. **One hundred and ninety-four objectives — a third of the entire course — sit at *Guided response***, the level at which a student performs while being shown. Sixty-five sit at *Mechanism*, where they perform unaided. That is worth pausing on: the curriculum concentrates its largest single block of objectives one rung *below* independent competence. Moving students from the first to the second is not a side effect of the practical hours; in this course it is the main work of them.

GOING ABOUT IT — FIVE LEVELS

Receiving, Responding, Valuing, Organisation, Characterisation — from paying attention when it matters, through treating something as worth doing unprompted, to it having become how the student works.

> **This course is nearly silent here too.** Six objectives in 577 carry an affective code. That is worth knowing rather than working around: professional habits — verifying material, recording the reading actually taken, declining to sign off on what was not checked — are being formed across these four semesters whether or not the curriculum codes them, and a postgraduate is precisely the person in whom they matter. If you want evidence of them you will have to plan for it, because the syllabus does not ask. Part B therefore shows two ladders per module; the six affective objectives appear in their own bands in the units that hold them.

THE DOMAINS DO NOT MOVE TOGETHER

An objective at Mechanism may need only level 1 thinking if the procedure is followed by rote. Another at Set may need level 4 thinking if choosing the right equipment requires judgement. The two ceilings move independently across this course, and the modules print both for that reason. Read both before planning, and before choosing a modular assessment.

PART A · CHAPTER 3

The second axis — what kind of knowledge an objective works on

The 2001 revision added a second question. The first is: what thinking must the student do? The second is: *what kind of knowledge is it about?*

- **Factual** — terminology, names, specific details. Learned by rehearsal and self-testing.
- **Conceptual** — classifications, principles, relationships, the reason a thing is so. Learned by explaining it to somebody.
- **Procedural** — methods, techniques, criteria for choosing between them. Learned by carrying them out.
- **Metacognitive** — knowing about your own learning. Learned by being made to look back at how you worked.

> **This course is overwhelmingly Procedural.** Procedural knowledge dominates, conceptual follows, and factual knowledge is a small minority. That is the profile of a laboratory and workshop course rather than a reading course, and it lines up with the 50% psychomotor figure. The practical consequence is blunt: **reading a procedure is not learning it**, and a student who has only read will look identical to one who has practised right up until the moment they are asked to perform.

WHY IT CHANGES THE TEACHING

Two objectives at the same level can need entirely different sessions. One asks for the names in a group; another asks for the steps of a procedure in order. Rehearsal fixes the first. The second will not hold until the student has done it, because a remembered list of steps and a performable procedure are not the same possession. This is why “they know it but they cannot do it” is such a common complaint — usually the objective was Procedural and the teaching was pitched as though it were Factual.

In Part B every objective is labelled with its kind of knowledge, read from the words of the objective itself. Where an objective works on two kinds, both are named. Where the wording supports a reading without settling it, the label carries a small circle (°), so you can see a judgement was made and overrule it.

PART A · CHAPTER 4

Where Bloom already sits in your curriculum

Bloom is not something this book adds. It has been in the curriculum since publication, in a column most teachers have never had pointed out.

COLUMN 3E

Open Table 3. The fifth column is headed *Domain/Sub Domain*. Every one of the 577 session learning objectives carries a code there: CK, CC, CAP, CAN, CS, CE for thinking; PSY-SET, PSY-GUD, PSY-MEC, PSY-ADT, PSY-ORG for doing; AFT-REC through AFT-CHR for how the student goes about the work. The full key is Appendix C1.

Table 3 in this curriculum is organised by **paper first, then semester** — all four semesters of Paper 1, then all four of Paper 2, and so on. Part B reorders it by semester, which is how a department actually teaches, so Module 1 and Module 2 sit together in Semester III rather than being separated by three semesters of Paper 1.

THE COLUMNS AROUND IT

- **3A — Course outcome.** Which of the course-level outcomes the objective serves.
- **3C — Notional learning hours.** How long the objective is expected to take.
- **3D — Lecture, practical training or experiential learning.** The mode, and for the last two a reference into the activity tables. Chapter 6 is about this column.
- **3G — Teaching and learning methods.**

> **A word about Column 3F, which this book does not use.** Column 3F is headed *Level*, but it carries a different scheme — Knows, Knows how, Shows how, Does — which is Miller's pyramid, not Bloom. It answers a different question: not what kind of thinking is needed, but how authentic the evidence has to be. It has its own companion volume. In *this* book, "level" always means the Bloom level in Column 3E. Where you read "level 5" here, it never means "Shows how".

MODULE LEARNING OBJECTIVES SIT ABOVE SESSION OBJECTIVES

Each module states its own learning objectives, and these are what the modular assessment tests. They are pitched at their own levels, which do not always match the session objectives beneath them. Part B shows each module's objectives with the level of their verb, so you can see at a glance whether the module assessment is aimed higher or lower than the sessions that feed it.

FOUR PAPERS, AND WHAT THEY COVER

- **Paper 1** — Rasashastra (Iatrochemistry): Principles and Practices
- **Paper 2** — Exposition of Ayurvedic Pharmaceutics
- **Paper 3** — Standardization and clinical Utility of Ayurvedic formulations
- **Paper 4** — Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

Each paper carries two modules per semester, so eight modules per paper across the four semesters, and thirty-two in all.

PART A · CHAPTER 5

Verbs — the working end of the taxonomy

A level is an idea; a verb is what a student meets. **An objective is taught and examined at the level of its verb**, whatever else is written beside it.

- **Level 1, Knowledge / Remember** — define, enlist, list, name, recall, recite, state, identify, recognise, observe, record.
- **Level 2, Comprehension / Understand** — describe, explain, discuss, illustrate, classify, differentiate, outline, present.
- **Level 3, Application / Apply** — apply, demonstrate, perform, prepare, calculate, measure, conduct, practise, operate, handle.
- **Level 4, Analysis / Analyse** — analyse, compare, examine, determine, categorise, correlate, review, explore, investigate, search.
- **Level 5 in 1956, Synthesis** — design, formulate, develop, construct, devise, synthesise, integrate, optimise. (Level 6, Create, in the 2001 scheme.)
- **Level 6 in 1956, Evaluation** — evaluate, justify, appraise, critique, assess, judge, defend, interpret. (Level 5 in the 2001 scheme.)

The affective domain has its own verbs, and they are not thinking verbs: *appreciate, value, accept, comply, respond, defend, advocate, adhere, display, practise.*

> **Where a verb and a code disagree, follow the verb — and know you are choosing.** The verb is what appears on the paper, so it wins by default. That is fine provided it is a decision rather than an accident. If the verb asks less than the code, you have room to teach higher than the syllabus requires. If it asks more, plan for the verb. Part B tells you, objective by objective, which case you are in.

WRITING YOUR OWN

Decide the level first, then take a verb from that level, then check the verb can be assessed by the tool you have in mind. One verb, one level, one objective. Where you find yourself writing “identify, demonstrate and evaluate” on a single line, you have three objectives at three levels and they will need three different checks.

PART A · CHAPTER 6

The one-to-two-to-three split — lecture, practical training, experiential learning

This is the feature of postgraduate study that most decides what students actually reach, and it has no equivalent in the undergraduate course. Every credit is thirty hours, and the curriculum fixes how those hours are spent: **five hours of lecture, ten of practical training, thirteen of experiential learning, and two of modular assessment** — a ratio of roughly one to two to three.

Across four semesters and 64 credits that comes to **1920 notional hours**: 1792 of teaching and 128 of modular assessment. Two thirds of this course happens with the teacher not talking, and it stays that way for two full years.

THE THREE KINDS OF HOUR DO NOT REACH THE SAME LEVELS

- **Lecture** puts material in front of students and lets them retrieve it. It builds level 1 well and level 2 poorly, and on its own it builds nothing above level 2. **273 of the 577 objectives are set as lecture** — the largest single mode by count, though the smallest by hours.
- **Practical training** moves a student from performing with help to performing unaided. That single transition is what these hours are for, and it is slow. **141 objectives** are set here, each pointing at one of the 141 practical training activities.
- **Experiential learning** hands over an unfamiliar situation and requires a judgement, a decision or a product. This is where the upper levels are built and where the modular assessment lives. **163 objectives** are set here, against 163 activities.

> **The commonest failure is silent demotion.** An objective set as experiential learning and then delivered as a lecture has been quietly moved down two levels, and nothing in the timetable will show it. The students will have been told about the thing instead of doing it. Part B prints the kind of hour against every objective for this reason: it is the clearest statement the curriculum makes about how a thing is meant to be learned.

THE ACTIVITY TABLES

Every practical and experiential objective carries a reference — 1.1, 8.2 and so on — into the curriculum's activity tables, which describe what the student is to do. There are **141 practical training activities and 163 experiential ones**, 304 in all. Appendices C6 and C7 reproduce every one with the level it is cited to serve, so you can see what a reference points at without turning back to the syllabus.

PART A · CHAPTER 7

Choosing how to teach, level by level

The commonest planning error is to choose the method by the hour rather than by the level. What follows is what each level needs.

LEVEL 1 — KNOWLEDGE / REMEMBER

Exposure, then rehearsal, then retrieval practice. Make students retrieve without looking, then check. In this course level 1 is very small — seven objectives in 577 — so it should be taking almost none of your contact time. If it is taking more, something pitched higher is being taught as recall.

LEVEL 2 — COMPREHENSION / UNDERSTAND

Students must handle the material, not receive it again: sort it, contrast it, put it in their own order, explain it to a classmate. **152 objectives sit here**, the largest cognitive group in the course, and lecture alone will not deliver them.

LEVEL 3 — APPLICATION / APPLY

Work examples through together, then hand over a situation the student has not seen. The hand-over is the teaching.

LEVEL 4 — ANALYSIS / ANALYSE

Give students two things and ask how they differ and why. Analysis needs something to set against something else.

LEVEL 5 IN 1956, SYNTHESIS

Used five times across the four semesters. It means the student assembles something that was not handed to them — a protocol, a specification, a procedure. Set the criteria in advance and mark against them.

LEVEL 6 — EVALUATION

Give something to judge and criteria to judge it against, then require the criteria to be stated. Small-group critique works well. This is the level most often taught as though it were level 2 — explaining the accepted view rather than reaching one.

THE DOING LADDER

- **Set** — show the setup, name every item, then have students lay it out themselves.
- **Guided response** — demonstrate once at speed, once slowly, then let them copy while you correct in the moment.
- **Mechanism** — repeated supervised practice with your help withdrawn a little each time. **Most practical hours belong here.**
- **Adaptation** — change one condition and let them work out the adjustment.
- **Origination** — used four times, in Semesters III, V and VI. The student devises a procedure that was never demonstrated. Reaching it needs the earlier levels genuinely secure, which is why it appears late.

PART A · CHAPTER 8

Choosing how to check — the modular assessment and the university examination

A check is fair when it can only be passed by a student who has reached the level, and can be passed by any student who has. Most unfair checks fail the first half.

- **Level 1** — short questions asking for a list, a name, a definition. Many of them, spread out.
- **Level 2** — questions requiring the material rearranged. If it can be answered word-for-word from notes, it has slipped to level 1.
- **Level 3** — an unseen case. A familiar case measures memory.
- **Level 4** — questions asking for the parts and how they relate.
- **Level 5** — something the student produced, marked against criteria set in advance.
- **Level 6** — give the marks for the reasoning and the criteria, not for the facts repeated inside the answer.

DOING OBJECTIVES CAN ONLY BE CHECKED BY WATCHING

A viva shows whether a student knows a procedure and can explain it. It cannot show whether they can carry it out, and a fluent student who has never handled the material will pass it comfortably. With **294 doing objectives** — half this course — this matters more here than anywhere else in the programme. The evidence has to be seen: a checklist ticked while the student works, an OSPE station, an observed performance. A checklist of the steps you will look for, written before the session, is enough.

THE TWO ASSESSMENTS THIS COURSE CARRIES

Formative — the modular assessment. Every module carries one, worth 25 marks per credit and using two of its hours per credit. That is 400 marks a semester and **1600 marks across the four**. The curriculum requires a different method in each module, with a record of the pattern used.

Summative — the university examination, at the end of Semester VI: four papers of 100 marks and a practical examination of 400, **800 marks in all**. It examines two years of work at once, which has one consequence worth planning for from Semester III onward: **anything certified only by observation during a practical session will not be re-examinable at the end.** If a student's competence at Mechanism or Adaptation was seen but never recorded, there is nothing to carry forward.

> **Choose the modular method against the module's ceiling, not by rotation.** A module whose ceiling is Origination or Synthesis needs an assessment that lets a student produce something. One that stops at Application does not, and an elaborate instrument there wastes everyone's time. The modules in Part B print both ceilings for exactly this decision.

> **Keep the record as you go, and keep it usable.** Across 124 units and 304 activities, a log book and a set of checklists are the only things that will still exist in Semester VI. Decide at the start of each module what artefact each doing objective will leave behind — a signed checklist, a log entry, a dated record — and the summative examination becomes a matter of drawing on evidence rather than of memory.

PART A · CHAPTER 9

How to use a module in Part B

Part B has thirty-two modules, ordered by semester and then by paper, so the two modules a department teaches together sit together. Each begins on its own page and carries its units inside it.

WHAT IS ON THE MODULE

1. **The heading** — module number, semester, paper, and the paper's title.
2. **The strip** — credits, units, thinking and doing counts, formative marks.
3. **Where the hours go** — the lecture, practical and experiential split for this module, with what each kind of hour must deliver here.
4. **What the module asks students to do** — both ladders across all its units. Empty levels are shown as empty, because knowing what a module does *not* ask for is as useful as knowing what it does.
5. **Module learning objectives**, each with the level of its verb. These are what the modular assessment tests.
6. **The units**, each as a strip: its title, objective count, hours and both ceilings, then its syllabus points as the curriculum numbers them, then its objectives grouped by level.
7. **Two panels** — one for the teacher, one to give to students.

READING A UNIT STRIP

Inside each strip the objectives are gathered under coloured bands, one band per level, thinking before doing. Every objective shows its verb, its kind of knowledge, and its kind of hour. Where an objective's verb sits at a different level from its code, a note in the same cell says which to teach to — and the verb is what a student will meet in the examination.

THREE WAYS TO USE IT

Planning a session. Find the unit, take one band at a time, and teach upward. Check the method against Chapter 7 and the kind of hour against Chapter 6.

Designing the modular assessment. Start from the module's two ceilings and its module learning objectives, then choose an instrument that can reach them.

Planning for Semester VI. Use the doing ladder to see which objectives will need a recorded artefact, and decide what that artefact is before the module starts.

> **What this book does not do.** It does not rewrite the curriculum and it does not tell you what to teach — every objective in Part B appears in the curriculum's own words. What it adds is the taxonomy layer: what kind of learning each objective asks for, and what follows for teaching and checking.

PART B

The modules

Thirty-two modules, ordered by semester and then by paper so the modules a department teaches together sit together. Each carries its units inside it, and every objective appears in the curriculum's own words, gathered under its level so a session can be planned around one level at a time, with the kind of hour printed against each. Chapter 9 explains the layout; after that the units are meant to be opened at whichever module you are about to teach.

1. **Semester III** — Modules 1, 2, 9, 10, 17, 18, 25, 26 across the four papers
2. **Semester IV** — Modules 3, 4, 11, 12, 19, 20, 27, 28
3. **Semester V** — Modules 5, 6, 13, 14, 21, 22, 29, 30
4. **Semester VI** — Modules 7, 8, 15, 16, 23, 24, 31, 32

↑ Contents

MODULE 1 · SEMESTER 3 · PAPER 1

Chronological Development and Taxonomy of Rasa Dravyas

Rasashastra (Iatrochemistry) — Principles and Practices

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	4	11	8	50
60 notional hours	19 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

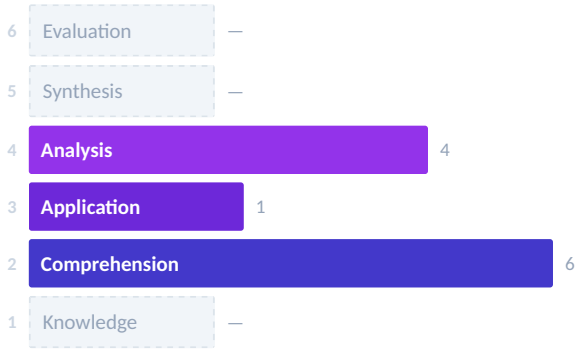
WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Discuss the philosophical relevance and chronological development of Rasashastra,	*discuss*	L2 Comprehension / Understand
2	Integrate the development of Iatrochemistry with modern chemistry	*integrate*	L5 Synthesis / Create
3	Differentiate Rasadravyas as per different ancient classics, and Mendeleev Periodic Table	*differentiate*	L4 Analysis / Analyse
4	Discuss the Sandigdha, Anupalabdha Rasadravyas, and authenticate ideal substitutes	*discuss*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Chronological Development and Overview of Rasashastra

5 obj · 9 h · Analysis / Adaptation

1.1 Philosophical Relevance of Rasashastra
1.2 Chronological Development of Rasashastra
1.3 Integration of the development of Iatrochemistry with modern chemistry

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Discuss the philosophical relevance of Rasashastra and historical evidence till the 6th century AD.	*discuss*	Conceptual	Lecture 1h
Describe the chronological development of Rasashastra from the 6th Century AD to the medieval period.	*describe*	Conceptual ◦	Lecture 1h
Integrate the development of Iatrochemistry with modern chemistry. — teach at level 5, the verb *integrate* is what will be asked	*integrate*	Procedural ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss the chronological development of Rasashastra after the medieval period. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 4 · ADAPTATION — 1			
Construct a chronological tree of Rasashastra and its integration with other medical traditions.	*construct*	Procedural	Experiential 5h

Unit 2 · Taxonomy of Rasadravyas

5 obj · 13 h · Analysis / Guided response

2.1 Taxonomy of Rasadravyas
2.2 Concepts and justification about division of Rasadravyas
2.3 Mendeleeve periodic table

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Justify the division of Rasadravyas. — teach at level 6, the verb *justify* is what will be asked	*justify*	Conceptual ◦	Lecture 1h
Discuss the critics of the classification of Rasadravya.	*discuss*	Conceptual	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2			
Elaborate classification of Rasadravyas as per different ancient classics. — teach at level 2, the verb *elaborate* is what will be asked	*elaborate*	Conceptual	Lecture 1h
Identify and analyse selected metals and minerals with the periodic table elements. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Experiential 5h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the application of the Mendeleev Periodic table in mapping Rasadravyas.	*demonstrate*	Procedural	Practical 5h

Unit 3 · Geological/mineralogical identification of Rasadravyas

6 obj · 22 h · Analysis / Mechanism

3.1 Grahya-Agrahyata of Rasadravyas
3.2 Geological/Mineralogical identification of Rasadravyas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss classical parameters for differentiating acceptable and non-acceptable Rasadravya. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Procedural	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Apply geological/mineralogical characters for the identification of Rasadravya. — teach at level 3, the verb *apply* is what will be asked	*apply*	Procedural	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the comparison and analyze the classification of Rasa Dravyas.	*demonstrate*	Procedural	Practical 5h
Demonstrate Geological/Mineralogical identification of Rasadravyas.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 2			
Recognise Rasadravya based on their Graahyatva-Agraahyatva.	*recognise*	Procedural	Experiential 5h
Identify Rasadravya using geological/mineralogical characters.	*identify*	Procedural	Experiential 5h

Unit 4 · Overview of Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya

3 obj · 12 h · Comprehension / Guided response

4.1 Sandigdha (Controversial)
4.3 Pratinidhi (Substitute) Rasadravya
4.2 Anupalabdha (Not Available) and Krutrima (Artificial) Rasadravya

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 1 · SET — 1			
Identify and analyse Sandigdha and Anuplabhdha Rasadravyas and propose suitable alternatives.	*identify*	Procedural	Experiential 6h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstratie the Krutrima and Pratinidhi Rasadravyas.	*demonstrate*	Procedural	Practical 5h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 4 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 2 · SEMESTER 3 · PAPER 1

Puta and Samskara (Transformative processes of drug manufacturing)

Rasashastra (Iatrochemistry) — Principles and Practices

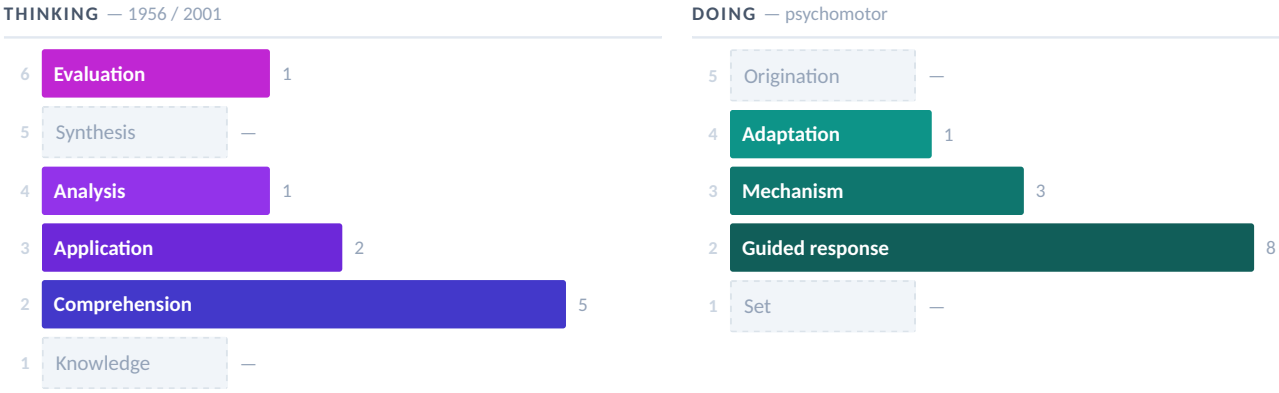
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	6	9	12	50
60 notional hours	21 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	9	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	8	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the definition, significance, construction, and design of traditional Puta	*describe*	L2 Comprehension / Understand
2	Elaborate on the different theories and laws of thermodynamics with traditional Puta	*elaborate*	L2 Comprehension / Understand
3	Describe the principles and significance of Jarana, Amritikarana, Lohitikarana, Satva, and Druti procedures	*describe*	L2 Comprehension / Understand
4	Integrate the principles of classical drug manufacturing (Samskara-transformative process) with modern techniques	*integrate*	L5 Synthesis / Create

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Puta and Law of thermodynamics

6 obj · 18 h · Evaluation / Mechanism

1.1 Definition, Significance and Classification of Puta

1.2 Laws of thermodynamics, Thermochemistry, Griffith theory, Collision theory, etc

1.3 Heating devices and fuels, with detailed knowledge about temperature, duration, designing and construction of different Puta

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Explore the traditional concept of Puta, discussing its definition, significance, and classification in the context of Ayurvedic pharmaceutical practices. — teach at level 4, the verb *explore* is what will be asked	*explore*	Conceptual	Lecture 1h
Explain the laws of thermodynamics (Zeroth, First, Second, Third), the law of thermochemistry, Griffith's theory, and Collision theory.	*explain*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Examine the modern applications and relevance of Puta in contemporary pharmaceutical processing, integrating traditional knowledge with modern technology. — teach at level 4, the verb *examine* is what will be asked	*examine*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate and compare the temperature pattern and total duration of different traditional Puta with EMF.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 2			
Perform comparison of Traditional and Modern Puta Methods: Observing and Analyzing Temperature Patterns and Duration.	*perform*	Procedural	Experiential 5h
Explore Thermal Dynamics in Traditional Puta Systems: An of Fuel Variations, Temperature Patterns, and Thermodynamic Principles.	*explore*	Conceptual + Procedural ◦	Experiential 5h

Unit 2 · Concept, understanding, application, SOPs and chemical changes of Jarana, Amritikarana and Lohitikarana

4 obj · 16 h · Analysis / Guided response

2.1 Principle and Significance of

2.2 Application of Amritikarana and Lohitikarana procedures in bhasma preparation

2.3 Standard Operating Procedures (SOP) and Chemical Changes of Jarana, Amritikarana, and Lohitikarana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss the fundamental principles and significance of Jarana, Amritikarana, and Lohitikarana procedures in the preparation of bhasmas. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual + Procedural ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the art of Gandhaka Jarana: A Process, Understanding its Role and Significance in Rasashastra.	*demonstrate*	Procedural	Practical 5h
Demonstrate Amritikarana and Lohitikarana: Traditional Methods for Mica Processing in Rasashastra.	*demonstrate*	Procedural	Practical 5h
Perform an experiential Investigation of Amritikarana and Lohitikarana Procedures Exploring Safety, Efficacy, and Principles of Transformation.	*perform*	Procedural	Experiential 4h

Unit 3 · Concept, understanding, application, SOPs (Standard Operating Procedures) and chemical changes of Satva and Druti procedures

4 obj · 9 h · Application / Guided response

3.2 Application and analysis of Satva and Druti procedures

3.3 SOP and chemical changes of Satva and Druti

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss the principles, significance, and applications of Satva procedures within the context of Ayurvedic pharmacy and modern pharmaceutical practices. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
Describe the principles, significance, and apply Druti procedures within the context of Ayurvedic pharmacy. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the procedure of Satvapatana.	*demonstrate*	Procedural	Practical 5h
Perform and analyse the Satva Patana procedure.	*perform*	Procedural	Experiential 2h

Unit 4 · Samskara (Principles of Transformative processes of drug manufacturing)

2 obj · 3 h · Comprehension / Adaptation

4.1 Application of Toya Sannikarsha (Processing with water) and Agni

4.2 Shaucha and Manthana (churning)

4.3 Vasana (process of infusing aroma), and Bhavana (Effect of Levigation / Trituration)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Integrate and apply classical Ayurvedic manufacturing principles, including water, heat and combined processing, cleaning, churning, proximity, and trituration effects. — teach at level 5, the verb *integrate* is what will be asked	*integrate*	Conceptual	Lecture 1h
DOING — LEVEL 4 · ADAPTATION — 1			
Apply Time-Tested Principles of Classical Drug Manufacturing: by integrating Toya Sannikarsha, Agni Sannikarsha, Toyagni Sannikarsha, Shaucha, and Manthana to Optimize Pharmaceutical Quality, Safety, and Efficacy.	*apply*	Procedural	Experiential 2h

Unit 5 · Influence of Desha (place) and Kala(time and seasonal factors) of drug qualities

2 obj · 3 h · Comprehension / Mechanism

5.1 Influence of Desha (geographical area and place of

5.2 Influence of Kala (processing time, season, and maturity of herbs) on drug quality

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Description of the influence of Desha (geographical area) and Kala (time and seasonal factors) on the quality of Ayurvedic drugs, including the impact of geographical location, climate, and seasonal variations on drug efficacy and safety.	*description*	—	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Perform A Hands-on Investigation of Geographical and Temporal Influences on drug 7, CO 8 quality: Desha–Kala in Pharmaceutical Manufacturing.	*perform*	Procedural	Experiential 2h

Unit 6 · Kalaprakarsha- Processing with/Effect of specific time exposure and Bhajana (processing vessel)

3 obj · 7 h · Comprehension / Guided response

5.1 Knowledge of Kalaprakarsha (Time taken

5.2 Knowledge of Bhajana (vessels used for manufacturing)

5.3 Environmental and Temporal Influences on Drug Quality and Processing

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the concepts of Kalaprakarsha (processing with specific time exposure) and Bhajana (processing vessels) in Ayurvedic pharmaceutical processing, including the significance of processing time, vessel material, and other temporal factors on drug quality and efficacy.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate and analyze the role of Vasana and Bhavana to Optimize Pharmaceutical Quality, Safety, and Efficacy.	*demonstrate*	Procedural	Experiential 4h
Demonstrate the effect of Kalaprakarsha and Bhajana in Pharmaceutical Processing Analyzing Processes and Findings to Optimize Quality, and stability.	*demonstrate*	Procedural	Experiential 2h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 9 lecture, 4 practical, 8 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 9 · SEMESTER 3 · PAPER 2

Scientific fundamentals of Ayurvedic Pharmaceutical preparations

Exposition of Ayurvedic Pharmaceutics

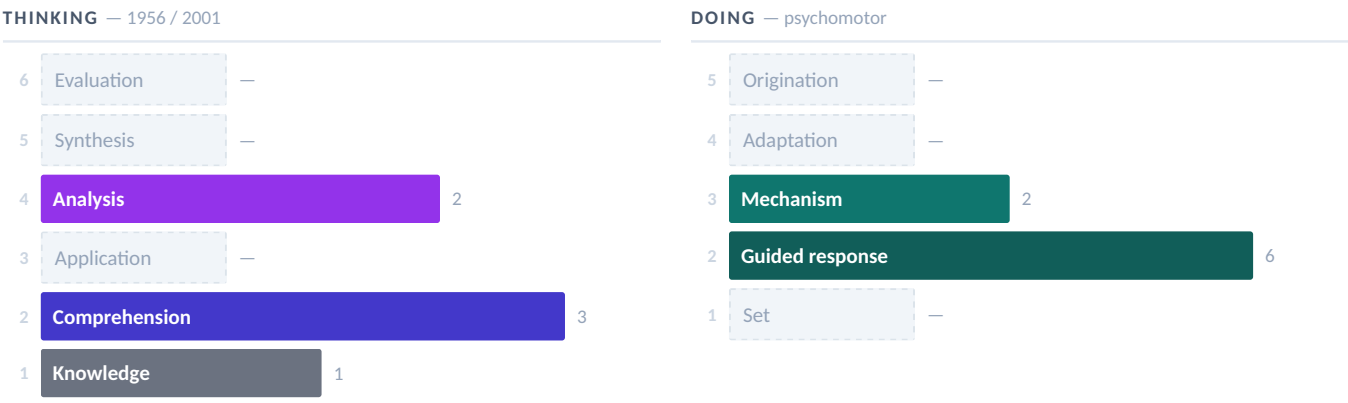
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	4	6	8	50
60 notional hours	15 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	5	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the classical Principles of drug processing and the scientific basis of Panchavidha Kashaya Kalpana	*describe*	L2 Comprehension / Understand
2	Evaluate Identity, Purity, and Quality of Raw Materials Used in Ayurveda Pharmaceutics	*evaluate*	L6 Evaluation / Evaluate
3	Conduct pharmaceutical processing of drugs with rationality	*conduct*	L3 Application / Apply
4	Analyze Ayurvedic Classical Dosage Forms and their integration with modern scientific validation	*analyze*	L4 Analysis / Analyse

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Classical Principles of drug processing and scientific basis of Panchavidha Kashaya Kalpana

3 obj · 6 h · Comprehension / Mechanism

1.1 Bhaishajya Synonyms and significance of its
1.2 Classical principles of processing
1.3 Scientific review of Panchavidha Kashaya Kalpana and Upakalpana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the rationality behind the selection of Panchavidha Kashaya Kalpana and compare classical and modern drug extraction techniques.	*discuss*	Procedural	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Prepare unique formulation from the category of Panchavidha Kashaya Kalpana.	*prepare*	Procedural	Experiential 3h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate processing of unique Panchavidha Kashaya Kalpana with their ingredients proportion of water and procedural details.	*demonstrate*	Procedural	Practical 2h

Unit 2 · Identity, Purity and Quality of Raw materials used in Ayurveda Pharmaceutics

3 obj · 12 h · Comprehension / Guided response

2.1 Classical knowledge about Identity, Purity, Quality of Raw
2.2 Collection of Raw material- GACP, QAQC parameters
2.3 Adulteration and Contamination
2.4 Substitute drugs (Pratinidhi Dravya)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe classical and contemporary parameters for quality assurance of raw drugs, as well as adulteration and contamination of the raw material.	*describe*	Procedural	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate application of QAQC parameters used for quality control of herbal drugs.	*demonstrate*	Procedural	Practical 4h
Identify and apply various tools and techniques used for quality control of raw materials and appraise the classical rules of the collection of raw material.	*identify*	Procedural	Experiential 6h

Unit 3 · Pharmaceutical processing of drugs with its rationality

3 obj · 13 h · Analysis / Mechanism

3.1 Processing of Drugs- Classical methodology
3.2 Simple and unique remedies from classical texts prepared with various processing methods
3.3 Rationality behind the Processing Method of Classical Dosage Forms
3.4 Siddhilakshanani (End-point assessment of pharmaceutical process) and its significance

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the classical and contemporary methods used for the processing of drugs including the assessment of the End-Point(Siddhi Lakshanani) of the dosage form, C_L.	*describe*	Procedural	Lecture 2h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Interpret significance of understanding the proportion of ingredients of formulations quoted in the classics. — teach at level 6, the verb *interpret* is what will be asked	*interpret*	Conceptual	Experiential 6h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate various formulations by describing their ingredients and understanding the uniqueness of their proportion.	*demonstrate*	Procedural	Practical 5h

Unit 4 · Ayurvedic Classical Dosage Forms and their integration with modern scientific validation

6 obj · 25 h · Analysis / Guided response

4.1 Classification of the source of raw material and the
4.2 Multiple factors responsible for Process optimization

4.3 Key qualities of good pharmaceutical products to ensure safety, efficacy, and authenticity

4.4 Research updates about the Integration of traditional wisdom with modern scientific validation

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Recognize the factors responsible for a change in the concentration of the prepared drug and assess the extracted phytochemicals with the help of TLC and other methods of analysis.	*recognize*	Procedural	Lecture 3h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Describe multiple factors responsible for the process optimization of Ayurvedic pharmaceutical preparations and illustrate key qualities of good pharmaceutical products to ensure safety, efficacy, and authenticity. — teach at level 2, the verb *describe* is what will be asked	*describe*	Procedural ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the extraction of phytochemicals in the prepared dosage form.	*demonstrate*	Procedural	Practical 4h
Perform the process of Extraction of Phytochemicals in Ayurvedic formulations.	*perform*	Procedural	Experiential 5h
Demonstrate the kwatha preparation considering its process optimization.	*demonstrate*	Procedural	Practical 5h
GOING ABOUT IT — LEVEL 3 · VALUING — 1			
Review Research Updates in the field of Ayurvedic pharmaceutics along with modern scientific validation methods.	*review*	Conceptual + Procedural ◦	Experiential 6h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 5 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 10 · SEMESTER 3 · PAPER 2

Principles of pharmaceutical formulation

Exposition of Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	5	6	50
60 notional hours	12 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	3	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Principles of Samskara (transformative processes of drug manufacturing) in relation to drug development

2 obj · 6 h · Application / Guided response

1.1 Qualities of drugs/Procedures
1.2 Classical concept of Samskara and expected alterations in qualities
1.3 Key role of Samskara in pharmacological and therapeutic modification of the drug
1.4 Modifications in samskaras in current practices and their consequences

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the applied principles of samskara involved in the preparation of various kalpana. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual + Procedural °	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Illustrate the role of samskaras in the changes observed during the preparation of various dosage forms according to classical texts.	*illustrate*	Procedural	Practical 4h

Unit 2 · Rationality behind Composition of Formulations

2 obj · 6 h · Analysis / Guided response

2.1 Combinations of ingredients (desirable/ inadmissible)
2.2 Rational integrity behind the combination of ingredients
2.3 Aushadhi Gana (therapeutic groups) or Varga (categories) of medicinal herbs and their role in formulations
2.4 Combinations based on predominant pharmacological actions, qualities, or therapeutic usage of the ingredients

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Analyze the purpose of composition and combinations of ingredients in classical formulations.	*analyze*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate combinations of various classical formulations and justify the rationale behind the ingredients.	*demonstrate*	Procedural	Practical 4h

Unit 3 · Principles and designing of various dosage forms

2 obj · 8 h · — / Guided response

3.1 Principles of dosage form design
3.2 Pharmaceutical Considerations
3.3 Formulation Considerations
3.4 Biopharmaceutical Considerations
3.5 Pharmacokinetic Considerations

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate various designs of dosage forms, considering all fundamentals of pharmacology.	*demonstrate*	Procedural	Practical 6h
GOING ABOUT IT — LEVEL 3 · VALUING — 1			
Justify the Principles of dosage form design as per modern pharmaceutical science.	*justify*	Conceptual	Lecture 2h

Unit 4 · Principles of Prakshep dravyas (Pharmaceutical Additives)

3 obj · 14 h · Comprehension / Mechanism

4.1 Prakshepa Dravya- Need, Importance, Proportion (matra)
4.2 Pharmaceutical Excipients

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Identify the selection and relevance of Prakshepa dravya added to various dosage forms and describe Pharmaceutical excipients. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate and analyze the importance of pharmaceutical excipients.	*demonstrate*	Procedural	Practical 6h
DOING — LEVEL 3 · MECHANISM — 1			
Illustrate various Prakshepa Dravyas and their proportions stated in the classics.	*illustrate*	Procedural	Experiential 6h

Unit 5 · Principles of Potency and Efficacy enhancement

3 obj · 22 h · Synthesis / Mechanism

5.1 Classical principles of Potency and efficacy enhancement (Shaktyutkarsha)
5.2 Methods of Potency and Efficacy Enhancement in Ayurvedic Classics
5.3 Methods of Potency and Efficacy Enhancement in Modern Pharmaceutics

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Describe and compare various techniques used in the classics for Shaktyutkarsha (Potency enhancement). — teach at level 2, the verb *describe* is what will be asked	*describe*	Procedural	Lecture 2h
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1			
Investigate the principles for improving potency and efficacy in modern pharmaceutics. — teach at level 4, the verb *investigate* is what will be asked	*investigate*	Conceptual	Experiential 10h
DOING — LEVEL 3 · MECHANISM — 1			
Justify the need for Shaktyutkarsha (Potency enhancement)in drug preparation and Advocate techniques of potency enhancement used in Ayurvedic pharmaceutics.	*justify*	Procedural	Experiential 10h

FOR THE TEACHER

- **This module tops out at Synthesis on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 4 practical, 3 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 17 · SEMESTER 3 · PAPER 3

Guidelines for administration of Rasoushadhi in clinical Practice

Standardization and clinical Utility of Ayurvedic formulations

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	11	11	50
60 notional hours	22 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	7	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Discuss the concept of Rasa Chikitsa and explain the clinical superiority of Rasoushadhi.	*discuss*	L2 Comprehension / Understand
2	Illustrate the concept, need, and procedure of Kshetrikarana and its significance in the safe administration of Rasoushadhi.	*illustrate*	L2 Comprehension / Understand
3	Discuss appropriate Pathya apathya during Rasa Yoga Sevana.	*discuss*	L2 Comprehension / Understand
4	Elaborate on the toxicological profile of Rasoushadhi, including Ashuddha, Apakwa Bhasma Sevana Janya Dosha, and its management with safe prescription	*elaborate*	L2 Comprehension / Understand
5	Discuss heavy metal toxicity and its management with chelation therapy.	*discuss*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Concept of Rasa chikitsa and significance of Rasoushadhi

4 obj · 11 h · Application / Guided response

1.1 Perspectives on the Notion of Rasa
1.2 Significance of Rasa Chikitsa
1.3 Various treatment modalities of Ayurveda

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Analyse the term Rasa from the Sanskrit and Ayurvedic literatures and discuss the significance of Rasoushadhi. — teach at level 4, the verb *analyse* is what will be asked	*analyse*	Factual + Conceptual °	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss various treatment modalities used in the Ayurvedic system of medicine. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual °	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the rational prescription for a selected disease.	*demonstrate*	Procedural	Practical 3h
Apply the principles of Rasa Chikitsa in the clinical practice/ alloted case.	*apply*	Procedural	Experiential 6h

Unit 2 · Concept of Kshetrikarana and its significance prior to administration of Rasoushadhi

4 obj · 11 h · Application / Guided response

2.1 Etymology and Concept of Kshetrikarana
2.2 Kshetrikarana for Optimising Medicine Efficacy
2.3 Procedure of Kshetrikarana (Panchakarma & Pancha Shodhana)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Define kshetrikarana and discuss the concept of kshetrikarana. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual °	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the need and benefits of Kshetrikarana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual °	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the need and role of Kshetrikarana.	*demonstrate*	Procedural	Practical 3h
Demonstrate the procedure and benefits of kshetrikarana.	*demonstrate*	Procedural	Experiential 6h

Unit 3 · Pathya -Apathya during Rasayoga Sevana

5 obj · 12 h · Application / Mechanism

3.1 Pathya and Apathya Ahara, Vihara, and Manasika Vikara during Rasayoga Sevana

3.2 Importance of Pathya-Apathya during Rasayoga Sevana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Define Pathya and Apathya ahara and describe its importance. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the application of the concept of what pathya and apathya vihara with respect to the Manasika vikara. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Enlist and demonstrate common pathya and apathya Ahara during rasa yoga sevana.	*enlist*	Procedural	Practical 2h
Enlist and demonstrate the common pathya-apathya Vihara during rasa yoga sevana.	*enlist*	Procedural	Practical 2h
DOING — LEVEL 3 · MECHANISM — 1			
Generate document depicting the outcome of compliance and non-compliance with the pathya apathya guidelines.	*generate*	Procedural	Experiential 6h

Unit 4 · Safety and toxicity of Rasoushadhi

5 obj · 14 h · Evaluation / Guided response

4.1 Safety considerations of Rasoushadhi

4.2 Ashuddha and Apakwa Bhasma Sevana Janya Vikara and their management

4.3 Safety and prescription trends of Rasoushadhi

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the safety considerations of Rasa oushadhies in detail. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Analyse the consequences of Ashuddha or Apakwa Bhasma sevana Dosha, along with the treatment.	*analyse*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Evaluate the prescription errors containing Rasoushadhis.	*evaluate*	Conceptual ◦	Experiential 4h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate a case study of signs and symptoms of rasa vikaras and their management.	*demonstrate*	Procedural	Practical 6h
Assess and comapre Ashuddha and Shuddha Bhasma effects in the clinical practice.	*assess*	Procedural	Practical 2h

Unit 5 · Heavy metal toxicity and chelation therapy

4 obj · 8 h · Comprehension / Mechanism

5.1 The mechanism of heavy metal toxicity
5.2 Symptoms of heavy metal toxicity
5.3 Factors influencing heavy metal toxicity
5.4 Concept of chelation therapy and risks, and benefits

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss in detail the sources, pathways, and factors influencing heavy metal toxicity.	*discuss*	Conceptual	Lecture 1h
Discuss chelating agents and their role in managing heavy metal toxicity.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate identification of heavy metal toxicity.	*demonstrate*	Procedural	Practical 2h
DOING — LEVEL 3 · MECHANISM — 1			
Evalute the effect of your chelation therapy in the management of our heavy metal toxicity.	*evaluate*	Procedural	Experiential 4h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 7 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 18 · SEMESTER 3 · PAPER 3

Prayogika Bhaishajya Kalpana(Clinical application of Pharmaceutical products)

Standardization and clinical Utility of Ayurvedic formulations

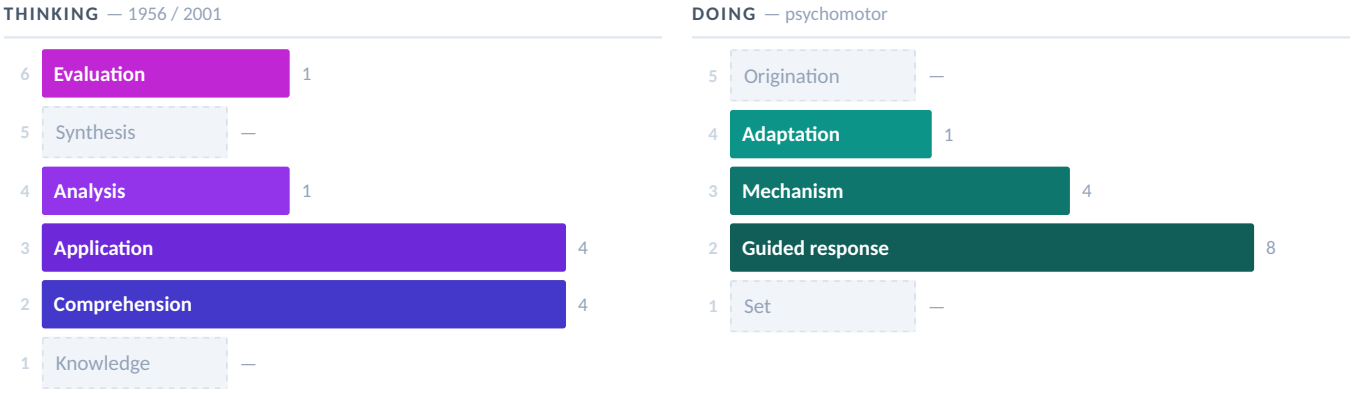
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	4	10	13	50
60 notional hours	24 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	8	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Aushadha Sevana Kala (Time of administration of drug)

7 obj · 15 h · Application / Adaptation

1.1 Classical concept of time of administration
1.2 Biological rhythms and chronobiology
1.3 Chrono-pharmaceutics and chronopharmaceutical drug delivery systems
1.4 Principles of chronopharmacology
1.5 Benefits of chronotherapeutics

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss in detail the concept of Oushadha Sevana Kala in Ayurveda.	*discuss*	Conceptual	Lecture 1h
Elaborate the principles of chrono pharmaceutical drug delivery systems and their benefits.	*elaborate*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss biological rhythms and chronobiology and their significance in drug delivery. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dynamics.	*demonstrate*	Procedural	Practical 3h
Demonstrate chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dynamics.	*demonstrate*	Procedural	Experiential 3h
Illustrate the modulation of the action of the dosage form by chronotherapeutic administration in Ayurveda.	*illustrate*	Procedural	Experiential 3h
DOING — LEVEL 4 · ADAPTATION — 1			
Demonstrate Modulation of dosage form action by chronotherapeutic administration in Ayurveda.	*demonstrate*	Procedural	Practical 3h

Unit 2 · Concept and scientific basis of Anupana

6 obj · 16 h · Evaluation / Mechanism

2.1 Classical concepts of Anupana in Ayurveda
2.2 Role of Anupana in bioavailability
2.3 Anupana for targeted drug delivery and distribution
2.4 Role of Anupana in the mitigation of adverse effects and toxicity
2.5 Anupana for synergistic effect

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss in detail the concept of anupana in Ayurveda. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Explain the role of Anupana in bioavailability and targeted drug delivery. — teach at level 2, the verb *explain* is what will be asked	*explain*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate appropriate Anupana for the selected dosage form in different clinical conditions.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 3 · MECHANISM — 2			
Enlist appropriate Anupanas for various formulations in different clinical conditions.	*enlist*	Procedural	Experiential 3h
Describe the role of Anupana in the modulation of drug tolerance and safety.	*describe*	Conceptual + Procedural	Experiential 4h
GOING ABOUT IT — LEVEL 3 · VALUING — 1			
Demonstrate drug response through synergistic Anupana integration.	*demonstrate*	Conceptual	Experiential 4h

Unit 3 · Matra (Posology)

5 obj · 12 h · Application / Guided response

3.1 Definition, importance, and clinical significance
3.2 Factors affecting drug dose
3.3 Posology according to modern pharmacology
3.4 Calculation of paediatric dose

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Elaborate on the concept, importance, and clinical significance of Matra.	*elaborate*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the factors affecting drug dose in detail. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the calculation of paediatric dose using different formulae.	*demonstrate*	Procedural	Practical 3h
Demonstrate the Patient-centric evaluation of Matra.	*demonstrate*	Procedural	Practical 3h
Demonstrate the utility of posology principles from an ayurvedic and modern pharmacology perspective.	*demonstrate*	Procedural	Experiential 4h

Unit 4 · Concept of drug repurposing

6 obj · 13 h · Application / Mechanism

4.1 Concept of drug repurposing
4.2 Systematic approaches to drug repurposing
4.3 Strategies of drug repurposing
4.4 Methodologies of drug repurposing

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the concept, need, and key aspects of drug repurposing.	*describe*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Describe systematic approaches to drug repurposing. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
Discuss strategies and methodologies of drug repurposing. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the analysis of classical ayurvedic drugs for novel indications.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 2			
Evaluate novel indications for classical Ayurvedic formulations.	*evaluate*	Procedural	Experiential 3h
Demonstrate the Reverse pharmacological approach for drug repurposing in Ayurveda.	*demonstrate*	Procedural	Experiential 2h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 6 practical, 8 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 25 · SEMESTER 3 · PAPER 4

Artificial intelligence in drug development and Pharmacogenetics; Pharmacogenomics

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	10	10	50
60 notional hours	20 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	5	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the role of Artificial Intelligence in drug development	*describe*	L2 Comprehension / Understand
2	Demonstrate the basics of machine learning and its applications	*demonstrate*	L3 Application / Apply
3	Discuss 3D-Printed Drug Delivery Systems Based on Natural Products	*discuss*	L2 Comprehension / Understand
4	Justify clinical application of pharmacogenetics	*justify*	L6 Evaluation / Evaluate
5	Describe the Current status and future perspective of Pharmacogenomics	*describe*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Role of Artificial intelligence in drug development

4 obj · 11 h · Application / Mechanism

1.1 Introduction to AI in drug development
1.2 AI, machine-based intelligence learning
1.3 Role of AI in Drug discovery and Drug development

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Define AI and discuss its importance in drug development. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the Role of AI in Drug Discovery and Drug development with a suitable example. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the basics of machine learning and its application.	*demonstrate*	Procedural	Practical 4h
DOING — LEVEL 3 · MECHANISM — 1			
Perform De Novo Drug. Design Using Generative Models.	*perform*	Procedural	Experiential 5h

Unit 2 · Types of software used in Artificial intelligence

3 obj · 6 h · Application / Origination

2.1 Different types of software used in AI
2.2 Data mining and analysing medical records, medication management
2.3 Recent updates

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Describe different types of software used in AI. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss Data mining and analysing medical records, medication management, with recent updates. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 5 · ORIGINATION — 1			
Demonstrate prediction of drug likeliness using machine learning.	*demonstrate*	Procedural	Practical 4h

Unit 3 · 3D printed drug delivery system

5 obj · 16 h · Application / Adaptation

3.1 Approaches of 3D printing in current drug delivery
3D printing in personalized drug delivery
3D-Printed Drug Delivery Systems Based on Natural Products

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the fundamentals and explore the materials and technology used in 3D printing.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the concept of personalised drug delivery, formulations, and natural products. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate different 3D printers used in the drug delivery system.	*demonstrate*	Procedural	Practical 4h
DOING — LEVEL 4 · ADAPTATION — 2			
Demonstrate the comparison of traditional dosage forms versus 3D printed ones regarding the structure and dissolution.	*demonstrate*	Procedural	Practical 4h
Analyse ADMET properties using AI tools.	*analyse*	Procedural	Experiential 6h

Unit 4 · Pharmacogenetics and Pharmacogenomics

4 obj · 11 h · Application / Guided response

4.1 Introduction to Pharmacogenetics and Pharmacogenomics along with its importance
4.2 Factors influencing drug response and ADE (Adverse drug events)
4.3 Methods of pharmacogenetics testing

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Appraise key concepts and terminologies and explore the role of Pharmacogenetics and Pharmacogenomics. — teach at level 6, the verb *appraise* is what will be asked	*appraise*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Justify clinical application of pharmacogenetics, benefits, and challenges. — teach at level 6, the verb *justify* is what will be asked	*justify*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate to predict Drug-Target Interaction.	*demonstrate*	Procedural	Practical 4h
Identify potential adverse effects of drug candidates using dataset and tools.	*identify*	Procedural	Experiential 5h

Unit 5 · Pharmacogenomics in drug discovery

4 obj · 12 h · Comprehension / Mechanism

5.1 Benefits of Pharmacogenomics
5.2 Drug Repurposing
5.3 Basis of gene-drug association
5.4 Current status and future perspective of Pharmacogenomics
5.5 Ethical concern

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss pharmacogenomics data contributes to the identification and validation of drug targets.	*discuss*	Conceptual ◦	Lecture 1h
Discuss the current status and future perspectives of pharmacogenomics, including ethical concerns.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 2			
Extract potential drug-disease relationship by using through Literature /data Mining.	*extract*	Conceptual + Procedural ◦	Experiential 5h
Interpret Pharmacokinetics and Toxicology Modelling using appropriate AI tools and methods.	*interpret*	Procedural	Experiential 5h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Origination on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 5 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Origination.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 26 · SEMESTER 3 · PAPER 4

Pharmacology

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	11	5	50
60 notional hours	16 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	2	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the basics of pharmacology, including Pharmacodynamics and pharmacokinetics	*describe*	L2 Comprehension / Understand
2	Describe Pharmacokinetics, therapeutic index, and drug interactions	*describe*	L2 Comprehension / Understand
3	Discuss the pharmacology of drugs acting on different systems, like the Central and Peripheral nervous systems	*discuss*	L2 Comprehension / Understand
4	Describe Adverse Drug Reactions and Emergency Pharmacology	*describe*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Introduction to Pharmacology

3 obj · 12 h · Comprehension / Guided response

1.1 Introduction to Pharmacology
1.2 Pharmacodynamics
1.3 Dose-response relationship and factors influencing drug action

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Define and classify pharmacology and describe the principles and mechanisms of drug action, receptor theories, and classification of rece. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual	Lecture 1h
Discuss dose response relationship, and factors influencing drug action.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate pharmacological experiments using videos or simulation software.	*demonstrate*	Procedural	Practical 10h

Unit 2 · Pharmacokinetics, therapeutic index, and drug interactions

3 obj · 12 h · Application / Adaptation

2.1 Pharmacokinetics (absorption, distribution, metabolism, excretion)
2.2 Therapeutic index and combined effects of drugs
2.3 Adverse drug reaction
2.4 Drug interactions (pharmacokinetic and pharmacodynamic)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss therapeutic index, drug interactions, and adverse drug reactions.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the basic principles of pharmacokinetics (absorption, distribution, metabolism 7, CO 8 excretion). — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 4 · ADAPTATION — 1			
Demonstrate the effect of drugs on animals by simulated experiments.	*demonstrate*	Procedural	Practical 10h

Unit 3 · Pharmacology of drugs that act on the peripheral nervous system

3 obj · 9 h · Synthesis / —

3.1 Organization and function of ANS (Autonomic Nervous System)
3.2 Neurohumoral transmission, co-transmission, and classification of neurotransmitters
3.3 Parasympathomimetics, Parasympatholytics, Sympathomimetics, Sympatholytics
3.4 Neuromuscular blocking agents and skeletal muscle relaxants (peripheral)
3.5 Local anesthetic agents

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss the organisation and function of ANS (Autonomic Nervous System) and describe Neurohumoral transmission, co-transmission, and classification of neurotransmitters.	*discuss*	Conceptual	Lecture 1h
Enlist Parasympathomimetics, Parasympatholytics, Sympathomimetics, sympatholytics Neuromuscular blocking agents, and skeletal muscle relaxants (peripheral) and Local anaesthetic agents. — teach at level 1, the verb *enlist* is what will be asked	*enlist*	Factual ◦	Lecture 1h
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1			
Analyse the mechanism of action of drugs acting on the Cardiovascular system through a case study scenario. — teach at level 4, the verb *analyse* is what will be asked	*analyse*	Conceptual ◦	Experiential 7h

Unit 4 · Pharmacology of drugs that act on the central nervous system I

3 obj · 9 h · Application / Mechanism

4.1 Neurohumoral transmission in the C.N.S. special emphasis on the importance of
4.2 General anaesthetics and pre-anaesthetics
4.3 Sedatives, hypnotics, and centrally acting muscle relaxants
4.4 Anti-epileptics

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Describe Neurohumoral transmission in the C.N.S and illustrate General anaesthetics and pre-anaesthetics. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Categorize drugs into classes such as Sedatives, hypnotics, and centrally acting muscle relaxants, and discuss anti-epileptics. — teach at level 4, the verb *categorize* is what will be asked	*categorize*	Conceptual ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Analyse the mechanism of action of drugs acting on the respiratory system through a case study scenario.	*analyse*	Procedural	Experiential 7h

Unit 5 · Pharmacology of drugs that act on the central nervous system II

4 obj · 14 h · Comprehension / Mechanism

5.1 Psychopharmacological agents: Antipsychotics, antidepressants, anti-anxiety
5.2 Drugs used in Parkinson's disease and Alzheimer's disease
5.3 CNS stimulants and nootropics
5.4 Drug addiction, drug abuse, tolerance, and dependence

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss Psychopharmacological agents and enlist drugs used in Parkinson's disease and Alzheimer's disease.	*discuss*	Conceptual ◦	Lecture 1h
Differentiate CNS stimulants and nootropics, and discuss drug addiction, drug abuse tolerance, and dependence. — teach at level 4, the verb *differentiate* is what will be asked	*differentiate*	Conceptual ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 2			
Analyse Adverse Drug Reaction in a polypharmacy patient through a case scenario.	*analyse*	Procedural	Experiential 6h
Discuss Emergency Pharmacology through a case scenario of anaphylaxis management.	*discuss*	Procedural	Experiential 6h

FOR THE TEACHER

- **This module tops out at Synthesis on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 2 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 3 · SEMESTER 4 · PAPER 1

Parada (Mercury) and Its Processing: Ancient wisdom and Scientific advances

Rasashastra (Iatrochemistry) — Principles and Practices

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	4	8	9	50
60 notional hours	17 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	9	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	3	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the ancient and contemporary perspectives of Parada (Mercury)	*describe*	L2 Comprehension / Understand
2	Perform and analyse the Samanya Shodhana, Vishesha Shodhana, and procedures of Parada processing (Ashta Samskara)	*perform*	L3 Application / Apply
3	Discuss the Physicochemical properties and pharmacology of mercury and mercurial compounds	*discuss*	L2 Comprehension / Understand
4	Describe the mercury poisoning and its management by the ancient and contemporary approaches	*describe*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Parada: Ancient and Contemporary Perspectives on Its Shodhana and Underlying Principles

8 obj · 22 h · Comprehension / Mechanism

1.1 Ancient & Contemporary knowledge of Parada/
1.2 Parada Dosha, Shodhana(Purification) of Parada and its significance
1.3 Principles of Parada Jarana and Parada Murcchana
1.4 Utility of Parada Bandha

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Describe the ancient and contemporary perspectives of Parada (Mercury).	*describe*	Conceptual ◦	Lecture 1h
Discuss the processes related to Parada and their significance.	*discuss*	Conceptual + Procedural ◦	Lecture 1h
Describe the scientific basis of Parada Jarana, Murcchana, and Bandha.	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Discuss the rationale behind Samanya and Vishesha Shodhana.	*discuss*	Conceptual + Procedural ◦	Lecture 1h
Demonstrate Samanya and Vishesha Shodhana procedures of Parada.	*demonstrate*	Procedural	Practical 7h
Perform the Parada Jarana, Parada Murchana and Bandhana processes.	*perform*	Procedural	Practical 5h
Perform the Samanya Shodhana of Parada.	*perform*	Procedural	Experiential 3h
DOING — LEVEL 3 · MECHANISM — 1			
Perform the Vishesha Shodhana procedures of Parada.	*perform*	Procedural	Experiential 3h

Unit 2 · Parada Ashta Samskara and Ashtadasha Samskara

4 obj · 21 h · Comprehension / Guided response

2.1 Parada Samskara with research updates
2.2 Pharmaceutical equipment design for Parada Ashta Samskara
2.3 Hands-on experience of Parada Ashta Samskara

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe and discuss the utility of Ashtadasha Samskara.	*describe*	Conceptual ◦	Lecture 2h
Describe the design of pharmaceutical equipment to carry out Parada Ashta Samskara.	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the Parada Samskara procedures- Swedana, Mardana, Murchana, and Utthapana.	*demonstrate*	Procedural	Practical 8h
Elaborate and perform the Parada Ashta Samskara procedures- Patana, Bodhana Niyamana and Deepana.	*elaborate*	Procedural	Experiential 10h

Unit 3 · Physico-chemical proprties and pharmacology of mercury and mercurial compounds

2 obj · 6 h · Analysis / Mechanism

3.1 Mercury metallurgy
3.2 Stoichiometry of mercury
3.3 Pharmacodynamics and pharmacokinetics of mercury and mercurial products

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Describe and elaborate on the metallurgy and stoichiometry of mercury. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Compare and analyze the Pharmacodynamics and pharmacokinetics of mercury and mercurial products.	*compare*	Procedural	Experiential 5h

Unit 4 · Mercury toxicity, ADR and its management

3 obj · 7 h · Analysis / Guided response

4.1 Mercury poisoning and its types

4.2 Ancient and Contemporary knowledge about the Treatment of Mercury poisoning

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the antidotes for mercury poisoning and its possible treatment by ancient and contemporary knowledge. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Describe the mercury poisoning and analyze its mechanism. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Analyse mercury toxicity and ADR incidences.	*analyse*	Procedural	Experiential 5h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 9 lecture, 3 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 4 · SEMESTER 4 · PAPER 1

Maharasa Dravya, its processing and utility

Rasashastra (Iatrochemistry) — Principles and Practices

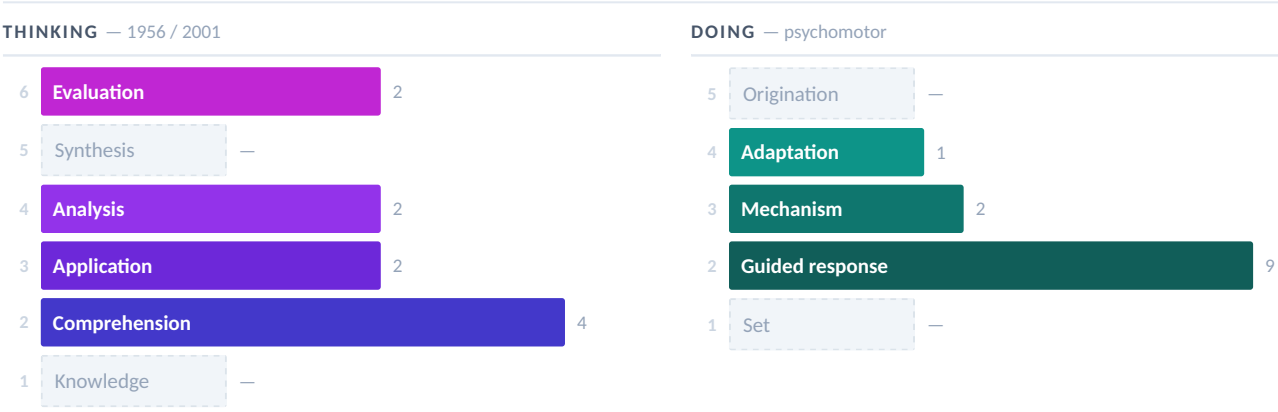
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	10	12	50
60 notional hours	22 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	6	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Discuss the characterization of the samples of Maharasa Dravyas	*discuss*	L2 Comprehension / Understand
2	Perform various pharmaceutical procedures like Shodhana, Marana, etc.	*perform*	L3 Application / Apply
3	Perform the physicochemical analysis with the current scientific updates	*perform*	L3 Application / Apply
4	Discuss the pharmacology and therapeutic application of Maharasa Dravya	*discuss*	L2 Comprehension / Understand
5	Describe the new metal, minerals, procedures, and principles related to Maharasa Dravyas	*describe*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Maharasa Dravyas with special reference to Abhraka and Vaikranta

8 obj · 18 h · Analysis / Adaptation

1.1 Comparative analysis of Maharasa Dravyas mentioned in various texts
1.2 Characterization of Maharasa dravyas
1.3 Abhraka
1.4 Vaikranta

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe and critically discuss the different Maharasa Dravyas mentioned in the classics.	*describe*	Conceptual ◦	Lecture 1h
Describe the significance of Abhraka Pancha Samskara.	*describe*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the mineralogical identity of Vaikranta. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Analyze the technique of characterization for Maharasa Dravyas.	*analyze*	Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Analyze and Verify Maharasa Dravya Samples.	*analyze*	Procedural	Practical 2h
Perform Shodhana, Marana, and other pharmaceutical procedures related to Abhraka and Vaikranta.	*perform*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 1			
Analyze the pharmacotherapeutic attributes of Abhraka Bhasma and Vaikranta Bhasma.	*analyze*	Procedural	Experiential 5h
DOING — LEVEL 4 · ADAPTATION — 1			
Analyse the pharmacology and clinical applications of Abhraka and Vaikranta.	*analyse*	Procedural	Experiential 2h

Unit 2 · Principles and Processing of Makshika and Vimala

3 obj · 11 h · Comprehension / Guided response

2.1 Critical understanding of Makshika (Copper Pyrite)
2.2 Detailed understanding of Vimala (Iron Pyrite)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the therapeutic applications of Makshika and Vimala Bhasma, along with the antidotes for toxicity/ADR caused by them.	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Perform Shodhana, Marana, and other pharmaceutical procedures related to Makshika and Vimala.	*perform*	Procedural	Practical 5h
Perform the Shodhana of Makshika Bhasma and Vimala Bhasma.	*perform*	Procedural	Experiential 5h

Unit 3 · Principles and Processing of Shilajit and Sasyaka

3 obj · 10 h · Analysis / Guided response

3.1 Shilajatu (Black Bitumen)
3.2 Sasyaka(Bornite)/Tuttha(Blue Vitriol)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Describe and analyze the therapeutic applications of Shilajatu and Sasyaka/Tuttha, along with the antidotes for toxicity/ADR caused by them. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Perform Shodhana, Marana, and other pharmaceutical procedures related to Shilajatu and Sasyaka/Tuttha.	*perform*	Procedural	Practical 3h
Perform the Shodhana of Shilajit and Sasyaka/Tuttha.	*perform*	Procedural	Experiential 6h

Unit 4 · Principles and Processing of Chapala and Kharpara, along with physico-chemical analysis, with research updates of all Maharasa Dravyas

4 obj · 9 h · Application / Mechanism

4.2 Kharpara

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the therapeutic applications of Chapala and Kharpara Bhasma, along with the antidotes for toxicity/ADR caused by them. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Perform Shodhana, Marana, and other pharmaceutical procedures related to Chapala/Kharpara.	*perform*	Procedural	Practical 2h
Demonstrate and perform physico-chemical analysis of Bhasma / Shodhita products of Maharasa Dravya category.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 3 · MECHANISM — 1			
Explore and perform the pharmaceutical procedure, like Shodhana, Marana for Chapala/ Kharpara, using authentic or substitute drugs.	*explore*	Procedural	Experiential 3h

Unit 5 · Utility of new metals and minerals

4 obj · 8 h · Evaluation / Guided response

5.1 New metals/ minerals that align with Maharasa Dravyas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the utility of new metals and minerals that align with Maharasa Dravyas.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 2			
Explore Shodhana process for the new metal/ minerals that align with Maharasa Dravyas. — teach at level 4, the verb *explore* is what will be asked	*explore*	Procedural ◦	Lecture 1h
Describe Marana process for the new metal/ minerals that align with Maharasa Dravyas. — teach at level 2, the verb *describe* is what will be asked	*describe*	Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Identify and analyze New Metallic/Mineral Resources that align with Maharasa Dravyas Principles.	*identify*	Conceptual + Procedural ◦	Experiential 5h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 6 practical, 6 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 11 · SEMESTER 4 · PAPER 2

Mana Paribhasha (Metrology in Ayurvedic Pharmaceutical preparation)

Exposition of Ayurvedic Pharmaceutics

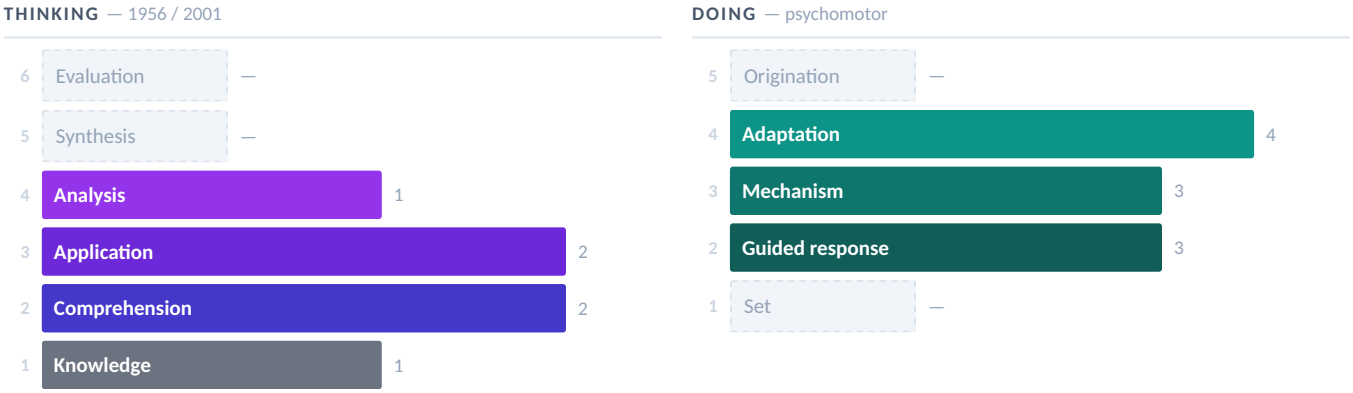
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	6	10	50
60 notional hours	16 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	6	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	5	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Mana Paribhasha (Classical system of measurement)

4 obj · 12 h · Analysis / Adaptation

1.1 Importance of accurate measurements in proportions of ingredients

1.2 Importance of accurate measurements in dosage and administration of drugs

1.3 Types of Mana and details of classical units used for measurement (Weight/ Volume/ Length/ Time)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Interpret the application of Mana in classical formulations with a variety of examples. — teach at level 6, the verb *interpret* is what will be asked	*interpret*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Recall classical knowledge of Mana Paribhasha with a comparison of various classical texts. EC, LS, S. — teach at level 1, the verb *recall* is what will be asked	*recall*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Apply the Knowledge of mana Paribhasha in the preparation of various formulations.	*apply*	Procedural	Practical 4h
DOING — LEVEL 4 · ADAPTATION — 1			
Compare and judge the differences in the Mana mentioned by different classics.	*compare*	Procedural	Experiential 6h

Unit 2 · Standard Units of Measurement

3 obj · 6 h · Knowledge / Adaptation

2.1 AYUSH Unit Converter

2.2 Utility of converted units in the preparation of Classical formulations

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Describe standard units of weights and measures and AYUSH Unit Converter. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Preparation of charts of unit converter (Ayurvedic Mana- Metric).	*preparation*	Procedural	Practical 2h
DOING — LEVEL 4 · ADAPTATION — 1			
Demonstrate preparation of classical formulations prepared with converted Units.	*demonstrate*	Procedural	Experiential 3h

Unit 3 · Volumetric Vs Gravimetric Methods

3 obj · 6 h · Application / Guided response

3.1 Key differences between Volumetric and gravimetric methods of measurement for a variety of drugs

3.2 Application of volumetric method for preparation of various classical formulations

3.3 Variation in proportion by using both methods and alteration in outcome

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Define the terms Volumetric methods and Gravimetric methods and apply the methods of both measurements to Ayurvedic Pharmaceutical Preparations. — teach at level 1, the verb *define* is what will be asked	*define*	Factual + Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Apply and demonstrate the volumetric method for the preparation of classical formulations.	*apply*	Procedural	Practical 2h
Justify and interpret differences occurring in the final product prepared by varying the weight of the ingredients once taken by volume and once taken by mass(Gravimetric Method).	*justify*	Procedural	Experiential 3h

Unit 4 · Rationality of Proportion of Ingredients in Ayurvedic formulations

3 obj · 16 h · Comprehension / Adaptation

4.1 Factors accountable for deciding the proportion of ingredients of the

4.2 Profile of Raw Material (Fresh/ dry, Soft/ hard, part used, therapeutic properties, compatibility, etc.)

4.3 Rules for Duplication of ingredients in the same formulation

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the classical rules and principles for the proportion of ingredients based on raw material and its nature/ Qualities.	*describe*	Conceptual	Lecture 3h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate the skill of preparing formulations with a diverse range of raw materials.	*demonstrate*	Procedural	Practical 6h
DOING — LEVEL 4 · ADAPTATION — 1			
Analyze and justify the proportions of ingredients of classical formulations.	*analyze*	Procedural	Experiential 7h

Unit 5 · Bulk production of Ayurvedic Formulations

3 obj · 16 h · Comprehension / Adaptation

5.1 Modifications expected for bulk production (in Material, Machine, Method, Mechanism)

5.2 Material- Type, proportion, effect of temperature pattern

5.3 Machine used for bulk manufacturing and automated drug manufacturing

5.4 Mechanism of machines used for bulk production and its merits and demerits

5.5 Methods used for bulk production in Ayurvedic Pharmacies and probable alterations in the qualities of the end products

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Enlist the machines, heating devices for bulk production. and describe their methods currently used by Ayurvedic pharmacies f. — teach at level 1, the verb *enlist* is what will be asked	*enlist*	Procedural	Lecture 3h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate the use of machinery used for large-scale production in Ayurvedic pharmacies, with their mechanism.	*demonstrate*	Procedural	Practical 6h
DOING — LEVEL 4 · ADAPTATION — 1			
Debate on merits and demerits of bulk production and the adaptation of advanced machines in Ayurvedic Pharmacies.	*debate*	Procedural	Experiential 7h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 6 lecture, 5 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 12 · SEMESTER 4 · PAPER 2

Process validation Protocol

Exposition of Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	3	4	6	50
60 notional hours	10 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	4	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	2	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Pharmaceutical Preparation process: Classical Dosage forms

6 obj · 41 h · Evaluation / Guided response

1.1 Primary Dosage forms- Panchavidha Kashaya Kalpana and their Upakalpana

1.2 Secondary Dosage Forms- Avaleha Kalpana, Lavana Kalpana, Sneha Kalpana, Guti-vati-guggulu kalpana, Sandhana Kalpana, Arka-Kalpana, Kshara-kalpana

1.3 Topical Dosage forms- Lepa kalpana, Malahara Kalpana, Upanaha Kalpana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the SOP of Primary dosage forms, Secondary dosage forms, and topical dosage forms.	*describe*	Procedural ◦	Lecture 3h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Assess the significance of the Process Validation Protocol.	*assess*	Conceptual + Procedural ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Demonstrate the development of a process validation protocol for Ayurvedic dosage forms.	*demonstrate*	Procedural	Practical 10h
Develop a process validation protocol for Ayurvedic dosage forms.	*develop*	Procedural	Practical 10h
Interpret the documentation related to Standard Operating Procedures (SOPs) used in pharmacies.	*interpret*	Procedural	Experiential 6h
Prepare unique classical formulations following constructed SOP.	*prepare*	Procedural	Experiential 10h

Unit 2 · Pharmaceutical Preparation process: Recent advances of Dosage forms

3 obj · 10 h · Comprehension / Guided response

2.1 Principles of processes viz. Expression, Extraction, Maceration,

2.2 Solid Dosage forms- Powders, Pills, Tablets, Capsules, Soft gel capsules, Gummies, Lozenges, Granules

2.3 Liquid Dosage Forms- Emulsions, Suspensions, Lotions, Liniments, Elixir, Syrup, Nasal Drops, Gargle

2.4 Semi-solid dosage forms- Ointment, Gel, Paste, Cream, Plasters, Suppositories

2.5 Gaseous dosage forms- Aerosols, Nasal Spray, Inhalers

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe principles of different dosage forms, viz. Extraction, Maceration, Percolation Infusion, Distillation, Active principle isolation, Particle size reduction, Separation techniques, and Mixing techniques.	*describe*	Conceptual	Lecture 2h
Discuss Pharmaceutical Preparation Process of Solid, semisolid, liquid, and gaseous dosage forms.	*discuss*	Procedural	Lecture 3h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Prepare different novel modern dosage forms of the given drug.	*prepare*	Procedural	Experiential 5h

Unit 3 · Novel Dosage and drug delivery Forms

1 obj · 5 h · — / Guided response

3.1 Importance and Need of Different Dosage Forms in Drug Delivery Systems and Treatment Modalities

3.2 Definition, significance, technologies driving Novel dosage forms, applications, and challenges

3.3 Modification of different dosage forms

3.4 Potential areas for future research and development

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Analyze, justify, and demonstrate the importance and need of the preparation of Novel dosage forms.	*analyze*	Procedural	Experiential 5h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Guided response on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 4 lecture, 2 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Guided response.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 19 · SEMESTER 4 · PAPER 3

SOP and clinical application of Chaturvidha Rasayana

Standardization and clinical Utility of Ayurvedic formulations

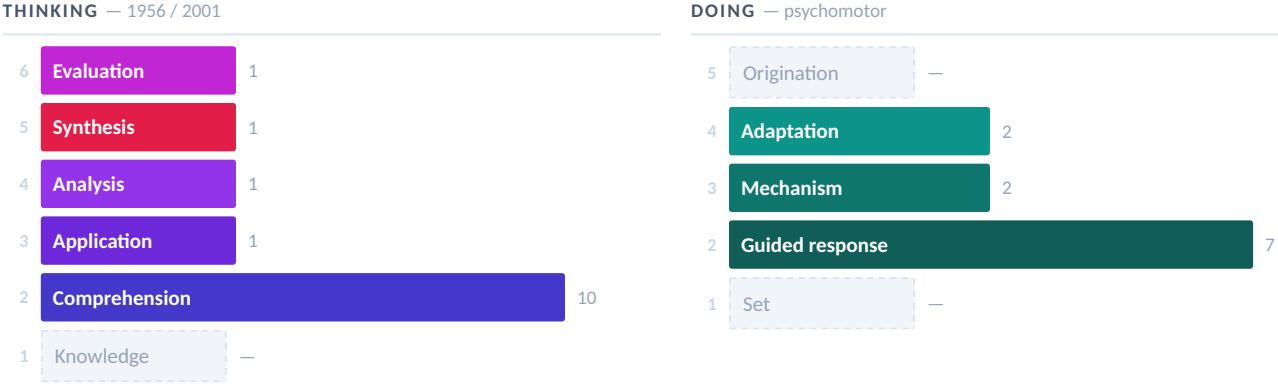
CREDITS	UNITS	THINKING	DOING	FORMATIVE
3	4	14	11	75
90 notional hours	25 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	15 h	14	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	30 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	39 h	7	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 6 hours carry the modular assessment, worth 75 marks, bringing the module to its 90 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe SOP, significance, and clinical application of Khalveeya Rasayana	*describe*	L2 Comprehension / Understand
2	Discuss SOP, significance, mechanism of action, clinical applications with Pathya-Apathya of Parpati Kalpana	*discuss*	L2 Comprehension / Understand
3	Describe SOP, role of Gandhaka Drava paka, significance, and clinical attributes of Pottali Rasayana	*describe*	L2 Comprehension / Understand
4	Discuss SOP, Significance, role of Kramagni Paka, and clinical application of Kupipakwa Rasayana	*discuss*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Standard Operating Procedure (SOP) and Clinical Application of Khalveeya Rasayana

5 obj · 17 h · Application / Adaptation

1.1 Definition and significance of Khalveeya Rasayana
1.2 Standard Operating Procedure for the preparation of Khalveeya Rasayana
1.3 Role of bhavana in potentiating Khalveeya rasayana
1.4 Clinical application of Khalveeya Rasayana with examples

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the unique features, types, and significance of parpati kalpana.	*describe*	Conceptual	Lecture 2h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the role of bhavana (pharmacodynamics of trituration) in potentiating kharaleeya rasayana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the method of preparation and SOP of Kharaleeya Rasayana.	*demonstrate*	Procedural	Practical 7h
Demonstrate the impact of Bhavana (A bioactivation tool) in Kharaleeya Rasayana.	*demonstrate*	Procedural	Experiential 3h
DOING — LEVEL 4 · ADAPTATION — 1			
Appraise the utility of kharaleeya rasayana to clinical case practice.	*appraise*	Procedural	Experiential 4h

Unit 2 · SOP and clinical application of Parpati Rasayana

6 obj · 15 h · Analysis / Adaptation

2.1 Definition and significance of Parpati
2.2 Standard Operating Procedure for the preparation of Parpati with process validation
2.3 Mechanism of action of Parpati kalpana
2.4 Pathyapathya in Parpati Prayoga and its significance
2.5 Clinical application of Parpati Kalpa with examples

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Describe the unique features, types, and significance of parpati kalpana.	*describe*	Conceptual	Lecture 1h
Discuss SOP and quality control parameters of Parpati Kalpa.	*discuss*	Procedural	Lecture 1h
Discuss in detail the pathya-apathya, matra, prayoga of parpati kalpana.	*discuss*	Conceptual	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Review and analyze research updates on Parpati Kalpana.	*review*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the SOP of parpati kalpana.	*demonstrate*	Procedural	Practical 7h
DOING — LEVEL 4 · ADAPTATION — 1			
Evaulate the pharmacodynamic rationale and clinical application of parpati kalpana.	*evaluate*	Conceptual + Procedural	Experiential 4h

Unit 3 · SOP and clinical application of Pottali Rasayana

7 obj · 25 h · Evaluation / Mechanism

3.1 Definition and significance of Pottali rasayana
3.2 Standard Operating Procedure for the preparation of Pottali Kalpa with process validation
3.3 Role of Gandhaka drava paka in the preparation Pottali rasayana
3.4 Special clinical attributes of Pottali with examples

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Define pottali and discuss in detail the concept, types, and significance of pottali rasayana. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual	Lecture 1h
Discuss the standardization and quality control parameters for Pottali Kalpa.	*discuss*	Procedural ◦	Lecture 1h
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1			
Discuss in detail the various classical methods of preparation of pottali kalpana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Procedural	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Evalaute the safety and efficacy of Pottali Rasayana.	*evaluate*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the method of preparation and SOP of pottali kalpana.	*demonstrate*	Procedural	Practical 8h
DOING — LEVEL 3 · MECHANISM — 2			
Prepare Pottali following SOP.	*prepare*	Procedural	Experiential 8h
Analyse the pharmacological profile of the ingredients, bhavana dravya, and the pharmaceutical procedure adopted in achieving therapeutic precision.	*analyse*	Procedural	Experiential 5h

Unit 4 · SOP and clinical application of Kupipakwa Rasayana

7 obj · 27 h · Comprehension / Guided response

4.1 Definition and significance of Kupipakwa rasayana
4.2 Standard Operative Procedure for the preparation of Kupipakwa Rasayana with process validation
4.3 Role of Kramagni paka in preparation of Kupipakwa Rasayana
4.4 Chemistry of Kupipakwa rasayana
4.5 Clinical application of Kupipakwa Rasayana Kalpa with examples

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 4			
Discuss the concept and significance of Kupipakwa Rasayana.	*discuss*	Conceptual	Lecture 1h
Describe in detail the role of Kramagni Paka in the preparation of Kupipakwa Rasayana.	*describe*	Conceptual + Procedural ◦	Lecture 1h
Describe the role of gandhaka jarana in the context of sagandha kupipakwa rasayana.	*describe*	Conceptual	Lecture 1h
Discuss the thermo–chemistry of kupipakwa rasayana.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the method of preparation and SOP of Kupipakwa Rasayana.	*demonstrate*	Procedural	Practical 8h
Prepare the selected Kupipakwa Rasayana.	*prepare*	Procedural	Experiential 10h
Demonstrate the scientific rationality for the selection of specific Kupipakwa Rasayana in the given scenario.	*demonstrate*	Procedural	Experiential 5h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 14 lecture, 4 practical, 7 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 20 · SEMESTER 4 · PAPER 3

SOP and Clinical application of Ayaskriti, Louha, and Mandura Kalpa

Standardization and clinical Utility of Ayurvedic formulations

CREDITS	UNITS	THINKING	DOING	FORMATIVE
1	3	5	7	25
30 notional hours	12 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	5 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	10 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	13 h	3	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 2 hours carry the modular assessment, worth 25 marks, bringing the module to its 30 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the concept of Ayaskriti, with SOP, clinical applications, and safety concerns of Ayaskriti	*describe*	L2 Comprehension / Understand
2	Discuss the concept, SOP, clinical applications, and key chemical considerations of Louha Kalpa	*discuss*	L2 Comprehension / Understand
3	Discuss the concept, SOP, and clinical applications of Mandura Kalpa	*discuss*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · SOP and clinical application of Ayaskriti

3 obj · 8 h · Comprehension / Mechanism

1.1 Ayaskriti
1.2 Standard Operating Procedure for the preparation of Ayaskriti
1.3 Types (forms) of ayaskriti
1.4 Method of administration and clinical utility of Ayaskriti
1.5 Safety concerns of Ayaskriti

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss in detail the concept of Ayaskriti, its types, method of administration, and safety concerns.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the SOP and preparation of Ayaskriti.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 3 · MECHANISM — 1			
Evaluate the role of Ayaskriti.as a multi-target formulation.	*evaluate*	Conceptual + Procedural ◦	Experiential 4h

Unit 2 · SOP and clinical application of Louha kalpa

4 obj · 9 h · Comprehension / Adaptation

2.1 Pharmaceutical vistas of Louha Kalpas
2.2 Key chemical considerations in Louha Kalpa
2.3 Pharmacology and clinical utility of Louha kalpa

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss the concept of Louha Kalpa.	*discuss*	Conceptual	Lecture 1h
Discuss the key chemical considerations in Louha Kalpa.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the SOP and preparation of Louha Kalpa.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 4 · ADAPTATION — 1			
Apply Louha Kalpa by integrating through textual insights with real word cases for the therapeutic purpose.	*apply*	Procedural	Experiential 4h

Unit 3 · SOP and Clinical application of Mandura Kalpa

5 obj · 11 h · Comprehension / Mechanism

3.1 Concept of mandura kalpa and its SOP
3.2 Types of mandura kalpa based on the method of preparation
3.3 Significance of go mutra in mandura kalpa
3.4 Clinical application of mandura kalpa

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss the concept and pharmaceutical vistas of Mandura Kalpa.	*discuss*	Conceptual	Lecture 1h
Describe chemistry and pharmacology associated with structure and preparation of Mandura Kalpas.	*describe*	Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the SOP and preparation of Mandura Kalpa.	*demonstrate*	Procedural	Practical 2h
Demonstrate the clinical application of Mandura Kalpa.	*demonstrate*	Procedural	Practical 2h
DOING — LEVEL 3 · MECHANISM — 1			
Evaluate the role of Gomutra in pharmaceutical processing and as a bio enhancer.	*evaluate*	Conceptual + Procedural ◦	Experiential 5h

FOR THE TEACHER

- **This module tops out at Comprehension on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 4 practical, 3 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 27 · SEMESTER 4 · PAPER 4

Network pharmacology & Reverse Pharmacology

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	2	10	6	50
60 notional hours	16 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	2	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Netwok Pharmacology

8 obj · 28 h · Analysis / Mechanism

1.1 Fundamental knowledge of Network pharmacology
1.2 Network pharmacology concepts
1.3 Computational methods
1.4 Applications
1.5 Research updates on Network pharmacology

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Define network pharmacology and differentiate it from traditional pharmacology.	*define*	Factual ◦	Lecture 1h
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Recognize key network concepts like nodes (genes/proteins/drugs) and edges(interactions). — teach at level 1, the verb *recognize* is what will be asked	*recognize*	Factual ◦	Lecture 1h
Describe network construction from herbal compounds.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss various applications and research updates on Network pharmacology. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss and analyze Research updates on Reverse pharmacology. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate network pharmacology applications with the help of a video.	*demonstrate*	Procedural	Practical 10h
DOING — LEVEL 3 · MECHANISM — 2			
Practice Network Pharmacology.	*practice*	Procedural	Experiential 10h
Carry out a mini-project on network pharmacology.	*carry*	Procedural	Experiential 3h

Unit 2 · Reverse Pharmacology

8 obj · 28 h · Analysis / Mechanism

2.1 Introduction, Definition of Reverse Pharmacology, History, and Scope
2.2 Drug Discovery Current Scenario, Challenges, and Future Perspective
2.3 Reverse pharmacology for herbal drug-based modern medicines
2.4 Research updates on Reverse pharmacology

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Describe the definition and history of reverse pharmacology and discuss its Scope.	*describe*	Factual ◦	Lecture 1h
Elaborate current scenario, challenges, and future perspectives of reverse pharmacology.	*elaborate*	Conceptual ◦	Lecture 1h
Assess reverse pharmacology leverages traditional medicine and clinical experience to accelerate drug development. — teach at level 6, the verb *assess* is what will be asked	*assess*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Apply Reverse pharmacology for herbal-based modern medicines.	*apply*	Procedural ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss and analyze Research updates on Reverse pharmacology. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate reverse pharmacology applications with the help of a video.	*demonstrate*	Procedural	Practical 10h
DOING — LEVEL 3 · MECHANISM — 2			
Present observations of analysis of Reverse Pharmacology in selected cases.	*present*	Procedural	Experiential 10h
Validate Reverse pharmacology with a systems biology approach.	*validate*	Procedural	Experiential 3h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 2 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 28 · SEMESTER 4 · PAPER 4

Pharmaceutical Quality Assurance

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

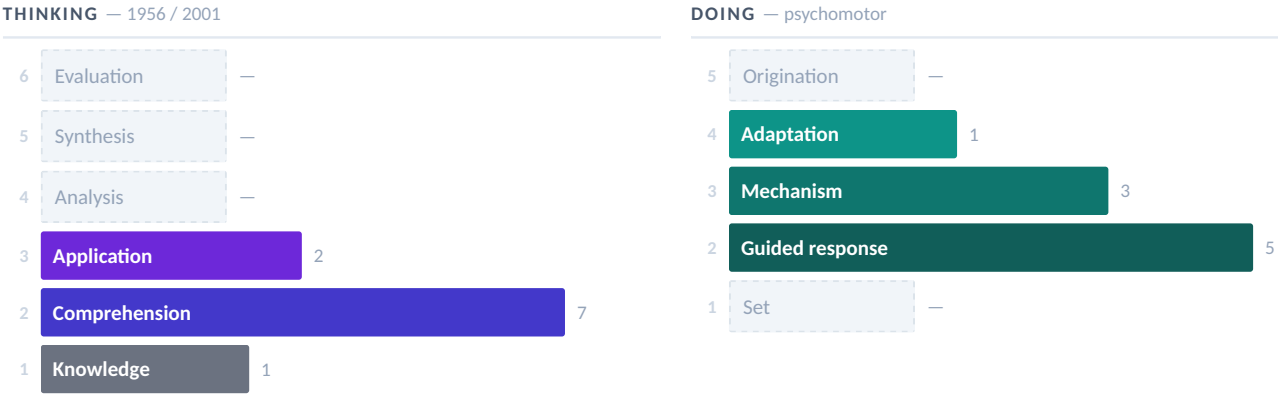
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	10	9	50
60 notional hours	19 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe Quality Assurance and Quality Management concepts	*describe*	L2 Comprehension / Understand
2	Explain Personnel Responsibilities	*explain*	L2 Comprehension / Understand
3	Discuss types of Containers, Packing materials	*discuss*	L2 Comprehension / Understand
4	Discuss Good Laboratory Practices	*discuss*	L2 Comprehension / Understand
5	Justify document maintenance in the pharmaceutical industry	*justify*	L6 Evaluation / Evaluate

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Quality Assurance and Quality Management concepts

6 obj · 19 h · Application / Guided response

1.1 Introduction to Quality Assurance and Quality Management concepts
1.2 Definition and concept of Quality control,
1.3 Quality assurance and GMP
1.4 Total Quality Management (TQM)
1.5 ICH Guidelines
1.6 Brief overview of Pharmaceutical industry guidelines for Quality, Safety, Efficacy and Multidisciplinary (QSEM)
1.7 Overview of ISO 9000 & ISO14000
1.8 Steps for registration of NABL accreditation

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Define and elaborate the concept of Quality Assurance and Quality Management as per. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual ◦	Lecture 1h
Describe Total Quality Management (TQM) and ICH Guidelines, QSEM, and ISO 9000.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Illustrate the Steps for registration of NABL accreditation. — teach at level 2, the verb *illustrate* is what will be asked	*illustrate*	Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the process of mock GMP Inspection.	*demonstrate*	Procedural	Practical 5h
Demonstrate GMP Hierarchy and Responsibilities.	*demonstrate*	Procedural	Practical 5h
Identify the Root cause, affected personnel roles, and suggest corrective and preventive measures through Case Study Discussion on GMP Violation.	*identify*	Procedural	Experiential 6h

Unit 2 · Personnel Responsibilities

3 obj · 8 h · Comprehension / Adaptation

2.1 Organization and personnel
2.2 Personnel responsibilities, training, hygiene, and personal records
2.3 Premises: Design, construction, and plant layout, maintenance, sanitation, environmental control, utilities, and maintenance of sterile areas, control of contamination
2.4 Equipment and raw materials: Equipment selection, purchase specifications, maintenance, and purchase specifications and maintenance of stores for raw materials

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe Organizational structure and personnel management.	*describe*	Conceptual ◦	Lecture 1h
Discuss design, construction, and maintenance of Premises.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 4 · ADAPTATION — 1			
Document a personal record of pharmacy personnel.	*document*	Procedural	Experiential 6h

Unit 3 · Containers, Packing materials

4 obj · 13 h · Comprehension / Mechanism

3.1 Types of Packaging
3.2 Packaging materials
3.3 Container types and closures
3.4 Packaging and stability
3.5 Regulatory aspects
3.6 Quality Control
3.7 Smart Packaging

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss selection and maintenance of equipment, purchase specifications of raw materials, and maintenance of stores.	*discuss*	Conceptual ◦	Lecture 1h
Discuss Types of containers and packaging, Regulatory aspects, their Quality Control and Smart Packaging.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the identification and Classification of packaging materials.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 1			
Inspect Packaging Materials to check their compliance with quality specifications.	*inspect*	Procedural	Experiential 6h

Unit 4 · Good Laboratory Practices

3 obj · 8 h · Knowledge / Mechanism

4.1 GLP Principles and Requirements
4.2 GLP documentation
4.3 Regulatory compliance
4.4 Audit procedures

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Elaborate GLP with its Principles and Requirements, documentation, Regulatory compliance, Audit procedures. — teach at level 2, the verb *elaborate* is what will be asked	*elaborate*	Conceptual + Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate SOP Writing for different equipment.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 1			
Perform calibration of Analytical Instruments.	*perform*	Procedural	Experiential 2h

Unit 5 · Document maintenance in pharmaceutical industry

3 obj · 8 h · Application / Mechanism

4.1 Documentation related to Grievances and Redressal
4.2 Complaints and evaluation of complaints, Handling of returned goods, recalling, and waste disposal
4.3 Document maintenance in the pharmaceutical industry: Batch Formula Record, Master Formula Record, SOP, Quality audit, Quality Review, Quality documentation,

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Define Complaints, their evaluation, product recall, and waste disposal. — teach at level 1, the verb *define* is what will be asked	*define*	Factual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe document maintenance in the pharmaceutical industry. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Conduct QA Documentation audit.	*conduct*	Procedural	Experiential 6h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 4 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 5 · SEMESTER 5 · PAPER 1

Critical appraisal of Uparasa Dravya and Sadharana Rasa Dravya

Rasashastra (Iatrochemistry) — Principles and Practices

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	2	10	14	50
60 notional hours	24 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	8	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe Uparasa and Sadharana Rasa Dravyas with their classical references.	*describe*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Principle and Processing of Uparasa Dravya

12 obj · 28 h · Application / Adaptation

1.1 Uparasa Dravyas mentioned in various texts
1.2 Characterization of Uparasa Dravyas
1.3 Pharmaceutical processes of Uparasa Dravya
1.4 Pharmacology and clinical application of Uparasa Dravya

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 4			
Describe he different Uparasa Dravyas mentioned in the classics.	*describe*	Conceptual ◦	Lecture 1h
Discuss the controversial drugs under Uparasa category.	*discuss*	Conceptual ◦	Lecture 1h
Discuss the mineralogical characterization of Uparasa.	*discuss*	Conceptual ◦	Lecture 1h
Discuss research updates on Uparasa drugs.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Recognise the therapeutic applications and toxico-kinetics of Uparasa Dravyas. — teach at level 1, the verb *recognise* is what will be asked	*recognise*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 5			
Demonstrate the authentication of Uparasa Dravyas.	*demonstrate*	Procedural	Practical 2h
Demonstrate Shodhana, Marana and other pharmaceutical procedures for Uparasa.	*demonstrate*	Procedural	Practical 4h
Demonstrate Quality Control testing for the products of Uparasa Dravya.	*demonstrate*	Procedural	Practical 4h
Perform Quality Control tests for the products of Uparasa Dravya.	*perform*	Procedural	Experiential 3h
Prepare formulations of Uparasa Dravya.	*prepare*	Procedural	Experiential 5h
DOING — LEVEL 3 · MECHANISM — 1			
Review articles on the toxicity and ADRs of Haratala and Manahshila and interpret.	*review*	Procedural	Experiential 2h
DOING — LEVEL 4 · ADAPTATION — 1			
Identify and authenticate Uparasa Dravyas.	*identify*	Procedural	Experiential 3h

Unit 2 · Principle and Processing of Sadharana Rasa

12 obj · 28 h · Evaluation / Adaptation

2.1 Sadharana rasa Dravyas mentioned in various texts
2.2 Characterization of Sadharana Rasa Dravyas
2.3 Pharmaceutical processes of Sadharana rasa Dravya
2.4 Pharmacology and clinical application of Sadharana rasa Dravya

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe and critically discuss the different Sadharana Rasa Dravyas mentioned in the classics.	*describe*	Conceptual ◦	Lecture 1h
Recognise the therapeutic applications and toxico-kinetics of Sadharana Rasa Dravyas. — teach at level 1, the verb *recognise* is what will be asked	*recognise*	Factual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Analyze the controversial drugs under Sadharana Rasa category. — teach at level 4, the verb *analyze* is what will be asked	*analyze*	Conceptual ◦	Lecture 1h
Discuss the mineralogical characterization of Sadharana Rasa. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Discuss research updates on Sadharana Rasa drugs. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 5			
Demonstrate authentication of Sadharana Rasa Dravya.	*demonstrate*	Procedural	Practical 2h
Demonstratie Shodhana, Marana and other pharmaceutical procedures for Sadharana.	*demonstrate*	Procedural	Practical 4h
Demonstrate Quality Control testing for the products of Sadharana Rasa Dravya.	*demonstrate*	Procedural	Practical 4h
Identify and authenticate Sadharana Rasa Dravyas.	*identify*	Procedural	Experiential 3h
Assemble the utility of Sadharana Dravyas in the clinical practice through a Field Survey of manufacturers and clinicians.	*assemble*	Procedural	Experiential 4h
DOING — LEVEL 4 · ADAPTATION — 2			
Perform Quality Control tests for the products of Sadharana Rasa Dravya.	*perform*	Procedural	Experiential 3h
Prepare and present the formulations of Sadharana Rasa Dravya.	*prepare*	Procedural	Experiential 3h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 6 practical, 8 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 6 · SEMESTER 5 · PAPER 1

Critical appraisal of Dhatu and Updhatu

Rasashastra (Iatrochemistry) — Principles and Practices

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	2	10	14	50
60 notional hours	24 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

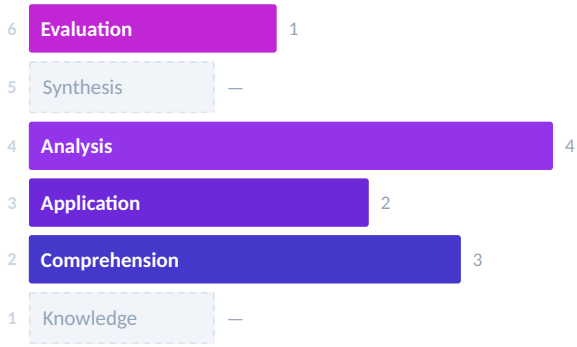
WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	8	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe classical references, properties, and classifications of Dhatu and Upadhatu.	*describe*	L2 Comprehension / Understand
2	Perform Shodhana, Marana, and other pharmaceutical procedures specific to Dhatu and Upadhatu Dravyas.	*perform*	L3 Application / Apply
3	Analyze physico-chemical characteristics of processed Dhatu and Upadhatu with scientific and classical correlation.	*analyze*	L4 Analysis / Analyse
4	Evaluate toxicological and therapeutic properties of commonly used Dhatu and Upadhatu formulations.	*evaluate*	L6 Evaluation / Evaluate

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Principle and Processing of Dhatu / Metals

12 obj · 28 h · Analysis / Mechanism

1.1 Dhatus mentioned in Rasa classics
1.2 Classification and characterization of metals
1.3 Pharmaceutical processing (Shodhana, Marana, etc.)
1.4 Therapeutic uses, toxicity, and safety of metallic preparations

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe classical Dhatu Dravyas.	*describe*	Conceptual °	Lecture 1h
Discuss the controversial metals in Ayurveda texts.	*discuss*	Conceptual °	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Identify therapeutic applications and toxico-kinetics of Dhatu. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual °	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2			
Characterize Dhatu Dravyas mineralogically and pharmaceutically.	*characterize*	—	Lecture 1h
Compile and interpret research and regulatory updates on metal-based Ayurvedic drugs.	*compile*	Conceptual °	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 6			
Demonstrate the authentication of Dhatu Dravya.	*demonstrate*	Procedural	Practical 2h
Perform pharmaceutical processing (Shodhana & Marana) of metals.	*perform*	Procedural	Practical 4h
Demonstrate Quality control of metallic preparations.	*demonstrate*	Procedural	Practical 4h
Perform Quality assessment of Dhatu Bhasma.	*perform*	Procedural	Experiential 3h
Prepare formuations using Dhatu Bhasma.	*prepare*	Procedural	Experiential 5h
Identify and analyze Adverse drug reactions (ADRs) due to metals.	*identify*	Procedural	Experiential 2h
DOING — LEVEL 3 · MECHANISM — 1			
Identify and document classical Dhatus.	*identify*	Procedural	Experiential 3h

Unit 2 · Principle and Processing of Upadhatu

12 obj · 28 h · Evaluation / Adaptation

2.1 Classical analysis of Upadhatu

2.2 Classification, sources, and characteristics of Upadhatu

2.3 Pharmaceutical processing (Shodhana, Marana, etc.)of Upadhatu

2.4 Pharmacological actions and clinical application of Upadhatu

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe and classify Upadhatu from Rasashastra references.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss controversial and newly interpreted Upadhatu. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2			
Analyze Upadhatu mineralogically and pharmaceutically.	*analyze*	Conceptual ◦	Lecture 1h
Review and interpret findings of research and current practices involving Upadhatu.	*review*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Recognize therapeutic applications and possible ADRs of Upadhatu. — teach at level 1, the verb *recognize* is what will be asked	*recognize*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 6			
Demonstrate the Identification of Upadhatu samples.	*demonstrate*	Procedural	Practical 2h
Demonstrate Shodhana and Marana of Upadhatu.	*demonstrate*	Procedural	Practical 4h
Demonstrate the Quality control tests for sub-metallic formulations.	*demonstrate*	Procedural	Practical 4h
Identify the Upadhatu samples.	*identify*	Procedural	Experiential 3h
Perform Quality Control tests for the products of Upadhatu.	*perform*	Procedural	Experiential 3h
Analyze and interpret findings of the utility of Dhatu and Upadhatu in clinical practice through a Field survey with Manufacturers or Vaidyas.	*analyze*	Procedural	Experiential 4h
DOING — LEVEL 4 · ADAPTATION — 1			
Prepare Upadhatu formulations.	*prepare*	Procedural	Experiential 3h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 6 practical, 8 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 13 · SEMESTER 5 · PAPER 2

Krutanna (Ayurvedic Food Preparations)

Exposition of Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	4	5	10	50
60 notional hours	15 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	5	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Classification of food substances and Formulations

3 obj · 12 h · Comprehension / Guided response

1.1 Ayurvedic dietary principles and Ahara Dravya Varga (Groups of food substance)

1.2 Aahara -mahabhaishajya: Significance of food substances used as medicine

1.3 Unique classical formulations prepared using Food substances belonging to various groups and their utility in treatment

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe Ayurvedic dietary principles and the significance of food substances used as medicine.	*describe*	Conceptual	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the method of preparation of distinctive formulations using Aahara Dravya from different groups (Vargas).	*demonstrate*	Procedural	Practical 4h
Prepare distinctive formulations using Aahara Dravya from different groups (Vargas).	*prepare*	Procedural	Experiential 6h

Unit 2 · Customized food for treatment

6 obj · 24 h · Comprehension / Guided response

2.1 Disease-specific Ayurvedic Aahara- Therapeutic utility

2.2 Samsarjana krama- Manda, Peya, Vilepi, Yavagu, Yusha, Odana, Krushara, Mansarasa- Process validation Protocol for all formulations

2.3 Personalised prescription of Aushada Siddha Aahara for Shamana and during panchakarma

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe disease-specific diet formulations.	*describe*	Conceptual ◦	Lecture 2h
Recite paribhasha and describe the importance of Yusha, Yavagu, Odana, and Mamsarasa. — teach at level 1, the verb *recite* is what will be asked	*recite*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Demonstrate preparation of therapeutic diet formulations.	*demonstrate*	Procedural	Practical 4h
Identify and prepare disease specific Ayurvedic Ahara.	*identify*	Procedural	Experiential 6h
Demonstrate the method of preparation of various Aahara Kalpana.	*demonstrate*	Procedural	Practical 4h
Develop the disease-specific Personalised prescription of Aushadhi Siddha Ahara.	*develop*	Procedural	Experiential 6h

Unit 3 · Ayurvedic Food Industry

3 obj · 10 h · Knowledge / Guided response

3.1 Market survey about Ayurvedic food products and research updates

3.2 Global needs and designing of modified food formulations

3.3 Semi-processed food/ semi-cooked food, RTC- ready to cook / RTD-Ready to drink Foods and drinks

3.4 Regulation Standards for Ayurvedic Food Preparation (Ayurveda Aahara Regulations- 2021), FSSAI Regulations

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Describe the regulation standards for Ayurvedic food preparations- FSSAI (Ayurveda Aahara regulations). — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the preparation of novel Ayurvedic food products.	*demonstrate*	Procedural	Practical 4h
Collect the data and interpret the findings through a market survey to explore the available Ayurvedic food products and assess the current trends in the Ayurvedic food industry.	*collect*	Procedural	Experiential 4h

Unit 4 · Food processing technology traditional and modern methods

3 obj · 10 h · Analysis / Adaptation

4.1 Traditional Methods- Steaming, Boiling, Drying, Salting, Smoking, Frying,

4.2 Modern Methods- Heating, Freeze Drying, Vacuum Drying, Canning, Juicing, Fermentation, Modified atmosphere packaging, Pasteurisation, Smoking, Pulse electric

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Differentiate between classical and contemporary food processing techniques and support the current and emerging trends in the Ayurvedic food industry.	*differentiate*	Procedural ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate various food processing techniques used in current Ayurvedic food Industries.	*demonstrate*	Procedural	Practical 4h
DOING — LEVEL 4 · ADAPTATION — 1			
Analyze Functional Food and Modified Food Formulations in Ayurvedic Food Industry.	*analyze*	Procedural	Experiential 4h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 5 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 14 · SEMESTER 5 · PAPER 2

Ayurvedic Formulations for Holistic Wellbeing, Vitality and Body Care

Exposition of Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	2	4	7	50
60 notional hours	11 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	4	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	3	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Rasayana-Personalized Rejuvenation Formulations

5 obj · 24 h · Comprehension / Mechanism

1.1 Key aspects of Rasayana Formulations

1.2 General Rasayana and Organ-specific/Target-specific Rasayana- Varnya, Tvachya, Kanthya, Swarya, Keshya, Hrudya, Medhya- Formulations

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe the mode of action of Rasayana Formulations.	*describe*	Conceptual ◦	Lecture 2h
Describe classical Rasayana and Vajikarana Formulations.	*describe*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the role of Ayurvedic Rasayana formulations in the maintenance of health and Vajikarana Formulations for better progeny(Suprajanana).	*demonstrate*	Procedural	Experiential 5h
Select appropriate Ayurvedic Formulations for the fitness of specific body parts and analyze their role in the maintenance of health.	*select*	Conceptual + Procedural ◦	Experiential 7h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate the preparation method of Rasayana Drugs following validated SOP.	*demonstrate*	Procedural	Practical 8h

Unit 2 · Ayurvedic based cosmetic preparation

6 obj · 32 h · Evaluation / Mechanism

2.1 Body/ Hair/ Lip/ Oral/Skin/ Foot care products- Preparation methods and application in Ayurvedic

2.2 Preparation process of Lip balm, lipsticks, shampoos, cold cream and vanishing cream, toothpaste, hair dyes, Sunscreen lotion, Face pack, Mask, Toners, herbal soaps,

2.3 Current and Emerging trends – Ayurvastram, Ayurvedic footwear, Eye mask

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the role of Ayurvedic formulations in Cosmetic preparations.	*discuss*	Conceptual	Lecture 3h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Evaluate the Global need for body care products in the healthcare sector.	*evaluate*	Conceptual ◦	Lecture 3h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the preparation of Modified Cosmeceuticals.	*demonstrate*	Procedural	Practical 6h
Analyze different cosmetics available in the market.	*analyze*	Procedural	Experiential 7h
Design and prepare novel Ayurvedic cosmeceuticals.	*design*	Procedural	Experiential 7h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate preparation of Herbal Cosmetics.	*demonstrate*	Procedural	Practical 6h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 4 lecture, 3 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 21 · SEMESTER 5 · PAPER 3

Clinical application of Rasoushadhi according to Srotas and in selected challenging clinical conditions

Standardization and clinical Utility of Ayurvedic formulations

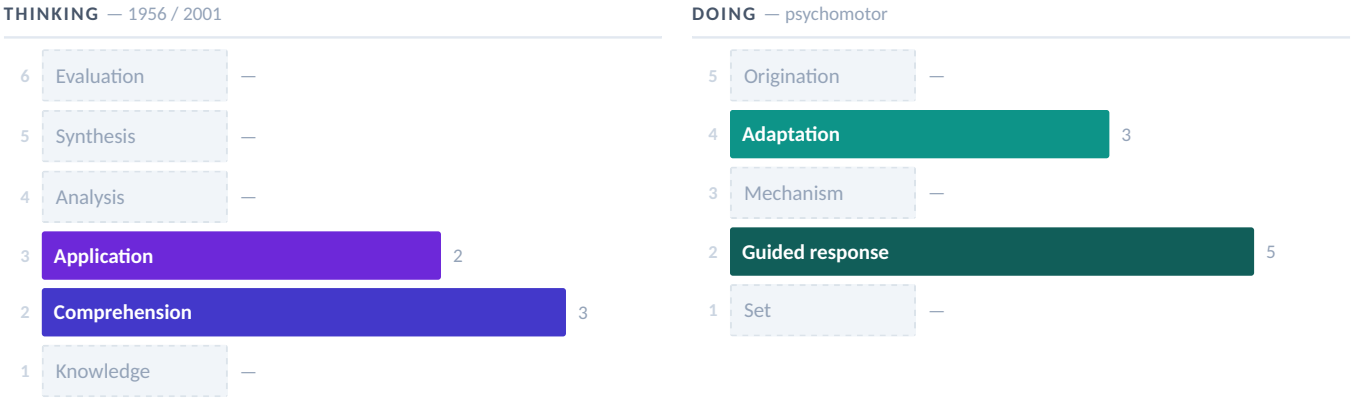
CREDITS	UNITS	THINKING	DOING	FORMATIVE
1	3	5	8	25
30 notional hours	13 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	5 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	10 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	13 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 2 hours carry the modular assessment, worth 25 marks, bringing the module to its 30 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Clinical application of Rasoushadhis specific to Srotas

4 obj · 8 h · Application / Adaptation

1.1 Concept and rationality of selection and application of Rasoushadhi according to Srotas

1.2 Clinical application of Rasouashdhi in disorders of different Srotas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss Srotas-specific Rasoushadhi, tailored Ayurvedic formulations for individual biological channels.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Enlist and discuss Rasoushadhi for different Srotas, explaining their therapeutic efficacy and mechanism of action. — teach at level 1, the verb *enlist* is what will be asked	*enlist*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the clinical application of Rasoushadhi in selected Srotas, illustrating their practical utility and therapeutic outcomes in addressing specific health conditions.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 4 · ADAPTATION — 1			
Illustrate the therapeutic application of Rasoushadhi in different Srotas.	*illustrate*	Procedural	Experiential 3h

Unit 2 · Clinical application of Rasoushadhi in emergency conditions

4 obj · 9 h · Comprehension / Adaptation

2.1 Role of Rasoushadhis in Managing Medical Emergencies

2.2 Specific Rasa Yogas among Chaturvidha Rasayana kalpas for Managing Medical Emergencies

2.3 Application of Pottali Kalpa in Emergency Management

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss the unique properties of Rasoushadhi, enabling its use in clinical emergencies.	*discuss*	Conceptual ◦	Lecture 1h
Enlist specific Rasa Yogas for managing medical emergencies and discuss their rationality, highlighting the therapeutic mechanisms and clinical applications. — teach at level 1, the verb *enlist* is what will be asked	*enlist*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the clinical applicability of Rasoushadhi in Medical Emergencies.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 4 · ADAPTATION — 1			
Apply the appropriate Rasa Oushadhi in a alloted /selected Medical Emergency conditions.	*apply*	Procedural	Experiential 4h

Unit 3 · Rasoushadhi in specific, clinically challenging conditions

5 obj · 11 h · Application / Adaptation

3.1 Rational application of Rasoushadhi in cancer

3.2 Application of Rasoushadhi in neurodevelopmental conditions

3.3 Utility of Rasoushadhi in neurodegenerative conditions

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the role of Rasoushadhi in clinically challenging conditions like cancers neurodevelopmental and neurodegenerative disorders. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the role of Rasoushadhi in clinically challenging conditions.	*demonstrate*	Procedural	Practical 2h
Demonstrate the role of Rasoushadhi as Rasayana in clinically challenging conditions.	*demonstrate*	Procedural	Practical 2h
Evalaute the role of Rasoushadhi in clinically challenging conditions such as Cancer.	*evaluate*	Conceptual + Procedural ◦	Experiential 3h
DOING — LEVEL 4 · ADAPTATION — 1			
Explore the role of Rasoushadhi in clinically challenging conditions such as neurodevelopmental and neurodegenerative diseases.	*explore*	Conceptual + Procedural ◦	Experiential 3h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 4 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 22 · SEMESTER 5 · PAPER 3

Yogas from different Kalpanas

Standardization and clinical Utility of Ayurvedic formulations

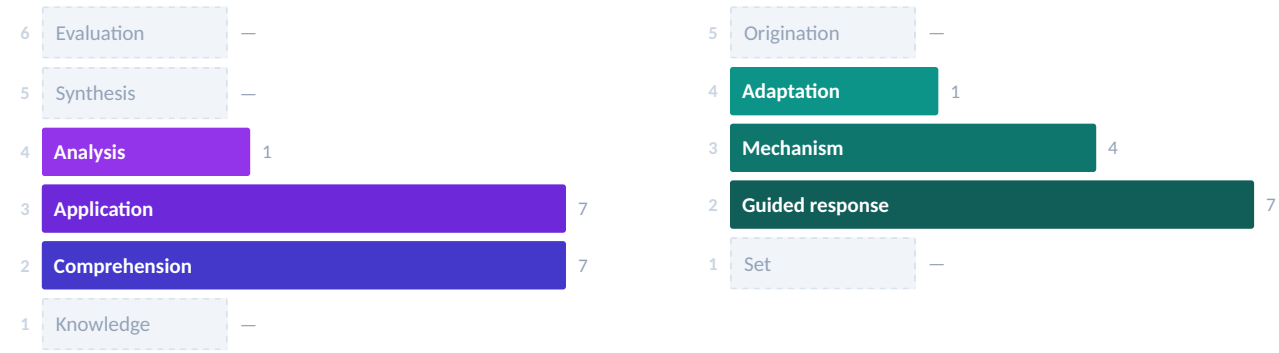
CREDITS	UNITS	THINKING	DOING	FORMATIVE
3	5	15	12	75
90 notional hours	27 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	15 h	15	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	30 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	39 h	6	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 6 hours carry the modular assessment, worth 75 marks, bringing the module to its 90 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · SOP of Vati, Gutika, Guggulu and Varti Kalpas and their therapeutic applications

6 obj · 20 h · Application / Mechanism

1.1 SOP of selected Vati preparation including packing,
1.2 SOP regarding preparation of selected formulations from Guggulu Kalpana, including packing, storage, and therapeutic applications
1.3 Types of Varti and their SOP
1.4 Clinical applications of different Varti
1.5 Suppositories and pharmaceutical modifications of Ayurvedic suppositories

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss modern aspects of solid dosage forms.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Describe the therapeutic significance of vati kalpana. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
Summarize the therapeutic significance of Guggulu Kalpa and Varti Kalpana. — teach at level 2, the verb *summarize* is what will be asked	*summarize*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the preparation of Vati, Guggulu Kalpa, and their modified modern dosage forms.	*demonstrate*	Procedural	Practical 5h
Design an S.O.P for the preparation of Ayurvedic suppository specifying its clinical utility based on Varti Kalpana.	*design*	Procedural	Practical 4h
DOING — LEVEL 3 · MECHANISM — 1			
Analyse the role of Gugulu Kalpana in various clinical conditions by selecting a specific Gulggulu Yoga.	*analyse*	Conceptual + Procedural ◦	Experiential 8h

Unit 2 · SOP and Clinical applications of Avaleha and Khanda kalpana

5 obj · 16 h · Application / Mechanism

2.1 SOP of Avaleha preparation with suitable Leha yogas
2.2 Therapeutic application of different varieties of Avaleha yogas
2.3 SOP of Khanda kalpana with suitable Khanda Yogas
2.4 Therapeutic application of different varieties of Khanda yogas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss modern semisolid dosage forms.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss the therapeutic significance of Avaleha kalpana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
Discuss therapeutic significance of Paka and Khanda kalpana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the role of pharmaceutical additives and Samyak Paka of Avaleha.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 1			
Analyse the role of selected leha yoga in different clinical conditions.	*analyse*	Conceptual + Procedural ◦	Experiential 8h

Unit 3 · SOP of Sneha Kalpana and its clinical significance

5 obj · 17 h · Application / Adaptation

3.1 Different categories of Sneha with special reference to therapeutic application
3.2 Relevance of Avarthita Taila, Taila murchana and Taila Paka with special reference to critical endpoints
3.3 SOP of Sneha kalpana in selected formulations

3.4 Therapeutic application of selected Ghrita yogas and Taila yogas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Identify the similarities of Sneha Kalpa with liposomal drug delivery system. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss the concept of Sneha Kalpana with special reference to critical end points (Sneha Paka) and its pharmaceutical and therapeutical significance. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
Discuss and compare the significance of Murchana of Katu Taila, Tila Taila, Eranda Taila and Ghrita with special reference to analytical profile and pharmacotherapeutics. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate S.O.P for a selected Sneha Kalpa which is widely used and determine its analytical profile.	*demonstrate*	Procedural	Practical 6h
DOING — LEVEL 4 · ADAPTATION — 1			
Analyze different stages of Sneha Paka.	*analyze*	Procedural	Experiential 8h

Unit 4 · SOP and clinical applications of Arka and Dravakam in various disorders

5 obj · 14 h · Analysis / Mechanism

4.1 SOP for preparing selected Arkas
4.2 Analytical profiling of various yogas from Arka Kalpana
4.3 SOP of preparing various Dravakas
4.4 Analytical Profiling of selected formulations of Dravaka
4.5 Therapeutic applications and method of administration of Dravaka

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Desribe therapeutic significance of Arka Kalpana and Dravaka.	*describe*	Conceptual	Lecture 1h
Discuss different distillation techniques used in pharmacy practice.	*discuss*	Procedural ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Elaborate Dravaka Kalpa with a critical correlation with acidic and alkaline preparations. — teach at level 2, the verb *elaborate* is what will be asked	*elaborate*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate S.O.P for the preparation of a selected Arka and Dravaka and create their analytical profile.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 1			
Illusrate the clinical application of selected Arka/Dravaka Kalpa.	*illustrate*	Procedural	Experiential 6h

Unit 5 · SOP and therapeutic applications of Sandhana Kalpana

6 obj · 17 h · Application / Mechanism

5.1 Different and novel fermentation techniques
5.2 SOP of preparing different types of Formulations in Sandhana Kalpana
5.3 Therapeutic applications of selected yogas from Sandhana Kalpana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe chemistry behind classical fermentation techniques.	*describe*	Procedural	Lecture 1h
Discuss industrial fermentation techniques.	*discuss*	Procedural	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Elaborate therapeutic significance of Sandhana kalpana. — teach at level 2, the verb *elaborate* is what will be asked	*elaborate*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate S.O.P for preparing a selected sandhana Yoga and determine the analytical standards.	*demonstrate*	Procedural	Practical 5h
Record the dose, duration, indications, and clinical outcome of the various formulation i.e. Sandhana Kalpanas in the prescribed patients.	*record*	Procedural	Experiential 4h
DOING — LEVEL 3 · MECHANISM — 1			
Perform Comparative analysis of Asava Arishta Samples having different percentage and types of sugar.	*perform*	Procedural	Experiential 5h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 15 lecture, 6 practical, 6 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 29 · SEMESTER 5 · PAPER 4

Pharmacy Practice

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	3	10	8	50
60 notional hours	18 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the Hospital Pharmacy and the Community Pharmacy with their organization	*describe*	L2 Comprehension / Understand
2	Discuss Therapeutic Drug Monitoring	*discuss*	L2 Comprehension / Understand
3	Discuss the Pharmacy and Therapeutic Committee	*discuss*	L2 Comprehension / Understand
4	Enlist the emergency drug list	*enlist*	L1 Knowledge / Remember

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Hospital Pharmacy and Community Pharmacy

7 obj · 26 h · Comprehension / Mechanism

1.1 Hospital and its organization
1.2 Definition, Classification of Hospital
1.3 Hospital pharmacy and its organization
1.4 Functions of hospital pharmacy
1.5 Community Pharmacy
1.6 Organization and structure of retail and wholesale drug stores

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Define and classify Hospital Pharmacy, and discuss its organization. — teach at level 1, the verb *define* is what will be asked	*define*	Factual ◦	Lecture 1h
Describe Community Pharmacy.	*describe*	Conceptual ◦	Lecture 1h
Discuss the Organization and structure of retail and wholesale drug stores.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate Drug Information and Case Analysis.	*demonstrate*	Procedural	Practical 5h
Demonstrate Prescription Handling and Dispensing.	*demonstrate*	Procedural	Practical 5h
Identify Adverse drug reaction in selected case.	*identify*	Procedural	Experiential 7h
DOING — LEVEL 3 · MECHANISM — 1			
Appraise about Basic Health Screening at community setup.	*appraise*	Procedural	Experiential 6h

Unit 2 · Therapeutic Drug Monitoring

7 obj · 17 h · Evaluation / Adaptation

2.1 Drug distribution system in a hospital
2.2 Dispensing of drugs to inpatients, types of drug distribution systems, charging policy and labelling, dispensing of drugs to ambulatory patients, and dispensing of
2.3 Hospital formulary
2.4 Therapeutic drug monitoring
2.5 Factors to be considered during the Therapeutic Drug Monitoring
2.6 Medication adherence
2.7 Patient medication history interview
2.8 Community pharmacy management
2.9 Financial, materials, staff, and infrastructure requirements

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss about Hospital formulary and Therapeutic drug monitoring.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 3			
Describe the Drug distribution system in a hospital. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
Discuss Dispensing of drugs to inpatients, types of drug distribution systems, charging policy, and labelling, dispensing of drugs to ambulatory patients, and dispensing of controlled drugs. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
Analyse Financial, materials, staff, and infrastructure requirements. — teach at level 4, the verb *analyse* is what will be asked	*analyse*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Evaluate Medication Adherence.	*evaluate*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Perform Data Interpretation & Pharmacokinetic Calculations.	*perform*	Procedural	Practical 5h
DOING — LEVEL 4 · ADAPTATION — 1			
Discuss Multidisciplinary Integration for Therapeutic Drug Monitoring.	*discuss*	Procedural	Experiential 7h

Unit 3 · Pharmacy and therapeutic committee

4 obj · 13 h · Comprehension / Origination

3.1 Introduction to Pharmacy and Therapeutic Committee

3.2 Drug information services

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe Pharmacy and Therapeutic Committee along with Drug Information Services.	*describe*	Conceptual ◦	Lecture 1h
Enlist emergency drugs used in pharmacy. — teach at level 1, the verb *enlist* is what will be asked	*enlist*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate Formulary Review and Management.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 5 · ORIGINATION — 1			
Appraise and. Develop Standard Therapeutic Guidelines.	*appraise*	Procedural	Experiential 6h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Origination on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 4 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Origination.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 30 · SEMESTER 5 · PAPER 4

Pharmaceutical Microbiology

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

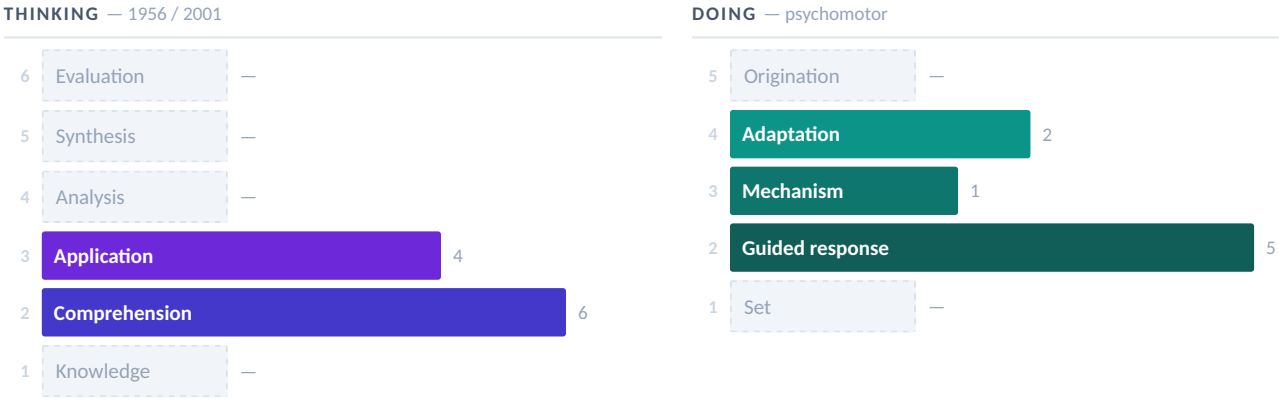
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	3	10	8	50
60 notional hours	18 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Introduction to microbes with applications of sterilization

5 obj · 12 h · Application / Guided response

1.1 Identification of bacteria using staining techniques (simple, Gram's & Acid-fast

1.2 Study of principles, procedures, merits, demerits, and applications of physical, chemical, gaseous, radiation, and mechanical methods of sterilization

1.3 Evaluation of the efficiency of sterilization methods

1.4 Equipment employed in large-scale sterilization and Sterility indicators

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Define and classify microorganisms (microbes) and explain their general characteristics. — teach at level 1, the verb *define* is what will be asked	*define*	Factual ◦	Lecture 1h
Describe the concept of sterilization, its importance, and differentiate various sterilization methods.	*describe*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the beneficial and harmful roles of microbes in health, industry, and the environment. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate dry heat sterilization method.	*demonstrate*	Procedural	Practical 3h
Conduct dry heat sterilization.	*conduct*	Procedural	Experiential 6h

Unit 2 · Sources of contamination

6 obj · 12 h · Application / Adaptation

2.1 Designing an aseptic area

2.2 Study of different sources of contamination in an aseptic area and methods of prevention, clean area classification

2.3 Principles and methods of different microbiological assay methods

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Define contamination and explain its significance in aseptic and cleanroom environments. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual ◦	Lecture 1h
Discuss the impact of contamination and describe methods used to minimize or prevent contamination.	*discuss*	Procedural ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Recognise common sources of contamination in pharmaceutical and microbiological settings and discuss each source contributes to microbial and particulate contamination. — teach at level 1, the verb *recognise* is what will be asked	*recognise*	Factual ◦	Lecture 1h
Discuss the principles and methods of different microbiological assay methods. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual + Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the Aseptic Technique Practice for the transfer of microbes.	*demonstrate*	Procedural	Practical 2h
DOING — LEVEL 4 · ADAPTATION — 1			
Perform Simulation for Aseptic Practices.	*perform*	Procedural	Experiential 6h

Unit 3 · Types of spoilage and use of antimicrobial agents

7 obj · 32 h · Application / Adaptation

3.1 Types of spoilage, factors affecting the microbial spoilage of pharmaceutical products,

3.2 Preservation of pharmaceutical products using antimicrobial agents, evaluation of microbial stability of formulations

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Define spoilage, explain its significance in pharmaceuticals, food, and cosmetics, and discuss the factors that contribute to spoilage. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual ◦	Lecture 1h
Describe the signs of spoilage.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Classify different types of antimicrobial agents (e.g., preservatives, disinfectants antibiotics) based on their mode of action and explain their antimicrobial mechanisms. — teach at level 2, the verb *classify* is what will be asked	*classify*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate Personnel Contamination Assessment.	*demonstrate*	Procedural	Practical 7h
Demonstrate Different Types of Spoilage.	*demonstrate*	Procedural	Practical 8h
DOING — LEVEL 3 · MECHANISM — 1			
Conduct Real-life case studies on spoilage outbreaks and preservative failures.	*conduct*	Procedural	Experiential 7h
DOING — LEVEL 4 · ADAPTATION — 1			
Observe and detect Spoilage of Pharmaceutical Product in Real Time.	*observe*	Procedural	Experiential 7h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 4 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 7 · SEMESTER 6 · PAPER 1

Sudha Varga, Sikata Varga, Visha Varga Dravyas

Rasashastra (Iatrochemistry) — Principles and Practices

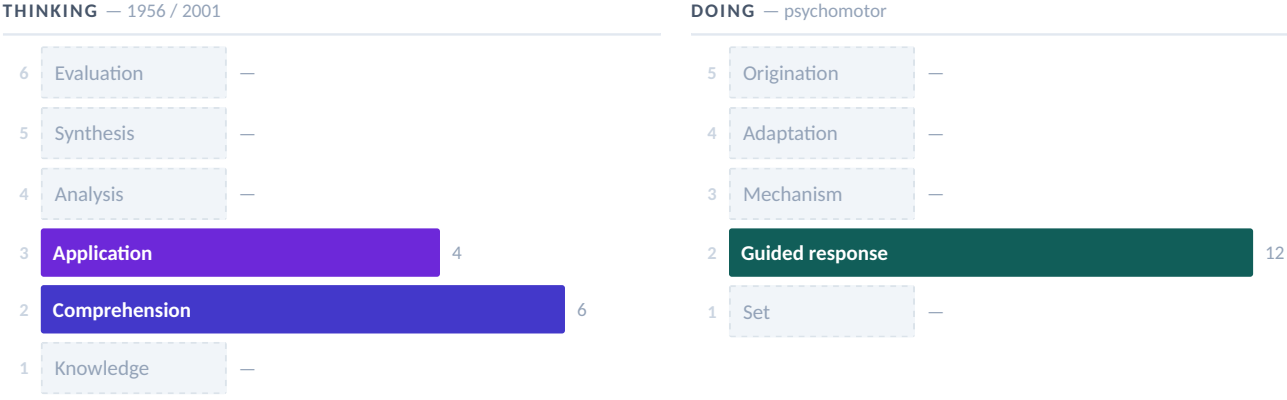
CREDITS	UNITS	THINKING	DOING	FORMATIVE
3	3	10	12	75
90 notional hours	22 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	15 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	30 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	39 h	6	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 6 hours carry the modular assessment, worth 75 marks, bringing the module to its 90 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Identify and classify the drugs of Sudha Varga, Sikata Varga, and Visha Varga.	*identify*	L1 Knowledge / Remember
2	Perform pharmaceutical processes, including Shodhana and Marana, for Sudha Varga and Sikata Varga drugs, and Shodhana for Visha Varga drugs.	*perform*	L3 Application / Apply
3	Describe the physico-chemical properties of Sudha Varga, Sikata Varga, and Visha Varga drugs.	*describe*	L2 Comprehension / Understand
4	Discuss the pharmacology and therapeutic applications of Sudha Varga, Sikata Varga, and Visha Varga drugs.	*discuss*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Characterization, Pharmaceutical Processing, and Clinical Applications of Sudha Varga Dravya

7 obj · 28 h · Application / Guided response

1.1 Classification and Identification of Sudha
1.2 Pharmaceutical Processing of Sudha Varga Dravya
1.3 Quality Control Methods for Sudha Varga Dravya
1.4 Pharmacology, Therapeutic Applications, and Research Updates on Sudha Varga Dravya

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the list of Sudha varga Dravya mentioned in various classics and their mineralogical correlation.	*discuss*	Factual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss the pharmacological and therapeutic applications of Sudha Varga Dravya. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 2h
Discuss research updates on characterization, pharmaceutical processing, QC, and pharmacotherapeutics of Sudhavarga Dravya. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Demonstrate Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya.	*demonstrate*	Procedural	Practical 6h
Demonstrate the quality testing procedures for Bhasma/Pishti of Sudha vargeeya Dravya.	*demonstrate*	Procedural	Practical 4h
Perform Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya.	*perform*	Procedural	Experiential 7h
Perform quality testing procedures for Bhasma/Pishti of Sudha Vargeeya Dravya.	*perform*	Procedural	Experiential 6h

Unit 2 · Characterization, pharmaceutical processing, and clinical applications of Sikata Varga Dravya

7 obj · 28 h · Application / Guided response

2.1 Identification of Sikata Varga dravyas
2.2 Shodhana, marana, and other pharmaceutical processes of Sikata varga Dravya as per the SOP laid down
2.3 Quality control methods for the drugs of Sikata Varga Dravya
2.4 Pharmacology, therapeutic application, and research updates on Sikata Varga Dravya)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss the list of Sikata varga Dravya mentioned in various classics and their mineralogical correlation.	*discuss*	Factual ◦	Lecture 1h
Discuss research updates on characterization, pharmaceutical processing, QC, and pharmacotherapeutics of Sikata Varga Dravya.	*discuss*	Conceptual ◦	Lecture 2h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the pharmacology and therapeutic applications of Sikata Varga Dravya. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Demonstrate Shodhana, marana, and Pishti Nirmana of Sikata varga Dravya.	*demonstrate*	Procedural	Practical 6h
Demonstrate quality testing procedures for Bhasma/Pishti of Sikata vargeeya Dravya.	*demonstrate*	Procedural	Practical 4h
Perform Shodhana, marana and Pishti Nirmana of Sikata varga Dravya.	*perform*	Procedural	Experiential 7h
Perform quality testing procedures for Bhasma/Pishti of Sikata vargeeya Dravya.	*perform*	Procedural	Experiential 6h

Unit 3 · Pharmaceutical processing, phytopharmacology and clinical applications of Visha Varga dravya

8 obj · 28 h · Application / Guided response

3.1 Drugs of Visha and Upavisha categories
3.2 Shodhana of Visha-Upavisha Dravya based on different references
3.3 Phytochemical changes after the process of Shodhana
3.4 Clinical application and therapeutics of Visha Varga

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Discuss and compare the drugs of the Visha and Upavisha categories from different classics.	*discuss*	Conceptual ◦	Lecture 1h
Describe the identification of herbal, animal, and mineral drugs mentioned under the Visha-Upavisha category based on organoleptic and physico-chemical characters.	*describe*	Conceptual ◦	Lecture 1h
Discuss the phytopharmacology of herbal drugs mentioned under the Visha-Upavisha category.	*discuss*	Conceptual ◦	Lecture 2h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe formulations and therapeutic applications of Visha Dravya. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Demonstrate the Shodhana Process of Visha Dravya.	*demonstrate*	Procedural	Practical 6h
Demonstrate the analytical testing process for Shodhita Visha dravya.	*demonstrate*	Procedural	Practical 4h
Perform Shodhana of Visha Dravyas as per different classics.	*perform*	Procedural	Experiential 7h
Perform analytical testing process for Shodhita Visha dravya.	*perform*	Procedural	Experiential 6h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Guided response on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 6 practical, 6 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Guided response.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 8 · SEMESTER 6 · PAPER 1

Appraisal of Ratna and Uparatna and therapeutic significance of Pishti and Druti Kalpana

Rasashastra (Iatrochemistry) — Principles and Practices

CREDITS	UNITS	THINKING	DOING	FORMATIVE
1	2	5	8	25
30 notional hours	13 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	5 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	10 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	13 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 2 hours carry the modular assessment, worth 25 marks, bringing the module to its 30 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Identify and explore precious and semiprecious stones classified under Ratna and Uparatna Varga.	*identify*	L1 Knowledge / Remember
2	Prepare Pishti Kalpa adhering to Standard Operating Procedures (SOPs).	*prepare*	L3 Application / Apply
3	Prepare and pharmaceutically modify Gandhaka Druti and Gandhaka Taila Kalpas.	*prepare*	L3 Application / Apply
4	Analyze the pharmacology and therapeutic applications of selected Pishti, Gandhaka Druti, and Gandhaka Taila Kalpas.	*analyze*	L4 Analysis / Analyse

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Mineralogy, Crystal chemistry,and Pharmacotherapeutics of Ratna-Uparatna Varga

7 obj · 15 h · Comprehension / Mechanism

1.1 Precious stones
1.2 Uparatna and semiprecious stones
1.3 Crystal chemistry and mineralogical properties of Ratna-Uparatna
1.4 Pharmacology and therapeutic applications of Ratna and Uparatnas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Discuss the characteristics of precious stones with special reference to the mineralogical identification of Ratnas.	*discuss*	Conceptual ◦	Lecture 1h
Describe different semiprecious stones mentioned in Rasashastra texts under the category of Uparatna.	*describe*	Conceptual ◦	Lecture 1h
Discuss crystal chemistry, pharmacology and therapeutic applications of Ratna and Uparatnas.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate mineralogical identification of precious and semiprecious stones under Ratna and Uparatna varga.	*demonstrate*	Procedural	Practical 3h
Demonstrate SOP of Pishti Kalpana of Ratna-Uparatna.	*demonstrate*	Procedural	Practical 3h
Identify the precious and semiprecious stones under Ratna and Uparatna varga using keys for mineralogical identification.	*identify*	Procedural	Experiential 3h
DOING — LEVEL 3 · MECHANISM — 1			
Identify substitutes for Ratnavarga and Uparatna varga drugs.	*identify*	Procedural	Experiential 3h

Unit 2 · SOP and therapeutic applications of Pishti, Gandhaka Druti and Gandhaka Taila kalpana

6 obj · 13 h · Application / Guided response

2.1 Pishti Kalpana: Concept, SOP, and Therapeutic
2.2 Classical concept of Druti
2.2 Gandhaka Druti and Gandhaka Taila Kalpana: Concept, SOP, and Therapeutic Applications
2.3 Pharmaceutical Modifications of Gandhaka Druti and Gandhaka Taila

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss the concept, significance, and therapeutic value of Pishti kalpas. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
Discuss the concept of Druti mentioned in Rasa classics and explore the therapeutic value of Gandhaka Druti and Gandhaka Taila. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Demonstrate different methods of Gandhaka Druti.	*demonstrate*	Procedural	Practical 2h
Demonstrate Pharmaceutical modification of Gandhaka Druti and Gandhaka Taila.	*demonstrate*	Procedural	Practical 2h
Prepare Pishti of Selected Ratna or Uparatna.	*prepare*	Procedural	Experiential 3h
Prepare Gandhaka Druti using different media.	*prepare*	Procedural	Experiential 4h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 4 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 15 · SEMESTER 6 · PAPER 2

Disease condition specific selection of formulation

Exposition of Ayurvedic Pharmaceutics

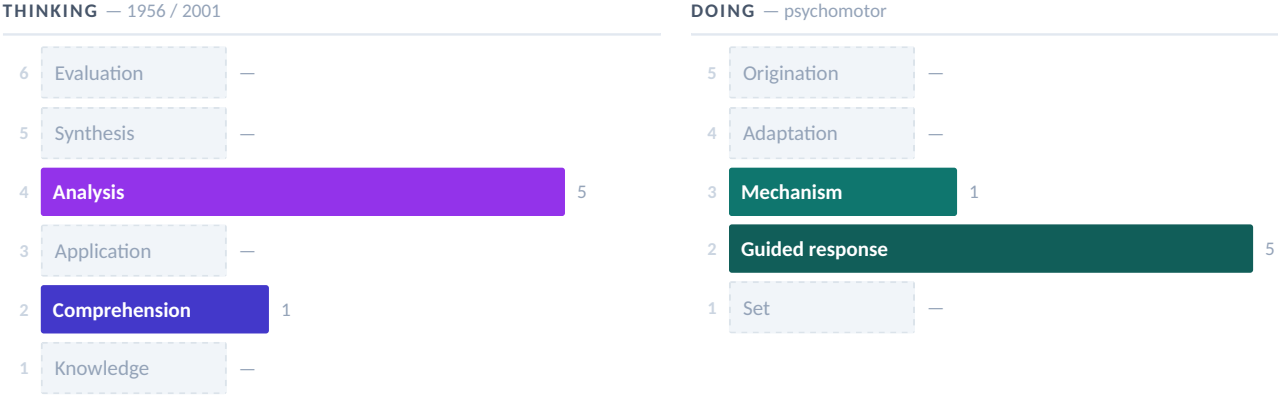
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	6	6	50
60 notional hours	12 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	6	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	3	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	3	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Application of Yukti Pramana in Formulation Selection

2 obj · 2 h · Analysis / —

1.1 Disease Category and Determination of multiple factors involved in diagnosis and

1.2 Strategic selection of drugs and dosage forms

1.3 Stage-specific selection (Rogavastha and Bhaishajya Kalpana)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2			
Appraise the rational relationship between Roga and Bhaishajya Kalpana based on classical references. — teach at level 6, the verb *appraise* is what will be asked	*appraise*	Conceptual	Lecture 1h
Analyse and justify the selection of formulation according to the Rogavastha (stage of disease).	*analyse*	Conceptual	Lecture 1h

Unit 2 · Selection of formulations for Speciality practice in Ayurveda

3 obj · 12 h · Analysis / Guided response

2.1 Speciality practice in Ayurveda

2.2 Alignment of formulations with the specialty branch of clinical Practice

2.3 Drugs with ideal guna, virya, vipaka, and of appropriate dosage forms for the patient and pathology

2.4 Evidence-based drugs and dosage form selection for specialty practice

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Describe different branches /specialty practices in Ayurveda and justify the need for specific dosage forms for effective clinical outcomes. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the essential formulations needed for the diverse branches of Ayurveda.	*demonstrate*	Procedural	Practical 5h
Identify different dosage forms used by local health practitioners. through filed based survey.	*identify*	Procedural	Practical 5h

Unit 3 · Routes of Drug Administration

2 obj · 12 h · Analysis / Mechanism

3.1 Classical and conventional concepts of Drug Administration

3.2 Routes of Drug Administration – Need and Relevance

3.3 Factors in the selection of drug and dosage form based on route of administration

3.4 Matching Dosage Forms to Pharmacological Needs by Selected Route

3.5 Properties of formulations and their effect on Pharmacodynamics

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Analyze the formulations and routes of drug administration, and justify the relevance of each route in pharmacology.	*analyze*	Conceptual ◦	Lecture 2h
DOING — LEVEL 3 · MECHANISM — 1			
Analyze the key characteristics of dosage forms that meet the needs of various routes of drug administration.	*analyze*	Procedural	Experiential 10h

Unit 4 · Designing formulations for Shodhana and Shamana Chikitsa

2 obj · 12 h · Analysis / Guided response

4.1 Ideal characteristics of drugs as per classics (Matravat Bheshaja lakshanani)

4.2 Principles of Shodhana chikitsa (Purificatory/detoxifying therapy) and actions of formulations used for shodhana - Langhana, Doshanirhana, srotahshodhana

4.3 Principles of Shamana chikitsa (palliative treatment) and actions of formulations used for Shamana- Deepana, Pachana, Stambhana, Rasayana, etc

4.4 Evidence-based dosage forms and formulations for common diseases

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Compare dosage forms and formulations used for shodhana and shamana chikitsa.	*compare*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the method of prepartion of frequently used formulation for Shodhana and Shamana Chikitsa of common diseases.	*demonstrate*	Procedural	Practical 10h

Unit 5 · Innovation in Ayurvedic Formulations

3 obj · 18 h · Comprehension / Guided response

5.1 Need and current trends in the industry of Ayurvedic Pharmaceutics

5.2 Innovative Formulations-Transdermal patches, Nebulizers, Eye masks, etc

5.3 Relevance and consequences of innovation done by the R&D department of Ayurvedic Pharmacies

5.4 Knowledge, Attitude, and Practice (KAP) studies on various aspects related to Ayurvedic Formulations available in the market

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss challenges associated with recent modifications and innovations in Ayurvedic formulations, specifically focusing on transdermal patches and nebulizers.	*discuss*	Conceptual	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the method of nnovative formulations that enhance well-being and redefine healthcare.	*demonstrate*	Procedural	Experiential 10h
Demonstrate the essential competencies needed to develop R&D department of Ayurvedic Pharmacy.	*demonstrate*	Procedural	Experiential 6h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 6 lecture, 3 practical, 3 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 16 · SEMESTER 6 · PAPER 2

Current trends in Ayurvedic Pharmaceutical Industry.

Exposition of Ayurvedic Pharmaceutics

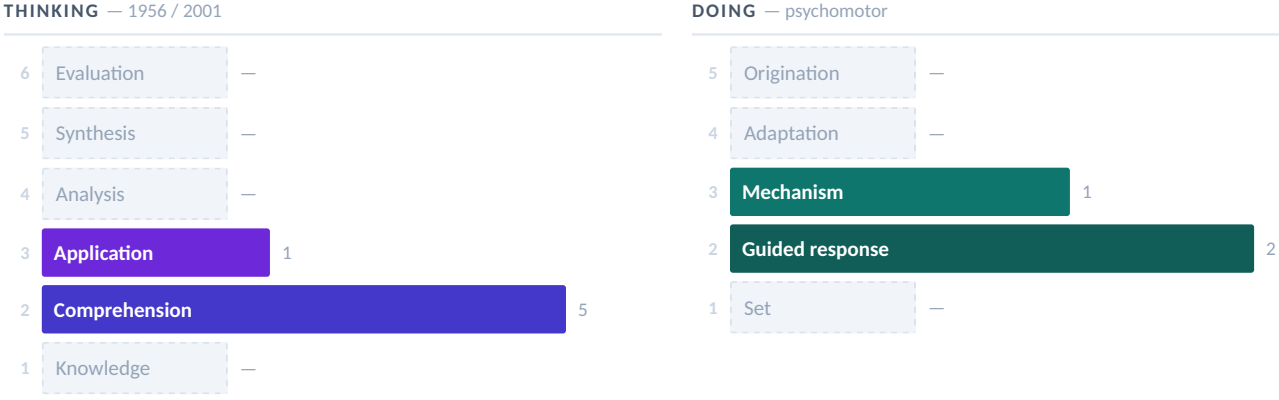
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	3	6	3	50
60 notional hours	12 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	6	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	3	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	3	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Key Competencies Required for Employment in Pharmacy

6 obj · 22 h · Comprehension / Mechanism

1.1 Orientation about current trends in the Ayurvedic Pharmaceutical Industry

1.2 Ayurvedic Pharmacy Industry (India & Global Overview): Market size, growth, and key importing countries

1.3 Ayurvedic Products and Market Stakeholders

1.4 Consumer Preferences and Distribution Channels

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 4			
Discuss current trends in Ayurvedic pharma industry.	*discuss*	Conceptual ◦	Lecture 1h
Discuss key competencies required for employment in the Ayurvedic pharmacy.	*discuss*	Conceptual ◦	Lecture 1h
Describe career opportunities in Ayurvedic pharma industries.	*describe*	Conceptual ◦	Lecture 1h
Discuss key competencies required for employment in the Ayurvedic pharmacy.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Analyze the Market Status of Ayurvedic Pharmacy Industry.	*analyze*	Procedural	Practical 8h
DOING — LEVEL 3 · MECHANISM — 1			
Assess real-life experience of the market for the growing popularity of Ayurvedic products.	*assess*	Procedural	Experiential 10h

Unit 2 · Internship/ Visits to GMP certified Pharmacies

3 obj · 16 h · Comprehension / —

2.1 Factory set up and daily Work flow

2.2 Hygiene checks & Cleaning

2.3 Inventory Management, material management and Production

2.4 Packaging and labelling

2.5 Documentation (batch records) and QC record

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the sections and working flow of GMP-certified pharmacy.	*describe*	Conceptual ◦	Lecture 2h
GOING ABOUT IT — LEVEL 3 · VALUING — 1			
Observe and appraise the operational management of Ayurvedic Pharmacy.	*observe*	Factual ◦	Practical 6h
GOING ABOUT IT — LEVEL 4 · ORGANISATION — 1			
Interpret benefits of field visit and Internship.	*interpret*	Conceptual	Experiential 8h

Unit 3 · Quality Assurance and Quality Control (QAQC) Laboratory Training

3 obj · 18 h · Application / Guided response

3.1 Information about Leading Ayurvedic Research Labs in India
3.2 Regulatory & Certification Bodies for QAQC of Ayurvedic Formulations
3.3 Laboratory set up and daily Work flow
3.4 LIMS (Laboratory Information Management System)
3.5 Instrument Calibration, Sample testing and analysis, Record keeping

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe modern parameters used for quality assurance and quality control of Ayurvedic Drugs. — teach at level 2, the verb *describe* is what will be asked	*describe*	Procedural	Lecture 4h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate and assist quality control practical to display quality standards of Ayurvedic formulations.	*demonstrate*	Procedural	Practical 6h
GOING ABOUT IT — LEVEL 3 · VALUING — 1			
Justify the utility of LIMS(Laboratory Information Management System) in Ayurvedic research laboratories and the workflow of the laboratory.	*justify*	Conceptual	Experiential 8h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 6 lecture, 3 practical, 3 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 23 · SEMESTER 6 · PAPER 3

SOP and clinical applications of Churna Kalpana(Powder dosage forms)

Standardization and clinical Utility of Ayurvedic formulations

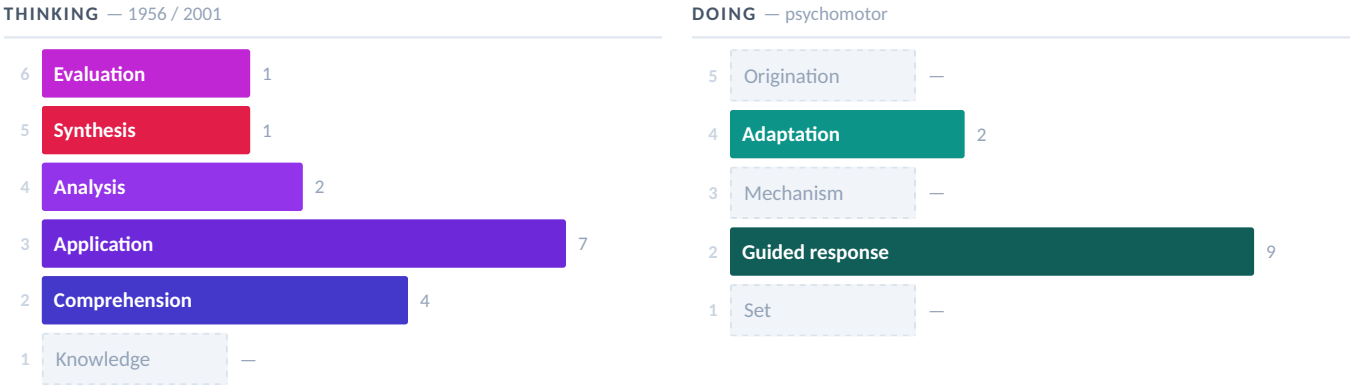
CREDITS	UNITS	THINKING	DOING	FORMATIVE
3	5	15	11	75
90 notional hours	26 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	15 h	15	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	30 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	39 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 6 hours carry the modular assessment, worth 75 marks, bringing the module to its 90 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · SOP and clinical utility of Churna Kalpana

4 obj · 15 h · Application / Adaptation

1.1 Standardisation of various Churna yogas with Standard Operating Procedure
1.2 Therapeutic application of various Churna yogas
1.3 Relevance of Churna kalpana in therapeutics

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss the importance of particle size of different types of Churna concerning its pharmaceutical and therapeutic significance. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
Describe the pharmacological and therapeutic significance of Churna kalpana. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the SOP and quality analysis of a selected Churna yoga.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 4 · ADAPTATION — 1			
Evaluate the role of a selected Churna in a specific clinical condition.	*evaluate*	Conceptual + Procedural ◦	Experiential 8h

Unit 2 · Standardization, analytical profiling, and clinical application of Kshara

5 obj · 16 h · Application / Guided response

2.1 Relevance of Kshara kalpana and Kshara sutra in therapeutics
2.2 SOP of preparation of Paneeya Kshara and Pratisaraneeya Kshara
2.3 SOP of preparation of Kshara Sutra
2.4 Analytical profiling of Kshara

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss Kshara Kalpana, its sources, and evaluate the chemical composition.	*discuss*	Conceptual ◦	Lecture 1h
Discuss the techniques for the preparation of Kshara Sutra.	*discuss*	Procedural	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the therapeutic application of Paneeya Kshara and Prathisaraneeya Kshara. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the preparation of a selected Paneeya Kshara and Pratisaraneeya Kshara and their quality analysis.	*demonstrate*	Procedural	Practical 5h
Demonstrate the therapeutic applications of Kshara and Ksharasutra in a given selected.	*demonstrate*	Procedural	Experiential 8h

Unit 3 · Lavana Kalpana with special reference to pharmaceutico-analytical aspects and therapeutic attributes

5 obj · 16 h · Application / Guided response

3.1 Standard Operating
3.2 Analytical Profiling of Lavanas
3.3 Therapeutic Applications of Selected Formulations in Lavana Kalpana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe salt-based formulations (Lavana Kalpa) mentioned in different texts.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss chemical profiling and standardization of Lavana Kalpa. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
Illustrate the probable mechanism of action and clinical significance of Lavana Kalpa. — teach at level 2, the verb *illustrate* is what will be asked	*illustrate*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the preparation of a selected Lavana formulation following different methods and analytical profiling.	*demonstrate*	Procedural	Practical 5h
Demonstrate the role of Lavana Kalpana in various clinical conditions.	*demonstrate*	Procedural	Experiential 8h

Unit 4 · Satva Kalpana SOP and clinical significance

5 obj · 15 h · Evaluation / Guided response

4.1 SOP for preparing selected Satvas
4.2 Analytical profiling of various yogas from Satva Kalpana
4.3 Therapeutic application of Satva kalpana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the uniqueness and pharmaceutical relevance of Satva kalpana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss SOP and analytical profiling of Satva Kalpana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Procedural ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Evaluate the clinical significance of Satva Kalpana.	*evaluate*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the preparation of a selected Satva and perform its analytical profiling.	*demonstrate*	Procedural	Practical 5h
Demonstrate the role of a selected satva in a specified disease condition.	*demonstrate*	Procedural	Experiential 7h

Unit 5 · Therapeutic attributes of Masi Kalpana

7 obj · 22 h · Synthesis / Adaptation

5.1 Masi from plant and animal origin
5.2 Pharmacology of Masi
5.3 Activated charcoal in therapeutics
5.4 Clinical applications of Masi

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the therapeutic significance of various Masi Kalpas.	*discuss*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Evaluate the clinical significance of activated charcoal. — teach at level 6, the verb *evaluate* is what will be asked	*evaluate*	Conceptual	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss the analytical profiling of Mashi Kalpas. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1			
Discuss SOP for the preparation of Masi by following different classical and advanced methods IS. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Procedural	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the preparation of Mashi and interpret its analytical profiling.	*demonstrate*	Procedural	Practical 5h
Demonstrate pharmaceutical modification of Masi Kalpana.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 4 · ADAPTATION — 1			
Evaluate the Role of Malaharas as a Topical application in Skin disease.	*evaluate*	Conceptual + Procedural ◦	Experiential 8h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 15 lecture, 6 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 24 · SEMESTER 6 · PAPER 3

Therapeutic significance of Topical dosage forms

Standardization and clinical Utility of Ayurvedic formulations

CREDITS	UNITS	THINKING	DOING	FORMATIVE
1	3	5	6	25
30 notional hours	11 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	5 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	10 h	3	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	13 h	3	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 2 hours carry the modular assessment, worth 25 marks, bringing the module to its 30 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the therapeutic significance of various topical dosage forms.	*describe*	L2 Comprehension / Understand
2	Evaluate the challenges in current topical applications of Ayurveda.	*evaluate*	L6 Evaluation / Evaluate
3	Design and develop novel topical dosage forms while preserving their effectiveness.	*design*	L5 Synthesis / Create
4	Integrate clinical knowledge to develop topical applications for specific areas, such as skin disorders, burns, wounds, and pain relief.	*integrate*	L5 Synthesis / Create
5	Access databases to develop strategies for topical applications in cosmeceuticals, with special regard to safety.	*access*	L? —

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Ophthalmic, Otic, and Buccal dosage forms and their clinical applications1.SOP for Ophthalmic, Otic and buccal dosage forms

4 obj · 10 h · Application / Guided response

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss classical and novel dosage forms for ophthalmic, otic, and buccal use.	*discuss*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe therapeutic significance and clinical applications of dosage forms for ophthalmic, otic, and buccal use. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate various Netra Kalpa and their clinical applications.	*demonstrate*	Procedural	Practical 3h
Demonstrate the role of Ayurvedic ophthalmic preparations in various eye diseases.	*demonstrate*	Procedural	Experiential 5h

Unit 2 · Standardisation of Lepa and Malahara preparations and its application in various disorders

3 obj · 6 h · Application / Guided response

2.1 SOP for the preparation of various Lepas
2.2 Innovative approaches including Patches, Bandages, Bands
2.3 Clinical applications of Lepa and Malahara kalpa

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss classical topical dosage forms like Lepa, Malahara.	*discuss*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe novel dosage forms for topical application, along with their regulatory guidelines. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Prepare a novel dosage form using one of the classical lepa Yoga.	*prepare*	Procedural	Practical 4h

Unit 3 · Standard Operating Procedure (SOP) for Upanaha, Dhupa, Dhumapana, and Avachurnana Kalpa and their therapeutic applications

4 obj · 12 h · Application / Origination

3.2 Anti-microbial activity of Dhupa
3.5 Clinical application of various Dhumapana yogas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the therapeutic significance of Upanaha, Dhupa, Dhumapana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the preparation of a selected Upanaha which is useful in treating Shopha.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate the role of Malahara as a Topical application in Skin disease.	*demonstrate*	Procedural	Experiential 3h
DOING — LEVEL 5 · ORIGINATION — 1			
Demonstrate the role of lepa Yogas in managing various medical needs by innovative approaches.	*demonstrate*	Procedural	Experiential 5h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Origination on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 3 practical, 3 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Origination.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 31 · SEMESTER 6 · PAPER 4

Regulatory Acts and Rules applicable to Ayurvedic Drugs

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
3	6	15	16	75
90 notional hours	31 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	15 h	15	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	30 h	7	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	39 h	9	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 6 hours carry the modular assessment, worth 75 marks, bringing the module to its 90 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Chapter I of the Drugs and Cosmetics Act, 1940 and Rules 1945

3 obj · 6 h · Comprehension / Mechanism

1.1 Introduction to 'Drugs and Cosmetic Act, 1940; and 1945' and definitions

1.2 Prescribed Books on Ayurveda, Siddha, and Unani Drugs: The First Schedule

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe the Scope and Applicability of the Drugs and Cosmetics Act and Key Definitions.	*describe*	Conceptual °	Lecture 1h
Discuss the Legal importance of Drug Regulation and enlist prescribed books of ASU from the first Schedule of D&C Act.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Analyse case law on the Factory Act.	*analyse*	Procedural	Experiential 4h

Unit 2 · Chapter II - Regulatory Bodies

3 obj · 9 h · Comprehension / Mechanism

2.1 The Drugs Technical Advisory Board (DTAB)
2.2 The Central Drugs Laboratory (CDL)
2.3 The Drugs Consultative Committee (DCC)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Identify key regulatory authorities and develop foundational knowledge of regulatory frameworks. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Record their observations regarding hypothetical issue through Mock composition exercise of demonstration of the roles of DTAB (Drug Technical Advisory Board) composition, including representatives from DGHS, DCGI, pharmacologists.	*record*	Procedural	Practical 4h
DOING — LEVEL 3 · MECHANISM — 1			
Prepare a technical report on Adverse Drug Reactions (ADRs) for submission to DTAB.	*prepare*	Procedural	Experiential 4h

Unit 3 · Chapters III, IV, and IV-A - Regulation of Drugs and Cosmetics

8 obj · 18 h · Application / Mechanism

3.1 Chapter III: Standards of quality
3.2 Misbranded, adulterated, and spurious drugs and cosmetics
3.3 Prohibition of import of certain drugs or cosmetics
3.4 Manufacturing, Labelling, and Sale of Drugs and Cosmetics (Chapter IV-A)
3.5 Standards of quality for drugs and cosmetics (Sections 16 and 16A)
3.6 Prohibition of manufacture and sale of certain drugs and cosmetics (Sections 18 and 19)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 4			
Identify legal restrictions and prohibitions for misbranded and spurious drugs and. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Lecture 1h
Describe manufacturing, labelling, and sale provisions, including penalties and seizures (Chapter IV-A).	*describe*	Conceptual ◦	Lecture 1h
Justify the importance of standards of quality for drugs and cosmetics (Sections 16 and. — teach at level 6, the verb *justify* is what will be asked	*justify*	Conceptual	Lecture 1h
Discuss the prohibition of the manufacture and sale of certain drugs and cosmetics.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Summarize the provisions of Chapter III regarding the import of drugs and cosmetics. — teach at level 2, the verb *summarize* is what will be asked	*summarize*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Prepare mock import license applications.	*prepare*	Procedural	Practical 4h
Demonstrate case scenarios of misbranded/spurious drugs and cosmetics.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 1			
Identify misbranded and spurious drugs/cosmetics.	*identify*	Procedural	Experiential 4h

Unit 4 · Schedules to Drugs and Cosmetics Rules related to ASU drugs and cosmetics

5 obj · 16 h · Application / Mechanism

4.1 Good Manufacturing Practices (GMP) for ASU Medicines

4.2 Prescription and Safety Requirements (Schedules H, G, X) and Pharmacological Information (Schedule Y)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the provision of Schedule T and Schedule E (1).	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the requirements of Schedules H, G, X (prescription and safety), and Schedule Y (pharmacological information). — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the formulation of a proprietary formula and apply for a license.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 2			
Apply for GMP certification.	*apply*	Procedural	Experiential 4h
Prepare a new formula and apply for a mock license.	*prepare*	Procedural	Experiential 5h

Unit 5 · Pharmaceutical observation of drugs and cosmetics ASU drugs

3 obj · 10 h · Comprehension / Mechanism

5.1 Shelf life and stability of ASU drugs: recent amendments

5.2 Use of Preservatives in ASU drugs

5.3 Packaging and labelling requirements for ASU drugs

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe Shelf life of ASU drugs: recent amendments, Preservatives, and. Discuss Packaging and labelling technique.	*describe*	Procedural ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 2			
Evaluate label compliance and identify shortcomings in given labels.	*evaluate*	Procedural	Experiential 5h
Design and develop a label for a hypothetical ASU product, ensuring compliance with regulatory requirements.	*design*	Procedural	Experiential 4h

Unit 6 · Miscellaneous Regulatory Acts related to drugs and cosmetics

9 obj · 25 h · Comprehension / Mechanism

6.1 Narcotic and Psychotropic Substances Act 1985
6.2 Factories Act, 1948
6.3 Industries Act, 1951
6.4 Biological Diversity Act 2002
6.5 Consumer Protection Act 1986
6.6 Drugs and Magic Remedies Act, 1954
6.7 Medicinal and Toilet Preparations Act, 1955
6.8 Opium Act, 1857
6.9 Pharmacy Act 1948
6.10 Recent Amendments in D & C Act concerning Ayurvedic Drug Industry

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 4			
Describe the key provisions of the Narcotic and Psychotropic Substances Act, 1985 Factories Act, 1948, and Industries Act, 1951.	*describe*	Conceptual ◦	Lecture 1h
Discuss the Biological Diversity Act 2002, Consumer Protection Act 1986, and Drugs and Magic Remedies Act, 1954.	*discuss*	Conceptual ◦	Lecture 1h
Discuss Medicinal and Toilet Preparations Act, 1955, Opium Act, 1857, and Pharmacy.	*discuss*	Conceptual ◦	Lecture 1h
Describe Recent Amendments in D & C Act concerning Ayurvedic Drug Industry.	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate Landmark Judgments under the NDPS Act.	*demonstrate*	Procedural	Practical 4h
Demonstrate Case Law on the Factory Act.	*demonstrate*	Procedural	Practical 4h
Demonstrate Case Study under the Drugs and Magic Remedies Act.	*demonstrate*	Procedural	Practical 4h
DOING — LEVEL 3 · MECHANISM — 2			
Analyse Landmark Judgments under the NDPS Act.	*analyse*	Procedural	Experiential 5h
Prepare Case Studies and Report related to Objectionable Advertisements.	*prepare*	Procedural	Experiential 4h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 15 lecture, 7 practical, 9 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 32 · SEMESTER 6 · PAPER 4

Entrepreneurship and Start-Ups

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
1	4	6	8	25
30 notional hours	14 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

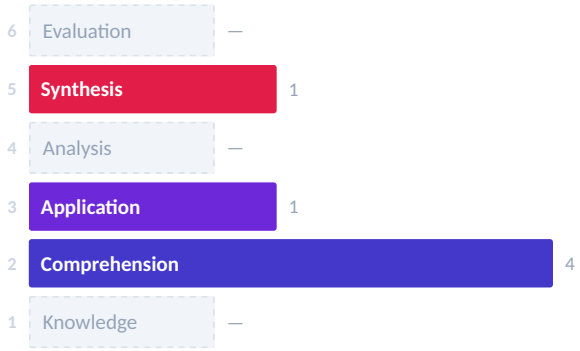
WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	5 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	10 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	13 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 2 hours carry the modular assessment, worth 25 marks, bringing the module to its 30 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Define Entrepreneurship	*define*	L1 Knowledge / Remember
2	Identify Entrepreneurial Traits	*identify*	L1 Knowledge / Remember
3	Describe the Entrepreneurial Process	*describe*	L2 Comprehension / Understand
4	Enlist Different Types of Entrepreneurships	*enlist*	L1 Knowledge / Remember
5	Recognize the opportunities of entrepreneurship and start-ups	*recognize*	L1 Knowledge / Remember

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Entrepreneurship and Startup

3 obj · 4 h · Application / Mechanism

1.1 Introduction to Entrepreneurship and Start-ups
1.2 MSMED Act and its Process of Registration and Role of Department for Promotion of Industry and Internal Trade(DPIIT) and Bharat Startup Knowledge Access
1.3 Types of Enterprises and Role of Enterprises in National and Global Economy
1.4 Government Policies, Schemes, and Institutional Support for Entrepreneurship

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Identify the role of DPIIT (Department for Promotion of Industry and Internal Trade) BHASKAR (Bharat Startup Knowledge Access Registry) in the promotion of entrepreneurship and startups. — teach at level 1, the verb *identify* is what will be asked	*identify*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe entrepreneurship, the startup life cycle, the MSME Act, and the process of registration. — teach at level 2, the verb *describe* is what will be asked	*describe*	Procedural ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Create an elevator pitch for a startup idea.	*create*	Procedural	Experiential 2h

Unit 2 · Process of entrepreneurship development

3 obj · 8 h · Comprehension / Origination

2.1 Need and the process of entrepreneurship development
2.2 Dynamics of Entrepreneurial Motivation
2.3 Developing Entrepreneurial Competencies

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the dynamism of Entrepreneurial Motivation and key entrepreneurial competencies.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the process of development of a basic business model as per the template.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 5 · ORIGINATION — 1			
Conduct and document market research for a start-up, including survey design and analysis.	*conduct*	Procedural	Experiential 4h

Unit 3 · Project Proposal for new Invention/startup

2 obj · 3 h · Comprehension / Adaptation

3.1 Challenges and the feasibility report of the Startups
3.2 Planning, resource mobilisation, and implementation of a start-up project

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Identify the challenges and feasibility report of the start-up. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Lecture 1h
DOING — LEVEL 4 · ADAPTATION — 1			
Carry out A Case Study-Based Role-Play, navigating a Startup's Early Days.	*carry*	Procedural	Experiential 2h

Unit 4 · Intellectual Property Rights and Patents

6 obj · 13 h · Synthesis / Mechanism

4.1 Introduction to IPR and Patents
4.2 Indian and international patent laws, proposed amendments as applicable to herbal/natural products and processes
4.3 Procedure for patent filing
4.4 Geographical Indication and Copyright

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Define IPR, describe the purpose and importance of IPR, and the laws governing patent. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1			
Identify the patentability of a product. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Experiential 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the process of patent filing.	*demonstrate*	Procedural	Practical 3h
Demonstrate a case scenario of the new molecule patent.	*demonstrate*	Procedural	Practical 2h
Demonstrate a case scenario of patenting the formulation process.	*demonstrate*	Procedural	Practical 2h
DOING — LEVEL 3 · MECHANISM — 1			
Design an application to Patent the formulation process.	*design*	Procedural	Experiential 3h

FOR THE TEACHER

- **This module tops out at Synthesis on the thinking ladder and Origination on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 4 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Origination.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

PART C

Reference

Nine appendices, all worked out from the curriculum's own objectives, so the counts here and the counts in Part B are the same numbers. C1 and C2 decode anything in the syllabus document. C3 compares every module and unit at a glance. C6 and C7 reproduce all three hundred and four practical and experiential activities with the level each serves, so a reference like "Experiential Learning 8.2" can be looked up without turning back to the syllabus.

1. **C1** The sixteen curriculum codes, decoded
2. **C2** The two schemes side by side
3. **C3** Level profile of every module and unit
4. **C4** Verb index — every verb the curriculum uses, and its level
5. **C5** Kinds of knowledge across the semester
6. **C6** All 141 practical training activities, mapped
7. **C7** All 163 experiential learning activities, mapped
8. **C8** Where the hours go, module by module
9. **C9** Sources and further reading

↑ Contents

PART C · APPENDIX C1

The sixteen curriculum codes, decoded

Every code that appears in Column 3E of Table 3, with the domain it belongs to, the level it sits at, and how many of the 577 objectives carry it. Keep this beside the syllabus document.

CODE	DOMAIN	LEVEL	NAME	TIMES USED
CK	Thinking	1	Knowledge / Remember	7
CC	Thinking	2	Comprehension / Understand	152
CAP	Thinking	3	Application / Apply	69
CAN	Thinking	4	Analysis / Analyse	32
CS	Thinking	5	Synthesis / Create	5
CE	Thinking	6	Evaluation / Evaluate	12
PSY-SET	Doing	1	Set	1
PSY-GUD	Doing	2	Guided response	194
PSY-MEC	Doing	3	Mechanism	65
PSY-ADT	Doing	4	Adaptation	30
PSY-ORG	Doing	5	Origination	4
AFT-REC	Going about it	1	Receiving	not used
AFT-RES	Going about it	2	Responding	not used
AFT-VAL	Going about it	3	Valuing	5
AFT-SET	Going about it	4	Organisation	1
AFT-CHR	Going about it	5	Characterisation	not used

Sixteen codes are defined; 13 are used here. Unlike the undergraduate curriculum, this course **does** use Synthesis and Origination, and every semester reaches Evaluation.

PART C · APPENDIX C2

The two schemes side by side

The six thinking levels in both schemes, with what a student actually does at each.

LEVEL	BLOOM 1956, AS PRINTED IN THE CURRICULUM	REVISED 2001	WHAT THE STUDENT DOES
Level 1	**Knowledge**	**Remember**	recalls it
Level 2	**Comprehension**	**Understand**	restates it in their own arrangement
Level 3	**Application**	**Apply**	uses it on something new
Level 4	**Analysis**	**Analyse**	takes it apart and shows how the parts relate
Level 5	**Synthesis**	**Evaluate** — the schemes differ here	makes something that was not given
Level 6	**Evaluation**	**Create** — the schemes differ here	judges it against stated criteria

The top two levels are the only place the schemes disagree. This book follows the 1956 numbering, because that is what the curriculum prints in Column 3E.

PART C · APPENDIX C3

Level profile of every module and unit

All thirty-two modules and one hundred and twenty-four units in one view. The fastest way to see where a semester's weight sits and to check that a modular assessment matches its ceiling.

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
SEMESTER 3						
PAPER 1 · MODULE 1 — CHRONOLOGICAL DEVELOPMENT AND TAXONOMY OF RASA DRAVYAS — 19 OBJ, 56 H, CEILING ANALYSIS / ADAPTATION						
1	1	Chronological Development and Overview of Rasashastra	5	9	Analysis	Adaptation
1	2	Taxonomy of Rasadravyas	5	13	Analysis	Guided response
1	3	Geological/mineralogical identification of Rasadravyas	6	22	Analysis	Mechanism
1	4	Overview of Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya	3	12	Comprehension	Guided response
PAPER 1 · MODULE 2 — PUTA AND SAMSKARA (TRANSFORMATIVE PROCESSES OF DRUG MANUFACTURING) — 21 OBJ, 56 H, CEILING EVALUATION / ADAPTATION						
2	1	Puta and Law of thermodynamics	6	18	Evaluation	Mechanism
2	2	Concept, understanding, application, SOPs and chemical changes of Jarana, Amritikarana and Lohitikarana	4	16	Analysis	Guided response
2	3	Concept, understanding, application, SOPs (Standard Operating Procedures) and chemical changes of Satva and Druti procedures	4	9	Application	Guided response
2	4	Samskara (Principles of Transformative processes of drug manufacturing)	2	3	Comprehension	Adaptation
2	5	Influence of Desha (place) and Kala(time and seasonal factors) of drug qualities	2	3	Comprehension	Mechanism
2	6	Kalaprakarsha- Processing with/Effect of specific time exposure and Bhajana (processing vessel)	3	7	Comprehension	Guided response
PAPER 2 · MODULE 9 — SCIENTIFIC FUNDAMENTALS OF AYURVEDIC PHARMACEUTICAL PREPARATIONS — 15 OBJ, 56 H, CEILING ANALYSIS / MECHANISM						
9	1	Classical Principles of drug processing and scientific basis of Panchavidha Kashaya Kalpana	3	6	Comprehension	Mechanism
9	2	Identity, Purity and Quality of Raw materials used in Ayurveda Pharmaceutics	3	12	Comprehension	Guided response
9	3	Pharmaceutical processing of drugs with its rationality	3	13	Analysis	Mechanism
9	4	Ayurvedic Classical Dosage Forms and their integration with modern scientific validation	6	25	Analysis	Guided response
PAPER 2 · MODULE 10 — PRINCIPLES OF PHARMACEUTICAL FORMULATION — 12 OBJ, 56 H, CEILING SYNTHESIS / MECHANISM						
10	1	Principles of Samskara (transformative processes of drug manufacturing) in relation to drug development	2	6	Application	Guided response
10	2	Rationality behind Composition of Formulations	2	6	Analysis	Guided response
10	3	Principles and designing of various dosage forms	2	8	—	Guided response
10	4	Principles of Prakshep dravyas (Pharmaceutical Additives)	3	14	Comprehension	Mechanism

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
10	5	Principles of Potency and Efficacy enhancement	3	22	Synthesis	Mechanism
PAPER 3 · MODULE 17 — GUIDELINES FOR ADMINISTRATION OF RASOUSHADHI IN CLINICAL PRACTICE — 22 OBJ, 56 H, CEILING EVALUATION / MECHANISM						
17	1	Concept of Rasa chikitsa and significance of Rasoushadhi	4	11	Application	Guided response
17	2	Concept of Kshetrikarana and its significance prior to administration of Rasoushadhi	4	11	Application	Guided response
17	3	Pathya -Apathya during Rasayoga Sevana	5	12	Application	Mechanism
17	4	Safety and toxicity of Rasoushadhi	5	14	Evaluation	Guided response
17	5	Heavy metal toxicity and chelation therapy	4	8	Comprehension	Mechanism
PAPER 3 · MODULE 18 — PRAYOGIKA BHAISHAJYA KALPANA(CLINICAL APPLICATION OF PHARMACEUTICAL PRODUCTS) — 24 OBJ, 56 H, CEILING EVALUATION / ADAPTATION						
18	1	Aushadha Sevana Kala (Time of administration of drug)	7	15	Application	Adaptation
18	2	Concept and scientific basis of Anupana	6	16	Evaluation	Mechanism
18	3	Matra (Posology)	5	12	Application	Guided response
18	4	Concept of drug repurposing	6	13	Application	Mechanism
PAPER 4 · MODULE 25 — ARTIFICIAL INTELLIGENCE IN DRUG DEVELOPMENT AND PHARMACOGENETICS; PHARMACOGENOMICS — 20 OBJ, 56 H, CEILING APPLICATION / ORIGINATION						
25	1	Role of Artificial intelligence in drug development	4	11	Application	Mechanism
25	2	Types of software used in Artificial intelligence	3	6	Application	Origination
25	3	3D printed drug delivery system	5	16	Application	Adaptation
25	4	Pharmacogenetics and Pharmacogenomics	4	11	Application	Guided response
25	5	Pharmacogenomics in drug discovery	4	12	Comprehension	Mechanism
PAPER 4 · MODULE 26 — PHARMACOLOGY — 16 OBJ, 56 H, CEILING SYNTHESIS / ADAPTATION						
26	1	Introduction to Pharmacology	3	12	Comprehension	Guided response
26	2	Pharmacokinetics, therapeutic index, and drug interactions	3	12	Application	Adaptation
26	3	Pharmacology of drugs that act on the peripheral nervous system	3	9	Synthesis	—
26	4	Pharmacology of drugs that act on the central nervous system I	3	9	Application	Mechanism
26	5	Pharmacology of drugs that act on the central nervous system II	4	14	Comprehension	Mechanism
SEMESTER 4						
PAPER 1 · MODULE 3 — PARADA (MERCURY) AND ITS PROCESSING: ANCIENT WISDOM AND SCIENTIFIC ADVANCES — 17 OBJ, 56 H, CEILING ANALYSIS / MECHANISM						
3	1	Parada: Ancient and Contemporary Perspectives on Its Shodhana and Underlying Principles	8	22	Comprehension	Mechanism
3	2	Parada Ashta Samskara and Ashtadasha Samskara	4	21	Comprehension	Guided response
3	3	Physico-chemical proprties and pharmacology of mercury and mercurial compounds	2	6	Analysis	Mechanism

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
3	4	Mercury toxicity, ADR and its management	3	7	Analysis	Guided response
PAPER 1 · MODULE 4 — MAHARASA DRAVYA, ITS PROCESSING AND UTILITY — 22 OBJ, 56 H, CEILING EVALUATION / ADAPTATION						
4	1	Maharasa Dravyas with special reference to Abhraka and Vaikranta	8	18	Analysis	Adaptation
4	2	Principles and Processing of Makshika and Vimala	3	11	Comprehension	Guided response
4	3	Principles and Processing of Shilajit and Sasyaka	3	10	Analysis	Guided response
4	4	Principles and Processing of Chapala and Kharpara, along with physico-chemical analysis, with research updates of all Maharasa Dravyas	4	9	Application	Mechanism
4	5	Utility of new metals and minerals	4	8	Evaluation	Guided response
PAPER 2 · MODULE 11 — MANA PARIBHASHA (METROLOGY IN AYURVEDIC PHARMACEUTICAL PREPARATION) — 16 OBJ, 56 H, CEILING ANALYSIS / ADAPTATION						
11	1	Mana Paribhasha (Classical system of measurement)	4	12	Analysis	Adaptation
11	2	Standard Units of Measurement	3	6	Knowledge	Adaptation
11	3	Volumetric Vs Gravimetric Methods	3	6	Application	Guided response
11	4	Rationality of Proportion of Ingredients in Ayurvedic formulations	3	16	Comprehension	Adaptation
11	5	Bulk production of Ayurvedic Formulations	3	16	Comprehension	Adaptation
PAPER 2 · MODULE 12 — PROCESS VALIDATION PROTOCOL — 10 OBJ, 56 H, CEILING EVALUATION / GUIDED RESPONSE						
12	1	Pharmaceutical Preparation process: Classical Dosage forms	6	41	Evaluation	Guided response
12	2	Pharmaceutical Preparation process: Recent advances of Dosage forms	3	10	Comprehension	Guided response
12	3	Novel Dosage and drug delivery Forms	1	5	—	Guided response
PAPER 3 · MODULE 19 — SOP AND CLINICAL APPLICATION OF CHATURVIDHA RASAYANA — 25 OBJ, 84 H, CEILING EVALUATION / ADAPTATION						
19	1	Standard Operating Procedure (SOP) and Clinical Application of Khalveeya Rasayana	5	17	Application	Adaptation
19	2	SOP and clinical application of Parpati Rasayana	6	15	Analysis	Adaptation
19	3	SOP and clinical application of Pottali Rasayana	7	25	Evaluation	Mechanism
19	4	SOP and clinical application of Kupipakwa Rasayana	7	27	Comprehension	Guided response
PAPER 3 · MODULE 20 — SOP AND CLINICAL APPLICATION OF AYASKRITI, LOUHA, AND MANDURA KALPA — 12 OBJ, 28 H, CEILING COMPREHENSION / ADAPTATION						
20	1	SOP and clinical application of Ayaskriti	3	8	Comprehension	Mechanism
20	2	SOP and clinical application of Louha kalpa	4	9	Comprehension	Adaptation
20	3	SOP and Clinical application of Mandura Kalpa	5	11	Comprehension	Mechanism
PAPER 4 · MODULE 27 — NETWORK PHARMACOLOGY & REVERSE PHARMACOLOGY — 16 OBJ, 56 H, CEILING ANALYSIS / MECHANISM						
27	1	Netwok Pharmacology	8	28	Analysis	Mechanism
27	2	Reverse Pharmacology	8	28	Analysis	Mechanism
PAPER 4 · MODULE 28 — PHARMACEUTICAL QUALITY ASSURANCE — 19 OBJ, 56 H, CEILING APPLICATION / ADAPTATION						
28	1	Quality Assurance and Quality Management concepts	6	19	Application	Guided response
28	2	Personnel Responsibilities	3	8	Comprehension	Adaptation

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
28	3	Containers, Packing materials	4	13	Comprehension	Mechanism
28	4	Good Laboratory Practices	3	8	Knowledge	Mechanism
28	5	Document maintenance in pharmaceutical industry	3	8	Application	Mechanism
SEMESTER 5						
PAPER 1 · MODULE 5 — CRITICAL APPRAISAL OF UPARASA DRAVYA AND SADHARANA RASA DRAVYA — 24 OBJ, 56 H, CEILING EVALUATION / ADAPTATION						
5	1	Principle and Processing of Uparasa Dravya	12	28	Application	Adaptation
5	2	Principle and Processing of Sadharana Rasa	12	28	Evaluation	Adaptation
PAPER 1 · MODULE 6 — CRITICAL APPRAISAL OF DHATU AND UPDHATU — 24 OBJ, 56 H, CEILING EVALUATION / ADAPTATION						
6	1	Principle and Processing of Dhatu / Metals	12	28	Analysis	Mechanism
6	2	Principle and Processing of Upadhatu	12	28	Evaluation	Adaptation
PAPER 2 · MODULE 13 — KRUTANNA (AYURVEDIC FOOD PREPARATIONS) — 15 OBJ, 56 H, CEILING ANALYSIS / ADAPTATION						
13	1	Classification of food substances and Formulations	3	12	Comprehension	Guided response
13	2	Customized food for treatment	6	24	Comprehension	Guided response
13	3	Ayurvedic Food Industry	3	10	Knowledge	Guided response
13	4	Food processing technology traditional and modern methods	3	10	Analysis	Adaptation
PAPER 2 · MODULE 14 — AYURVEDIC FORMULATIONS FOR HOLISTIC WELLBEING, VITALITY AND BODY CARE — 11 OBJ, 56 H, CEILING EVALUATION / MECHANISM						
14	1	Rasayana-Personalized Rejuvenation Formulations	5	24	Comprehension	Mechanism
14	2	Ayurvedic based cosmetic preparation	6	32	Evaluation	Mechanism
PAPER 3 · MODULE 21 — CLINICAL APPLICATION OF RASOUSHADHI ACCORDING TO SROTAS AND IN SELECTED CHALLENGING CLINICAL CONDITIONS — 13 OBJ, 28 H, CEILING APPLICATION / ADAPTATION						
21	1	Clinical application of Rasoushadhis specific to Srotas	4	8	Application	Adaptation
21	2	Clinical application of Rasoushadhi in emergency conditions	4	9	Comprehension	Adaptation
21	3	Rasoushadhi in specific, clinically challenging conditions	5	11	Application	Adaptation
PAPER 3 · MODULE 22 — YOGAS FROM DIFFERENT KALPANAS — 27 OBJ, 84 H, CEILING ANALYSIS / ADAPTATION						
22	1	SOP of Vati, Gutika, Guggulu and Varti Kalpas and their therapeutic applications	6	20	Application	Mechanism
22	2	SOP and Clinical applications of Avaleha and Khanda kalpana	5	16	Application	Mechanism
22	3	SOP of Sneha Kalpana and its clinical significance	5	17	Application	Adaptation
22	4	SOP and clinical applications of Arka and Dravakam in various disorders	5	14	Analysis	Mechanism
22	5	SOP and therapeutic applications of Sandhana Kalpana	6	17	Application	Mechanism
PAPER 4 · MODULE 29 — PHARMACY PRACTICE — 18 OBJ, 56 H, CEILING EVALUATION / ORIGINATION						
29	1	Hospital Pharmacy and Community Pharmacy	7	26	Comprehension	Mechanism
29	2	Therapeutic Drug Monitoring	7	17	Evaluation	Adaptation
29	3	Pharmacy and therapeutic committee	4	13	Comprehension	Origination

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
PAPER 4 · MODULE 30 — PHARMACEUTICAL MICROBIOLOGY — 18 OBJ, 56 H, CEILING APPLICATION / ADAPTATION						
30	1	Introduction to microbes with applications of sterilization	5	12	Application	Guided response
30	2	Sources of contamination	6	12	Application	Adaptation
30	3	Types of spoilage and use of antimicrobial agents	7	32	Application	Adaptation
SEMESTER 6						
PAPER 1 · MODULE 7 — SUDHA VARGA, SIKATA VARGA, VISHA VARGA DRAVYAS — 22 OBJ, 84 H, CEILING APPLICATION / GUIDED RESPONSE						
7	1	Characterization, Pharmaceutical Processing, and Clinical Applications of Sudha Varga Dravya	7	28	Application	Guided response
7	2	Characterization, pharmaceutical processing, and clinical applications of Sikata Varga Dravya	7	28	Application	Guided response
7	3	Pharmaceutical processing, phytopharmacology and clinical applications of Visha Varga dravya	8	28	Application	Guided response
PAPER 1 · MODULE 8 — APPRAISAL OF RATNA AND UPARATNA AND THERAPEUTIC SIGNIFICANCE OF PISHTI AND DRUTI KALPANA — 13 OBJ, 28 H, CEILING APPLICATION / MECHANISM						
8	1	Mineralogy, Crystal chemistry,and Pharmacotherapeutics of Ratna-Uparatna Varga	7	15	Comprehension	Mechanism
8	2	SOP and therapeutic applications of Pishti, Gandhaka Druti and Gandhaka Taila kalpana	6	13	Application	Guided response
PAPER 2 · MODULE 15 — DISEASE CONDITION SPECIFIC SELECTION OF FORMULATION — 12 OBJ, 56 H, CEILING ANALYSIS / MECHANISM						
15	1	Application of Yukti Pramana in Formulation Selection	2	2	Analysis	—
15	2	Selection of formulations for Speciality practice in Ayurveda	3	12	Analysis	Guided response
15	3	Routes of Drug Administration	2	12	Analysis	Mechanism
15	4	Designing formulations for Shodhana and Shamana Chikitsa	2	12	Analysis	Guided response
15	5	Innovation in Ayurvedic Formulations	3	18	Comprehension	Guided response
PAPER 2 · MODULE 16 — CURRENT TRENDS IN AYURVEDIC PHARMACEUTICAL INDUSTRY. — 12 OBJ, 56 H, CEILING APPLICATION / MECHANISM						
16	1	Key Competencies Required for Employment in Pharmacy	6	22	Comprehension	Mechanism
16	2	Internship/ Visits to GMP certified Pharmacies	3	16	Comprehension	—
16	3	Quality Assurance and Quality Control (QAQC) Laboratory Training	3	18	Application	Guided response
PAPER 3 · MODULE 23 — SOP AND CLINICAL APPLICATIONS OF CHURNA KALPANA(POWDER DOSAGE FORMS) — 26 OBJ, 84 H, CEILING EVALUATION / ADAPTATION						
23	1	SOP and clinical utility of Churna Kalpana	4	15	Application	Adaptation
23	2	Standardization, analytical profiling, and clinical application of Kshara	5	16	Application	Guided response
23	3	Lavana Kalpana with special reference to pharmaceutico-analytical aspects and therapeutic attributes	5	16	Application	Guided response
23	4	Satva Kalpana SOP and clinical significance	5	15	Evaluation	Guided response
23	5	Therapeutic attributes of Masi Kalpana	7	22	Synthesis	Adaptation

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
PAPER 3 · MODULE 24 — THERAPEUTIC SIGNIFICANCE OF TOPICAL DOSAGE FORMS — 11 OBJ, 28 H, CEILING APPLICATION / ORIGINATION						
24	1	Ophthalmic, Otic, and Buccal dosage forms and their clinical applications1.SOP for Ophthalmic, Otic and buccal dosage forms	4	10	Application	Guided response
24	2	Standardisation of Lepa and Malahara preparations and its application in various disorders	3	6	Application	Guided response
24	3	Standard Operating Procedure (SOP) for Upanaha, Dhupa, Dhumapana, and Avachurnana Kalpa and their therapeutic applications	4	12	Application	Origination
PAPER 4 · MODULE 31 — REGULATORY ACTS AND RULES APPLICABLE TO AYURVEDIC DRUGS — 31 OBJ, 84 H, CEILING APPLICATION / MECHANISM						
31	1	Chapter I of the Drugs and Cosmetics Act, 1940 and Rules 1945	3	6	Comprehension	Mechanism
31	2	Chapter II - Regulatory Bodies	3	9	Comprehension	Mechanism
31	3	Chapters III, IV, and IV-A - Regulation of Drugs and Cosmetics	8	18	Application	Mechanism
31	4	Schedules to Drugs and Cosmetics Rules related to ASU drugs and cosmetics	5	16	Application	Mechanism
31	5	Pharmaceutical observation of drugs and cosmetics ASU drugs	3	10	Comprehension	Mechanism
31	6	Miscellaneous Regulatory Acts related to drugs and cosmetics	9	25	Comprehension	Mechanism
PAPER 4 · MODULE 32 — ENTREPRENEURSHIP AND START-UPS — 14 OBJ, 28 H, CEILING SYNTHESIS / ORIGINATION						
32	1	Entrepreneurship and Startup	3	4	Application	Mechanism
32	2	Process of entrepreneurship development	3	8	Comprehension	Origination
32	3	Project Proposal for new Invention/startup	2	3	Comprehension	Adaptation
32	4	Intellectual Property Rights and Patents	6	13	Synthesis	Mechanism

Ordered by semester and then paper, as a department teaches it, rather than by paper as Table 3 prints it. Both ceilings are shown because they move independently.

PART C · APPENDIX C4

Verb index — every verb the curriculum uses, and its level

The opening verb of all 577 objectives, sorted by the level that verb belongs to. The verb is what a student meets, which makes this the most practical table in the book.

LEVEL	NAME	VERBS THE CURRICULUM USES, WITH HOW OFTEN
Level 1	**Knowledge** Remember	identify 26, define 14, enlist 8, recognise 4, recognize 3, observe 2, record 2, recall 1, recite 1
Level 2	**Comprehension** Understand	discuss 110, describe 79, elaborate 8, illustrate 7, explain 2, summarize 2, present 1, classify 1
Level 3	**Application** Apply	demonstrate 137, perform 30, prepare 17, apply 9, conduct 4, carry 2
Level 4	**Analysis** Analyse	analyse 15, analyze 15, explore 5, review 4, compare 3, differentiate 2, examine 1, compile 1, investigate 1, select 1, categorize 1
Level 5	**Synthesis** Create	design 4, integrate 2, develop 2, construct 1, generate 1, create 1
Level 6	**Evaluation** Evaluate	evaluate 15, justify 7, interpret 5, appraise 5, assess 4, validate 1
Affective	**Attitude verbs**	debate 1, practice 1

Choose the level first, then take a verb from that row. Where an objective's verb sits at a different level from its code, Part B says which to pitch to.

PART C · APPENDIX C5

Kinds of knowledge across the course

The second axis of the 2001 scheme, counted across all 577 objectives.

KIND OF KNOWLEDGE	OBJECTIVES	HOW IT IS LEARNED
Factual	28	Rehearsal and self-testing. Only ten objectives here — this is not a memorisation course.
Conceptual	237	Explaining it aloud to somebody without notes.
Procedural	335	Carrying it out, more than once. **The dominant kind in this course**, and the reason reading cannot substitute for the practical hours.
Metacognitive	not used	Being made to look back at how you worked. Not asked for here.

An objective working on two kinds is counted under both, so the total exceeds 577. The kind is read from the wording of each objective; where the wording does not settle it, the label carries a small circle (°).

PART C · APPENDIX C6

All 141 practical training activities, mapped

Table 4 of the curriculum, with the level each activity is cited to serve. Practical training is where a student moves from performing with help to performing unaided.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 1 — CHRONOLOGICAL DEVELOPMENT AND TAXONOMY OF RASA DRAVYAS			
1.1	Demonstration of the application of the Mendeleev Periodic table in mapping Rasadravyas	Doing 2 · Guided response	The teacher will allot each student at least one category from the classification of Rasadravya or one drug from each category. Each student will map minerals/Metals to the Periodic Table and search the elemental properties of minerals/metals mentioned in Rasashastra. Students will locate these metals on the Mendeleev Periodic Table, noting their atomic numbers, groups, and periods. Each student will compare and analyze the traditional name with the Periodic table, along with other properties, etc.
1.2	Demonstration of different Rasadravyas mentioned in different classics and analysis of the classification of Rasa Dravyas	Doing 2 · Guided response	The teacher will demonstrate different Rasadravyas mentioned in different classics based on classical Grahya- Agrahya Lakshanas, compare them, and analyze. The teacher helps the students to identify similarities, differences, and the evolution of thought in Rasashastra and its dravyas.
1.3	Geological Mineralogical Identification of Rasadravyas	Doing 2 · Guided response	The teacher will demonstrate the identification of selected Rasadravya using geological/ mineralogical characters.
1.4	Demonstration of Krutrima and Pratinidhi Rasadravyas	Doing 2 · Guided response	The teacher will demonstrate the preparation of a Krutrima Rasadravya and discuss the steps involved, the rationale behind the process, and the therapeutic benefits of the resulting product. The teacher will encourage students to reflect on the challenges of identifying Krutima and Pratinidhi substances and the significance of these classifications in ensuring the integrity of Rasashastra formulations.
MODULE 2 — PUTA AND SAMSKARA (TRANSFORMATIVE PROCESSES OF DRUG MANUFACTURING)			
2.1	Demonstration and comparison of the temperature pattern and total duration of different	Doing 2 · Guided response	—
2.2	Demonstration of the art of Gandhaka Jarana: A Practical Exploration of the Rasa Jarana Process, Understanding its Role and Significance in Rasashastra	Doing 2 · Guided response	The teacher will demonstrate the Gandhaka Jarana procedure using a suitable method and explain the role and significance of Rasa Jarana. Students will then conduct the practical and present their findings at the end, highlighting their understanding of the process.
2.3	Demonstration of Amritikarana and Lohitikarana: Traditional Methods for Mica Processing in Rasashastra	Doing 2 · Guided response	The teacher will demonstrate the procedures of Abhraka Amritikarana and Abhraka Lohitikarana using a selected method. Students will observe, record their observations, and present their findings and insights at the end of the practical.
2.4	Demonstration and Observation of Satva Patana	Doing 2 · Guided response	The teacher will select any one ore sample used in Rasashastra and demonstrate the procedure of Satva Patana. The students will observe and present their findings at the end of the practical.
MODULE 3 — PARADA (MERCURY) AND ITS PROCESSING: ANCIENT WISDOM AND SCIENTIFIC ADVANCES			
3.1	Demonstration of Samanya and Vishesh Shodhana procedures of Parada	Doing 2 · Guided response	The teacher will demonstrate to each student the Samanya and Vishesha Shodhana of Parada by one reference. The teacher will help the students to observe the changes in the Parada before and after Shodhana, along with different stages of the process.
3.2	Parada Jarana, Parada Murchana and Bandhana processes	Doing 2 · Guided response	The teacher will demonstrate the Parada Jarana, Parada Murchana Bandhana by any one method. The students will note down the observations and record the findings

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
3.3	Demonstration of the Parada Samskara procedures- Swedana, Mardana, Murchana, and Utthapana	Doing 2 · Guided response	The teacher will demonstrate the selected procedures of Parada Samskara like, Swedana, Mardana, Murchana, and Utthapana. The observations, like changes in the Parada, total duration, temperature range, and yield, will be noted down by the students. Following this, the students will perform these Samskaras under the direct observation of the teacher, record their findings, and discuss.
MODULE 4 — MAHARASA DRAVYA, ITS PROCESSING AND UTILITY			
4.1	Analysis and Verification of Maharasa Dravya Samples. The students will identify and aut	Doing 2 · Guided response	—
4.2	Application of Shodhana, Marana, and Other Pharmaceutical Procedures for Abhraka and Vaikranta	Doing 2 · Guided response	The students will perform various procedures, such as Shodhana, Dhanyabhrakikarana, and Marana, related to Abhraka/Vaikranta under the teacher's supervision, following a selected method. Relevant observations and yield will be recorded, presented, and discussed in class. The discussion will include the topics like, basis of each type of sodhana/marana or each bhavana Dravya etc
4.3	Practical Application of Shodhana, Marana, and Other Pharmaceutical Procedures for Makshika and Vimala	Doing 2 · Guided response	The students will perform various procedures, such as Shodhana and Marana, related to Makshika and Vimala, using a selected method. Relevant observations and yield will be recorded, presented, and discussed in class.
4.4	Practical Application of Shodhana, Marana, and Other Pharmaceutical Procedures for Shilajatu and Sasyaka/ Tuttha	Doing 2 · Guided response	The students will perform various procedures, such as Shodhana and Marana, related to Shilajatu and Sasyaka/Tuttha, using a selected method. Relevant observations and yield will be recorded, presented, and discussed in class.
4.5	Practical Application of Shodhana, Marana, and Other Pharmaceutical Procedures for Chapala/Kharpara	Doing 2 · Guided response	The students will perform various procedures, such as Shodhana and Marana, related to Chapala/Kharpara, using a selected method. Relevant observations and yield will be recorded, presented, and discussed in class.
4.6	Physico-Chemical Characterization of Bhasma/Shodhita Products in Maharasa Dravya categor	Doing 2 · Guided response	—
MODULE 5 — CRITICAL APPRAISAL OF UPARASA DRAVYA AND SADHARANA RASA DRAVYA			
5.1	Authentication of Uparasa Dravyas	Doing 2 · Guided response	The teacher will select a drug from Uparasa category and will demonstrate the identification and authentication of the selected drug using classical Grahya Lakshana and mineralogical references with cross-verification through reference samples
5.2	Demonstration of Shodhana, Marana and other pharmaceutical procedures for Uparasa	Doing 2 · Guided response	The students will perform various procedures such as Shodhana, Marana, Pishti Nirmana etc related to Uparasa Dravyas under teacher's direct supervision, following a selected method. Relevant observations and yield will be recorded, presented, and discussed in class. The students may be allotted a selected Shodhana/Marana procedure of Uparasa Dravya mentioned in UG syllabus. The students will demonstrate the practical to UG students and discuss
5.3	Demonstration of Quality Control testing for the products of Uparasa Dravya	Doing 2 · Guided response	The teacher will select a processed drug from Uparasa category, such as a Shodhita Dravya, Bhasma or Pishti and will demonstrate the quality testing procedure using organoleptic, physico-chemical parameters and instrumental methods. The students will observe the procedures, record the findings, present the same and discuss
5.4	Identification and authentication of Sadharana Rasa Dravyas	Doing 2 · Guided response	The teacher will select a drug from Sadharana Rasa category and will demonstrate the identification and authentication of the selected drug using classical Grahya Lakshana, mineralogical characters and physico-chemical properties. The sample will be cross-verified with the reference samples.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
5.5	Demonstration of Shodhana, Marana and other pharmaceutical procedures for Sadharana Rasa	Doing 2 · Guided response	The students will perform various procedures such as Shodhana, Marana, Satva Patana etc related to Sadharana Rasa Dravyas under teacher's direct supervision, following a selected method. Relevant observations and yield will be recorded, presented, and discussed in class. The students may be allotted a selected Shodhana/Marana procedure of Sadharana Rasa Dravya mentioned in UG syllabus. The students will demonstrate the practical to UG students and discuss
5.6	Demonstration of Quality Control testing for the products of Sadharana Rasa Dravya	Doing 2 · Guided response	The teacher will select a processed drug from Sadharana Rasa category, such as a Shodhita Dravya, Bhasma and will demonstrate the quality testing procedure using organoleptic, physico-chemical parameters and instrumental methods. The students will observe the procedures, record the findings, present the same and discuss
MODULE 6 — CRITICAL APPRAISAL OF DHATU AND UPDHATU			
6.1	Demonstration of authentication of Dhatu Dravya	Doing 2 · Guided response	The teacher will select a Dhatu (metal) and demonstrate the identification and authentification of the selected Dhatu/metal using both classical and modern identification techniques using Grahya Lakshana and mineralogical methods. The students will not the observations and discuss
6.2	Pharmaceutical processing (Shodhana & Marana) of metals	Doing 2 · Guided response	The teacher will select a Dhatu(metal) and explain the pharmaceuticl process for that metal like, Shodhana and Marana. The students will perform these pharmaceutical procedures under the guidance of the teacher. Later, the students will demonstrate the relevant pharmaceutical process of a metal to the UG students as per the NCISM syllabus, describe and discuss under the direct sipervision of the teacher
6.3	Quality control of metallic preparations	Doing 2 · Guided response	The teacher will select a processed drug from Dhatu category, such as a Shodhita Dravya or Bhasma and will demonstrate the quality testing procedure using organoleptic, physico-chemical parameters and instrumental methods. The students will observe the procedures, record the findings, present the same and discuss
6.4	Identification of Upadhatu samples	Doing 2 · Guided response	The teacher will select a drug from Upadhatu category and will demonstrate the identification and authentication of the selected drug using classical Grahya Lakshana and mineralogical references with cross-verification through reference samples
6.5	Shodhana and Marana of Upadhatu	Doing 2 · Guided response	The students will perform various procedures such as Shodhana and Marana related to Upadhatu Dravyas under teacher's direct supervision, following a selected method. Relevant observations and yield will be recorded, presented, and discussed in class. The students may be allotted a selected Shodhana/Marana procedure of UpardhatuDravya mentioned in UG syllabus. The students will demonstrate the practical to UG students and discuss
6.6	Demonstrate the Quality control of sub-metallic formulations	Doing 2 · Guided response	The teacher will select a processed drug from Upadhatu category, such as a Shodhita Dravya, Bhasma or a formulation containing Upadhatu Bhasma and will demonstrate the quality testing procedure using organoleptic, physico-chemical parameters and instrumental methods. The students will observe the procedures, record the findings, present the same and discuss
MODULE 7 — SUDHA VARGA, SIKATA VARGA, VISHA VARGA DRAVYAS			
7.1	Demonstration of Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya	Doing 2 · Guided response	The teacher will select a drug from Sudha Varga and explain the Standard Operating Procedure (SOP) for Shodhana, Marana, and Pishti Nirmana based on a selected classical reference. Under direct supervision, students will then carry out Shodhana, Marana, and Pishtikarana of the selected drug, following the SOP. Students will record their observations and engage in a discussion.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
7.2	Demonstration of the quality testing procedures for Bhasma/Pishti of Sudha vargeeya Dravya	Doing 2 · Guided response	The teacher will select a sample of Bhasma/Pishti from Sudha Varga Dravya and demonstrate quality testing methods for the product, covering organoleptic, physicochemical parameters, and NPS tests. Students will observe the demonstration and record their observations.
7.3	Demonstration of Shodhana, marana and Pishti Nirmana of Sikata varga Dravya	Doing 2 · Guided response	The teacher will select a drug from Sikata Varga category and explain the Standard Operating Procedure (SOP) for Shodhana, Marana, and Pishti Nirmana based on a selected classical reference. Under direct supervision, students will then carry out Shodhana, Marana, and Pishtikarana of the selected drug, following the SOP. Students will record their observations and engage in a discussion.
7.4	Demonstration of the quality testing procedures for Bhasma/Pishti of Sikata vargeeya Dravya	Doing 2 · Guided response	The teacher will select a sample of Bhasma/Pishti from Sikata Varga Dravya and demonstrate quality testing methods for the product, covering organoleptic, physicochemical parameters, and NPS tests. Students will observe the demonstration and record their observations.
7.5	Demonstration of Shodhana Process of Visha Dravya	Doing 2 · Guided response	The teacher will select a Visha Dravya and demonstrate one of its Shodhana processes, explaining the Standard Operating Procedure (SOP) involved. Students will then perform the Shodhana process under the teacher's direct supervision, adhering to the SOP.
7.6	Demonstration of the analytical testing process for Shodhita Visha dravya	Doing 2 · Guided response	The teacher will select a Visha Dravya and demonstrate the testing of the sample before and after Shodhana, using parameters such as organoleptic characters, physico-chemical parameters, qualitative phytochemical tests, and Thin Layer Chromatography (TLC).
MODULE 8 — APPRAISAL OF RATNA AND UPARATNA AND THERAPEUTIC SIGNIFICANCE OF PISHTI AND DRUTI KALPANA			
8.1	Mineralogical identification of precious and semiprecious stones under Ratna and Uparatna varga	Doing 2 · Guided response	The teacher will select a precious or semi-precious stone and demonstrate its identification using identification keys. The teacher will then discuss the physico-chemical properties and crystallography of the identified specimen.
8.2	Demonstration of SOP of Pishti Kalpana of Ratna-Uparatna	Doing 2 · Guided response	The teacher will select a drug from Ratna/Upartna Varga and instruct students to prepare its Pishti under their supervision and guidance. Students will then prepare the Pishti according to the Standard Operating Procedure (SOP) and present their observations.
8.3	Demonstration of different methods of Gandhaka Druti and Gandhaka taila	Doing 2 · Guided response	Under the direct supervision of the teacher, students will prepare Gandhaka Druti and Gandhaka Taila using various classical and modified methods as instructed. The students will then discuss their observations, note the physico-chemical characteristics, and explore the therapeutic applications of the prepared formulations.
8.4	Demonstration of the Pharmaceutical modification of Gandhaka Druti and Gandhaka Taila	Doing 2 · Guided response	The teacher will demonstrate the pharmaceutical modification of Gandhaka Druti/Taila into ointment/cream using suitable excipients. The students will observe the procedure and prepare the same under the guidance of the teacher.
MODULE 9 — SCIENTIFIC FUNDAMENTALS OF AYURVEDIC PHARMACEUTICAL PREPARATIONS			
9.1	Demonstration of unique Panchavidha Kashaya Kalpana	Doing 3 · Mechanism	The teacher will instruct the students to explore the classical references for unique Panchavidha Kashaya Kalpana like, Agnisiddha and Anagni Siddha Swarasa, Saptavidha Kashaya Kalpana, Sagni and Niragni Hima Kalpana etc along with the commentaries, and outline all steps of each preparation. The students will perform the fselected practical under the observation and guidance of the teacher. Each student will perform a different formulation, record their observations, and discuss

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
9.2	Demonstration of the application of QAQC parameters used for quality control of herbal drugs	Doing 2 · Guided response	The teacher will demonstrate the quality testing protocol of raw materials like organoleptic characters, estimation of foreign matter, bulk density, tapped density, extractive values, ash values, specific gravity, and refractive indexetc selecting an appropriate raw drug sample. The students will carry out the practical using different samples under the guidance of the teacher, record their findings, and compare them with the pharmacopoeial standards
9.3	Demonstration of formulations to understand uniqueness of proportion of the ingredients	Doing 3 · Mechanism	—
9.4	Assessment of Phytochemicals in the prepared dosage form by TLC The faculty will select	Doing 2 · Guided response	—
9.5	Demonstration of Kwatha Kalpana considering its process optimization	Doing 2 · Guided response	The faculty will choose a specific kwatha from a classical text for each student. Each student will demonstrate the methodical preparation of allotted kwatha. Everyone will observe the changes that occur by varying the proportions of water, temperature, and time. To explore various factors that affect the optimization of the process, the same kwatha preparation will be conducted multiple times to note the differences. Each student will be given drugs with different hardness and asked to prepare Kashaya using different ratio of water. The students will prepare kashaya samples, compare the time taken for the preparation, comparatively analyse them using organoleptic paramaters and basic physico-chemical parameters and present their findings.
MODULE 10 — PRINCIPLES OF PHARMACEUTICAL FORMULATION			
10.1	Demonstration of Samskara and alterations in the qualities observed during the preparati	Doing 2 · Guided response	—
10.2	Demonstration of formulations prepared with specific Aushadhi Gana and various classes(	Doing 2 · Guided response	—
10.3	Demonstration of Dosage Form	Doing 2 · Guided response	Design. Faculty will arrange a group discussion on the Principles of dosage form design in Modern Pharmaceutical Science 1) Pharmaceutical and Formulation considerations for dosage form design 2) Biopharmaceutical and Pharmacokinetic considerations for dosage form design Each student will develop and demonstrate any 2 formulations(for example- Mouthwash from Triphala decoction, Tooth paste from Bakula Pushpa churna, Ointment from Vrana Shodhana Taila, Cream from Kunkumadi taila, etc.)
10.4	Demonstration and analysis of the importance of preservatives, additives, excipients, fi	Doing 2 · Guided response	—
MODULE 11 — MANA PARIBHASHA (METROLOGY IN AYURVEDIC PHARMACEUTICAL PREPARATION)			
11.1	Application of Knowledge of mana Paribhasha in the preparation of various formulations	Doing 2 · Guided response	The students will design a chart of classical manaparibhasha applied to Ayurvedic Pharmaceutical preparation of Various dosage forms. They will study the following examples for evaluating the use of mana paribhasha in the preparation of the following classical formulations. Accordingly, each student will prepare a selected formulation from the list given below or any other relevant formulation under the direct observation of the teacher. 1.Bhagottara churna 2.Bahushal Guda 3.Yogaraja guggulu 4.Kutajavaleha 5.Kasisadya ghruta 6.Kutajarishta 7.Lauhadi Rasayana - critically appraise the measurements mentioned for Ayurveda 8. Pathya Ghruta

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
11.2	Unit Conversion for Pautava and Druvaya Mana	Doing 3 · Mechanism	Each student will study the Unit Converter for Pautava and Druvaya Mana Each student will select any 10 Dry samples of various sources and observe their mana based on unit conversion. A discussion will be held on the observations and application of this information.
11.3	Preparation of Formulations using the Volumetric Method of Measurement	Doing 2 · Guided response	Faculty will describe the differences between volumetric and gravimetric methods of measurement for various drugs, allowing students to explore the application of the volumetric method in preparing classical formulations. Preparation of Amruta Kwatha, Bala Kwatha, Kutaja Kwatha, Sharavari Kwatha, and Bhrungaraja Kwatha will be done using the volumetric method for measurement. Faculty will elaborate on the difference in the total amount of water required for the formulations when water is measured using the volumetric method. Verify the difference observed between the total amount of water required for the formulation when water is taken by the volumetric method.
11.4	Preparing formulations with a diverse range of raw materials. Faculty will elaborate on	Doing 3 · Mechanism	—
11.5	Demonstration of the machinery used for large-scale production in Ayurvedic pharmacies, with their mechanism	Doing 3 · Mechanism	The department will arrange a visit to a pharmacy where bulk production is carried out using advanced machinery. OR A video demonstration will be arranged to cover the assembly and operation of the equipment used for bulk production. Each student will-Observe and understand the working of these machines, along with understanding their working principle. Operate these machines in the teaching Pharmacy unit. Assess the modifications and impacts of bulk production on the final product. Explain the structure and operation of all these machines. Discuss with production unit personnel about the advantages and disadvantages of the machines used in Ayurvedic pharmacies.
MODULE 12 — PROCESS VALIDATION PROTOCOL			
12.1	Develop Process validation Protocol and demonstrate the preparation of formulations foll	Doing 2 · Guided response	—
12.2	Develop Process validation Protocol and demonstrate the preparation of formulations following SOP Each	Doing 2 · Guided response	Student will conduct practical sessions for undergraduate students(Minimum 3 for each PG student) to demonstrate, observe, assist, and perform the following practical as part of their NCISM curriculum. 1) Arka- Gomutra arka 2) Kshara-Palasha kshara 3) Lepa- Dashanga lepa 4) Malahara- Gairikadya malahara 5) Upanaha- Atasi upanaha For Netra Kalpana- Seka, Ashchyotana, Pindi, Bidal, Tarpana, Anjana, Putapaka- (Observe in Shalakya Netra department)
MODULE 13 — KRUTANNA (AYURVEDIC FOOD PREPARATIONS)			
13.1	Formulations from Classics prepared with aahara Dravya	Doing 2 · Guided response	The teacher will instruct each student to refer the classical texts for ingredients and proportions of all the formulations listed here. The students will prepare any 4 selected formulations following SOP. under the guidance of teacher. 1) Toya varga- Shadaga Paniya, Ushnodaka 2) Kshira varga- Shunthi Kshirapaka, Takrarishta, Tila navaneeta yoga, Panchagavya ghruta 3) Ikshu varga- Sharkara kalpana, Guda chatushtaya gutika 4) Madhu varga- Madhvasava, Madhu shukta 5) Taila Varga- masha Saindhava Taila, hingwadya Taila 6) Madya Varga - Yava Sura, Mardika, Varuni

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
13.2	Demonstration of therapeutic diet formulations	Doing 2 · Guided response	The teacher will instruct the students to refer the classical texts and explore disease specific diet formulations, understand their classical description and specific therapeutic utility. The students will prepare any 4 formulations following SOP under the supervision of the teacher. They may select the formulations from the list mentioned below or any other formulations as instructed by the teacher. 1) Jvara- - (Charaka Chi. 3/ 178 to 196) 2) Rakta pitta Tarpana- (Cha. Chi 4/33-39) 3) Aushadhi Siddha Yavagu - (Cha. Su. 2- summarize all 28 Yavagu) Prepare Gavedhuka Yavagu if material is available. 4) Amavata- Sapta mushtika Yusha- (Sharangadhara Ma. Kh. 2/164) 5) Chhardi - Draksha/ Kapittha Panaka (Ch.chi 20/36) 6) Atisara- Appraise formulations of Yavagu, Vilepi, Khada, Yusha, Rasaudana and their administration in different clinical conditions of atisara.(Ch.chi 19/23-41)
13.3	Demonstrate Protocol for the selected Aahara Kalpana-Each student will Demonstrate/ Obse	Doing 2 · Guided response	—
13.4	Demonstration of the preparation of novel Ayurvedic food products	Doing 2 · Guided response	The teacher will demonstrate selected novel Ayurvedic food products such as biscuits/cookies, candy/chocolate and granules and explain the SOP. The students will observe the procedure and prepare the same under the direct observation and guidance of the teacher. Based on the ingrdients the students will calculate the nutritional values of the products prepared
13.5	Demonstration of food processing techniques used in current Ayurvedic food Industries	Doing 2 · Guided response	The department will arrange a visit to the nearest food industry after getting prior permission. The students will observe the food processing technology, quality check, packaging etc. They will lcritically assess the potential effects of processing techniques on food quality. OR A visit will be arranged to the Ayurvedic pharmaceutical unit that prepares nutraceutical products, where the students will learn the entire process of manufacture, quality check, packaging etc. OR. Visit Ayurvedic food manufacturing units, ensuring that at least one is GMP-certified and FSSAI-approved, and observe various food processing OR A video demonstration of a food processing unit including various techniques, machinery, QC methods, packaging, research etc. The students will observe the video, record their findings, present them, and discuss. The students will visit the factory where freeze drying and vacuum drying, and other drying technologies are used, and understand the technique or a video demonstration of the same may be arranged. Students will review ready-to-cook (RTC) and ready-to-drink (RTD) Ayurvedic foods in the market.
MODULE 14 — AYURVEDIC FORMULATIONS FOR HOLISTIC WELLBEING, VITALITY AND BODY CARE			
14.1	Demonstration of Rasayana Formulations. Each student will prepare a selected Rasayana product under the supervision of the teacher, following the SOP. The teacher will allot any of the following products or any other Rasayana formulation. Chyavanprash Chaturvidha Triphala Rasayana Vrushya payasa yoga Apatyakara Ghruta Every year	Doing 3 · Mechanism	Faculty can allot different formulations selected from the classical texts.
14.2	Demonstration of Body care products. The teacher will demonstrate the preparation of a s	Doing 3 · Mechanism	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
14.3	Demonstration of the preparation of Modified Cosmeceuticals	Doing 2 · Guided response	The teacher will ask each student will search e-resources to assess the expectations of end users from the herbal cosmetics Industry. Based on the information collected, each student will design at least one their own innovative modified Cosmeceutical formulation under the guidance of the teacher developing an SOP. The students will note down the challenges faced, corrective actions done during the preparation, present their findings and discuss
MODULE 15 — DISEASE CONDITION SPECIFIC SELECTION OF FORMULATION			
15.1	Demonstration of formulations used in various branches of Ayurveda. Each student will prepare at least one formulation from each group under the direct observation of the teacher: 1.Kaumarbhrtya- Balachaturbhadra Rasa, Preenana modaka 2. Striroga- Pushyanuga Churna, PhalaGhruta 3.Garbhini- Garbhapal Rasa, Formulations from Masanumasiki Garbhini paricharya 4. Sutika- Bala taila, stanyashodhana Yusha 5. Shalyatantra- Kshara Sutra, Vranashodhan Taila, Murivenna 6. Shalakya Tantra- Bilwa taila / Irimedadi taila / Chandrodaya Varti 7.Kayachikitsa- Yavani shadava Churna, Vasaghruta 8. Panchakarma- Niruha Basti Samyojana, Pradhaman Nasya Yogas, Ghreya Virechana Yoga 9.Rasayana-Bhallataka Taila, Medhya Vachayoga 10.Vajikarana- Vrishya Payasa Yoga, Shatavari Ghruta	Doing 2 · Guided response	The teacher will explain and discuss the unique combination of ingredients, their proportions, SOP, endpoint, Matra, Anupana, and time for administration, probable mode of action, therapeutic efficacy, and relevant research articles
15.2	Survey the different dosage forms used by local health practitioners. The teacher will i	Doing 2 · Guided response	—
15.3	Demonstration of frequently used formulation for Shodhana and Shamana Chikitsa of common	Doing 2 · Guided response	—
MODULE 16 — CURRENT TRENDS IN AYURVEDIC PHARMACEUTICAL INDUSTRY.			
16.1	Market Status of Ayurvedic Pharmacy Industry	Doing 2 · Guided response	Students will-Conduct surveys, interviews, and focus groups with consumers, retailers, manufacturers, and healthcare professionals. Resourses Utilize market reports, industry publications, and academic research to gather data and insights.
16.2	Internship / Visits to GMP-certified Ayurvedic pharmacy Units(Minimum 5 units- Small/ Medium/ Large)	Affective · Valuing	The department will arrange a visit/training (internship) in a GMP-certified pharmacy. The students will observe the daily work patterns in all sections of the Pharmacy and get an on-hand training in various sections (based on the availability). They will also learn about inventory management and the documentation pattern of the Pharmacy. They will submit the report after the training session.
16.3	Perform any 5 QC practicals on formulation prepared by yourself. Faculty should demonstr	Doing 2 · Guided response	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 17 — GUIDELINES FOR ADMINISTRATION OF RASOUSHADHI IN CLINICAL PRACTICE			
17.1	Demonstration of the rational prescription for a selected disease	Doing 2 · Guided response	The teacher will create a case scenario of a clinical condition and demonstrate the clinical prescription of treatment modalities of Ayurveda, such as Shodhana, Shamana, Rasayana, and Vajikarana. Based on the demonstration, students will follow the same procedure in another disease, present the same, and discuss with a rationale.
17.2	Demonstration of the need and role of Kshetrikarana	Doing 2 · Guided response	The teacher will demonstrate the identification of the patients requiring kshetrikarana and explain why kshetrikarana is required in that case; Students will observe and record the agni- dosha- dushya and readiness for the administration of rasaoushadhi in that case. Based on the demonstration, each student will be given a different case scenario. Students will analyse the case and explore the requirement of Kshetrikarana in that case and design the protocol of Kshetrikarana.
17.3	Demonstration of common pathya and apathya Ahara during rasa yoga sevana	Doing 2 · Guided response	The teacher will provide a list of common Ayurvedic Pathya and Apathya Ahara based on classical guidelines. The teacher will assign each student a different list of Rasayoga. The students will prepare charts for the given rasa yogas and suggest appropriate pathya-apathya Ahara accordingly.
17.4	Demonstration of the common pathya-apathya Vihara during rasa yoga sevana	Doing 2 · Guided response	The teacher will provide a list of common Ayurvedic Pathya and Apathya Vihara based on classical guidelines, such as sleep, daily regimen, seasonal modifications etc. The teacher will assign each student a different list of Rasayoga. The students will prepare charts for the given rasa yogas and suggest appropriate pathya- apathya Vihara accordingly
17.5	Determination of signs and symptoms of rasa vikaras and their management	Doing 2 · Guided response	The teacher will demonstrate a case study of the patient experiencing side effects of Rasa Oushadhies. The students will identify and record the key signs and symptoms and analyse them. Accordingly, they will prepare at least one formulation for the management of the side effects.
17.6	Comparative assessment of Ashuddha and Shuddha Bhasma effects	Doing 2 · Guided response	The teacher will create two sets of case scenarios, one with properly prepared Bhasma and the other with improperly prepared Bhasma. Students will present variations in clinical outcomes and adverse reactions.
17.7	Identification of heavy metal toxicity	Doing 2 · Guided response	The teacher will give the clinical description of a case scenario with suspected heavy metal toxicity. The student will identify the heavy metal responsible for the toxic symptoms.
MODULE 18 — PRAYOGIKA BHAISHAJYA KALPANA(CLINICAL APPLICATION OF PHARMACEUTICAL PRODUCTS)			
18.1	Demonstration of chrono therapeutic alignment of Oushadha Sevana Kala with patient–disea	Doing 2 · Guided response	—
18.2	Demonstration of Modulation of dosage form action by chronotherapeutic administration in Ayurveda	Doing 4 · Adaptation	The teacher will select a dosage form and demonstrate the suitable Oushadha Sevana Kala for the selected dosage form based on classical guidelines of Oushadha Sevana Kala and Chronotherapeutics.
18.3	Demonstration of appropriate Anupana for the selected dosage form in different clinical	Doing 2 · Guided response	—
18.4	Demonstration of the calculation of paediatric dose using different formulae. The teache	Doing 2 · Guided response	—
18.5	Demonstration of the Patient-centric evaluation of Matra. The teacher will demonstrate m	Doing 2 · Guided response	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
18.6	Re-analysing classical ayurvedic drugs for novel indications	Doing 2 · Guided response	The teacher will select a classical Ayurvedic formulation and list out the ingredients and therapeutic indications of the formulation. With the available modern research articles on the pharmacological insights of the formulation's ingredients, the teacher will propose and justify a potential new therapeutic use for that formulation.
MODULE 19 — SOP AND CLINICAL APPLICATION OF CHATURVIDHA RASAYANA			
19.1	DOAP of preparation and SOP of Kharaleeya Rasayana. The teacher will select a Khalveeya	Doing 2 · Guided response	—
19.2	Demonstration of the method of preparation based on SOP of parpati kalpana. The teacher	Doing 2 · Guided response	—
19.3	Demonstration of the method of preparation and SOP of pottali kalpana. The teacher will	Doing 2 · Guided response	—
19.4	Demonstration of the method of preparation and SOP of Kupipakwa Rasayana. The teacher wi	Doing 2 · Guided response	—
MODULE 20 — SOP AND CLINICAL APPLICATION OF AYASKRITI, LOUHA, AND MANDURA KALPA			
20.1	Demonstration of the SOP and preparation of Ayaskriti. The teacher will describe the com	Doing 2 · Guided response	—
20.2	Demonstration of the SOP and preparation of Louha Kalpa. The teacher will select a Louha	Doing 2 · Guided response	—
20.3	Demonstration of the SOP and preparation of Mandura Kalpa.. The teacher will select a Ma	Doing 2 · Guided response	—
20.4	Clinical application of mandura kalpa. The teacher will demonstrate the clinical applica	Doing 2 · Guided response	—
MODULE 21 — CLINICAL APPLICATION OF RASOUSHADHI ACCORDING TO SROTAS AND IN SELECTED CHALLENGING CLINICAL CONDITIONS			
21.1	Demonstration of the clinical application of Rasoushadhi in selected Srotas, illustratin	Doing 2 · Guided response	—
21.2	Demonstration of the clinical applicability of Rasoushadhi in Medical Emergencies	Doing 2 · Guided response	The teacher will select a particular emergency condition and identify the applicable Rasa Oushadhi from Chaturvidha Rasayana. The teacher will then select a specific formulation and instruct the students to prepare it under their direct supervision. Students will record their observations and participate in a discussion. Following this, the teacher will demonstrate the rationality and probable mode of action of the selected formulation, drawing on classical properties, pharmacology of individual ingredients, and the method of preparation. Additionally, the teacher will analyze the findings in relation to published research works.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
21.3	Demonstration of role of Rasoushadhi in clinically challenging conditions	Doing 2 · Guided response	The teacher will select a relevant Rasoushadhi applicable in clinically challenging conditions, such as neurodevelopmental and neurodegenerative diseases. The teacher will then demonstrate the clinical application of the selected formulation, including the suitable Anupana (vehicle) and route of administration. Furthermore, the teacher will discuss the probable mechanism of action of the selected formulation, drawing on the classical properties and pharmacology of its ingredients. OR The teacher will demonstrate the therapeutic application of selected Rasoushadhies in managing specific, clinically challenging conditions within their respective outpatient departments (OPDs).
21.4	Demonstration of the role of Rasoushadhi as Rasayana in clinically challenging condition	Doing 2 · Guided response	—
MODULE 22 — YOGAS FROM DIFFERENT KALPANAS			
22.1	Demonstration of the preparation of Vati, Guggulu Kalpa, and their modified modern dosage forms	Doing 2 · Guided response	The teacher will select one formulation each from Vati and Guggulu categories. The students will prepare the selected formulations under the direct supervision of the teacher. Under the guidance of the teacher, the students will modify the selected formulation into tablet/capsule form using suitable excipients, present their observations, and discuss. Then the teacher will demonstrate the clinical utility of the prepared suppository based on pharmacological properties, classical descriptors, and the possible mode of action based on available published works.
22.2	Designing an S.O.P for the preparation of Ayurvedic suppository specifying its clinical utility based on Varti Kalpana	Doing 2 · Guided response	The Teacher will select a Varti formulation and instruct the students to prepare the formulation under the direct supervision of the teacher. The students will modify the Varti into modern dosage form suppository, using suitable excipient and mould under the guidance of the teacher. Then the Teacher will demonstrate the clinical utility of the prepared suppository based on pharmacological properties, classical descriptors, and the possible mode of action based on available published works.
22.3	Demonstration of the role of pharmaceutical additives and Samyak Paka of Avaleha	Doing 2 · Guided response	The teacher will assign each student to prepare an Avaleha, Paka, and Khanda Yoga and mark the critical end point based on Paka Lakshanas. The students will differentiate the Paka Lakshanas of Avaleha, Paka, and Khanda Kalpana and present their findings. The teacher will instruct the students to assess the authenticity and genuineness of the spices and condiments used in the preparation by comparing them with the pharmacopeial standards. Then the teacher will demonstrate how the pharmaceutical additives and leha paaka influence the clinical activity based on classical descriptors and using published research works.
22.4	Demonstration of S.O.P for a selected Sneha Kalpa which is widely used and determine its analytical profile	Doing 2 · Guided response	The teacher will assign each student a different Sneha Kalpa. The students will prepare the given formulation under the direct supervision of the teacher. The students will record the findings like, temperature changes, critical end points etc. The students will be instructed to analyze the prepared Sneha Kalpa using relevant quality control parameters and present their findings. Then the Teacher will demonstrate how the analytical profiling done matches with the pharmacopeial standards to ensure the quality of the Sneha Kalpa, which in turn will influence the clinical activity.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
22.5	Demonstration of S.O.P for the preparation of a selected Arka and Dravaka and create their analytical profile. The teacher will assign each student to prepare an arka and dravaka each. The students will prepare the given products following the SOP under the direct supervision of the teacher. The teacher will instruct the students to analyse the prepared products following relevant quality control parameters. The students will carry out the analysis accordingly under the guidance of the teacher and present their observations. Then the	Doing 2 · Guided response	Teacher will demonstrate how the pharmacopeial standards and classical descriptors help in determining the quality of the product, which in turn will influence the clinical activity.
22.6	Demonstration of S.O.P for preparing a selected sandhana Yoga and determine the analytical standards	Doing 2 · Guided response	The Teacher will select a Sandhana Yoga (Madhura and Amla) and assign the students to prepare the given formulation following SOP under the direct supervision of the teacher. The students will analyse the product after the completion of given formulation using relevant analytical parameters, including alcohol percentage. Then the teacher will demonstrate how the percentage of alcohol is significant in post-marketing surveillance. Note: It is expected that each batch will prepare a different Sandhana Kalpana, and the analysis will be carried out here only after the completion of fermentation.
MODULE 23 — SOP AND CLINICAL APPLICATIONS OF CHURNA KALPANA(POWDER DOSAGE FORMS)			
23.1	Demonstration of the SOP and quality analysis of a selected Churna yoga	Doing 2 · Guided response	The teacher will select different Churna Yogas according to the particle size (Sukshma churna, Yavakuta churna etc). The teacher will allot the selected Churna Kalpa to the students. The students will authenticate raw materials in the selected Yoga and will apply processing techniques like powdering, sieving using an appropriate mesh, and homogenous mixing of ingredients. Then the students will carry out the quality analysis of the prepared Churna following appropriate organoleptic, physico-chemical, particle size estimation, and other relevant parameters under the guidance of the teacher. The findings will be compared with the pharmacopoeial standards
23.2	Demonstration of the preparation of a selected Paneeya Kshara and Pratisaraneeya Kshara and their quality analysis	Doing 2 · Guided response	The teacher will assign students to collect and authenticate raw materials required for Kshara preparation. Students will apply processing techniques as per the provided SOP. Samples will be collected at the Paneeya Kshara and Pratisaraneeya Kshara stages. Under the teacher's guidance, collected samples will undergo quality testing using relevant parameters. The results will be compared with pharmacopeial standards to ensure compliance.
23.3	Demonstration of the preparation of a selected Lavana formulation following different methods and analytical profiling	Doing 2 · Guided response	The teacher will select a Lavana Kalpa and instruct students to prepare the product following the standard operating procedure (SOP). A Lavana Kalpa with multiple preparation methods (e.g., Sajala and Nirjala Narikela Lavana) will be chosen. Under the teacher's guidance, students will prepare the Lavana Kalpa, record observations, present, and discuss their findings. Additionally, students will analyze the prepared Lavana Kalpa using relevant quality parameters and compare the results with pharmacopeial standards. Alternatively, students may demonstrate the practical preparation of Lavana Kalpa to undergraduate students, as per the syllabus, under the teacher's supervision.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
23.4	Demonstration of the preparation of a selected Satva and performing its analytical profiling	Doing 2 · Guided response	The teacher will assign the students to collect and authenticate raw materials needed for the preparation of the selected Satva. Under the teacher's supervision, students will apply standard processing techniques for its preparation. The students will then perform various analytical tests as per pharmacopeial standards and create an analytical profile of the taken Satva.
23.5	Demonstration of the preparation of Masi and its analytical profiling	Doing 2 · Guided response	The teacher will select a Masi Kalpa prepared by different methods (Antardhuma and Bahirdhuma). Students will prepare Masi using these methods as per the SOP, guided by the teacher. Under the teacher's supervision, students will analyze the prepared Masi samples using relevant quality parameters, present their findings, and discuss.
23.6	Demonstration of the pharmaceutical modification of Masi Kalpana	Doing 2 · Guided response	The teacher will select a Masi Kalpa and demonstrate its modification into modern topical dosage forms, such as ointments, creams, and gels, using suitable bases and excipients. Students will observe the process and prepare the products under the teacher's supervision, then present their observations and discuss
MODULE 24 — THERAPEUTIC SIGNIFICANCE OF TOPICAL DOSAGE FORMS			
24.1	Demonstration of various Netra Kalpa and their clinical applications	Doing 2 · Guided response	The teacher will assign each student a unique Netra Kalpa to prepare under guidance. Students will then analyze its pharmacotherapeutic properties, identifying its effectiveness in eye diseases, and explain its probable mode of action based on pharmacological properties, classical descriptors, and research updates
24.2	Preparation of a novel dosage form using one of the classical lepa Yoga	Doing 2 · Guided response	The teacher will assign the students to authenticate raw materials in the selected Yoga. Then, the students will identify the pharmacotherapeutic properties of the selected Yoga, listing its ingredients, proportion, method of preparation, dose, mode of application, and explain its probable mode of action based on pharmacological properties, classical descriptors, and available research updates. The teacher will demonstrate the conversion of Lepa into a bandage using a suitable base. The students will prepare the same following the teacher's guidelines. Pharmaceutical modification of the classical Lepa into other novel forms like cream, ointment, gel, or liniment can also be selected for practical training.
24.3	Preparation of a selected Upanaha which is useful in treating Shopha	Doing 2 · Guided response	The teacher will assign the students to authenticate raw materials in the selected Yoga. Then, the students will identify the pharmacotherapeutic properties of the selected Yoga, which makes it effective as a bandage, and list the ingredients, proportion, method of preparation, dose, and mode of application. They will also explain the probable mode of action based on pharmacological properties, classical descriptors, and available research updates. Furthermore, students will prepare the formulation and create an analytical profile of the selected Yoga as per the Pharmacopeia standards to standardize the formulation.
MODULE 25 — ARTIFICIAL INTELLIGENCE IN DRUG DEVELOPMENT AND PHARMACOGENETICS; PHARMACOGENOMICS			
25.1	Demonstration of training in the basics of machine learning and its application	Doing 2 · Guided response	The teacher/experts in the field will give a Brief demo or guided tour of AI used cases (target identification, virtual screening)
25.2	Prediction of drug likeliness using machine learning	Doing 5 · Origination	Teacher/ experts will instruct/guide the scholars to load a dataset of small molecules (Ex. SMILES strings with molecular descriptors and bio-activity)
25.3	Demonstration of different 3D printers used in the drug delivery system	Doing 3 · Mechanism	The teacher/expert in the field will conduct a video demonstration of 3D printers used in the drug delivery system. Students will observe, record their findings, and present for discussion

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
25.4	Comparison of traditional dosage forms versus 3D printed ones regarding the structure and dissolution	Doing 4 · Adaptation	Teacher/experts in the field will conduct a video demonstration on the Comparison of traditional dosage forms versus 3D printed ones regarding the structure and dissolution
25.5	Demonstration of the prediction of Drug-Target Interaction	Doing 2 · Guided response	The teacher or expert in the field has to demonstrate Drug-Target Interaction Prediction by using tools like Swiss Target Prediction (Web-based), STITCH (Web-based, Binding DB (Database), Python + scikit-learn or Deep Purpose (for coding-based practical), etc.
MODULE 26 — PHARMACOLOGY			
26.1	Demonstration of pharmacological experiments using videos or simulation software	Doing 2 · Guided response	The teacher will demonstrate selected five pharmacological experiments like, study of different routes of drug administration in mice/rats, study of effect of hepatic microsomal enzyme inducers on the phenobarbitone sleeping time in mice, effect of drugs on ciliary motility of frog oesophagus, effect of drugs on rabbit eye, effects of skeletal muscle relaxants using rota-rod apparatus using videos or simulation software. Students will observe, discuss, and record the demonstration
26.2	Demonstrate the effect of drugs on animals by simulated experiments	Doing 4 · Adaptation	The teacher will demonstrate five selected pharmacological experiments like, effect of drugs on locomotor activity using actophotometer, anticonvulsant effect of drugs by MES and PTZ method, study of stereotype and anti-catatonic activity of drugs on rats/mice, study of anxiolytic activity of drugs using rats/mice and study of local anaesthetics by different methods. Students will observe, discuss and record the demonstration
MODULE 27 — NETWORK PHARMACOLOGY & REVERSE PHARMACOLOGY			
27.1	Demonstration of network pharmacology applications with the help of a video	Doing 2 · Guided response	The teacher will select a drug and demonstrate network pharmacology applications of the selected drug, with the help of a video demonstration, covering topics like polypharmacology, biological networks, computational approaches, drug repurposing, and systems pharmacology. The students will observe and record findings
27.2	Demonstration of reverse pharmacology applications with the help of a video	Doing 2 · Guided response	The teacher will select a drug and demonstrate reverse pharmacology applications of the selected drug, with the help of a video demonstration including the steps (1 hour for each step) like, Bioactivity Screening, Target Identification, Mechanism of Action Studies, Systems Biology, Clinical Observations, Computational Modelling, Animal Models, Safety and Toxicity Assessment, Ethnopharmacology and Phytochemical Analysis
MODULE 28 — PHARMACEUTICAL QUALITY ASSURANCE			
28.1	Demonstration of the process of mock GMP Inspection	Doing 2 · Guided response	The teacher will demonstrate the process of a mock GMP inspection by assigning different roles to the students, as QA officers and pharmacy personnel. The students will perform audits, check documentation, cleanliness, personnel hygiene, equipment logs etc
28.2	Demonstration of GMP Hierarchy and Responsibilities	Doing 2 · Guided response	The teacher will demonstrate GMP Hierarchy and assign Responsibilities, such as Quality Assurance (QA), Quality Control (QC), Production, Warehousing, Maintenance, Regulatory Affairs, Top Management, etc, to each student. The students will present their responsibilities, experiences, and challenges
28.3	Demonstration of Identification and Classification of Packaging Materials	Doing 2 · Guided response	The teacher will demonstrate a range of containers and packaging samples (e.g., glass bottles, blisters, plastic containers, aluminum foil, cartons) to the students, and the students will classify them as primary or secondary packaging and identify material type (HDPE, glass, PVC, etc.) and Record suitability for different dosage forms

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
28.4	Demonstration for SOP Writing of different equipment	Doing 2 · Guided response	The teacher will demonstrate SOP for an equipment or a lab process (e.g., pH measurement, reagent preparation), and the students will be asked to write a Standard Operating Procedure (SOP) of the demonstrated instrument
MODULE 29 — PHARMACY PRACTICE			
29.1	Drug Information and Case Analysis	Doing 2 · Guided response	The teacher will demonstrate a systematic approach to drug information queries using primary, secondary, and tertiary sources. The teacher will interpret lab reports to optimize drug therapy in real patient cases.
29.2	Prescription Handling and Dispensing	Doing 2 · Guided response	The teacher will demonstrate Prescription Handling, including legal compliance, checking for completeness, and identifying potential drug-drug interactions.
29.3	Data Interpretation and Pharmacokinetic Calculations	Doing 2 · Guided response	The teacher will select a formulation and demonstrate pharmacokinetic formulas, including volume of distribution, clearance, elimination constant, half-life, and calculation of loading and maintenance doses using nomograms and tailored regimens
29.4	Formulary Review and Management	Doing 2 · Guided response	The teacher will demonstrate selected case-based drug monographs, including discussion of cost, efficacy, and safety, with examples. The students will follow the guidelines and review different monographs and present.
MODULE 30 — PHARMACEUTICAL MICROBIOLOGY			
30.1	Demonstration of dry heat sterilization method	Doing 2 · Guided response	The teacher will select glassware contaminated with microbes based on the swab test and demonstrate the sterilization of the contaminated glassware using dry heat. The teacher will then demonstrate sterility testing of the glassware after dry heat sterilization. Additionally, the teacher will compare dry heat with moist heat (autoclaving) methods, discussing their applications and limitations. Students will observe the demonstration and then perform the dry heat sterilization process independently as part of experiential learning. OR A video demonstration of the dry heat sterilization method will be arranged. The students will observe the video, record the findings, present the same, and discuss
30.2	Demonstration of Aseptic Technique Practice for the transfer of microbes	Doing 2 · Guided response	The teacher will demonstrate proper aseptic techniques, including the use of sterile gloves, flame sterilization, and safe working practices near a Bunsen burner or in a laminar flow cabinet. The teacher will also demonstrate the aseptic transfer of microbes from one medium to another. OR A video demonstration of the aseptic technique practice in the microbiology lab will be arranged. The students will observe the video, record the findings, present the same, and discuss
30.3	Demonstration of Personnel Contamination Assessment	Doing 2 · Guided response	The teacher will demonstrate how personnel can be a source of contamination. Student volunteers will be asked to press their unwashed hands onto nutrient agar plates. The results will be compared with plates touched after hand washing or sanitizing. The plates will be incubated, and the differences in colony growth will be observed. The teacher will then explain the critical role of hand hygiene in contamination control. OR A video demonstration on personnel contamination assessment in a laboratory setting will be arranged, showcasing proper attire, hand hygiene, and awareness of potential contamination sources. Students will observe the video, record their findings, and participate in a class discussion.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
30.4	Demonstration of Different Types of Spoilage	Doing 2 · Guided response	The teacher will display spoiled food or pharmaceutical preparations. The teacher will then demonstrate the examination of food samples (e.g., bread, milk, fruit juice) or pharmaceutical preparations stored under different conditions (room temperature, refrigerated, exposed to air). Students will be instructed to note changes such as discoloration, mold growth, gas bubbles, off-odors, and texture changes. Over several days, students will record their observations and understand how microbial spoilage manifests physically and chemically.
MODULE 31 — REGULATORY ACTS AND RULES APPLICABLE TO AYURVEDIC DRUGS			
31.1	Recording their observations regarding hypothetical issue through Mock composition exercise of demonstration of the roles of DTAB (Drug Technical Advisory Board) composition, including representatives from DGHS, DCGI, pharmacologists	Doing 2 · Guided response	The teacher will demonstrate the roles of DTAB (Drug Technical Advisory Board) composition, including representatives from DGHS, DCGI, pharmacologists, and others, and conduct a mock meeting to discuss a hypothetical issue, such as the approval of a new drug or banning a drug. The students will participate in the mock meeting, following the teacher's guidelines, and record their observations.
31.2	Mock Import License Application	Doing 2 · Guided response	The teacher will demonstrate filling a sample Form 10D for importing an ASU product, such as a herbal capsule, and show mock supporting documents to be attached, including the Free Sale Certificate and lab test report. The students will record their observations and engage in a class discussion.
31.3	Demonstration of case scenarios of Misbranded/spurious drugs and cosmetics	Doing 2 · Guided response	The teacher will demonstrate case scenarios related to misbranded or spurious drugs/cosmetics, explaining how to identify misbranded products based on misleading labelling and spurious products through fictitious company names and other deceptive tactics. The students will record their findings and observations.
31.4	Application for a license of a proprietary formula	Doing 2 · Guided response	The teacher will demonstrate how to formulate a treatment protocol for a selected disease, utilizing Schedule I and published literature. The teacher will also explain the process of applying for a license and showcase the necessary documents and study reports required for the application.
31.5	Demonstration of Landmark Judgments under the NDPS Act	Doing 2 · Guided response	The teacher will select 1-2 landmark judgments under the NDPS Act and demonstrate how courts have interpreted key provisions, such as Section 27 (consumption) and Section 21 (possession). The discussion will focus on crucial aspects, including quantity differentiation (small vs. commercial), presumption of guilt, and bail provisions under Section 37.
31.6	Case Law demonstration on the Factory Act	Doing 2 · Guided response	The teacher will present a notable case study related to labor law violations, focusing on issues such as worker safety, overtime pay disputes, or factory licensing. Students will document the case in a standard format and engage in a class discussion.
31.7	Case Study Demonstration under the Drugs and Magic Remedies Act	Doing 2 · Guided response	The teacher will demonstrate a case study illustrating the application of the Drugs and Magic Remedies Act.
MODULE 32 — ENTREPRENEURSHIP AND START-UPS			
32.1	Demonstration of the process of development of a basic business model as per the template	Doing 2 · Guided response	Students will design a new startup concept, develop a comprehensive business plan, create a prototype, and pitch their idea. They will discuss, brainstorm innovative ideas, and select one idea to develop further. Each student will prepare a written business plan, including market research, financial projections, and marketing strategies. They will also create a prototype or minimum viable product to demonstrate their idea. Students will then prepare a persuasive elevator pitch to present their startup idea to investors, and finally, they will present their pitches and prototypes to the class, receiving feedback and guidance

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
32.2	Demonstration of patent filing	Doing 2 · Guided response	The teacher will demonstrate filing a provisional patent in a real-life or simulated setting.
32.3	Demonstration of a scenario: The New Molecule Patent	Doing 2 · Guided response	The teacher will demonstrate a scenario where an R&D team in a mid-sized pharmaceutical company has discovered a promising new molecule or new chemical entity (NCE) for treating a rare disease condition. The demonstration will cover the process of patent filing, including timing, scope, strategy, disclosure of information, risks, and challenges.
32.4	Demonstration of a case scenario of patenting the formulation process	Doing 2 · Guided response	The teacher will demonstrate a scenario where an Ayurveda pharmaceutical company has developed a modified dosage form featuring a novel delivery mechanism, utilizing traditional ingredients. The demonstration will focus on patenting the formulation process, excluding the herbs themselves. Additionally, the teacher will explore utility model options (where applicable), and create an IP protection roadmap, encompassing design patents, trademarks, and copyrights for packaging and branding.

141 activities. The level shown is that of the objective in Table 3 which cites the activity, so it tells you what the activity is being asked to certify.

PART C · APPENDIX C7

All 163 experiential learning activities, mapped

Table 5 of the curriculum. These are the largest block of hours in the semester and where the upper levels are built; the modular assessment is drawn from them.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 1 — CHRONOLOGICAL DEVELOPMENT AND TAXONOMY OF RASA DRAVYAS			
1.1	Preparing a chronological tree of Rasashastra and its integration with other medical traditions	Doing 4 · Adaptation	Each student will be asked to create a chronological timeline highlighting key texts and developments in Rasashastra. They will compare and contrast excerpts from classical texts to understand the evolution of alchemical practices. Students will investigate the influence of Rasashastra on contemporary Ayurvedic practices and its integration with other medical traditions. Each student will present the findings and discuss
1.2	Integration of the selected metals and minerals with the periodic table elements	Thinking 4 · Analysis	The teacher will allot each student at least one category from the classification of Rasadravya or one drug from each category. Each student will map minerals/Metals to the Periodic Table and search the elemental properties of minerals/metals mentioned in Rasashastra. Students will locate these metals on the Mendeleev Periodic Table, noting their atomic numbers, groups, and periods. Each student will compare and analyze the traditional name with the Periodic table, along with other properties, etc.
1.3	Recognise Rasadravya based on their Graahyatva-Agraahyatva	Doing 3 · Mechanism	The teacher will allot each student at least one category from the classification of Rasadravya or one drug from each category. The student will identify the Rasadravya based on classical characters and find out the ideal characters for acceptance (Grahya)
1.4	Identification of Rasadravya using geological/mineralogical characters	Doing 3 · Mechanism	The teacher will allot each student at least one category from the classification of Rasadravya or one drug from each category. The student will identify the Rasadravya based on geological/mineralogical characteristics
1.5	Exploring Sandigdha and Anuplabhdha Rasadravyas and propose suitable alternatives	Doing 1 · Set	Each student will collect different samples of Sandigdha Rasadravyas and identify them with labelling. Student will also explore present scenarios where certain Rasadravyas are absent or unavailable, with proposed alternative substances or modifications to compensate for the absence, drawing from their knowledge of Rasashastra principles
MODULE 2 — PUTA AND SAMSKARA (TRANSFORMATIVE PROCESSES OF DRUG MANUFACTURING)			
2.1	Hands-on Comparison of Traditional and Modern Puta Methods: Observing and Analyzing Temperature Patterns and Duration	Doing 3 · Mechanism	Each student will go through the different laws of thermodynamics and relate with traditional Puta and understand the concept of heatiing and energy. Student will analyse and monitor the temperature at various points within the Puta and Sharava Samputa. Then compare the observed data with theoretical predictions based on thermodynamic principles.
2.2	Exploring Thermal Dynamics in Traditional Puta Systems: An Experiential Investigation of Fuel Variations, Temperature Patterns, and Thermodynamic Principles	Doing 3 · Mechanism	The teacher will assign a unique Puta to each student, asking them to utilize different fuel materials and observe the resulting temperature patterns. Students will present their findings using graphical representations. Additionally, students will explore the laws of thermodynamics and relate them to traditional Puta, gaining insight into heating and energy concepts. They will also analyze and monitor temperature fluctuations at various points within the Puta and Sharava Samputa.
2.3	Perform an experiential Investigation of Amritikarana and Lohitikarana Procedures, Explo	Doing 2 · Guided response	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
2.4	Performing and analyzing the Satva Patana procedure. The teacher will allot each student	Doing 2 · Guided response	—
2.5	Applcation of Time-Tested Principles of Classical Drug Manufacturing: by integrating Toy	Doing 4 · Adaptation	—
2.6	Exploring the Interplay of Desha and Kala on Pharmaceutical Quality: A Hands-on Investigation of Geographical and Temporal Influences on Drug Manufacturing, Analyzing Processes and Findings to Enhance Quality, Safety, and Efficacy	Doing 3 · Mechanism	Each student will research and conduct practicals investigating the impact of Desha (geographical location and manufacturing site) and Kala (processing time, seasonal variations, and herb maturity) on pharmaceutical quality. Following hands-on experience, students will analyze the processes and findings, correlating them with manufacturing principles to enhance quality, safety, and efficacy, and present their conclusions in a class presentation.
2.7	Develop Mastery of Classical Drug Manufacturing: Integrating Time-Tested Principles of Vasana and Bhavana to Optimize Pharmaceutical Quality, Safety, and Efficacy	Doing 2 · Guided response	Each student will search for and perform practicals related to Vasana (infusion of aroma and effect of proximity), and Bhavana (effect of levigation/trituration). After the procedure, they will analyze each process and findings, correlating them with different principles of manufacturing to enhance quality, safety, and efficacy, and present their findings in a class presentation.
2.8	Demonstration of the effect of Kalaprakarsha and Bhajana in Pharmaceutical Processing, Analyzing Processes and Findings to Optimize Quality, and stability	Doing 2 · Guided response	Each student will conduct practical on Kalaprakarsha (investigating the effect of specific time exposure) and Bhajana (examining the influence of processing vessels). Following hands-on experience, students will analyze each process, correlate findings with manufacturing principles, and present their insights in a class presentation to enhance understanding of quality, stability and efficacy.
MODULE 3 — PARADA (MERCURY) AND ITS PROCESSING: ANCIENT WISDOM AND SCIENTIFIC ADVANCES			
3.1	Samanya Shodhana of Parada	Doing 2 · Guided response	The teacher will allot each student to perform the Samanya Shodhana of Parada following references other than the one that is not taught during practical hours. The students will record the observations and present their findings, and discuss
3.2	Vishesha Shodhana procedures of Parada	Doing 3 · Mechanism	The teacher will allot each student to perform the Vishesha Shodhana of Parada following references other than the one that is not taught during practical hours. The students will record the observations and present their findings, and discuss
3.3	Performance of the Parada Ashta Samskara procedures- Patana, Bodhana, Niyamana, and Deepana	Doing 2 · Guided response	Each student or group will practically explore and perform the Samskaras of Parada like, Patana, Bodhana, Niyamana, and Deepana. The observations, like changes in the Parada, total duration, temperature range, and yield, will be noted down and presented in class.
3.4	Comparison and analysis between the Pharmacodynamics and pharmacokinetics of mercury and mercurial products	Doing 3 · Mechanism	The teacher will allot each student different mercurial products/ compounds and ask them to explore and compare the pharmacodynamic and pharmacokinetic behaviour of allotted products/compounds using both classical Ayurvedic theory and modern biomedical models like Rasa Panchaka and modern concepts of pharmacodynamics and pharmacokinetics. The students will also explore available published data. They will present their findings and discuss.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
3.5	Analysis of mercury toxicity and ADR incidences	Doing 2 · Guided response	The teacher will ask each student to search and record any case of mercury toxicity among OPD/IPD patients using mercurial products, explore the probable toxicokinetics, and suggest its management using relevant classical antidotes and modern therapeutics. OR The teacher will allot each student a case scenario of mercurial toxicity. The student will be asked to explore the toxico-kinetics and suggest its management using relevant classical antidotes and modern therapeutics.
MODULE 4 — MAHARASA DRAVYA, ITS PROCESSING AND UTILITY			
4.1	Pharmacotherapeutic attributes of Abhraka Bhasma and Vaikranta Bhasma	Doing 3 · Mechanism	The teacher will assign each student a specific topic related to the pharmacological, toxicokinetic, and clinical attributes and applications of Abhraka and Vaikranta. Students will search and gather information from classical texts, published scientific literature, and case studies. They will then present their findings and engage in a class discussion
4.2	Critical Analysis of Abhraka and Vaikranta: Pharmacology and Clinical Applications	Doing 4 · Adaptation	The teacher will assign each student a specific topic related to the pharmacological, toxicokinetic, and clinical attributes and applications of Abhraka and Vaikranta. Students will search and gather information from classical texts, published scientific literature, and case studies. They will then present their findings and engage in a class discussion
4.3	Explore and perform the Makshika Bhasma and Vimala Bhasma	Doing 2 · Guided response	Each student will prepare Makshika/Vimala Bhasma using at least one method. The student will record their observations and findings, and then present their results in class. The teacher will provide feedback and recent updates on the subject
4.4	Exploration and performance of Shodhana and other pharmaceutical procedures of Shilajit and Sasyaka/ Tuttha	Doing 2 · Guided response	Each student will perform at least one method of Shodhana/Marana for Shilajit and Sasyaka/Tuttha. The student will record their observations and findings, and then present their results in class. The teacher will provide feedback and recent updates on the subject.
4.5	Explore and perform the pharmaceutical procedure, like Shodhana, Marana for Chapala/Khar	Doing 3 · Mechanism	—
4.6	Exploring New Metallic/Mineral Resources that align with Maharasa Dravyas Principles. Ea	Doing 2 · Guided response	—
MODULE 5 — CRITICAL APPRAISAL OF UPARASA DRAVYA AND SADHARANA RASA DRAVYA			
5.1	Identify and perform an authentication process of Uparasa Dravyas	Doing 4 · Adaptation	The teacher will allot each student with a different drug from Uparasa category. The students will identify and authenticate the given drug based on classical Grahya Lakshana and mineralogical references with cross-verification through reference samples, The students will present their findings and discuss The teacher will allot each student with different drug from Uparasa category. The teacher will instruct each student to search the relevant database regarding the mineralogical characters, physico-chemical properties and crystallography(wherever applicable) of the given samples. The students will explore, note down the findings, present the same and discuss.
5.2	Performing Quality Control tests for the products of Uparasa Dravya	Doing 2 · Guided response	The teacher will allot each student with different processed drug from Uparasa category, such as a Shodhita Dravya, Bhasma or Pishti. The students will test the given sample following organoleptic, physico-chemical parameters and instrumental methods. The students will record the findings, present the same and discuss

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
5.3	Preparation and presentation of formulations of Uparasa Dravya	Doing 2 · Guided response	The teacher will allot each student with a different formulation of Uparasa Dravyas such as, Gandhaka Rasayana, Kasisa Bhasma etc. The students will explore the reference, Standard Manufacturing Process and will prepare the given formulation accordingly. The students will present their findings in the class and discuss the steps, challenges, and outcomes
5.4	Critical Review of toxicity and ADRs of Haratala and Manahshila	Doing 3 · Mechanism	The teacher will instruct each student to explore real or hypothetical case reports involving ADRs or Toxicity of the products containing Haratala and Manahshila. The students will carry out literature review from classics and modern toxicology texts, critically discuss the findings and suggest preventive and remedial measures
5.5	Identify and authenticate Sadharana Rasa Dravyas	Doing 2 · Guided response	The teacher will allot each student with a different drug from Sadharana rasa category. The students will identify and authenticate the given drug based on classical Grahya Lakshana and mineralogical references with cross-verification through reference samples, The students will present their findings and discuss The teacher will allot each student with different drug from Sadharana Rasa. The teacher will instruct each student to search the relevant database regarding the mineralogical characters, physico-chemical properties and crystallography(wherever applicable) of the given samples. The students will explore, note down the findings, present the same and discuss.
5.6	Performing Quality Control tests for the products of Sadharana Rasa Dravya	Doing 4 · Adaptation	The teacher will allot each student with different processed drug from Sadharana Rasa category, such as a Shodhita Dravya, Bhasma or Pishti. The students will test the given sample following organoleptic, physico-chemical parameters and instrumental methods. The students will record the findings, present the same and discuss
5.7	Preparation and presentation of formulations of Sadharana Rasa Dravya	Doing 4 · Adaptation	The teacher will allot each student with a different formulation of Sadharana Rasa Dravyas. The students will explore the reference, Standard Manufacturing Process and will prepare the given formulation accordingly. The students will present their findings in the class and discuss the steps, challenges, and outcomes
5.8	Field Survey: Manufacturer/clinicians using Sadharana Dravyas	Doing 2 · Guided response	The teacher will instruct each student to explore the Sadharana Rasa Dravya used by manufacturers/clinicians/traditional healers. The students will interact, and collect the information regarding Sadharana Rasa Dravyas. They will compile the data and present
MODULE 6 — CRITICAL APPRAISAL OF DHATU AND UPDHATU			
6.1	Identification & documentation of classical Dhatus	Doing 3 · Mechanism	The teacher will allot each student with a different drug from Dhatu category. The students will identify and authenticate the given drug based on classical Grahya Lakshana and mineralogical references with cross-verification through reference samples, The students will present their findings and discuss
6.2	Quality assessment of Dhatu Bhasma	Doing 2 · Guided response	The teacher will allot each student with different Dhatu Bhasma (Loha Bhasma, tamra Bhasma etc). The students will test the given sample following classical Bhasma pareeksha, organoleptic, physico-chemical parameters and instrumental methods. The students will record the findings, present the same and discuss
6.3	Formulation preparation using Dhatu Bhasma	Doing 2 · Guided response	The teacher will allot each student with a different formulation containing Dhatu Bhasma(e.g.Navayasa Louha). The students will explore the reference, Standard Manufacturing Process and will prepare the given formulation accordingly. The students will present their findings in the class and discuss the steps, challenges, and outcomes

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
6.4	Critical appraisal of ADRs due to metals	Doing 2 · Guided response	The teacher will instruct each student to explore real or hypothetical case reports involving ADRs or Toxicity of the products containing metals. The students will carry out literature review from classics and modern toxicology texts, critically discuss the findings and suggest preventive and remedial measures
6.5	Identification of Upadhatu samples	Doing 2 · Guided response	The teacher will allot each student with a different drug from Upadhatu category. The students will identify and authenticate the given drug based on classical Grahya Lakshana and mineralogical references with cross-verification through reference samples, The students will present their findings and discuss
6.6	Performing Quality Control tests for the products of Upadhatu	Doing 2 · Guided response	The teacher will allot each student with different processed drug from Upardhatu category, such as a Shodhita Dravya or Bhasma. The students will test the given sample following organoleptic, physico-chemical parameters and instrumental methods. The students will record the findings, present the same and discuss
6.7	Preparation of Upadhatu formulations	Doing 4 · Adaptation	The teacher will allot each student with a different formulation of Upadhatu Dravyas. The students will explore the reference, Standard Manufacturing Process and will prepare the given formulation accordingly. The students will present their findings in the class and discuss the steps, challenges, and outcomes
6.8	Field survey: Manufacturers or Vaidyas using Dhatu/Upadhatu	Doing 2 · Guided response	The teacher will instruct each student to explore the formulations of Dhatu/Upadhatu used by manufacturers/clinicians. The students will interact, and collect the information regarding Dhatu/Upadhatu Dravyas. They will carry out the Case documentation on sourcing, processing, and therapeutic applications
MODULE 7 — SUDHA VARGA, SIKATA VARGA, VISHA VARGA DRAVYAS			
7.1	Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya	Doing 2 · Guided response	The teacher will assign a different Sudha Varga drug to each student and instruct them to carry out Shodhana, Marana, and Pishtikarana of the allotted drug. Students will conduct the practical following the Standard Operating Procedure (SOP), record their observations, present their findings, and engage in a discussion.
7.2	Quality testing procedures for Bhasma/Pishti of Sudha Vargeeya Dravya	Doing 2 · Guided response	The teacher will assign each student a unique sample of Bhasma/ Pishti from Sudha Varga Dravya. Students will conduct quality testing of the assigned product, evaluating organoleptic, physicochemical parameters, and NPS tests. They will observe, record, and present their findings, followed by a class discussion.
7.3	Shodhana, marana and Pishti Nirmana of Sikata varga Dravya	Doing 2 · Guided response	The teacher will assign a different drug from Sikata Varga category to each student. Students will then carry out Shodhana, Marana, and Pishtikarana of their allotted drug, following the Standard Operating Procedure (SOP). They will record their observations, present their findings, and engage in a class discussion
7.4	Quality testing procedures for Bhasma/Pishti of Sikata vargeeya Dravya	Doing 2 · Guided response	The teacher will assign each student a unique sample of Bhasma/ Pishti from Sikata Varga Dravya. Students will then conduct quality testing of their assigned product, evaluating organoleptic, physicochemical parameters, and NPS tests. They will observe, record, and present their findings, followed by a class discussion.
7.5	Performing Shodhana of Visha Dravyas as per different classics	Doing 2 · Guided response	The teacher will assign a different Visha Dravya to each student. Students will then carry out Shodhana of their assigned drug, following the Standard Operating Procedure (SOP). They will present their observations and engage in a class discussion. Alternatively, the teacher may assign the same drug to all students and instruct them to perform Shodhana using different classical methods.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
7.6	Performing analytical testing process for Shodhita Visha dravya	Doing 2 · Guided response	The teacher will assign each student a different Visha Dravya, providing both unprocessed (Ashuddha) and processed (Shodhita) samples. Students will then conduct comparative testing of the given samples, evaluating parameters such as organoleptic characters, physico-chemical properties, qualitative phytochemical tests, and Thin Layer Chromatography (TLC).
MODULE 8 — APPRAISAL OF RATNA AND UPARATNA AND THERAPEUTIC SIGNIFICANCE OF PISHTI AND DRUTI KALPANA			
8.1	Identification of the precious and semiprecious stones under Ratna and Uparatna varga using keys for mineralogical identification	Doing 2 · Guided response	The teacher will randomly assign a precious or semi-precious stone to each student. Students will then identify their assigned stone using identification keys, present their findings, and discuss the physico-chemical properties and crystallography of the identified specimen.
8.2	Identification of substitutes for Ratnavarga and Uparatna varga drugs	Doing 3 · Mechanism	The teacher will assign each student a different precious or semi-precious stone and instruct them to identify at least one substitute from the museum and demonstrate the method of identification of the selected substitute. The students will present their findings and discuss the physico-chemical properties, crystallography, and pharmacology of the selected specimen.
8.3	Preparation of the Pishti of Selected Ratna or Uparatna	Doing 2 · Guided response	The teacher will assign each student a different drug from Ratna/ Upartna Varga and instruct them to prepare the Pishti using various media for trituration. Students will then prepare the Pishti according to the Standard Operating Procedure (SOP) and present their observations.
8.4	Preparation of Gandhaka Druti using different media	Doing 2 · Guided response	The teacher will instruct each student to prepare Gandhaka Druti using various media (Tila Taila, Ghrita, Sarshapa taila) and different proportions. Students will then prepare the products, compare their physico-chemical properties, present their observations, and discuss
MODULE 9 — SCIENTIFIC FUNDAMENTALS OF AYURVEDIC PHARMACEUTICAL PREPARATIONS			
9.1	Preparation of unique formulation from the category of Panchavidha Kashaya Kalpana	Doing 2 · Guided response	Each student will enlist unique examples of formulations (minimum 10) of Panchavidha Kashaya kalpana from classical texts, and will enlist the steps of their preparation. All students will discuss to evaluate all dimensions of the formulations, viz. ingredients, water proportion, time required, therapeutic efficacy, samskaras performed, and the difference between various classical paribhasha. Each student will prepare at least one formulation from each group, record the observations, present the same, and discuss. Each student will be allotted with different formulation and the students will explore preparing the given formulation with different methods with due importance to heat, proportion etc and prepare the formulations, note their observations, present the same with scientific rationale.
9.2	Standardization of raw material used in Bhaishajya Kalpana. Each student will refer to a	Doing 2 · Guided response	—
9.3	Interpretation of the significance of understanding the proportion of ingredients of formulations quoted in the classics	Thinking 4 · Analysis	Each student will compile 6 examples of classical formulations of different dosage forms on the basis of the uniqueness of ingredients and their proportions. The faculty will discuss and explain its significance. Each student will demonstrate preparation of one formulation, present the findings and discuss the rationality behind the proportion of the ingredients
9.4	Performing the extraction of phytochemicals in Ayurvedic formulations. All students will	Doing 2 · Guided response	—
9.5	Review of Research Updates in the field of Ayurvedic pharmaceutics along with modern sci	Affective · Valuing	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 10 — PRINCIPLES OF PHARMACEUTICAL FORMULATION			
10.1	Illustration of various Prakshepa Dravyas and their proportions stated in the classics. All students will summarize the list of Prakshepa dravyas in the library session and assess the identity, purity, quality, and guna karma of the following Prakshepa dravyas-1.Madhu 2. Sita/ sharkara 3.Guda 4.Kshara 5.Lavana 6.Ghruta 7.Taila 8.Churna- Jiraka, Shunthi,Hingu, Pippali, Trikatu, Chaturjata 9.Kalka 10.Drava- Jala, Ksheera, Madya, Mutra 11.Guggulu 12. Shilajit	Doing 3 · Mechanism	Faculty will allot 2 or 3 prak?epa dravya for studying, and each student will submit an essay based on the findings. Each student will prepare at least 2 Formulations from the following (or any other) to understand the importance of Prakshepa Dravya-1.Brahmi Svarasa 2.Vasa kvatha with different combinations 3.Draksha Phanta 4.Rasona Kalka 5.Narikela Khandapaka 6.Khadirarishta
10.2	Classical techniques of Shaktyutkarsha/ Potency enhancement	Doing 3 · Mechanism	The faculty will plan a workshop where all the students will demonstrate the following or any relevant practical-1) Ghana Nirmana-Kutaja Ghana 2) Bhavana- Shatabhavita Amalaki Rasayana 3) Punaravartita Paka-Anu Taila Nirmana 4) Sura Nirmana-Rasona Sura(Chakradatta, Bhaishjya Ratnavali- Aamavata) 5) Addition of Yogavahi Dravya- addition of Pippali/Kajjali/ Rasasindura Students will compile classical examples of formulations that utilize shaktyutkarsha techniques. A brainstorming session will be conducted following the faculty's concluding remarks.
10.3	Evaluation of Methods used to improve potency and efficacy in modern pharmaceutics A bra	Thinking 5 · Synthesis	—
MODULE 11 — MANA PARIBHASHA (METROLOGY IN AYURVEDIC PHARMACEUTICAL PREPARATION)			
11.1	Mana Paribhasaha from classics and its importance for preparing classical formulations	Doing 4 · Adaptation	—
11.2	Preparation of Classical Formulations by applying knowledge of converted units Group act	Doing 4 · Adaptation	—
11.3	Application of classical knowledge of Mana paribhsha with understanding its effect on pr	Doing 2 · Guided response	—
11.4	Significance of proportion of ingredients . Brainstorming sessions will be conducted wit	Doing 4 · Adaptation	—
11.5	Assessment of advanced tools and techniques used currently for manufacturing Ayurvedic f	Doing 4 · Adaptation	—
MODULE 12 — PROCESS VALIDATION PROTOCOL			
12.1	Studying the documentation related to Standard Operating Procedures (SOPs) used in pharm	Doing 2 · Guided response	—
12.2	Construct a protocol for unique classical formulations and prepare the selected product	Doing 2 · Guided response	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
12.3	Preparation of	Doing 2 · Guided response	Prepare different novel modern dosage forms The faculty will organize a workshop focused on practical training in modern dosage forms. OR will arrange an educational visit to a nearby GMP-certified drug pharmacy or a well-regarded pharmacy college. This visit will provide an opportunity to observe and assist in the preparation of modern dosage forms. OR will incorporate video recordings to train participants on the Standard Operating Procedures (SOP) for preparing modern dosage forms. This should include, Demonstrate SOP for-1. Solid dosage forms - Powders, Pills, Capsules, Tablets, Capsules, Lozenges. 2. Liquid dosage forms - Emulsions, Suspensions, Lotions, Liniments, Syrups, Nasal Drops, Gargle. 3. Semi-solid dosage forms - Ointments, Suppositories, Gel, Paste, Cream. Plasters, Suppositories. 4. Gaseous dosage forms- Aerosols, nasal Spray, Inhalers.
12.4	Overview of Novel Dosage Forms. After conducting a thorough search by all students in the library and on mobile platforms for relevant articles, the	Doing 2 · Guided response	Faculty will arrange a departmental seminar on the topic: "Overview of Novel Dosage Forms." The discussion will focus on the ongoing evolution of healthcare and how the development of advanced formulations is crucial for delivering safe and effective treatments to patients.
MODULE 13 — KRUTANNA (AYURVEDIC FOOD PREPARATIONS)			
13.1	Critical appraisal of unique formulations prepared from Aahara dravya	Doing 2 · Guided response	The teacher will instruct students to study the Ahara Varga from classical texts in depth, accompanied by commentary, for better understanding. Each student will search the formulations from the E-samhita portal or manually from the printed Samhitas. Each student will prepare any 4 (one from each group mentioned below) selected formulations following SOP. 1) Shuka dhanya varga- Vatya manda, Tandulodaka 2) Shimbi Dhanya Varga- Masura Ghruta, Mudgadi Ghruta (Yogaratnakara Striroga) 3) Mamsa Varga- Mayura Ghruta(Ch. chi 26/156-158)Tittiri Putapaka(Sushruta. Uttara 40/84,85), Mushika taila (Sushruta/ Kshudraroga/ Gudabhramsha chikitsa (Su.chi. 20/61-63)Pishita odana(Su. U.43/20) 4) Shaka Varga- Patoladi Ghruta, Surana Vataka, Mulaka Taila, Kushmandavaleha 5) Pushpa varga- Shriphala kusuma vati, Lavangarishta(Gada-nigraha) , Palash Pushpasava. 6) Phala Varga- Dadimavaleha, Draksharishta, Kharjuradi Mantha, Bilvavaleha, Kapitthashtaka Churna. 7) Lavana Varga- Hingu sauvarchala ghruta, Vallabha Ghruta(Yogaratnakara, Hrudroga) Bhaskar Lavana Churna. 8) Spices- Pancha kola Phanta, Eladi Guti, Hingvashtaka Chrna, Haridra Khanda, Jirakarishta
13.2	Disease specific Ayurvedic Ahara	Doing 2 · Guided response	Faculty will convey to the students the plan of the studyas follows-1. Select three Adhyayas from Charaka Samhita Chikitsa Sthana and three Adhyayas from Sushruta Samhita Uttarsthana. 2. Distinguish and list the Aahara dravyas (food substances) and dosage forms used to treat the specified diseases. 3. Interpret the rationale behind the combinations of the ingredients used in these treatments. 4. Explain the correlation between the stages of the disease (Rogavastha) and the recommended therapeutic diet. 5. Appraise the classical descriptions of the vyadhi-pratyanika (disease-specific) and Lakshana-pratyanika (symptom-specific) therapeutic diet recommendations. 6. Demonstrate any four preparations from your compilation of dietary yogas based on the Charaka and Sushruta Samhita. OR The teacher will instruct the students to visitthe hospital and review the prescribed diet for the diseases the students have studied. The students will note down the dietary formulations and present their findings with the method of preparation and rationality of prescription.
13.3	Personalised prescription for Aushadhi Siddha Ahara. The teacher will instruct the stude	Doing 2 · Guided response	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
13.4	Exploratory Survey about the Ayurvedic Food Industry and Ayurvedic Food products availab	Doing 2 · Guided response	—
13.5	Analysis of Functional Food and Modified Food Formulations in Ayurvedic Food Industry. T	Doing 4 · Adaptation	—
MODULE 14 — AYURVEDIC FORMULATIONS FOR HOLISTIC WELLBEING, VITALITY AND BODY CARE			
14.1	Role of Ayurvedic Rasayana Vajikarana Formulations	Doing 2 · Guided response	The students will compile classical formulations having Rasayana and Vajikarana action, classify them based on various criteria, Viz. Dhatu Karya, Avayava Karya, Vyadhikshamattva Vardhana etc. They will present their findings in front of the peers. The contents of presentation should include, the rationality behind the selection and combination of Ingredients and the proportion of ingredients in the selected formulation, justification of the role of Rasayana based various mechanisms of action and the role of Vajikarana formulations in Suprajanana. The students will refer relavant research articles published in peer reviewed journals
14.2	Role of Ayurvedic Formulations in the wellness of specific body parts	Doing 2 · Guided response	The teacher will allot each student with different target specific Rasayana Dravya such as, Hridya, Varnya, Tvachya etc. Each student will explore the principle of designing classical Formulations and their specific target and compile a list of formulations having a specific target action. The faculty will distribute 2 formulations to each student from the following list(or any relevant formulations) 1.Varnya - Kunkumadi Taila, Kanaka Taila(Charaka samhita) 2.Tvachya- Eladi taila, Chandan bala Lakshadi taila, Nalpamaradi taila (sahasrayoga) 3.Kanthya- Khadiradi Vati, Kanthasudharaka vati 4.Keshya - Bhrungaraja Taila 5.Hrudya- Prabhakara Vati 6.Medhya- Brahmi Vati
14.3	Market survey for Ayurvedic Body care products. Students will be instructed to conduct a	Doing 2 · Guided response	—
14.4	Designing and preparing novel Ayurvedic Cosmeceuticals. The teacher will allot each stud	Doing 2 · Guided response	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 15 — DISEASE CONDITION SPECIFIC SELECTION OF FORMULATION			
15.1	Understanding the routes of drug administration is crucial, as it highlights the importance of well-designed dosage forms that improve patient outcomes and therapeutic effectiveness. The teacher will instruct each student to conduct the following UG curriculum teaching class and revise knowledge about the route of drug administration.They will discuss the advantages and disadvantages of each aushadhi prayoga marga and probable mode of action after administration of the following dosage forms-1. Mukha (Oral Cavity): Vati, Gutika, Churna, Asava, Arishta, Kashaya, Avaleha, Khanda, Sneha (Ghrita/Taila), 2. Nasa(Nasal Route)- Dosage form used: Churna, taila, swarasa, arka 3. Karna (Through Ear)- Taila, Ghrita 4. Akshi (Through Eyes) - Ghrita, Taila, 5. Twak (Through Skin)- Lepa, Alepa, Pralepa, Malahara,upanaha,pradeha, abhyanga, udvartana • Shirodhara - Takra • Abhyanga- Sahacharadi Taila • Asthti Bhagna- Murivenna Taila • Vrana- Jatyadi Taila • Smashru – Shankha Bhasma • Kesha Ghanata- Bringaraja Taila • Akala Palita – Hasthi Danta Masi • Indralupta – Icchabhedi Rasa • Lomashatana – Lomashatana Lepa 6. Guda (Anal Route)- Dosage forms - Vartis, taila, ghrita, kalka, churna, kashaya 7. Mutra marga (Through urethra)- Uttara Basti with dosage forms- Taila, ghrita 8. Yoni marga (Through vagina)- Yoni Dharana, Yoni Dhavana, Yoni Pichu, Yoni Dhoopana	Doing 3 · Mechanism	Each student will prepare a suitable dosage form from the list mentioned below, present their findings, and discuss: 1.Nasya Kalpana- Navana, Avapidaka, Dhuma, Pratimarsha, Pradhamana 2.Mukha Kalpana- Kavala, Gandusha,P ratisaran,manjan 3.Basti kalpana- Yapana, Anuvasana, Karshana, Sneha, Bruhana, Vajikara, Rasayana, Matra 4.Netra Kalpana- Seka,Aschyotana,Pinda,Bidaal,Tarpana,putapaka, 5.Nabhipurana- Bindu Taila (Sha. Sam. Ma. Kh. 9/61-66) 6.Other bahya Kalpana- Lepa,Dhara,Bandha,Dhupana,Agnikarma,Kshara karma,Avagaha,Pinda, Drava sveda, Taap sveda,Vrana karma,jalauka, Paadabhyanga Evaluate the role and responsibility of effective dosage form design for the aforementioned Ayurvedic treatment modalities. Refer to relevant articles to enhance your understanding and organize a departmental presentation on this topic.
15.2	Hands-on training focusing on innovative formulations	Doing 2 · Guided response	The faculty will instruct each student to prepare different innovative formulations such as transdermal patches, nebulizers, and eye masks. They will discuss the necessity and current trends in innovative formulations within the field of Ayurvedic pharmaceutics, review modern pharmaceutical literature, consult experts, or watch videos to understand the SOP for transdermal patches, nebulizers, eyemasks, and other innovative formulations. Each student will prepare the allotted innovative product, present the observations, and discuss the challenges faced
15.3	Learning about the R&D department of Ayurvedic Pharmacy. The students will be posted to	Doing 2 · Guided response	—
MODULE 16 — CURRENT TRENDS IN AYURVEDIC PHARMACEUTICAL INDUSTRY.			
16.1	Assessment of the real-life experience of the market for the growing popularity of Ayurv	Doing 3 · Mechanism	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
16.2	Internship in the Ayurvedic Pharmacy. Each student should-Understand current trends in t	Affective · Organisation	—
16.3	QAQC Laboratory Training	Affective · Valuing	The students will visit the QAQC laboratory and observe and understand the working pattern of the Laboratory. The students will interact with the concerned experts and obtain updated information about Leading Drug research laboratories / CCRAS drug research units, regulatory and certification bodies for QAQC of Ayurvedic Formulations. The students will be able to justify the need and importance of QAQC in the field of Ayurvedic Pharmaceutics. The students will also observe the Calibration of instruments used for analysis. They will prepare a report of their experience, present the same, and discuss.
MODULE 17 — GUIDELINES FOR ADMINISTRATION OF RASOUSHADHI IN CLINICAL PRACTICE			
17.1	Exploring the Rasa Chikitsa through field visits. The teacher will ask the students to e	Doing 2 · Guided response	—
17.2	Exploring the procedure and benefits of kshetrikarana	Doing 2 · Guided response	The teacher will instruct each student to visit Panchakarma unit to observe the process of Kshetrikarana. They will record their observations and present different steps of Kshetrikarana and discuss the benefits
17.3	Outcome of compliance and non-compliance with the pathya apathya guidelines	Doing 3 · Mechanism	The students will observe case records of the patients undergoing Rasayoga sevana and document the benefits of compliance with prescribed Pathya-Apathya measures. Students will be assigned case studies where deviation from Pathya-Apathya resulted in suboptimal Rasayoga effect. Students will analyse the cases, identify key factors, and present how the patient outcome was influenced by non-adherence to Pathya-Apathya
17.4	Evaluation of prescription errors containing Rasoushadhis	Thinking 6 · Evaluation	The teacher will create sample prescriptions of Rasoushadhi with prescription errors. The students will analyse them for potential errors such as overdose, improper combinations, etc, and they will present their findings and discuss.
17.5	Analysis of chelation therapy in heavy metal toxicity	Doing 3 · Mechanism	Students will present a case scenario of heavy metal toxicity with a description of clinical signs and symptoms, and suggest treatment strategies with suitable chelating agents. They will also explore classical methods of detoxification and compare them with contemporary chelation practices.
MODULE 18 — PRAYOGIKA BHAISHAJYA KALPANA(CLINICAL APPLICATION OF PHARMACEUTICAL PRODUCTS)			
18.1	Exploring chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dyna	Doing 2 · Guided response	—
18.2	Exploring the modulation of the action of the dosage form by chronotherapeutic administr	Doing 2 · Guided response	—
18.3	Enlisting appropriate Anupanas for various formulations in different clinical conditions	Doing 3 · Mechanism	—
18.4	Optimising drug response through synergistic Anupana integration. Each student will anal	Affective · Valuing	—
18.5	Exploring the role of Anupana in the modulation of drug tolerance and safety. The studen	Doing 3 · Mechanism	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
18.6	Exploring posology principles from an ayurvedic and modern pharmacology perspective. The	Doing 2 · Guided response	—
18.7	Exploring novel indications for classical Ayurvedic formulations	Doing 3 · Mechanism	Each student will be allotted five formulations, and students will evaluate the new therapeutic use of that formulation. With the available modern research articles on the pharmacological insights of the formulation's ingredients, the students will propose and justify a potential new therapeutic use for that formulation
18.8	Exploring the Reverse pharmacological approach for drug repurposing in Ayurveda	Doing 3 · Mechanism	The students will explore various scientific articles for successful clinical case studies of classical ayurvedic formulations for new indications (COVID, lifestyle disorders, etc). Students will create a pharmacological profile of the ingredients of the formulation from the ayurvedic text and the modern context. The students will link the clinical success of the formulation to the pharmacological profile of the ingredients of the formulation.
MODULE 19 — SOP AND CLINICAL APPLICATION OF CHATURVIDHA RASAYANA			
19.1	Validation of utility of kharaleeya rasayana to clinical case practice. Each student wil	Doing 4 · Adaptation	—
19.2	Demonstration of the impact of Bhavana – A bioactivation tool in Kharaleeya Rasayana. Ea	Doing 2 · Guided response	—
19.3	The pharmacodynamic rationale and clinical application of parpati kalpana. Each student	Doing 4 · Adaptation	—
19.4	Preparation of Pottali following SOP The teacher will assign each student a different po	Doing 3 · Mechanism	—
19.5	Analysis of the pharmacological profile of the ingredients, bhavana dravya, and the phar	Doing 3 · Mechanism	—
19.6	Preparation of selected Kupipakwa Rasayana	Doing 2 · Guided response	The teacher will assign one Sagandha and one Nirgandha Kupipakwa Rasayana for the group of students (other than the ones prepared in practical). The students will prepare the allotted formulation following SOP, record their findings, present the same and discuss
19.7	Demonstration of the scinetific rationality for the selection of specific Kupipakwa Rasa	Doing 2 · Guided response	—
MODULE 20 — SOP AND CLINICAL APPLICATION OF AYASKRITI, LOUHA, AND MANDURA KALPA			
20.1	Decoding the role of Ayaskriti. as a multi-target formulation. The students will list ou	Doing 3 · Mechanism	—
20.2	Application of Louha Kalpa by integrating through textual insights with real word cases	Doing 4 · Adaptation	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
20.3	Gomutra as a medium of pharmaceutical processing and as a bio enhancer – "Re-evaluating its role in mandura kalpa." The teacher will instruct each student to carry out pharmaceutical processes like Shodhana where Gomutra is used. The students will carry out the practical, record their observations, and present them and discuss	Doing 3 · Mechanism	The students will list out mandura kalpas containing gomutra. They will critically evaluate how go mutra functions as a bio enhancer in the preparation of the mandura kalpa, enhancing therapeutic potency, absorption, and clinical efficacy from both ayurvedic and modern pharmacological perspectives. The students will also refer to the scientific articles which substantiate the role of gomutra in attaining therapeutic benefits.
MODULE 21 — CLINICAL APPLICATION OF RASOUSHADHI ACCORDING TO SROTAS AND IN SELECTED CHALLENGING CLINICAL CONDITIONS			
21.1	Therapeutic application of Rasoushadhi in different Srotas Rasoushadhi applicable to the	Doing 4 · Adaptation	—
21.2	Exploring the clinical utility of Rasoushadhi in Medical Emergencies	Doing 4 · Adaptation	The teacher will select certain emergency conditions and instruct the students to explore and list the relevant Rasoushadhi from Chaturvidha Rasayana. Each student will then be assigned a different formulation from the listed Rasoushadhi to prepare, record their observations, and present their findings. Following this, students will delve into the rationality and probable mode of action of their allotted formulations, drawing on the classical properties and pharmacology of the ingredients. Additionally, they will analyze the findings of relevant scientific articles published on their assigned formulations.
21.3	Exploring the role of Rasoushadhi in clinically challenging conditions such as Cancer. T	Doing 2 · Guided response	—
21.4	The role of Rasoushadhi in clinically challenging conditions such as neurodevelopmental	Doing 4 · Adaptation	—
MODULE 22 — YOGAS FROM DIFFERENT KALPANAS			
22.1	Analysis of the role of Gugulu Kalpana in various clinical conditions by selecting a specific Gulggulu Yoga. The	Doing 3 · Mechanism	Teacher will select different gugulu preparations and ask the student to select one Guggulu formulation and assign them to list the therapeutic actions of that formulation. Also, the student will analyse how the formulation is effective in those clinical conditions by analysing the possible mode of action, discussing the pharmacological activity of each ingredient, method of preparation, and published research works. OR The teacher will allot each student a different Vati/ Guggulu Kalpa and instruct them to explore the prescriptions containing those formulations in OPD/IPD files of the Hospital. The students will note the dose, Anupana, duration of administration, clinical indication, and therapeutic outcome, present their observations, and discuss
22.2	Analysis of the role of selected leha yoga in different clinical conditions. The	Doing 3 · Mechanism	Teacher will assign each student a particular Avaleha/Paka/Khanda Yoga, and the student will analyse how the selected Avaleha Yoga works in different clinical conditions and demonstrate the mode of action based on classical descriptors, pharmacology of individual ingredients, and published research works. OR The teacher will instruct each student to explore the prescription containing Avaleha/Paka/Khanda. The students will note down the dose, Anupana, duration, clinical indication, and therapeutic outcome. They will present their findings and discuss

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
22.3	Analysis of different stages of Sneha Paka	Doing 4 · Adaptation	The teacher will assign each student a different Sneha Kalpa and will ask them to carry out Sneha Paka. The students will carry out the Sneha Paka till Dagdha Paka and collect the samples at different stages of Sneha Paka (Ama, Mridu, Madhyama, Khara, and Dagdha). The students will comparatively analyse the samples of different Paka stages and present their findings. Based on the analytical parameters, therapeutic significance of the samples will be discussed.
22.4	Illustration of the clinical application of selected Arka/Dravaka Kalpa	Doing 3 · Mechanism	The Teacher will assign different Arka/Dravaka formulations to each student. The students will explore the clinical indications of the allotted dosage form and discuss the dose, clinical application, and mechanism of action based on classical and pharmacological descriptors of the ingredients. They will refer relevant publications (case studies/case reports), present the same, and discuss.
22.5	Comparative analysis of Asava Arishta Samples having different percentage and types of sugar	Doing 3 · Mechanism	The teacher will allot each student a different Asava/Arishta sample containing different proportions of sweetening substances and different types of sweeteners like Sharkara, Sita, Guda, madhu or their combinations. The students will analyse the given samples for alcohol percentage, total, reducing and non reducing sugars, present their findings and discuss
22.6	Record the dose, duration, indications, and clinical outcome of the various formulation	Doing 2 · Guided response	—
MODULE 23 — SOP AND CLINICAL APPLICATIONS OF CHURNA KALPANA(POWDER DOSAGE FORMS)			
23.1	Evaluating the role of a selected Churna in a specific clinical condition	Doing 4 · Adaptation	The teacher will allot each student a Churna yoga. The students will explore the reference, note down the classical indications. The students will anayse how each component ingredient helps in the management of said clinical condition. Also, the students will plot the possible mode of action by analysing the pharmacology, classical descriptors, dose, anupana, time of administration, Pharmacotherapeutic properties of ingredients, and their role in the mentioned disease condition and research updates. OR The teacher will allot each student a Churna yoga and instruct them to explore the prescriptions containing those formulations in OPD/IPD files of the Hospital. The students will record the dose, duration, indications, and clinical outcome, present their observations and discuss.
23.2	Clinical applications of Kshara and Ksharasutra	Doing 2 · Guided response	The teacher will allot each student a Kshara and Ksharasutra yoga. Students will explore the reference, note clinical indications, and analyze how Kshara Sutra manages conditions like Arsha and Bhagandara. They will also examine the mechanism of action of the allotted Kshara in specific disease conditions. Furthermore, students will plot the possible mode of action by analyzing pharmacology, classical descriptors, dose, anupana, administration time, pharmacotherapeutic properties of ingredients for Kshara Kalpa, and the mechanism of action of Ksharasutra in conditions like Arsha, Bhagandara, incorporating research updates. OR The teacher will allot each student to explore a prescription containing Kshara (oral or Pratisarana) and a Kshara Sutra yoga in OPD/IPD files of the Hospital. Students will record the dose, duration, indications, and clinical outcome, present their observations, and discuss.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
23.3	Demonstration of role of Lavana Kalpana in various clinical conditions	Doing 2 · Guided response	The teacher will allot each student a different yoga from Lavana Kalpana. Students will explore the reference, note clinical indications, and analyze how each component ingredient contributes to managing the specified disease. They will plot the possible mode of action by analyzing pharmacology, classical descriptors, dose, anupana, administration time, pharmacotherapeutic properties of ingredients, and their role in specific diseases, incorporating research updates. OR The teacher will instruct the students to explore a prescription containing Lavana Kalpa, from OPD/IPD files of the Hospital. The students will record the dose, duration, indications, and clinical outcome, present their observations, and discuss.
23.4	Role of a selected satva in a specified disease condition	Doing 2 · Guided response	The teacher will allot each student a yoga from Satva Kalpana. Students will explore their clinical indications and plot the possible mode of action by analyzing pharmacology, classical descriptors, dose, anupana, administration time, pharmacotherapeutic properties of ingredients, and their role in specific diseases, incorporating research updates. OR The teacher will instruct the students to explore prescriptions containing Guduchi Satva in various clinical conditions. Students will explore such prescriptions in OPD/IPD files of the Hospital, record the dose, duration, indications, and clinical outcome, present their observations, and discuss.
23.5	Role of Malaharas as a Topical application in Skin disease, The teacher will allot each	Doing 4 · Adaptation	—
MODULE 24 — THERAPEUTIC SIGNIFICANCE OF TOPICAL DOSAGE FORMS			
24.1	Exploring the role of Ayurvedic ophthalmic preparations in various eye diseases	Doing 2 · Guided response	The teacher will allot each student a different Netra Kalpa indicated for various eye diseases. Students will analyze how each component ingredient contributes to managing the indicated eye diseases. They will plot the possible mode of action by analyzing the pharmacology, classical descriptors, mode and time of administration, pharmacotherapeutic properties of ingredients, and their role in specific eye diseases, incorporating research updates. OR The teacher will instruct the students to search for prescriptions in OPD/IPD files for different marketed eye drops. Students will record the product name, ingredients, dose, duration, indications, and clinical outcome, then present their observations and discuss.
24.2	Demonstration of the role of lepa Yogas in managing various medical needs by innovative approaches	Doing 5 · Origination	The teacher will allot each student a selected Lepa yoga, indicated for external application in various diseases. Students will analyze how each component ingredient alleviates symptoms of the specific disease condition. They will plot the possible mode of action by analyzing pharmacology, classical descriptors, mode and medium of application, pharmacotherapeutic properties of ingredients, and discuss research updates.
24.3	The role of Malahara as a Topical application in Skin disease. The teacher will allot ea	Doing 3 · Mechanism	—
MODULE 25 — ARTIFICIAL INTELLIGENCE IN DRUG DEVELOPMENT AND PHARMACOGENETICS; PHARMACOGENOMICS			
25.1	Performing De Novo Drug	Doing 3 · Mechanism	Design Using Generative Models Each scholar is given with different assignment and is made to experience De Novo Drug Design Using Generative Models using appropriate tools and methods. Their findings are presented to peers and discussed
25.2	Analysis of ADMET properties using AI tools	Doing 4 · Adaptation	Each scholar is given with different drug molecule and is made to experience using pre-trained models or public web tools (Ex, Swiss ADME, pkCSM) to predict pharmacokinetic profile and compare predictions to real-world outcomes from literature

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
25.3	Identification of potential adverse effects of drug candidates using a dataset and tools	Doing 2 · Guided response	Each scholar is given with different natural drug molecule and is made to experience Side Effect Prediction following a suitable method, dataset, and tools. The outcome will be presented and discussed
25.4	Extraction of potential drug-disease relationship by using through Literature /data Mining	Doing 3 · Mechanism	Each scholar is asked to carry out Literature Mining for Drug Repurposing using appropriate tools and methods. They are asked to use NLP to extract potential drug-disease relationships, present their findings, and discuss
25.5	Interpretion of Pharmacokinetics and Toxicology Modelling using appropriate AI tools and methods	Doing 3 · Mechanism	Each scholar is asked to use appropriate tools and methods to experience Pharmacokinetics and Toxicology Modelling. Predicted ADMET properties are presented and discussed
MODULE 26 — PHARMACOLOGY			
26.1	Analysis of the mechanism of action of drugs acting on the Cardiovascular system through a case study scenario	Thinking 5 · Synthesis	The teacher will create a case scenario of cardiovascular disease, like a 60-year-old man who presents with chest pain and is diagnosed with stable angina. His blood pressure is 160/95 mm Hg. Now, discuss the mechanism of action and rationale for using nitrates, beta-blockers, and calcium channel blockers. Students will explore the role of antihypertensive agents (e.g., ACE inhibitors, diuretics) and recognize potential drug interactions and side effects.
26.2	Analysis of the mechanism of action of drugs acting on the Respiratory system through a case	Doing 3 · Mechanism	The teacher will create a case scenario of respiratory disease, such as a 28-year-old woman has a history of asthma with recurrent wheezing, and dyspnoea. She is currently experiencing an exacerbation. Now, differentiate between short-acting and long-acting bronchodilators, and explain the role of corticosteroids in asthma management. The teacher may create a case scenario of diseases pertaining to different systems for every batch.
26.3	Analyse Adverse Drug Reaction in a polypharmacy patient through a case scenario	Doing 3 · Mechanism	The teacher will create a case scenario of Adverse Drug Reaction in a polypharmacy patient, supposing a 70-year-old patient who is on digoxin, furosemide, and amlodipine. He complains of nausea, confusion, and visual disturbances. Students will be instructed to focus on Digoxin toxicity, Electrolyte imbalances (e.g., hypokalaemia from furosemide), Pharmacokinetics in elderly patients, and will try to find out the points like factors that predispose the elderly to ADRs, management of digoxin toxicity, and lab monitoring
26.4	Discussion on Emergency Pharmacology through a case scenario of anaphylaxis management	Doing 3 · Mechanism	The teacher will create a case scenario of an emergency condition about different systems, like the cardiovascular system, respiratory system, etc. Students will be asked to find out a first-line treatment/ management for the said anaphylaxis condition
MODULE 27 — NETWORK PHARMACOLOGY & REVERSE PHARMACOLOGY			
27.1	Practice of Network Pharmacology	Doing 3 · Mechanism	The teacher will give each student selected hands-on activities and approaches in network pharmacology that can involve a variety of hands-on activities and approaches that help deepen understanding of drug interactions, mechanisms of action, and the overall pharmacological landscape. The activities can be, simulation exercises OR, field trips, OR, interdisciplinary Workshops OR, Online Courses and MOOCs. The student will perform/attend one or more activities spanning not less than 10 hours. After learning, students will present their experience and discuss
27.2	A mini-project on network pharmacology	Doing 3 · Mechanism	The teacher will allot each student a different herb. Students will be asked to select a major pharmacological activity of the given herb and follow a systems biology approach to validate reverse pharmacology. After the completion of the project, students will present the outcomes and discuss

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
27.3	Presentation of observations of analysis of Reverse Pharmacology in selected cases. The	Doing 3 · Mechanism	—
27.4	Interpret Reverse pharmacology with a systems biology approach. The teacher will allot e	Doing 3 · Mechanism	—
MODULE 28 — PHARMACEUTICAL QUALITY ASSURANCE			
28.1	Identification of the Root cause, affected personnel roles, and suggest corrective and preventive measures through Case Study Discussion on GMP Violation	Doing 2 · Guided response	The teacher will create a case scenario of GMP Violation due to Poor Personnel Management. The students will be provided with a real or fictional GMP warning letter or case study involving personnel errors. The students will identify the Root cause, affected personnel roles, and suggest corrective and preventive measures
28.2	Documentation of a personal record of pharmacy personnel	Doing 4 · Adaptation	The teacher will ask the students to create a mock record of an employee of the pharmacy, and it will be cross-checked by another student for errors
28.3	Inspection of Packaging Materials to check their compliance with quality specifications	Doing 3 · Mechanism	The teacher will give mock purchase orders, specification sheets, and provide some packaging material samples (some compliant, some defective). The students will inspect the materials, match them against specifications (dimensions, clarity, print quality), and approve or reject with reasons
28.4	Calibration of Analytical Instruments	Doing 3 · Mechanism	The students will perform the calibration of different analytical instruments, pH meter, an Analytical balance, UV-Vis spectrophotometer
28.5	QA Documentation audit	Doing 3 · Mechanism	The teacher will allot each student mock documents like Batch Manufacturing Records (BMR), deviations, Change control forms, etc. The students will audit the given documents for errorsapprove/reject documentation and justify their decisions
MODULE 29 — PHARMACY PRACTICE			
29.1	Identification of Adverse drug reaction in selected case and .documentmentation cases of Adverse Drug Reaction	Doing 2 · Guided response	The students will be posted in the hospital. They will document a minimum of 10 patient cases concerning anupana, sahapana, pathya-apathya, kala maryada, and ADRs, if any. They may also document the cases retrieved from Hospital records.
29.2	Experience about Basic Health Screening at community screening	Doing 3 · Mechanism	Students will perform hands-on tests using community pharmacy equipment in a community setting, on a minimum of 10 patients, including measurements of blood pressure using a sphygmomanometer, capillary blood glucose using a glucometer, lung function using a peak flow meter and incentive spirometer, pulse oximetry to measure SPO?, and BMI assessment
29.3	Discuss on Multidisciplinary Integration for TDM(Therapeutic Drug Monitoring) decision making	Doing 4 · Adaptation	Students will interact with clinicians and laboratory personnel, communicating Therapeutic Drug Monitoring (TDM) recommendations, discussing laboratory results, and planning monitoring strategies. Additionally, students will conduct case-based ward rounds, incorporating lab values, patient status, and polypharmacy considerations into TDM decision-making.
29.4	Development of Standard Therapeutic Guidelines	Doing 5 · Origination	Students will draft protocols for common conditions, such as hypertension, based on evidence-based medicine. These protocols will undergo peer review by a committee, which will evaluate and refine the guidelines before their formal adoption.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 30 — PHARMACEUTICAL MICROBIOLOGY			
30.1	Conduction of Dry heat sterilization practice	Doing 2 · Guided response	Each student will be assigned an article contaminated with microbes. The students will then sterilize the articles using the dry heat method in a hot air oven. After sterilization, students will test for sterility by inoculating the nutrient broth and observing for growth. The recorded findings will be presented and discussed.
30.2	Simulation Activity for Aseptic Practices	Doing 4 · Adaptation	The students will simulate activities such as preparing sterile media, gowning, and handling materials in an aseptic area. Peers will observe and provide feedback on potential contamination risks. The students will then discuss how behaviour and protocols can reduce contamination risks.
30.3	Real-life case studies on spoilage outbreaks and preservative failures	Doing 3 · Mechanism	The teacher will assign each student a spoiled food item, a pharmaceutical preparation, or a scenario. Students will then identify the type of spoilage, determine the probable cause of the spoilage, and suggest ways in which contamination could have been prevented.
30.4	Observing and detecting Spoilage of Pharmaceutical Product in Real Time	Doing 4 · Adaptation	The teacher will provide each student with two samples of the same dosage form (one fresh and one old) of a pharmaceutical product, such as Panchavidha Kashaya Kalpana, and their suitable Upakalpana. Students will then assess both samples for spoilage using organoleptic and basic physicochemical parameters. They will record their observations on the comparative stability of the fresh and old samples. By applying theoretical knowledge to practical scenarios, students will highlight the importance of microbial limit tests in quality control of pharmaceutical products.
MODULE 31 — REGULATORY ACTS AND RULES APPLICABLE TO AYURVEDIC DRUGS			
31.1	Case Law analysis on the Factory Act	Doing 3 · Mechanism	The teacher will instruct each student to explore a case study related to Worker safety violations/Overtime pay disputes/Factory licensing issues. The students will present the case in a standard format.
31.2	Technical Report to DTAB: Reporting Adverse Drug Reactions (ADRs)	Doing 3 · Mechanism	The teacher will assign students to prepare a mock technical report on the adverse effects of a new cosmetic product, adhering to the technical and regulatory guidelines outlined in the Drugs and Cosmetics (D&C) Act. Students will present their reports, which will be reviewed and verified by the teacher for accuracy and compliance.
31.3	Identification of Misbranded/ Spurious Drugs and Cosmetics	Doing 3 · Mechanism	Students will be tasked with collecting and compiling real-life case studies on misbranded or spurious drugs and cosmetics from published literature or electronic resources. They will analyze each case to identify and document the key parameters that classify a product as misbranded or spurious. Finally, students will present their findings and engage in a class discussion to share insights and perspectives.
31.4	Application for GMP certification	Doing 3 · Mechanism	The students will be instructed to compile all the mock forms required to get GMP certification and try to fill out the forms. The teacher will discuss - How to get the licence for single herb/herbo-mineral compound/extract, and how to apply for a Loan licence. Students will record the findings, discuss with the teacher
31.5	Development and Licensing of Proprietary Herbal/ Ayurvedic Formulations	Doing 3 · Mechanism	The teacher will assign each student a different disease and instruct them to formulate a herbal/ Ayurvedic formula for the given condition. Students will then develop the necessary documents to support a license application for their formulated product. The teacher will review and discuss the submissions, providing conclusions and feedback

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
31.6	Label Compliance Check	Doing 3 · Mechanism	The teacher will distribute various Ayurvedic, Siddha, and Unani (ASU) drug labels (authentic or printed) among the students. Each student will assess their assigned label for compliance with the requirements outlined in Rule 161, documenting their findings. The students will then discuss their observations with the teacher
31.7	Product Label Designing as per D&C Act	Doing 3 · Mechanism	The teacher will assign each student a different dosage form and instruct them to design a product label in compliance with the Drugs and Cosmetics (D&C) Act. Students will create their labels, present them to the class, and engage in a discussion with the teacher to receive feedback and guidance.
31.8	Analysis of Landmark Judgments under the NDPS Act	Doing 3 · Mechanism	The teacher will instruct each student to explore a landmark judgment under the NDPS Act, evaluating court interpretations of key provisions (Sections 27, 21, etc.) and focusing on quantity differentiation, presumption of guilt, and bail provisions under Section 37.
31.9	Objectionable Advertisements: Case Studies and Report	Doing 3 · Mechanism	The students will be instructed to explore the cases coming under the Drug and Magic Remedies (Objectionable Advertisements) Act, 1954. Each student will explore a case, present the observations, and report the same in standard format to the respective pharmacovigilance cell.
MODULE 32 — ENTREPRENEURSHIP AND START-UPS			
32.1	Creating an elevator pitch for a startup idea	Doing 3 · Mechanism	Students will design a new startup concept, develop a comprehensive business plan, create a prototype, and pitch their idea. They will discuss, brainstorm innovative ideas, and select one idea to develop further. Each student will prepare a written business plan, including market research, financial projections, and marketing strategies. They will also create a prototype or minimum viable product to demonstrate their idea. Students will then prepare a persuasive elevator pitch to present their startup idea to investors, and finally, they will present their pitches and prototypes to the class, receiving feedback and guidance.
32.2	Conducting market research for a start-up, including survey design and analysis	Doing 5 · Origination	Each student will conduct a market survey for a startup, then design a startup proposal based on the survey findings, including logical analysis and recommendations. Students will record and present their proposals, demonstrating their understanding of market research and its application in startup development
32.3	Navigating a Startup's Early Days: A Case Study-Based Role-Play	Doing 4 · Adaptation	The teacher will assign each student a case study of a startup in its early days. Provide each student with a set of role-play scenarios related to the startup's early days, including pitching to investors, managing conflicts between co-founders, handling customer complaints, and deciding on a marketing strategy. After the role-play, each student will be asked to reflect on their experiences and discuss questions such as what challenges they faced, how they overcame these challenges, and what they learned from the experience. Finally, ask each student to present their findings and insights to the class
32.4	Designing of an application to Patent the formulation process	Doing 3 · Mechanism	The teacher will allot each student a different scenario related to pharmaceutical modification. The student will design and develop a patent application for the formulation process
32.5	Identification of the patentability of a product	Thinking 5 · Synthesis	The teacher will allot each student a drug/product containing the ingredient mentioned in Ayurvedic classics. Students will be asked to check the patentability of the product, present their findings, and discuss

163 activities. The level shown is that of the objective in Table 3 which cites the activity, so it tells you what the activity is being asked to certify.

PART C · APPENDIX C8

Where the hours go, module by module

The lecture, practical and experiential hours of every module, computed from Table 3 and reconciled against each module's notional hours. Read alongside Chapter 6.

M	MODULE	CR	LECTURE	PRACTICAL	EXPERIENTIAL	ASSESS	TOTAL
SEMESTER 3							
1	Chronological Development and Taxonomy of Rasa Dravyas	2	10 h	20 h	26 h	4 h	60 h
2	Puta and Samskara (Transformative processes of drug manufacturing)	2	10 h	20 h	26 h	4 h	60 h
9	Scientific fundamentals of Ayurvedic Pharmaceutical preparations	2	10 h	20 h	26 h	4 h	60 h
10	Principles of pharmaceutical formulation	2	10 h	20 h	26 h	4 h	60 h
17	Guidelines for administration of Rasoushadhi in clinical Practice	2	10 h	20 h	26 h	4 h	60 h
18	Prayogika Bhaishajya Kalpana(Clinical application of Pharmaceutical products)	2	10 h	20 h	26 h	4 h	60 h
25	Artificial intelligence in drug development and Pharmacogenetics; Pharmacogenomics	2	10 h	20 h	26 h	4 h	60 h
26	Pharmacology	2	10 h	20 h	26 h	4 h	60 h
SEMESTER 4							
3	Parada (Mercury) and Its Processing: Ancient wisdom and Scientific advances	2	10 h	20 h	26 h	4 h	60 h
4	Maharasa Dravya, its processing and utility	2	10 h	20 h	26 h	4 h	60 h
11	Mana Paribhasha (Metrology in Ayurvedic Pharmaceutical preparation)	2	10 h	20 h	26 h	4 h	60 h
12	Process validation Protocol	2	10 h	20 h	26 h	4 h	60 h
19	SOP and clinical application of Chaturvidha Rasayana	3	15 h	30 h	39 h	6 h	90 h
20	SOP and Clinical application of Ayaskriti, Louha, and Mandura Kalpa	1	5 h	10 h	13 h	2 h	30 h
27	Network pharmacology & Reverse Pharmacology	2	10 h	20 h	26 h	4 h	60 h
28	Pharmaceutical Quality Assurance	2	10 h	20 h	26 h	4 h	60 h
SEMESTER 5							
5	Critical appraisal of Uparasa Dravya and Sadharana Rasa Dravya	2	10 h	20 h	26 h	4 h	60 h
6	Critical appraisal of Dhatu and Updhatu	2	10 h	20 h	26 h	4 h	60 h
13	Krutanna (Ayurvedic Food Preparations)	2	10 h	20 h	26 h	4 h	60 h
14	Ayurvedic Formulations for Holistic Wellbeing, Vitality and Body Care	2	10 h	20 h	26 h	4 h	60 h
21	Clinical application of Rasoushadhi according to Srotas and in selected challenging clinical conditions	1	5 h	10 h	13 h	2 h	30 h

M	MODULE	CR	LECTURE	PRACTICAL	EXPERIENTIAL	ASSESS	TOTAL
22	Yogas from different Kalpanas	3	15 h	30 h	39 h	6 h	90 h
29	Pharmacy Practice	2	10 h	20 h	26 h	4 h	60 h
30	Pharmaceutical Microbiology	2	10 h	20 h	26 h	4 h	60 h
SEMESTER 6							
7	Sudha Varga, Sikata Varga, Visha Varga Dravyas	3	15 h	30 h	39 h	6 h	90 h
8	Appraisal of Ratna and Uparatna and therapeutic significance of Pishti and Druti Kalpana	1	5 h	10 h	13 h	2 h	30 h
15	Disease condition specific selection of formulation	2	10 h	20 h	26 h	4 h	60 h
16	Current trends in Ayurvedic Pharmaceutical Industry.	2	10 h	20 h	26 h	4 h	60 h
23	SOP and clinical applications of Churna Kalpana(Powder dosage forms)	3	15 h	30 h	39 h	6 h	90 h
24	Therapeutic significance of Topical dosage forms	1	5 h	10 h	13 h	2 h	30 h
31	Regulatory Acts and Rules applicable to Ayurvedic Drugs	3	15 h	30 h	39 h	6 h	90 h
32	Entrepreneurship and Start-Ups	1	5 h	10 h	13 h	2 h	30 h

Course totals: lecture **320 h**, practical **640 h**, experiential **832 h**, modular assessment **128 h** — **1920 notional hours** over 64 credits. The split is fixed by the credit framework at one part lecture to two parts practical to three parts experiential, so two thirds of this course happens with the teacher not talking.

PART C · APPENDIX C9

Sources and further reading

The primary sources for the taxonomies, the curriculum this book is built on, and a note on what was and was not altered.

THE TAXONOMIES

- Bloom, B. S. (ed.), *Taxonomy of Educational Objectives. Handbook I: Cognitive Domain.* New York: Longmans, Green, 1956. The original, and the scheme Column 3E uses.
- Anderson, L. W. and Krathwohl, D. R. (eds.), *A Taxonomy for Learning, Teaching and Assessing.* New York: Longman, 2001. The revision: verb names, the swapped top two levels, and the knowledge axis.
- Krathwohl, D. R., Bloom, B. S. and Masia, B. B., *Taxonomy of Educational Objectives, Handbook II: Affective Domain.* New York: David McKay, 1964.
- Simpson, E. J., *The Classification of Educational Objectives in the Psychomotor Domain.* Washington: Gryphon House, 1972; and Dave, R. H., in R. J. Armstrong (ed.), *Developing and Writing Behavioral Objectives*, 1970.
- Krathwohl, D. R., 'A Revision of Bloom's Taxonomy: An Overview', *Theory into Practice*, 41(4), 2002.

THE CURRICULUM

- National Commission for Indian System of Medicine, Board of Ayurveda. *Curriculum for MD/MS Ayurveda, Semesters III to VI — Rasashastra and Bhaishajya Kalpana (AYPG-RB).* New Delhi: NCISM. Applicable from the 2024–25 batch. Tables 2, 3, 4 and 5 are the source of every objective, code, hour, mode and activity in this book; Table 6 is the source of the assessment figures in Chapter 8.

A NOTE ON HOW THIS BOOK WAS MADE

Every objective in Part B is reproduced in the curriculum's own words. Plain-English spelling errors have been corrected and sentence punctuation lost to the source document's line wrapping has been restored. No Sanskrit or technical term has been altered, and no subject content has been added, removed or reworded.

Extracting this curriculum needed several repairs peculiar to its layout: domain codes split across line breaks and interleaved with the level beside them, hours joined to the end of an objective's own full stop, activity references torn apart mid-word, syllabus points glued to the unit titles they follow, and the activity description tables which sit inside Table 3 and had to be separated from the objectives around them. Table 3 is also printed paper-first rather than semester-first, and has been reordered here.

The result reconciles against the curriculum's own credit framework at three levels: 577 objectives accounting for 1792 teaching hours, plus 128 hours of modular assessment, giving the **1920 notional hours** of 64 credits; each of the four semesters independently totalling 480 hours; and 304 activities matching the 141 practical and 163 experiential entries of Tables 4 and 5, with every reference in Table 3 resolving to one of them.

The counts and tables in Part C are generated from the same parsed data as Part B, so the two cannot disagree. Where a judgement had to be made — chiefly in reading the kind of knowledge from an objective's wording — it is made from the verb and its grammatical object, and marked with a small circle (°) where the wording supports a reading without settling it.

A COMPETENCY-ASSESSMENT HANDBOOK

Miller's Pyramid for Rasashastra & Bhaishajya Kalpana

Module-by-module competency mapping, assessment design and usage guide for AYPG-RB — Rasashastra and Bhaishajya Kalpana MD/MS Ayurveda, Semesters III to VI

Curriculum	NCISM, Board of Ayurveda — four semesters, four papers, applicable from the 2024–25 batch
Coverage	32 modules · 124 units · 577 session learning objectives · 141 practical training and 163 experiential learning activities
Framework	Miller (1990) four tiers, with the Cruess *Is* tier (2016), mapped onto NCISM Table 3, Column 3F

64	1920	1600	800
CREDITS	NOTIONAL HOURS	FORMATIVE MARKS	UNIVERSITY EXAM

Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026 · For departmental use in curriculum delivery, assessment design and paper setting.

⌂ HOME

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PART B — SEMESTER 3

Module 1 Chronological Development and Taxonomy of Rasa Dravyas 19 obj · Does 11

Module 2 Puta and Samskara (Transformative processes of drug manufacturing) 21 obj · Does 14

Module 9 Scientific fundamentals of Ayurvedic Pharmaceutical preparations 15 obj · Does 18

Module 10 Principles of pharmaceutical formulation 12 obj · Does 21

Module 17 Guidelines for administration of Rasoushadhi in clinical Practice 22 obj · Does 24

Module 18 Prayogika Bhaishajya Kalpana(Clinical application of Pharmaceutical products) 24 obj · Does 27

Module 25 Artificial intelligence in drug development and Pharmacogenetics; Pharmacogenomics 20 obj · Does 30

Module 26 Pharmacology 16 obj · Does 33

PART B — SEMESTER 4

Module 3 Parada (Mercury) and Its Processing: Ancient wisdom and Scientific advances 17 obj · Does 36

Module 4 Maharasa Dravya, its processing and utility 22 obj · Does 39

Module 11 Mana Paribhasha (Metrology in Ayurvedic Pharmaceutical preparation) 16 obj · Does 42

Module 12 Process validation Protocol 10 obj · Shows how 45

Module 19 SOP and clinical application of Chaturvidha Rasayana 25 obj · Does 48

Module 20 SOP and Clinical application of Ayaskriti, Louha, and Mandura Kalpa 12 obj · Does 51

Module 27 Network pharmacology & Reverse Pharmacology 16 obj · Does 54

Module 28 Pharmaceutical Quality Assurance 19 obj · Does 57

PART B — SEMESTER 5

Module 5 Critical appraisal of Uparasa Dravya and Sadharana Rasa Dravya 24 obj · Does 60

Module 6 Critical appraisal of Dhatu and Updhatu 24 obj · Does 63

Module 13 Krutanna (Ayurvedic Food Preparations) 15 obj · Does 66

Module 14 Ayurvedic Formulations for Holistic Wellbeing, Vitality and Body Care 11 obj · Shows how 69

Module 21 Clinical application of Rasoushadhi according to Srotas and in selected challenging clinical conditions 13 obj · Does 72

Module 22 Yogas from different Kalpanas 27 obj · Does 75

Module 29 Pharmacy Practice 18 obj · Does 79

Module 30 Pharmaceutical Microbiology 18 obj · Does 82

PART B — SEMESTER 6

Module 7 Sudha Varga, Sikata Varga, Visha Varga Dravyas 22 obj · Does 85

Module 8 Appraisal of Ratna and Uparatna and therapeutic significance of Pishti and Druti Kalpana 13 obj · Does 88

Module 15 Disease condition specific selection of formulation 12 obj · Shows how 91

Module 16 Current trends in Ayurvedic Pharmaceutical Industry. 12 obj · Does 94

Module 23 SOP and clinical applications of Churna Kalpana(Powder dosage forms) 26 obj · Does 97

Module 24 Therapeutic significance of Topical dosage forms 11 obj · Shows how 101

Module 31 Regulatory Acts and Rules applicable to Ayurvedic Drugs 31 obj · Does 104

Module 32 Entrepreneurship and Start-Ups 14 obj · Does 108

Every unit is listed with its ceiling in Appendix C2.

PART C — REFERENCE

C1 The four tiers, and what certifies each 112

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C3 The instruments this curriculum names, by tier 118

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PART A

Foundations

Five short chapters. The first explains what Miller's pyramid is and, just as importantly, what it is not. The second shows that it is already inside your curriculum, in a column most teachers have never had pointed out. The third gives the three rules of constructive alignment, the fourth sets out which instrument can certify which tier, and the fifth explains how to read a unit in Part B. Read straight through once; after that, use the modules.

1. **Miller's pyramid** — what it is, and what it is not
2. **The pyramid is already inside your curriculum** — reading Column 3F
3. **Constructive alignment** — the three rules
4. **The assessment toolkit**, tier by tier
5. **How to use a module in Part B**

↑ Contents

PART A · CHAPTER 1

Miller's pyramid — what it is, and what it is not

In 1990 George Miller, a medical educator, published a single diagram that has shaped assessment ever since. He was answering a question every examiner faces: *what does this test actually prove?*

His answer was a pyramid of four tiers, each demanding evidence more authentic than the one below.

- **Knows** — the student can recall it.
- **Knows how** — the student can reason with it, apply it to a case on paper.
- **Shows how** — the student can do it when someone is watching, in a set-up arranged for the purpose.
- **Does** — the student does it in real practice, unprompted, when nobody is checking.

The tiers are cumulative but not interchangeable. A student who reaches *Does* necessarily passed through the others; a student who has only been tested at *Knows* has been told nothing about the tiers above. That asymmetry is the whole point. **Evidence from a lower tier never certifies a higher one.**

WHAT THE PYRAMID IS NOT

It is not a taxonomy of thinking. It says nothing about whether a task is easy or hard, or whether it calls for recall or judgement. A *Knows* objective can be intellectually demanding and a *Does* objective can be routine. Bloom's taxonomy answers "what kind of thinking is this?"; Miller answers "how authentic must the evidence be?" The two are independent axes, and this curriculum carries both — Bloom in one column, Miller in another. Each has its own companion volume.

It is also not a ranking of worth. A curriculum that stops at *Shows how* for a dangerous procedure is not being lazy; it is making a judgement about supervision and safety. Reading the ceilings as a scoreboard is the commonest misuse of the pyramid.

> **The Cruess addition.** In 2016 Richard and Sylvia Cruess proposed a fifth tier above *Does*: **Is** — professional identity. Not what the student does but who they have become. It cannot be examined in a hall, and it is not coded anywhere in your curriculum. It is worth naming because some of what you certify at *Does* — a safety briefing recorded honestly, a result reported when it was inconvenient — is really evidence of *Is*. Where a unit in Part B says so, that is what it means.

WHY IT MATTERS HERE

These four semesters sit higher on the pyramid than anything before them: 289 of the 577 objectives — exactly half — are at *Shows how* or *Does*, and 67 reach *Does*. Half of what this course certifies cannot be certified on paper, and the summative examination at the end of Semester VI examines two years at once. Whatever was seen but not recorded is gone by then.

PART A · CHAPTER 2

The pyramid is already inside your curriculum

Miller is not something this book adds to your curriculum. It has been there since publication, in a column most teachers have never had pointed out.

COLUMN 3F

Open Table 3. The column headed **Level** carries one of four values against every objective: *Know*, *Knows how*, *Shows how* or *Does*. Those are Miller's four tiers, and they are the curriculum's statement of how authentic the evidence must be before that objective can be called met.

Across the 577 objectives of these four semesters the distribution is:

TIER	OBJECTIVES	SHARE
Knows	21	3.6%
Knows how	267	46.3%
Shows how	222	38.5%
Does	67	11.6%

Total 577 objectives.

The shape is a postgraduate one. Recall has largely been left behind — only 21 objectives sit at *Knows* — and the weight has moved to reasoning and performance. The *Does* tier is substantial rather than token, which means the record book and log book are load-bearing, not administrative.

> **A word about the other column headed "Domain".** Beside the Level column sits another carrying codes like CK, CC, CAP, PSY-MEC and AFT-VAL. Those are Bloom's taxonomy and the psychomotor and affective domains — a different question entirely, and the subject of the companion volume. In *this* book, "level" and "tier" always mean Miller.

THE CEILING

Every module in this curriculum has a **ceiling**: the highest tier any of its objectives reaches. That single fact governs assessment more than anything else on the page. It tells you what evidence you must gather, and it tells you what you may not ask for. Part B prints it for every module.

PART A · CHAPTER 3

Constructive alignment — the three rules

Constructive alignment is the principle that what you teach, what you ask students to do, and what you accept as proof must all sit at the same tier. Stated as rules, it comes to three.

RULE 1 — THE TIER DECIDES THE INSTRUMENT

Not the topic, not the hours, not the marks. Once you know an objective sits at *Shows how*, the set of instruments that can certify it is fixed, and a written paper is not among them. Chapter 4 gives the toolkit tier by tier.

RULE 2 — NEVER ASSESS ABOVE THE CEILING

If the curriculum certifies a module to *Knows how*, do not build an OSPE station for it. You would be examining something the students were never taught to that standard, and the marks would measure prior exposure rather than your teaching. This rule protects students.

RULE 3 — NEVER CERTIFY A HIGHER TIER WITH LOWER-TIER EVIDENCE

This is the rule that gets broken, and it is broken in one direction. A *Does* objective assessed by viva has not been assessed. A *Shows how* objective assessed by a long answer has not been assessed. The student may pass, the record may look complete, and nothing has been established.

> **The asymmetry is the thing to remember.** You may always gather evidence from a tier *above* the one required — a student observed doing the thing has plainly also shown how. You may never accept evidence from a tier *below*. Every alignment error worth catching is a violation of that one sentence.

WORKING THE THREE RULES BACKWARDS

In practice the fastest way to plan a session is to start at the end. Ask what evidence you will hold at the finish — a script, a checklist, a log entry. That names the tier. The tier names the instrument, and the instrument names the teaching that must precede it. A session planned forwards from the syllabus tends to arrive at the assessment it can most easily set, which is almost always a written one.

PART A · CHAPTER 4

The assessment toolkit, tier by tier

Each tier admits a particular set of instruments. Using one from the wrong tier is the single commonest fault in assessment design, and it is invisible unless you look for it, because the paperwork is complete either way.

TIER	THE QUESTION IT ANSWERS	TEACHING THAT BUILDS IT	INSTRUMENTS THAT CERTIFY IT	EVIDENCE YOU KEEP
Knows	**Can the student recall it?**	Lecture, reading, recitation, retrieval practice.	MCQ, short answer, oral quiz, spotting.	Written script, quiz sheet.
Knows how	**Can the student reason with it?**	Worked cases, discussion, problem-based work; the student must reason aloud and be corrected.	Structured long answer, case-based questions, structured viva.	Reasoned written answer, viva score sheet.
Shows how	**Can the student do it when watched?**	Demonstration, then supervised practice with your help withdrawn a little at a time.	OSPE or OSCE station, observed performance against a checklist.	Checklist score, annotated artefact, station record.
Does	**Does the student do it unprompted, in real practice?**	Real work with real consequences, supervised at a distance rather than at the elbow.	Log book, record book, portfolio, observation over time, presentation with peer feedback.	Dated log entry, countersigned record, portfolio piece.

THE TRAP AT EACH TIER

- **Knows** — Spending long-answer marks here. A long answer cannot separate a student who knows the list from one who knows it better.
- **Knows how** — Setting a question answerable from memory. If the stem can be met by recall, the objective has silently dropped to Knows.
- **Shows how** — Certifying by viva. A fluent student who has never handled the material will pass a viva comfortably.
- **Does** — Certifying with a written test. Nothing written can show that a student does something unprompted when nobody is watching.

> **The instruments this curriculum names.** This curriculum sets a modular assessment in every module, worth 25 marks per credit and 1600 marks across the four semesters, and a university examination at the end of Semester VI: four papers of 100 marks and a practical examination of 400.

ON THE LOG BOOK AND RECORD BOOK

These are the only instruments in routine use that can carry *Does*-tier evidence, and they are usually treated as paperwork. A log filled in at the end of term is a piece of writing. The same log filled in as the work happens, dated and countersigned, is the only proof you will have that a student did something unprompted. Decide at the start of each module what artefact each *Does* objective will leave behind, and the record becomes evidence rather than an administrative task.

PART A · CHAPTER 5

How to use a module in Part B

Part B has thirty-two modules, each carrying its units inside it. Each begins on its own page and is laid out the same way, so once you have read one you can use any of them.

WHAT IS ON THE PAGE

1. **The heading and counters** — hours, objectives, marks, and the **ceiling**: the highest tier this module reaches. Read the ceiling first; it governs everything else.
2. **The session map** — the curriculum's own objectives with the tier each is rated to, so you can see at a glance where the weight sits.
3. **The pyramid for this module** — four rows, one per tier. Each names what a student must be able to do at that tier here, then how to build it, how to assess it, and what evidence you keep. Tiers the curriculum does not use are shown greyed, because knowing what is *not* certified matters as much as knowing what is.
4. **The ceiling statement** — a single line naming the highest tier and warning against assessing above it or certifying it with the wrong instrument.
5. **Two panels** — one for the teacher, one to give to students.

THREE WAYS TO USE IT

Planning a session. Start at the ceiling row of the pyramid and work down: that row tells you what the students must end up able to do, and the rows beneath tell you what has to be in place first.

Setting a paper. A written paper can honestly reach *Knows* and *Knows how* and no further. Use the session map to see which objectives those are; anything above must be certified elsewhere, and the module says where.

Auditing what you already do. Take your existing assessment for a module and ask which tier it can certify. If it is a written test and the ceiling is *Shows how* or *Does*, the gap is your finding.

> **What this book does not do.** It does not rewrite the curriculum and does not tell you what to teach — every objective in Part B appears in the curriculum's own words. What it adds is the Miller layer: what standard of evidence each objective demands, and which instruments can supply it.

PART B

The modules

Thirty-two modules, ordered by semester and then by paper so the modules a department teaches together sit together. Each carries its units inside it, with every objective gathered under the tier its Level column gives it and the kind of hour beside it. The pyramid block names what every tier asks for in that module, how to build it, how to assess it and what evidence to keep. Chapter 5 explains the layout.

1. **Semester III** — Modules 1, 2, 9, 10, 17, 18, 25, 26
2. **Semester IV** — Modules 3, 4, 11, 12, 19, 20, 27, 28
3. **Semester V** — Modules 5, 6, 13, 14, 21, 22, 29, 30
4. **Semester VI** — Modules 7, 8, 15, 16, 23, 24, 31, 32

↑ Contents

MODULE 1 · SEMESTER 3 · PAPER 1

Chronological Development and Taxonomy of Rasa Dravyas

2 credits | 60 notional hours | 4 units | **19 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	10	7	2
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Recognise Rasadravya based on their Graahyatva-Agraahyatva; Identify Rasadravya using geological/mineralogical characters.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Construct a chronological tree of Rasashastra and its integration with other medical traditions; Demonstrate the application of the Mendeleev Periodic table in mapping Rasadravyas; Identify and analyse selected metals and minerals with the periodic table elements; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
10 objectives

Discuss the philosophical relevance of Rasashastra and historical evidence till the 6th century AD; Describe the chronological development of Rasashastra from the 6th Century AD to the medieval period; Discuss the chronological development of Rasashastra after the medieval period; and 7 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Chronological Development and Overview of Rasashastra

5 obj · 9 h · ceiling Shows how

1.1 Philosophical Relevance of Rasashastra
1.2 Chronological Development of Rasashastra
1.3 Integration of the development of Iatrochemistry with modern chemistry

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Discuss the philosophical relevance of Rasashastra and historical evidence till the 6th century AD.	Lecture	1	Knows how
Describe the chronological development of Rasashastra from the 6th Century AD to the medieval period.	Lecture	1	Knows how
Discuss the chronological development of Rasashastra after the medieval period.	Lecture	1	Knows how
Integrate the development of Iatrochemistry with modern chemistry.	Lecture	1	Knows how
SHOWS HOW — 1			
Construct a chronological tree of Rasashastra and its integration with other medical traditions.	Experiential Learning	5	Shows how

Unit 2 · Taxonomy of Rasadravyas

5 obj · 13 h · ceiling Shows how

2.1 Taxonomy of Rasadravyas
2.2 Concepts and justification about division of Rasadravyas
2.3 Mendeleeve periodic table

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Elaborate classification of Rasadravyas as per different ancient classics.	Lecture	1	Knows how
Justify the division of Rasadravyas.	Lecture	1	Knows how
Discuss the critics of the classification of Rasadravya.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the application of the Mendeleev Periodic table in mapping Rasadravyas.	Practical Training	5	Shows how
Identify and analyse selected metals and minerals with the periodic table elements.	Experiential Learning	5	Shows how

Unit 3 · Geological/mineralogical identification of Rasadravyas

6 obj · 22 h · ceiling Does

3.1 Grahya-Agrahyata of Rasadravyas
3.2 Geological/Mineralogical identification of Rasadravyas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss classical parameters for differentiating acceptable and non-acceptable Rasadravya.	Lecture	1	Knows how
Apply geological/mineralogical characters for the identification of Rasadravya.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the comparison and analyze the classification of Rasa Dravyas.	Practical Training	5	Shows how
Demonstrate Geological/Mineralogical identification of Rasadravyas.	Practical Training	5	Shows how
DOES — 2			
Recognise Rasadravya based on their Graahyatva-Agraahyatva.	Experiential Learning	5	Does
Identify Rasadravya using geological/mineralogical characters.	Experiential Learning	5	Does

Unit 4 · Overview of Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya

3 obj · 12 h · ceiling Shows how

4.1 Sandigdha (Controversial)
4.2 Anupalabdha (Not Available) and Krutrima (Artificial) Rasadravya
4.3 Pratinidhi (Substitute) Rasadravya

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstratie the Krutrima and Pratinidhi Rasadravyas.	Practical Training	5	Shows how
Identify and analyse Sandigdha and Anuplabhdha Rasadravyas and propose suitable alternatives.	Experiential Learning	6	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **9 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 2 · SEMESTER 3 · PAPER 1

Puta and Samskara (Transformative processes of drug manufacturing)

2 credits | 60 notional hours | 6 units | **21 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	7	11	1
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	9	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	8	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Explore Thermal Dynamics in Traditional Puta Systems: An of Fuel Variations, Temperature Patterns, and Thermodynamic Principles.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
11 objectives

Demonstrate and compare the temperature pattern and total duration of different traditional Puta with EMF; Perform comparison of Traditional and Modern Puta Methods: Observing and Analyzing Temperature Patterns and Duration; Demonstrate the art of Gandhaka Jarana: A Process, Understanding its Role and Significance in Rasashastra; and 8 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
7 objectives

Explore the traditional concept of Puta, discussing its definition, significance, and classification in the context of Ayurvedic pharmaceutical practices; Explain the laws of thermodynamics (Zeroth, First, Second, Third), the law of thermochemistry, Griffith's theory, and Collision theory; Discuss the fundamental principles and significance of Jarana, Amritikarana, and Lohitikarana procedures in the preparation of bhasmas; and 4 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Examine the modern applications and relevance of Puta in contemporary pharmaceutical processing, integrating traditional knowledge with modern technology; Discuss the concepts of Kalaprakarsha (processing with specific time exposure) and Bhajana (processing vessels) in Ayurvedic pharmaceutical processing, including the significance of processing time, vessel material, and other temporal factors on drug quality and efficacy.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Puta and Law of thermodynamics

6 obj · 18 h · ceiling Does

1.1 Definition, Significance and Classification of Puta

1.2 Laws of thermodynamics, Thermochemistry, Griffith theory, Collision theory, etc

1.3 Heating devices and fuels, with detailed knowledge about temperature, duration, designing and construction of different Puta

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Examine the modern applications and relevance of Puta in contemporary pharmaceutical processing, integrating traditional knowledge with modern technology.	Lecture	1	Knows
KNOWS HOW — 2			
Explore the traditional concept of Puta, discussing its definition, significance, and classification in the context of Ayurvedic pharmaceutical practices.	Lecture	1	Knows how
Explain the laws of thermodynamics (Zeroth, First, Second, Third), the law of thermochemistry, Griffith's theory, and Collision theory.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate and compare the temperature pattern and total duration of different traditional Puta with EMF.	Practical Training	5	Shows how
Perform comparison of Traditional and Modern Puta Methods: Observing and Analyzing Temperature Patterns and Duration.	Experiential Learning	5	Shows how
DOES — 1			
Explore Thermal Dynamics in Traditional Puta Systems: An of Fuel Variations, Temperature Patterns, and Thermodynamic Principles.	Experiential Learning	5	Does

Unit 2 · Concept, understanding, application, SOPs and chemical changes of Jarana, Amritikarana and Lohitikarana

4 obj · 16 h · ceiling Shows how

2.1 Principle and Significance of

2.2 Application of Amritikarana and Lohitikarana procedures in bhasma preparation

2.3 Standard Operating Procedures (SOP) and Chemical Changes of Jarana, Amritikarana, and Lohitikarana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss the fundamental principles and significance of Jarana, Amritikarana, and Lohitikarana procedures in the preparation of bhasmas.	Lecture	2	Knows how
SHOWS HOW — 3			
Demonstrate the art of Gandhaka Jarana: A Process, Understanding its Role and Significance in Rasashastra.	Practical Training	5	Shows how
Demonstrate Amritikarana and Lohitikarana: Traditional Methods for Mica Processing in Rasashastra.	Practical Training	5	Shows how
Perform an experiential Investigation of Amritikarana and Lohitikarana Procedures Exploring Safety, Efficacy, and Principles of Transformation.	Experiential Learning	4	Shows how

Unit 3 · Concept, understanding, application, SOPs (Standard Operating Procedures) and chemical changes of Satva and Druti procedures

4 obj · 9 h · ceiling Shows how

3.2 Application and analysis of Satva and Druti procedures

3.3 SOP and chemical changes of Satva and Druti

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the principles, significance, and applications of Satva procedures within the context of Ayurvedic pharmacy and modern pharmaceutical practices.	Lecture	1	Knows how
Describe the principles, significance, and apply Druti procedures within the context of Ayurvedic pharmacy.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the procedure of Satvapatana.	Practical Training	5	Shows how
Perform and analyse the Satva Patana procedure.	Experiential Learning	2	Shows how

Unit 4 · Samskara (Principles of Transformative processes of drug manufacturing)

2 obj · 3 h · ceiling Shows how

4.1 Application of Toya Sannikarsha (Processing with water) and Agni

4.2 Shaucha and Manthana (churning)

4.3 Vasana (process of infusing aroma), and Bhavana (Effect of Levigation / Trituration)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Integrate and apply classical Ayurvedic manufacturing principles, including water, heat and combined processing, cleaning, churning, proximity, and trituration effects.	Lecture	1	Knows how
SHOWS HOW — 1			
Apply Time-Tested Principles of Classical Drug Manufacturing: by integrating Toya Sannikarsha, Agni Sannikarsha, Toyagni Sannikarsha, Shaucha, and Manthana to Optimize Pharmaceutical Quality, Safety, and Efficacy.	Experiential Learning	2	Shows how

Unit 5 · Influence of Desha (place) and Kala(time and seasonal factors) of drug qualities

2 obj · 3 h · ceiling Shows how

5.1 Influence of Desha (geographical area and place of

5.2 Influence of Kala (processing time, season, and maturity of herbs) on drug quality

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Description of the influence of Desha (geographical area) and Kala (time and seasonal factors) on the quality of Ayurvedic drugs, including the impact of geographical location, climate, and seasonal variations on drug efficacy and safety.	Lecture	1	Knows how
SHOWS HOW — 1			
Perform A Hands-on Investigation of Geographical and Temporal Influences on drug 7, CO 8 quality: Desha–Kala in Pharmaceutical Manufacturing.	Experiential Learning	2	Shows how

Unit 6 · Kalaprakarsha- Processing with/Effect of specific time exposure and Bhajana (processing vessel)

3 obj · 7 h · ceiling Shows how

5.1 Knowledge of Kalaprakarsha (Time taken

5.2 Knowledge of Bhajana (vessels used for manufacturing)

5.3 Environmental and Temporal Influences on Drug Quality and Processing

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss the concepts of Kalaprakarsha (processing with specific time exposure) and Bhajana (processing vessels) in Ayurvedic pharmaceutical processing, including the significance of processing time, vessel material, and other temporal factors on drug quality and efficacy.	Lecture	1	Knows

SHOWS HOW — 2			
Demonstrate and analyze the role of Vasana and Bhavana to Optimize Pharmaceutical Quality, Safety, and Efficacy.	Experiential Learning	4	Shows how
Demonstrate the effect of Kalaprakarsha and Bhajana in Pharmaceutical Processing Analyzing Processes and Findings to Optimize Quality, and stability.	Experiential Learning	2	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **12 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 9 · SEMESTER 3 · PAPER 2

Scientific fundamentals of Ayurvedic Pharmaceutical preparations

2 credits | 60 notional hours | 4 units | **15 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	5	6	2
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Demonstrate processing of unique Panchavidha Kashaya Kalpana with their ingredients proportion of water and procedural details; Review Research Updates in the field of Ayurvedic pharmaceutics along with modern scientific validation methods.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Prepare unique formulation from the category of Panchavidha Kashaya Kalpana; Identify and apply various tools and techniques used for quality control of raw materials and appraise the classical rules of the collection of raw material; Demonstrate various formulations by describing their ingredients and understanding the uniqueness of their proportion; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Discuss the rationality behind the selection of Panchavidha Kashaya Kalpana and compare classical and modern drug extraction techniques; Describe classical and contemporary parameters for quality assurance of raw drugs, as well as adulteration and contamination of the raw material; Demonstrate application of QAQC parameters used for quality control of herbal drugs; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Describe the classical and contemporary methods used for the processing of drugs including the assessment of the End-Point(Siddhi Lakshanani) of the dosage form, C_L; Recognize the factors responsible for a change in the concentration of the prepared drug and assess the extracted phytochemicals with the help of TLC and other methods of analysis.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Classical Principles of drug processing and scientific basis of Panchavidha Kashaya Kalpana

3 obj · 6 h · ceiling Does

1.1 Bhaishajya Synonyms and significance of its
1.2 Classical principles of processing
1.3 Scientific review of Panchavidha Kashaya Kalpana and Upakalpana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss the rationality behind the selection of Panchavidha Kashaya Kalpana and compare classical and modern drug extraction techniques.	Lecture	1	Knows how
SHOWS HOW — 1			
Prepare unique formulation from the category of Panchavidha Kashaya Kalpana.	Experiential Learning	3	Shows how
DOES — 1			
Demonstrate processing of unique Panchavidha Kashaya Kalpana with their ingredients proportion of water and procedural details.	Practical Training	2	Does

Unit 2 · Identity, Purity and Quality of Raw materials used in Ayurveda Pharmaceutics

3 obj · 12 h · ceiling Shows how

2.1 Classical knowledge about Identity, Purity, Quality of Raw
2.2 Collection of Raw material- GACP, QAQC parameters
2.3 Adulteration and Contamination
2.4 Substitute drugs (Pratinidhi Dravya)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe classical and contemporary parameters for quality assurance of raw drugs, as well as adulteration and contamination of the raw material.	Lecture	2	Knows how
Demonstrate application of QAQC parameters used for quality control of herbal drugs.	Practical Training	4	Knows how
SHOWS HOW — 1			
Identify and apply various tools and techniques used for quality control of raw materials and appraise the classical rules of the collection of raw material.	Experiential Learning	6	Shows how

Unit 3 · Pharmaceutical processing of drugs with its rationality

3 obj · 13 h · ceiling Shows how

3.1 Processing of Drugs- Classical methodology
3.2 Simple and unique remedies from classical texts prepared with various processing methods
3.3 Rationality behind the Processing Method of Classical Dosage Forms
3.4 Siddhilakshanani (End-point assessment of pharmaceutical process) and its significance

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the classical and contemporary methods used for the processing of drugs including the assessment of the End-Point(Siddhi Lakshanani) of the dosage form, C_L.	Lecture	2	Knows
KNOWS HOW — 1			
Interpret significance of understanding the proportion of ingredients of formulations quoted in the classics.	Experiential Learning	6	Knows how
SHOWS HOW — 1			
Demonstrate various formulations by describing their ingredients and understanding the uniqueness of their proportion.	Practical Training	5	Shows how

Unit 4 · Ayurvedic Classical Dosage Forms and their integration with modern scientific validation

6 obj · 25 h · ceiling Does

4.1 Classification of the source of raw material and the
4.2 Multiple factors responsible for Process optimization
4.3 Key qualities of good pharmaceutical products to ensure safety, efficacy, and authenticity

4.4 Research updates about the Integration of traditional wisdom with modern scientific validation

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Recognize the factors responsible for a change in the concentration of the prepared drug and assess the extracted phytochemicals with the help of TLC and other methods of analysis.	Lecture	3	Knows
KNOWS HOW — 1			
Describe multiple factors responsible for the process optimization of Ayurvedic pharmaceutical preparations and illustrate key qualities of good pharmaceutical products to ensure safety, efficacy, and authenticity.	Lecture	2	Knows how
SHOWS HOW — 3			
Demonstrate the extraction of phytochemicals in the prepared dosage form.	Practical Training	4	Shows how
Perform the process of Extraction of Phytochemicals in Ayurvedic formulations.	Experiential Learning	5	Shows how
Demonstrate the kwatha preparation considering its process optimization.	Practical Training	5	Shows how
DOES — 1			
Review Research Updates in the field of Ayurvedic pharmaceutics along with modern scientific validation methods.	Experiential Learning	6	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 10 · SEMESTER 3 · PAPER 2

Principles of pharmaceutical formulation

2 credits | 60 notional hours | 5 units | **12 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	6	5	1
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Justify the need for Shaktyutkarsha (Potency enhancement)in drug preparation and Advocate techniques of potency enhancement used in Ayurvedic pharmaceutics.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
5 objectives

Illustrate the role of samskaras in the changes observed during the preparation of various dosage forms according to classical texts; Demonstrate combinations of various classical formulations and justify the rationale behind the ingredients; Demonstrate various designs of dosage forms, considering all fundamentals of pharmacology; and 2 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
6 objectives

Describe the applied principles of samskara involved in the preparation of various kalpana; Analyze the purpose of composition and combinations of ingredients in classical formulations; Justify the Principles of dosage form design as per modern pharmaceutical science; and 3 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Principles of Samskara (transformative processes of drug manufacturing) in relation to drug development

2 obj · 6 h · ceiling Shows how

1.1 Qualities of drugs/Procedures
1.2 Classical concept of Samskara and expected alterations in qualities
1.3 Key role of Samskara in pharmacological and therapeutic modification of the drug
1.4 Modifications in samskaras in current practices and their consequences

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe the applied principles of samskara involved in the preparation of various kalpana.	Lecture	2	Knows how
SHOWS HOW — 1			
Illustrate the role of samskaras in the changes observed during the preparation of various dosage forms according to classical texts.	Practical Training	4	Shows how

Unit 2 · Rationality behind Composition of Formulations

2 obj · 6 h · ceiling Shows how

2.1 Combinations of ingredients (desirable/ inadmissible)
2.2 Rational integrity behind the combination of ingredients
2.3 Aushadhi Gana (therapeutic groups) or Varga (categories) of medicinal herbs and their role in formulations
2.4 Combinations based on predominant pharmacological actions, qualities, or therapeutic usage of the ingredients

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Analyze the purpose of composition and combinations of ingredients in classical formulations.	Lecture	2	Knows how
SHOWS HOW — 1			
Demonstrate combinations of various classical formulations and justify the rationale behind the ingredients.	Practical Training	4	Shows how

Unit 3 · Principles and designing of various dosage forms

2 obj · 8 h · ceiling Shows how

3.1 Principles of dosage form design
3.2 Pharmaceutical Considerations
3.3 Formulation Considerations
3.4 Biopharmaceutical Considerations
3.5 Pharmacokinetic Considerations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Justify the Principles of dosage form design as per modern pharmaceutical science.	Lecture	2	Knows how
SHOWS HOW — 1			
Demonstrate various designs of dosage forms, considering all fundamentals of pharmacology.	Practical Training	6	Shows how

Unit 4 · Principles of Prakshep dravyas (Pharmaceutical Additives)

3 obj · 14 h · ceiling Shows how

4.1 Prakshepa Dravya- Need, Importance, Proportion (matra)
4.2 Pharmaceutical Excipients

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Identify the selection and relevance of Prakshepa dravya added to various dosage forms and describe Pharmaceutical excipients.	Lecture	2	Knows how

SHOWS HOW — 2			
Demonstrate and analyze the importance of pharmaceutical excipients.	Practical Training	6	Shows how
Illustrate various Prakshepa Dravyas and their proportions stated in the classics.	Experiential Learning	6	Shows how

Unit 5 · Principles of Potency and Efficacy enhancement

3 obj · 22 h · ceiling Does

5.1 Classical principles of Potency and efficacy enhancement (Shaktyutkarsha)
5.2 Methods of Potency and Efficacy Enhancement in Ayurvedic Classics
5.3 Methods of Potency and Efficacy Enhancement in Modern Pharmaceutics

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe and compare various techniques used in the classics for Shaktyutkarsha (Potency enhancement).	Lecture	2	Knows how
Investigate the principles for improving potency and efficacy in modern pharmaceutics.	Experiential Learning	10	Knows how
DOES — 1			
Justify the need for Shaktyutkarsha (Potency enhancement)in drug preparation and Advocate techniques of potency enhancement used in Ayurvedic pharmaceutics.	Experiential Learning	10	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **6 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 17 · SEMESTER 3 · PAPER 3

Guidelines for administration of Rasoushadhi in clinical Practice

2 credits | 60 notional hours | 5 units | **22 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	10	9	3
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	7	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
3 objectives

Generate document depicting the outcome of compliance and non-compliance with the pathya apathya guidelines; Evaluate the prescription errors containing Rasoushadhis; Evalute the effect of your chelation therapy in the management of our heavy metal toxicity.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
9 objectives

Demonstrate the rational prescription for a selected disease; Apply the principles of Rasa Chikitsa in the clinical practice/ alloted case; Demonstrate the need and role of Kshetrikarana; and 6 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
10 objectives

Analyse the term Rasa from the Sanskrit and Ayurvedic literatures and discuss the significance of Rasoushadhi; Discuss various treatment modalities used in the Ayurvedic system of medicine; Define kshetrikarana and discuss the concept of kshetrikarana; and 7 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Concept of Rasa chikitsa and significance of Rasoushadhi

4 obj · 11 h · ceiling Shows how

1.1 Perspectives on the Notion of Rasa
1.2 Significance of Rasa Chikitsa
1.3 Various treatment modalities of Ayurveda

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Analyse the term Rasa from the Sanskrit and Ayurvedic literatures and discuss the significance of Rasoushadhi.	Lecture	1	Knows how
Discuss various treatment modalities used in the Ayurvedic system of medicine.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the rational prescription for a selected disease.	Practical Training	3	Shows how
Apply the principles of Rasa Chikitsa in the clinical practice/ alloted case.	Experiential Learning	6	Shows how

Unit 2 · Concept of Kshetrikarana and its significance prior to administration of Rasoushadhi

4 obj · 11 h · ceiling Shows how

2.1 Etymology and Concept of Kshetrikarana
2.2 Kshetrikarana for Optimising Medicine Efficacy
2.3 Procedure of Kshetrikarana (Panchakarma & Pancha Shodhana)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Define kshetrikarana and discuss the concept of kshetrikarana.	Lecture	1	Knows how
Discuss the need and benefits of Kshetrikarana.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the need and role of Kshetrikarana.	Practical Training	3	Shows how
Demonstrate the procedure and benefits of kshetrikarana.	Experiential Learning	6	Shows how

Unit 3 · Pathya -Apathya during Rasayoga Sevana

5 obj · 12 h · ceiling Does

3.1 Pathya and Apathya Ahara, Vihara, and Manasika Vikara during Rasayoga Sevana
3.2 Importance of Pathya-Apathya during Rasayoga Sevana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Define Pathya and Apathya ahara and describe its importance.	Lecture	1	Knows how
Describe the application of the concept of what pathya and apathya vihara with respect to the Manasika vikara.	Lecture	1	Knows how
SHOWS HOW — 2			
Enlist and demonstrate common pathya and apathya Ahara during rasa yoga sevana.	Practical Training	2	Shows how
Enlist and demonstrate the common pathya-apathya Vihara during rasa yoga sevana.	Practical Training	2	Shows how
DOES — 1			
Generate document depicting the outcome of compliance and non-compliance with the pathya apathya guidelines.	Experiential Learning	6	Does

Unit 4 · Safety and toxicity of Rasoushadhi

5 obj · 14 h · ceiling Does

4.1 Safety considerations of Rasoushadhi
4.2 Ashuddha and Apakwa Bhasma Sevana Janya Vikara and their management

4.3 Safety and prescription trends of Rasoushadhi

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the safety considerations of Rasa oushadhies in detail.	Lecture	1	Knows how
Analyse the consequences of Ashuddha or Apakwa Bhasma sevana Dosha, along with the treatment.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate a case study of signs and symptoms of rasa vikaras and their management.	Practical Training	6	Shows how
Assess and comapre Ashuddha and Shuddha Bhasma effects in the clinical practice.	Practical Training	2	Shows how
DOES — 1			
Evaluate the prescription errors containing Rasoushadhis.	Experiential Learning	4	Does

Unit 5 · Heavy metal toxicity and chelation therapy

4 obj · 8 h · ceiling Does

5.1 The mechanism of heavy metal toxicity
5.2 Symptoms of heavy metal toxicity
5.3 Factors influencing heavy metal toxicity
5.4 Concept of chelation therapy and risks, and benefits

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss in detail the sources, pathways, and factors influencing heavy metal toxicity.	Lecture	1	Knows how
Discuss chelating agents and their role in managing heavy metal toxicity.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate identification of heavy metal toxicity.	Practical Training	2	Shows how
DOES — 1			
Evalute the effect of your chelation therapy in the management of our heavy metal toxicity.	Experiential Learning	4	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **12 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 18 · SEMESTER 3 · PAPER 3

Prayogika Bhaishajya Kalpana(Clinical application of Pharmaceutical products)

2 credits | 60 notional hours | 4 units | **24 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	11	11	2
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	8	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Demonstrate Modulation of dosage form action by chronotherapeutic administration in Ayurveda; Describe the role of Anupana in the modulation of drug tolerance and safety.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
11 objectives

Demonstrate chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dynamics; Demonstrate chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dynamics; Illustrate the modulation of the action of the dosage form by chronotherapeutic administration in Ayurveda; and 8 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
11 objectives

Discuss in detail the concept of Oushadha Sevana Kala in Ayurveda; Discuss biological rhythms and chronobiology and their significance in drug delivery; Elaborate the principles of chrono pharmaceutical drug delivery systems and their benefits; and 8 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Aushadha Sevana Kala (Time of administration of drug)

7 obj · 15 h · ceiling Does

1.1 Classical concept of time of administration
1.2 Biological rhythms and chronobiology
1.3 Chrono-pharmaceutics and chronopharmaceutical drug delivery systems
1.4 Principles of chronopharmacology

1.5 Benefits of chronotherapeutics

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss in detail the concept of Oushadha Sevana Kala in Ayurveda.	Lecture	1	Knows how
Discuss biological rhythms and chronobiology and their significance in drug delivery.	Lecture	1	Knows how
Elaborate the principles of chrono pharmaceutical drug delivery systems and their benefits.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dynamics.	Practical Training	3	Shows how
Demonstrate chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dynamics.	Experiential Learning	3	Shows how
Illustrate the modulation of the action of the dosage form by chronotherapeutic administration in Ayurveda.	Experiential Learning	3	Shows how
DOES — 1			
Demonstrate Modulation of dosage form action by chronotherapeutic administration in Ayurveda.	Practical Training	3	Does

Unit 2 · Concept and scientific basis of Anupana

6 obj · 16 h · ceiling Does

2.1 Classical concepts of Anupana in Ayurveda
2.2 Role of Anupana in bioavailability
2.3 Anupana for targeted drug delivery and distribution
2.4 Role of Anupana in the mitigation of adverse effects and toxicity
2.5 Anupana for synergistic effect

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss in detail the concept of anupana in Ayurveda.	Lecture	1	Knows how
Explain the role of Anupana in bioavailability and targeted drug delivery.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate appropriate Anupana for the selected dosage form in different clinical conditions.	Practical Training	3	Shows how
Enlist appropriate Anupanas for various formulations in different clinical conditions.	Experiential Learning	3	Shows how
Demonstrate drug response through synergistic Anupana integration.	Experiential Learning	4	Shows how
DOES — 1			
Describe the role of Anupana in the modulation of drug tolerance and safety.	Experiential Learning	4	Does

Unit 3 · Matra (Posology)

5 obj · 12 h · ceiling Shows how

3.1 Definition, importance, and clinical significance
3.2 Factors affecting drug dose
3.3 Posology according to modern pharmacology
3.4 Calculation of paediatric dose

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Elaborate on the concept, importance, and clinical significance of Matra.	Lecture	1	Knows how
Discuss the factors affecting drug dose in detail.	Lecture	1	Knows how
Demonstrate the calculation of paediatric dose using different formulae.	Practical Training	3	Knows how

SHOWS HOW — 2			
Demonstrate the Patient-centric evaluation of Matra.	Practical Training	3	Shows how
Demonstrate the utility of posology principles from an ayurvedic and modern pharmacology perspective.	Experiential Learning	4	Shows how

Unit 4 · Concept of drug repurposing

6 obj · 13 h · ceiling Shows how

4.1 Concept of drug repurposing
4.2 Systematic approaches to drug repurposing
4.3 Strategies of drug repurposing
4.4 Methodologies of drug repurposing

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe the concept, need, and key aspects of drug repurposing.	Lecture	1	Knows how
Describe systematic approaches to drug repurposing.	Lecture	1	Knows how
Discuss strategies and methodologies of drug repurposing.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the analysis of classical ayurvedic drugs for novel indications.	Practical Training	5	Shows how
Evaluate novel indications for classical Ayurvedic formulations.	Experiential Learning	3	Shows how
Demonstrate the Reverse pharmacological approach for drug repurposing in Ayurveda.	Experiential Learning	2	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **13 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 25 · SEMESTER 3 · PAPER 4

Artificial intelligence in drug development and Pharmacogenetics; Pharmacogenomics

2 credits | 60 notional hours | 5 units | **20 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	11	7	2
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Perform De Novo Drug. Design Using Generative Models; Demonstrate prediction of drug likeliness using machine learning.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate the basics of machine learning and its application; Demonstrate different 3D printers used in the drug delivery system; Analyse ADMET properties using AI tools; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
11 objectives

Define AI and discuss its importance in drug development; Discuss the Role of AI in Drug Discovery and Drug development with a suitable example; Describe different types of software used in AI; and 8 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Role of Artificial intelligence in drug development

4 obj · 11 h · ceiling Does

1.1 Introduction to AI in drug development
1.2 AI, machine-based intelligence learning
1.3 Role of AI in Drug discovery and Drug development

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Define AI and discuss its importance in drug development.	Lecture	1	Knows how
Discuss the Role of AI in Drug Discovery and Drug development with a suitable example.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the basics of machine learning and its application.	Practical Training	4	Shows how
DOES — 1			
Perform De Novo Drug. Design Using Generative Models.	Experiential Learning	5	Does

Unit 2 · Types of software used in Artificial intelligence

3 obj · 6 h · ceiling Does

2.1 Different types of software used in AI
2.2 Data mining and analysing medical records, medication management
2.3 Recent updates

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe different types of software used in AI.	Lecture	1	Knows how
Discuss Data mining and analysing medical records, medication management, with recent updates.	Lecture	1	Knows how
DOES — 1			
Demonstrate prediction of drug likeliness using machine learning.	Practical Training	4	Does

Unit 3 · 3D printed drug delivery system

5 obj · 16 h · ceiling Shows how

3.1 Approaches of 3D printing in current drug delivery
3D printing in personalized drug delivery
3D-Printed Drug Delivery Systems Based on Natural Products

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe the fundamentals and explore the materials and technology used in 3D printing.	Lecture	1	Knows how
Discuss the concept of personalised drug delivery, formulations, and natural products.	Lecture	1	Knows how
Demonstrate the comparison of traditional dosage forms versus 3D printed ones regarding the structure and dissolution.	Practical Training	4	Knows how
SHOWS HOW — 2			
Demonstrate different 3D printers used in the drug delivery system.	Practical Training	4	Shows how
Analyse ADMET properties using AI tools.	Experiential Learning	6	Shows how

Unit 4 · Pharmacogenetics and Pharmacogenomics

4 obj · 11 h · ceiling Shows how

4.1 Introduction to Pharmacogenetics and Pharmacogenomics along with its importance

4.2 Factors influencing drug response and ADE (Adverse drug events)

4.3 Methods of pharmacogenetics testing

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Appraise key concepts and terminologies and explore the role of Pharmacogenetics and Pharmacogenomics.	Lecture	1	Knows how
Justify clinical application of pharmacogenetics, benefits, and challenges.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate to predict Drug-Target Interaction.	Practical Training	4	Shows how
Identify potential adverse effects of drug candidates using dataset and tools.	Experiential Learning	5	Shows how

Unit 5 · Pharmacogenomics in drug discovery

4 obj · 12 h · ceiling Shows how

5.1 Benefits of Pharmacogenomics

5.2 Drug Repurposing

5.3 Basis of gene-drug association

5.4 Current status and future perspective of Pharmacogenomics

5.5 Ethical concern

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss pharmacogenomics data contributes to the identification and validation of drug targets.	Lecture	1	Knows how
Discuss the current status and future perspectives of pharmacogenomics, including ethical concerns.	Lecture	1	Knows how
SHOWS HOW — 2			
Extract potential drug-disease relationship by using through Literature /data Mining.	Experiential Learning	5	Shows how
Interpret Pharmacokinetics and Toxicology Modelling using appropriate AI tools and methods.	Experiential Learning	5	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **9 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 26 · SEMESTER 3 · PAPER 4

Pharmacology

2 credits | 60 notional hours | 5 units | **16 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	11	1	2
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	2	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Analyse Adverse Drug Reaction in a polypharmacy patient through a case scenario; Discuss Emergency Pharmacology through a case scenario of anaphylaxis management.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
1 objective

Analyse the mechanism of action of drugs acting on the respiratory system through a case study scenario.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
11 objectives

Discuss dose response relationship, and factors influencing drug action; Demonstrate pharmacological experiments using videos or simulation software; Discuss therapeutic index, drug interactions, and adverse drug reactions; and 8 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Define and classify pharmacology and describe the principles and mechanisms of drug action, receptor theories, and classification of rece; Describe the basic principles of pharmacokinetics (absorption, distribution, metabolism 7, CO 8 excretion).

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Introduction to Pharmacology

3 obj · 12 h · ceiling Knows how

1.1 Introduction to Pharmacology
1.2 Pharmacodynamics
1.3 Dose-response relationship and factors influencing drug action

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Define and classify pharmacology and describe the principles and mechanisms of drug action, receptor theories, and classification of rece.	Lecture	1	Knows
KNOWS HOW — 2			
Discuss dose response relationship, and factors influencing drug action.	Lecture	1	Knows how
Demonstrate pharmacological experiments using videos or simulation software.	Practical Training	10	Knows how

Unit 2 · Pharmacokinetics, therapeutic index, and drug interactions

3 obj · 12 h · ceiling Knows how

2.1 Pharmacokinetics (absorption, distribution, metabolism, excretion)
2.2 Therapeutic index and combined effects of drugs
2.3 Adverse drug reaction
2.4 Drug interactions (pharmacokinetic and pharmacodynamic)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the basic principles of pharmacokinetics (absorption, distribution, metabolism 7, CO 8 excretion).	Lecture	1	Knows
KNOWS HOW — 2			
Discuss therapeutic index, drug interactions, and adverse drug reactions.	Lecture	1	Knows how
Demonstrate the effect of drugs on animals by simulated experiments.	Practical Training	10	Knows how

Unit 3 · Pharmacology of drugs that act on the peripheral nervous system

3 obj · 9 h · ceiling Knows how

3.1 Organization and function of ANS (Autonomic Nervous System)
3.2 Neurohumoral transmission, co-transmission, and classification of neurotransmitters
3.3 Parasympathomimetics, Parasympatholytics, Sympathomimetics, Sympatholytics
3.4 Neuromuscular blocking agents and skeletal muscle relaxants (peripheral)
3.5 Local anesthetic agents

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the organisation and function of ANS (Autonomic Nervous System) and describe Neurohumoral transmission, co-transmission, and classification of neurotransmitters.	Lecture	1	Knows how
Enlist Parasympathomimetics, Parasympatholytics, Sympathomimetics, sympatholytics Neuromuscular blocking agents, and skeletal muscle relaxants (peripheral) and Local anaesthetic agents.	Lecture	1	Knows how
Analyse the mechanism of action of drugs acting on the Cardiovascular system through a case study scenario.	Experiential Learning	7	Knows how

Unit 4 · Pharmacology of drugs that act on the central nervous system I

3 obj · 9 h · ceiling Shows how

4.1 Neurohumoral transmission in the C.N.S. special emphasis on the importance of
4.2 General anaesthetics and pre-anaesthetics
4.3 Sedatives, hypnotics, and centrally acting muscle relaxants
4.4 Anti-epileptics

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe Neurohumoral transmission in the C.N.S and illustrate General anaesthetics and pre-anaesthetics.	Lecture	1	Knows how
Categorize drugs into classes such as Sedatives, hypnotics, and centrally acting muscle relaxants, and discuss anti-epileptics.	Lecture	1	Knows how
SHOWS HOW — 1			
Analyse the mechanism of action of drugs acting on the respiratory system through a case study scenario.	Experiential Learning	7	Shows how

Unit 5 · Pharmacology of drugs that act on the central nervous system II

4 obj · 14 h · ceiling Does

5.1 Psychopharmacological agents: Antipsychotics, antidepressants, anti-anxiety
5.2 Drugs used in Parkinson's disease and Alzheimer's disease
5.3 CNS stimulants and nootropics
5.4 Drug addiction, drug abuse, tolerance, and dependence

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss Psychopharmacological agents and enlist drugs used in Parkinson's disease and Alzheimer's disease.	Lecture	1	Knows how
Differentiate CNS stimulants and nootropics, and discuss drug addiction, drug abuse tolerance, and dependence.	Lecture	1	Knows how
DOES — 2			
Analyse Adverse Drug Reaction in a polypharmacy patient through a case scenario.	Experiential Learning	6	Does
Discuss Emergency Pharmacology through a case scenario of anaphylaxis management.	Experiential Learning	6	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **3 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 3 · SEMESTER 4 · PAPER 1

Parada (Mercury) and Its Processing: Ancient wisdom and Scientific advances

2 credits | 60 notional hours | 4 units | **17 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	7	7	1
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	9	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	3	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Perform the Vishesha Shodhana procedures of Parada.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate Samanya and Vishesha Shodhana procedures of Parada; Perform the Parada Jarana, Parada Murchana and Bandhana processes; Perform the Samanya Shodhana of Parada; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
7 objectives

Describe the ancient and contemporary perspectives of Parada (Mercury); Discuss the rationale behind Samanya and Vishesha Shodhana; Describe the scientific basis of Parada Jarana, Murcchana, and Bandha; and 4 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Discuss the processes related to Parada and their significance; Describe and elaborate on the metallurgy and stoichiometry of mercury.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Parada: Ancient and Contemporary Perspectives on Its Shodhana and Underlying Principles

8 obj · 22 h · ceiling Does

1.1 Ancient & Contemporary knowledge of Parada/
1.2 Parada Dosha, Shodhana(Purification) of Parada and its significance
1.3 Principles of Parada Jarana and Parada Murcchana
1.4 Utility of Parada Bandha

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss the processes related to Parada and their significance.	Lecture	1	Knows
KNOWS HOW — 3			
Describe the ancient and contemporary perspectives of Parada (Mercury).	Lecture	1	Knows how
Discuss the rationale behind Samanya and Vishesha Shodhana.	Lecture	1	Knows how
Describe the scientific basis of Parada Jarana, Murcchana, and Bandha.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate Samanya and Vishesha Shodhana procedures of Parada.	Practical Training	7	Shows how
Perform the Parada Jarana, Parada Murchana and Bandhana processes.	Practical Training	5	Shows how
Perform the Samanya Shodhana of Parada.	Experiential Learning	3	Shows how
DOES — 1			
Perform the Vishesha Shodhana procedures of Parada.	Experiential Learning	3	Does

Unit 2 · Parada Ashta Samskara and Ashtadasha Samskara

4 obj · 21 h · ceiling Shows how

2.1 Parada Samskara with research updates
2.2 Pharmaceutical equipment design for Parada Ashta Samskara
2.3 Hands-on experience of Parada Ashta Samskara

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe and discuss the utility of Ashtadasha Samskara.	Lecture	2	Knows how
Describe the design of pharmaceutical equipment to carry out Parada Ashta Samskara.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the Parada Samskara procedures- Swedana, Mardana, Murchana, and Utthapana.	Practical Training	8	Shows how
Elaborate and perform the Parada Ashta Samskara procedures- Patana, Bodhana Niyamana and Deepana.	Experiential Learning	10	Shows how

Unit 3 · Physico-chemical proprties and pharmacology of mercury and mercurial compounds

2 obj · 6 h · ceiling Shows how

3.1 Mercury metallurgy
3.2 Stoichiometry of mercury
3.3 Pharmacodynamics and pharmacokinetics of mercury and mercurial products

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe and elaborate on the metallurgy and stoichiometry of mercury.	Lecture	1	Knows
SHOWS HOW — 1			
Compare and analyze the Pharmacodynamics and pharmacokinetics of mercury and mercurial products.	Experiential Learning	5	Shows how

Unit 4 · Mercury toxicity, ADR and its management

3 obj · 7 h · ceiling Shows how

4.1 Mercury poisoning and its types

4.2 Ancient and Contemporary knowledge about the Treatment of Mercury poisoning

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the mercury poisoning and analyze its mechanism.	Lecture	1	Knows how
Describe the antidotes for mercury poisoning and its possible treatment by ancient and contemporary knowledge.	Lecture	1	Knows how
SHOWS HOW — 1			
Analyse mercury toxicity and ADR incidences.	Experiential Learning	5	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 4 · SEMESTER 4 · PAPER 1

Maharasa Dravya, its processing and utility

2 credits | 60 notional hours | 5 units | **22 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	12	9	1
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	6	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Explore and perform the pharmaceutical procedure, like Shodhana, Marana for Chapala/Kharpara, using authentic or substitute drugs.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
9 objectives

Analyze and Verify Maharasa Dravya Samples; Perform Shodhana, Marana, and other pharmaceutical procedures related to Abhraka and Vaikranta; Perform Shodhana, Marana, and other pharmaceutical procedures related to Makshika and Vimala; and 6 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
12 objectives

Describe and critically discuss the different Maharasa Dravyas mentioned in the classics; Analyze the technique of characterization for Maharasa Dravyas; Describe the significance of Abhraka Pancha Samskara; and 9 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Maharasa Dravyas with special reference to Abhraka and Vaikranta

8 obj · 18 h · ceiling Shows how

1.1 Comparative analysis of Maharasa Dravyas mentioned in various texts
1.2 Characterization of Maharasa dravyas
1.3 Abhraka
1.4 Vaikranta

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 6			
Describe and critically discuss the different Maharasa Dravyas mentioned in the classics.	Lecture	1	Knows how
Analyze the technique of characterization for Maharasa Dravyas.	Lecture	1	Knows how
Describe the significance of Abhraka Pancha Samskara.	Lecture	1	Knows how
Discuss the mineralogical identity of Vaikranta.	Lecture	1	Knows how
Analyze the pharmacotherapeutic attributes of Abhraka Bhasma and Vaikranta Bhasma.	Experiential Learning	5	Knows how
Analyse the pharmacology and clinical applications of Abhraka and Vaikranta.	Experiential Learning	2	Knows how
SHOWS HOW — 2			
Analyze and Verify Maharasa Dravya Samples.	Practical Training	2	Shows how
Perform Shodhana, Marana, and other pharmaceutical procedures related to Abhraka and Vaikranta.	Practical Training	5	Shows how

Unit 2 · Principles and Processing of Makshika and Vimala

3 obj · 11 h · ceiling Shows how

2.1 Critical understanding of Makshika (Copper Pyrite)
2.2 Detailed understanding of Vimala (Iron Pyrite)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe the therapeutic applications of Makshika and Vimala Bhasma, along with the antidotes for toxicity/ADR caused by them.	Lecture	1	Knows how
SHOWS HOW — 2			
Perform Shodhana, Marana, and other pharmaceutical procedures related to Makshika and Vimala.	Practical Training	5	Shows how
Perform the Shodhana of Makshika Bhasma and Vimala Bhasma.	Experiential Learning	5	Shows how

Unit 3 · Principles and Processing of Shilajit and Sasyaka

3 obj · 10 h · ceiling Shows how

3.1 Shilajatu (Black Bitumen)
3.2 Sasyaka(Bornite)/Tuttha(Blue Vitriol)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe and analyze the therapeutic applications of Shilajatu and Sasyaka/Tuttha, along with the antidotes for toxicity/ADR caused by them.	Lecture	1	Knows how
SHOWS HOW — 2			
Perform Shodhana, Marana, and other pharmaceutical procedures related to Shilajatu and Sasyaka/Tuttha.	Practical Training	3	Shows how
Perform the Shodhana of Shilajit and Sasyaka/Tuttha.	Experiential Learning	6	Shows how

Unit 4 · Principles and Processing of Chapala and Kharpara, along with physico-chemical analysis, with research updates of all Maharasa Dravyas

4 obj · 9 h · ceiling Does

4.2 Kharpara

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe the therapeutic applications of Chapala and Kharpara Bhasma, along with the antidotes for toxicity/ADR caused by them.	Lecture	1	Knows how
SHOWS HOW — 2			
Perform Shodhana, Marana, and other pharmaceutical procedures related to Chapala/ Kharpara.	Practical Training	2	Shows how
Demonstrate and perform physico-chemical analysis of Bhasma / Shodhita products of Maharasa Dravya category.	Practical Training	3	Shows how
DOES — 1			
Explore and perform the pharmaceutical procedure, like Shodhana, Marana for Chapala/ Kharpara, using authentic or substitute drugs.	Experiential Learning	3	Does

Unit 5 · Utility of new metals and minerals

4 obj · 8 h · ceiling Shows how

5.1 New metals/ minerals that align with Maharasa Dravyas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe the utility of new metals and minerals that align with Maharasa Dravyas.	Lecture	1	Knows how
Explore Shodhana process for the new metal/ minerals that align with Maharasa Dravyas.	Lecture	1	Knows how
Describe Marana process for the new metal/ minerals that align with Maharasa Dravyas.	Lecture	1	Knows how
SHOWS HOW — 1			
Identify and analyze New Metallic/Mineral Resources that align with Maharasa Dravyas Principles.	Experiential Learning	5	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **10 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 11 · SEMESTER 4 · PAPER 2

Mana Paribhasha (Metrology in Ayurvedic Pharmaceutical preparation)

2 credits | 60 notional hours | 5 units | **16 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1 certified here	5 certified here	7 certified here	3 certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	6	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
3 objectives

Compare and judge the differences in the Mana mentioned by different classics; Analyze and justify the proportions of ingredients of classical formulations; Debate on merits and demerits of bulk production and the adaptation of advanced machines in Ayurvedic Pharmacies.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Apply the Knowledge of mana Paribhasha in the preparation of various formulations; Preparation of charts of unit converter (Ayurvedic Mana- Metric); Demonstrate preparation of classical formulations prepared with converted Units; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Interpret the application of Mana in classical formulations with a variety of examples; Describe standard units of weights and measures and AYUSH Unit Converter; Define the terms Volumetric methods and Gravimetric methods and apply the methods of both measurements to Ayurvedic Pharmaceutical Preparations; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Recall classical knowledge of Mana Paribhasha with a comparison of various classical texts. EC, LS, S.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Mana Paribhasha (Classical system of measurement)

4 obj · 12 h · ceiling Does

1.1 Importance of accurate measurements in proportions of ingredients

1.2 Importance of accurate measurements in dosage and administration of drugs

1.3 Types of Mana and details of classical units used for measurement (Weight/ Volume/ Length/ Time)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Recall classical knowledge of Mana Paribhasha with a comparison of various classical texts. EC, LS, S.	Lecture	1	Knows
KNOWS HOW — 1			
Interpret the application of Mana in classical formulations with a variety of examples.	Lecture	1	Knows how
SHOWS HOW — 1			
Apply the Knowledge of mana Paribhasha in the preparation of various formulations.	Practical Training	4	Shows how
DOES — 1			
Compare and judge the differences in the Mana mentioned by different classics.	Experiential Learning	6	Does

Unit 2 · Standard Units of Measurement

3 obj · 6 h · ceiling Shows how

2.1 AYUSH Unit Converter

2.2 Utility of converted units in the preparation of Classical formulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe standard units of weights and measures and AYUSH Unit Converter.	Lecture	1	Knows how
SHOWS HOW — 2			
Preparation of charts of unit converter (Ayurvedic Mana- Metric).	Practical Training	2	Shows how
Demonstrate preparation of classical formulations prepared with converted Units.	Experiential Learning	3	Shows how

Unit 3 · Volumetric Vs Gravimetric Methods

3 obj · 6 h · ceiling Shows how

3.1 Key differences between Volumetric and gravimetric methods of measurement for a variety of drugs

3.2 Application of volumetric method for preparation of various classical formulations

3.3 Variation in proportion by using both methods and alteration in outcome

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Define the terms Volumetric methods and Gravimetric methods and apply the methods of both measurements to Ayurvedic Pharmaceutical Preparations.	Lecture	1	Knows how
SHOWS HOW — 2			
Apply and demonstrate the volumetric method for the preparation of classical formulations.	Practical Training	2	Shows how
Justify and interpret differences occurring in the final product prepared by varying the weight of the ingredients once taken by volume and once taken by mass(Gravimetric Method).	Experiential Learning	3	Shows how

Unit 4 · Rationality of Proportion of Ingredients in Ayurvedic formulations

3 obj · 16 h · ceiling Does

4.1 Factors accountable for deciding the proportion of ingredients of the

4.2 Profile of Raw Material (Fresh/ dry, Soft/ hard, part used, therapeutic properties, compatibility, etc.)

4.3 Rules for Duplication of ingredients in the same formulation

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe the classical rules and principles for the proportion of ingredients based on raw material and its nature/ Qualities.	Lecture	3	Knows how
SHOWS HOW — 1			
Demonstrate the skill of preparing formulations with a diverse range of raw materials.	Practical Training	6	Shows how
DOES — 1			
Analyze and justify the proportions of ingredients of classical formulations.	Experiential Learning	7	Does

Unit 5 · Bulk production of Ayurvedic Formulations

3 obj · 16 h · ceiling Does

5.1 Modifications expected for bulk production (in Material, Machine, Method, Mechanism)

5.2 Material- Type, proportion, effect of temperature pattern

5.3 Machine used for bulk manufacturing and automated drug manufacturing

5.4 Mechanism of machines used for bulk production and its merits and demerits

5.5 Methods used for bulk production in Ayurvedic Pharmacies and probable alterations in the qualities of the end products

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Enlist the machines, heating devices for bulk production. and describe their methods currently used by Ayurvedic pharmacies f.	Lecture	3	Knows how
SHOWS HOW — 1			
Demonstrate the use of machinery used for large-scale production in Ayurvedic pharmacies, with their mechanism.	Practical Training	6	Shows how
DOES — 1			
Debate on merits and demerits of bulk production and the adaptation of advanced machines in Ayurvedic Pharmacies.	Experiential Learning	7	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **10 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 12 · SEMESTER 4 · PAPER 2

Process validation Protocol

2 credits | 60 notional hours | 3 units | **10 objectives** | 50 formative marks | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	6	4	—
not used	certified here	certified here	not used

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	4	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	2	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
not used here

This module does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
4 objectives

Develop a process validation protocol for Ayurvedic dosage forms; Interpret the documentation related to Standard Operating Procedures (SOPs) used in pharmacies; Prepare different novel modern dosage forms of the given drug; and 1 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
6 objectives

Describe the SOP of Primary dosage forms, Secondary dosage forms, and topical dosage forms; Assess the significance of the Process Validation Protocol; Demonstrate the development of a process validation protocol for Ayurvedic dosage forms; and 3 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this module cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Pharmaceutical Preparation process: Classical Dosage forms

6 obj · 41 h · ceiling Shows how

1.1 Primary Dosage forms- Panchavidha Kashaya Kalpana and their Upakalpana

1.2 Secondary Dosage Forms- Avaleha Kalpana, Lavana Kalpana, Sneha Kalpana, Guti-vati-guggulu kalpana, Sandhana Kalpana, Arka-Kalpana, Kshara-kalpana

1.3 Topical Dosage forms- Lepa kalpana, Malahara Kalpana, Upanaha Kalpana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Describe the SOP of Primary dosage forms, Secondary dosage forms, and topical dosage forms.	Lecture	3	Knows how
Assess the significance of the Process Validation Protocol.	Lecture	2	Knows how
Demonstrate the development of a process validation protocol for Ayurvedic dosage forms.	Practical Training	10	Knows how
Prepare unique classical formulations following constructed SOP.	Experiential Learning	10	Knows how
SHOWS HOW — 2			
Develop a process validation protocol for Ayurvedic dosage forms.	Practical Training	10	Shows how
Interpret the documentation related to Standard Operating Procedures (SOPs) used in pharmacies.	Experiential Learning	6	Shows how

Unit 2 · Pharmaceutical Preparation process: Recent advances of Dosage forms

3 obj · 10 h · ceiling Shows how

2.1 Principles of processes viz. Expression, Extraction, Maceration,

2.2 Solid Dosage forms- Powders, Pills, Tablets, Capsules, Soft gel capsules, Gummies, Lozenges, Granules

2.3 Liquid Dosage Forms- Emulsions, Suspensions, Lotions, Liniments, Elixir, Syrup, Nasal Drops, Gargle

2.4 Semi-solid dosage forms- Ointment, Gel, Paste, Cream, Plasters, Suppositories

2.5 Gaseous dosage forms- Aerosols, Nasal Spray, Inhalers

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe principles of different dosage forms, viz. Extraction, Maceration, Percolation Infusion, Distillation, Active principle isolation, Particle size reduction, Separation techniques, and Mixing techniques.	Lecture	2	Knows how
Discuss Pharmaceutical Preparation Process of Solid, semisolid, liquid, and gaseous dosage forms.	Lecture	3	Knows how
SHOWS HOW — 1			
Prepare different novel modern dosage forms of the given drug.	Experiential Learning	5	Shows how

Unit 3 · Novel Dosage and drug delivery Forms

1 obj · 5 h · ceiling Shows how

3.1 Importance and Need of Different Dosage Forms in Drug Delivery Systems and Treatment Modalities

3.2 Definition, significance, technologies driving Novel dosage forms, applications, and challenges

3.3 Modification of different dosage forms

3.4 Potential areas for future research and development

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
SHOWS HOW — 1			
Analyze, justify, and demonstrate the importance and need of the preparation of Novel dosage forms.	Experiential Learning	5	Shows how

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **4 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Shows how.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 19 · SEMESTER 4 · PAPER 3

SOP and clinical application of Chaturvidha Rasayana

3 credits | 90 notional hours | 4 units | **25 objectives** | 75 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	13	9	2
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	15 h	14	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	30 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	39 h	7	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 6 hours carry the modular assessment, worth 75 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Evaulate the pharmacodynamic rationale and clinical application of parpati kalpana; Prepare Pottali following SOP.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
9 objectives

Demonstrate the method of preparation and SOP of Kharaleeya Rasayana; Appraise the utility of kharaleeya rasayana to clinical case practice; Demonstrate the impact of Bhavana (A bioactivation tool) in Kharaleeya Rasayana; and 6 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
13 objectives

Describe the unique features, types, and significance of parpati kalpana; Discuss the role of bhavana (pharmacodynamics of trituration) in potentiating kharaleeya rasayana; Describe the unique features, types, and significance of parpati kalpana; and 10 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Review and analyze research updates on Parpati Kalpana.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Standard Operating Procedure (SOP) and Clinical Application of Khalveeya Rasayana

5 obj · 17 h · ceiling Shows how

1.1 Definition and significance of Khalveeya Rasayana
1.2 Standard Operating Procedure for the preparation of Khalveeya Rasayana
1.3 Role of bhavana in potentiating Khalveeya rasayana
1.4 Clinical application of Khalveeya Rasayana with examples

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the unique features, types, and significance of parpati kalpana.	Lecture	2	Knows how
Discuss the role of bhavana (pharmacodynamics of trituration) in potentiating kharaleeya rasayana.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the method of preparation and SOP of Kharaleeya Rasayana.	Practical Training	7	Shows how
Appraise the utility of kharaleeya rasayana to clinical case practice.	Experiential Learning	4	Shows how
Demonstrate the impact of Bhavana (A bioactivation tool) in Kharaleeya Rasayana.	Experiential Learning	3	Shows how

Unit 2 · SOP and clinical application of Parpati Rasayana

6 obj · 15 h · ceiling Does

2.1 Definition and significance of Parpati
2.2 Standard Operating Procedure for the preparation of Parpati with process validation
2.3 Mechanism of action of Parpati kalpana
2.4 Pathyapathya in Parpati Prayoga and its significance
2.5 Clinical application of Parpati Kalpa with examples

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Review and analyze research updates on Parpati Kalpana.	Lecture	1	Knows
KNOWS HOW — 3			
Describe the unique features, types, and significance of parpati kalpana.	Lecture	1	Knows how
Discuss SOP and quality control parameters of Parpati Kalpa.	Lecture	1	Knows how
Discuss in detail the pathya-apathya, matra, prayoga of parpati kalpana.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the SOP of parpati kalpana.	Practical Training	7	Shows how
DOES — 1			
Evaulate the pharmacodynamic rationale and clinical application of parpati kalpana.	Experiential Learning	4	Does

Unit 3 · SOP and clinical application of Pottali Rasayana

7 obj · 25 h · ceiling Does

3.1 Definition and significance of Pottali rasayana
3.2 Standard Operating Procedure for the preparation of Pottali Kalpa with process validation
3.3 Role of Gandhaka drava paka in the preparation Pottali rasayana
3.4 Special clinical attributes of Pottali with examples

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Define pottali and discuss in detail the concept, types, and significance of pottali rasayana.	Lecture	1	Knows how
Discuss in detail the various classical methods of preparation of pottali kalpana.	Lecture	1	Knows how
Discuss the standardization and quality control parameters for Pottali Kalpa.	Lecture	1	Knows how
Evalaute the safety and efficacy of Pottali Rasayana.	Lecture	1	Knows how

SHOWS HOW — 2			
Demonstrate the method of preparation and SOP of pottali kalpana.	Practical Training	8	Shows how
Analyse the pharmacological profile of the ingredients, bhavana dravya, and the pharmaceutical procedure adopted in achieving therapeutic precision.	Experiential Learning	5	Shows how
DOES — 1			
Prepare Pottali following SOP.	Experiential Learning	8	Does

Unit 4 · SOP and clinical application of Kupipakwa Rasayana

7 obj · 27 h · ceiling Shows how

4.1 Definition and significance of Kupipakwa rasayana
4.2 Standard Operative Procedure for the preparation of Kupipakwa Rasayana with process validation
4.3 Role of Kramagni paka in preparation of Kupipakwa Rasayana
4.4 Chemistry of Kupipakwa rasayana
4.5 Clinical application of Kupipakwa Rasayana Kalpa with examples

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Discuss the concept and significance of Kupipakwa Rasayana.	Lecture	1	Knows how
Describe in detail the role of Kramagni Paka in the preparation of Kupipakwa Rasayana.	Lecture	1	Knows how
Describe the role of gandhaka jarana in the context of sagandha kupipakwa rasayana.	Lecture	1	Knows how
Discuss the thermo–chemistry of kupipakwa rasayana.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the method of preparation and SOP of Kupipakwa Rasayana.	Practical Training	8	Shows how
Prepare the selected Kupipakwa Rasayana.	Experiential Learning	10	Shows how
Demonstrate the scientific rationality for the selection of specific Kupipakwa Rasayana in the given scenario.	Experiential Learning	5	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **11 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 75 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 20 · SEMESTER 4 · PAPER 3

SOP and Clinical application of Ayaskriti, Louha, and Mandura Kalpa

1 credits | 30 notional hours | 3 units | **12 objectives** | 25 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	5	6	1
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	5 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	10 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	13 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 2 hours carry the modular assessment, worth 25 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Evaluate the role of Gomutra in pharmaceutical processing and as a bio enhancer.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate the SOP and preparation of Ayaskriti; Evaluate the role of Ayaskriti.as a multi-target formulation; Demonstrate the SOP and preparation of Louha Kalpa; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Discuss in detail the concept of Ayaskriti, its types, method of administration, and safety concerns; Discuss the concept of Louha Kalpa; Discuss the key chemical considerations in Louha Kalpa; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · SOP and clinical application of Ayaskriti

3 obj · 8 h · ceiling Shows how

1.1 Ayaskriti
1.2 Standard Operating Procedure for the preparation of Ayaskriti
1.3 Types (forms) of ayaskriti
1.4 Method of administration and clinical utility of Ayaskriti
1.5 Safety concerns of Ayaskriti

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss in detail the concept of Ayaskriti, its types, method of administration, and safety concerns.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the SOP and preparation of Ayaskriti.	Practical Training	3	Shows how
Evaluate the role of Ayaskriti.as a multi-target formulation.	Experiential Learning	4	Shows how

Unit 2 · SOP and clinical application of Louha kalpa

4 obj · 9 h · ceiling Shows how

2.1 Pharmaceutical vistas of Louha Kalpas
2.2 Key chemical considerations in Louha Kalpa
2.3 Pharmacology and clinical utility of Louha kalpa

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the concept of Louha Kalpa.	Lecture	1	Knows how
Discuss the key chemical considerations in Louha Kalpa.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the SOP and preparation of Louha Kalpa.	Practical Training	3	Shows how
Apply Louha Kalpa by integrating through textual insights with real word cases for the therapeutic purpose.	Experiential Learning	4	Shows how

Unit 3 · SOP and Clinical application of Mandura Kalpa

5 obj · 11 h · ceiling Does

3.1 Concept of mandura kalpa and its SOP
3.2 Types of mandura kalpa based on the method of preparation
3.3 Significance of go mutra in mandura kalpa
3.4 Clinical application of mandura kalpa

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the concept and pharmaceutical vistas of Mandura Kalpa.	Lecture	1	Knows how
Describe chemistry and pharmacology associated with structure and preparation of Mandura Kalpas.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the SOP and preparation of Mandura Kalpa.	Practical Training	2	Shows how
Demonstrate the clinical application of Mandura Kalpa.	Practical Training	2	Shows how
DOES — 1			
Evaluate the role of Gomutra in pharmaceutical processing and as a bio enhancer.	Experiential Learning	5	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **7 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 25 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 27 · SEMESTER 4 · PAPER 4

Network pharmacology & Reverse Pharmacology

2 credits | 60 notional hours | 2 units | **16 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	9	4	2
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	2	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Practice Network Pharmacology; Carry out a mini-project on network pharmacology.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
4 objectives

Demonstrate network pharmacology applications with the help of a video; Demonstrate reverse pharmacology applications with the help of a video; Present observations of analysis of Reverse Pharmacology in selected cases; and 1 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Define network pharmacology and differentiate it from traditional pharmacology; Describe network construction from herbal compounds; Discuss and analyze Research updates on Reverse pharmacology; and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Recognize key network concepts like nodes (genes/proteins/drugs) and edges(interactions).

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Netwok Pharmacology

8 obj · 28 h · ceiling Does

1.1 Fundamental knowledge of Network pharmacology
1.2 Network pharmacology concepts
1.3 Computational methods
1.4 Applications
1.5 Research updates on Network pharmacology

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Recognize key network concepts like nodes (genes/proteins/drugs) and edges(interactions).	Lecture	1	Knows
KNOWS HOW — 4			
Define network pharmacology and differentiate it from traditional pharmacology.	Lecture	1	Knows how
Describe network construction from herbal compounds.	Lecture	1	Knows how
Discuss and analyze Research updates on Reverse pharmacology.	Lecture	1	Knows how
Discuss various applications and research updates on Network pharmacology.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate network pharmacology applications with the help of a video.	Practical Training	10	Shows how
DOES — 2			
Practice Network Pharmacology.	Experiential Learning	10	Does
Carry out a mini-project on network pharmacology.	Experiential Learning	3	Does

Unit 2 · Reverse Pharmacology

8 obj · 28 h · ceiling Shows how

2.1 Introduction, Definition of Reverse Pharmacology, History, and Scope
2.2 Drug Discovery Current Scenario, Challenges, and Future Perspective
2.3 Reverse pharmacology for herbal drug-based modern medicines
2.4 Research updates on Reverse pharmacology

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Describe the definition and history of reverse pharmacology and discuss its Scope.	Lecture	1	Knows how
Elaborate current scenario, challenges, and future perspectives of reverse pharmacology.	Lecture	1	Knows how
Assess reverse pharmacology leverages traditional medicine and clinical experience to accelerate drug development.	Lecture	1	Knows how
Apply Reverse pharmacology for herbal-based modern medicines.	Lecture	1	Knows how
Discuss and analyze Research updates on Reverse pharmacology.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate reverse pharmacology applications with the help of a video.	Practical Training	10	Shows how
Present observations of analysis of Reverse Pharmacology in selected cases.	Experiential Learning	10	Shows how
Validate Reverse pharmacology with a systems biology approach.	Experiential Learning	3	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **6 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 28 · SEMESTER 4 · PAPER 4

Pharmaceutical Quality Assurance

2 credits | 60 notional hours | 5 units | **19 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	9	6	3
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
3 objectives

Inspect Packaging Materials to check their compliance with quality specifications; Perform calibration of Analytical Instruments; Conduct QA Documentation audit.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate the process of mock GMP Inspection; Demonstrate GMP Hierarchy and Responsibilities; Identify the Root cause, affected personnel roles, and suggest corrective and preventive measures through Case Study Discussion on GMP Violation; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Define and elaborate the concept of Quality Assurance and Quality Management as per; Describe Total Quality Management (TQM) and ICH Guidelines, QSEM, and ISO 9000; Illustrate the Steps for registration of NABL accreditation; and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Elaborate GLP with its Principles and Requirements, documentation, Regulatory compliance, Audit procedures.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Quality Assurance and Quality Management concepts

6 obj · 19 h · ceiling Shows how

1.1 Introduction to Quality Assurance and Quality Management concepts
1.2 Definition and concept of Quality control,
1.3 Quality assurance and GMP
1.4 Total Quality Management (TQM)
1.5 ICH Guidelines
1.6 Brief overview of Pharmaceutical industry guidelines for Quality, Safety, Efficacy and Multidisciplinary (QSEM)

1.7 Overview of ISO 9000 & ISO14000

1.8 Steps for registration of NABL accreditation

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Define and elaborate the concept of Quality Assurance and Quality Management as per.	Lecture	1	Knows how
Describe Total Quality Management (TQM) and ICH Guidelines, QSEM, and ISO 9000.	Lecture	1	Knows how
Illustrate the Steps for registration of NABL accreditation.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the process of mock GMP Inspection.	Practical Training	5	Shows how
Demonstrate GMP Hierarchy and Responsibilities.	Practical Training	5	Shows how
Identify the Root cause, affected personnel roles, and suggest corrective and preventive measures through Case Study Discussion on GMP Violation.	Experiential Learning	6	Shows how

Unit 2 · Personnel Responsibilities

3 obj · 8 h · ceiling Shows how

2.1 Organization and personnel

2.2 Personnel responsibilities, training, hygiene, and personal records

2.3 Premises: Design, construction, and plant layout, maintenance, sanitation, environmental control, utilities, and maintenance of sterile areas, control of contamination

2.4 Equipment and raw materials: Equipment selection, purchase specifications, maintenance, and purchase specifications and maintenance of stores for raw materials

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe Organizational structure and personnel management.	Lecture	1	Knows how
Discuss design, construction, and maintenance of Premises.	Lecture	1	Knows how
SHOWS HOW — 1			
Document a personal record of pharmacy personnel.	Experiential Learning	6	Shows how

Unit 3 · Containers, Packing materials

4 obj · 13 h · ceiling Does

3.1 Types of Packaging

3.2 Packaging materials

3.3 Container types and closures

3.4 Packaging and stability

3.5 Regulatory aspects

3.6 Quality Control

3.7 Smart Packaging

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss selection and maintenance of equipment, purchase specifications of raw materials, and maintenance of stores.	Lecture	1	Knows how
Discuss Types of containers and packaging, Regulatory aspects, their Quality Control and Smart Packaging.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the identification and Classification of packaging materials.	Practical Training	5	Shows how
DOES — 1			
Inspect Packaging Materials to check their compliance with quality specifications.	Experiential Learning	6	Does

Unit 4 · Good Laboratory Practices

3 obj · 8 h · ceiling Does

4.1 GLP Principles and Requirements

4.2 GLP documentation

4.3 Regulatory compliance

4.4 Audit procedures

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Elaborate GLP with its Principles and Requirements, documentation, Regulatory compliance, Audit procedures.	Lecture	1	Knows
SHOWS HOW — 1			
Demonstrate SOP Writing for different equipment.	Practical Training	5	Shows how
DOES — 1			
Perform calibration of Analytical Instruments.	Experiential Learning	2	Does

Unit 5 · Document maintenance in pharmaceutical industry

3 obj · 8 h · ceiling Does

4.1 Documentation related to Grievances and Redressal

4.2 Complaints and evaluation of complaints, Handling of returned goods, recalling, and waste disposal

4.3 Document maintenance in the pharmaceutical industry: Batch Formula Record, Master Formula Record, SOP, Quality audit, Quality Review, Quality documentation,

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Define Complaints, their evaluation, product recall, and waste disposal.	Lecture	1	Knows how
Describe document maintenance in the pharmaceutical industry.	Lecture	1	Knows how
DOES — 1			
Conduct QA Documentation audit.	Experiential Learning	6	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **9 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 5 · SEMESTER 5 · PAPER 1

Critical appraisal of Uparasa Dravya and Sadharana Rasa Dravya

2 credits | 60 notional hours | 2 units | **24 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	10	12	2
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	8	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Prepare and present the formulations of Sadharana Rasa Dravya; Assemble the utility of Sadharana Dravyas in the clinical practice through a Field Survey of manufacturers and clinicians.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
12 objectives

Demonstrate the authentication of Uparasa Dravyas; Demonstrate Shodhana, Marana and other pharmaceutical procedures for Uparasa; Demonstrate Quality Control testing for the products of Uparasa Dravya; and 9 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
10 objectives

Describe he different Uparasa Dravyas mentioned in the classics; Discuss the controversial drugs under Uparasa category; Discuss the mineralogical characterization of Uparasa; and 7 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Principle and Processing of Uparasa Dravya

12 obj · 28 h · ceiling Shows how

1.1 Uparasa Dravyas mentioned in various texts
1.2 Characterization of Uparasa Dravyas
1.3 Pharmaceutical processes of Uparasa Dravya
1.4 Pharmacology and clinical application of Uparasa Dravya

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Describe he different Uparasa Dravyas mentioned in the classics.	Lecture	1	Knows how
Discuss the controversial drugs under Uparasa category.	Lecture	1	Knows how
Discuss the mineralogical characterization of Uparasa.	Lecture	1	Knows how
Recognise the therapeutic applications and toxico-kinetics of Uparasa Dravyas.	Lecture	1	Knows how
Discuss research updates on Uparasa drugs.	Lecture	1	Knows how
SHOWS HOW — 7			
Demonstrate the authentication of Uparasa Dravyas.	Practical Training	2	Shows how
Demonstrate Shodhana, Marana and other pharmaceutical procedures for Uparasa.	Practical Training	4	Shows how
Demonstrate Quality Control testing for the products of Uparasa Dravya.	Practical Training	4	Shows how
Identify and authenticate Uparasa Dravyas.	Experiential Learning	3	Shows how
Perform Quality Control tests for the products of Uparasa Dravya.	Experiential Learning	3	Shows how
Prepare formulations of Uparasa Dravya.	Experiential Learning	5	Shows how
Review articles on the toxicity and ADRs of Haratala and Manahshila and interpret.	Experiential Learning	2	Shows how

Unit 2 · Principle and Processing of Sadharana Rasa

12 obj · 28 h · ceiling Does

2.1 Sadharana rasa Dravyas mentioned in various texts
2.2 Characterization of Sadharana Rasa Dravyas
2.3 Pharmaceutical processes of Sadharana rasa Dravya
2.4 Pharmacology and clinical application of Sadharana rasa Dravya

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Describe and critically discuss the different Sadharana Rasa Dravyas mentioned in the classics.	Lecture	1	Knows how
Analyze the controversial drugs under Sadharana Rasa category.	Lecture	1	Knows how
Discuss the mineralogical characterization of Sadharana Rasa.	Lecture	1	Knows how
Recognise the therapeutic applications and toxico-kinetics of Sadharana Rasa Dravyas.	Lecture	1	Knows how
Discuss research updates on Sadharana Rasa drugs.	Lecture	1	Knows how

SHOWS HOW — 5			
Demonstrate authentication of Sadharana Rasa Dravya.	Practical Training	2	Shows how
Demonstratie Shodhana, Marana and other pharmaceutical procedures for Sadharana.	Practical Training	4	Shows how
Demonstrate Quality Control testing for the products of Sadharana Rasa Dravya.	Practical Training	4	Shows how
Identify and authenticate Sadharana Rasa Dravyas.	Experiential Learning	3	Shows how
Perform Quality Control tests for the products of Sadharana Rasa Dravya.	Experiential Learning	3	Shows how
DOES — 2			
Prepare and present the formulations of Sadharana Rasa Dravya.	Experiential Learning	3	Does
Assemble the utility of Sadharana Dravyas in the clinical practice through a Field Survey of manufacturers and clinicians.	Experiential Learning	4	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **14 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 6 · SEMESTER 5 · PAPER 1

Critical appraisal of Dhatu and Updhatu

2 credits | 60 notional hours | 2 units | **24 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	10	11	3
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	8	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
3 objectives

Identify and document classical Dhatus; Prepare formuations using Dhatu Bhasma; Prepare Upadhatu formulations.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
11 objectives

Demonstrate the authentication of Dhatu Dravya; Perform pharmaceutical processing (Shodhana & Marana) of metals; Demonstrate Quality control of metallic preparations; and 8 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
10 objectives

Describe classical Dhatu Dravyas; Discuss the controversial metals in Ayurveda texts; Characterize Dhatu Dravyas mineralogically and pharmaceutically; and 7 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Principle and Processing of Dhatu / Metals

12 obj · 28 h · ceiling Does

1.1 Dhatus mentioned in Rasa classics
1.2 Classification and characterization of metals
1.3 Pharmaceutical processing (Shodhana, Marana, etc.)
1.4 Therapeutic uses, toxicity, and safety of metallic preparations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Describe classical Dhatu Dravyas.	Lecture	1	Knows how
Discuss the controversial metals in Ayurveda texts.	Lecture	1	Knows how
Characterize Dhatu Dravyas mineralogically and pharmaceutically.	Lecture	1	Knows how
Identify therapeutic applications and toxico-kinetics of Dhatu.	Lecture	1	Knows how
Compile and interpret research and regulatory updates on metal-based Ayurvedic drugs.	Lecture	1	Knows how
SHOWS HOW — 5			
Demonstrate the authentication of Dhatu Dravya.	Practical Training	2	Shows how
Perform pharmaceutical processing (Shodhana & Marana) of metals.	Practical Training	4	Shows how
Demonstrate Quality control of metallic preparations.	Practical Training	4	Shows how
Perform Quality assessment of Dhatu Bhasma.	Experiential Learning	3	Shows how
Identify and analyze Adverse drug reactions (ADRs) due to metals.	Experiential Learning	2	Shows how
DOES — 2			
Identify and document classical Dhatus.	Experiential Learning	3	Does
Prepare formuations using Dhatu Bhasma.	Experiential Learning	5	Does

Unit 2 · Principle and Processing of Upadhatu

12 obj · 28 h · ceiling Does

2.1 Classical analysis of Upadhatu
2.2 Classification, sources, and characteristics of Upadhatu
2.3 Pharmaceutical processing (Shodhana, Marana, etc.)of Upadhatu
2.4 Pharmacological actions and clinical application of Upadhatu

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Describe and classify Upadhatu from Rasashastra references.	Lecture	1	Knows how
Discuss controversial and newly interpreted Upadhatu.	Lecture	1	Knows how
Analyze Upadhatu mineralogically and pharmaceutically.	Lecture	1	Knows how
Recognize therapeutic applications and possible ADRs of Upadhatu.	Lecture	1	Knows how
Review and interpret findings of research and current practices involving Upadhatu.	Lecture	1	Knows how

SHOWS HOW — 6			
Demonstrate the Identification of Upadhatu samples.	Practical Training	2	Shows how
Demonstrate Shodhana and Marana of Upadhatu.	Practical Training	4	Shows how
Demonstrate the Quality control tests for sub-metallic formulations.	Practical Training	4	Shows how
Identify the Upadhatu samples.	Experiential Learning	3	Shows how
Perform Quality Control tests for the products of Upadhatu.	Experiential Learning	3	Shows how
Analyze and interpret findings of the utility of Dhatu and Upadhatu in clinical practice through a Field survey with Manufacturers or Vaidyas.	Experiential Learning	4	Shows how
DOES — 1			
Prepare Upadhatu formulations.	Experiential Learning	3	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **14 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 13 · SEMESTER 5 · PAPER 2

Krutanna (Ayurvedic Food Preparations)

2 credits | 60 notional hours | 4 units | **15 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	4	6	4
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	5	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
4 objectives

Prepare distinctive formulations using Aahara Dravya from different groups (Vargas); Develop the disease-specific Personalised prescription of Aushadhi Siddha Ahara; Demonstrate the preparation of novel Ayurvedic food products; and 1 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate the method of preparation of distinctive formulations using Aahara Dravya from different groups (Vargas); Demonstrate preparation of therapeutic diet formulations; Identify and prepare disease specific Ayurvedic Ahara; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Describe Ayurvedic dietary principles and the significance of food substances used as medicine; Describe disease-specific diet formulations; Recite paribhasha and describe the importance of Yusha, Yavagu, Odana, and Mamsarasa; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Describe the regulation standards for Ayurvedic food preparations- FSSAI (Ayurveda Aahara regulations).

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Classification of food substances and Formulations

3 obj · 12 h · ceiling Does

1.1 Ayurvedic dietary principles and Ahara Dravya Varga (Groups of food substance)

1.2 Aahara -mahabhaishajya: Significance of food substances used as medicine

1.3 Unique classical formulations prepared using Food substances belonging to various groups and their utility in treatment

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe Ayurvedic dietary principles and the significance of food substances used as medicine.	Lecture	2	Knows how
SHOWS HOW — 1			
Demonstrate the method of preparation of distinctive formulations using Aahara Dravya from different groups (Vargas).	Practical Training	4	Shows how
DOES — 1			
Prepare distinctive formulations using Aahara Dravya from different groups (Vargas).	Experiential Learning	6	Does

Unit 2 · Customized food for treatment

6 obj · 24 h · ceiling Does

2.1 Disease-specific Ayurvedic Aahara- Therapeutic utility

2.2 Samsarjana krama- Manda, Peya, Vilepi, Yavagu, Yusha, Odana, Krushara, Mansarasa- Process validation Protocol for all formulations

2.3 Personalised prescription of Aushada Siddha Aahara for Shamana and during panchakarma

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe disease-specific diet formulations.	Lecture	2	Knows how
Recite paribhasha and describe the importance of Yusha, Yavagu, Odana, and Mamsarasa.	Lecture	2	Knows how
SHOWS HOW — 3			
Demonstrate preparation of therapeutic diet formulations.	Practical Training	4	Shows how
Identify and prepare disease specific Ayurvedic Ahara.	Experiential Learning	6	Shows how
Demonstrate the method of preparation of various Aahara Kalpana.	Practical Training	4	Shows how
DOES — 1			
Develop the disease-specific Personalised prescription of Aushadhi Siddha Ahara.	Experiential Learning	6	Does

Unit 3 · Ayurvedic Food Industry

3 obj · 10 h · ceiling Does

3.1 Market survey about Ayurvedic food products and research updates

3.2 Global needs and designing of modified food formulations

3.3 Semi-processed food/ semi-cooked food, RTC- ready to cook / RTD-Ready to drink Foods and drinks

3.4 Regulation Standards for Ayurvedic Food Preparation (Ayurveda Aahara Regulations- 2021), FSSAI Regulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Describe the regulation standards for Ayurvedic food preparations- FSSAI (Ayurveda Aahara regulations).	Lecture	2	Knows
DOES — 2			
Demonstrate the preparation of novel Ayurvedic food products.	Practical Training	4	Does
Collect the data and interpret the findings through a market survey to explore the available Ayurvedic food products and assess the current trends in the Ayurvedic food industry.	Experiential Learning	4	Does

Unit 4 · Food processing technology traditional and modern methods

3 obj · 10 h · ceiling Shows how

4.1 Traditional Methods- Steaming, Boiling, Drying, Salting, Smoking, Frying,

4.2 Modern Methods- Heating, Freeze Drying, Vacuum Drying, Canning, Juicing, Fermentation, Modified atmosphere packaging, Pasteurisation, Smoking, Pulse electric

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Differentiate between classical and contemporary food processing techniques and support the current and emerging trends in the Ayurvedic food industry.	Lecture	2	Knows how
SHOWS HOW — 2			
Demonstrate various food processing techniques used in current Ayurvedic food Industries.	Practical Training	4	Shows how
Analyze Functional Food and Modified Food Formulations in Ayurvedic Food Industry.	Experiential Learning	4	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **10 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 14 · SEMESTER 5 · PAPER 2

Ayurvedic Formulations for Holistic Wellbeing, Vitality and Body Care

2 credits | 60 notional hours | 2 units | **11 objectives** | 50 formative marks | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	4	7	—
not used	certified here	certified here	not used

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	4	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	3	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
not used here

This module does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate the preparation method of Rasayana Drugs following validated SOP; Demonstrate the role of Ayurvedic Rasayana formulations in the maintenance of health and Vajikarana Formulations for better progeny(Suprajanana); Select appropriate Ayurvedic Formulations for the fitness of specific body parts and analyze their role in the maintenance of health; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Describe the mode of action of Rasayana Formulations; Describe classical Rasayana and Vajikarana Formulations; Discuss the role of Ayurvedic formulations in Cosmetic preparations; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this module cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Rasayana-Personalized Rejuvenation Formulations

5 obj · 24 h · ceiling Shows how

1.1 Key aspects of Rasayana Formulations

1.2 General Rasayana and Organ-specific/Target-specific Rasayana- Varnya, Tvachya, Kanthya, Swarya, Keshya, Hrudya, Medhya- Formulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the mode of action of Rasayana Formulations.	Lecture	2	Knows how
Describe classical Rasayana and Vajikarana Formulations.	Lecture	2	Knows how
SHOWS HOW — 3			
Demonstrate the preparation method of Rasayana Drugs following validated SOP.	Practical Training	8	Shows how
Demonstrate the role of Ayurvedic Rasayana formulations in the maintenance of health and Vajikarana Formulations for better progeny(Suprajanana).	Experiential Learning	5	Shows how
Select appropriate Ayurvedic Formulations for the fitness of specific body parts and analyze their role in the maintenance of health.	Experiential Learning	7	Shows how

Unit 2 · Ayurvedic based cosmetic preparation

6 obj · 32 h · ceiling Shows how

2.1 Body/ Hair/ Lip/ Oral/Skin/ Foot care products- Preparation methods and application in Ayurvedic

2.2 Preparation process of Lip balm, lipsticks, shampoos, cold cream and vanishing cream, toothpaste, hair dyes, Sunscreen lotion, Face pack, Mask, Toners, herbal soaps,

2.3 Current and Emerging trends – Ayurvastram, Ayurvedic footwear, Eye mask

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the role of Ayurvedic formulations in Cosmetic preparations.	Lecture	3	Knows how
Evaluate the Global need for body care products in the healthcare sector.	Lecture	3	Knows how
SHOWS HOW — 4			
Demonstrate preparation of Herbal Cosmetics.	Practical Training	6	Shows how
Demonstrate the preparation of Modified Cosmeceuticals.	Practical Training	6	Shows how
Analyze different cosmetics available in the market.	Experiential Learning	7	Shows how
Design and prepare novel Ayurvedic cosmeceuticals.	Experiential Learning	7	Shows how

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **7 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Shows how.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 21 · SEMESTER 5 · PAPER 3

Clinical application of Rasoushadhi according to Srotas and in selected challenging clinical conditions

1 credits | 30 notional hours | 3 units | **13 objectives** | 25 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	5	7	1
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	5 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	10 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	13 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 2 hours carry the modular assessment, worth 25 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Explore the role of Rasoushadhi in clinically challenging conditions such as neurodevelopmental and neurodegenerative diseases.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate the clinical application of Rasoushadhi in selected Srotas, illustrating their practical utility and therapeutic outcomes in addressing specific health conditions; Illustrate the therapeutic application of Rasoushadhi in different Srotas; Demonstrate the clinical applicability of Rasoushadhi in Medical Emergencies; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Discuss Srotas-specific Rasoushadhi, tailored Ayurvedic formulations for individual biological channels; Enlist and discuss Rasoushadhi for different Srotas, explaining their therapeutic efficacy and mechanism of action; Discuss the unique properties of Rasoushadhi, enabling its use in clinical emergencies; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Clinical application of Rasoushadhis specific to Srotas

4 obj · 8 h · ceiling Shows how

1.1 Concept and rationality of selection and application of Rasoushadhi according to Srotas

1.2 Clinical application of Rasouashdhi in disorders of different Srotas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss Srotas-specific Rasoushadhi, tailored Ayurvedic formulations for individual biological channels.	Lecture	1	Knows how
Enlist and discuss Rasoushadhi for different Srotas, explaining their therapeutic efficacy and mechanism of action.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the clinical application of Rasoushadhi in selected Srotas, illustrating their practical utility and therapeutic outcomes in addressing specific health conditions.	Practical Training	3	Shows how
Illustrate the therapeutic application of Rasoushadhi in different Srotas.	Experiential Learning	3	Shows how

Unit 2 · Clinical application of Rasoushadhi in emergency conditions

4 obj · 9 h · ceiling Shows how

2.1 Role of Rasoushadhis in Managing Medical Emergencies

2.2 Specific Rasa Yogas among Chaturvidha Rasayana kalpas for Managing Medical Emergencies

2.3 Application of Pottali Kalpa in Emergency Management

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the unique properties of Rasoushadhi, enabling its use in clinical emergencies.	Lecture	1	Knows how
Enlist specific Rasa Yogas for managing medical emergencies and discuss their rationality, highlighting the therapeutic mechanisms and clinical applications.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the clinical applicability of Rasoushadhi in Medical Emergencies.	Practical Training	3	Shows how
Apply the appropriate Rasa Oushadhi in a alloted /selected Medical Emergency conditions.	Experiential Learning	4	Shows how

Unit 3 · Rasoushadhi in specific, clinically challenging conditions

5 obj · 11 h · ceiling Does

3.1 Rational application of Rasoushadhi in cancer

3.2 Application of Rasoushadhi in neurodevelopmental conditions

3.3 Utility of Rasoushadhi in neurodegenerative conditions

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss the role of Rasoushadhi in clinically challenging conditions like cancers neurodevelopmental and neurodegenerative disorders.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the role of Rasoushadhi in clinically challenging conditions.	Practical Training	2	Shows how
Demonstrate the role of Rasoushadhi as Rasayana in clinically challenging conditions.	Practical Training	2	Shows how
Evalaute the role of Rasoushadhi in clinically challenging conditions such as Cancer.	Experiential Learning	3	Shows how
DOES — 1			
Explore the role of Rasoushadhi in clinically challenging conditions such as neurodevelopmental and neurodegenerative diseases.	Experiential Learning	3	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 25 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 22 · SEMESTER 5 · PAPER 3

Yogas from different Kalpanas

3 credits | 90 notional hours | 5 units | **27 objectives** | 75 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	16	8	3
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	15 h	15	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	30 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	39 h	6	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 6 hours carry the modular assessment, worth 75 marks.

THE PYRAMID FOR THIS MODULE

Does
3 objectives

Analyse the role of selected leha yoga in different clinical conditions; Illusrate the clinical application of selected Arka/ Dravaka Kalpa; Perform Comparative analysis of Asava Arishta Samples having different percentage and types of sugar.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
8 objectives

Demonstrate the preparation of Vati, Guggulu Kalpa, and their modified modern dosage forms; Design an S.O.P for the preparation of Ayurvedic suppository specifying its clinical utility based on Varti Kalpana; Demonstrate the role of pharmaceutical additives and Samyak Paka of Avaleha; and 5 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
16 objectives

Describe the therapeutic significance of vati kalpana; Summarize the therapeutic significance of Guggulu Kalpa and Varti Kalpana; Discuss modern aspects of solid dosage forms; and 13 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · SOP of Vati, Gutika, Guggulu and Varti Kalpas and their therapeutic applications

6 obj · 20 h · ceiling Shows how

1.1 SOP of selected Vati preparation including packing,
1.2 SOP regarding preparation of selected formulations from Guggulu Kalpana, including packing, storage, and therapeutic applications
1.3 Types of Varti and their SOP
1.4 Clinical applications of different Varti
1.5 Suppositories and pharmaceutical modifications of Ayurvedic suppositories

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Describe the therapeutic significance of vati kalpana.	Lecture	1	Knows how
Summarize the therapeutic significance of Guggulu Kalpa and Varti Kalpana.	Lecture	1	Knows how
Discuss modern aspects of solid dosage forms.	Lecture	1	Knows how
Analyse the role of Gugulu Kalpana in various clinical conditions by selecting a specific Gulggulu Yoga.	Experiential Learning	8	Knows how
SHOWS HOW — 2			
Demonstrate the preparation of Vati, Guggulu Kalpa, and their modified modern dosage forms.	Practical Training	5	Shows how
Design an S.O.P for the preparation of Ayurvedic suppository specifying its clinical utility based on Varti Kalpana.	Practical Training	4	Shows how

Unit 2 · SOP and Clinical applications of Avaleha and Khanda kalpana

5 obj · 16 h · ceiling Does

2.1 SOP of Avaleha preparation with suitable Leha yogas
2.2 Therapeutic application of different varieties of Avaleha yogas
2.3 SOP of Khanda kalpana with suitable Khanda Yogas
2.4 Therapeutic application of different varieties of Khanda yogas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the therapeutic significance of Avaleha kalpana.	Lecture	1	Knows how
Discuss therapeutic significance of Paka and Khanda kalpana.	Lecture	1	Knows how
Discuss modern semisolid dosage forms.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the role of pharmaceutical additives and Samyak Paka of Avaleha.	Practical Training	5	Shows how
DOES — 1			
Analyse the role of selected leha yoga in different clinical conditions.	Experiential Learning	8	Does

Unit 3 · SOP of Sneha Kalpana and its clinical significance

5 obj · 17 h · ceiling Shows how

3.1 Different categories of Sneha with special reference to therapeutic application
3.2 Relevance of Avarthita Taila, Taila murchana and Taila Paka with special reference to critical endpoints
3.3 SOP of Sneha kalpana in selected formulations
3.4 Therapeutic application of selected Ghrita yogas and Taila yogas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the concept of Sneha Kalpana with special reference to critical end points (Sneha Paka) and its pharmaceutical and therapeutical significance.	Lecture	1	Knows how
Discuss and compare the significance of Murchana of Katu Taila, Tila Taila, Eranda Taila and Ghrita with special reference to analytical profile and pharmacotherapeutics.	Lecture	1	Knows how
Identify the similarities of Sneha Kalpa with liposomal drug delivery system.	Lecture	1	Knows how

SHOWS HOW — 2			
Demonstrate S.O.P for a selected Sneha Kalpa which is widely used and determine its analytical profile.	Practical Training	6	Shows how
Analyze different stages of Sneha Paka.	Experiential Learning	8	Shows how

Unit 4 · SOP and clinical applications of Arka and Dravakam in various disorders

5 obj · 14 h · ceiling Does

4.1 SOP for preparing selected Arkas
4.2 Analytical profiling of various yogas from Arka Kalpana
4.3 SOP of preparing various Dravakas
4.4 Analytical Profiling of selected formulations of Dravaka
4.5 Therapeutic applications and method of administration of Dravaka

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Desribe therapeutic significance of Arka Kalpana and Dravaka.	Lecture	1	Knows how
Discuss different distillation techniques used in pharmacy practice.	Lecture	1	Knows how
Elaborate Dravaka Kalpa with a critical correlation with acidic and alkaline preparations.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate S.O.P for the preparation of a selected Arka and Dravaka and create their analytical profile.	Practical Training	5	Shows how
DOES — 1			
Illusrate the clinical application of selected Arka/Dravaka Kalpa.	Experiential Learning	6	Does

Unit 5 · SOP and therapeutic applications of Sandhana Kalpana

6 obj · 17 h · ceiling Does

5.1 Different and novel fermentation techniques
5.2 SOP of preparing different types of Formulations in Sandhana Kalpana
5.3 Therapeutic applications of selected yogas from Sandhana Kalpana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe chemistry behind classical fermentation techniques.	Lecture	1	Knows how
Discuss industrial fermentation techniques.	Lecture	1	Knows how
Elaborate therapeutic significance of Sandhana kalpana.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate S.O.P for preparing a selected sandhana Yoga and determine the analytical standards.	Practical Training	5	Shows how
Record the dose, duration, indications, and clinical outcome of the various formulation i.e.Sandhana Kalpanas in the prescribed patients.	Experiential Learning	4	Shows how
DOES — 1			
Perform Comparative analysis of Asava Arishta Samples having different percentage and types of sugar.	Experiential Learning	5	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **11 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 75 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 29 · SEMESTER 5 · PAPER 4

Pharmacy Practice

2 credits | 60 notional hours | 3 units | **18 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	9	6	2
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Appraise about Basic Health Screening at community setup; Appraise and. Develop Standard Therapeutic Guidelines.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate Drug Information and Case Analysis; Demonstrate Prescription Handling and Dispensing; Identify Adverse drug reaction in selected case; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Describe Community Pharmacy; Discuss the Organization and structure of retail and wholesale drug stores; Describe the Drug distribution system in a hospital; and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Define and classify Hospital Pharmacy, and discuss its organization.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Hospital Pharmacy and Community Pharmacy

7 obj · 26 h · ceiling Does

1.1 Hospital and its organization
1.2 Definition, Classification of Hospital
1.3 Hospital pharmacy and its organization
1.4 Functions of hospital pharmacy
1.5 Community Pharmacy
1.6 Organization and structure of retail and wholesale drug stores

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Define and classify Hospital Pharmacy, and discuss its organization.	Lecture	1	Knows
KNOWS HOW — 2			
Describe Community Pharmacy.	Lecture	1	Knows how
Discuss the Organization and structure of retail and wholesale drug stores.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate Drug Information and Case Analysis.	Practical Training	5	Shows how
Demonstrate Prescription Handling and Dispensing.	Practical Training	5	Shows how
Identify Adverse drug reaction in selected case.	Experiential Learning	7	Shows how
DOES — 1			
Appraise about Basic Health Screening at community setup.	Experiential Learning	6	Does

Unit 2 · Therapeutic Drug Monitoring

7 obj · 17 h · ceiling Shows how

2.1 Drug distribution system in a hospital
2.2 Dispensing of drugs to inpatients, types of drug distribution systems, charging policy and labelling, dispensing of drugs to ambulatory patients, and dispensing of
2.3 Hospital formulary
2.4 Therapeutic drug monitoring
2.5 Factors to be considered during the Therapeutic Drug Monitoring
2.6 Medication adherence
2.7 Patient medication history interview
2.8 Community pharmacy management
2.9 Financial, materials, staff, and infrastructure requirements

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 5			
Describe the Drug distribution system in a hospital.	Lecture	1	Knows how
Discuss Dispensing of drugs to inpatients, types of drug distribution systems, charging policy, and labelling, dispensing of drugs to ambulatory patients, and dispensing of controlled drugs.	Lecture	1	Knows how
Discuss about Hospital formulary and Therapeutic drug monitoring.	Lecture	1	Knows how
Evaluate Medication Adherence.	Lecture	1	Knows how
Analyse Financial, materials, staff, and infrastructure requirements.	Lecture	1	Knows how
SHOWS HOW — 2			
Perform Data Interpretation & Pharmacokinetic Calculations.	Practical Training	5	Shows how
Discuss Multidisciplinary Integration for Therapeutic Drug Monitoring.	Experiential Learning	7	Shows how

Unit 3 · Pharmacy and therapeutic committee

4 obj · 13 h · ceiling Does

3.1 Introduction to Pharmacy and Therapeutic Committee 3.2 Drug information services

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe Pharmacy and Therapeutic Committee along with Drug Information Services.	Lecture	1	Knows how
Enlist emergency drugs used in pharmacy.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate Formulary Review and Management.	Practical Training	5	Shows how
DOES — 1			
Appraise and. Develop Standard Therapeutic Guidelines.	Experiential Learning	6	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 30 · SEMESTER 5 · PAPER 4

Pharmaceutical Microbiology

2 credits | 60 notional hours | 3 units | **18 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	9	5	3
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
3 objectives

Perform Simulation for Aseptic Practices; Conduct Real-life case studies on spoilage outbreaks and preservative failures; Observe and detect Spoilage of Pharmaceutical Product in Real Time.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
5 objectives

Demonstrate dry heat sterilization method; Conduct dry heat sterilization; Demonstrate the Aseptic Technique Practice for the transfer of microbes; and 2 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
9 objectives

Discuss the beneficial and harmful roles of microbes in health, industry, and the environment; Describe the concept of sterilization, its importance, and differentiate various sterilization methods; Define contamination and explain its significance in aseptic and cleanroom environments; and 6 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Define and classify microorganisms (microbes) and explain their general characteristics.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Introduction to microbes with applications of sterilization

5 obj · 12 h · ceiling Shows how

1.1 Identification of bacteria using staining techniques (simple, Gram's & Acid-fast

1.2 Study of principles, procedures, merits, demerits, and applications of physical, chemical, gaseous, radiation, and mechanical methods of sterilization

1.3 Evaluation of the efficiency of sterilization methods

1.4 Equipment employed in large-scale sterilization and Sterility indicators

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Define and classify microorganisms (microbes) and explain their general characteristics.	Lecture	1	Knows
KNOWS HOW — 2			
Discuss the beneficial and harmful roles of microbes in health, industry, and the environment.	Lecture	1	Knows how
Describe the concept of sterilization, its importance, and differentiate various sterilization methods.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate dry heat sterilization method.	Practical Training	3	Shows how
Conduct dry heat sterilization.	Experiential Learning	6	Shows how

Unit 2 · Sources of contamination

6 obj · 12 h · ceiling Does

2.1 Designing an aseptic area

2.2 Study of different sources of contamination in an aseptic area and methods of prevention, clean area classification

2.3 Principles and methods of different microbiological assay methods

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Define contamination and explain its significance in aseptic and cleanroom environments.	Lecture	1	Knows how
Recognise common sources of contamination in pharmaceutical and microbiological settings and discuss each source contributes to microbial and particulate contamination.	Lecture	1	Knows how
Discuss the impact of contamination and describe methods used to minimize or prevent contamination.	Lecture	1	Knows how
Discuss the principles and methods of different microbiological assay methods.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the Aseptic Technique Practice for the transfer of microbes.	Practical Training	2	Shows how
DOES — 1			
Perform Simulation for Aseptic Practices.	Experiential Learning	6	Does

Unit 3 · Types of spoilage and use of antimicrobial agents

7 obj · 32 h · ceiling Does

3.1 Types of spoilage, factors affecting the microbial spoilage of pharmaceutical products,

3.2 Preservation of pharmaceutical products using antimicrobial agents, evaluation of microbial stability of formulations

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Define spoilage, explain its significance in pharmaceuticals, food, and cosmetics, and discuss the factors that contribute to spoilage.	Lecture	1	Knows how
Describe the signs of spoilage.	Lecture	1	Knows how
Classify different types of antimicrobial agents (e.g., preservatives, disinfectants antibiotics) based on their mode of action and explain their antimicrobial mechanisms.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate Personnel Contamination Assessment.	Practical Training	7	Shows how
Demonstrate Different Types of Spoilage.	Practical Training	8	Shows how
DOES — 2			
Conduct Real-life case studies on spoilage outbreaks and preservative failures.	Experiential Learning	7	Does
Observe and detect Spoilage of Pharmaceutical Product in Real Time.	Experiential Learning	7	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 7 · SEMESTER 6 · PAPER 1

Sudha Varga, Sikata Varga, Visha Varga Dravyas

3 credits | 90 notional hours | 3 units | **22 objectives** | 75 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	10	10	2
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	15 h	10	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	30 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	39 h	6	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 6 hours carry the modular assessment, worth 75 marks.

THE PYRAMID FOR THIS MODULE

Does
2 objectives

Perform Shodhana of Visha Dravyas as per different classics; Perform analytical testing process for Shodhita Visha dravya.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
10 objectives

Demonstrate Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya; Demonstrate the quality testing procedures for Bhasma/Pishti of Sudha vargeeya Dravya; Perform Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya; and 7 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
10 objectives

Discuss the list of Sudha varga Dravya mentioned in various classics and their mineralogical correlation; Discuss the pharmacological and therapeutic applications of Sudha Varga Dravya; Discuss research updates on characterization, pharmaceutical processing, QC, and pharmacotherapeutics of Sudhavarga Dravya; and 7 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Characterization, Pharmaceutical Processing, and Clinical Applications of Sudha Varga Dravya

7 obj · 28 h · ceiling Shows how

1.1 Classification and Identification of Sudha
1.2 Pharmaceutical Processing of Sudha Varga Dravya
1.3 Quality Control Methods for Sudha Varga Dravya
1.4 Pharmacology, Therapeutic Applications, and Research Updates on Sudha Varga Dravya

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the list of Sudha varga Dravya mentioned in various classics and their mineralogical correlation.	Lecture	1	Knows how
Discuss the pharmacological and therapeutic applications of Sudha Varga Dravya.	Lecture	2	Knows how
Discuss research updates on characterization, pharmaceutical processing, QC, and pharmacotherapeutics of Sudhavarga Dravya.	Lecture	2	Knows how
SHOWS HOW — 4			
Demonstrate Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya.	Practical Training	6	Shows how
Demonstrate the quality testing procedures for Bhasma/Pishti of Sudha vargeeya Dravya.	Practical Training	4	Shows how
Perform Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya.	Experiential Learning	7	Shows how
Perform quality testing procedures for Bhasma/Pishti of Sudha Vargeeya Dravya.	Experiential Learning	6	Shows how

Unit 2 · Characterization, pharmaceutical processing, and clinical applications of Sikata Varga Dravya

7 obj · 28 h · ceiling Shows how

2.1 Identification of Sikata Varga dravyas
2.2 Shodhana, marana, and other pharmaceutical processes of Sikata varga Dravya as per the SOP laid down
2.3 Quality control methods for the drugs of Sikata Varga Dravya
2.4 Pharmacology, therapeutic application, and research updates on Sikata Varga Dravya)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the list of Sikata varga Dravya mentioned in various classics and their mineralogical correlation.	Lecture	1	Knows how
Describe the pharmacology and therapeutic applications of Sikata Varga Dravya.	Lecture	2	Knows how
Discuss research updates on characterization, pharmaceutical processing, QC, and pharmacotherapeutics of Sikata Varga Dravya.	Lecture	2	Knows how
SHOWS HOW — 4			
Demonstrate Shodhana, marana, and Pishti Nirmana of Sikata varga Dravya.	Practical Training	6	Shows how
Demonstrate quality testing procedures for Bhasma/Pishti of Sikata vargeeya Dravya.	Practical Training	4	Shows how
Perform Shodhana, marana and Pishti Nirmana of Sikata varga Dravya.	Experiential Learning	7	Shows how
Perform quality testing procedures for Bhasma/Pishti of Sikata vargeeya Dravya.	Experiential Learning	6	Shows how

Unit 3 · Pharmaceutical processing, phytopharmacology and clinical applications of Visha Varga dravya

8 obj · 28 h · ceiling Does

3.1 Drugs of Visha and Upavisha categories
3.2 Shodhana of Visha-Upavisha Dravya based on different references
3.3 Phytochemical changes after the process of Shodhana
3.4 Clinical application and therapeutics of Visha Varga

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 4			
Discuss and compare the drugs of the Visha and Upavisha categories from different classics.	Lecture	1	Knows how
Describe the identification of herbal, animal, and mineral drugs mentioned under the Visha-Upavisha category based on organoleptic and physico-chemical characters.	Lecture	1	Knows how
Discuss the phytopharmacology of herbal drugs mentioned under the Visha-Upavisha category.	Lecture	2	Knows how
Describe formulations and therapeutic applications of Visha Dravya.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the Shodhana Process of Visha Dravya.	Practical Training	6	Shows how
Demonstrate the analytical testing process for Shodhita Visha dravya.	Practical Training	4	Shows how
DOES — 2			
Perform Shodhana of Visha Dravyas as per different classics.	Experiential Learning	7	Does
Perform analytical testing process for Shodhita Visha dravya.	Experiential Learning	6	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **12 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 75 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 8 · SEMESTER 6 · PAPER 1

Appraisal of Ratna and Uparatna and therapeutic significance of Pishti and Druti Kalpana

1 credits | 30 notional hours | 2 units | **13 objectives** | 25 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	5	7	1
not used	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	5 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	10 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	13 h	4	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 2 hours carry the modular assessment, worth 25 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Identify substitutes for Ratnavarga and Uparatna varga drugs.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate mineralogical identification of precious and semiprecious stones under Ratna and Uparatna varga; Demonstrate SOP of Pishti Kalpana of Ratna-Uparatna; Identify the precious and semiprecious stones under Ratna and Uparatna varga using keys for mineralogical identification; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Discuss the characteristics of precious stones with special reference to the mineralogical identification of Ratnas; Describe different semiprecious stones mentioned in Rasashastra texts under the category of Uparatna; Discuss crystal chemistry, pharmacology and therapeutic applications of Ratna and Uparatnas; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Mineralogy, Crystal chemistry,and Pharmacotherapeutics of Ratna-Uparatna Varga

7 obj · 15 h · ceiling Does

1.1 Precious stones
1.2 Uparatna and semiprecious stones
1.3 Crystal chemistry and mineralogical properties of Ratna-Uparatna
1.4 Pharmacology and therapeutic applications of Ratna and Uparatnas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the characteristics of precious stones with special reference to the mineralogical identification of Ratnas.	Lecture	1	Knows how
Describe different semiprecious stones mentioned in Rasashastra texts under the category of Uparatna.	Lecture	1	Knows how
Discuss crystal chemistry, pharmacology and therapeutic applications of Ratna and Uparatnas.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate mineralogical identification of precious and semiprecious stones under Ratna and Uparatna varga.	Practical Training	3	Shows how
Demonstrate SOP of Pishti Kalpana of Ratna-Uparatna.	Practical Training	3	Shows how
Identify the precious and semiprecious stones under Ratna and Uparatna varga using keys for mineralogical identification.	Experiential Learning	3	Shows how
DOES — 1			
Identify substitutes for Ratnavarga and Uparatna varga drugs.	Experiential Learning	3	Does

Unit 2 · SOP and therapeutic applications of Pishti, Gandhaka Druti and Gandhaka Taila kalpana

6 obj · 13 h · ceiling Shows how

2.1 Pishti Kalpana: Concept, SOP, and Therapeutic
2.2 Classical concept of Druti
2.2 Gandhaka Druti and Gandhaka Taila Kalpana: Concept, SOP, and Therapeutic Applications
2.3 Pharmaceutical Modifications of Gandhaka Druti and Gandhaka Taila

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the concept, significance, and therapeutic value of Pishti kalpas.	Lecture	1	Knows how
Discuss the concept of Druti mentioned in Rasa classics and explore the therapeutic value of Gandhaka Druti and Gandhaka Taila.	Lecture	1	Knows how
SHOWS HOW — 4			
Demonstrate different methods of Gandhaka Druti.	Practical Training	2	Shows how
Demonstrate Pharmaceutical modification of Gandhaka Druti and Gandhaka Taila.	Practical Training	2	Shows how
Prepare Pishti of Selected Ratna or Uparatna.	Experiential Learning	3	Shows how
Prepare Gandhaka Druti using different media.	Experiential Learning	4	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 25 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 15 · SEMESTER 6 · PAPER 2

Disease condition specific selection of formulation

2 credits | 60 notional hours | 5 units | **12 objectives** | 50 formative marks | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	6	6	—
not used	certified here	certified here	not used

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	6	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	3	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
not used here

This module does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate the essential formulations needed for the diverse branches of Ayurveda; Identify different dosage forms used by local health practitioners. through filed based survey; Analyze the key characteristics of dosage forms that meet the needs of various routes of drug administration; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
6 objectives

Appraise the rational relationship between Roga and Bhaishajya Kalpana based on classical references; Analyse and justify the selection of formulation according to the Rogavastha (stage of disease); Describe different branches / specialty practices in Ayurveda and justify the need for specific dosage forms for effective clinical outcomes; and 3 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this module cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Application of Yukti Pramana in Formulation Selection

2 obj · 2 h · ceiling Knows how

1.1 Disease Category and Determination of multiple factors involved in diagnosis and
1.2 Strategic selection of drugs and dosage forms
1.3 Stage-specific selection (Rogavastha and Bhaishajya Kalpana)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Appraise the rational relationship between Roga and Bhaishajya Kalpana based on classical references.	Lecture	1	Knows how
Analyse and justify the selection of formulation according to the Rogavastha (stage of disease).	Lecture	1	Knows how

Unit 2 · Selection of formulations for Speciality practice in Ayurveda

3 obj · 12 h · ceiling Shows how

2.1 Speciality practice in Ayurveda
2.2 Alignment of formulations with the specialty branch of clinical Practice
2.3 Drugs with ideal guna, virya, vipaka, and of appropriate dosage forms for the patient and pathology
2.4 Evidence-based drugs and dosage form selection for specialty practice

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe different branches /specialty practices in Ayurveda and justify the need for specific dosage forms for effective clinical outcomes.	Lecture	2	Knows how
SHOWS HOW — 2			
Demonstrate the essential formulations needed for the diverse branches of Ayurveda.	Practical Training	5	Shows how
Identify different dosage forms used by local health practitioners. through filed based survey.	Practical Training	5	Shows how

Unit 3 · Routes of Drug Administration

2 obj · 12 h · ceiling Shows how

3.1 Classical and conventional concepts of Drug Administration
3.2 Routes of Drug Administration – Need and Relevance
3.3 Factors in the selection of drug and dosage form based on route of administration
3.4 Matching Dosage Forms to Pharmacological Needs by Selected Route
3.5 Properties of formulations and their effect on Pharmacodynamics

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Analyze the formulations and routes of drug administration, and justify the relevance of each route in pharmacology.	Lecture	2	Knows how
SHOWS HOW — 1			
Analyze the key characteristics of dosage forms that meet the needs of various routes of drug administration.	Experiential Learning	10	Shows how

Unit 4 · Designing formulations for Shodhana and Shamana Chikitsa

2 obj · 12 h · ceiling Shows how

4.1 Ideal characteristics of drugs as per classics (Matravat Bheshaja lakshanani)
4.2 Principles of Shodhana chikitsa (Purificatory/detoxifying therapy) and actions of formulations used for shodhana - Langhana, Doshanirhana, srotahshodhana
4.3 Principles of Shamana chikitsa (palliative treatment) and actions of formulations used for Shamana- Deepana, Pachana, Stambhana, Rasayana, etc
4.4 Evidence-based dosage forms and formulations for common diseases

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Compare dosage forms and formulations used for shodhana and shamana chikitsa.	Lecture	2	Knows how

SHOWS HOW — 1			
Demonstrate the method of prepartion of frequently used formulation for Shodhana and Shamana Chikitsa of common diseases.	Practical Training	10	Shows how

Unit 5 · Innovation in Ayurvedic Formulations

3 obj · 18 h · ceiling Shows how

5.1 Need and current trends in the industry of Ayurvedic Pharmaceutics

5.2 Innovative Formulations-Transdermal patches, Nebulizers, Eye masks, etc

5.3 Relevance and consequences of innovation done by the R&D department of Ayurvedic Pharmacies

5.4 Knowledge, Attitude, and Practice (KAP) studies on various aspects related to Ayurvedic Formulations available in the market

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss challenges associated with recent modifications and innovations in Ayurvedic formulations, specifically focusing on transdermal patches and nebulizers.	Lecture	2	Knows how
SHOWS HOW — 2			
Demonstrate the method of nnovative formulations that enhance well-being and redefine healthcare.	Experiential Learning	10	Shows how
Demonstrate the essential competencies needed to develop R&D department of Ayurvedic Pharmacy.	Experiential Learning	6	Shows how

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **6 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Shows how.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 16 · SEMESTER 6 · PAPER 2

Current trends in Ayurvedic Pharmaceutical Industry.

2 credits | 60 notional hours | 3 units | **12 objectives** | 50 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	6	4	1
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	10 h	6	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	20 h	3	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	26 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 4 hours carry the modular assessment, worth 50 marks.

THE PYRAMID FOR THIS MODULE

Does
1 objective

Assess real-life experience of the market for the growing popularity of Ayurvedic products.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
4 objectives

Analyze the Market Status of Ayurvedic Pharmacy Industry; Interpret benefits of field visit and Internship; Demonstrate and assist quality control practical to display quality standards of Ayurvedic formulations; and 1 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
6 objectives

Discuss key competencies required for employment in the Ayurvedic pharmacy; Describe career opportunities in Ayurvedic pharma industries; Discuss key competencies required for employment in the Ayurvedic pharmacy; and 3 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Discuss current trends in Ayurvedic pharma industry.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Key Competencies Required for Employment in Pharmacy

6 obj · 22 h · ceiling Does

1.1 Orientation about current trends in the Ayurvedic Pharmaceutical Industry
1.2 Ayurvedic Pharmacy Industry (India & Global Overview): Market size, growth, and key importing countries
1.3 Ayurvedic Products and Market Stakeholders
1.4 Consumer Preferences and Distribution Channels

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss current trends in Ayurvedic pharma industry.	Lecture	1	Knows
KNOWS HOW — 3			
Discuss key competencies required for employment in the Ayurvedic pharmacy.	Lecture	1	Knows how
Describe career opportunities in Ayurvedic pharma industries.	Lecture	1	Knows how
Discuss key competencies required for employment in the Ayurvedic pharmacy.	Lecture	1	Knows how
SHOWS HOW — 1			
Analyze the Market Status of Ayurvedic Pharmacy Industry.	Practical Training	8	Shows how
DOES — 1			
Assess real-life experience of the market for the growing popularity of Ayurvedic products.	Experiential Learning	10	Does

Unit 2 · Internship/ Visits to GMP certified Pharmacies

3 obj · 16 h · ceiling Shows how

2.1 Factory set up and daily Work flow
2.2 Hygiene checks & Cleaning
2.3 Inventory Management, material management and Production
2.4 Packaging and labelling
2.5 Documentation (batch records) and QC record

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the sections and working flow of GMP-certified pharmacy.	Lecture	2	Knows how
Observe and appraise the operational management of Ayurvedic Pharmacy.	Practical Training	6	Knows how
SHOWS HOW — 1			
Interpret benefits of field visit and Internship.	Experiential Learning	8	Shows how

Unit 3 · Quality Assurance and Quality Control (QAQC) Laboratory Training

3 obj · 18 h · ceiling Shows how

3.1 Information about Leading Ayurvedic Research Labs in India
3.2 Regulatory & Certification Bodies for QAQC of Ayurvedic Formulations
3.3 Laboratory set up and daily Work flow
3.4 LIMS (Laboratory Information Management System)
3.5 Instrument Calibration, Sample testing and analysis, Record keeping

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe modern parameters used for quality assurance and quality control of Ayurvedic Drugs.	Lecture	4	Knows how
SHOWS HOW — 2			
Demonstrate and assist quality control practical to display quality standards of Ayurvedic formulations.	Practical Training	6	Shows how
Justify the utility of LIMS(Laboratory Information Management System) in Ayurvedic research laboratories and the workflow of the laboratory.	Experiential Learning	8	Shows how

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **5 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 50 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 23 · SEMESTER 6 · PAPER 3

SOP and clinical applications of Churna Kalpana(Powder dosage forms)

3 credits | 90 notional hours | 5 units | **26 objectives** | 75 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
1	14	7	4
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	15 h	15	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	30 h	6	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	39 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 6 hours carry the modular assessment, worth 75 marks.

THE PYRAMID FOR THIS MODULE

Does
4 objectives

Evaluate the role of a selected Churna in a specific clinical condition; Demonstrate the preparation of a selected Paneeya Kshara and Pratisaraneeya Kshara and their quality analysis; Demonstrate the preparation of a selected Lavana formulation following different methods and analytical profiling; and 1 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
7 objectives

Demonstrate the SOP and quality analysis of a selected Churna yoga; Demonstrate the therapeutic applications of Kshara and Ksharasutra in a given selected; Demonstrate the role of Lavana Kalpana in various clinical conditions; and 4 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
14 objectives

Discuss the importance of particle size of different types of Churna concerning its pharmaceutical and therapeutic significance; Describe the pharmacological and therapeutic significance of Churna kalpana; Discuss Kshara Kalpana, its sources, and evaluate the chemical composition; and 11 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
1 objective

Discuss SOP for the preparation of Masi by following different classical and advanced methods IS.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · SOP and clinical utility of Churna Kalpana

4 obj · 15 h · ceiling Does

1.1 Standardisation of various Churna yogas with Standard Operating Procedure
1.2 Therapeutic application of various Churna yogas
1.3 Relevance of Churna kalpana in therapeutics

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the importance of particle size of different types of Churna concerning its pharmaceutical and therapeutic significance.	Lecture	1	Knows how
Describe the pharmacological and therapeutic significance of Churna kalpana.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the SOP and quality analysis of a selected Churna yoga.	Practical Training	5	Shows how
DOES — 1			
Evaluate the role of a selected Churna in a specific clinical condition.	Experiential Learning	8	Does

Unit 2 · Standardization, analytical profiling, and clinical application of Kshara

5 obj · 16 h · ceiling Does

2.1 Relevance of Kshara kalpana and Kshara sutra in therapeutics
2.2 SOP of preparation of Paneeya Kshara and Pratisaraneeya Kshara
2.3 SOP of preparation of Kshara Sutra
2.4 Analytical profiling of Kshara

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss Kshara Kalpana, its sources, and evaluate the chemical composition.	Lecture	1	Knows how
Describe the therapeutic application of Paneeya Kshara and Prathisaraneeya Kshara.	Lecture	1	Knows how
Discuss the techniques for the preparation of Kshara Sutra.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the therapeutic applications of Kshara and Ksharasutra in a given selected.	Experiential Learning	8	Shows how
DOES — 1			
Demonstrate the preparation of a selected Paneeya Kshara and Pratisaraneeya Kshara and their quality analysis.	Practical Training	5	Does

Unit 3 · Lavana Kalpana with special reference to pharmaceutico-analytical aspects and therapeutic attributes

5 obj · 16 h · ceiling Does

3.1 Standard Operating
3.2 Analytical Profiling of Lavanas
3.3 Therapeutic Applications of Selected Formulations in Lavana Kalpana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Describe salt-based formulations (Lavana Kalpa) mentioned in different texts.	Lecture	1	Knows how
Discuss chemical profiling and standardization of Lavana Kalpa.	Lecture	1	Knows how
Illustrate the probable mechanism of action and clinical significance of Lavana Kalpa.	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the role of Lavana Kalpana in various clinical conditions.	Experiential Learning	8	Shows how

DOES — 1			
Demonstrate the preparation of a selected Lavana formulation following different methods and analytical profiling.	Practical Training	5	Does

Unit 4 · Satva Kalpana SOP and clinical significance

5 obj · 15 h · ceiling Shows how

4.1 SOP for preparing selected Satvas
4.2 Analytical profiling of various yogas from Satva Kalpana
4.3 Therapeutic application of Satva kalpana

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 3			
Discuss the uniqueness and pharmaceutical relevance of Satva kalpana.	Lecture	1	Knows how
Discuss SOP and analytical profiling of Satva Kalpana.	Lecture	1	Knows how
Evaluate the clinical significance of Satva Kalpana.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the preparation of a selected Satva and perform its analytical profiling.	Practical Training	5	Shows how
Demonstrate the role of a selected satva in a specified disease condition.	Experiential Learning	7	Shows how

Unit 5 · Therapeutic attributes of Masi Kalpana

7 obj · 22 h · ceiling Does

5.1 Masi from plant and animal origin
5.2 Pharmacology of Masi
5.3 Activated charcoal in therapeutics
5.4 Clinical applications of Masi

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss SOP for the preparation of Masi by following different classical and advanced methods IS.	Lecture	1	Knows
KNOWS HOW — 3			
Discuss the analytical profiling of Mashi Kalpas.	Lecture	1	Knows how
Discuss the therapeutic significance of various Masi Kalpas.	Lecture	1	Knows how
Evaluate the clinical significance of activated charcoal.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate the preparation of Mashi and interpret its analytical profiling.	Practical Training	5	Shows how
Demonstrate pharmaceutical modification of Masi Kalpana.	Practical Training	5	Shows how
DOES — 1			
Evaluate the Role of Malaharas as a Topical application in Skin disease.	Experiential Learning	8	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **11 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 75 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 24 · SEMESTER 6 · PAPER 3

Therapeutic significance of Topical dosage forms

1 credits | 30 notional hours | 3 units | **11 objectives** | 25 formative marks | Ceiling: **Shows how**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
—	5	6	—
not used	certified here	certified here	not used

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	5 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	10 h	3	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	13 h	3	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 2 hours carry the modular assessment, worth 25 marks.

THE PYRAMID FOR THIS MODULE

Does
not used here

This module does not ask for Does. Nothing here needs to be certified at that standard.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
6 objectives

Demonstrate various Netra Kalpa and their clinical applications; Demonstrate the role of Ayurvedic ophthalmic preparations in various eye diseases; Prepare a novel dosage form using one of the classical lepa Yoga; and 3 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
5 objectives

Discuss classical and novel dosage forms for ophthalmic, otic, and buccal use; Describe therapeutic significance and clinical applications of dosage forms for ophthalmic, otic, and buccal use; Discuss classical topical dosage forms like Lepa, Malahara; and 2 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
not used here

This module does not ask for Knows. Nothing here needs to be certified at that standard.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Shows how. The top of this module cannot be certified by a written paper or a viva. It needs to be watched: an OSPE station or an observed performance against a checklist. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Ophthalmic, Otic, and Buccal dosage forms and their clinical applications1.SOP for Ophthalmic, Otic and buccal dosage forms

4 obj · 10 h · ceiling Shows how

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss classical and novel dosage forms for ophthalmic, otic, and buccal use.	Lecture	1	Knows how
Describe therapeutic significance and clinical applications of dosage forms for ophthalmic, otic, and buccal use.	Lecture	1	Knows how
SHOWS HOW — 2			
Demonstrate various Netra Kalpa and their clinical applications.	Practical Training	3	Shows how
Demonstrate the role of Ayurvedic ophthalmic preparations in various eye diseases.	Experiential Learning	5	Shows how

Unit 2 · Standardisation of Lepa and Malahara preparations and its application in various disorders

3 obj · 6 h · ceiling Shows how

2.1 SOP for the preparation of various Lepas
2.2 Innovative approaches including Patches, Bandages, Bands
2.3 Clinical applications of Lepa and Malahara kalpa

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss classical topical dosage forms like Lepa, Malahara.	Lecture	1	Knows how
Describe novel dosage forms for topical application, along with their regulatory guidelines.	Lecture	1	Knows how
SHOWS HOW — 1			
Prepare a novel dosage form using one of the classical lepa Yoga.	Practical Training	4	Shows how

Unit 3 · Standard Operating Procedure (SOP) for Upanaha, Dhupa, Dhumapana, and Avachurnana Kalpa and their therapeutic applications

4 obj · 12 h · ceiling Shows how

3.2 Anti-microbial activity of Dhupa
3.5 Clinical application of various Dhumapana yogas

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Discuss the therapeutic significance of Upanaha, Dhupa, Dhumapana.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the preparation of a selected Upanaha which is useful in treating Shopha.	Practical Training	3	Shows how
Demonstrate the role of lepa Yogas in managing various medical needs by innovative approaches.	Experiential Learning	5	Shows how
Demonstrate the role of Malahara as a Topical application in Skin disease.	Experiential Learning	3	Shows how

FOR THE TEACHER

- **The ceiling is Shows how, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **6 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 25 marks and must reach Shows how.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 31 · SEMESTER 6 · PAPER 4

Regulatory Acts and Rules applicable to Ayurvedic Drugs

3 credits | 90 notional hours | 6 units | **31 objectives** | 75 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	13	8	8
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	15 h	15	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	30 h	7	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	39 h	9	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 6 hours carry the modular assessment, worth 75 marks.

THE PYRAMID FOR THIS MODULE

Does
8 objectives

Analyse case law on the Factory Act; Prepare a technical report on Adverse Drug Reactions (ADRs) for submission to DTAB; Identify misbranded and spurious drugs/cosmetics; and 5 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
8 objectives

Record their observations regarding hypothetical issue through Mock composition exercise of demonstration of the roles of DTAB (Drug Technical Advisory Board) composition, including representatives from DGHS, DCGI, pharmacologists; Prepare mock import license applications; Demonstrate case scenarios of misbranded/spurious drugs and cosmetics; and 5 more.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
13 objectives

Describe the Scope and Applicability of the Drugs and Cosmetics Act and Key Definitions; Discuss the Legal importance of Drug Regulation and enlist prescribed books of ASU from the first Schedule of D&C Act; Identify key regulatory authorities and develop foundational knowledge of regulatory frameworks; and 10 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Summarize the provisions of Chapter III regarding the import of drugs and cosmetics; Discuss Medicinal and Toilet Preparations Act, 1955, Opium Act, 1857, and Pharmacy.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Chapter I of the Drugs and Cosmetics Act, 1940 and Rules 1945

3 obj · 6 h · ceiling Does

1.1 Introduction to 'Drugs and Cosmetic Act, 1940; and 1945' and definitions

1.2 Prescribed Books on Ayurveda, Siddha, and Unani Drugs: The First Schedule

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Describe the Scope and Applicability of the Drugs and Cosmetics Act and Key Definitions.	Lecture	1	Knows how
Discuss the Legal importance of Drug Regulation and enlist prescribed books of ASU from the first Schedule of D&C Act.	Lecture	1	Knows how
DOES — 1			
Analyse case law on the Factory Act.	Experiential Learning	4	Does

Unit 2 · Chapter II - Regulatory Bodies

3 obj · 9 h · ceiling Does

2.1 The Drugs Technical Advisory Board (DTAB)

2.2 The Central Drugs Laboratory (CDL)

2.3 The Drugs Consultative Committee (DCC)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Identify key regulatory authorities and develop foundational knowledge of regulatory frameworks.	Lecture	1	Knows how
SHOWS HOW — 1			
Record their observations regarding hypothetical issue through Mock composition exercise of demonstration of the roles of DTAB (Drug Technical Advisory Board) composition, including representatives from DGHS, DCGI, pharmacologists.	Practical Training	4	Shows how
DOES — 1			
Prepare a technical report on Adverse Drug Reactions (ADRs) for submission to DTAB.	Experiential Learning	4	Does

Unit 3 · Chapters III, IV, and IV-A - Regulation of Drugs and Cosmetics

8 obj · 18 h · ceiling Does

3.1 Chapter III: Standards of quality

3.2 Misbranded, adulterated, and spurious drugs and cosmetics

3.3 Prohibition of import of certain drugs or cosmetics

3.4 Manufacturing, Labelling, and Sale of Drugs and Cosmetics (Chapter IV-A)

3.5 Standards of quality for drugs and cosmetics (Sections 16 and 16A)

3.6 Prohibition of manufacture and sale of certain drugs and cosmetics (Sections 18 and 19)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Summarize the provisions of Chapter III regarding the import of drugs and cosmetics.	Lecture	1	Knows
KNOWS HOW — 4			
Identify legal restrictions and prohibitions for misbranded and spurious drugs and.	Lecture	1	Knows how
Describe manufacturing, labelling, and sale provisions, including penalties and seizures (Chapter IV-A).	Lecture	1	Knows how
Justify the importance of standards of quality for drugs and cosmetics (Sections 16 and.	Lecture	1	Knows how
Discuss the prohibition of the manufacture and sale of certain drugs and cosmetics.	Lecture	1	Knows how

SHOWS HOW — 2			
Prepare mock import license applications.	Practical Training	4	Shows how
Demonstrate case scenarios of misbranded/spurious drugs and cosmetics.	Practical Training	5	Shows how
DOES — 1			
Identify misbranded and spurious drugs/cosmetics.	Experiential Learning	4	Does

Unit 4 · Schedules to Drugs and Cosmetics Rules related to ASU drugs and cosmetics

5 obj · 16 h · ceiling Does

4.1 Good Manufacturing Practices (GMP) for ASU Medicines

4.2 Prescription and Safety Requirements (Schedules H, G, X) and Pharmacological Information (Schedule Y)

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the provision of Schedule T and Schedule E (1).	Lecture	1	Knows how
Discuss the requirements of Schedules H, G, X (prescription and safety), and Schedule Y (pharmacological information).	Lecture	1	Knows how
SHOWS HOW — 1			
Demonstrate the formulation of a proprietary formula and apply for a license.	Practical Training	5	Shows how
DOES — 2			
Apply for GMP certification.	Experiential Learning	4	Does
Prepare a new formula and apply for a mock license.	Experiential Learning	5	Does

Unit 5 · Pharmaceutical observation of drugs and cosmetics ASU drugs

3 obj · 10 h · ceiling Does

5.1 Shelf life and stability of ASU drugs: recent amendments

5.2 Use of Preservatives in ASU drugs

5.3 Packaging and labelling requirements for ASU drugs

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Describe Shelf life of ASU drugs: recent amendments, Preservatives, and. Discuss Packaging and labelling technique.	Lecture	1	Knows how
DOES — 2			
Evaluate label compliance and identify shortcomings in given labels.	Experiential Learning	5	Does
Design and develop a label for a hypothetical ASU product, ensuring compliance with regulatory requirements.	Experiential Learning	4	Does

Unit 6 · Miscellaneous Regulatory Acts related to drugs and cosmetics

9 obj · 25 h · ceiling Does

6.1 Narcotic and Psychotropic Substances Act 1985

6.2 Factories Act, 1948

6.3 Industries Act, 1951

6.4 Biological Diversity Act 2002

6.5 Consumer Protection Act 1986

6.6 Drugs and Magic Remedies Act, 1954

6.7 Medicinal and Toilet Preparations Act, 1955

6.8 Opium Act, 1857

6.9 Pharmacy Act 1948

6.10 Recent Amendments in D & C Act concerning Ayurvedic Drug Industry

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 1			
Discuss Medicinal and Toilet Preparations Act, 1955, Opium Act, 1857, and Pharmacy.	Lecture	1	Knows

KNOWS HOW — 3			
Describe the key provisions of the Narcotic and Psychotropic Substances Act, 1985 Factories Act, 1948, and Industries Act, 1951.	Lecture	1	Knows how
Discuss the Biological Diversity Act 2002, Consumer Protection Act 1986, and Drugs and Magic Remedies Act, 1954.	Lecture	1	Knows how
Describe Recent Amendments in D & C Act concerning Ayurvedic Drug Industry.	Lecture	1	Knows how
SHOWS HOW — 4			
Demonstrate Landmark Judgments under the NDPS Act.	Practical Training	4	Shows how
Demonstrate Case Law on the Factory Act.	Practical Training	4	Shows how
Demonstrate Case Study under the Drugs and Magic Remedies Act.	Practical Training	4	Shows how
Analyse Landmark Judgments under the NDPS Act.	Experiential Learning	5	Shows how
DOES — 1			
Prepare Case Studies and Report related to Objectionable Advertisements.	Experiential Learning	4	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **16 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 75 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

MODULE 32 · SEMESTER 6 · PAPER 4

Entrepreneurship and Start-Ups

1 credits | 30 notional hours | 4 units | **14 objectives** | 25 formative marks | Ceiling: **Does**

KNOWS	KNOWS HOW	SHOWS HOW	DOES
2	4	3	5
certified here	certified here	certified here	certified here

WHERE THE HOURS GO, AND WHAT EACH CAN CERTIFY

KIND OF HOUR	HOURS	OBJECTIVES	THE HIGHEST TIER THESE HOURS CAN HONESTLY CERTIFY
Lecture	5 h	5	**Knows how** at best. Lecture hours build recall and reasoning; they cannot show that a student can perform.
Practical Training	10 h	4	**Shows how.** Performance in a set-up arranged for the purpose, watched and recorded.
Experiential Learning	13 h	5	**Does.** Real work with real consequences — the only hours that can reach the top tier.

A further 2 hours carry the modular assessment, worth 25 marks.

THE PYRAMID FOR THIS MODULE

Does
5 objectives

Create an elevator pitch for a startup idea; Conduct and document market research for a start-up, including survey design and analysis; Carry out A Case Study-Based Role-Play, navigating a Startup's Early Days; and 2 more.

Build: Real work with real consequences, supervised at a distance rather than at the elbow. · **Assess:** Log book, record book, portfolio, observation over time, presentation with peer feedback. · **Evidence:** Dated log entry, countersigned record, portfolio piece.

Shows how
3 objectives

Demonstrate the process of patent filing; Demonstrate a case scenario of the new molecule patent; Demonstrate a case scenario of patenting the formulation process.

Build: Demonstration, then supervised practice with your help withdrawn a little at a time. · **Assess:** OSPE or OSCE station, observed performance against a checklist. · **Evidence:** Checklist score, annotated artefact, station record.

Knows how
4 objectives

Discuss the dynamism of Entrepreneurial Motivation and key entrepreneurial competencies; Demonstrate the process of development of a basic business model as per the template; Identify the challenges and feasibility report of the start-up; and 1 more.

Build: Worked cases, discussion, problem-based work; the student must reason aloud and be corrected. · **Assess:** Structured long answer, case-based questions, structured viva. · **Evidence:** Reasoned written answer, viva score sheet.

Knows
2 objectives

Describe entrepreneurship, the startup life cycle, the MSME Act, and the process of registration; Identify the role of DPIIT (Department for Promotion of Industry and Internal Trade) BHASKAR (Bharat Startup Knowledge Access Registry) in the promotion of entrepreneurship and startups.

Build: Lecture, reading, recitation, retrieval practice. · **Assess:** MCQ, short answer, oral quiz, spotting. · **Evidence:** Written script, quiz sheet.

Ceiling — Does. The top of this module is independent practice. It cannot be certified by any written test, and a viva will not reach it either. It needs a dated record of the student doing the thing unprompted. Do not assess above it either: that examines what was never taught to that standard.

THE UNITS, AND THEIR OBJECTIVES BY TIER

Unit 1 · Entrepreneurship and Startup

3 obj · 4 h · ceiling Does

1.1 Introduction to Entrepreneurship and Start-ups

1.2 MSMED Act and its Process of Registration and Role of Department for Promotion of Industry and Internal Trade(DPIIT) and Bharat Startup Knowledge Access

1.3 Types of Enterprises and Role of Enterprises in National and Global Economy

1.4 Government Policies, Schemes, and Institutional Support for Entrepreneurship

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS — 2			
Describe entrepreneurship, the startup life cycle, the MSME Act, and the process of registration.	Lecture	1	Knows
Identify the role of DPIIT (Department for Promotion of Industry and Internal Trade) BHASKAR (Bharat Startup Knowledge Access Registry) in the promotion of entrepreneurship and startups.	Lecture	1	Knows
DOES — 1			
Create an elevator pitch for a startup idea.	Experiential Learning	2	Does

Unit 2 · Process of entrepreneurship development

3 obj · 8 h · ceiling Does

2.1 Need and the process of entrepreneurship development

2.2 Dynamics of Entrepreneurial Motivation

2.3 Developing Entrepreneurial Competencies

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 2			
Discuss the dynamism of Entrepreneurial Motivation and key entrepreneurial competencies.	Lecture	1	Knows how
Demonstrate the process of development of a basic business model as per the template.	Practical Training	3	Knows how
DOES — 1			
Conduct and document market research for a start-up, including survey design and analysis.	Experiential Learning	4	Does

Unit 3 · Project Proposal for new Invention/startup

2 obj · 3 h · ceiling Does

3.1 Challenges and the feasibility report of the Startups

3.2 Planning, resource mobilisation, and implementation of a start-up project

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Identify the challenges and feasibility report of the start-up.	Lecture	1	Knows how
DOES — 1			
Carry out A Case Study-Based Role-Play, navigating a Startup's Early Days.	Experiential Learning	2	Does

Unit 4 · Intellectual Property Rights and Patents

6 obj · 13 h · ceiling Does

4.1 Introduction to IPR and Patents

4.2 Indian and international patent laws, proposed amendments as applicable to herbal/natural products and processes

4.3 Procedure for patent filing

4.4 Geographical Indication and Copyright

SESSION LEARNING OBJECTIVE	MODE	HRS	TIER
KNOWS HOW — 1			
Define IPR, describe the purpose and importance of IPR, and the laws governing patent.	Lecture	1	Knows how
SHOWS HOW — 3			
Demonstrate the process of patent filing.	Practical Training	3	Shows how
Demonstrate a case scenario of the new molecule patent.	Practical Training	2	Shows how
Demonstrate a case scenario of patenting the formulation process.	Practical Training	2	Shows how

DOES — 2			
Design an application to Patent the formulation process.	Experiential Learning	3	Does
Identify the patentability of a product.	Experiential Learning	2	Does

FOR THE TEACHER

- **The ceiling is Does, so plan the evidence first.** Decide now what you will be holding at the end — a script, a checklist, a dated log entry — because that decision determines the teaching, not the other way round.
- **8 objectives here cannot be certified on paper.** Whatever else the timetable contains, those need to be watched, and the watching has to be recorded at the time.
- **Write the checklist before the session.** Name the steps you will look for and how many attempts a student gets, so the mark reflects the student rather than the day.
- **The modular assessment is worth 25 marks and must reach Does.** Choose the instrument against that, not by rotation: a module whose ceiling is Does needs a record of independent work, and one that stops at Knows how does not.

FOR THE STUDENT

- **Know which tier each objective sits at.** Preparing for a written answer and preparing to be watched are different kinds of work, and only one of them can be done the night before.
- **Reaching "shows how" means doing it unaided.** Being able to follow along while someone guides you is a stage on the way, not the standard.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence, and it is the only evidence that reaches the top tier.

PART C

Reference

Four appendices, generated from the same data as Part B so the two cannot disagree. C1 decodes the tiers and names the instruments that certify each. C2 gives the ceiling of every module in one view — the fastest way to check that an assessment matches what it is meant to certify. C3 lists the instruments the curriculum itself names, sorted by the tier each can honestly reach. C4 records the sources and what was and was not altered.

1. **C1** The four tiers, and what certifies each
2. **C2** Ceiling of every module
3. **C3** The instruments this curriculum names, by tier
4. **C4** Sources, and a note on how this book was made

↑ Contents

PART C · APPENDIX C1

The four tiers, and what certifies each

Miller's four tiers with the count of objectives at each, the instruments that can certify them, and the evidence you keep. Nothing below a tier can certify it.

TIER	OBJECTIVES	INSTRUMENTS	EVIDENCE
Knows	21	MCQ, short answer, oral quiz, spotting.	Written script, quiz sheet.
Knows how	267	Structured long answer, case-based questions, structured viva.	Reasoned written answer, viva score sheet.
Shows how	222	OSPE or OSCE station, observed performance against a checklist.	Checklist score, annotated artefact, station record.
Does	67	Log book, record book, portfolio, observation over time, presentation with peer feedback.	Dated log entry, countersigned record, portfolio piece.

TIER	OBJECTIVES	SHARE
Knows	21	3.6%
Knows how	267	46.3%
Shows how	222	38.5%
Does	67	11.6%

Total 577 objectives.

PART C · APPENDIX C2

Ceiling of every module

The highest tier each module reaches. Read across: the ceiling names the evidence you must gather, and the tier above it names what you may not ask for.

M	U	UNIT	OBJ	CEILING
SEMESTER 3				
MODULE 1 — CHRONOLOGICAL DEVELOPMENT AND TAXONOMY OF RASA DRAVYAS — 19 OBJECTIVES, CEILING DOES				
1	1	Chronological Development and Overview of Rasashastra	5	Shows how
1	2	Taxonomy of Rasadravyas	5	Shows how
1	3	Geological/mineralogical identification of Rasadravyas	6	Does
1	4	Overview of Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya	3	Shows how
MODULE 2 — PUTA AND SAMSKARA (TRANSFORMATIVE PROCESSES OF DRUG MANUFACTURING) — 21 OBJECTIVES, CEILING DOES				
2	1	Puta and Law of thermodynamics	6	Does
2	2	Concept, understanding, application, SOPs and chemical changes of Jarana, Amritikarana and Lohitikarana	4	Shows how
2	3	Concept, understanding, application, SOPs (Standard Operating Procedures) and chemical changes of Satva and Druti procedures	4	Shows how
2	4	Samskara (Principles of Transformative processes of drug manufacturing)	2	Shows how
2	5	Influence of Desha (place) and Kala(time and seasonal factors) of drug qualities	2	Shows how
2	6	Kalaprakarsha- Processing with/Effect of specific time exposure and Bhajana (processing vessel)	3	Shows how
MODULE 9 — SCIENTIFIC FUNDAMENTALS OF AYURVEDIC PHARMACEUTICAL PREPARATIONS — 15 OBJECTIVES, CEILING DOES				
9	1	Classical Principles of drug processing and scientific basis of Panchavidha Kashaya Kalpana	3	Does
9	2	Identity, Purity and Quality of Raw materials used in Ayurveda Pharmaceutics	3	Shows how
9	3	Pharmaceutical processing of drugs with its rationality	3	Shows how
9	4	Ayurvedic Classical Dosage Forms and their integration with modern scientific validation	6	Does
MODULE 10 — PRINCIPLES OF PHARMACEUTICAL FORMULATION — 12 OBJECTIVES, CEILING DOES				
10	1	Principles of Samskara (transformative processes of drug manufacturing) in relation to drug development	2	Shows how
10	2	Rationality behind Composition of Formulations	2	Shows how
10	3	Principles and designing of various dosage forms	2	Shows how
10	4	Principles of Prakshep dravyas (Pharmaceutical Additives)	3	Shows how
10	5	Principles of Potency and Efficacy enhancement	3	Does
MODULE 17 — GUIDELINES FOR ADMINISTRATION OF RASOUSHADHI IN CLINICAL PRACTICE — 22 OBJECTIVES, CEILING DOES				
17	1	Concept of Rasa chikitsa and significance of Rasoushadhi	4	Shows how
17	2	Concept of Kshetrikarana and its significance prior to administration of Rasoushadhi	4	Shows how
17	3	Pathya -Apathya during Rasayoga Sevana	5	Does

M	U	UNIT	OBJ	CEILING
17	4	Safety and toxicity of Rasoushadhi	5	Does
17	5	Heavy metal toxicity and chelation therapy	4	Does
MODULE 18 — PRAYOGIKA BHAISHAJYA KALPANA(CLINICAL APPLICATION OF PHARMACEUTICAL PRODUCTS) — 24 OBJECTIVES, CEILING DOES				
18	1	Aushadha Sevana Kala (Time of administration of drug)	7	Does
18	2	Concept and scientific basis of Anupana	6	Does
18	3	Matra (Posology)	5	Shows how
18	4	Concept of drug repurposing	6	Shows how
MODULE 25 — ARTIFICIAL INTELLIGENCE IN DRUG DEVELOPMENT AND PHARMACOGENETICS; PHARMACOGENOMICS — 20 OBJECTIVES, CEILING DOES				
25	1	Role of Artificial intelligence in drug development	4	Does
25	2	Types of software used in Artificial intelligence	3	Does
25	3	3D printed drug delivery system	5	Shows how
25	4	Pharmacogenetics and Pharmacogenomics	4	Shows how
25	5	Pharmacogenomics in drug discovery	4	Shows how
MODULE 26 — PHARMACOLOGY — 16 OBJECTIVES, CEILING DOES				
26	1	Introduction to Pharmacology	3	Knows how
26	2	Pharmacokinetics, therapeutic index, and drug interactions	3	Knows how
26	3	Pharmacology of drugs that act on the peripheral nervous system	3	Knows how
26	4	Pharmacology of drugs that act on the central nervous system I	3	Shows how
26	5	Pharmacology of drugs that act on the central nervous system II	4	Does
SEMESTER 4				
MODULE 3 — PARADA (MERCURY) AND ITS PROCESSING: ANCIENT WISDOM AND SCIENTIFIC ADVANCES — 17 OBJECTIVES, CEILING DOES				
3	1	Parada: Ancient and Contemporary Perspectives on Its Shodhana and Underlying Principles	8	Does
3	2	Parada Ashta Samskara and Ashtadasha Samskara	4	Shows how
3	3	Physico-chemical proprties and pharmacology of mercury and mercurial compounds	2	Shows how
3	4	Mercury toxicity, ADR and its management	3	Shows how
MODULE 4 — MAHARASA DRAVYA, ITS PROCESSING AND UTILITY — 22 OBJECTIVES, CEILING DOES				
4	1	Maharasa Dravyas with special reference to Abhraka and Vaikranta	8	Shows how
4	2	Principles and Processing of Makshika and Vimala	3	Shows how
4	3	Principles and Processing of Shilajit and Sasyaka	3	Shows how
4	4	Principles and Processing of Chapala and Kharpara, along with physico-chemical analysis, with research updates of all Maharasa Dravyas	4	Does
4	5	Utility of new metals and minerals	4	Shows how
MODULE 11 — MANA PARIBHASHA (METROLOGY IN AYURVEDIC PHARMACEUTICAL PREPARATION) — 16 OBJECTIVES, CEILING DOES				
11	1	Mana Paribhasha (Classical system of measurement)	4	Does
11	2	Standard Units of Measurement	3	Shows how

M	U	UNIT	OBJ	CEILING
11	3	Volumetric Vs Gravimetric Methods	3	Shows how
11	4	Rationality of Proportion of Ingredients in Ayurvedic formulations	3	Does
11	5	Bulk production of Ayurvedic Formulations	3	Does
MODULE 12 — PROCESS VALIDATION PROTOCOL — 10 OBJECTIVES, CEILING SHOWS HOW				
12	1	Pharmaceutical Preparation process: Classical Dosage forms	6	Shows how
12	2	Pharmaceutical Preparation process: Recent advances of Dosage forms	3	Shows how
12	3	Novel Dosage and drug delivery Forms	1	Shows how
MODULE 19 — SOP AND CLINICAL APPLICATION OF CHATURVIDHA RASAYANA — 25 OBJECTIVES, CEILING DOES				
19	1	Standard Operating Procedure (SOP) and Clinical Application of Khalveeya Rasayana	5	Shows how
19	2	SOP and clinical application of Parpati Rasayana	6	Does
19	3	SOP and clinical application of Pottali Rasayana	7	Does
19	4	SOP and clinical application of Kupipakwa Rasayana	7	Shows how
MODULE 20 — SOP AND CLINICAL APPLICATION OF AYASKRITI, LOUHA, AND MANDURA KALPA — 12 OBJECTIVES, CEILING DOES				
20	1	SOP and clinical application of Ayaskriti	3	Shows how
20	2	SOP and clinical application of Louha kalpa	4	Shows how
20	3	SOP and Clinical application of Mandura Kalpa	5	Does
MODULE 27 — NETWORK PHARMACOLOGY & REVERSE PHARMACOLOGY — 16 OBJECTIVES, CEILING DOES				
27	1	Netwok Pharmacology	8	Does
27	2	Reverse Pharmacology	8	Shows how
MODULE 28 — PHARMACEUTICAL QUALITY ASSURANCE — 19 OBJECTIVES, CEILING DOES				
28	1	Quality Assurance and Quality Management concepts	6	Shows how
28	2	Personnel Responsibilities	3	Shows how
28	3	Containers, Packing materials	4	Does
28	4	Good Laboratory Practices	3	Does
28	5	Document maintenance in pharmaceutical industry	3	Does
SEMESTER 5				
MODULE 5 — CRITICAL APPRAISAL OF UPARASA DRAVYA AND SADHARANA RASA DRAVYA — 24 OBJECTIVES, CEILING DOES				
5	1	Principle and Processing of Uparasa Dravya	12	Shows how
5	2	Principle and Processing of Sadharana Rasa	12	Does
MODULE 6 — CRITICAL APPRAISAL OF DHATU AND UPDHATU — 24 OBJECTIVES, CEILING DOES				
6	1	Principle and Processing of Dhatu / Metals	12	Does
6	2	Principle and Processing of Upadhatu	12	Does
MODULE 13 — KRUTANNA (AYURVEDIC FOOD PREPARATIONS) — 15 OBJECTIVES, CEILING DOES				
13	1	Classification of food substances and Formulations	3	Does
13	2	Customized food for treatment	6	Does
13	3	Ayurvedic Food Industry	3	Does

M	U	UNIT	OBJ	CEILING
13	4	Food processing technology traditional and modern methods	3	Shows how
MODULE 14 — AYURVEDIC FORMULATIONS FOR HOLISTIC WELLBEING, VITALITY AND BODY CARE — 11 OBJECTIVES, CEILING SHOWS HOW				
14	1	Rasayana-Personalized Rejuvenation Formulations	5	Shows how
14	2	Ayurvedic based cosmetic preparation	6	Shows how
MODULE 21 — CLINICAL APPLICATION OF RASOUSHADHI ACCORDING TO SROTAS AND IN SELECTED CHALLENGING CLINICAL CONDITIONS — 13 OBJECTIVES, CEILING DOES				
21	1	Clinical application of Rasoushadhis specific to Srotas	4	Shows how
21	2	Clinical application of Rasoushadhi in emergency conditions	4	Shows how
21	3	Rasoushadhi in specific, clinically challenging conditions	5	Does
MODULE 22 — YOGAS FROM DIFFERENT KALPANAS — 27 OBJECTIVES, CEILING DOES				
22	1	SOP of Vati, Gutika, Guggulu and Varti Kalpas and their therapeutic applications	6	Shows how
22	2	SOP and Clinical applications of Avaleha and Khanda kalpana	5	Does
22	3	SOP of Sneha Kalpana and its clinical significance	5	Shows how
22	4	SOP and clinical applications of Arka and Dravakam in various disorders	5	Does
22	5	SOP and therapeutic applications of Sandhana Kalpana	6	Does
MODULE 29 — PHARMACY PRACTICE — 18 OBJECTIVES, CEILING DOES				
29	1	Hospital Pharmacy and Community Pharmacy	7	Does
29	2	Therapeutic Drug Monitoring	7	Shows how
29	3	Pharmacy and therapeutic committee	4	Does
MODULE 30 — PHARMACEUTICAL MICROBIOLOGY — 18 OBJECTIVES, CEILING DOES				
30	1	Introduction to microbes with applications of sterilization	5	Shows how
30	2	Sources of contamination	6	Does
30	3	Types of spoilage and use of antimicrobial agents	7	Does
SEMESTER 6				
MODULE 7 — SUDHA VARGA, SIKATA VARGA, VISHA VARGA DRAVYAS — 22 OBJECTIVES, CEILING DOES				
7	1	Characterization, Pharmaceutical Processing, and Clinical Applications of Sudha Varga Dravya	7	Shows how
7	2	Characterization, pharmaceutical processing, and clinical applications of Sikata Varga Dravya	7	Shows how
7	3	Pharmaceutical processing, phytopharmacology and clinical applications of Visha Varga dravya	8	Does
MODULE 8 — APPRAISAL OF RATNA AND UPARATNA AND THERAPEUTIC SIGNIFICANCE OF PISHTI AND DRUTI KALPANA — 13 OBJECTIVES, CEILING DOES				
8	1	Mineralogy, Crystal chemistry,and Pharmacotherapeutics of Ratna-Uparatna Varga	7	Does
8	2	SOP and therapeutic applications of Pishti, Gandhaka Druti and Gandhaka Taila kalpana	6	Shows how
MODULE 15 — DISEASE CONDITION SPECIFIC SELECTION OF FORMULATION — 12 OBJECTIVES, CEILING SHOWS HOW				
15	1	Application of Yukti Pramana in Formulation Selection	2	Knows how
15	2	Selection of formulations for Speciality practice in Ayurveda	3	Shows how

M	U	UNIT	OBJ	CEILING
15	3	Routes of Drug Administration	2	Shows how
15	4	Designing formulations for Shodhana and Shamana Chikitsa	2	Shows how
15	5	Innovation in Ayurvedic Formulations	3	Shows how
MODULE 16 — CURRENT TRENDS IN AYURVEDIC PHARMACEUTICAL INDUSTRY. — 12 OBJECTIVES, CEILING DOES				
16	1	Key Competencies Required for Employment in Pharmacy	6	Does
16	2	Internship/ Visits to GMP certified Pharmacies	3	Shows how
16	3	Quality Assurance and Quality Control (QAQC) Laboratory Training	3	Shows how
MODULE 23 — SOP AND CLINICAL APPLICATIONS OF CHURNA KALPANA(POWDER DOSAGE FORMS) — 26 OBJECTIVES, CEILING DOES				
23	1	SOP and clinical utility of Churna Kalpana	4	Does
23	2	Standardization, analytical profiling, and clinical application of Kshara	5	Does
23	3	Lavana Kalpana with special reference to pharmaceutico-analytical aspects and therapeutic attributes	5	Does
23	4	Satva Kalpana SOP and clinical significance	5	Shows how
23	5	Therapeutic attributes of Masi Kalpana	7	Does
MODULE 24 — THERAPEUTIC SIGNIFICANCE OF TOPICAL DOSAGE FORMS — 11 OBJECTIVES, CEILING SHOWS HOW				
24	1	Ophthalmic, Otic, and Buccal dosage forms and their clinical applications1.SOP for Ophthalmic, Otic and buccal dosage forms	4	Shows how
24	2	Standardisation of Lepa and Malahara preparations and its application in various disorders	3	Shows how
24	3	Standard Operating Procedure (SOP) for Upanaha, Dhupa, Dhumapana, and Avachurnana Kalpa and their therapeutic applications	4	Shows how
MODULE 31 — REGULATORY ACTS AND RULES APPLICABLE TO AYURVEDIC DRUGS — 31 OBJECTIVES, CEILING DOES				
31	1	Chapter I of the Drugs and Cosmetics Act, 1940 and Rules 1945	3	Does
31	2	Chapter II - Regulatory Bodies	3	Does
31	3	Chapters III, IV, and IV-A - Regulation of Drugs and Cosmetics	8	Does
31	4	Schedules to Drugs and Cosmetics Rules related to ASU drugs and cosmetics	5	Does
31	5	Pharmaceutical observation of drugs and cosmetics ASU drugs	3	Does
31	6	Miscellaneous Regulatory Acts related to drugs and cosmetics	9	Does
MODULE 32 — ENTREPRENEURSHIP AND START-UPS — 14 OBJECTIVES, CEILING DOES				
32	1	Entrepreneurship and Startup	3	Does
32	2	Process of entrepreneurship development	3	Does
32	3	Project Proposal for new Invention/startup	2	Does
32	4	Intellectual Property Rights and Patents	6	Does

Module ceilings are shown in the coloured band; unit ceilings in the rows beneath. A unit may sit below its module's ceiling, and most do.

PART C · APPENDIX C3

The instruments this curriculum names, by tier

Every assessment abbreviation that appears in the curriculum, sorted by the highest tier it can honestly certify.

HIGHEST TIER IT REACHES	INSTRUMENTS THE CURRICULUM NAMES	WHAT IT CAN AND CANNOT SHOW
Knows	MCQ, Spotting, T-OBT	Recall only. Cannot show reasoning or performance.
Knows how	LAQ, SAQ, Structured viva, ABQ, Case study	Reasoning on paper or aloud. Cannot show that the student can perform.
Shows how	OSPE, Practical examination, Checklist	Performance when watched. The lowest instrument that settles a doing objective.
Does	Log book, Record book, Portfolio, Presentation	Independent practice over time. The only evidence that reaches the top tier.

Counts are how many objectives name each instrument. An instrument may always be used below its tier; it may never be used above it.

PART C · APPENDIX C4

Sources, and a note on how this book was made

The primary sources for the framework, the curriculum this book is built on, and a note on what was and was not altered.

THE FRAMEWORK

- Miller, G. E., 'The assessment of clinical skills / competence / performance', *Academic Medicine*, 65(9), 1990. The original four-tier paper.
- Cruess, R. L., Cruess, S. R. and Steinert, Y., 'Amending Miller's Pyramid to Include Professional Identity Formation', *Academic Medicine*, 91(2), 2016. The proposed fifth tier, *Is*.
- Biggs, J., 'Enhancing teaching through constructive alignment', *Higher Education*, 32, 1996. The source of the alignment principle in Chapter 3.

A NOTE ON HOW THIS BOOK WAS MADE

Every objective in Part B is reproduced in the curriculum's own words. Plain-English spelling errors have been corrected and sentence punctuation lost to the source document's line wrapping has been restored. No Sanskrit or technical term has been altered, and no subject content has been added, removed or reworded.

National Commission for Indian System of Medicine, Board of Ayurveda. *Curriculum for MD/MS Ayurveda, Semesters III to VI — Rasashastra and Bhaishajya Kalpana (AYPG-RB).* New Delhi: NCISM. Tables 2, 3, 4, 5 and 6 are the source of every objective, tier, hour and mark in this book. All 577 objectives and 304 activities reconcile against the curriculum's own credit framework: 1792 teaching hours plus 128 of modular assessment give the 1920 notional hours of 64 credits, and each semester independently totals 480.

The counts and tables in Part C are generated from the same parsed data as Part B, so the two cannot disagree.